

The Biomolecular Design Curriculum

Third Edition — every computational method for proteins, DNA and RNA, the AI research upstream of it, and the engineering that makes it fast

A self-study curriculum

October 4, 2026

The Biomolecular Design Curriculum

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- Flagged as unverifiable

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The Biomolecular Design Curriculum

Third Edition — every computational method for proteins, DNA and RNA, the AI research upstream of it, and the engineering that makes it fast

A self-study curriculum · Compiled October 2026

0.1 What changed, and why

From the first edition to the second

The first edition was a six-month survey, and it was too coarse in three ways.

It pointed at playlists instead of lectures. A link to a 71-video course is not a curriculum; it is a deferral of the actual work of deciding what to watch and in what order. The second edition enumerates individual lectures with titles and runtimes, so that every study session has a specific named target.

It was thin on applications. It named the major design methods but did not treat them individually. Most significant methods now get their own entry with a recorded talk — preferably by a first author — paired with the exact paper that talk narrates.

It was missing entire method families. Coarse-graining, Monte Carlo in all its forms, Brownian and Langevin dynamics, polymer theory, normal modes and elastic networks, continuum electrostatics, docking, QM/MM, machine-learned interatomic potentials, the reconstruction algorithms behind cryo-EM and crystallography, integrative modeling, sequence and evolutionary algorithms. These are now organized as a **Method Atlas** aimed at the whole space rather than the fashionable corner of it.

From the second edition to this one

The second edition was audited against its own claims, and it failed several. This edition is the repair.

It had, in about ten places, done the exact thing it criticized. Jeffrey Gray's course, the PLUMED Masterclass, the forty-three-talk free energy workshop, the geometric deep learning courses — all referenced as important and all left as bare playlist links. They are enumerated now.

Three Atlas families were reading lists rather than video sections. Continuum and mesoscale methods had zero lectures, integrative modeling had four, free energy had six. Those are filled.

RNA was badly under-served despite Rhiju Das being one of the ten scientists the document is built around. The secondary-structure algorithms, chemical probing, ribozymes, riboswitches and RNA design now get proper treatment, along with an evidence-based answer to why RNA structure prediction trails protein structure prediction.

The cross-references were broken. Capstone numbers cited throughout Parts II and III pointed at projects that did not match what Part V defined, and one referenced a capstone that did not exist. All eleven are now named, specified, and reachable from the family that cites them.

Performance engineering was absent entirely — and that was the largest gap. Part IV is new, and it exists because making molecular computation faster is a genuine source of original contribution, not plumbing. Anton

was an ASIC. FlashAttention introduced no new mathematics. ColabFold's speedup came from replacing the sequence search, not the network. The open problems in this area are unusually tractable for one person with a few GPUs.

Two things are now generated rather than written, so they cannot drift out of date: the runtime budget in Section 0.9 and the index in Appendix D, both rebuilt from the assembled text on every build.

On what is deliberately not here. There is no wet-lab or experimental protocol module. The measurement side appears only where it changes how you read a number — what a deep mutational scan actually reports, what an SEC trace means, why a stability dataset is skewed toward destabilizing mutations. That is a scoping decision, not an oversight.

0.2 The central thesis, unchanged

The ten scientists this curriculum is built around — David Baker, Possu Huang, Tanja Kortemme, Phil Bradley, John Jumper, David E. Shaw, Greg Bowman, Pranam Chatterjee, Daniel Zuckerman, Rhiju Das — do not practice a single discipline. They practice three that share a subject matter and disagree, quietly and politely, about what counts as truth.

Culture One — Physics-first. *Shaw, Zuckerman, Bowman; historically Karplus and Levitt.* The ground truth is the Boltzmann distribution. A molecule is a high-dimensional energy landscape and the problem is sampling it correctly. A method is judged on whether it is **statistically correct**: does it converge, is it unbiased, what is the error bar. The cardinal sin is an unconverged result. The weakness is that a perfectly converged simulation of a slightly wrong Hamiltonian is a perfectly converged wrong answer.

Culture Two — Engineering-first. *Baker, Kortemme, Huang, Bradley, and the Rosetta community.* The ground truth is the tube. A method is judged on whether the thing it designs **folds, binds or catalyzes when made**. Energy functions are empirical scoring devices with free parameters fit to whatever makes designs work. Success rate is the metric; the cardinal sin is a beautiful method with no experimental validation. The weakness is that a function fit to successes explains very little about the failures.

Culture Three — Learning-first. *Jumper, Chatterjee, modern Das, and nearly every group founded after 2020.* The ground truth is the data distribution. The PDB is a sample and the problem is to learn it well enough to generalize. A method is judged on **held-out benchmark performance**; the cardinal sin is test set leakage. The weakness is that a model which interpolates the PDB has no defined behavior off the manifold of natural proteins — which is exactly where design lives.

A fourth culture supplies the ground truth all three depend on: **the experimentalists**, whose deep mutational scanning, display technologies, cryo-EM and high-throughput biochemistry turn design into data.

Why this is the organizing principle

Nearly every major methodological advance of the last fifteen years came from someone **carrying an idea across one of these boundaries**.

Advance	Imported from	Into	What the carry accomplished
Markov state models	Learning (clustering, Markov chains)	Physics	Short trajectories became long-timescale kinetics
Replica exchange	Physics (tempering)	Engineering (Rosetta sampling)	Escaped local minima in design search
Rosetta energy refits (REF15)	Physics (explicit solvation, electrostatics)	Engineering	A more transferable empirical function
AlphaFold2 triangle attention	Physics (geometric consistency)	Learning	The right inductive bias for distances
Boltzmann generators	Learning (normalizing flows)	Physics	Equilibrium sampling with no trajectory
ProteinMPNN	Learning (graph neural networks)	Engineering	A learned packer replacing a hand-tuned one
RFdiffusion	Learning (denoising diffusion)	Engineering	Backbone generation became a sampling problem
Deep mutational scanning as supervision	Experiment (high-throughput assays)	Learning	A non-PDB training signal
Cryptic pocket discovery	Physics (enhanced sampling)	Engineering (drug design)	Druggable sites no structure showed
cryoDRGN / 3DFlex	Learning (latent variable models)	Experiment (cryo-EM)	Continuous ensembles from particle stacks
Machine-learned force fields	Learning (equivariant networks)	Physics	Near-QM accuracy at MM cost
AlphaFold-derived ensembles	Learning	Physics	Conformational states without simulation
GFlowNets	Learning (RL/flow networks)	Design	Sampling diverse high-reward sequences

Read that table as a job description. **Your contribution, when it comes, will most likely be an unexchanged idea** — something routine in one of these cultures that nobody has yet carried into another. The Method Atlas in Part II exists to give you the complete inventory of what each culture already knows, because you cannot notice a missing exchange between fields you have only surveyed.

0.3 How this document is organized

Part I — Foundations. The mathematics, physics, computer science, chemistry and experimental grounding. Enumerated lecture by lecture. This is the part that does not go out of date.

Part II — The Method Atlas. The core of this edition. Every family of computational method for proteins, DNA and RNA, organized by what problem it solves rather than by who invented it. Each family gets: the problem statement, theory lectures, practical lectures, canonical papers, the software that implements it, and the open problems in it.

Part III — Upstream AI Research. Vision, architecture research, training at scale, post-training and reinforcement learning, reasoning, deep learning theory, and interpretability — the literature that produces the ideas Part II inherits two years later. It closes with fourteen specific transfers between the two fields, in both directions.

Part IV — Performance Engineering. How molecular computation actually gets fast: GPU and kernel programming, accelerators and compilers, the internals of simulation engines, and the systems engineering of structure-prediction models. This part exists because making something ten times faster is a real contribution and is treated in most curricula as if it were plumbing. It is not. Anton is a hardware-software codesign result; FlashAttention is an algorithm with no new mathematics; ColabFold’s speedup came from replacing the search, not the network. **The open problems here are unusually tractable for one person with a few GPUs.**

Part V — Contribution. Sixty-three derivation checkpoints, ten capstone projects, the five generators by which new methods actually get invented, and the permanent seminar diet.

Part VI — Appendices. The standing archives enumerated talk by talk (Appendix A), the ten named courses enumerated lecture by lecture (B), an honest register of what is missing from the public record (C), the progress tracker (D), and a generated index of every method, model and package in the document (E).

0.4 Conventions

Every entry gives what it can verify and flags what it cannot.

- **Lecture tables** list individual videos: number, exact title, runtime, link. Where a course has natural units, that structure is preserved.
- **Tier tags** indicate how to watch: **A** foundation (work it, with problem sets), **B** method (reimplement it), **C** talk (interrogate it, watch the Q&A), **D** cultural (context and taste, 2x, no notes).
- **Atlas entries** follow a fixed shape: *Problem · Theory · Practice · Papers · Software · Open*.
- **Talk+paper pairs** give the recorded talk and the exact paper it narrates.
- Items that could not be verified are marked **[unverified]** with a search string rather than given a possibly-dead link. Known absences are recorded in Appendix D rather than silently omitted — a gap in the public record is information too.

Runtimes and lecture counts were read from live pages at compile time, not estimated. They drift; the titles are the durable identifier.

0.5 The pairing principle

The organizing device of this curriculum is that **no talk stands alone**. Each significant video is paired with the specific paper it narrates, consumed as a unit, in a fixed order, with a specific question held in mind.

Video and text fail in opposite directions and the failures cancel. A **talk** gives you the author’s own sense of which part was hard — almost never what the paper’s structure suggests — the motivation that was true at the time rather than the one reverse-engineered for publication, and the Q&A, where objections are voiced by people under no obligation to be diplomatic. A **paper** gives you the precise claim with its conditions and error bars, and the methods section, where the thirty undocumented decisions that make it work are hiding.

Watch first, read second, then write the gap.

1. Watch at the tier-appropriate speed. Take the three-line note: claim, evidence, what-would-make-this-wrong.
2. Read the paper properly, methods section included.

3. Write two or three sentences on **what the paper contains that the talk omitted, and what the talk said that the paper does not.**

Step three takes five minutes and is the entire point. That delta is composed of things that are true, known to practitioners, and unpublishable: failure modes, the hyperparameter that matters more than the architecture, the dataset quirk, the reviewer who demanded a figure that obscured the result. After forty of these you will notice the same gaps recurring within a research group, which is what it feels like to understand a culture from the inside.

0.6 How to watch, by tier

Tier A — foundations. Watch at 1x with a notebook. Pause before every derivation and attempt the next step. Afterward, close everything and write the central result from memory. Do the problem sets; a statistical mechanics course without them is entertainment. One lecture per sitting.

Tier B — methods. Watch once at 1.25x for the shape of the idea. Write one paragraph on what problem it solves and what it assumes — if you cannot state the assumption you did not understand it. Implement the smallest possible version from scratch before looking at the real code. Then read the real implementation and catalog every difference; the differences are the accumulated scar tissue of what does not work.

Tier C — talks. Watch at 1.5x. Three lines: claim, evidence, falsifier. **Watch the Q&A, always** — it is where real disagreements surface and the part everyone skips. Log every open problem a speaker names as unsolved.

Tier D — cultural. 2x, no notes. The purpose is knowing who the field is and what it has already tried and abandoned, which is what stops you spending a year reinventing something that failed in 2009 for reasons nobody published.

The three numbers

Monthly, record: **lines added to the open-problems file** (should grow 20–40 a month), **derivation checkpoints completed unaided**, and **things you built that run**. If video hours are high and these are flat, stop watching and start building.

0.7 Suggested orderings

There is no single correct path through a multi-year reference. Four that work:

The physics-deficit path (likeliest fit given your background). Part I mathematics and statistical mechanics at full depth, then Atlas families A through I in order, then Part III. You already run the tools; this makes you able to derive them.

The learning-first path. Part I mathematics and CS, Atlas N (deep learning for molecules), then O, P, Q, then back-fill A through I as each becomes necessary. Faster to the frontier, weaker foundations, and you will feel the gap the first time a reviewer asks about convergence.

The atlas-sweep path. Read every Atlas entry's *Problem* and *Open* sections first — perhaps forty pages — before watching anything. This gives you the map of the whole field in a week and lets you choose where to go deep. **This is the recommended first move regardless of which path you then take.**

The depth-first path. Pick one Atlas family, watch everything in it, read every paper, do every derivation, build the tool. Then move. Slower, but it produces genuine expertise in a sub-area rather than uniform competence — and genuine expertise in one sub-area is what gets you taken seriously.

0.8 Prerequisites, honestly assessed

This assumes undergraduate chemistry and biology, fluency in Python, the ability to read a structural biology paper, and practical experience running molecular dynamics and structure prediction. Based on your project history you have all of this, and much of the practical content in Atlas families B, E and O will be review you can accelerate past.

It does not assume, and therefore teaches: measure-theoretic probability at a working level, statistical mechanics derived rather than recalled, the mathematics of equivariance, the internals of the deep learning methods you currently use as tools, and the algorithmic computer science that makes any of it fast. Those are the gaps between someone who can run RFDiffusion and someone who could have invented it.

0.9 How long this is, honestly

3,271 individual videos. About 3,512 hours of timed runtime, plus the enumerated courses whose lectures are listed without durations. Call it three thousand hours.

That number is not a target. It is the size of the field’s recorded output in the areas this document covers, and stating it plainly is more useful than pretending there is a schedule. At ten hours a week it is roughly six years; at twenty it is three. **Nobody should watch all of it, and the document is not built on the assumption that anyone will.**

Part	Timed hours	Timed items
Part I — Foundations	253	238
Part II — The Method Atlas	1,124	1,252
Part III — Upstream AI Research	948	915
Part IV — Performance Engineering	436	394
Part V — Contribution	58	66
Part VI — Appendices	694	801
Total	3,512	3,666

What to actually do with it

The 150-hour core. If you do nothing else, do these: the matrix-calculus course in B.1, Kardar’s statistical mechanics in B.2, the classical prediction course enumerated in Atlas L.1, the BUILD block of Atlas B and Atlas S, and the paired-reading tables for Atlas P, Q and R. That is about a hundred and fifty hours and it is the difference between running tools and understanding them.

The reference mode. Everything else is a reference. When you hit a method you do not understand, open its family, watch the two or three talks flagged in bold, read the paired paper, and move on. The Atlas is organized by problem precisely so that this works.

The standing diet. One talk a week from Part III, permanently, chosen from the archives in Appendix A. The point is not to keep up. It is to keep the vocabulary live so that when a transfer is available you recognize it.

The thing that actually converts hours into ability is none of the above. It is the sixty-three derivation checkpoints and the nineteen BUILD blocks. A hundred hours of watching with ten checkpoints completed beats a thousand hours of watching with none, and it is not close.

Part I — Foundations

B.0 What this part is for

The distinction this part creates is between someone who can *apply* methods and someone who can *derive* them. Those look similar on a CV and are entirely different in practice. The applier reads a new paper and learns what the method does. The deriver reads the same paper, sees which lemma is doing the work, notices it holds under weaker assumptions than the authors used, and has a new method by Friday.

Everything here is enumerated lecture by lecture with runtimes, so each study session has a specific named target. The counts are real — they were read from live playlists at compile time — and where a course has gaps or mislabeled items, that is noted inline rather than left for you to discover at 11pm.

Six modules:

	Module	What it gives you
B.1	Mathematics	Linear algebra, matrix calculus, probability, stochastic processes, information theory
B.2	Statistical mechanics	Equilibrium and nonequilibrium, Langevin, Fokker-Planck, linear response
B.3	Computer science	Algorithms, combinatorial optimization, graphical models, parallel computing, ML systems
B.4	Geometry and symmetry	Manifolds, Lie groups, representation theory, equivariant architectures
B.5	Chemistry and structural biology	What the molecules actually are, and what a structure actually is
B.6	Deep learning foundations	Backpropagation through transformers, generative modeling, built from scratch

Order matters less than depth. The one hard dependency: **B.4 before any equivariant architecture work**, and **B.2 before Atlas families D, E and F**

B.1 — Mathematics

B.1.1 Linear algebra and matrix calculus

MIT 18.065 — Matrix Methods in Data Analysis, Signal Processing and Machine Learning

Gilbert Strang (MIT) · 36 videos · ~28 hours · Tier A

The applied counterpart to 18.06. If you have had a linear algebra course, start here instead. Lectures 7, 13, 17 and 21–27 are the ones that pay off directly in molecular machine learning.

#	Lecture	Length	Link
—	Course Introduction	7:04	▶
—	Interview with Gilbert Strang on Teaching Matrix Methods	8:07	▶
1	The Column Space of A Contains All Vectors Ax	52:15	▶
2	Multiplying and Factoring Matrices	48:26	▶
3	Orthonormal Columns in Q Give $Q'Q = I$	49:24	▶
4	Eigenvalues and Eigenvectors	48:56	▶
5	Positive Definite and Semidefinite Matrices	45:27	▶
6	Singular Value Decomposition (SVD)	53:34	▶
7	Eckart-Young: The Closest Rank k Matrix to A	47:16	▶
8	Norms of Vectors and Matrices	49:21	▶
9	Four Ways to Solve Least Squares Problems	49:51	▶
10	Survey of Difficulties with $Ax = b$	49:36	▶
11	Minimizing $\ x\ $ Subject to $Ax = b$	50:22	▶
12	Computing Eigenvalues and Singular Values	49:28	▶
13	Randomized Matrix Multiplication	52:24	▶
14	Low Rank Changes in A and Its Inverse	50:34	▶
15	Matrices $A(t)$ Depending on t, Derivative = dA/dt	50:52	▶
16	Derivatives of Inverse and Singular Values	43:08	▶
17	Rapidly Decreasing Singular Values	50:34	▶
18	Counting Parameters in SVD, LU, QR, Saddle Points	49:00	▶
19	Saddle Points Continued, Maxmin Principle	52:13	▶
20	Definitions and Inequalities	55:01	▶
21	Minimizing a Function Step by Step	53:45	▶
22	Gradient Descent: Downhill to a Minimum	52:44	▶

#	Lecture	Length	Link
23	Accelerating Gradient Descent (Use Momentum)	49:02	▶
24	Linear Programming and Two-Person Games	53:34	▶
25	Stochastic Gradient Descent	53:03	▶
26	Structure of Neural Nets for Deep Learning	53:17	▶
27	Backpropagation: Find Partial Derivatives	52:38	▶
30	Completing a Rank-One Matrix, Circulants	49:53	▶
31	Eigenvectors of Circulant Matrices: Fourier Matrix	52:37	▶
32	ImageNet is a CNN; The Convolution Rule	47:19	▶
33	Neural Nets and the Learning Function	56:07	▶
34	Distance Matrices, Procrustes Problem	29:17	▶
35	Finding Clusters in Graphs	34:49	▶
36	Alan Edelman and Julia Language	38:11	▶

Lectures 28 and 29 were never published. The count of 36 is real; the numbering is not consecutive.

Why lectures 7, 13 and 17 matter to you specifically. Eckart-Young is the theorem that says the best low-rank approximation is the truncated SVD — which is the mathematical justification for every attempt to compress the pair representation in a structure predictor. Lecture 17 on rapidly decreasing singular values tells you *when* that compression will work. Given that your AF3-chromatin work is memory-bound on the triangle operations, this is not abstract: it is the theory of whether the thing you want to do is possible.

Lecture 34, on distance matrices and the Procrustes problem, is the direct mathematical ancestor of structural superposition and of the Kabsch algorithm you use every time you compute an RMSD.

MIT 18.S096 — Matrix Calculus for Machine Learning and Beyond

Steven Johnson & Alan Edelman (MIT) · 17 videos · ~14 hours · Tier A

The highest-leverage item in this module and the one almost nobody takes. Matrix calculus is what lets you derive adjoint and reverse-mode gradients through a simulation or an implicit solver — the skill needed to differentiate through molecular dynamics, a Poisson-Boltzmann solve, or a docking inner loop. Differentiable simulation is one of the cleanest open directions in this field and essentially nobody entering it from biology has the calculus to attempt it.

#	Lecture	Length	Link
1.1	Introduction and Motivation	57:42	▶
1.2	Derivatives as Linear Operators	48:28	▶
2.1	Derivatives in Higher Dimensions: Jacobians and Matrix Functions	1:13:57	▶
2.2	Vectorization of Matrix Functions	30:18	▶
3.1	Kronecker Products and Jacobians	53:05	▶
3.2	Finite-Difference Approximations	51:10	▶
4.1	Gradients and Inner Products in Other Vector Spaces	1:03:49	▶
4.2	Nonlinear Root Finding, Optimization, and Adjoint Gradient Methods	44:26	▶
5.1	Derivative of Matrix Determinant and Inverse	28:03	▶
5.2	Forward Automatic Differentiation via Dual Numbers	36:02	▶
5.3	Differentiation on Computational Graphs	32:46	▶
6.1	Adjoint Differentiation of ODE Solutions	58:21	▶
6.2	Calculus of Variations and Gradients of Functionals	42:32	▶
7.1	Derivatives of Random Functions	1:06:18	▶
7.2	Second Derivatives, Bilinear Forms, and Hessian Matrices	46:09	▶
8.1	Derivatives of Eigenproblems	36:37	▶
8.2	Automatic Differentiation on Computational Graphs	1:05:29	▶

Lecture 6.1 is the single most important video in this module. The adjoint method for ODEs is how you differentiate through a trajectory without storing every intermediate state, and it is the mathematical core of neural ODEs, of differentiable MD, and of training anything where the forward pass is a simulation.

Lecture 8.1 on derivatives of eigenproblems is the one to watch if you ever want to optimize a normal-mode spectrum — which is Atlas family I's most obvious unexploited design objective.

Supporting

Course	Instructor	Scale	Link
Essence of Linear Algebra	3Blue1Brown	16 videos, 3 hr	Playlist
MIT 18.06 Linear Algebra	Gilbert Strang	36 videos, 28 hr	Playlist
Numerical Linear Algebra (Trefethen & Bau)	Martijn Anthonissen (TU Eindhoven)	38 videos, 13 hr	Playlist

Do 3Blue1Brown first — three hours, and it builds the geometric intuition for change of basis that makes SE(3) frame algebra readable rather than magical. *Use the MIT OCW playlist for 18.06; the widely circulated alternative is a third-party re-upload.*

B.1.2 Probability, statistics and stochastic processes

Course	Instructor	Scale	Link	
Stat 110: Probability	Joe Blitzstein (Harvard)	35 lectures, 27 hr	Playlist	essential
MIT 6.041SC Probabilistic Systems Analysis	John Tsitsiklis (MIT)	76 videos, 31 hr	Playlist	essential
MIT 18.650 Statistics for Applications	Philippe Rigollet (MIT)	22 videos, 28 hr	Playlist	essential
Statistical Rethinking 2023	Richard McElreath (MPI-EVA)	20 lectures, 26 hr	Playlist · site	essential
MIT 6.262 Discrete Stochastic Processes	Robert Gallager (MIT)	25 lectures, 33 hr	Playlist	essential
Probabilistic Machine Learning	Philipp Hennig (Tübingen)	25 videos, 35 hr	Playlist	essential
Gaussian Process Summer School 2024	Neil Lawrence et al.	8 lectures	Playlist	optional

Gallager 6.262 pays off most unexpectedly. Rigorous renewal and Markov chain theory is where you learn to *prove* ergodicity and detailed balance for a sampler you invented — and inventing samplers is a large fraction of what methodological contribution looks like in both the physics and the learning cultures.

McElreath is the one to watch for taste rather than technique. Generative- model-first thinking and posterior predictive checking is the discipline that makes a new benchmark measure the causal claim you intend rather than a correlate of it.

B.1.3 Information theory

David MacKay — Information Theory, Pattern Recognition and Neural Networks

University of Cambridge · 16 lectures · ~18 hours · Tier A

One of the great lecture series in any subject. MacKay died in 2016; the lectures remain, and they are directly relevant in a way that is easy to miss. Entropy, typicality and the minimum-description-length view give you the

language to argue that a sequence-design model has learned real biophysical constraint rather than dataset bias — which is the central unsettled question of Atlas family Q.

#	Lecture	Length	Link
1	Introduction to Information Theory	1:01:51	▶
2	Entropy and Data Compression I: Introduction to Compression, Inf. Theory and Entropy	51:09	▶
3	Entropy and Data Compression II: Shannon's Source Coding Theorem, The Bent Coin Lottery	51:01	▶
4	Entropy and Data Compression III: Source Coding Theorem, Symbol Codes	56:58	▶
5	Entropy and Data Compression IV: Source Coding Theorem, Symbol Codes	1:02:48	▶
6	Noisy Channel Coding I: Inference and Information Measures for Noisy Channels	54:42	▶
7	Noisy Channel Coding II: The Capacity of a Noisy Channel	46:54	▶
8	Noisy Channel Coding III: The Noisy-Channel Coding Theorem	1:08:59	▶
9	A Noisy Channel Coding Gem, and an Introduction to Bayesian Inference I	48:36	▶
10	Bayesian Inference II: Inference of Parameters and Models	1:15:53	▶
11	Approximating Probability Distributions I: Clustering as an Inference Problem	56:56	▶
12	Approximating Probability Distributions II: Monte Carlo Methods I	1:23:48	▶
13	Approximating Probability Distributions III: Monte Carlo II, Slice Sampling	1:47:57	▶
14	Approximating Probability Distributions IV: Variational Methods	46:35	▶
15	Data Modelling with Neural Networks I: Feedforward Networks, Capacity of a Neuron	1:27:16	▶
16	Data Modelling with Neural Networks II: Content-Addressable Memories	1:36:33	▶

Lectures 12 and 13 double as the best introduction to Monte Carlo in this entire curriculum — watch them before Atlas family C.

Supporting	Instructor	Scale	Link
Information Theory, Coding and Cryptography	NPTEL / IIT Delhi	40 videos, 30 hr	Playlist
Stanford EE274 Data Compression	Weissman & Chandak	18 lectures, 24 hr	Playlist

MIT 6.441 has no video; the NPTEL course is the substitute for a full theorem-proof treatment.

B.1.4 BUILD — Module B.1

1. **Implement PCA and TICA from scratch** on one of your own trajectories. Articulate why TICA finds slow coordinates where PCA finds high-variance ones.
2. **Implement Metropolis-Hastings** for a nontrivial target and *prove* — on paper, then verify numerically — that your proposal and acceptance rule satisfy detailed balance.
3. **Reverse-mode differentiation through a simulation.** Differentiate a three-step gradient descent on a toy energy function with respect to the energy parameters, by hand and then with autograd. Then do it with the adjoint method and compare memory usage. This is 18.S096 lecture 6.1 made concrete and it is the seed of differentiable simulation.
4. **Compute per-position entropy of a protein family alignment.** Compare to burial from a structure. What is the mutual information between conservation and burial, and what does the residual tell you?

Derivation checkpoints due: 56 (backpropagation and the adjoint method).

B.1.5 Paired reading — Module B.1

Watch this	Then read this	Hold this question
Strang 18.065 Lec 7 (Eckart-Young)	Any efficient-attention or memory-reduction paper for structure models	Which block in a structure predictor is low-rank, and has anyone exploited it?
Strang 18.065 Lec 34 (Procrustes)	Kabsch 1976	Your RMSD calculation is solving a Procrustes problem. What does the SVD step do?
Johnson & Edelman 6.1 (adjoint ODE)	Chen et al. 2018, <i>Neural Ordinary Differential Equations</i>	What is the memory cost of the adjoint method, and when is checkpointing better?
Johnson & Edelman 8.1 (eigenproblems)	Any normal-mode analysis paper	Could you optimize a protein's normal-mode spectrum by gradient descent?
Gallager 6.262 (Markov chains)	Prinz et al. 2011	What must hold for a coarse-grained chain to remain Markovian?
Rigollet 18.650	Any design paper reporting a success rate	What is the confidence interval on 3/50? Is it ever reported?
McElreath, <i>Statistical Rethinking</i>	Kapoor & Narayanan 2023	Draw the causal diagram for a design benchmark. Where does leakage enter?
MacKay Lec 1–5	Shannon 1948	Perplexity measures compression. What would show the compression is biophysical?
MacKay Lec 12–13	Metropolis et al. 1953; Neal on slice sampling	Detailed balance is sufficient, not necessary. What is necessary?
Hennig, <i>Probabilistic ML</i>	Guo et al., <i>On calibration of modern neural networks</i>	Is pLDDT calibrated? Design the experiment that settles it.

B.2 — Statistical Mechanics

B.2.0 Why this module differentiates you

The field has a large surplus of people who understand transformers and a shortage of people who understand the Boltzmann distribution well enough to derive things with it.

Every quantity you care about — a binding affinity, a stability, a population, a rate — is an ensemble average or a free energy difference. Scoring functions approximate free energies. Diffusion models learn score functions that are, up to temperature, forces. Normalizing flows trained to sample equilibrium are the explicit bridge between the two fields. If you can derive statistical mechanics rather than recall it, the entire generative-model literature reads differently: you see the physics hiding inside machine learning papers, which is exactly the vantage point from which the cross-cultural carries become visible.

Roughly 100 hours if you do all three courses. Worth every one.

B.2.1 MIT 8.333 — Statistical Mechanics of Particles

Mehran Kardar (MIT) · 26 lectures · ~36 hours · Tier A

The canonical graduate course, and the exact language in which partition functions, free energies and every MD and Monte Carlo estimator are written. **Do the problem sets** — OCW has them with solutions, and a statistical mechanics course without problem sets is entertainment rather than training.

Unit I — Thermodynamics

#	Lecture	Length	Link
1	Thermodynamics Part 1	1:26:25	▶
2	Thermodynamics Part 2	1:23:38	▶
3	Thermodynamics Part 3	1:23:44	▶
4	Thermodynamics Part 4	1:18:53	▶

Unit II — Probability

#	Lecture	Length	Link
5	Probability Part 1	1:21:30	▶
6	Probability Part 2	1:24:53	▶

Unit III – Kinetic Theory of Gases

#	Lecture	Length	Link
7	Kinetic Theory of Gases Part 1	1:18:39	▶
8	Kinetic Theory of Gases Part 2	1:15:56	▶
9	Kinetic Theory of Gases Part 3	1:25:36	▶
10	Kinetic Theory of Gases Part 4	1:25:18	▶
11	Kinetic Theory of Gases Part 5	1:22:23	▶

Unit IV – Classical Statistical Mechanics

#	Lecture	Length	Link
12	Classical Statistical Mechanics Part 1	1:25:38	▶
13	Classical Statistical Mechanics Part 2	1:22:35	▶
14	Classical Statistical Mechanics Part 3	1:25:17	▶

Unit V – Interacting Particles

#	Lecture	Length	Link
15	Interacting Particles Part 1	1:25:42	▶
16	Interacting Particles Part 2	1:22:06	▶
17	Interacting Particles Part 3	1:23:18	▶
18	Interacting Particles Part 4	1:24:22	▶
19	Interacting Particles Part 5	1:19:18	▶

Units VI–VII – Quantum Statistical Mechanics and Ideal Quantum Gases

Optional for biomolecular work; included for completeness.

#	Lecture	Length	Link
20	Quantum Statistical Mechanics Part 1	1:23:32	▶
21	Quantum Statistical Mechanics Part 2	1:23:48	▶
22	Ideal Quantum Gases Part 1	1:20:31	▶
23	Ideal Quantum Gases Part 2	1:23:42	▶
24	Ideal Quantum Gases Part 3	1:23:42	▶
25	Ideal Quantum Gases Part 4	1:22:55	▶
26	Ideal Quantum Gases Part 5	1:21:05	▶

The protein-relevant core is lectures 1–19. Units VI and VII matter only if you move toward QM/MM or electronic structure. Unit V (interacting particles) is where the cluster expansion and the mean-field

approximation appear, and those are the ancestors of every implicit-solvent and mean-field DCA method you will meet later.

B.2.2 V. Balakrishnan — Physical Applications of Stochastic Processes

NPTEL / IIT Madras · 29 lectures · ~28 hours · Tier A

This is the course nobody tells you to watch and the one that pays off most. Langevin dynamics, Fokker-Planck and first-passage times derived in full — the machinery directly beneath thermostats, diffusion coefficients, enhanced-sampling estimators, *and* the stochastic differential equation formulation of diffusion models. Watching it before Atlas family N makes that material roughly twice as legible.

#	Lecture	Length	Link
1	Discrete probability distributions (Part 1)	1:02:41	▶
2	Discrete probability distributions (Part 2)	54:54	▶
3	Continuous random variables	56:50	▶
4	Central Limit Theorem	1:00:54	▶
5	Stable distributions	1:08:37	▶
6	Stochastic processes	1:00:20	▶
7	Markov processes (Part 1)	54:12	▶
8	Markov processes (Part 2)	1:02:21	▶
9	Markov processes (Part 3)	52:22	▶
10	Birth-and-death processes	51:20	▶
11	Continuous Markov processes	57:36	▶
12	Langevin dynamics (Part 1)	57:53	▶
13	Langevin dynamics (Part 2)	51:46	▶
14	Langevin dynamics (Part 3)	58:39	▶
15	Langevin dynamics (Part 4)	57:15	▶
16	Itô and Fokker-Planck equations for diffusion processes	47:42	▶
17	Level-crossing statistics of a continuous random process	54:33	▶
18	Diffusion of a charged particle in a magnetic field	59:56	▶
19	Power spectrum of noise	53:38	▶
20	Elements of linear response theory	1:01:50	▶
21	Random pulse sequences	57:33	▶
22	Dichotomous diffusion	1:07:31	▶
23	First passage time (Part 1)	59:50	▶
24	First passage time (Part 2)	1:03:14	▶
25	First passage and recurrence in Markov chains	1:06:21	▶
26	Recurrent and transient random walks	1:11:49	▶
27	Non-Markovian random walks	51:31	▶
28	Statistical aspects of deterministic dynamics (Part 1)	54:36	▶
29	Statistical aspects of deterministic dynamics (Part 2)	1:01:54	▶

If you watch only six lectures from this course, watch 12 through 16 and 23. Lectures 12–16 give you Langevin through Fokker-Planck in sequence, which is the derivation that makes thermostats, Brownian dynamics and score-based diffusion all one subject. Lecture 23 on first-passage time is the theory of rates, and therefore of every kinetics claim in Atlas family I.

B.2.3 V. Balakrishnan — Nonequilibrium Statistical Mechanics

NPTEL / IIT Madras · 36 lectures · ~33 hours · Tier A

The deeper treatment. Fluctuation-dissipation and response theory let you prove whether a biased simulation still reports equilibrium observables — which is the question underneath every enhanced-sampling method in Atlas family D.

#	Lecture	Length	Link
1	Recapitulation of equilibrium statistical mechanics	50:36	▶
2	The Langevin model (Part 1)	52:25	▶
3	The Langevin model (Part 2)	51:20	▶
4	The Langevin model (Part 3)	57:12	▶
5	The Langevin model (Part 4)	49:57	▶
6	Linear response theory (Part 1)	47:55	▶
7	Linear response theory (Part 2)	51:08	▶
8	Linear response (Part 3)	50:24	▶
9	Linear response (Part 4)	49:40	▶
10	Linear response (Part 5)	49:05	▶
11	Linear response (Part 6)	49:19	▶
12	Linear response theory (Part 7)	52:16	▶
13	Quiz 1 — questions and answers	51:17	▶
14	Linear response theory (Part 8)	48:23	▶
15	Linear response theory (Part 9)	56:45	▶
16	The dynamic mobility	52:21	▶
17	Fokker-Planck equations (Part 1)	50:22	▶
18	Fokker-Planck equations (Part 2)	52:36	▶
19	Fokker-Planck equations (Part 3)	1:00:26	▶
20	The generalized Langevin equation (Part 1)	1:00:14	▶
21	The generalized Langevin equation (Part 2)	56:27	▶
22	Diffusion in a magnetic field	51:17	▶
23	The Boltzmann equation for a dilute gas (Part 1)	57:24	▶
24	The Boltzmann equation for a dilute gas (Part 2)	54:58	▶
25	The Boltzmann equation for a dilute gas (Part 3)	52:12	▶
26	The Boltzmann equation for a dilute gas (Part 4)	56:23	▶
27	The Boltzmann equation for a dilute gas (Part 5)	53:51	▶
28	Quiz 2 — questions and answers	59:06	▶
29	Critical phenomena (Part 1)	1:06:56	▶
30	Critical phenomena (Part 2)	1:03:44	▶

#	Lecture	Length	Link
31	Critical phenomena (Part 3)	1:04:57	▶
32	Critical phenomena (Part 4)	57:26	▶
33	Critical phenomena (Part 5)	1:00:27	▶
34	Critical phenomena (Part 6)	51:11	▶
35	Critical phenomena (Part 7)	53:47	▶
36	The Wiener process (standard Brownian motion)	1:03:10	▶

Lecture 20, the generalized Langevin equation, is the one to mark. It contains the memory kernel — the term that coarse-grained models almost always discard, and whose discarding is why coarse-grained dynamics is wrong even when coarse-grained structure is right. That is derivation checkpoint 29 and it is the theoretical heart of Atlas family F.

B.2.4 Supporting courses

Course	Instructor	Scale	Link
Statistical Mechanics (Theoretical Minimum)	Leonard Susskind (Stanford)	10 × ~2 hr	Playlist
MIT 5.60 Thermodynamics & Kinetics	Nelson & Bawendi (MIT)	36 × 50 min	OCW
MIT 8.334 Statistical Mechanics II: Fields	Mehran Kardar	26 lectures, 35 hr	Playlist
Statistical Mechanics: Algorithms and Computations	Werner Krauth (ENS Paris)	10 modules	Coursera, free audit
Physical Biology of the Cell	Rob Phillips (Caltech)	100 videos	Playlist
Statistical Mechanics of Soft Matter	Jonathan Selinger (Kent State)	41 videos	Playlist
Nonequilibrium Physics: Stochastic Dynamics & Field Theories	LMU Munich, SS 2025	40 videos, 30 hr	Playlist
Brownian Motion and Stochastic Differential Equations	K. Hlyniana	10 lectures, 13 hr	Playlist
Large Deviation Theory in Statistical Physics	ICTS program (Touchette, Mallick, Lelièvre)	66 talks, 57 hr	Playlist

Krauth is the one to do if you only do one computational item — Monte Carlo, detailed balance and sampling implemented in downloadable Python by someone who thinks carefully about correctness. **The LMU nonequilibrium course is the modern path-integral treatment** and is the right language for score-based and flow-matching generative models of conformations; it is the most direct bridge in this module to Atlas family N.

B.2.5 Statistical honesty: the Zuckerman thread

Daniel Zuckerman is the field's conscience on convergence and error estimation. His best material is **text rather than video**, which is unusual in this curriculum and worth accommodating rather than working around.

Resource	Type	Link
Introduction to Weighted Ensemble Simulation	video, 31 min	▶
Recent Advances in Weighted Ensemble Simulation	video, 36 min	▶
Simple trajectory physics for path sampling and improving Markov models	video, 33 min	▶
Physical Lens on the Cell	free online book	physicallensonthecell.org
Basic Nonequilibrium Physics	free online book	OSF
Statistical Biophysics Blog	articles with exercises	statisticalbiophysicsblog.org

B.2.6 BUILD — Module B.2

1. **Derive the Boltzmann distribution** from maximum entropy under a mean-energy constraint. State precisely what temperature is in that derivation.
2. **Implement Metropolis Monte Carlo** for a 2D Ising model. Verify detailed balance numerically. Measure the heat capacity and locate the transition.
3. **Implement a Langevin integrator** for a 1D double well. Measure populations against analytic Boltzmann weights. Then deliberately break the fluctuation-dissipation relation — change friction without changing noise — and watch the thermodynamics go wrong.
4. **Error estimation.** For item 3, implement block averaging and compute a correlation time. Produce an honest error bar. Then answer Zuckerman's question: how would you know if this were not converged?
5. **The bridge exercise.** Write the Fokker-Planck equation for your Langevin system. Then write the reverse-time SDE of a diffusion model. Identify every corresponding term. Keep this page; you will need it in Atlas family N.

Derivation checkpoints due: 7, 29, 36, 37.

B.2.7 Paired reading — Module B.2

Watch this	Then read this	Hold this question
Kardar Units I–II	Any text on ensemble equivalence	When do canonical and microcanonical disagree, and does it ever matter for a protein?
Kardar Unit IV	Zwanzig 1954	FEP is exact. Why does it fail in practice, and what quantity governs the failure?
Kardar Unit V (interacting particles)	Any implicit-solvent or mean-field DCA paper	Both are mean-field approximations. What exactly is being averaged over?
Balakrishnan SP 12–16	Any Langevin thermostat paper	Which thermostats preserve correct dynamics, and which only the correct distribution?
Balakrishnan SP 23–24	Kramers 1940	Derive the rate from first-passage theory. Where does the barrier height enter?
Balakrishnan NESM 6–12	Any Green-Kubo derivation	Transport coefficients from equilibrium fluctuations — what is the protein analogue?
Balakrishnan NESM 20–21	Mori-Zwanzig formalism; any coarse-graining paper	The memory kernel is what CG models discard. What does discarding it cost?
Balakrishnan NESM 36	Any SDE text on Wiener processes	Write the Wiener process. Now write the forward diffusion of a DDPM. Compare.
Krauth, <i>Algorithms and Computations</i>	Metropolis et al. 1953	Detailed balance is sufficient but not necessary. What is necessary?
LMU nonequilibrium course	Song et al. 2021, <i>Score-based generative modeling through SDEs</i>	The field theory and the generative model use the same formalism. Map the terms.
Zuckerman, weighted ensemble	Zuckerman & Chong 2017	WE is unbiased. What exactly does that mean, and what is still not guaranteed?
Zuckerman, any talk	Grossfield & Zuckerman 2009	Read before publishing any simulation. Which of its tests does your workflow pass?
Phillips, <i>Physical Biology of the Cell</i>	Any MWC allostery paper	Allostery as a two-state model — could a design method target that directly?

B.3 — Computer Science

B.3.0 The filter

This is not a computer science degree. It is the specific algorithmic, geometric, systems and inference material that someone inventing molecular methods actually needs, and it was entirely absent from the first edition.

A warning before the list: **the full essential set here is around 590 hours, which is larger than the rest of Part I combined.** Computer science is where a methods person over-invests most easily, because every course looks relevant and most are only adjacent. Section B.3.9 gives a ~130-hour core path. Take that first, and return to the rest only when a specific problem demands it.

B.3.1 Algorithms for sequences and structures

Course	Instructor	Scale	Link	
Algorithms for DNA Sequencing	Ben Langmead (JHU)	55 videos, 6.6 hr	Playlist	essential
Computational Geometry	Philipp Kindermann (Trier)	56 videos, 8.0 hr	Playlist	essential
MIT 6.851 Advanced Data Structures	Erik Demaine (MIT)	22 lectures, 29.9 hr	Playlist	essential
MIT 6.849 Geometric Folding Algorithms	Erik Demaine (MIT)	40 videos, 42.8 hr	Playlist	essential
CMU 15-458 Discrete Differential Geometry	Keenan Crane (CMU)	22 lectures, 23.8 hr	Playlist · site	essential
MIT 6.046J Design and Analysis of Algorithms	Demaine, Devadas, Lynch	34 videos, 39.5 hr	Playlist	optional
MIT 6.006 Introduction to Algorithms	Ku, Solomon, Demaine	32 videos, 35.3 hr	Playlist	optional
CENG 773 Computational Geometry	METU OCW	29 lectures, 22.8 hr	Playlist	optional
Algorithms in Computational Biology and Sequence Analysis	NPTEL / IISc	48 lectures, 27.5 hr	Playlist	optional

Langmead's course has the densest hours in this entire document. Fifty-five short videos covering k-mer indices, Boyer-Moore, pigeonhole filtering, edit distance DP, overlap and De Bruijn graphs — the exact machinery you must have reimplemented before you can modify an MMseqs2 or Foldseek-class search, or write a new aligner. Six and a half hours for all of it.

Kindermann's computational geometry course is the sleeper. Convex hulls, sweep-line intersection, range trees and k-d trees, Voronoi diagrams, Delaunay triangulation, 3D convex hulls. That is literally every geometric primitive in a packing, SASA, or pocket-detection pipeline — Voronoi and Delaunay for protein packing and cavity

detection, k-d trees for neighbor lists, 3D hulls for molecular envelopes. Eight hours, and it replaces a shelf of folklore.

MIT 6.849 is the unexpected one. Demaine on linkage reachability and rigidity is the rigorous theory of the object a protein backbone actually *is*, and the course contains protein-folding-specific lectures. Loop closure, inverse kinematics and rigidity analysis all come from here.

Crane's discrete differential geometry gives you curvature, the Laplace-Beltrami operator and discrete exterior calculus on triangle meshes — the correct formalism for molecular surfaces, solvent-excluded-surface curvature features, and the geodesic descriptors MaSIF uses in Atlas I.

B.3.2 Randomized, sketching and approximation algorithms

Course	Instructor	Scale	Link	
Harvard CS229r Algorithms for Big Data	Jelani Nelson (Harvard)	25 lectures, 34.4 hr	Playlist · notes	essential
Harvard CS224 Advanced Algorithms	Jelani Nelson	25 lectures, 35.5 hr	Playlist	essential
Stanford CS261 A Second Course in Algorithms	Tim Roughgarden	20 lectures, 26.2 hr	Playlist	optional
Approximation Algorithms	Rasmus Pagh (Copenhagen)	15 videos, 4.8 hr	Playlist	optional

CS229r individual lectures — the first twelve are the ones that matter

#	Topic	Length	Link
1	Tail bounds; Morris approximate counting	1:26:05	▶
2	Distinct elements	1:23:57	▶
3	Turnstile streaming, AMS sketch, Johnson-Lindenstrauss lemma	1:22:13	▶
4	ℓ_p norm estimation	1:21:03	▶
5	Johnson-Lindenstrauss lower bound	1:23:03	▶
6	JL lower bound wrap-up; heavy hitters	1:25:47	▶
7	CountMin, CountSketch, insertion-only improvements	1:19:26	▶
8	Continuous monitoring; chaining	1:22:02	▶
9	Fast Johnson-Lindenstrauss Transform	1:23:12	▶
10	Sparse Johnson-Lindenstrauss Transform	1:18:47	▶
11	Randomized linear algebra: approximate matmul, subspace embeddings	1:19:20	▶
12	Randomized linear algebra: regression, low-rank approximation	1:30:06	▶
13	Compressive sensing, RIP, ℓ_1 minimization	1:21:48	▶
14	Iterative hard thresholding, model-based compressive sensing	1:21:38	▶
15	Sparse recovery using sparse matrices	1:23:56	▶
16	Sparse Fourier Transform	1:23:02	▶
18	ℓ_p sampling, graph sketching	1:23:14	▶
20	Lower bounds (communication complexity)	1:19:58	▶

Johnson-Lindenstrauss is the theorem behind every embedding-based protein search, and lectures 11 and 12 are the provable machinery for random-projection featurization and low-rank approximation. If you ever claim that a compressed structure descriptor preserves distances, this is where the claim is justified or refuted. Lectures 13–16 on compressive sensing are also the theory behind sparse-sampling cryo-EM and NMR reconstruction.

Stanford CS168, frequently recommended for this material, has Canvas-only video; the notes and projects are public but there is no recording.

B.3.3 Parallel computing and GPU programming

Course	Instructor	Scale	Link	
MIT 6.172 Performance Engineering of Software Systems	Leiserson & Shun (MIT)	23 lectures, 29.8 hr	Playlist	essential
Stanford CS149 Parallel Computing	Fatahalian & Olukotun	19 lectures, 24.3 hr	Playlist	essential
UIUC ECE408 Programming Massively Parallel Processors	Wen-mei Hwu (UIUC)	25 lectures, 32.2 hr	Playlist	essential
OLCF/NVIDIA CUDA Training Series	OLCF / NVIDIA	13 sessions, 19.2 hr	Playlist · exercises	essential
AUB Programming Massively Parallel Processors	Izzat El Hajj (AUB)	24 lectures, 28.3 hr	Playlist	optional
UIUC ECE508 Parallel Algorithm Techniques	Wen-mei Hwu	29 lectures, 21.0 hr	Playlist	optional
Parallel Programming (node-level performance engineering)	NHR@FAU (Hager/Wellein)	13 lectures, 12.5 hr	Playlist	optional
NVIDIA Modern CUDA C++ Programming Class	NVIDIA Developer	3 parts, 4.1 hr	Playlist	optional
GPU MODE lecture series	community, ongoing	84 lectures, 107.5 hr	Archive playlist · gpumode.com	reference

MIT 6.172 — the lectures that matter most

#	Lecture	Length	Link
1	Introduction and Matrix Multiplication	1:00:21	▶
2	Bentley Rules for Optimizing Work	1:20:10	▶
6	Multicore Programming	1:16:46	▶
8	Analysis of Multithreaded Algorithms	1:17:34	▶
9	What Compilers Can and Cannot Do	1:18:46	▶
10	Measurement and Timing	1:21:28	▶
14	Caching and Cache-Efficient Algorithms	1:18:23	▶
15	Cache-Oblivious Algorithms	1:21:47	▶
16	Nondeterministic Parallel Programming	1:22:12	▶
18	Domain Specific Languages and Autotuning	1:11:22	▶
22	Graph Optimization	1:18:40	▶
23	High Performance in Dynamic Languages	1:25:44	▶

6.172 is the single highest-leverage skill for a methods person: turning a correct-but-slow prototype into something fifty to five hundred times faster. Lectures 14, 15 and 22 are the ones a molecular simulation person will reread; lecture 10 on measurement is the one that prevents you from optimizing the wrong thing, which is the usual failure. Lectures 19–20 are a chess project and are skippable.

B.3.4 Systems for machine learning

Course	Instructor	Scale	Link	
CMU 10-414/714 Deep Learning Systems	Kolter & Chen	24 lectures, 23.3 hr	dlsyscourse.org · mirror	essential
Stanford CS336 Language Modeling from Scratch (2026)	Liang & Hashimoto	18 lectures, 24.2 hr	Playlist	essential
Stanford CS336 (2025 edition)	Liang & Hashimoto	17 lectures, 22.3 hr	Playlist · assignments	essential
MIT 6.5940 EfficientML.ai	Song Han (MIT)	45 videos, 54 hr	Playlist · efficientml.ai	essential
Intro to Triton Series	community	6 videos, 1.0 hr	Playlist	optional
GPU MODE × ScaleML: Frontiers in LM Systems and Algorithms	GPU MODE	5 talks, 8.0 hr	Playlist	optional

CMU 10-414 individual lectures — you build an autodiff framework from nothing

#	Lecture	Length	Link
1	Introduction and Logistics	57:55	▶
2	ML Refresher / Softmax Regression	1:20:01	▶
3a	“Manual” Neural Networks, Part I	53:39	▶
3b	“Manual” Neural Networks, Part II	47:41	▶
4	Automatic Differentiation	1:03:35	▶
5	Automatic Differentiation Implementation	1:05:57	▶
6	Fully Connected Networks, Optimization, Initialization	1:26:56	▶
7	Neural Network Abstractions	59:59	▶
8	Neural Network Library Implementation	55:49	▶
9	Normalization and Regularization	1:19:30	▶
10	Convolutional Networks	1:08:31	▶
11	Hardware Acceleration	45:22	▶
12	GPU Acceleration	44:20	▶
13	Hardware Acceleration Implementation	50:09	▶
14	Implementing Convolutions	1:19:42	▶
15	Training Large Models	46:37	▶
18	Sequence Modeling and Recurrent Networks	1:10:48	▶
20	Transformers and Attention	1:10:16	▶
21	Transformer Implementation	1:01:13	▶
23	Model Deployment	42:53	▶
24	ML Compilation and Deployment Implementation	36:37	▶

This is the course that converts “I use PyTorch” into “I can add a new differentiable operator, a new layer type, or a new backend.” That is the precondition for implementing an SE(3)-equivariant or geometric operator that does not already exist — which is what inventing a new molecular architecture actually requires. Pair every lecture with its notebook from dlsyscourse.org.

CS336 individual lectures (2026 edition)

#	Lecture	Length	Link
1	Overview, Tokenization	1:19:22	▶
2	PyTorch, einops	1:17:25	▶
3	Architectures	1:29:14	▶
4	Attention Alternatives	1:26:21	▶
5	GPUs, TPUs	1:18:39	▶
6	Kernels, Triton, XLA	1:26:41	▶
7	Parallelism I	1:21:03	▶
8	Parallelism II	1:20:11	▶
9	Scaling Laws I	1:17:57	▶
10	Inference	1:25:30	▶
11	Scaling Laws II	1:17:04	▶
12	Evaluation	1:18:34	▶
13	Data: Sources, Datasets	1:22:02	▶
14	Data II	1:24:46	▶
15	Mid/Post-Training	1:19:55	▶
16	Post-Training: RLVR	1:15:51	▶
17	Alignment, Multimodality	1:17:40	▶
18	Guest Lecture: Dan Fu	1:11:41	▶

B.3.5 Probabilistic graphical models and inference

Course	Instructor	Scale	Link	
CMU 10-708 Probabilistic Graphical Models	Eric Xing (CMU)	28 lectures, 35.8 hr	Playlist	essential
Probabilistic Machine Learning	Philipp Hennig (Tübingen)	25 lectures, 35.3 hr	Playlist	essential
Numerics of Machine Learning	Hennig et al.	14 lectures, 19.2 hr	Playlist	optional

CMU 10-708 is quietly one of the most relevant courses in this document and almost nobody in protein design takes it. Belief propagation, junction trees and variational inference are the formal home of message-passing side-chain packing — tree-reweighted BP and dual decomposition were applied to packing years ago and then largely forgotten. Direct coupling analysis is a graphical model. ProteinMPNN is a graphical model with learned potentials. Seeing all three as the same object is a real advantage, and this is where you acquire it.

Hennig's *Numerics of ML* treats linear solves, ODE integration and optimization as inference problems with error bars — directly relevant to uncertainty-aware integrators and to Atlas family A's surrogate models.

B.3.6 Learning theory and optimization

Course	Instructor	Scale	Link	
Stanford CS229M / STATS214 Machine Learning Theory	Tengyu Ma (Stanford)	20 lectures, 28.3 hr	Playlist	essential
Stanford EE364A Convex Optimization I	Stephen Boyd (Stanford)	18 lectures, 23.7 hr	Playlist	essential

Full treatment of the optimization material, including combinatorial optimization and the NP-hardness of packing, is in Module B.4.

B.3.7 Research software engineering

Course	Host	Scale	Link	
CodeRefinery Workshop	Nordic e-Infrastructure	18 sessions, 17.6 hr	Playlist · lessons	essential
Tuesday Tools and Techniques for HPC	CodeRefinery	7 sessions, 7.3 hr	Playlist	optional

This is the practice layer that separates a method people cite from a script that gets lost. Git branching for research, automated testing of numerical code, documentation, reproducible environments, packaging and licensing — taught specifically to scientists. It is the single best free research-software curriculum, and it is the prerequisite for Capstone VI.

Software Carpentry's lessons are canonical but are written material, not a recorded course.

B.3.8 Large-scale data, search, and complexity

Course	Instructor	Scale	Link	
CMU 15-721 Advanced Database Systems	Andy Pavlo (CMU)	22 lectures, 27.2 hr	Playlist	essential
CMU 15-445 Intro to Database Systems (lec 1–12 only)	Andy Pavlo	25 lectures, 33.7 hr	Playlist	optional
Introduction to HDF5	Quincey Koziol (HDF Group / LBNL)	1:17:10	▶	essential
Parallel HDF5	Quincey Koziol	40:01	▶	essential
Scalable HDF5	Quincey Koziol	1:13:42	▶	essential
Tutorial: Zarr	OGC	1:00:32	▶	optional
The Beauty of Zarr	Sanket Verma (PyData Global)	28:24	▶	optional
Billion-scale Approximate Nearest Neighbor Search	Yusuke Matsui (CVPR 2020 tutorial)	47:36	▶	essential
Neural Search in Action	CVPR tutorial	3:12:55	▶	optional
Information Retrieval and Web Search	Dwaipayan Roy	60 videos, 24.8 hr	Playlist	optional
MIT 6.890 Algorithmic Lower Bounds: Fun with Hardness Proofs	Erik Demaine (MIT)	23 lectures, 31.1 hr	Playlist	essential
MIT 18.404J Theory of Computation (lec 13–20)	Michael Sipser (MIT)	25 lectures, 32.3 hr	Playlist	optional
Parameterized Complexity	NPTEL / IIT Madras	43 videos, 30.2 hr	Playlist	optional

Matsui’s ANN tutorial is forty-eight minutes that determine whether a structure-embedding search over 200 million predicted structures is feasible on one machine. IVE, product quantization, HNSW, and the memory/recall/latency trade-off surface. Watch it, then watch the Foldseek and Steinegger talks in Atlas N, and you will see the 3Di alphabet and k-mer prefiltering as specific instances of the general machinery.

Demaine’s 6.890 teaches you to *construct* a reduction, not cite one — so you can prove your new formulation of threading, packing or design is hard, or better, find the restriction under which it is not. Parameterized complexity is the follow-on that explains why side-chain packing is tractable when the interaction graph has small treewidth, which is the regime real proteins occupy.

B.3.9 The core path — 130 hours, not 590

If the CS section must fit inside a sane budget, this is the subset that earns its place.

	Course	Hours	Why it is in
1	Langmead, Algorithms for DNA Sequencing	6.6	The sequence-algorithms spine; cheapest hours in the document
2	Kindermann, Computational Geometry	8.0	Voronoi, Delaunay, k-d trees for packing and neighbor search
3	CMU 10-414 Deep Learning Systems	23.3	Build your own autodiff and CUDA backend
4	CS336 lectures 1–11	16	Kernels, parallelism, scaling
5	MIT 6.172 (skipping 19–20)	27	The hundredfold-speedup skill
6	Harvard CS229r lectures 1–12	16	JL, sketching, randomized linear algebra
7	Matsui ANN + Foldseek/Steinegger talks	2.7	The retrieval bridge
8	HDF5 (Koziol) + CMU 15-721 columnar lectures	8	Large-data I/O
9	CodeRefinery	17.6	The practices that make any of it reusable

Everything else in this module is a reference shelf.

B.3.10 BUILD — Module B.3

1. **Implement a k-mer index and an exact Smith-Waterman** from scratch. Then implement the pigeonhole filter and measure the speedup. You have now written the core of BLAST.
2. **Implement a k-d tree neighbor list** and compare to cell lists for a Lennard-Jones system. Measure where each wins as a function of density.
3. **Build a Voronoi tessellation of a protein** and use it to compute packing density and detect cavities. Compare to fpocket’s output.
4. **Work through CMU 10-414 to the point where you have written reverse-mode autodiff and a CUDA kernel.** Then implement one SE(3)-equivariant operation in your own framework. That is the capability this whole module exists for.
5. **Profile one of your own production scripts** with the 6.172 methodology. Find the actual bottleneck — it will not be where you think — and fix it. Report the speedup.

B.3.11 Paired reading — Module B.3

Watch this	Then read this	Hold this question
Langmead, index and alignment modules	Needleman & Wunsch 1970; Smith & Waterman 1981; Gotoh 1982	Derive the affine-gap recursion. What is the complexity?
Langmead, De Bruijn graphs	Pevzner, Tang & Waterman 2001	Assembly as an Eulerian path. Where does the analogy break on real data?
Kindermann, Voronoi and Delaunay	Richards 1974 (protein packing); any fpocket paper	A pocket is a geometric object. State its definition precisely.
Demaine 6.849, linkages	Any loop-closure or kinematic-closure paper	A backbone is a linkage. What does rigidity theory say about its reachable set?
Crane, Discrete Differential Geometry	Gainza et al. 2020 (MaSIF)	Curvature on a molecular surface. Which discrete operator, and why that one?
Nelson CS229r Lec 3, 5	Johnson & Lindenstrauss 1984; Achlioptas 2003	How low can an embedding go before distances distort? Apply the bound to a protein embedding.
Nelson CS229r Lec 11–12	Any randomized SVD paper	Approximate matmul with a guarantee. Where could this cut a structure model's cost?
Leiserson 6.172 Lec 14–15	Frigo et al. 1999 (cache-oblivious algorithms)	Why does a cache-oblivious layout beat a tuned one? Apply to a neighbor list.
Leiserson 6.172 Lec 10	Any benchmarking methodology paper	How do you measure a speedup honestly? What does your current method get wrong?
Hwu ECE408	Kirk & Hwu, <i>Programming Massively Parallel Processors</i>	Write a tiled matmul. Then write a tiled PME or attention kernel.
Kolter & Chen 10-414 Lec 4–5	Baydin et al. 2018, <i>Automatic differentiation in machine learning: a survey</i>	Implement reverse-mode AD. Then differentiate through an ODE solver.
Kolter & Chen 10-414 Lec 11–13	—	Add a GPU backend. Then add one equivariant op that does not exist in PyTorch.
CS336 Lec 6, <i>Kernels, Triton, XLA</i>	Dao et al. 2022, FlashAttention	IO-awareness beat FLOP-reduction. What is the molecular analogue?
Xing 10-708, belief propagation	Any TRBP or dual-decomposition packing paper; then Dauparas et al. 2022	Write ProteinMPNN's factor graph. What inference is it doing?
Xing 10-708, variational inference	Morcos et al. 2011 (DCA)	DCA is inference in a Potts model. What does mean-field cost you?
Matsui, <i>Billion-scale ANN</i>	van Kempen et al. 2024 (Foldseek)	Foldseek turns structure search into sequence search. What is the 3Di alphabet, formally?
Demaine 6.890	Desmet et al. 1992 (DEE); Pierce & Winfree 2002	Construct the reduction showing packing is NP-hard. Then find a tractable restriction.
Pavlo 15-721, columnar storage	Any Parquet/Arrow specification	Your trajectory format is row-major. What would columnar buy you?
Koziol, HDF5	—	Chunking and compression determine slice latency. Benchmark your own trajectory I/O.
CodeRefinery, testing	Wilson et al. 2014, <i>Best Practices for Scientific Computing</i>	Write one test that would catch a silent numerical regression in your pipeline.

B.4 Chemistry and Structural Biology

The module most computational people skip, and the one whose absence shows.

You can build a generative model of protein backbones without knowing what a hydrogen bond costs in kcal/mol, what pKa shifts do in a buried active site, or why a designed interface with perfect shape complementarity still has micromolar affinity. You will just be unable to tell, when your design fails, whether the model was wrong or the chemistry was always going to refuse.

This module is the vocabulary for arguing with a wet-lab collaborator, and the single best predictor of whether a computational person's designs get made.

B.4.1 Physical chemistry of macromolecules

Course / resource	Instructor	Note
Biophysical Chemistry (full course)	Erik Lindahl (KTH/Stockholm)	► — thermodynamics, kinetics and simulation taught by someone who writes the software
MIT 5.60 Thermodynamics & Kinetics	MIT OCW	Classical, rigorous, free
Physical Biology of the Cell lectures	Rob Phillips (Caltech)	Order-of-magnitude biology; the best training in estimation in all of biology
Molecular Biophysics lectures	NPTEL / various	—

Required reading, and these are short: Dill & Bromberg, *Molecular Driving Forces* — the best textbook in this entire curriculum, and the one to work through with a pencil. Phillips et al., *Physical Biology of the Cell* — read it for the estimation culture, not the content.

Derivation checkpoint 1. Compute, from scratch, the free energy cost of burying a charged residue in a protein core. Use a Born solvation estimate and a reasonable interior dielectric. Then explain why that number is both roughly right and systematically wrong, and what the protein actually does about it.

Checkpoint 2. Estimate the entropic cost of restraining a single rotatable bond on binding. Then multiply by the number of rotatable bonds in a 14-residue macrocycle and ask yourself what cyclization is buying.

B.4.2 The forces, one at a time

The whole module compresses to understanding five contributions and their magnitudes, because every scoring function in Atlas A and L is an attempt to approximate them:

Force	Rough magnitude	What usually goes wrong
Hydrophobic effect	~ 25 cal/mol/ $\text{\AA}^2$ of buried nonpolar surface	It is entropic at room temperature and the temperature dependence is almost never modeled
Hydrogen bonds	1–5 kcal/mol, but ~ 0 net in water	The desolvation penalty nearly cancels the bond; this is why H-bond counting overpredicts affinity
Electrostatics	Huge, and screened	Dielectric is not a constant; buried pairs behave nothing like surface pairs
Van der Waals	Small individually, decisive collectively	Packing density, not pair energies, is what distinguishes real cores
Conformational entropy	0.5–1.5 kcal/mol per restrained side chain	Omitted from nearly every design score function

Papers: Chandler 2005, *Nature* 437:640 (interfaces and the driving force of hydrophobic assembly) · Baldwin 2007, *J Mol Biol* 371:283 · Pace et al. 2014, *J Biol Chem* 289:19165 (contribution of hydrogen bonding to stability) · Fleming & Rose 2005, *Protein Sci* 14:1911 · Dill 1990, *Biochemistry* 29:7133 (dominant forces in protein folding — the review to read first).

The most useful single fact in this module. A hydrogen bond in water is worth close to nothing net, because forming it requires breaking two bonds to water. Design scoring functions that reward hydrogen bonds therefore systematically overvalue polar interfaces. **This is a known, quantified, decades-old result, and it is still visibly present in design output** — which is exactly why interfaces that score beautifully bind weakly.

B.4.3 Protein structure and folding

Resource	Note
IPD lecture 1: Intro + review of protein structure (41:06)	▶
IPD lecture 2: Protein Geometry (1:09:31)	▶
Branden & Tooze, <i>Introduction to Protein Structure</i>	The fold atlas; skim, do not read
Petsko & Ringe, <i>Protein Structure and Function</i>	Short and excellent
Energy landscape theory	→ Atlas H.5 (Onuchic, Wolynes, Dill, Chan)

Papers: Ramachandran et al. 1963, *J Mol Biol* 7:95 · Anfinsen 1973, *Science* 181:223 · Richardson 1981, *Adv Protein Chem* 34:167 (the anatomy and taxonomy of protein structure — still the best single document on folds) · Levinthal 1969 · Bryngelson & Wolynes 1987, *PNAS* 84:7524 (the funnel).

Derivation checkpoint 3. Explain the Ramachandran plot from steric clashes alone — which regions are forbidden, by which atoms, and why glycine and proline differ. Then explain why a rotamer library exists at all, which is **checkpoint 4**, and why the backbone-dependent version is better, which is the beginning of Atlas L.

B.4.4 Enzymology and catalysis

Resource	Note
Fersht, <i>Structure and Mechanism in Protein Science</i>	The canonical text; chapters on transition state theory and catalysis are essential
MIT 5.08J / Harvard enzymology lectures	OCW
Enzyme design talks	→ Atlas R.5

Papers: Pauling 1946, *Chem Eng News* 24:1375 (transition state stabilization — two pages) · Warshel et al. 2006, *Chem Rev* 106:3210 (electrostatic basis for enzyme catalysis) · Bruice 2002, *Acc Chem Res* 35:139 (near attack conformations) · Kamerlin & Warshel 2010, *Proteins* 78:1339 (at the dawn of the 21st century: is dynamics the missing link?) · Blomberg et al. 2013, *Nature* 503:418.

The Warshel-versus-dynamics argument is worth following carefully, because it is the clearest case in biochemistry of a well-posed mechanistic question that resisted resolution for twenty years. Warshel's position — catalysis is electrostatic preorganization, and “dynamics” adds nothing beyond transition-state theory — is the one to understand first, and the one most design papers implicitly assume without saying so.

Derivation checkpoint 5. Write down the rate enhancement a perfectly preorganized active site can provide, from transition-state theory, and compare it to the enhancements achieved by designed enzymes in Atlas R.5. The gap is roughly ten orders of magnitude and nobody fully knows why.

B.4.5 Nucleic acids

Resource	Note
Bloomfield, Crothers & Tinoco, <i>Nucleic Acids</i>	The reference
SantaLucia nearest-neighbor thermodynamics	The basis of every melting-temperature calculation
RNA structure and design	→ Atlas Q.5

Papers: SantaLucia 1998, *PNAS* 95:1460 (unified nearest-neighbor parameters) · Turner & Mathews 2010, *NAR* 38:D280 (NNDB) · Tinoco & Bustamante 1999, *J Mol Biol* 293:271 (how RNA folds) · Leontis & Westhof 2001, *RNA* 7:499 (base-pair classification — the vocabulary RNA structure papers assume you have).

Derivation checkpoint 6. Compute the melting temperature of a 14-mer duplex from nearest-neighbor parameters by hand. Then explain why the same machinery works far less well for RNA secondary structure, and what that implies for the difficulty gap between RNA and protein structure prediction you will meet in Atlas Q.5.

B.4.6 Experimental methods — enough to read a figure honestly

You do not need to run these. You need to know what their error bars mean, because every structure, affinity and stability number you train on came from one of them.

Method	What it actually measures	What it cannot tell you
X-ray crystallography	Electron density, averaged over the crystal	Anything about the solution ensemble; crystal contacts can lock conformations
Cryo-EM	A density map from averaged particles	Local resolution varies enormously; flexible regions vanish
NMR	Ensembles consistent with restraints	Size limited; restraints are averages over states
SPR / BLI	Association and dissociation kinetics	Surface immobilization changes avidity
ITC	Enthalpy directly — the only method that does	Needs lots of material
DSF / thermal shift	A melting temperature	Not ΔG at your temperature of interest
Yeast display / deep mutational scanning	Relative enrichment	Not affinity; expression confounds everything

Papers: Fowler & Fields 2014, *Nat Methods* 11:801 (deep mutational scanning) · Rodrigues & Bonvin 2014 (on interpreting interface data) · Wlodawer et al. 2008, *FEBS J* 275:1 (protein crystallography for non-crystallographers — read before you trust a PDB entry).

The most important row in that table is the last one. Deep mutational scanning data — which trains ProteinGym, ThermoMPNN and half of Atlas Q — is enrichment in a selection, not a thermodynamic quantity. **Expression, stability and binding are confounded in a single number**, and models trained on it learn the confound. Tsuboyama’s mega-scale stability dataset is valuable precisely because it separates them.

B.4.7 Paired reading — B.4

Watch this	Then read this	Hold this question
Lindahl, <i>Biophysical Chemistry</i>	Dill & Bromberg, <i>Molecular Driving Forces</i> , ch. 1–10	Work the problems. This is the one textbook to do properly.
Phillips, <i>Physical Biology of the Cell</i>	The book's estimation chapters	Estimate three quantities in your own project before computing them.
—	Dill 1990, <i>Biochemistry</i> 29:7133	Rank the forces by magnitude. Which does your score function model worst?
—	Chandler 2005, <i>Nature</i> 437:640	Why is the hydrophobic effect length-scale dependent?
—	Pace et al. 2014, <i>J Biol Chem</i> 289:19165	What is a hydrogen bond worth in water, net?
IPD lectures 1–2	Richardson 1981, <i>Adv Protein Chem</i> 34:167	Learn twenty folds by sight. It pays off constantly.
—	Anfinsen 1973, <i>Science</i> 181:223 then Bryngelson & Wolynes 1987, <i>PNAS</i> 84:7524	Sequence determines structure. What did the funnel add?
Atlas R.5 enzyme talks	Warshel et al. 2006, <i>Chem Rev</i> 106:3210	Preorganization versus dynamics. Take a side and defend it.
—	Pauling 1946 (two pages)	Then compare designed enzyme rate enhancements to the theoretical ceiling.
—	SantaLucia 1998, <i>PNAS</i> 95:1460	Compute a T_m by hand. Why is RNA harder?
—	Fowler & Fields 2014, <i>Nat Methods</i> 11:801	What is a DMS score, physically? What is confounded in it?
—	Wlodawer et al. 2008, <i>FEBS J</i> 275:1	Before trusting a PDB entry: what do you check?

Part II — The Method Atlas

A.0 What this part is

Every computational method for understanding proteins, DNA and RNA, organized by the problem it solves.

The organizing claim is that the field looks fragmented but is not. Underneath the hundreds of named tools there are perhaps nineteen genuine families of method, and almost every tool is one family's answer to one question. Once you can see the families, three things become possible that are not possible otherwise: you can place a new paper within an hour of it appearing, you can tell whether a claimed advance is a new family or a refinement within one, and — the point of this document — you can notice which exchanges between families have not yet happened.

A caution about the organization. Methods are grouped by *what they compute*, not by who built them or which software implements them. That means Rosetta appears in five places and AlphaFold in four. This is deliberate. Organizing by software teaches you tools; organizing by problem teaches you the field.

A.1 The nineteen families

	Family	The question it answers
A	Potential energy functions	What is the energy of this arrangement of atoms?
B	Molecular dynamics	How does this system move in time?
C	Monte Carlo	How do I sample this distribution without dynamics?
D	Enhanced sampling and rare events	How do I reach states that are too rare to see?
E	Free energy calculation	What is the free energy difference between two states?
F	Coarse-graining and reduced models	What can I throw away and still be right?
G	Continuum and mesoscale methods	What if I treat solvent, membrane or chromatin as a continuum?
H	Polymer theory and minimal models	What does a chain do, independent of chemistry?
I	Normal modes, networks and dynamics analysis	What are the meaningful motions, and what do they couple?
J	Structure determination algorithms	How does experimental data become coordinates?
K	Integrative and hybrid modeling	How do I combine data of different kinds and resolutions?
L	Classical structure prediction	How do I get a structure from a sequence without a network?
M	Docking and virtual screening	What binds where, and how tightly?
N	Sequence analysis and evolution	What does the evolutionary record say about this molecule?
O	Deep learning foundations for molecules	What architectures respect molecular structure?
P	Learned structure prediction	How do I get a structure from a sequence with a network?
Q	Sequence and biological language models	What is learnable from sequence alone, at scale?
R	Generative design	How do I produce a molecule that does not yet exist?
S	Benchmarking and assessment	How would I know if any of this were true?

Families A through I are the physics-first culture. J and K are the experimental interface. L through N are the classical computational-biology tradition. O through R are the learning-first culture. S is the discipline that keeps all of them honest, and it is the one most often skipped.

A.2 Reading an Atlas entry

Each family entry has a fixed shape so you can navigate without re-reading.

Problem. The question stated precisely, including what is being approximated and what is assumed.

Theory. Lectures that derive the method. Tier A or B. These are where the derivation checkpoints live.

Practice. Lectures and tutorials that run it. Tier B.

Talks. Research-level talks on the frontier of this family. Tier C.

Papers. The canonical references, with the one-or-two marked **[landmark]** that you should read twice.

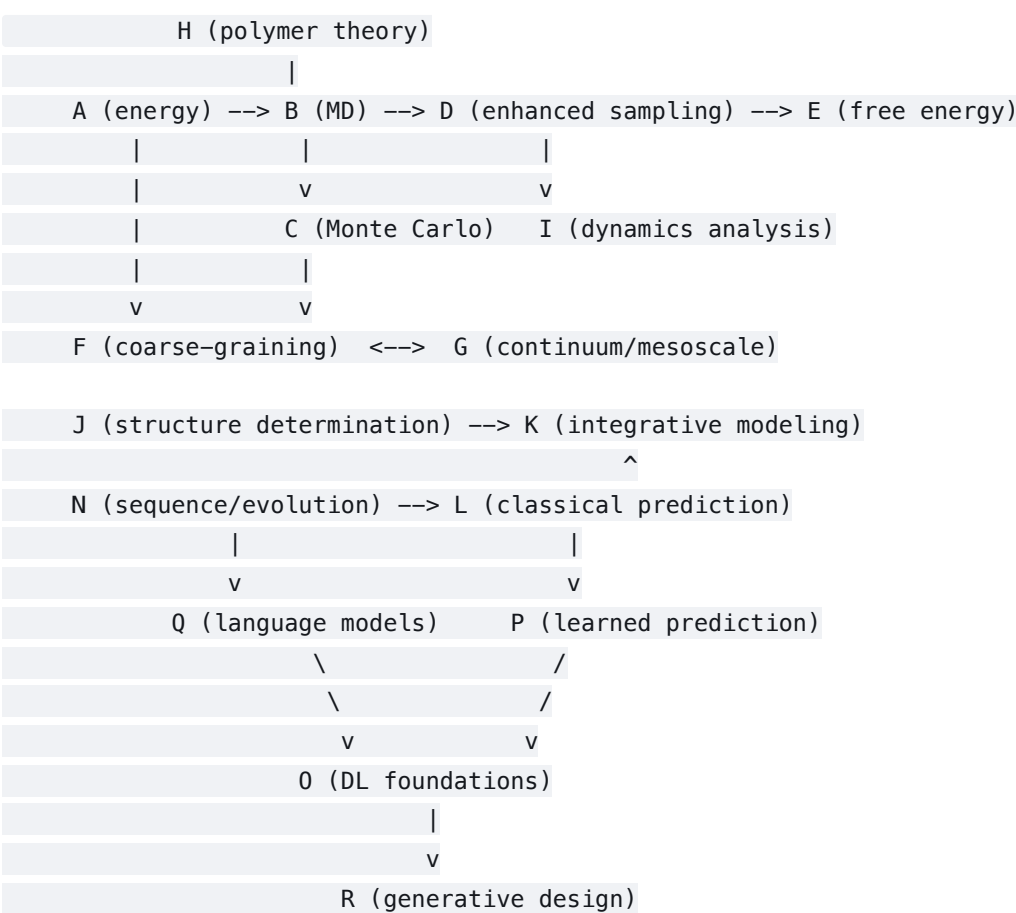
Software. What actually implements this, so you can read source.

Open. What is unsolved in this family, as its own practitioners state it. These feed the open-problems file and are where contributions come from.

The **Open** sections are the most valuable part of this document and the part most likely to be wrong within two years. Treat them as a snapshot of October 2026 and update them as you watch.

A.3 The dependency structure

Families are not independent. The practical order of study is roughly:



	v
S (assessment)	<-- applies to everything

Three dependencies are worth stating explicitly because they are commonly violated:

You cannot understand E without D, and you cannot understand D without B and C. People routinely run free energy calculations without understanding the sampling underneath, which is why so many published $\Delta\Delta G$ values are confidently wrong.

You cannot evaluate P or R without S. A structure predictor or a design method is a claim about generalization, and claims about generalization are meaningless without a defensible split. This is the single most common failure in the current literature.

You cannot usefully contribute to R without A. Generative design models produce molecules that must obey physics they were never told about. The groups producing the highest experimental success rates are the ones that kept a physical energy function somewhere in the loop.

A.4 Where the unexchanged ideas are

A working list, as of compilation, of exchanges between families that have not happened or have barely started. These are not predictions that they will work; they are observations that nobody has seriously tried. Each is a candidate project and each maps to a capstone in Part III.

From	To	The unexchanged idea
D, E	R	Convergence and error estimation for generative design. The physics culture's single greatest export; the learning culture has essentially not imported it.
D	R	Treat reverse-diffusion sampling as a rare-event problem and apply weighted ensemble or path sampling to it, rather than to a physical trajectory.
F	O	Learned coarse-graining as an architectural prior for structure models, rather than as a separate simulation technique.
H	R	Polymer-theoretic constraints as a design objective — designing for chain statistics rather than for a single backbone.
I	R	Design against an ensemble objective or a normal-mode spectrum rather than a static structure.
J, K	P	Chemical probing, HDX and crosslinking data as differentiable likelihood terms inside a learned structure model, the way MSA depth is used now.
B, D	Q	Simulation-generated training data off the PDB manifold, to teach models about the region where design actually operates.
N	S	Homology-aware and epistasis-aware splits as a standard, rather than per-paper improvisation.
C	R	Proper MCMC over design space with a defined stationary distribution, instead of greedy or temperature-sampled decoding.
G	R	Continuum electrostatics as a differentiable layer, so designs can be optimized for solubility and pKa directly.
A (ML potentials)	B, E	Machine-learned potentials at biomolecular scale, which would collapse the force-field accuracy ceiling that bounds every number in families B through E.
S	R	A blind, adversarial, CASP-style assessment for de novo design. It does not exist.

The last row is the largest single piece of missing infrastructure in this field, and it is the one most likely to make a career.

Atlas A — Potential Energy Functions

Problem. Given the positions of every atom, what is the energy? Everything downstream — dynamics, sampling, free energy, design scoring — is an operation on this function. **The accuracy of the potential is the ceiling on every number produced by families B through E**, and it is a ceiling the field has been pressing against for twenty years.

There are four tiers: empirical molecular mechanics, continuum solvation added on top, quantum mechanics for the small region where chemistry happens, and — new and genuinely disruptive — machine-learned potentials that aim at quantum accuracy at empirical cost.

A.1 Empirical force fields

Covered principally in Module B.6 and Atlas B; the construction side lives here.

Title	Speaker	Scale	Link
Open Force Field Workshops (2019–2026)	OpenFF / OMSF	10 workshop playlists	Channel
Introduction to Classical Force Fields	Emad Tajkhorshid (UIUC TCBG)	1:33:42	▶
Introduction to Force Field Toolkit (ffTK)	Christopher Mayne (UIUC)	1:08:24	▶
BioExcel #58: CHARMM Force Field Development History and Features	BioExcel CoE	1:07:56	▶
Electrostatics: Fixed Point Charges and Polarization	OpenFF	1:00:12	▶
Accuracy of the AMOEBA Force Field in Binding Free Energy Simulations	Jay Ponder (WashU)	30:24	▶
Strategies for ab initio Biomolecular Force Field Development	David Cerutti	58:42	▶
Automated Optimization of the CHARMM Lipid Force Field	Andreas Krämer	1:03:55	▶
Fragmenting molecules for QC torsion drives	Chaya Stern	57:06	▶
Graph Nets for partial charge prediction	Yuanqing Wang	51:18	▶
Machine-learned MM force fields from large-scale QC data	Ken Takaba	23:08	▶
Parameterization of AMBER ff19SB	Chuan Tian	50:32	▶
Data Curation: The Forgotten Practice in the Era of AI	Pankaj Daga	59:02	▶

Open Force Field is where force-field construction is taught publicly — torsion fragmentation, training-set selection, Bayesian parameterization, and the move toward graph-neural-network parameter assignment. If you ever want to *fit* rather than consume a force field, this is the only free curriculum.

Ponder’s AMOEBA talk is the honest assessment of what polarizability actually buys in binding free energies, which is less than its cost would suggest — a useful calibration on the general claim that more physics is always better.

A.2 Electrostatics and continuum solvation

Title	Speaker	Length	Link
CompChem 06.04: Solvation Models — Continuum (Implicit) Solvent Electrostatics	Chris Cramer (Minnesota)	23:11	▶
CompChem 06.01: Solvation Models — Chemical Phenomena	Chris Cramer	19:00	▶
CECAM Advances in Electrostatics: boundary elements	Christopher Cooper (UTFSM)	1:08:00	▶
CECAM Advances in Electrostatics: treecodes	Leighton Wilson (Michigan)	57:00	▶
The Multi-Faceted Roles of Electrostatics in Biomolecular Simulation	Protein Electrostatics seminar	1:01:00	▶
How does pH affect the distribution of ions around proteins?	Protein Electrostatics seminar	1:06:00	▶
Modeling electrostatics and polarization in biomolecules	Protein Electrostatics seminar	52:00	▶
Generalized Born and implicit solvent	Alexey Onufriev (Virginia Tech)	35:00	▶
Using 3D structure to predict protein-protein and protein-ligand interactions	Barry Honig (Columbia)	40:00	▶
Constant-pH Methods in Biomolecular Simulations, Class 1	Fernando Barroso da Silva (ICTP-SAIFR)	1:08:00	▶
Constant pH molecular dynamics in GROMACS	BioExcel CoE	19:05	▶
TABI-PB 2.0 boundary-element solver	Protein Electrostatics	50:00	▶

Cramer's 23-minute lecture is the cleanest derivation anywhere of why a continuum dielectric is a legitimate approximation and exactly where it breaks. Watch it before anything else in this subsection.

Onufriev is the generalized Born authority, and the key point is one most people get wrong: **GB is a specific analytic approximation to Poisson-Boltzmann, not an independent theory.** Knowing that tells you precisely which of its failures are fixable and which are structural.

Constant-pH MD is the correct answer to the pKa problem and the one almost everyone skips. If your designed protein has a buried ionizable residue — and peptide binders routinely do — a fixed protonation state is an assumption you have not tested.

A.3 Quantum mechanics and QM/MM

Title	Speaker	Length	Link
Hybrid QM/MM — Day 1 Lectures	CECAM	4:04:00	▶
Hybrid QM/MM — Day 2 Lectures	CECAM	4:03:00	▶
QM/MM Best Practice: Kick-off webinar	BioExcel CoE	1:36:00	▶
QM/MM Best Practice: Accurate structures and energies with QM/MM	BioExcel CoE	1:23:00	▶
QM/MM Best Practice: Chemical accuracy in enzyme catalytic mechanisms	BioExcel CoE	1:23:00	▶
QM/MM Best Practice: Studies on enzyme-catalysed reactions	BioExcel CoE	1:11:00	▶
QM/MM Best Practice: Validation of DFT functionals in QM/MM	BioExcel CoE	53:00	▶
BioExcel #51: Multiscale QM/MM with the GROMACS/CP2K interface	BioExcel CoE	51:00	▶
Modelling Enzymes with QM/MM	CCPBioSim	42:00	▶
Introduction to QM/MM, Part 1	BioExcel CoE	30:23	▶
QM/MM Methods and development	Andres Cisneros (BAGIM)	1:00:00	▶
Introduction to DFTB	Bálint Aradi (Bremen)	31:00	▶
Extended Tight Binding (GFN-xTB)	Sebastian Ehlert (Grimme group)	26:00	▶
Online-ICNI Symposium	Stefan Grimme (Bonn)	41:00	▶
Introduction to Density Functional Theory	David Sherrill (Georgia Tech)	52:00	▶
Basis Sets, part 1	David Sherrill (Georgia Tech)	34:00	▶
MIT OCW: Modern Electronic Structure Theory — Basis Sets	MIT OCW	50:00	▶
Nobel Lecture 2013	Arieh Warshel (USC)	28:00	▶

The two CECAM days are **over eight hours of formal QM/MM theory** — embedding schemes, link atoms, boundary treatment, mechanical versus electrostatic embedding — and nothing else free comes close to that depth.

The DFT-functional validation lecture will save you real time: it is a direct empirical answer to “which functional do I actually use for a barrier height,” which is otherwise a question you resolve by burning a month.

Warshel's Nobel lecture is the origin of both QM/MM and the empirical valence bond method, from the person who invented them. It pairs with Atlas family R's enzyme design material: electrostatic preorganization is the hypothesis every enzyme design objective is implicitly betting on.

A.4 Machine-learned interatomic potentials — the disruptive tier

Why this subsection matters more than its length suggests. If machine-learned potentials reach biomolecular scale with transferable accuracy, they collapse the ceiling that bounds every number in Atlas families B through E. That is the largest single thing that could happen to computational biophysics this decade, and it is currently happening in materials science rather than in biology.

Title	Speaker	Length	Link
Tutorial: How to Train and Use MACE Interatomic Potentials, Part 1	Ilyes Batatia (Cambridge)	1:06:00	▶
Tutorial: How to Train and Use MACE, Part 2	Ilyes Batatia	50:00	▶
Foundation Models for Materials and Molecules	Gábor Csányi (Cambridge)	58:00	▶
Machine Learning Meets Quantum Chemistry	Klaus-Robert Müller (TU Berlin)	51:00	▶
Challenges for General-Purpose MLFFs	Alexandre Tkatchenko (Luxembourg)	51:00	▶
MACE: Higher Order Equivariant Message Passing NNs	Valence Labs	1:22:00	▶
Local Equivariant Representations for Large-Scale Atomistic Dynamics	Batzner & Musaelian (Harvard)	1:26:00	▶
The state of neural network interatomic potentials	Justin Smith (ANI author)	41:00	▶
Uncertainty-aware ML models of many-body atomic interactions	Boris Kozinsky (Harvard)	57:00	▶
Learning how to break symmetry with symmetry-preserving networks	Tess Smidt (MIT)	59:00	▶
AIMNet2: A Robust Neural Network Potential	Roman Zubatyuk (CMU)	40:00	▶
AI Solutions for Computational and Organic Chemistry	Olexandr Isayev (CMU)	1:05:00	▶

Batzner and Musaelian's talk is the conceptual centerpiece: it explains why *strict* equivariance buys extreme data efficiency, and why Allegro's strict locality buys scale. That argument — a symmetry constraint as a substitute for data — is the single most transferable idea from this subsection into protein design, where data is the binding constraint.

Tkatchenko’s “Challenges” talk is the necessary skepticism. It asks what “foundation model” means for a potential and whether current benchmarks measure transferability at all. Watch it immediately after Csányi’s foundation-model talk and notice that they disagree.

AIMNet2 is the one currently usable for drug-like molecules, including charged species — which is why it, rather than the materials-focused models, is the one to try on your own systems.

A.5 BUILD — Atlas A

1. **Implement a Lennard-Jones plus Coulomb energy function** from scratch for a small molecule. Add a simple reaction-field or generalized-Born correction. Compare solvation free energies to a PB solver.
2. **Write the Poisson-Boltzmann equation** and solve it numerically in 1D for a charged plane. Derive the Debye length from your solution and check against the analytic result.
3. **Run a constant-pH simulation** on a protein with a buried ionizable residue. Compare the computed pKa to the fixed-protonation assumption your pipelines normally make. Quantify the error that assumption introduces.
4. **Train a small machine-learned potential** on a dataset of your own QM calculations — your CAPN5 DFT work is directly usable here. Measure how it degrades off the training manifold. That degradation curve is the transferability question of Section A.4, measured on your own data.

Derivation checkpoints due: 1, 2, 3, 4, 5, 34, 35, 36.

A.6 Paired reading — Atlas A

Watch this	Then read this	Hold this question
Tajkhorshid, <i>Classical Force Fields</i>	Best et al. 2012 (CHARMM36); Maier et al. 2015 (ff14SB)	For each term, name the experimental or QM data it was fit to.
OpenFF workshops	Any OpenFF parameterization paper	How is a torsion parameter determined, and how transferable is it?
Ponder, <i>AMOEBA accuracy</i>	Ponder et al. 2010, JPCB 114:2549	What does polarizability buy, and is it worth the cost?
Cramer, <i>Continuum solvent electrostatics</i>	Honig & Nicholls 1995, Science 268:1144	When is a continuum dielectric legitimate, and what is the dielectric constant of a protein?
Cooper / Wilson, CECAM electrostatics	Baker et al. 2001, PNAS 98:10037 (APBS)	How is PB actually solved at protein scale, and what is the discretization error?
Onufriev, <i>Generalized Born</i>	Still et al. 1990, JACS 112:6127; Onufriev et al. 2004, Proteins 55:383	GB approximates PB. Derive the approximation and state when it fails.
Barroso / BioExcel, <i>Constant-pH MD</i>	Swails, York & Roitberg 2014, JCTC 10:1341	A fixed protonation state is an assumption. How large is the error?
CECAM QM/MM Days 1–2	Warshel & Levitt 1976, JMB 103:227; Senn & Thiel 2009, Angew Chem 48:1198	Compare mechanical and electrostatic embedding. What does the link atom break?
BioExcel, <i>DFT functional validation</i>	Any QM/MM barrier-height benchmark	Which functional for a barrier height, and what is the residual error?
Grimme / Ehlert, <i>xTB</i>	Bannwarth, Ehlert & Grimme 2019, JCTC 15:1652 (GFN2-xTB)	Semi-empirical at near-DFT accuracy. What is sacrificed?
Warshel, Nobel Lecture	Warshel & Weiss 1980, JACS 102:6218 (EVB)	Electrostatic preorganization is the mechanism. What design objective follows?
Batatia, <i>MACE tutorial</i>	Batatia et al. 2022, NeurIPS (MACE); Kovács et al. 2023 (MACE-OFF23)	Train one. Then measure its error on a molecule outside the training set.
Batzner & Musaelian	Batzner et al. 2022, Nat Commun 13:2453 (NequIP); Musaelian et al. 2023 (Allegro)	Strict equivariance buys data efficiency. Quantify how much.
Csányi, <i>Foundation Models</i> vs. Tkatchenko, <i>Challenges</i>	Both sides' position papers	Write the strongest version of each case. Which benchmark would settle it?
Zubatyuk, <i>AIMNet2</i>	Anstine, Zubatyuk & Isayev 2024/25	Charged species break most MLIPs. How does AIMNet2 handle them?
Smidt, <i>Breaking symmetry</i>	Thomas, Smidt et al. 2018 (tensor field networks)	When should a model be allowed to break a symmetry it was given?

Atlas B — Molecular Dynamics

Problem. Integrate Newton’s equations for a biomolecule in explicit solvent and get a trajectory you are willing to defend.

The hardest part of MD is not running it. It is knowing what the trajectory means, and the honest answer for most published simulations is **less than the figures imply**. This family therefore spends as much space on convergence and error as it does on integrators.

B.1 Foundations — the integrator, the thermostat, the box

Course / talk	Speaker / host	Length	Link
Biophysical Chemistry 2016 — complete, 13 lectures (Appendix B.2)	Erik Lindahl (KTH/Stockholm)	playlist	▶
Molecular Dynamics Simulation (lecture series)	Lindahl	~50 min ea	▶
Introduction to Molecular Dynamics	Mark Tuckerman (CECAM / IAS)	~1 hr	see B.8
MD simulations of biomolecules — foundations	Alex MacKerell (CHARMM workshops)	~1 hr	see B.8
GROMACS tutorials, full playlist	Justin Lemkul	varies	▶
Rosetta: Refining structures with minimization and Monte Carlo	RosettaCommons	39:13	▶

Required reading, in this order. Frenkel & Smit, *Understanding Molecular Simulation*, chapters 4 and 6 (the integrator and the thermostat); Tuckerman, *Statistical Mechanics: Theory and Molecular Simulation*, chapter 3 (Liouville operator, the symplectic property, why velocity Verlet is not an arbitrary choice); Allen & Tildesley chapter 1.

Derivation checkpoint 7. Derive velocity Verlet from the Trotter factorization of the Liouville propagator. Then explain, without hand-waving, why a symplectic integrator has bounded energy error over long times while a higher-order non-symplectic one drifts. Almost nobody who runs MD daily can do this, and it is the reason the field uses a second-order method in 2026.

The thermostat trap. Berendsen is not a thermostat — it does not sample the canonical ensemble, and it famously produces the “flying ice cube.” Nosé-Hoover is canonical but not ergodic for a single harmonic oscillator, which is why chains exist. Langevin is canonical and ergodic but changes the dynamics. **Derivation checkpoint 8: write down which observables are safe under each and why**, and then go and check what your own last three simulations used.

B.2 Anton and the long-timescale program

Talk	Speaker	Length	Link
Millisecond-Scale Molecular Dynamics Simulations	D. E. Shaw	~60 min	search "D. E. Shaw" Anton molecular dynamics lecture
Biomolecular Simulation at Long Timescales	D. E. Shaw Research	varies	▶
Large-scale conformational transition in a membrane transporter	Trebesch / Fakharzadeh, Tajkhorshid lab (Broad MIA)	1:34:25	▶

Papers, and read them as a sequence: Shaw et al. 2010, *Science* 330:341 (Anton, atomic-level characterization of folding) · Lindorff-Larsen et al. 2011, *Science* 334:517 (How fast-folding proteins fold) · Shaw et al. 2014, SC'14 (Anton 2) · Shaw et al. 2021 (Anton 3) · Dror et al. 2012, *Annu Rev Biophys* 41:429 (biomolecular simulation review).

What Anton actually proved. Not that proteins fold — we knew that. It proved that *a force field extrapolated far beyond its fitting data stays approximately right*, which nobody had the right to expect. The 2011 paper folded twelve proteins and got the right structures with the right relative rates. Read the supplement: the places where it failed are the places where the force field's limitations show, and those are still the limitations today.

Then ask the question the paper does not: **if a special-purpose machine is the answer to timescale, why did the field spend the next decade building learned surrogates instead?** The answer involves cost, access, and the fact that one millisecond of one protein is not a research program. That tension is the subject of Atlas D and of Capstone IV.

B.3 Force fields in practice — and the parts that are wrong

Talk	Speaker / host	Length	Link
Force field development and validation	MacKerell lab / CHARMM workshop	~1 hr	see B.8
Benchmark and Critical Evaluation for ML Force Fields	Xiang Fu (Valence)	1:00:38	▶
Improving Speed and Accuracy of NN Interatomic Potentials	Valence	54:24	▶
MACE: Higher Order Equivariant Message Passing for Force Fields	Valence	1:22:55	▶
Allegro: Local Equivariant Representations for Large-Scale Dynamics	Musaelian (Harvard, Valence)	1:09:17	▶
NequIP (earlier session)	Batzner & Musaelian	1:26:43	▶
Learning QM Using Embedding, Symmetric Polynomials and Composition	Gábor Csányi (Valence)	38:34	▶
Learning ML Interatomic Potentials	De Fabritiis (ML4DD)	51:39	▶
Orb-v3: atomistic simulation at scale	Duignan & Vandenhaute (Valence)	1:13:35	▶
Transformers Discover Molecular Structure Without Graph Priors	Kreiman (Berkeley, Valence)	1:04:09	▶
Stability-Aware Boltzmann Estimator (StABLE) Training	Sanjeev Raja (Valence)	1:22:10	▶
Scalable simulation of electrolytes with quantum chemical accuracy	Duignan (Valence)	1:12:12	▶
AquaGen: generative models at MD precision on thousands of atoms	Valence	1:16:27	▶

Papers: Best et al. 2012, *JCTC* 8:3257 (CHARMM36) · Huang et al. 2017, *Nat Methods* 14:71 (**CHARMM36m — the IDP correction**) · Maier et al. 2015, *JCTC* 11:3696 (ff14SB) · Tian et al. 2020, *JCTC* 16:528 (ff19SB) · Robustelli et al. 2018, *PNAS* 115:E4758 (**a99SB-disp, the honest IDP comparison**) · Batzner et al. 2022, *Nat Commun* 13:2453 (NequIP) · Musaelian et al. 2023, *Nat Commun* 14:579 (Allegro) · Batatia et al. 2022, *NeurIPS* (MACE) · Fu et al. 2023, TMLR (**the ML force field benchmark that shows low force MAE does not imply stable simulation**).

The single most load-bearing fact in this subsection. Fu et al. showed that ML potentials with excellent force errors produce *unstable* trajectories — they blow up, or they drift into unphysical configurations, on timescales long enough to matter. Force MAE is a proxy, and it is a bad one. StABLE training (Raja) is one attempt at a fix.

This is the cleanest generator of new methods in the Atlas. The pattern — “the standard metric for this learned object is not the quantity you care about” — recurs in Atlas R (sequence recovery), Atlas M (docking RMSD without PoseBusters), Atlas Q (ProteinGym means over stratified splits) and Atlas P (pLDDT as a proxy for correctness). **Derivation checkpoint 9 asks you to state the general form of that failure.**

In practice: the water model is a force field choice. TIP3P is used because CHARMM and AMBER were parameterized against it, not because it is good — it has the wrong dielectric constant and the wrong diffusion coefficient. TIP4P-Ew and OPC are better and change your results. If your system is an IDP or has large solvent-exposed loops, this is not a detail.

B.4 Running MD correctly — the practitioner layer

Topic	Resource	Link
GROMACS complete tutorials	Justin Lemkul, 8 systems	▶
GROMACS performance tuning	GROMACS workshop talks (BioExcel)	▶
BioExcel Building Blocks: interoperable simulation workflows	RosettaCommons Workflows	▶
OpenMM	OpenMM workshop recordings	▶
MDAnalysis	MDAnalysis UGM talks	▶
Setup and parameterization	CHARMM-GUI tutorials	▶

The practitioner’s checklist that papers never print. Hydrogen mass repartitioning buys you $1.7\times$ at 4 fs — and silently breaks if your water model’s oxygen mass was repartitioned too. Periodic image artifacts in a rhombic dodecahedron are not fixed by any ordering of `trjconv` flags when you have two chains plus a ligand. `-pbc nojump` splits a receptor that crosses the boundary; `-fit` re-images a bound peptide into the void. Every one of these has cost someone a month.

BUILD, and do this before anything else in this family. Take one of your own completed trajectories. Compute the density. If it is near 926 kg/m^3 instead of ~ 1028 , your water was light — a repartitioned oxygen mass under HMR — and the trajectory is wrong. This exact failure has been found in production force field directories. **Checking is one `gmx energy` call.**

B.5 Convergence, error bars, and what a trajectory does not tell you

This is the most important subsection in Atlas B and the one most often skipped.

Talk	Speaker / host	Length	Link
Statistical analysis and error estimation in MD	Michael Shirts (Alchemy / CECAM)	~1 hr	see B.8
Big Data for Protein Dynamics	Hannah Wayment-Steele (MLSB 2025)	33:47	▶
Systematic Analysis of Conformational Ensembles with PENSE	Vögele (Stanford, Valence)	50:33	▶
Boltzmann Weighted Ensembles	Pratyush Tiwary (MLSB 2025)	31:12	▶

Papers: Grossfield & Zuckerman 2009, *Annu Rep Comput Chem* 5:23 (Quantifying uncertainty and sampling quality) [read this one twice] · Chodera 2016, *JCTC* 12:1799 (automatic equilibration detection) · Zuckerman & Chong 2017, *Annu Rev Biophys* 46:43 (weighted ensemble) · Shirts & Chodera 2008, *JCP* 129:124105 (MBAR) · Flyvbjerg & Petersen 1989, *JCP* 91:461 (block averaging) · Romo & Grossfield 2011, *JCTC* 7:2464.

Daniel Zuckerman's thread, which is one of your ten, runs straight through here. His position, compressed: *RMSD plateauing is not convergence*. A trajectory can look flat for a microsecond and be stuck in one basin. The quantities that do diagnose sampling are block-averaged observables with honest error bars, multiple independent runs started from different seeds, and overlap measures between replicate ensembles.

Derivation checkpoint 10. Take a trajectory you already own. Compute the statistical inefficiency and effective sample size of three observables: total energy, radius of gyration, and a specific side-chain dihedral. They will differ by orders of magnitude. Write down what that means for the error bar on any claim you have made from that trajectory.

Checkpoint 11. Explain precisely why RMSF computed from a single trajectory is not an estimate of equilibrium fluctuation unless the trajectory is ergodic on the relevant timescale — and state the test that would tell you whether it is.

B.6 Specialized regimes

Regime	Talk / resource	Link
Membranes	CHARMM-GUI membrane builder tutorials; MARTINI workshops	▶
Constant-pH MD	CECAM constant-pH workshops	see B.8
QM/MM	CECAM QM/MM school, ~8 hrs	see Atlas A.3
Polarizable force fields	AMOEBA / Drude workshop talks	see B.8
Nucleic acids	OL15/OL21 parameterization talks (Šponer, Otyepka)	—
Enhanced-sampling MD	→ Atlas D	—
Coarse-grained MD	→ Atlas F	—

Papers for nucleic acids specifically, because the Atlas is otherwise protein-heavy and this is where DNA/RNA simulation lives: Zgarbová et al. 2011, *JCTC* 7:2886 (χOL3 for RNA) · Galindo-Murillo et al. 2016, *JCTC* 12:4114

(OL15 for DNA) · Šponer et al. 2018, *Chem Rev* 118:4177 (RNA structural dynamics — the honest review of what RNA force fields get wrong).

B.7 BUILD — Atlas B

1. **Run the same system three times** with three different random seeds, 100 ns each. Not one run — three.
2. **Compute block-averaged error bars** on three observables across the replicates. Plot the error bar as a function of block size and find the plateau.
3. **Compute the overlap** between the three ensembles in a reduced space (PCA on backbone, or a few collective variables). Report it as a number.
4. **Write the one page.** Which of your observables converged in 100 ns, which did not, and what would it have taken? Put a number on the shortfall.

Derivation checkpoints due: 7, 8, 9, 10, 11.

B.8 Where the workshop recordings live

Several resources above are workshop series rather than single videos. These are the standing archives for this family:

Archive	What it has	Link
CECAM	Schools on QM/MM, enhanced sampling, coarse-graining, free energy	cecam.org
BioExcel CoE	GROMACS, HADDOCK, PMX workshops and performance tuning	▶
MolSSI	Software-engineering-for-simulation schools	molssi.org
Alchemistry / Chodera lab	The 2018 free-energy workshop, 43 talks	→ Atlas E
PRACE / PATC	HPC-for-simulation training	—
Valence Labs	The ML-potential talks listed in B.3	▶

B.9 Paired reading — Atlas B

Watch this	Then read this	Hold this question
Lindahl, <i>Biophysical Chemistry</i>	Frenkel & Smit ch. 4, 6	Derive velocity Verlet from Trotter. Why second order and not fourth?
Lindahl, thermostats	Tuckerman ch. 4	Which thermostats sample the canonical ensemble, and which merely control T?
Shaw, <i>Millisecond MD</i>	Shaw et al. 2010, <i>Science</i> 330:341	Special-purpose silicon. What question did it make askable?
—	Lindorff-Larsen et al. 2011, <i>Science</i> 334:517	Read the supplement. Which proteins failed, and what does that say about the force field?
Mackerell, force fields	Huang et al. 2017, <i>Nat Methods</i> 14:71 (CHARMM36m)	What was wrong with CHARMM36 for IDPs, and how was it diagnosed?
—	Robustelli et al. 2018, <i>PNAS</i> 115:E4758	Four force fields on folded and disordered proteins. Who wins where?
Valence, <i>MACE</i>	Batatia et al. 2022, <i>NeurIPS</i>	Higher-order equivariant messages. What does body-order buy?
Musaelian, <i>Allegro</i>	Batzner et al. 2022, <i>Nat Commun</i> 13:2453	Strictly local versus message-passing. What scales, and what is lost?
Fu, <i>ML force field benchmark</i>	Fu et al. 2023, <i>TMLR</i>	Low force MAE, unstable simulation. Write the general form of this failure.
Raja, <i>StABIE training</i>	The StABIE paper	How do you train against a simulation observable rather than against forces?
Kreiman, <i>Transformers without graph priors</i>	The paper	If a transformer rediscovers the geometry, what was the inductive bias worth?
Shirts, <i>Error estimation</i>	Grossfield & Zuckerman 2009, <i>Annu Rep Comput Chem</i> 5:23	Compute statistical inefficiency for your own trajectory.
Tiary, <i>Boltzmann weighted ensembles</i>	Zuckerman & Chong 2017, <i>Annu Rev Biophys</i> 46:43	A sample is not a sample from the right distribution. How would you test?
Broad MIA, <i>Membrane transporter transition</i>	The accompanying paper	A large conformational change in explicit membrane. How was it sampled?

Atlas C — Monte Carlo and Stochastic Sampling

Problem. You want expectations under a distribution you can evaluate up to a constant but cannot sample from directly. Molecular dynamics is one answer — follow a trajectory and hope it is ergodic. Monte Carlo is the other, and it is more general: construct a Markov chain whose stationary distribution is the one you want, and let it run. **Almost every method in this document is a Monte Carlo method wearing different clothes**, including Rosetta’s design protocol, simulated annealing in NMR refinement, and the reverse process of a diffusion model.

This family is badly underrepresented in biomolecular curricula, which is why so many practitioners can run a Metropolis move but cannot say what would make it incorrect.

C.1 Fundamentals

Werner Krauth — Statistical Mechanics: Algorithms and Computations

ENS Paris · Tier A

The spine of this family. Krauth builds Metropolis from a pebble game, so detailed balance and rejection arrive as consequences rather than incantations.

Fragility note. These individual videos sit on a third-party re-upload channel rather than an institutional one, so they could disappear. The durable route is the free-to-audit [Coursera course](#); the ICTP lectures below are the institutionally-hosted fallback.

#	Lecture	Length	Link
1	Introduction to Monte Carlo algorithms	28:08	▶
2	Hard disks: from Classical to Statistical Mechanics	35:59	▶
4	Sampling and Integration: from Gaussians to Maxwell and Boltzmann	32:20	▶
8	Ising model: from enumeration to Cluster Monte Carlo	28:49	▶
9	Dynamical Monte Carlo and the Faster-than-the-Clock approach	25:07	▶
10	The Alpha and the Omega of Monte Carlo (convergence, error bars)	32:28	▶
T1	Tutorial: Exponential convergence and the 3×3 pebble game	32:39	▶
T8	Tutorial: Heat bath algorithm, coupling of Markov chains	19:21	▶

Tutorial 8's coupling-of-Markov-chains material is the cleanest free explanation of exact sampling anywhere — Propp-Wilson coupling from the past, which terminates with a sample guaranteed to be from the stationary distribution rather than approximately so.

Krauth at ICTP — same material, graduate pace, roughly double the time, with autocorrelation and error analysis done more rigorously: [I](#) (1:33:56) · [II](#) (2:05:19) · [III](#) (2:03:39)

Why high-dimensional sampling needs MCMC at all

Title	Speaker	Length	Link
MacKay Lec 12: Monte Carlo Methods I	David MacKay (Cambridge)	1:23:48	▶
MacKay Lec 13: Monte Carlo Methods II, Slice Sampling	David MacKay	1:47:57	▶
Markov Chain Monte Carlo	Iain Murray (Edinburgh, MLSS Africa)	1:17:50	▶
Introduction to MCMC for Deep Learning	Iain Murray (IPAM)	1:01:50	▶
Pseudo-Marginal Slice Sampling	Iain Murray	32:08	▶
A Gentle Introduction to Markov chain Monte Carlo	Dootika Vats (IIT Kanpur)	1:08:45	▶

MacKay gives the best motivation anywhere for *why* importance sampling and rejection sampling die in high dimension — the fact that makes MCMC necessary rather than merely convenient. **Murray is the clearest living explainer of why detailed balance is sufficient but not necessary**, and of what burn-in does and does not fix.

Vats is the authority on honest MCMC standard errors. Her lecture is where effective sample size stops being a number a library prints and becomes something you can defend in a referee reply.

Course modules

Title	Instructor	Length	Link
Lec 34: Markov chain algorithm, equilibrium and detailed balance	NPTEL / IIT Roorkee	38:04	▶
Lec 35: Metropolis algorithm, periodic boundary conditions	NPTEL / IIT Roorkee	45:46	▶
MIT 3.320 Lec 17: Monte Carlo Simulations	Ceder & Marzari (MIT)	1:14:31	▶
MIT 3.320 Lec 18: Monte Carlo Simulation II	Ceder & Marzari	1:15:27	▶
Lec 30: Introduction to Markov Chain Monte Carlo	NPTEL / IIT Kanpur	24:30	▶
Computational Physics: Monte Carlo methods	Morten Hjorth-Jensen (Oslo/MSU)	1:28:30	▶

Sokal’s Cargèse lectures on autocorrelation and critical slowing down are the canonical treatment and have no video. His recorded Newton Institute lectures are on combinatorics, not MC algorithms — do not substitute them. Read the written notes.

C.2 Advanced MCMC

Hamiltonian Monte Carlo

Title	Speaker	Length	Link
New Monte Carlo Methods Based on Hamiltonian Dynamics	Radford Neal (Toronto)	1:05:31	▶
Efficient Bayesian inference with HMC, Part 1	Michael Betancourt (MLSS Iceland)	1:29:43	▶
HMC and Stan, Part 2	Michael Betancourt	42:26	▶
Unravelling A Geometric Conspiracy	Michael Betancourt (LOGML)	58:06	▶
Scalable Bayesian Inference with HMC	Michael Betancourt	53:20	▶
No-U-Turn Sampler (NIPS 2011)	Matthew D. Hoffman	20:51	▶
The No-U-Turn Sampler (2024)	Nawaf Bou-Rabee (Rutgers)	58:31	▶
An Introduction to HMC for Sampling	Simons Institute	1:10:16	▶
Lec 36: Hamiltonian Monte Carlo	NPTEL / IIT Kanpur	40:29	▶

Neal presenting his own method is the highest-value item here. And note what HMC actually is: a molecular dynamics integrator used as an MCMC proposal, with a Metropolis correction for integration error. The physics culture invented the dynamics; the statistics culture noticed it was a proposal distribution. That is a completed cross-cultural carry, and knowing it happened once makes the next one easier to see.

Nested sampling, Wang-Landau, tempering

Title	Speaker	Length	Link
Computing Bayes in Big Spaces	John Skilling (MaxEnt 2011)	43:23	▶
Big spaces: Quantification	John Skilling (MLSS Africa)	1:33:17	▶
A Brief Introduction to Nested Sampling	Joshua Speagle (Toronto)	38:02	▶
Nested Sampling from scratch	Johannes Buchner (MPE Garching)	1:00:00	▶
State of the art in nested sampling and MCMC	Johannes Buchner	1:46:52	▶
Replica-Exchange Wang-Landau Sampling, Lec 1	David P. Landau (Georgia)	45:00	▶
Replica-Exchange Wang-Landau Sampling, Lec 2	Ying-Wai Li (LANL)	55:44	▶
Replica-Exchange Wang-Landau Sampling, Lec 3	Ying-Wai Li	53:20	▶
Recent advances in accelerated Monte Carlo algorithms	Ying-Wai Li	54:55	▶
Theory of simulated tempering and replica exchange	Massimiliano Bonomi (PASI 2012)	1:02:31	▶
Non-Reversible Parallel Tempering	Alexandre Bouchard-Côté (UBC)	33:35	▶
Non-Reversible Parallel Tempering: a Scalable Scheme	Saifuddin Syed (Oxford)	58:21	▶

Landau presenting Wang-Landau is the draw. Nested sampling is the one method here that computes the *evidence* — the normalizing constant — which is exactly the partition function. A physicist and a Bayesian statistician are computing the same object and almost never say so to each other.

Cluster algorithms, non-reversible MCMC, sequential MC

Title	Speaker	Length	Link
Reversible and irreversible Markov chains in statistical physics, Lec 1	Werner Krauth	1:04:16	▶
Reversible and irreversible Markov chains, Lec 2	Werner Krauth	51:55	▶
Fast irreversible Markov chains in statistical physics	Werner Krauth	1:10:34	▶
Swendsen-Wang on the Mean-Field Potts Model	Simons Institute	45:49	▶
Asymmetric momentum sampling in event-chain Monte Carlo	Michael Faulkner (Bristol)	53:03	▶
The coordinate sampler: a non-reversible Gibbs-like MCMC	Christian P. Robert (Paris-Dauphine)	32:45	▶
Tutorial on Sequential Monte Carlo methods	Anthony Lee (Bristol)	1:02:58	▶
Particle Markov chain Monte Carlo, Part 1	Fredrik Lindsten (Linköping)	1:16:17	▶

The event-chain and lifted-MCMC thread is the one nobody teaches and the one most likely to transfer.

These chains violate detailed balance deliberately — they satisfy the weaker global balance condition — and get order-of-magnitude speedups on hard-particle and polymer systems as a result. If you ever wondered what “sufficient but not necessary” buys you, this is the answer, in production.

C.3 Monte Carlo in molecular systems

Title	Speaker	Length	Link
Lec 52: Extension of canonical MC to other ensembles (GCMC)	NPTEL / IIT Roorkee	33:55	▶
Lec 53: Monte Carlo in Gibbs and semi-grand canonical ensembles	NPTEL / IIT Roorkee	24:37	▶
ChEn 513: Gibbs Ensemble Monte Carlo	Thomas Knotts (BYU)	1:27:50	▶
Lec 23: Monte Carlo simulations of polymer chains	NPTEL / IIT Roorkee	46:58	▶
Introduction to Atomic Simulations by Metropolis Monte Carlo	Mathieu Bauchy (UCLA)	2:36:28	▶
L21: Kinetic Monte Carlo	Peter Kratzer (Duisburg-Essen)	53:16	▶
MIT 10.34 Lec 34: Stochastic Chemical Kinetics 1	Green & Swan (MIT)	53:57	▶
MIT 10.34 Lec 35: Stochastic Chemical Kinetics 2 (tau-leaping)	Green & Swan	47:32	▶
Dan Gillespie presenting the Gillespie Algorithm	Daniel T. Gillespie	39:37	▶
Refining biomolecular structures with minimization and Monte Carlo	RosettaCommons	39:13	▶
Rotamer Libraries and Side-chain Packing	Brian Weitzner (Gray course, JHU)	1:05:04	▶
StepWise Monte Carlo for modeling and design of RNA and protein	Rhiju Das (Stanford)	28:20	▶
Cryo-BIFE: Bayesian Inference of Free Energy profiles	Pilar Cossio (Flatiron)	37:34	▶

The two Rosetta items are the ones to study hardest. The first is the Monte-Carlo-plus-Minimization protocol; the second is the simulated-annealing side-chain packer underneath every design run you have ever launched. Then ask the question that Atlas R's open problems hinge on: *what is the stationary distribution of Rosetta's design protocol?* It is not the Boltzmann distribution of the energy function, because the annealing schedule and the minimization steps break detailed balance. Nobody has characterized it properly.

Gillespie explaining why his algorithm is *exact* rather than an ODE approximation has unusual primary-source value.

Configurational bias Monte Carlo has no dedicated free video. The polymer-chain and Gibbs-ensemble lectures set up the Rosenbluth chain-growth logic; read Siepmann & Frenkel 1992 and Frenkel & Smit ch. 13.

C.4 Rare-event and path sampling

Title	Speaker	Length	Link
Transition path sampling of complex activated processes	Peter Bolhuis (Amsterdam, ICTP)	1:21:36	▶
Understanding emergent rare event behaviour in high-dimensional systems	Peter Bolhuis	1:09:02	▶
TPS and quantitative mechanistic hypothesis testing	Baron Peters (IPAM)	1:09:37	▶
Path Sampling and OpenPathSampling	David W. H. Swenson (ENS Lyon)	1:37:43	▶
∞ RETIS: exchanging replicas with unequal cost	Titus S. van Erp (NTNU)	36:21	▶
Forward Flux Sampling	Sapna Sarupria (Minnesota)	23:08	▶
Introduction to Weighted Ensemble Simulation	Daniel M. Zuckerman (OHSU)	31:14	▶
Recent Advances in Weighted Ensemble Simulation	Daniel M. Zuckerman	35:59	▶
Weighted ensemble simulations of long-timescale dynamics	Lillian T. Chong (Pittsburgh)	56:39	▶
Introduction to WESTPA	Matthew C. Zwier	48:44	▶
Rare Events: Theory	Eric Vanden-Eijnden (Courant/NYU)	59:53	▶
Rare Events: Numerical Implementation (string method)	Eric Vanden-Eijnden	1:08:10	▶
Exploring Complex Reaction Pathways I (string with swarms)	Mahmoud Moradi (Arkansas)	1:24:20	▶
Adaptive multilevel splitting for molecular dynamics	Arnaud Guyader (Sorbonne)	1:05:08	▶
Modeling Transition States (NEB / CI-NEB)	Shyue Ping Ong (UCSD)	37:10	▶
Lecture 14: The Committor Function	Pietro Faccioli (Milano-Bicocca)	37:29	▶

The committor is the exact reaction coordinate — the probability of reaching the product before the reactant — and every collective variable anyone has ever hand-picked is an approximation to it. Faccioli's lecture is short and it reframes the entire enhanced-sampling enterprise.

Milestoning has no free video lecture. Read Faradjian & Elber 2004 and Elber 2020.

C.5 Monte Carlo meets machine learning — the live frontier

This is the most important subsection in Atlas C for someone trying to make a contribution, because the exchange is happening right now and is incomplete.

Title	Speaker	Length	Link
ML for Sampling Probability Distributions in Lattice Field Theory	Phiala Shanahan (MIT)	1:05:34	▶
Building symmetries into generative flow models	Phiala Shanahan	1:11:50	▶
Normalizing Flows for Lattice Gauge Theory, Part 1	Gurtej Kanwar (Bern)	1:32:03	▶
Towards Flow-based MCMC for Lattice Gauge Theory with Fermions	Danilo Rezende (DeepMind)	1:16:22	▶
Boltzmann Generators	Frank Noé (FU Berlin / MSR)	22:39	▶
Deep Generative Learning for Physics Many-Body Systems	Frank Noé (IPAM)	1:01:56	▶
Stochastic Normalizing Flows	Frank Noé	15:58	▶
Equivariant flow matching	Leon Klein (FU Berlin)	1:08:07	▶
Flow Annealed Importance Sampling Bootstrap	Midgley & Stimper	1:33:26	▶
Generalizing HMC with Neural Networks (L2HMC)	Jascha Sohl-Dickstein (Google Brain)	46:55	▶
Enhancing MCMC with Deep Learning	Eric Vanden-Eijnden (Courant)	46:05	▶
Adaptive Monte Carlo with Normalizing Flows	Marylou Gabrié (NYU/Flatiron)	1:00:37	▶
Diffusion models for sampling	Arnaud Doucet (Oxford/DeepMind)	42:43	▶
Non-equilibrium transport and tilt matching for sampling	Michael Albergo (Harvard/NYU)	38:53	▶

Watch Gabrié and Vanden-Eijnden as a pair. They explain why a learned global proposal plus a local MCMC kernel beats either alone, and — crucially — they name the mode-collapse failure mode honestly. The lattice field theory community (Shanahan, Kanwar, Rezende) has been doing flow-based sampling with exactness guarantees for years, largely unread by the biomolecular community.

FAB is the one to study for a design application. It trains a flow on a Boltzmann target *with no samples, only an energy function* — which is precisely the situation for any designed protein that does not exist yet.

C.6 Quantum and Bayesian Monte Carlo

Title	Speaker	Length	Link
Introduction to Monte Carlo	David Ceperley (UIUC)	1:03:42	▶
Variational Monte Carlo	David Ceperley	57:17	▶
Diffusion Monte Carlo, Part 1	David Ceperley	54:54	▶
Natural QMC Computation of Excited States (FermiNet)	David Pfau (DeepMind)	25:39	▶
Neural-network wave functions for quantum chemistry (PauliNet)	Jan Hermann (FU Berlin/MSR)	50:41	▶
Approximating Many-Electron Wave Functions using Neural Networks	Matthew Foulkes (Imperial)	50:02	▶

Bayesian computation and diagnostics

Title	Speaker	Length	Link
Statistical Rethinking 2023 Lec 08: MCMC	Richard McElreath (MPI-EVA)	1:16:17	▶
BDA Lec 5.2: Convergence diagnostics, R-hat and ESS	Aki Vehtari (Aalto)	37:39	▶
BDA Lec 6.1: HMC, NUTS, dynamic HMC and diagnostics	Aki Vehtari	49:59	▶
A Few of My Favorite Diagnostics	Aki Vehtari (PyMCon)	58:54	▶
Practical pre-asymptotic diagnostic of Monte Carlo estimates	Aki Vehtari	50:14	▶
PSIS-LOO and K-fold cross-validation	Aki Vehtari	50:50	▶
Bayesian Workflow	Andrew Gelman (Columbia)	1:02:48	▶
Problems I'd like to solve in Stan	Andrew Gelman	36:14	▶
How does Stan work?	Bob Carpenter (Flatiron)	55:47	▶
Debugging Bayesian Inference	PyMC Labs	54:41	▶
The state of Bayesian workflows in JAX	Colin Carroll (Google)	31:14	▶

Vehtari's two BDA lectures are the ones that justify the R-hat and ESS numbers you will otherwise copy blindly. The molecular simulation community has no equivalent of rank-normalized R-hat, and importing it is a small, concrete, immediately useful contribution available to anyone who watches these two hours.

C.7 BUILD — Atlas C

1. **Implement Metropolis for a 2D Ising model.** Verify detailed balance numerically. Measure the autocorrelation time as you approach the critical temperature and watch critical slowing down happen.

2. **Implement the Wolff cluster algorithm** on the same system. Measure the autocorrelation time again. The speedup is the lesson.
3. **Implement HMC** for a 2D Gaussian with strong correlation. Tune the step size and trajectory length. Then implement NUTS and observe what the automatic tuning buys you.
4. **Compute R -hat and effective sample size** for one of your own production MD trajectories, treating independent replicas as chains. Most MD papers would fail this test. See whether yours does.
5. **The research exercise.** Characterize the stationary distribution of a Rosetta fixed-backbone design run empirically: run it many times from the same input, histogram the sequences, and compare to the Boltzmann distribution of the score function at the annealing temperature. Write up the discrepancy. Nobody has published this carefully.

Derivation checkpoints due: 11, 12, 13, 14, 15, 16.

C.8 Paired reading — Atlas C

Watch this	Then read this	Hold this question
Krauth Lec 1	Metropolis et al. 1953	Derive the acceptance ratio. Where does the normalizing constant cancel?
Krauth Lec 8	Swendsen & Wang 1987; Wolff 1989	Why does a cluster move beat a single-spin flip near criticality?
Krauth Lec 9	Bortz, Kalos & Lebowitz 1975	Rejection-free MC changes the time variable. What is the new clock?
Krauth Tutorial 8	Propp & Wilson 1996	Coupling from the past terminates with an exact sample. How?
Krauth, <i>Irreversible Markov chains</i>	Bernard, Krauth & Wilson 2009; Michel et al. 2014	These violate detailed balance deliberately. What replaces it?
MacKay Lec 12	MacKay, <i>Information Theory</i> , ch. 29–30	Why do importance and rejection sampling fail in high dimension?
Murray, <i>MCMC</i>	Neal 1993, <i>Probabilistic Inference Using MCMC</i>	What exactly does burn-in fix, and what does it not?
Vats, <i>Gentle Introduction</i>	Vats, Flegal & Jones 2019, <i>Biometrika</i> 106:321	Compute a defensible ESS for your own trajectory.
Neal, <i>HMC</i>	Neal 2011, <i>Handbook of MCMC</i> ch. 5	HMC is MD as a proposal. What does the Metropolis step correct for?
Betancourt, Part 1	Betancourt 2017, arXiv:1701.02434	Why does HMC work geometrically, and when does it fail?
Hoffman, <i>NUTS</i>	Hoffman & Gelman 2014, <i>JMLR</i> 15:1593	What does the no-U-turn criterion detect?
Skilling, <i>Computing Bayes</i>	Skilling 2006, <i>Bayesian Analysis</i> 1:833	Nested sampling computes the evidence. That is a partition function. Why does nobody say so?
Landau, <i>Wang-Landau</i>	Wang & Landau 2001, <i>PRL</i> 86:2050	It converges to the density of states. What is the protein-design analogue?
Bonomi, <i>Tempering and REX</i>	Sugita & Okamoto 1999; Marinari & Parisi 1992	Derive the replica-count scaling. Why is it fatal for large systems?
Weitzner, <i>Side-chain packing</i>	Desmet et al. 1992; Dunbrack & Karplus 1993	DEE is exact; annealing is not. Why does production use annealing?
RosettaCommons, <i>MC with minimization</i>	Li & Scheraga 1987; Rohl et al. 2004	Minimization inside a MC move breaks detailed balance. What is sampled instead?
Gillespie, his own algorithm	Gillespie 1977, <i>J Phys Chem</i> 81:2340	Why is SSA exact where an ODE is not?
Kratzer, <i>Kinetic Monte Carlo</i>	Voter 2007	kMC needs a rate catalogue. Who supplies it, and what if it is incomplete?
Bolhuis, <i>TPS</i>	Dellago, Bolhuis, Csajka & Chandler 1998, <i>JCP</i> 108:1964	TPS samples trajectory space. What is the measure on that space?
Faccioli, <i>Committor</i>	Bolhuis et al. 2002, <i>Annu Rev Phys Chem</i> 53:291	The committor is the exact reaction coordinate. Why can we never compute it?
Vanden-Eijnden, <i>Rare Events: Theory</i>	E & Vanden-Eijnden 2006, <i>J Stat Phys</i> 123:503	What does transition path theory add beyond sampling paths?

Watch this	Then read this	Hold this question
Sarupria, <i>Forward Flux Sampling</i>	Allen, Warren & ten Wolde 2005, <i>PRL</i> 94:018104	FFS needs no equilibrium. When is that decisive?
Noé, <i>Boltzmann Generators</i>	Noé, Olsson, Köhler & Wu 2019, <i>Science</i> 365:eaaw1147	Where does exactness come from, and what is the catch?
Noé, <i>Stochastic Normalizing Flows</i>	Wu, Köhler & Noé 2020, NeurIPS	Interleaving flow layers and MCMC steps. Why does that help?
Midgley & Stimper, <i>FAB</i>	Midgley et al. 2023, ICLR	Training on an energy with no samples. Apply this to a designed protein.
Gabrielé, <i>Adaptive MC with flows</i>	Gabrielé, Rotskoff & Vanden-Eijnden 2022, <i>PNAS</i> 119:e2109420119	Global proposal plus local kernel. Why does neither suffice alone?
Sohl-Dickstein, <i>L2HMC</i>	Levy, Hoffman & Sohl-Dickstein 2018, ICLR	A learned MCMC kernel that stays correct. What constrains the learning?
Vehtari, BDA 5.2	Vehtari et al. 2021, <i>Bayesian Analysis</i> 16(2)	Apply rank-normalized R-hat to replica MD. Does your ensemble pass?
Gelman, <i>Bayesian Workflow</i>	Gelman et al. 2020, arXiv:2011.01808	What would a “simulation workflow” paper look like, by analogy?
Carpenter, <i>How does Stan work?</i>	Carpenter et al. 2017, <i>JSS</i> 76(1)	Constrained-to-unconstrained transforms and log-Jacobians. Where do these appear in molecular sampling?

Atlas D — Enhanced Sampling

Problem. The event you care about takes a millisecond. Your simulation reaches a microsecond. Buy the three orders of magnitude without lying about the ensemble.

Every method here is a trade: you distort the sampling, then correct for the distortion. **The correction is the method.** When you read a talk in this family, the question is always *what exactly is being reweighted, and is the reweighting unbiased?*

D.1 The taxonomy — learn this before any individual method

Enhanced sampling splits cleanly in two, and conflating them is the most common error in the field:

Collective-variable methods bias along a chosen low-dimensional coordinate — umbrella sampling, metadynamics, OPES, adaptive biasing force, steered MD. They are powerful and they are only as good as the CV. **If your CV misses the slow degree of freedom, the method converges confidently to the wrong answer**, and nothing in the output warns you.

CV-free methods avoid choosing — replica exchange, simulated tempering, weighted ensemble, Markov state models built from many short runs, transition path sampling. They pay for it in cost or in complexity.

Derivation checkpoint 21. Write down, for each of umbrella sampling, metadynamics, replica exchange and weighted ensemble, (a) what distribution is actually sampled, (b) the exact reweighting that recovers the Boltzmann distribution, and (c) the condition under which that reweighting fails. If you can do this for all four, you understand this family.

D.2 Collective-variable methods

Method	Talk / resource	Speaker / host	Link
Umbrella sampling, WHAM	Alchemy and CECAM free-energy schools	—	→ Atlas E
Metadynamics	PLUMED Masterclass 21.4 — all 43 sessions in Appendix B.3	PLUMED consortium	▶
Metadynamics theory	Bussi / Laio / Parrinello lecture recordings	CECAM	cecam.org
OPES (On-the-fly Probability Enhanced Sampling)	PLUMED Masterclass 22.3 (Appendix B.3)	Michele Invernizzi	▶
Adaptive biasing force	NAMD / Chipot tutorials	Chipot lab	—
ML collective variables	Accelerate Atomistic Simulations, Sampling, and Dynamics	Pratyush Tiwary (ML4DD Day 2)	▶
ML collective variables	Boltzmann Weighted Ensembles	Tiwary (MLSB 2025)	▶
Differentiable sampling	Differentiable Simulations for Enhanced Sampling of Rare Events	Gómez-Bombarelli (Valence)	▶
Differentiable sampling	second session	Martin Šípka (Valence)	▶

Papers: Torrie & Valleau 1977, *J Comput Phys* 23:187 (umbrella sampling) · Kumar et al. 1992, *J Comput Chem* 13:1011 (WHAM) · Laio & Parrinello 2002, *PNAS* 99:12562 (metadynamics) · Barducci et al. 2008, *PRL* 100:020603 (well-tempered metadynamics) · Invernizzi & Parrinello 2020, *JPCL* 11:2731 (OPES) · Darve et al. 2008, *JCP* 128:144120 (ABF) · Tiwary & Berne 2016, *PNAS* 113:2839 (SGOOP — learning the CV) · Bonati et al. 2021, *PNAS* 118:e2113533118 (deep-LDA CVs) · Ribeiro et al. 2018, *JCP* 149:072301 (RAVE).

The PLUMED Masterclass — enumerated in full in Appendix B.3 — is the single best hands-on resource in this entire Atlas family and it is free. Twenty-plus sessions, each with code, data and exercises, taught by the people who wrote the software. If you do nothing else in Atlas D, do Masterclasses 21.01 through 21.06 and 22.03.

A practical note from the field: many conda PLUMED builds **do not include the OPES module**. If `opes_metad` is missing, you will need to build PLUMED from source against your MD engine. Budget half a day.

D.3 Replica exchange and tempering

Method	Resource	Link
Temperature REMD	GROMACS / CECAM REMD tutorials	→ B.8
Hamiltonian REMD, REST2	PLUMED Masterclass 22.10 (Appendix B.3)	▶
Solute tempering	REST2 tutorials	—

Papers: Sugita & Okamoto 1999, *Chem Phys Lett* 314:141 (REMD) · Wang, Friesner & Berne 2011, *JPCB* 115:9431 (REST2) · Rosta & Hummer 2009, *JCP* 131:165102 (the paper showing REMD's efficiency gain is smaller than usually claimed) · Nymeyer 2008, *JCTC* 4:626.

Read Rosta & Hummer before you run REMD. The naive argument — more replicas means faster barrier crossing — does not survive a careful analysis of round-trip times. REMD helps, but the scaling with system size is unfavorable because the number of replicas needed grows with the square root of the number of degrees of freedom. For a solvated protein, that is a lot of replicas.

D.4 Path and trajectory-space methods

Method	Talk	Speaker / host	Length	Link
Weighted ensemble	WESTPA 2015 workshop + Zuckerman's own talks — Appendix B.9	Zwier, Zuckerman, Chong	—	▶
Transition path sampling	CECAM rare-events schools	Bolhuis, Dellago	—	→ B.8
Action-Minimization Meets Generative Modeling: Transition Path Sampling	Sanjeev Raja (Valence)	1:04:36	▶	
Diffusion Models on Sampling Rare Events	Chenru Duan (Valence)	48:08	▶	
Milestoning, forward flux	Elber / Allen lecture recordings	—	—	

Papers: Huber & Kim 1996, *Biophys J* 70:97 (weighted ensemble, the original) · Zuckerman & Chong 2017, *Annu Rev Biophys* 46:43 (the review to read) · Zwier et al. 2015, *JCTC* 11:800 (WESTPA) · Bolhuis et al. 2002, *Annu Rev Phys Chem* 53:291 (TPS) · Allen et al. 2009, *J Phys Condens Matter* 21:463102 (forward flux) · Faradjian & Elber 2004, *JCP* 120:10880.

Weighted ensemble is Zuckerman's method and it is the most under-appreciated technique in this family. It is *unbiased* — no reweighting approximation, no CV-dependent distortion of the ensemble — because it runs unmodified dynamics and only manipulates trajectory *weights* through splitting and merging. The cost is bookkeeping, and WESTPA does the bookkeeping.

Derivation checkpoint 22. Show that weighted ensemble's resampling leaves the ensemble average of any observable unbiased. Then explain why this is a fundamentally different kind of guarantee than metadynamics' asymptotic convergence — and why that difference matters for a rate constant.

D.5 Markov state models — many short runs instead of one long one

Talk / resource	Speaker / host	Link
PyEMMA workshops — all 29 videos enumerated in Appendix B.10	Noé group	▶
MSMBuilder / Folding@home	Pande / Bowman lineage	—
Accelerating Cryptic Pocket Discovery With Deep Learning (PocketMiner)	Greg Bowman (Valence)	▶
Cryptic Pocket Discovery Using AlphaFold and Markov State Modelling	Valence	▶
VAMPnets and deep MSMs	Noé group talks	—

Papers: Prinz et al. 2011, *JCP* 134:174105 (MSMs: the definitive construction paper) · Husic & Pande 2018, *JACS* 140:2386 (MSM review) · Bowman et al. 2009, *Methods* 49:197 · Bowman & Geissler 2012, *PNAS* 109:11681 (equilibrium fluctuations reveal cryptic pockets) · Mardt et al. 2018, *Nat Commun* 9:5 (VAMPnets) · Wu & Noé 2020, *J Nonlinear Sci* 30:23 (variational approach to Markov processes) · Meller et al. 2023, *Nat Commun* 14:1177 (PocketMiner).

Greg Bowman’s thread, one of your ten, is here. The MSM argument is that you do not need one long trajectory — you need many short ones that collectively visit the states and the transitions between them, plus an estimator that stitches them into a kinetic model. Folding@home is that argument at planetary scale.

The payoff that matters for design: **cryptic pockets**. Bowman & Geissler showed that equilibrium fluctuations open binding sites invisible in the crystal structure. PocketMiner then learned to predict where those are without running the simulation — which is exactly the carry across cultures the whole curriculum is about: a physics-first result becoming a learning-first predictor.

Derivation checkpoint 23. State the implied-timescale test and explain what it diagnoses. Then explain why a *lag time* is needed at all and what you lose by making it longer. Checkpoint 24: explain why an MSM’s slowest timescale is a lower bound, not an estimate.

D.6 The learned-sampler frontier

This is where Atlas D meets Part III, and it is the most active research area in the whole physics-first culture.

Talk	Speaker / host	Length	Link
Designing Losses for Data-Free Training of Normalizing Flows on Boltzmann Distributions	Valence	1:07:25	▶
Rigid Body Flows for Sampling Molecular Crystal Structures	Jonas Köhler (Valence)	1:24:24	▶
Timewarp: Transferable Acceleration of MD by Time-Coarsened Dynamics	Klein & Foong (Valence)	54:35	▶
Timewarp (extended)	Valence	1:14:35	▶
Scalable Emulation of Protein Equilibrium Ensembles (BioEmu)	Valence	1:17:02	▶
AlphaFold Meets Flow Matching (AlphaFlow)	Bowen Jing (Valence)	56:30	▶
Consistent Sampling and Simulation: MD with Energy-Based Diffusion Models	Valence	43:48	▶
PepFlow: Direct Conformational Sampling From Peptide Energy Landscapes	Abdin (Valence)	55:42	▶
Converging Advances to Accelerate Molecular Simulation	Max Welling (Valence)	1:14:04	▶

Papers: Noé et al. 2019, *Science* 365:eaaw1147 (Boltzmann generators) [landmark] · Klein et al. 2023, NeurIPS (Timewarp) · Lewis et al. 2025, *Science* 389:eadv9817 (BioEmu) · Jing et al. 2024, ICML (AlphaFlow) · Zheng et al. 2024, *Nat Mach Intell* 6:558 (DiG) · Köhler et al. 2020, ICML (equivariant flows) · Midgley et al. 2023, ICLR (flow annealed importance sampling bootstrap).

The question that decides this entire subsection. A Boltzmann generator promises samples from the equilibrium distribution without dynamics. Does it deliver, or does it deliver samples that look like the training distribution?

The honest answer in 2026: **for small systems with good training data, yes, with importance weights that can be checked. For a new protein with no simulation data, the guarantee degrades to “it generalizes as well as the training set covers.”** That is a real advance and it is not the same claim.

Derivation checkpoint 25 asks you to write down the importance weight for a normalizing-flow sampler and identify exactly where the Boltzmann guarantee enters and where it can be lost. **Capstone IV** asks you to test it.

D.7 BUILD — Atlas D

1. **Pick a system with a known slow transition** — a ligand unbinding, a loop flip, a side-chain rotamer switch with a real barrier.

2. **Sample it three ways:** long unbiased MD, metadynamics along a CV you choose, and either weighted ensemble or an MSM from many short runs.
3. **Compare the free energy profiles.** They will not agree. Find out why.
4. **Deliberately choose a bad CV** for the metadynamics run — one that misses the slow coordinate — and show what the output looks like. It will look converged. That is the lesson.
5. **Write the one page.** What would have told you the CV was wrong, using only the metadynamics output?

Derivation checkpoints due: 21, 22, 23, 24, 25.

D.8 Paired reading — Atlas D

Watch this	Then read this	Hold this question
PLUMED Masterclass 21.01–06	Laio & Parrinello 2002, <i>PNAS</i> 99:12562	What exactly is deposited, and what is the converged bias?
PLUMED Masterclass 22.03 (OPES)	Invernizzi & Parrinello 2020, <i>JPCL</i> 11:2731	What does OPES fix about well-tempered metadynamics?
Tiwary, <i>ML collective variables</i>	Tiwary & Berne 2016, <i>PNAS</i> 113:2839	Learning a CV from data. What prevents it from learning a fast coordinate?
Gómez-Bombarelli, <i>Differentiable simulations</i>	The accompanying paper	Backprop through a simulation. Where does the gradient become useless?
GROMACS REMD tutorial	Sugita & Okamoto 1999 then Rosta & Hummer 2009, <i>JCP</i> 131:165102	How many replicas for your system? Is the gain worth it?
WESTPA tutorials	Zuckerman & Chong 2017, <i>Annu Rev Biophys</i> 46:43	Why is WE unbiased where metadynamics is asymptotic?
Raja, <i>Action minimization for TPS</i>	The paper + Bolhuis et al. 2002	Paths, not states. What object is being sampled?
PyEMMA 2017 workshop (Appendix B.10)	Prinz et al. 2011, <i>JCP</i> 134:174105	Implied timescales. What does the plateau actually prove?
Bowman, <i>PocketMiner</i>	Bowman & Geissler 2012, <i>PNAS</i> 109:11681 then Meller et al. 2023, <i>Nat Commun</i> 14:1177	A physics result became a learned predictor. What was kept and what was discarded?
Valence, <i>Boltzmann flows</i>	Noé et al. 2019, <i>Science</i> 365:eaaw1147	Write the importance weight. Where can the guarantee be lost?
Klein & Foong, <i>Timewarp</i>	Klein et al. 2023, NeurIPS	Learning the propagator rather than the distribution. What transfers?
Valence, <i>BioEmu</i>	Lewis et al. 2025, <i>Science</i> 389:eadv9817	Emulated equilibrium ensembles. What was the validation, and is it enough?
Jing, <i>AlphaFlow</i>	Jing et al. 2024, ICML	AF2 as a generative prior over conformations. Is the output Boltzmann?
Tiwary, <i>Boltzmann Weighted Ensembles</i>	Compare to BioEmu and AlphaFlow	State the physics-first objection to all of D.6 in one sentence.

Atlas E — Free Energy

Problem. Compute a binding affinity, a solvation free energy, or a $\Delta\Delta G$ of mutation that an experimentalist would bet on.

This family has the most mature error theory in computational biology and the widest gap between what is possible and what is typically done. The methods are rigorous. The practice frequently is not.

E.1 The standing resource

The 2018 free energy workshop — 43 talks, every one enumerated in Appendix B.4. This is the densest single archive on free energy calculation anywhere, covering theory, methods, software, and a blunt assessment of what works. Several of the talks are by people who build the methods arguing with each other about protocol, which is the most instructive content in the field.

Archive	Contents	Link
Alchemistry.org	Method documentation, best practices, the workshop	alchemistry.org
Chodera lab channel	The 2018 workshop talks	▶
Open Force Field Initiative	Parameterization and benchmarking talks	▶
D3R / SAMPL challenges	Blind prediction results and post-mortems	drugdesigndata.org

SAMPL is the CASP of free energy, and almost nobody outside the field knows it exists. Blind challenges on host-guest binding, solvation free energies and partition coefficients, with results published whether good or bad. The post-mortem papers are the most honest writing in computational chemistry.

E.1b The recorded literature, filled

The second edition pointed at the Chodera workshop without enumerating it. All 43 talks are now in Appendix B.4. What follows is the rest of the family.

Alchemical foundations

Talk	Speaker	Len	Link
Free energies: What we've learned about how to estimate them	Michael Shirts (CU Boulder, MBAR author)	30:13	▶
Estimator variance and convergence diagnosis	Michael Shirts	29:49	▶
A guide to multistate reweighting — the MBAR tutorial proper	Michael Shirts	36:27	▶
Teaching free energy calculations to learn	John Chodera (MSKCC)	29:22	▶
Developing and Using FE Calculations to Guide Lead Optimization	David Mobley (UC Irvine)	1:00:03	▶
Introduction to Free Energy Calculations	Christophe Chipot (CNRS/UIUC)	56:09	▶
A General Overview of Free Energy Methods	Christophe Chipot	1:30:38	▶
Accurate Calculation of Protein-Ligand Binding Energies	Christophe Chipot	1:40:42	▶
Introduction to Free-Energy Calculations, Part 1	Christophe Chipot	1:31:10	▶
Introduction to Free-Energy Calculations, Part 2	Christophe Chipot	1:32:55	▶
Geometrical Free Energy Methods (the CV/PMF route)	Giacomo Fiorin (NIH, Colvars author)	2:25:35	▶
Accelerating Convergence with replica-exchange alchemy	Wei Jiang (Argonne)	1:11:26	▶
Introduction to free energy calculations (GROMACS-flavoured)	BioExcel	55:25	▶
A rigorous statistical-mechanics-first derivation	Leandro Martínez (UNICAMP)	1:34:46	▶
Statistical-mechanical basis of binding free energy	Benoit Roux (Chicago)	33:43	▶
λ -dynamics and multisite λ -dynamics	Charles Brooks III (Michigan)	30:56	▶
Constant-pH and charge-change alchemy	Thomas Simonson (Polytechnique)	30:44	▶
Soft-core potentials and the end-point singularity	Darrin York (Rutgers)	26:16	▶
Reducing free energy simulations to the bare essentials	Stefan Boresch (Vienna)	23:34	▶

Talk	Speaker	Len	Link
Polarizable force fields in alchemy (AMOEBA)	Jay Ponder (WashU)	30:23	▶
Binding free energy calculations with non-equilibrium alchemy	Vytautas Gapsys (MPI-NAT)	52:01	▶
Free Energy Calculations with BioSimSpace	CCPBioSim	19:45	▶
MDAnalysis and alchemlyb — not fooling yourself with your own estimator	Richard Gowers (OpenFE)	35:31	▶
Open Free Energy: Open Source Alchemy	Hannah Baumann (OpenFE)	22:03	▶
Engineering FEP science in the open	Julien Michel (Edinburgh)	31:18	▶
Free energy of solids: the 1983 CECAM workshop retrospective	CECAM	3:17:06	▶

Shirts's thirty minutes is the best overview talk in this family. Start there. Then his MBAR tutorial, then Chipot's long survey.

Industrial practice, and the skeptics

Talk	Speaker	Len	Link
FEP and Drug Discovery: Where We've Been and Where We'd Like to Go	Derek Lowe (Novartis, <i>In the Pipeline</i>)	35:09	▶
Promises and Limitations: Free Energy Methods in Drug Discovery Projects	Xin Yan (Merck)	26:51	▶
The FEP+ author on the FEP+ benchmark	Lingle Wang (Schrödinger)	35:36	▶
Updated FEP+ results	Lingle Wang	30:08	▶
IFD-MD and FEP+ with AlphaFold inputs	Márton Vass (Schrödinger)	55:23	▶
Physics-Driven Discovery of SNX281	Woody Sherman (Silicon Therapeutics)	54:06	▶
From Concept to Clinic: a non-CDN STING agonist	Woody Sherman (Psivant)	29:15	▶
Free Energy Calculations from Butane to COVID-19	William Jorgensen (Yale)	36:33	▶
Absolute binding free energy for pose prediction	David Minh (IIT)	23:25	▶
Active learning for absolute binding free energies	Steven Jerome (Broad)	23:11	▶
Large-scale alchemical screening with GROMACS/pmx	BioExcel #63	54:29	▶

Lowe's talk is the most useful skeptical hour in this family — a working medicinal chemist on what FEP has and has not delivered. **Pair it immediately with Lingle Wang's FEP+ advocacy.** Yan's is the industrial retrospective on prospective failures. **Jorgensen's is the field's founder narrating forty years of it**, having run the first FEP on a biomolecular system.

Blind challenges

D3R 2019 Community Discussion on the Future of Blinded Prediction Challenges, 1:09:32 (▶) — **the single most valuable blind-challenge retrospective available**, the field arguing openly about what the challenges actually measured · the 2018 version (▶) · Chodera's submission post-mortem (▶) · Gapsys under blind conditions (▶) · Mobley on SAMPL (▶) · **Patrick Walters on why most reported blind-challenge rankings are not statistically significant** (▶) · **the SAMPL6 pKa challenge, where nearly every method failed informatively** (▶).

Nonequilibrium methods

Christopher Jarzynski himself — *Scaling down the laws of thermodynamics*, 1:14:06 (▶) · *Nonequilibrium Statistical Mechanics II*, a proper lecture-course treatment (▶) · strong-coupling corrections, which matter when the “system” is a ligand in a binding site (▶) · nanoscale thermodynamics (▶).

Also: the LMU biophysics lecture covering **both Jarzynski and Crooks properly** (▶) · a two-hour derivation (▶) · **Giovanni Bussi connecting the fluctuation theorems to practical steered MD** — the bridge the other talks skip (▶).

$\Delta\Delta G$ of mutation

pmx from its authors — Gapsys and de Groot, BioExcel #4, 1:02:18 (▶) · the current pmx (▶) · the teaching version (▶) · applications to protein stability and resistance mutations (▶) · **the one honest, non-promotional MM-GBSA tutorial**, Brandon Havranek (Drexel) (▶).

Two gaps that matter, both critical rather than foundational. There is **no good critical MM-PBSA lecture on YouTube** — that search space is almost entirely uncritical tutorial-mill content. Use **Genheden & Ryde 2015, *Expert Opin Drug Discov* 10:449** and Hou et al. 2011, *JCIM* 51:69.

And there is **no talk at all on the $\Delta\Delta G$ anti-symmetry failure**; the only FoldX videos are promotional. That literature is **Usmanova et al. 2018, *Bioinformatics* 34:3653**, Pucci et al. 2018, *Bioinformatics* 34:3659, and Caldararu et al. 2021, *Sci Rep* 11:9234. **Derivation checkpoint 28 is this gap.**

E.2 The estimators

Topic	Resource	Link
Thermodynamic integration, FEP, BAR, MBAR	Alchemistry workshop talks	▶
Free energy perturbation theory	CECAM free energy schools	→ Atlas B.8
FEAT: Free energy Estimators with Adaptive Transport	Du & He (Valence)	▶
Alchemical pathways in practice	BioExcel PMX workshop recordings	▶

Papers, and this is a short canonical list: Zwanzig 1954, *JCP* 22:1420 (FEP) · Kirkwood 1935, *JCP* 3:300 (thermodynamic integration) · **Bennett 1976, *J Comput Phys* 22:245 (BAR)** · **Shirts & Chodera 2008, *JCP* 129:124105 (MBAR) [the one to internalize]** · Crooks 1999, *PRE* 60:2721 (the fluctuation theorem) · **Jarzynski 1997, *PRL* 78:2690** · Pohorille, Jarzynski & Chipot 2010, *JPCB* 114:10235 (good practices).

Derivation checkpoint 26. Derive BAR from the Crooks fluctuation theorem. Then show that MBAR is its multi-state generalization and that it is the minimum-variance unbiased estimator given the samples you have. This is the single most elegant derivation in the Atlas and it takes an afternoon.

Checkpoint 27. Explain why the Zwanzig exponential average is correct in expectation but useless in practice — the variance argument, with the overlap integral written down explicitly.

E.3 Binding free energies

Topic	Talk / resource	Link
Absolute binding free energy	Alchemistry workshop; Gilson/Mobley talks	▶
Relative binding free energy (RBFE)	Open Force Field, FEP+ benchmark talks	▶
Boltz-2: affinity prediction	Valence	▶
Quantitative Affinity Data at Scale: the Data Bottleneck	Murakowska (BPDMC)	▶
MM-GBSA and endpoint methods	—	see note below

Papers: Gilson et al. 1997, *Biophys J* 72:1047 (the statistical-thermodynamic basis) · **Mobley & Gilson 2017, *Annu Rev Biophys* 46:531 (binding free energy calculations — the review)** · Wang et al. 2015, *JACS* 137:2695 (FEP+ on congeneric series) · Cournia et al. 2017, *JCIM* 57:2911 · **Schindler et al. 2020, *JCIM* 60:5457 (large-scale industrial FEP benchmark)** · Genheden & Ryde 2015, *Expert Opin Drug Discov* 10:449 (MM-PBSA/GBSA review).

On MM-GBSA, since you use it. It is fast and it is not a free energy calculation. It omits the entropy term or approximates it badly, it is extremely sensitive to the solute dielectric, and its correlation with experiment across *unrelated* ligands is typically poor. Where it works is **ranking within a congeneric series with consistent protocol** — which is a real and useful thing, just not the thing the name suggests. Genheden & Ryde are honest about this; read them before quoting a number.

Alchemical FEP is the method that actually predicts affinity, at roughly 1 kcal/mol RMSE on well-behaved congeneric series after substantial setup effort. The Schindler benchmark is what that looks like at industrial scale.

E.4 $\Delta\Delta G$ of mutation and protein stability

Resource	Note
PMX / BioExcel tutorials	Alchemical mutation setup end-to-end
FoldX, Rosetta ddg_monomer	Empirical, fast, and systematically biased toward destabilizing
ThermoMPNN tutorial (RosettaCommons, 42:39)	▶
Mega-scale stability (Tsuboyama, ML4PE, 50:57)	▶
Antibody $\Delta\Delta G$ data requirements (Hummer, BPDMC, 1:14:43)	▶

Papers: Gapsys et al. 2016, *Chem Sci* 7:4728 (PMX, accurate alchemical $\Delta\Delta G$) · Steinbrecher et al. 2017, *J Mol Biol* 429:948 · Tsuboyama et al. 2023, *Nature* 620:434 (the 500,000-measurement dataset that changed what is trainable) · Dieckhaus et al. 2024, *PNAS* 121:e2314853121 (ThermoMPNN) · Nisthal et al. 2019, *PNAS* 116:16367 · Pancotti et al. 2022, *Brief Bioinform* 23:bbab555 (the anti-symmetry audit — most $\Delta\Delta G$ predictors fail the simple test that $\Delta\Delta G(A \rightarrow B) = -\Delta\Delta G(B \rightarrow A)$).

The anti-symmetry test is the cheapest and most damning benchmark in this family. A predictor should give exactly opposite answers for a forward and reverse mutation. Most do not, because they were trained on a dataset that is itself asymmetric — experimentalists report destabilizing mutations far more often.

Derivation checkpoint 28: explain how a training-set asymmetry becomes a model asymmetry, and design the correction. That is a publishable idea stated as a homework problem.

E.5 BUILD — Atlas E

1. **Run a relative binding free energy calculation** on a small congeneric series with known experimental affinities. Three or four ligands is enough.
2. **Compute the overlap** between neighboring lambda windows. If any pair has poor overlap, your result is wrong regardless of how small the error bar looks.
3. **Estimate with BAR and with the Zwanzig exponential average** from the same data. Compare both the values and the variances.
4. **Run the same series through MM-GBSA** and plot both against experiment.
5. **Write the one page.** Report RMSE and Spearman for each method, and state what you would need to halve the error.

Derivation checkpoints due: 26, 27, 28.

E.6 Paired reading — Atlas E

Watch this	Then read this	Hold this question
Alchemy workshop, estimators	Bennett 1976 then Shirts & Chodera 2008	Derive BAR from Crooks. Then show MBAR generalizes it.
Alchemy workshop, overlap	Pohorille et al. 2010, <i>JPCB</i> 114:10235	Write the overlap integral. At what value is a result untrustworthy?
Du & He, <i>FEAT</i>	The FEAT paper	Learned transport between thermodynamic states. What replaces lambda windows?
Mobley/Gilson talks	Mobley & Gilson 2017 , <i>Annu Rev Biophys</i> 46:531	What are the standard-state and symmetry corrections, and when do they bite?
FEP+ benchmark talks	Schindler et al. 2020 , <i>JCIM</i> 60:5457	Industrial scale, honest numbers. What is the actual RMSE distribution?
Valence, <i>Boltz-2 affinity</i>	The Boltz-2 paper	A learned affinity head. Against what data, and what is the generalization gap?
Murakowska , <i>Affinity data bottleneck</i>	Compare to Schindler 2020	Is the limit the method or the measurements?
ThermoMPNN tutorial	Tsuboyama et al. 2023 , <i>Nature</i> 620:434	500k stabilities. What became trainable that was not before?
—	Pancotti et al. 2022 , <i>Brief Bioinform</i> 23:bbab555	Run the anti-symmetry test on any $\Delta\Delta G$ predictor you use.
SAMPL post-mortems	Any SAMPL challenge overview paper	Blind prediction. What systematically went wrong, across groups?

Atlas F — Coarse-Graining and Reduced Models

Problem. An all-atom simulation of a nucleosome is 10^6 particles and reaches microseconds. A chromosome is 10^9 base pairs and reorganizes over minutes. No amount of hardware closes that gap. Coarse-graining asks what you can throw away and still be right — and, crucially, **about what you can still be right**. A coarse-grained model that reproduces structure almost always gets dynamics wrong, and the reason is a specific term in a specific equation that most practitioners have never seen.

This family was entirely missing from the first edition. It should not have been: it is the only systematic theory of approximation in the whole document, and it is the closest thing the physics culture has to a theory of what a learned representation is allowed to discard.

F.1 The theory — why coarse-graining works and where it fails

Voth and multiscale coarse-graining

Title	Speaker	Length	Link
Theory of Coarse-graining	Gregory Voth (Chicago)	58:45	▶
Theory of Ultra-Coarse-Graining	Gregory Voth	1:08:56	▶
Molecular Modeling: A Window to the Biochemical World	Gregory Voth	1:06:25	▶

The first is **the single most important theory lecture in this family**: the variational / force-matching derivation of multiscale coarse-graining, which shows that the correct CG potential is the many-body potential of mean force obtained by integrating out the discarded degrees of freedom. Everything else here is a tractable approximation to that object.

Force matching and iterative Boltzmann inversion — the PASI 2012 school

Title	Speaker	Length	Link
Force matching fundamentals	Monica Lamm (Iowa State)	1:02:50	▶
Fundamentals of the Iterative Boltzmann Inversion	Roland Faller (UC Davis)	52:14	▶
Using the IBI for polymers	Roland Faller	43:07	▶
Developing numerical potentials and their state point dependence	Roland Faller	50:24	▶
From high to low resolution: the mapping problem	Marco Giulini	29:31	▶

Faller's state-point-dependence lecture states the central weakness of all structure-based coarse-graining plainly: a potential fitted at one temperature and density is not valid at another. **That is a transferability failure with an**

exact analogue in machine learning — a model fitted to one distribution does not transfer to another — and the CG community has thought about it far longer.

Giulini's lecture treats *choice of mapping* as the real open problem via mapping entropy, which is the question nobody in representation learning asks either.

Mori-Zwanzig and the memory kernel

Title	Speaker	Length	Link
Entropic Barriers and Dynamical Coarse-Graining via Mori-Zwanzig	Xingjie Li	25:43	▶
CM4: Mori-Zwanzig	George Karniadakis (Brown)	1:18:30	▶
Mori-Zwanzig Formulation / Model order reduction	CRUNCH Group, Brown	2:02:11	▶
Generalized Langevin Equations from MD simulations	Laura Scalfi	57:28	▶

This is derivation checkpoint 29 and it is the intellectual core of the family. Projecting out degrees of freedom produces exactly three terms: a mean force, a memory kernel, and a noise. Virtually every coarse-grained model keeps the first and discards the other two. That is why CG dynamics is wrong even when CG structure is right, and why you cannot read kinetics off a MARTINI trajectory. Watch Xingjie Li first for the short version, then Karniadakis for the derivation.

Machine-learned coarse-graining — Clementi

Title	Speaker	Length	Link
Coarse-graining classical and quantum systems	Cecilia Clementi (FU Berlin/Rice)	56:52	▶
Coarse-graining for classical and quantum systems (GGI)	Cecilia Clementi	37:47	▶
Learning molecular models from simulation and experimental data	Cecilia Clementi (IPAM)	54:53	▶
Designing molecular models by ML and experimental data	Cecilia Clementi	54:42	▶
Clementi, TCBG seminar (CGnet/CGSchNet results)	Cecilia Clementi	1:00:37	▶
Designing molecular models with ML and experimental data (IPAM)	Cecilia Clementi	46:18	▶
ML Coarse-Grained Models with Graph Neural Networks	Computational Biomedicine	1:12:21	▶
HITS-SIMPLAIX Colloquium: protein dynamics and ML	Cecilia Clementi	50:45	▶

Learned mappings and backmapping

Title	Speaker	Length	Link
Physics in and out of machine learning for molecular simulations	Tristan Bereau (Amsterdam)	52:34	▶
Tractable Mapping Entropy and Generative Backmapping via Split-Flows	Tristan Bereau	55:30	▶
Chemically Transferable Generative Backmapping of CG Proteins	Soojung Yang (MIT)	1:06:07	▶
Coarse-graining autoencoders and evolutionary learning	Rafael Gómez-Bombarelli (MIT)	36:42	▶
Generative Coarse-Graining of Molecular Conformations	Wujie Wang (MIT)	1:53:41	▶
Simulate Time-integrated CG Molecular Dynamics with Geometric ML	Xiang Fu (MIT)	1:40:25	▶
Utilizing ML for Scale Bridging: atomistic to CG and back	CaSToRC	1:20:39	▶

Xiang Fu's talk is the one that does something genuinely new: rather than learning a CG force field and integrating it, it learns CG *dynamics* with large timesteps directly — circumventing the memory-kernel problem rather than solving it. Whether that is legitimate is a live question and exactly the kind of thing worth having an opinion about.

F.2 The models in practice

MARTINI

Title	Speaker	Length	Link
Perspective on the Martini model: the road to simulating entire cells	Siewert-Jan Marrink (Groningen)	40:22	▶
BioExcel #24: Perspective on the Martini Force Field	BioExcel CoE	1:19:47	▶
BioExcel #84: Simulating whole cells with Martini	BioExcel CoE	1:01:50	▶
Coarse Grain (full workshop session)	Manuel Melo (MoBioChem)	2:02:57	▶
Key aspects when modelling amphiphiles with MARTINI	Germán Pérez-Sánchez	43:00	▶
OpenMM + Martini3: Simulating Proteins in Membrane Systems	Neurosnap	6:52	▶
Martinize.py: Generate Martini Protein Topology	Ubeiden Samboni	11:21	▶
Increasing complexity in simulations of biological membranes	D. Peter Tieleman (Calgary)	1:16:01	▶
Tieleman, TCBG seminar (Martini 3 validation)	D. Peter Tieleman	1:04:31	▶
TCBG 2003: Coarse Grained Modeling of Lipid Phases	UIUC TCBG	48:00	▶

Structure-based (Gō) models and the funnel lineage

Title	Speaker	Length	Link
Structure-based model workshop, part 1 (SMOG)	Paul Whitford (Northeastern)	58:35	▶
Structure-based model workshop, part 2	Paul Whitford	49:31	▶
Funnels and Landscapes in protein folding: a historical perspective	José Onuchic (Rice)	1:05:51	▶
Funnels, cooperativity, desolvation, and enthalpic barriers	Hue Sun Chan (Toronto)	1:00:52	▶

Watch Onuchic and then immediately watch Chan. Onuchic explains why structure-based models work — minimal frustration licenses a funnel. Chan is the principal critique: naive Gō models get cooperativity and desolvation barriers wrong because they build the answer into the potential. That disagreement is the most instructive hour in this family.

AWSEM, UNRES, SIRAH and the rest

Title	Speaker	Length	Link
Coarse-Grained Modeling of Biomolecules: A Brief History and Overview	Garegin Papoian (Maryland)	51:19	▶
Structure and Dynamics of Nucleosomes from Atomistic and CG Simulations	Garegin Papoian	1:01:02	▶
SIRAH: a CG force field for biomolecular simulations	Sergio Pantano (Pasteur Montevideo)	1:04:48	▶
Introduction to coarse-grained modeling for biological systems	L. Silva (SIRAH)	17:43	▶
Geometry-consistent expressions for energy terms (UNRES)	Adam Liwo (Gdańsk)	42:57	▶
Template-based prediction using coarse-grained UNRES	Adam Liwo	43:31	▶
CABS-flex 2.0: predicting structural ensembles	Bernal Institute	26:49	▶
Protein folding simulation using SURPASS	Bio Comp Pictures	3:37	▶
Coarse-Grained Simulation of Proteins to Capture Conformational Change	IIT Madras	1:11:55	▶

Papoian's history lecture is **the best opening item in this family** — a survey of the whole coarse-graining enterprise by someone who built one of its major models.

Adaptive resolution and multiscale coupling

Title	Speaker	Length	Link
Adaptive resolution simulation methods for soft matter	Kurt Kremer (MPI-P Mainz)	52:58	▶
BioExcel #20: Adaptive resolution methods in soft matter	BioExcel CoE	1:00:48	▶
BioExcel #19: Hybrid MM/Coarse-Grained approaches	BioExcel CoE	58:46	▶
Coarse-grained / multiscale simulation in NAMD, with validation	NanoBio Node	56:28	▶

AdResS lets resolution vary across space with particles changing representation as they move — the molecular analogue of adaptive mesh refinement, and an idea with no counterpart whatsoever in protein machine learning.

F.3 Mesoscale and continuum

Membrane elasticity — the Deserno course

Title	Speaker	Length	Link
Membrane Elasticity and Thermodynamics I	Markus Deserno (CMU)	1:33:32	▶
Membrane Elasticity and Thermodynamics II	Markus Deserno	1:40:11	▶
Membrane Elasticity and Thermodynamics III	Markus Deserno	1:43:25	▶
A Physicist's View on Biological Membranes (single-lecture version)	Markus Deserno	1:14:13	▶
Coarse-grained Simulation Studies of membranes	Markus Deserno (Nordita)	56:36	▶
Helfrich Energy: From Heisenberg Spins to Mean Curvature	PhysicsOfLifeLMU	45:41	▶
Physics approaches of biological membranes	Patricia Bassereau (Institut Curie)	1:08:10	▶
Coarse-Graining Phase Separation: Flory-Huggins to Ginzburg-Landau	PhysicsOfLifeLMU	48:20	▶

Deserno's three-part course is the best membrane theory available free — from differential geometry to the Helfrich Hamiltonian, properly derived. The LMU Flory-Huggins-to-Ginzburg-Landau lecture is **the cleanest worked example of coarse-graining as a concept anywhere in this document**: a lattice model becomes a continuum field theory, and you can see exactly what was discarded.

Dissipative particle dynamics and lattice Boltzmann

Title	Speaker	Length	Link
Lec 31: Dissipative Particle Dynamics	IIT Madras	39:54	▶
Mesoscale Modeling of Soft Matter with DPD	ATOMS UFRJ	1:19:41	▶
DPD Simulation of Red Blood Cells	Bruce Caswell	1:02:02	▶
DL_MESO (hands-on)	CCPBioSim	1:15:06	▶
Lattice Boltzmann for simple fluids 1/4	Ignacio Pagonabarraga (EPFL)	19:45	▶
Lattice Boltzmann for simple fluids 2/4	Ignacio Pagonabarraga	23:09	▶
Lattice Boltzmann for simple fluids 3/4	Ignacio Pagonabarraga	33:24	▶
Lattice Boltzmann for simple fluids 4/4	Ignacio Pagonabarraga	35:52	▶
Introduction to Lattice Boltzmann Method	ESPresSo	1:03:56	▶

F.4 BUILD — Atlas F

1. **Implement iterative Boltzmann inversion** on a simple liquid. Match the radial distribution function. Then change the temperature by 20% and measure how badly the potential transfers. Plot it. That plot is the transferability problem, and it has an exact analogue in every protein model you use.
2. **Implement force matching** on the same system and compare the resulting potential to the IBI one. They will differ. Explain why in terms of what each objective is actually minimizing.
3. **Measure the memory kernel.** Take an atomistic trajectory, define a CG coordinate, and extract the friction kernel. Show that it is not a delta function. That is checkpoint 29, made real.
4. **The research exercise.** Take any learned coarse-grained model (CGnet-style) and ask what Mori-Zwanzig says it is missing. Write two pages on whether adding a learned memory kernel is tractable. Nobody has done this properly and it is a clean thesis.

Derivation checkpoints due: 28, 29, 30.

F.5 Paired reading — Atlas F

Watch this	Then read this	Hold this question
Voth, <i>Theory of Coarse-graining</i>	Izvekov & Voth 2005, JPC B 109:2469; Noid et al. 2008, JCP 128:244114	Derive the force-matching objective. What is it a projection of?
Voth, <i>Ultra-Coarse-Graining</i>	Dama et al. 2013, JCTC 9:2466	What breaks when CG sites must change internal state?
Faller, <i>IBI fundamentals</i>	Reith, Pütz & Müller-Plathe 2003, JCC 24:1624	IBI inverts an RDF to a potential. Why is that inversion not unique?
Faller, <i>State point dependence</i>	Louis 2002, <i>Beware of density dependent pair potentials</i>	State the transferability failure formally. What is the ML analogue?
Giulini, <i>The mapping problem</i>	Giulini et al. 2020, JCTC 16:6795	Mapping entropy scores a choice of CG sites. Could you optimize a mapping?
Xingjie Li / Karniadakis, <i>Mori-Zwanzig</i>	Zwanzig 1961, Phys Rev 124:983; Mori 1965	Derive the three terms. Which does your favourite CG model keep?
Scalfi, <i>GLE from MD</i>	Jung, Hanke & Schmid 2017, JCTC 13:2481	How do you actually measure a memory kernel from a trajectory?
Clementi, <i>Coarse-graining classical and quantum</i>	Wang et al. 2019, ACS Cent Sci 5:755 (CGnets)	The loss is force matching. What guarantees thermodynamic consistency?
Clementi, <i>Learning from simulation and experiment</i>	Olsson et al. 2017, PNAS 114:8265	How do you constrain a CG model with experiment rather than simulation?
Fu, <i>Time-integrated CG dynamics</i>	Fu et al. 2023, TMLR, arXiv:2204.10348	Learning dynamics directly sidesteps the memory kernel. Is that legitimate?
Yang, <i>Generative backmapping</i>	Yang & Gómez-Bombarelli 2023, arXiv:2303.01569	Backmapping is a conditional generative problem. What makes it ill-posed?
Marrink, <i>Perspective on Martini</i>	Marrink et al. 2007, JPC B 111:7812; Souza et al. 2021, Nat Methods 18:382	Four heavy atoms per bead. What chemistry is lost at that resolution?
Whitford, <i>SMOG workshop</i>	Noel et al. 2016, PLoS Comput Biol 12:e1004794	A Gō model builds the native state into the potential. What can it still predict?
Onuchic, <i>Funnels</i>	Bryngelson, Onuchic, Socci & Wolynes 1995, Proteins 21:167	Derive the condition for minimal frustration.
Chan, <i>Funnels, cooperativity, desolvation</i>	Chan et al. 2011, Annu Rev Phys Chem 62:301	Name three things Gō models get wrong and say why.
Papoian, <i>Brief History</i>	Kmiecik et al. 2016, Chem Rev 116:7898	Place five CG models on a resolution-vs-transferability axis.
Liwo, <i>Geometry-consistent energy terms</i>	Liwo et al. 2001, JCP 115:2323	UNRES derives terms as cumulant expansions of the PMF. What does that buy?
Kremer, <i>Adaptive resolution</i>	Praprotnik, Delle Site & Kremer 2005, JCP 123:224106	Particles change representation as they move. What must be conserved?
Deserno I–III	Helfrich 1973, Z Naturforsch C 28:693	Derive the Helfrich Hamiltonian. What is the Gaussian modulus and why is it usually ignored?
LMU, <i>Flory-Huggins to Ginzburg-Landau</i>	Cahn & Hilliard 1958, JCP 28:258	Trace exactly what is discarded going from lattice to field theory.

Watch this	Then read this	Hold this question
Pagonabarraga, LBM 1–4	Dünweg & Ladd 2009, Adv Polym Sci 221:89	LBM solves hydrodynamics on a lattice. When do biomolecules need it?

Atlas G — Continuum and Mesoscale

Problem. The object is too big for atoms and too structured for a continuum average: cells, condensates, crowded cytoplasm, reaction-diffusion, hydrodynamics.

This family was a reading list in the second edition and that was a failure. It is now filled. It remains the least familiar family to protein designers and therefore the most likely source of an idea nobody in your field has imported.

G.1 Brownian and Langevin dynamics, and association kinetics

An honest framing first. The founding work here is from 1978 to 1984 and predates video by decades. Ermak, McCammon, Northrup and Allison left no lectures. What exists is the modern descendants, and they are good.

Talk	Speaker	Len	Link
Effects of crowding on binding and enzyme reaction rates	Gideon Schreiber (Weizmann)	1:06:51	▶
Diffusional association and residence times	Rebecca Wade (HITS, author of SDA)	32:25	▶
Protein Dynamics in Cellular Environments	Rommie Amaro (UCSD)	57:47	▶
Generalized Langevin Equations from MD simulations	Laura Scalfi (Sorbonne/FU Berlin)	57:28	▶
Life at Low Reynolds Number	Raymond Goldstein (Cambridge)	15:03	▶
Batchelor Prize Lecture — low-Reynolds biological fluid dynamics	Raymond Goldstein	48:55	▶

Schreiber's talk is the one that matters for a designer. He defined the experimental side of protein-protein association kinetics, and this talk joins crowding directly to *kon*. **Scalfi's derives the generalized Langevin equation and memory kernel from atomistic trajectories** — the rigorous version of the hand-wave that gets you from Newton to overdamped Langevin, and the direct companion to derivation checkpoint 29.

Papers: Ermak & McCammon 1978, *JCP* 69:1352 · Northrup, Allison & McCammon 1984, *JCP* 80:1517 · Schreiber, Haran & Zhou 2009, *Chem Rev* 109:839 · Huber & McCammon 2010, *Comput Phys Commun* 181:1896 (BrownDye) · Sharp, Fine & Honig 1987, *Science* 236:1460 (SOD electrostatic steering) · Zwanzig 1961, *Phys Rev* 124:983.

Why a designer should care. Your binder's *kon* is a diffusion problem, not a structure problem, and electrostatic steering can buy two orders of magnitude. **Schreiber's review is the best document on this and the design literature barely cites it.** That is one of the twelve unexchanged ideas in the Atlas introduction.

G.2 Reaction-diffusion and particle-based cell simulation

Talk	Speaker	Len	Link
MCell: A 3D Monte Carlo Model of a Virtual Cell	Terry Sejnowski (Salk)	11:06	▶
How to build a synapse from molecules, membranes and Monte Carlo	Thomas Bartol (Salk)	20:02	▶
Introduction to MCell4	Salk CNL	7:14	▶
MMBioS Cell Modeling Workshop, Day 1	MMBioS (Pittsburgh/CMU/Salk)	2:07:43	▶
MMBioS Cell Modeling Workshop, Day 2	MMBioS	1:14:38	▶
Smoldyn inside VCell	James Schaff (UConn Health)	30:57	▶
VCell tutorial	Ann Cowan (UConn Health)	1:03:26	▶
Multiscale and Multiphysics Modeling in VCell	Leslie Loew (UConn Health)	1:14:07	▶
Bridging Molecular to Cellular Scales	Leslie Loew	1:00:26	▶
Reaction Kinetics and Diffusion — when diffusion limits a rate	Leslie Loew	46:09	▶
Macromolecular Assembly (NERDSS)	Margaret Johnson (Johns Hopkins)	1:03:30	▶
The Gillespie Algorithm, by Gillespie	Daniel T. Gillespie	39:37	▶
Stochastic Chemical Kinetics 1 (chemical master equation)	MIT OCW	53:57	▶
Stochastic Chemical Kinetics 2 (SSA and tau-leaping)	MIT OCW	47:32	▶
Stochastic Simulation of Models in the Life Sciences	Samuel Isaacson (BU)	1:33:41	▶
Reaction-Diffusion Operator Splitting on Tetrahedral Meshes	Erik De Schutter (OIST, STEPS)	44:47	▶
Stochastic Simulations	Mendes & Slepchenko (UConn)	56:00	▶
COPASI tutorial	Pedro Mendes (UConn)	1:04:38	▶

Gillespie explaining his own algorithm is a genuinely rare primary-source recording. Isaacson's tutorial is the one to take seriously — a mathematician on the RDME and its convergence pathologies, which most biologists using these codes have never confronted. **De Schutter's talk names the operator- splitting error that silently corrupts mesh-based reaction-diffusion.**

Papers: Gillespie 1977, *J Phys Chem* 81:2340 · Gillespie 2001, *JCP* 115:1716 (tau-leaping) · Andrews & Bray 2004, *Phys Biol* 1:137 (Smoldyn) · Kerr et al. 2008, *SIAM J Sci Comput* 30:3126 (MCell) · Isaacson 2009, *SIAM J Appl Math* 70:77 · Hepburn et al. 2012, *BMC Syst Biol* 6:36 (STEPS).

Honest gaps: Steven Andrews has no Smoldyn lecture, and ReaDDy has none either. The Schaff talk is the substitute for Smoldyn; for ReaDDy read Hoffmann, Fröhner & Noé 2019, *PLoS Comput Biol* 15:e1006830.

G.3 Whole-cell modeling

Talk	Speaker	Len	Link
Whole cell simulation with Lattice Microbes, Part I	Zan Luthey-Schulten (UIUC)	1:11:19	▶
Whole cell simulation with Lattice Microbes, Part II	Zan Luthey-Schulten	1:19:11	▶
Introduction to Simulations of Bacterial Cells (best starting point)	Zan Luthey-Schulten	1:11:36	▶
4D simulations of a growing minimal bacterial cell	Zan Luthey-Schulten	1:02:23	▶
4D simulations of a growing and dividing cell	Zan Luthey-Schulten	22:06	▶
Whole-cell modeling — a long-form interview	Luthey-Schulten with Wieczór	1:24:04	▶
How to build a computer model of a cell	Markus Covert (Stanford)	28:30	▶
A Computational Whole-Cell Model of <i>M. genitalium</i> (the original)	Markus Covert	35:37	▶
Whole-Cell Modeling — a dialogue	Serrano (CRG) & Covert (Stanford)	1:11:55	▶
Simulations of Ribosome Biogenesis at Whole-Cell Level	Tyler Earnest (UIUC)	21:39	▶
Multi-scale Simulations of Whole Yeast Cells	Tyler Earnest (UIUC)	19:56	▶
Integrative illustration and modeling of JCVI-syn3A	David Goodsell (Scripps)	22:56	▶
MD of the JCVI-syn3A Cell Envelope	Mert Bozoflu (JCVI/UIUC)	15:37	▶
Simulating whole cells with Martini — the coarse-grained route	BioExcel #84	1:01:50	▶

The Serrano–Covert dialogue is the honest one — two whole-cell groups discussing where the approach has failed to scale. And the Martini webinar marks a genuine methodological fork: coarse-grained MD versus RDME as two incompatible routes to the same object.

Papers: Thornburg et al. 2022, *Cell* 185:345 (a whole minimal cell) · Karr et al. 2012, *Cell* 150:389 (the first whole-cell model) · Roberts, Stone & Luthey-Schulten 2013, *J Comput Chem* 34:245 (Lattice Microbes) · Earnest et al. 2018, *Biophys J* 114:473 · Johnson et al. 2015, *Nat Methods* 12:85 (cellPACK).

Capstone VI is built on the gap these talks expose. Read Thornburg and ask what it would take to put a *designed* protein into that model. Enumerate every parameter it needs; mark which are computable today; quantify how wrong the computable ones are. Most of them you cannot compute at all.

G.4 Crowding and the cytoplasm

Talk	Speaker	Len	Link
CASP SIG on Modeling Conformational Ensembles	Wodak (SickKids) & Michael Feig (Michigan State)	59:16	▶
Crowding effects on fibrillar self-assembly	Divya Nayar (IIT Delhi)	27:46	▶
Measuring cytoplasmic crowding with GEMs	Liam Holt (NYU Langone)	1:17:37	▶

Holt's talk gives you the data any crowding simulation has to match, which is the right order to learn this in.

Papers: Ellis 2001, *Trends Biochem Sci* 26:597 · McGuffee & Elcock 2010, *PLoS Comput Biol* 6:e1000694 (atomistic cytoplasm) · Yu et al. 2016, *eLife* 5:e19274 (100-million-atom *Mycoplasma* cytoplasm) · Nawrocki et al. 2019, *PCCP* 21:876 · Delarue et al. 2018, *Cell* 174:338 (GEMs) · Minton 2006, *J Cell Sci* 119:2863.

Honest gap: Adrian Elcock has no recorded lecture, and neither does R. John Ellis. The two people whose work defines this subject are absent from video entirely. **The design-relevant point stands regardless:** every affinity you compute is at infinite dilution and the cell is at 300 mg/mL.

G.5 Condensates and phase separation

The Dewpoint Kitchen Table Talk series is the richest single archive here, and Pappu's contribution is a genuine three-parter.

Talk	Speaker	Len	Link
Pappu, Part 1 — polymer-physics foundations	Rohit Pappu (WashU/JHU)	1:07:09	▶
Pappu, Part 2 — stickers and spacers	Rohit Pappu	1:09:48	▶
Pappu, Part 3 — material properties, aging, and what simulation cannot predict	Rohit Pappu	1:07:05	▶
Compact 20-minute version	Rohit Pappu	20:11	▶
Emergent properties of condensates	Tanja Mittag (St. Jude)	44:58	▶
Liquid Phase Separation in Living Cells	Clifford Brangwynne (Princeton)	46:04	▶
Multiphase Liquid Behavior of the Nucleus	Clifford Brangwynne	38:08	▶
Using Light to Study and Control Phase Behavior (optoDroplets)	Clifford Brangwynne	34:40	▶
MBL Friday Evening Lecture (the most complete)	Clifford Brangwynne	1:20:21	▶
Cell organization by LLPS	Michael Rosen (UTSW)	1:00:13	▶
Multiscale Dynamics of Charged Protein Condensates	Jeetain Mittal (Texas A&M)	29:59	▶
Simulations of Intrinsically Disordered Proteins	Robert Best (NIH)	1:01:07	▶
From disordered proteins to complexes and assemblies	Robert Best	1:35:21	▶
Conformational ensembles of IDRs (CALVADOS)	Kresten Lindorff-Larsen (Copenhagen)	1:02:59	▶
Driving forces in condensates from atomistic simulations	BioExcel #97	1:01:35	▶
Sequence-to-ensemble prediction	Alex Holehouse (WashU)	1:28:52	▶
Analytical statistical mechanics of condensates	Jeremy Schmit (Kansas State)	1:10:35	▶
Prediction of small-molecule partitioning into condensates	Jerelle Joseph (Princeton)	33:55	▶
Field-theoretic approaches to sequence-dependent phase separation	Hue Sun Chan (Toronto)	29:08	▶
Conformational entropy of IDPs in condensates	Ned Wingreen (Princeton)	32:26	▶
Role of condensates in protein aggregation	Tuomas Knowles (Cambridge)	45:40	▶
Condensates in stress responses and disease	Simon Alberti (TU Dresden)	45:47	▶

Talk	Speaker	Len	Link
Active processes in condensates	Frank Jülicher (MPI-PKS)	30:36	▶
Continuum models for elastic phenomena around condensates	David Zwicker (MPI-DS)	32:14	▶
LLPS in and out of equilibrium	Alexander Grosberg (NYU)	1:03:01	▶
Single-molecule dynamics inside coacervates	Ben Schuler (Zurich)	41:10	▶
Stress-induced mRNP condensation	Allan Drummond (Chicago)	46:35	▶
Condensates and Phase Separation in Biology, Session 1	Sharp, Gladfelter, Kornberg	1:40:01	▶

Joseph's talk is the most design-relevant one here: can you predict whether your compound enters a droplet? **Jülicher's is the physics that distinguishes a living condensate from an oil drop** — non-equilibrium droplet theory.

Papers: Banani et al. 2017, *Nat Rev Mol Cell Biol* 18:285 · Dignon, Zheng, Kim, Best & Mittal 2018, *PLoS Comput Biol* 14:e1005941 (the HPS coarse-grained model most LLPS simulation rests on) · Martin et al. 2020, *Science* 367:694 (stickers and spacers) · Tesei et al. 2021, *PNAS* 118:e2111696118 (CALVADOS) · Brangwynne et al. 2009, *Science* 324:1729 · Li et al. 2012, *Nature* 483:336 (multivalency) · Zwicker et al. 2017, *Nat Phys* 13:408 · Lin, Forman-Kay & Chan 2016, *PRL* 117:178101.

G.6 Continuum electrostatics and mesoscale hydrodynamics

The [@ProteinElectrostatics](#) channel is a dedicated seminar series and is effectively the only sustained video resource on Poisson-Boltzmann anywhere.

Talk	Speaker	Len	Link
The Multi-Faceted Roles of Electrostatics in Biomolecular Simulation	Walter Rocchia (IIT)	1:01:08	▶
Modeling electrostatics and polarization in biomolecules	Ray Luo (UC Irvine, Amber PB solver)	52:11	▶
Where the PB picture of ion atmospheres fails	António Baptista (ITQB)	1:06:40	▶
Boundary-element PB (PyGBe)	Christopher Cooper (UTFSM)	32:05	▶
CECAM workshop on Advances in Electrostatics	Christopher Cooper	1:08:55	▶
Treecode-accelerated PB solvers	Weihua Geng (SMU)	42:07	▶
Beyond PB into Poisson-Nernst-Planck	Robert Krasny (Michigan)	50:07	▶
ddCOSMO / ddPCM domain decomposition	Michele Nottoli (Stuttgart)	46:27	▶
Solvation Models: Continuum (Implicit) Solvent Electrostatics	Christopher Cramer (Minnesota)	23:10	▶
Using 3D structure to predict interactions	Barry Honig (Columbia)	40:57	▶

Mesoscale and hydrodynamics: DL_MESO — DPD and lattice Boltzmann in one framework, taught to a biosimulation audience, Michael Seaton (STFC), 1:15:05 ([▶](#)) · Lattice Boltzmann for biomedical flow ([▶](#)) · **Sauro Succi, *Boltzmann and The Lattice*** — the conceptual foundation by the person who wrote the book ([▶](#)) · Mesoscale modeling of soft matter with DPD ([▶](#)) · **DPD simulation of red blood cells**, Bruce Caswell (Brown) ([▶](#)) · **Stokesian Dynamics**, Rishabh More (Stanford) ([▶](#)) · **Modeling Complex and Active Fluids**, Michael Shelley (Flatiron CCB), 1:33:39 ([▶](#)).

Papers: Baker et al. 2001, *PNAS* 98:10037 (APBS) · Jurrus et al. 2018, *Protein Sci* 27:112 · Honig & Nicholls 1995, *Science* 268:1144 · Onufriev & Case 2019, *Annu Rev Biophys* 48:275 (the implicit-solvent critique) · Groot & Warren 1997, *JCP* 107:4423 (DPD) · Brady & Bossis 1988, *Annu Rev Fluid Mech* 20:111.

Honest gap: Nathan Baker has no APBS lecture — his channel contains only four silent ten-second animations. Read Baker et al. 2001 and Jurrus et al. 2018 instead.

G.7 Paired reading — Atlas G

Watch this	Then read this	Hold this question
Scalfi, <i>Generalized Langevin Equations</i>	Zwanzig 1961; Ermak & McCammon 1978	Derive overdamped Langevin. What timescale separation is assumed?
Schreiber, <i>Crowding and reaction rates</i>	Schreiber, Haran & Zhou 2009, <i>Chem Rev</i> 109:839	How much kon can electrostatic steering buy, and can you design for it?
Gillespie, on his own algorithm	Gillespie 1977, <i>J Phys Chem</i> 81:2340	Derive the SSA. When does tau-leaping break it?
Isaacson, <i>Stochastic Simulation</i>	Isaacson 2009, <i>SIAM J Appl Math</i> 70:77	The RDME has convergence pathologies. Name them.
De Schutter, <i>Operator Splitting</i>	Hepburn et al. 2012	What error does splitting introduce, and how would you detect it?
Luthey-Schulten, <i>Bacterial Cells</i>	Thornburg et al. 2022, <i>Cell</i> 185:345	What parameters does a whole-cell model need that you cannot compute?
Covert & Serrano, <i>dialogue</i>	Karr et al. 2012, <i>Cell</i> 150:389	Where has whole-cell modeling failed to scale, and why?
Holt, <i>Measuring crowding with GEMs</i>	Ellis 2001; Yu et al. 2016, <i>eLife</i> 5:e19274	Quantify how much crowding shifts a binding equilibrium.
Pappu, Parts 1–3	Pappu et al. 2023, <i>Chem Rev</i> 123:8945	Which sequence features drive phase separation? Are they designable?
Mittal, <i>Charged condensates</i>	Dignon et al. 2018, <i>PLoS Comput Biol</i> 14:e1005941	What does the HPS model assume, and where does it fail?
Lindorff-Larsen, <i>CALVADOS</i>	Tesei et al. 2021, <i>PNAS</i> 118	How is a CG model for IDRs actually parameterized?
Joseph, <i>Small-molecule partitioning</i>	Her paper	Can you predict whether your compound enters a droplet?
Jülicher, <i>Active processes</i>	Zwicker et al. 2017, <i>Nat Phys</i> 13:408	What distinguishes a living condensate from an oil drop?
Luo, <i>Electrostatics and polarization</i>	Baker et al. 2001, <i>PNAS</i> 98:10037	Where does Poisson-Boltzmann fail, and what replaces it there?
Baptista, <i>pH and ion distributions</i>	Onufriev & Case 2019	Implicit solvent versus explicit ions. What is lost?
Seaton, <i>DL_MESO</i>	Groot & Warren 1997	When does DPD beat atomistic MD, and what conserves hydrodynamics?

Atlas H — Polymer Theory and Minimal Models

Problem. What does a chain do, independent of its chemistry? A protein is a heteropolymer, DNA is a semiflexible polymer, chromatin is a polymer of polymers. Before any of the specific interactions matter, the fact of being a connected chain already determines a great deal — the distribution of end-to-end distances, the entropic cost of stretching, the scaling of size with length, the dynamics of relaxation. **Most of the intuition people call “biophysical” is really polymer physics they absorbed without deriving.**

This family is the shortest route to being able to estimate things. It is also where the intrinsically disordered proteome lives, which is most of the proteome.

H.1 The Grosberg-Rubinstein course

ICAM-I2CAM · 5 sessions · ~7.5 hours · Tier A

Taught by the authors of the two standard textbooks. This is the single best polymer physics course freely available.

#	Lecture	Length	Link
I	Ideal chains, freely jointed chain, random walk statistics	1:35:37	▶
II	Real chains, excluded volume, Flory theory, scaling exponents	1:34:03	▶
III	Semidilute solutions, blobs, correlation length	1:24:11	▶
IV	Dynamics: Rouse, Zimm, entanglement, reptation	1:33:28	▶
V	Additional session, topological constraints	1:29:49	▶

Supplementary	Speaker	Length	Link
Polymer Physics lecture 2: Real polymer chain	Michael Rubinstein (Università di Corsica)	1:23:34	▶
Polymer Entanglements: the Unsolved Problem of Polymer Physics	Michael Rubinstein (NYU colloquium)	1:13:09	▶

H.2 The LMU derivation course

The most mathematically serious treatment found — polymer theory done as field theory, which is the language that connects it to Atlas family N.

#	Lecture	Length	Link
4	Fluctuating Manifolds and Polymer Physics: Continuous Limit of Random Walks	55:05	▶
5	Conformational Entropy, Entropic Forces and the Blob Scaling Argument	39:33	▶
6	From Discrete Polymers to Continuous Field Theory: Langevin Elasticity and Path Integrals	54:02	▶
8	Continuous Wormlike Chains: Symmetry Arguments and Conformational Statistics	41:16	▶
10	Solving Polymer Path Integrals with Quantum Mechanics: Spherical Harmonics	47:56	▶
11	The Quantum Mechanics of DNA Stretching: Stark Effect and Harmonic Oscillators	38:38	▶
13	The Lattice Gas Model, Mean-Field Theory, Mixing Entropy, Flory-Huggins	44:08	▶

Lectures 10 and 11 derive the force-extension curve that every optical-tweezer paper reports, by mapping the worm-like chain onto a quantum problem. It is a genuinely beautiful piece of physics and it makes the WLC stop being a formula you look up.

H.3 Biology-facing treatments

Title	Speaker	Length	Link
Entropic elasticity of DNA	Rob Phillips (Caltech, PBoC)	44:53	▶
Flory-Huggins theory	Rob Phillips (Caltech)	21:18	▶
Entropic springs and persistence length	NPTEL / IIT Bombay	27:54	▶
Energy of DNA/actin configurations: the worm-like chain, curvature energy	Ranjith Padinhateeri (IIT Bombay)	31:08	▶
Single-molecule dynamics and interactions of disordered proteins	Ben Schuler (Zurich)	38:03	▶

Schuler's talk is the one that matters most for your work. It is the definitive demonstration that intrinsically disordered proteins obey polymer scaling laws, measured by single-molecule FRET. If you ever want to design or predict anything about an IDP, this is the framework — and the AlphaFold-shaped hole in the disordered proteome is exactly where Atlas P fails and Atlas H succeeds.

The NPTEL polymer physics course

IIT Roorkee · 60 lectures · [Full playlist](#)

The highest-value individual lectures:

#	Lecture	Length	Link
04	Random Walk Models of Single Chain I: end-to-end distance	30:52	▶
07	Models of semiflexible chains (Kratky-Porod), Part I	33:57	▶
11	Entropic Elasticity, Bead-Spring Model, simulations of random walks	36:15	▶
17	Virial expansion, Flory theory for good solvent	33:16	▶
18	Flory theory for bad solvent, self-similarity and fractals	27:24	▶
35	Lattice models of polymeric systems	21:58	▶
41	Rouse Model I	22:48	▶
45	Problems in Rouse Model, Hydrodynamic Interactions	30:31	▶
46	Zimm Model I	30:47	▶
50	Kuhn's Theory of Rubber Elasticity	32:24	▶
54	Models for entangled polymeric systems I (reptation)	33:29	▶

H.4 Polymer models of chromatin — the ICTP school

Spring College on the Physics of Complex Systems · 8 lectures · ~10 hours

#	Lecture	Length	Link
1	Polymer Physics of Chromosome Folding	1:57:41	▶
2	Equilibrium vs non-equilibrium chromosome states	1:21:59	▶
3	Topological constraints and the crumpled globule	1:35:41	▶
4	Hi-C contact maps as polymer observables	1:35:50	▶
5	Loop extrusion and TAD formation	1:34:01	▶
6a	Loop extrusion continued, part 1	48:00	▶
6b	Loop extrusion continued, part 2	34:11	▶
7	Phase separation and compartmentalization	1:27:08	▶
8	Closing session	1:00:38	▶
Also	Speaker	Length	Link
Strings-and-Binders-Switch model of chromatin	Mattia Conte / Nicodemi group	30:23	▶
Chromatin dynamics from single-locus trajectories and polymer models	David Holcman (ENS)	21:52	▶
Knots and Coarse-grained Models for DNA, Polymers and Proteins	ICTP QLS	46:14	▶

H.5 Minimal models of folding — lattice proteins and the random energy model

Title	Speaker	Length	Link
MIT 6.849 Lec 21: HP Model and Interlocked Chains	Erik Demaine (MIT OCW)	1:09:46	▶
MLCB24 Lec 09: Protein Folding Algorithms	Manolis Kellis (MIT)	1:20:21	▶
Provost Lecture: Pathways	Ken Dill (Stony Brook)	51:20	▶
Ken Dill, TCBG seminar	Ken Dill	1:17:00	▶
Protein Biophysics in the Cellular Context	Eugene Shakhnovich (Harvard)	1:35:22	▶
Statistical mechanics of protein evolution	Eugene Shakhnovich	36:09	▶
Exploring the Protein Folding Funnel Landscape I	ICAM-I2CAM	1:02:33	▶
Exploring the Protein Folding Funnel Landscape II	ICAM-I2CAM	1:02:13	▶
Protein Folding and Functions, Electron Transfer, Molecular Motors I	José Onuchic (Rice)	1:24:50	▶
Protein Sequence Evolution	José Onuchic	1:31:41	▶
Energy landscapes, order and disorder, and coevolution	José Onuchic	1:05:59	▶
Frustration in proteins (minicourse)	Diego Ferreira (Buenos Aires)	1:02:40	▶
Modeling selection on protein folding stability	Ugo Bastolla (Madrid)	42:32	▶
Semi-flexible Lattice Model for Crambin: a REWL study	Alfred Farris	40:55	▶

Demaine’s HP model lecture is the unexpected highlight. A theoretical computer scientist treating the lattice protein model with algorithmic rigour, including approximation-ratio results for folding. It is the cleanest available demonstration that protein folding, stated minimally, is a combinatorial optimization problem with provable structure.

H.6 BUILD — Atlas H

1. **Simulate an ideal chain and a self-avoiding walk.** Measure the scaling exponent for each. Recover $\nu = 1/2$ and $\nu \approx 0.588$. Then do it in two dimensions and explain the difference.
2. **Derive and then numerically verify the worm-like chain force-extension relation.** Fit a persistence length to published DNA stretching data.
3. **Implement the HP lattice model** on a 2D square lattice. Enumerate all conformations for a 16-mer. Count how many sequences have a unique ground state. That number is designability, and it is checkpoint 23 made concrete.

4. **The disorder exercise.** Take an IDP sequence you care about. Predict its radius of gyration from polymer scaling using sequence composition. Compare to any available SAXS or FRET measurement. Then ask what AlphaFold predicted and what that prediction could possibly have meant.

Derivation checkpoints due: 31, 32, 33.

H.7 Paired reading — Atlas H

Watch this	Then read this	Hold this question
Grosberg & Rubinstein I	Rubinstein & Colby, <i>Polymer Physics</i> , ch. 2	Derive the end-to-end distribution of an ideal chain. Why Gaussian?
Grosberg & Rubinstein II	Flory 1953; de Gennes 1979	Derive $\nu = 3/5$ from the Flory argument. Why is the exact value 0.588?
Grosberg & Rubinstein IV	Doi & Edwards 1986; de Gennes 1971, JCP 55:572	Rouse vs Zimm: what physical difference produces different exponents?
LMU Lec 8	Kratky & Porod 1949; Marko & Siggia 1995, <i>Macromolecules</i> 28:8759	Derive the WLC force-extension relation.
LMU Lec 10–11	Marko & Siggia 1995	The WLC maps to a quantum rotor. What is the correspondence?
LMU Lec 13	Flory 1942, JCP 10:51; Huggins 1942	Derive the Flory-Huggins free energy. Where does the χ parameter come from?
Phillips, <i>Entropic elasticity</i>	Bustamante et al. 1994, <i>Science</i> 265:1599	DNA is an entropic spring. At what force does enthalpy take over?
Schuler, <i>Disordered proteins</i>	Schuler et al. 2016, <i>Annu Rev Biophys</i> 45:207	IDPs obey polymer scaling. What does that make a “structure prediction” of one?
NPTEL Lec 41, 45–46	Rouse 1953; Zimm 1956	Hydrodynamic screening changes the exponent. Derive both.
ICTP chromosome folding 1–3	Rosa & Everaers 2008; Grosberg et al. 1993 (crumpled globule)	Why is an equilibrium globule the wrong model for a chromosome?
ICTP chromosome folding 5	Fudenberg et al. 2016, <i>Cell Rep</i> 15:2038	Loop extrusion is a non-equilibrium active process. What does it predict that equilibrium cannot?
Conte, <i>Strings and Binders</i>	Barbieri et al. 2012, <i>PNAS</i> 109:16173	A competing model fits the same Hi-C data. How would you distinguish them?
Demaine, <i>HP Model</i>	Lau & Dill 1989, <i>Macromolecules</i> 22:3986	Enumerate designability on a 2D lattice. What does the distribution look like?
Dill, <i>Pathways</i>	Dill & MacCallum 2012, <i>Science</i> 338:1042	Fifty years on: which parts of the folding problem are actually solved?
Shakhnovich, <i>Protein Biophysics in Cellular Context</i>	Shakhnovich & Gutin 1993, <i>PNAS</i> 90:7195	Derive the random energy model condition for a folding sequence.
Onuchic, <i>Funnels</i>	Bryngelson & Wolynes 1987, <i>PNAS</i> 84:7524	Spin glasses and folding: what is the order parameter?
Ferreiro, <i>Frustration</i>	Ferreiro, Komives & Wolynes 2014, <i>Q Rev Biophys</i> 47:285	Localized frustration is functional. Could you design it deliberately?
Bastolla, <i>Selection on folding stability</i>	Bastolla et al. 2005, <i>Proteins</i> 58:22	Stability constrains sequence evolution. What does that predict for a designed protein?

Atlas I — Dynamics Analysis, Allostery, Networks and Pockets

Problem. A trajectory is a enormous pile of coordinates. Which projection of it means something? Which motions are functional and which are thermal noise? How does a perturbation at one site propagate to another? And where, on a surface that looks featureless in the crystal structure, is there a pocket that opens for a fraction of the time?

This family is the analysis layer, and it is where most published figures in computational biophysics come from. **It is also where most of the irreproducibility lives**, because almost none of these analyses come with an error bar.

I.1 Allostery: theory

Title	Speaker	Length	Link
MWC model	Rob Phillips (Caltech, BE/APh 161)	24:33	▶
MWC model of chemoreceptors	Rob Phillips	22:54	▶
Allostery, no physicists allowed	Rob Phillips	8:55	▶
Opening Lecture at the European Conference on ALLODD	Jean-Pierre Changeux (Institut Pasteur)	1:10:23	▶

Phillips's course is the only place online that **derives** MWC rather than drawing the T/R cartoon. Changeux is the living author of the 1965 model assessing it sixty years on — rare, and worth the hour.

Evolutionary coupling and sectors

Title	Speaker	Length	Link
The Evolutionary “Design” of Protein Machines	Rama Ranganathan (Chicago)	58:35	▶
The Evolutionary Design of Proteins (NITMB, 2025)	Rama Ranganathan	1:10:57	▶
Evolutionary “Design” of Proteins (Aspen, 2019)	Rama Ranganathan	48:52	▶
Part 1: What is Protein Design?	Rama Ranganathan (iBiology)	27:06	▶
Part 3: Protein Function and Adaptability	Rama Ranganathan (iBiology)	45:15	▶

Watch Aspen 2019 then NITMB 2025 and notice how much the sector claim has been walked back and re-grounded over six years. That trajectory is itself the lesson: a strong structural claim meeting sustained scrutiny.

Elastic network models and ANM-based allostery

Title	Speaker	Length	Link
Network models in biology	Ivet Bahar (Pitt / Stony Brook)	1:23:01	▶
EMBO Lecture: Confluence of structure-based models	Ivet Bahar	48:58	▶
Elastic network models, targeting hinges for drug discovery (interview)	Ivet Bahar	39:33	▶
Network Models In Biology: From Molecular Machinery To Chromosomal Dynamics	Ivet Bahar	1:10:57	▶
Leveraging Hi-C with Elastic Network Models	Ivet Bahar	23:38	▶
ProDy (hands-on)	CCPBioSim	54:34	▶
ProDy (second treatment)	SBGrid	55:32	▶
Physical properties of folded peptide chains (domain decomposition NMA)	Konrad Hinsen (CNRS Orléans)	43:53	▶
Chromosomal dynamics predicted by an elastic network model	She Zhang (ISMB 2018)	6:35	▶
Computational Biophysics Workshop Day 1, Part 2 (ENM theory)	MMBioS	1:10:07	▶

The unexploited idea. An elastic network model gives you a normal-mode spectrum from a single structure in seconds. Every generative design method produces structures. **Nobody designs for a target normal-mode spectrum**, even though Johnson and Edelman's lecture on derivatives of eigenproblems (Module B.1) gives you the gradient you would need. That is row five of the unexchanged-ideas table and it is unusually tractable.

Dynamic allostery, networks, information-theoretic coupling

Title	Speaker	Length	Link
Dissecting Allosteric Mechanisms and Protein Interactions	Amnon Horovitz (Weizmann)	1:06:41	▶
A Nonequilibrium Approach to Allosteric Communication	Peter Hamm (Zurich)	1:25:24	▶
Allostery under the lens of molecular dynamics simulations	Julien Michel (Edinburgh)	58:15	▶
Insight into protein allostery from designed mechanical networks	Andrea Liu (Penn)	29:36	▶
Predicting allosteric networks in protein complexes	NIMBioS	50:38	▶
Dynamic network analysis of protein structural change	Aydin Wells (Notre Dame)	25:47	▶
Identifying Allosteric Sites Through Network Analysis	RSG-Türkiye	34:52	▶
Mapping the energetic and allosteric landscapes of protein binding domains	André Faure (CRG Barcelona)	28:56	▶
Allostery in DNA drives phenotype switching	Hagen Hofmann (Weizmann)	1:12:31	▶

Michel's hour is the best available on what MD can and cannot tell you about allosteric coupling, and he is honest about convergence.

Faure's talk is the one that should unsettle you. Double deep mutational scanning measures allosteric coupling *experimentally*, at scale, and the structure-plus-MD predictions do not do well against it. That is a benchmark existing in the wild that almost nobody in the computational allostery community tests against — which makes it an opportunity.

Vincent Hilser, whose ensemble allosteric model is the conceptual heart of entropic allostery, has no verifiable recorded research talk. Read Motlagh, Wrabl, Li & Hilser 2014, Nature 508:331 instead. Same for Sandor Vajda on FTMap and Heather Carlson on MixMD.

I.2 Pocket and site detection

Title	Speaker	Length	Link
Accelerating Cryptic Pocket Discovery With Deep Learning	Greg Bowman (Penn)	37:23	▶
Greg Bowman, TCBG seminar	Greg Bowman	1:09:45	▶
MSMs, excited states and cryptic pockets	Greg Bowman	33:23	▶
Translational Allostery: From Cryptic Binding Pockets to Latent Networks	The Protein Society	2:27:00	▶
Drugging the Undruggable: Detecting and Ranking Cryptic Pockets	OpenEye / Cadence	46:46	▶
Investigating cryptic binding sites and allostery	ALLODD	58:51	▶
Characterization of allosteric pockets in a protein-membrane environment	ALLODD	1:05:13	▶
Design and synthesis of covalent allosteric probes	ALLODD	1:09:17	▶
SILCS	CCPBioSim (MacKerell method)	1:14:34	▶
SILCS-WATER: the role of water in protein-ligand binding	SilcsBio	56:17	▶
Automating Structure-Based Design Using Fragment Hotspot Maps	Chris Radoux (CCDC)	34:56	▶
Identifying Druggable Pockets for Protein Ensembles	Lane Votapka (Amaro lab)	20:19	▶
CASP-16 Predictor talk: ClusPro	Dima Kozakov (Stony Brook)	20:02	▶

Bowman's Valence Labs talk is the PocketMiner talk and the one to assign. **Given your own cosolvent-MD and cryptic-pocket work, this subsection is the most directly operational in the entire Atlas** — SILCS, MixMD and fragment hotspot maps are three routes to the same physics, and knowing which assumptions differ between them is the difference between running a tool and choosing one.

I.3 Interfaces and hot spots

Title	Speaker	Length	Link
I, biochemist: Automation and AI in the lab	Tanja Kortemme (UCSF)	49:41	▶
Protein-Protein Interfaces	RosettaCommons	7:49	▶
How FoldX works, with examples	FoldX team	16:26	▶
Computational Protein Design: Design of Protein-Protein Interactions	Julia Shifman (Hebrew University)	42:32	▶
Deciphering interaction fingerprints from molecular surfaces (MaSIF)	Correia lab method, journal club	1:12:13	▶
Massively parallel discovery of peptides to inhibit protein interactions	BPDMC	1:12:44	▶
Structural and functional determinants inferred from deep mutational scans	Priyanka Bajaj	32:57	▶
Predicting folding stability and aggregation from large-scale experiments	BPDMC	1:00:30	▶
Affinity Maturation walkthrough (Rosetta Workshop 2021)	Meiler Lab	33:40	▶
New Capabilities in Computational Protein Design	The Protein Society	2:44:17	▶

MaSIF is worth particular attention: it treats the molecular surface as a geometric object and learns fingerprints on it with geometric deep learning — the most direct existing bridge between Atlas family I and Atlas family O, and the method that later produced de novo interactions from learned surface fingerprints.

I.4 Trajectory analysis — and its pitfalls

Title	Host	Length	Link
Intro to MD Trajectory Analysis using MDAnalysis	MDAnalysis project	1:40:34	▶
Intro to MDAnalysis Workshop	MDAnalysis project	2:09:37	▶
MDAnalysis Streaming Online Workshop	MDAnalysis project	2:56:58	▶
Intro to MDAnalysis and Molecular Nodes	MDAnalysis project	3:33:02	▶
Amber (covers cpptraj)	SBGrid	1:11:39	▶
VMD	SBGrid	1:23:25	▶
Visualization and analysis with VMD	David Winogradoff (UIUC)	53:18	▶
Quick setup and analysis of NAMD simulations with QwikMD	Rafael Bernardi (Auburn)	55:15	▶

Dimensionality reduction and Markov state models — the PyEMMA course

The single best free course on this material anywhere.

#	Lecture	Length	Link
2018-1	Introduction to Markov state models	1:07:35	▶
2017-3	Featurization and TICA	36:16	▶
2018-7	The variational approach	32:25	▶
2018-3	MSM estimation and validation	1:01:40	▶
2018-8	How to select models with the VAMP score	21:18	▶
2021-2	MSM theory	44:32	▶
2021-5	Introduction to VAMPNets	36:45	▶
—	TICA, TCCA and time-autoencoders	32:38	▶
—	Deep learning for molecular kinetics	37:32	▶
—	Intro to Trajectory Analysis and MSMs	32:58	▶

Full 2017, 2018 and 2021 listings are in Atlas family B's enumeration.

Convergence — assign these FIRST

Title	Speaker	Length	Link
Testing Molecular Simulation Results for Physical Validity	Michael Shirts (Colorado)	28:10	▶
Reproducibility, Convergence and Accuracy in Biomolecular Simulation	NCSA / Blue Waters	17:29	▶
Convergence and Reproducibility in MD Simulations of Nucleic Acids	NCSA / Blue Waters	20:03	▶
Reproducible molecular simulations with Python	Iván Pulido (Chodera lab)	30:37	▶

The question to hold. RMSF appears in essentially every MD paper in this curriculum, including your own peptide-RMSF figures. It is a single number per residue, extracted from a trajectory that is almost never shown to be converged, after alignment to a reference that is itself a choice. Shirts's twenty-eight minutes are the antidote. Before believing any RMSF figure, ask for the block-averaged error bar.

I.5 Intrinsically disordered proteins

Title	Speaker	Length	Link
Charge-rich disordered proteins	Rohit Pappu (WashU/Hopkins)	1:16:31	▶
Dewpoint Kitchen Table Talk, Part 1	Rohit Pappu	1:07:10	▶
Dewpoint Kitchen Table Talk, Part 2	Rohit Pappu	1:09:49	▶
Dewpoint Kitchen Table Talk, Part 3	Rohit Pappu	1:07:05	▶
Interpreting experiments using simulations	Kresten Lindorff-Larsen (Copenhagen)	1:27:01	▶
Lindorff-Larsen, TCBG seminar	Kresten Lindorff-Larsen	1:13:58	▶
From disordered proteins to complexes and assemblies	Robert Best (NIH)	1:35:22	▶
HITS Colloquium: simulations of IDPs	Robert Best	1:01:07	▶
BioExcel #98: Conformational ensembles of IDRs and IDPs	BioExcel CoE	1:02:59	▶
Current issues with MD force fields for IDPs	Brigita Urbanc (Drexel)	29:46	▶
The IDRome: Conformations of the Human Disordered Proteome	Giulio Tesei (Copenhagen)	2:19	▶
Single-molecule dynamics and interactions of disordered proteins	Ben Schuler (Zurich)	38:03	▶
Probing the rapid chain dynamics of disordered proteins and nucleic acids	Ben Schuler	55:18	▶
It takes tau to tangle	Elizabeth Rhoades (Penn)	1:10:54	▶
Small-molecule binding to intrinsically disordered proteins	Cambridge CMD	19:01	▶

The Dewpoint three-parter is the most complete recorded statement of Pappu's position anywhere, and it is conversational rather than a conference talk — he says things there he would not put in a paper. Urbanc's talk names which IDP force fields fail and how, which is information that is otherwise expensive to acquire.

I.6 Condensates and phase separation

Title	Speaker	Length	Link
Phase Separation of Multivalent Proteins	Rohit Pappu	56:49	▶
Computational Approaches to Protein Condensates	The Protein Society	1:46:03	▶
Emergent properties of condensates and their effects on protein function	Tanja Mittag (St. Jude, KITP)	44:59	▶
Instruct Biennial 2024	Tanja Mittag	34:54	▶
Biomolecular condensates across the Tree of Life	Agnes Toth-Petroczy (MPI-CBG, KITP)	39:36	▶
Multiscale modelling of liquid-like chromatin organisation	Rosana Collepardo-Guevara (Cambridge)	51:04	▶
From Wiggles to Flow: Multiscale Dynamics of Charged Protein Condensates	Jeetain Mittal (Texas A&M)	29:59	▶
Principles of structure/function in biomolecular condensates	Jeremy Schmit (Kansas State)	28:51	▶
Liquid Phase Separation in Living Cells	Cliff Brangwynne (Princeton/HHMI)	46:05	▶
Condensates at the nexus of cellular stress, disease and aging	Simon Alberti (TU Dresden)	1:08:29	▶
Measuring the liquid structure of condensates in live cells	Josh Riback (Baylor)	37:43	▶
Predictive models of sequence determinants of nuclear condensates	HHMI Janelia	17:11	▶
Droplets everywhere: cell organization by LLPS	NIH WALIS	1:00:13	▶

I.7 Membrane proteins and large assemblies

Title	Speaker	Length	Link
Structure and Dynamics of Biological Membranes: Lipid-Protein Interactions	Erik Lindahl (KTH)	2:07:23	▶
Application of molecular dynamics to biomolecular modeling	Emad Tajkhorshid (UIUC)	1:37:29	▶
Revealing the Structural Basis of GPCR Signaling	Ron Dror (Stanford)	1:08:43	▶
Unraveling C-type inactivation in potassium channels	Benoît Roux (Chicago)	58:44	▶
TSRC Workshop on Protein Dynamics	Benoît Roux	49:47	▶
Roux, Free Energy Workshop 2018	Benoît Roux	33:43	▶
On MD modeling, force fields, and membrane channels (interview)	Benoît Roux	31:09	▶
Water and Ions in Membrane Nanopores and Channels	Mark Sansom (Oxford)	48:14	▶
Origins of CHARMM-GUI (interview)	Wonpil Im (Lehigh)	44:00	▶
Whole cell simulation with Lattice Microbes, Part I	Zaida Luthey-Schulten (UIUC)	1:11:20	▶
Whole cell simulation with Lattice Microbes, Part II	Zaida Luthey-Schulten	1:19:13	▶
4D simulations of a growing minimal bacterial cell	Zaida Luthey-Schulten	1:02:24	▶
Freezing ribosomes, wet proteins, flying peptides	Helmut Grubmüller (MPI Göttingen)	1:39:58	▶
Atomistic Simulation of Biomolecular Function: Ligand Binding Heterogeneity	Helmut Grubmüller	1:29:07	▶
Juan Perilla, TCBG seminar (HIV capsid)	Juan Perilla (Delaware)	1:00:53	▶
Nanotechnology with molecular simulation	Aleksei Aksimentiev (UIUC)	47:25	▶
Protein Dynamics in Cellular Environments	Rommie Amaro (UCSD)	57:48	▶
EMBL Keynote: In Situ Dynamics Reveal Unseen Vulnerabilities of Viral Glycoproteins	Rommie Amaro	47:58	▶

Title	Speaker	Length	Link
Computational Microscopy for In Situ Molecular Dynamics	Rommie Amaro	1:16:49	▶

Amaro's glycan work is the clearest demonstration that a structure without its glycan shield is a different molecule — directly relevant if you ever model a glycoprotein surface as a design target.

I.8 BUILD — Atlas I

1. **Implement the Gaussian network model from scratch.** Compute B-factors from the pseudo-inverse of the contact matrix and correlate with crystallographic B-factors for ten proteins. You will get $r \approx 0.6$. Explain the residual.
2. **Implement the anisotropic network model** and visualize the lowest modes. Compare to the principal components of a real MD trajectory of the same protein. Measure the overlap.
3. **Community network analysis.** Build a dynamical network from a trajectory using generalized correlation, partition it, and identify the optimal paths between two sites. Then do the honest part: repeat on an independent replica and see whether you get the same communities.
4. **The convergence audit.** Take one of your own published RMSF figures. Recompute it with block averaging and error bars. Recompute it with a different alignment reference. Report how much the conclusion moves.

Derivation checkpoints due: 39, 40, 41, 42.

I.9 Paired reading — Atlas I

Watch this	Then read this	Hold this question
Phillips, <i>MWC model</i>	Monod, Wyman & Changeux 1965, JMB 12:88	Derive the MWC saturation curve. What are its three parameters?
Changeux, ALLODD lecture	Cooper & Dryden 1984, Eur Biophys J 11:103	Allostery without conformational change. How would you detect it computationally?
Ranganathan, Aspen then NITMB	Lockless & Ranganathan 1999; Halabi et al. 2009, Cell 138:774	What has survived of the sector claim, and what was retracted?
Bahar, <i>Network models in biology</i>	Atilgan et al. 2001, Biophys J 80:505; Bahar et al. 2010, Chem Rev 110:1463	Derive the GNM. Why does a single-parameter model predict B-factors at all?
Bahar interview	Atilgan & Atilgan 2009, PLoS Comput Biol 5:e1000544	Perturbation-response scanning. Could it be run in reverse, as a design objective?
Michel, <i>Allostery under MD</i>	Tsai & Nussinov 2014; Wodak et al. 2019, Structure 27:566	Which allosteric signals survive a convergence test?
Faure, <i>Energetic and allosteric landscapes</i>	Faure et al. 2022, Nature 604:175	Double DMS measures coupling directly. How well does any prediction do against it?
Luthey-Schulten lineage (network analysis)	Sethi et al. 2009, PNAS 106:6620; Melo et al. 2020, JCP 153:134104	What does generalized correlation capture that linear correlation does not?
Bowman, <i>Cryptic Pocket Discovery</i>	Meller et al. 2023, Nat Commun 14:1177 (PocketMiner)	A cryptic pocket is a low-population excited state. What population is druggable?
Radoux, <i>Fragment Hotspot Maps</i>	Brenke et al. 2009 (FTMap); Kozakov et al. 2015	Three routes to hotspots. Which assumptions differ?
CCPBioSim, <i>SILCS</i>	Guvench & MacKerell 2009, PLoS Comput Biol 5:e1000435; Ghanakota & Carlson 2016	Cosolvent MD gives a free energy map. What is the reference state?
MaSIF journal club	Gainza et al. 2020, Nat Methods 17:184; Gainza et al. 2023, Nature 617:176	Surface fingerprints bypass sequence and backbone. What is lost?
Kortemme, ASBMB	Kortemme & Baker 2002, PNAS 99:14116	Computational alanine scanning: what physical model underlies the hot spot prediction?
PyEMMA 2017-3, <i>TICA</i>	Pérez-Hernández et al. 2013, JCP 139:015102	Why does TICA find slow coordinates where PCA finds high-variance ones?
PyEMMA 2018-3, <i>estimation and validation</i>	Prinz et al. 2011, JCP 134:174105	Derive why the lag time must exceed the fastest resolved process.
PyEMMA, <i>VAMPNets</i>	Mardt et al. 2018, Nat Commun 9:5	A learned featurization with a variational guarantee. What is being bounded?
Shirts, <i>Testing Simulation Results for Physical Validity</i>	Grossfield et al. 2019, LiveCoMS 1:5067	Apply every test to your own trajectory. Which does it fail?
Pappu, Dewpoint 1–3	Das & Pappu 2013, PNAS 110:13392; Choi, Holehouse & Pappu 2020	Charge patterning (κ) predicts dimensions. What else does sequence predict?
Urbanc, <i>IDP force fields</i>	Robustelli, Piana & Shaw 2018, PNAS 115:E4758; Huang et al. 2017, Nat Methods 14:71	Which force fields over-compact IDPs, and how was that diagnosed?
Tesei, <i>IDRome</i>	Tesei et al. 2024, Nature 626:897; Tesei et al. 2021 (CALVADOS)	A coarse-grained model for the whole disordered proteome. What does it get

Watch this	Then read this	Hold this question
		wrong?
Schuler, <i>Disordered proteins</i>	Schuler et al. 2016, <i>Annu Rev Biophys</i> 45:207	smFRET gives a distance distribution. What must a model output to be comparable?
Mittag / Pappu, condensates	Martin et al. 2020, <i>Science</i> 367:694 (stickers and spacers)	Aromatic patterning sets phase behaviour. Could you design a condensate?
Roux, <i>C-type inactivation</i>	Bernèche & Roux 2001, <i>Nature</i> 414:73	Free energy of ion conduction. What made this tractable where binding is not?
Amaro, <i>Viral glycoproteins</i>	Casalino et al. 2020, <i>ACS Cent Sci</i> 6:1722	Glycans shield most of the surface. What does that do to epitope prediction?
Luthey-Schulten, <i>Lattice Microbes</i>	Thornburg et al. 2022, <i>Cell</i> 185:345	A whole minimal cell. What is the coarsest level that still predicts behaviour?

Atlas J — Structure Determination Algorithms

Problem. Experimental data — diffraction intensities, particle images, NOE cross-peaks, scattering curves — do not contain coordinates. Turning them into coordinates is an inverse problem, usually underdetermined, always noisy, and solved by algorithms that embed strong priors. **Every structure you have ever trained a model on is the output of one of these algorithms, and the prior is baked into the answer.** A computational person who does not know which parts of a PDB entry are measurement and which are prior will systematically over-trust their training data.

This family is where the learning-first culture is weakest, and the weakness is consequential. A model trained on coordinates inherits every assumption in this section without being told.

J.1 Cryo-EM reconstruction

Maximum likelihood and the Bayesian framework

Title	Speaker	Length	Link
Cryo-EM17 lecture 06: Image refinement in 2D and 3D	Sjors Scheres (MRC-LMB)	53:46	▶
Cryo-EM Image Processing (2024)	Sjors Scheres	52:16	▶
Reconstruction of Cryo-EM Images of Proteins to Atomic Resolution	Sjors Scheres (EPFL Imaging)	47:18	▶
EM Image Formation and Single-particle Reconstruction	Fred Sigworth (Yale)	1:31:22	▶
Algorithms and foundational math, Part II	Fred Sigworth (NCCAT)	56:29	▶
CryoEM 1.1: Two Astonishing Phenomena	Fred Sigworth	13:32	▶
Cryo-EM Workshop (the dissenting view)	Marin van Heel	38:42	▶

The Scheres lecture is the canonical derivation of empirical-Bayes refinement: **marginalize over unknown orientations rather than assigning them, and learn the regularization from the data instead of hand-tuning a filter.** That move — treating a nuisance parameter as something to integrate out rather than estimate — is one of the most transferable ideas in this entire document.

Watch van Heel afterward specifically because he disagrees. Pedagogically that is worth more than a second agreeing lecture.

Image formation, CTF and Fourier theory

Title	Speaker	Length	Link
Cryo-EM14 lecture 2: Image formation, Fourier analysis and CTF theory	Sjors Scheres	1:00:52	▶
Cryo-EM17 lecture 03: Image formation, Fourier analysis, CTF	Paula da Fonseca	45:15	▶
Part 3: CTF Correction	Grant Jensen (Caltech)	31:44	▶
Fourier transforms in cryo-EM: convolution, Gaussians, FSC, FFT	Pavel Afanasyev (ETH)	59:45	▶
Basics of single-particle cryo-EM image processing	Pavel Afanasyev	1:00:03	▶
Cryo-EM17 lecture 01: Cryo-EM past, present and future	Richard Henderson	59:38	▶

Henderson's lecture contains the Rosenthal-Henderson B-factor plot, which predicts how many particles you need for a given resolution. It is the information-theoretic limit of the experiment and it is the kind of result that should exist for design methods and does not.

Resolution and validation

Title	Speaker	Length	Link
Theory of CryoEM SPA 9: Resolution	Carlos Oscar Sorzano (CNB-CSIC)	33:39	▶
Theory of CryoEM SPA 1: Basics of image processing operations	Carlos Oscar Sorzano	1:39:06	▶
Theory of CryoEM SPA 7: Validation	Carlos Oscar Sorzano	32:43	▶
3DEM map-model validation using Q-scores	Greg Pintilie (Stanford)	1:00:28	▶
Practical validation in cryo-EM (and common pitfalls)	Oliver Clarke (Columbia)	1:15:31	▶
Cryo-EM map validation	Kako Stapleton (Glasgow/eBIC)	41:26	▶

Clarke's talk is a catalogue of the specific ways reconstruction algorithms produce confident garbage — model bias, Einstein-from-noise, mask artifacts. Watch it and then ask what the equivalent catalogue would be for a generative design model. Nobody has written that one.

Particle picking with deep learning

Title	Speaker	Length	Link
Detection of objects in cryo-EM micrographs using geometric deep learning	Tristan Bepler (NYSBC)	42:58	▶
Topaz (developer walkthrough)	SBGrid	44:49	▶
crYOLO	SBGrid	39:14	▶
AI methods for picking, refinement and model building	Jianlin Cheng (Missouri)	59:53	▶

Bepler frames picking as **positive-unlabeled learning** — the key insight that your labels are incomplete rather than negative. That reframing generalizes directly to protein design datasets, where “not reported to bind” is routinely and wrongly treated as “does not bind.”

Heterogeneity — the frontier

Title	Speaker	Length	Link
Resolving flexibility and heterogeneity with 3D Variability Analysis	Ali Punjani (Structura/Toronto)	1:28:55	▶
CryoSPARC 3D Flexible Refinement Tutorial	Structura	1:22:11	▶
New Developments for SPA Data Processing in CryoSPARC	Ali Punjani	28:41	▶
Cryo-EM Workshop (ab initio branch-and-bound)	Ali Punjani	30:40	▶
New methods in cryoDRGN: cryoDRGN-ET and cryoDRGN-AI	Ellen Zhong (Princeton)	58:14	▶
ML for determining protein structure and dynamics from cryo-EM images	Ellen Zhong (IPAM)	48:30	▶
Reconstructing continuous distributions of 3D protein structure (MLCB 2019)	Ellen Zhong	15:44	▶
Algorithms for Biomolecular Structures at Proteome Scale (2026)	Ellen Zhong	58:24	▶
CryoDRGN (developer deep dive)	SBGrid	1:02:55	▶
Contextual Conformational Variability using Deep Learning	Steven Ludtke (Baylor)	41:06	▶
FaNaC1 and Continuous Heterogeneity (case study)	Structura	1:28:45	▶

This subsection is the most important in Atlas J for your purposes. A particle stack contains an ensemble; 3DVA, 3DFlex and cryoDRGN are three different answers to the question of how to extract it. 3DVA is probabilistic

PCA in volume space. 3DFlex learns a deformation field in a canonical frame, so flexibility becomes warping rather than a mixture of rigid volumes. cryoDRGN is a VAE with a coordinate-based decoder.

These are MSM-style reasoning with learned representations, arrived at independently by the cryo-EM community. **Nobody has properly connected them to Atlas family I.** That connection — what is the relationship between a cryoDRGN latent coordinate and a TICA coordinate, and can one be used to validate the other — is an unclaimed project.

Motion correction and the inverse problem

Title	Speaker	Length	Link
Bayesian Particle-Polishing in Detail	Jasenko Zivanov (Basel)	31:39	▶
Cryo-EM Workshop (compact version)	Jasenko Zivanov	23:42	▶
CryoSPARC v4.4: Reference-Based Motion Correction	Structura	17:22	▶
Mathematics of Cryo-Electron Microscopy (2025)	Amit Singer (Princeton)	59:26	▶
Mathematics for cryo-EM — ICM 2018 plenary	Amit Singer	50:35	▶
RECOVAR: regularized covariance estimation for heterogeneity	Amit Singer	34:36	▶
Method of moments in cryo-EM	Joe Kileel	48:37	▶
Hyper-molecules: continuous heterogeneity before deep learning	Roy Lederman (Yale)	33:08	▶
Imaging biomolecules and high-resolution SPA	Alberto Bartesaghi (Duke)	41:23	▶
Machine learning for SPA by cryo-EM	Carlos Oscar Sorzano	40:12	▶

Singer's ICM plenary is the most mathematically rigorous item in this family. Kileel's method of moments — reconstruction without ever estimating orientations, by matching low-order moments — is algebraically elegant and worth an hour even if you never use it, because it demonstrates that the obvious formulation of a problem is not always the necessary one.

Symmetry, strategy, maps and models

Title	Speaker	Length	Link
Cryo-EM17 lecture 07: Data processing strategy	Rafael Fernandez-Leiro	47:27	▶
Processing difficult datasets in cryo-EM	Kelly Nguyen (MRC-LMB)	44:43	▶
RELION 5 (Blush regularization, DynaMight)	Sjors Scheres	28:28	▶
RELION 4.0 (subtomogram averaging unified)	MRC-LMB	33:48	▶
Map post-processing with DeepEMhancer	Ruben Sanchez-Garcia (Oxford)	17:21	▶
DeepEMhancer (long form, failure modes)	SBGrid	43:20	▶
Cryo-EM map density modification and model building	Tom Terwilliger (Phenix)	38:56	▶
ModelAngelo: automated atomic modelling and protein identification	Kiarash Jamali (MRC-LMB)	20:03	▶
ModelAngelo (architecture detail)	SBGrid	38:10	▶
Phenix automated model building (map_to_model)	Phenix	1:23:26	▶
Cryo-EM17 lecture 08: Atomic modeling and validation	Alan Brown (Harvard)	49:59	▶
Refinement and validation using TEMPy2	Maya Topf (Birkbeck)	39:54	▶
Seven years of ISOLDE: model building as restrained MD	Tristan Croll	33:52	▶

Pair DeepEMhancer with Terwilliger. One is a neural sharpener trained to map raw reconstructions to locally sharpened targets, with a frank account of the hallucination risk; the other is maximum-likelihood density modification carried over from crystallography, using half-map statistics as an error model. Same goal, learned versus principled. That pairing is the whole learning-versus-physics argument in miniature, on a problem small enough to see clearly.

ModelAngelo's striking result is that a graph network consuming density, sequence and geometry **can identify which protein is in the map by searching a proteome**. That is structure-based search, and it is the cryo-EM community inventing Foldseek's problem independently.

Tomography

Title	Speaker	Length	Link
Cryo-EM17 lecture 09: Tomography	John Briggs (MRC-LMB)	40:07	▶
High-throughput algorithms for subtomogram averaging	Mikhail Kudryashev	46:46	▶
Cryo-ET and subtomogram averaging	Giulia Zanetti (Birkbeck)	22:46	▶
Dynamo (geometry-constrained alignment)	SBGrid	46:49	▶
Representation learning for picking in cryo-ET with TomoTwin	MPI Dortmund	27:31	▶

J.2 Crystallographic algorithms

The phase problem

Title	Speaker	Length	Link
X-ray Crystallography: Phase Problem Part I	IIT Roorkee	58:31	▶
Direct Methods: Phase Determination in Crystallography	BioXFEL	23:25	▶
Phase problem, direct methods, Patterson method	Vidya-mitra	45:42	▶

Molecular replacement

Title	Speaker	Length	Link
Molecular Replacement and cryo-EM docking	Randy Read (Cambridge)	22:30	▶
Information content in molecular replacement	Randy Read (CCP4 SW2019)	27:32	▶
Trueblood Award Lecture 2022	Airlie McCoy (Cambridge)	55:04	▶
Solve your structure with MR at warp speed	Airlie McCoy	33:06	▶
Phasing and Molecular Replacement	Andrea Thorn (ThornLab)	50:27	▶
Phaser (algorithm walkthrough)	SBGrid	43:42	▶

Read’s “information content” talk is the one to steal from. It reframes molecular replacement as an information problem — expected log-likelihood gain as a predictor of success *before you run anything*. The protein design analogue would be a quantity that predicts, before synthesis, whether a design campaign will succeed. No such quantity exists. That is a thesis.

McCoy's warp-speed talk covers how AlphaFold models with per-residue error estimates get weighted into the search — one of the cleanest existing examples of a learned model's uncertainty being used correctly downstream.

Experimental phasing, density modification, refinement

Title	Speaker	Length	Link
Pattersons, locating heavy atoms, and SHELXC/D/E	Andrea Thorn	1:05:16	▶
Experimental Phasing	Andrea Thorn (ThornLab)	54:15	▶
Advanced SHELXC/D/E	ThornLab	37:32	▶
Experimental Phasing (SIR/SAD/MAD)	BioXFEL	1:00:55	▶
Map Improvement, Averaging, and Density Modification	Kevin Cowtan (York)	1:04:42	▶
Mini-lecture: Density modification	Phenix	4:49	▶
Refinement	Garib Murshudov (MRC-LMB)	1:10:09	▶
Refinement, X-ray and cryo-EM	Pavel Afonine (Phenix)	1:00:19	▶
Refinement	Andrea Thorn (ThornLab)	54:36	▶
CC*: linking crystallographic model and data quality	SBGrid	50:08	▶
Densities in X-ray crystallography and electron microscopy	Garib Murshudov	30:34	▶
Twinning	ThornLab	44:48	▶
PDB-REDO as data source for refinement and validation	Robbie Joosten (NKI)	20:04	▶
cctbx.xfel: software for serial crystallography	SBGrid	43:47	▶
DIALS	SBGrid	46:02	▶

Cowtan's lecture contains the phase-combination bookkeeping that stops you from double-counting information — a discipline the ML community has no equivalent of and badly needs when combining MSA, template and structural priors.

The CC* lecture matters disproportionately. Karplus and Diederichs showed that model quality bounds data quality and vice versa, which is the statistically defensible replacement for arbitrary resolution cutoffs. If you filter PDB training data by “quality,” this lecture tells you what you are actually doing.

Multiconformer models and room-temperature crystallography

Title	Speaker	Length	Link
Automated Multiconformer Model Building (qFit)	Fraser/van den Bedem lineage, ML4PE	57:53	▶
qFit (developer treatment)	SBGrid	1:00:17	▶
TSRC Workshop on Protein Dynamics	James Fraser (UCSF)	44:09	▶
Multitemperature data and diffuse scattering	BioCAT / BioXFEL	59:54	▶
CASP SIG on Modeling Conformational Ensembles	Wankowicz & Fraser	1:14:26	▶
Refactoring the B-factor	Nick Pearce (VU Amsterdam)	16:40	▶
Modeling structural heterogeneity in crystallography and cryo-EM	Rutgers IQB	37:27	▶

qFit's algorithm is worth knowing in detail: **sample backbone and side-chain alternates, then select a sparse subset by quadratic and mixed-integer programming** so that you fit the density without fitting the noise. That is a sparse model selection problem, and the same formulation would apply directly to selecting a sub-ensemble from a generative model's samples. Nobody has made that carry.

The Wankowicz and Fraser CASP talk asks how you would even *score* a predicted ensemble against experiment. That is the open problem that gates every ensemble prediction method in Atlas family P.

J.3 NMR structure calculation

The thinnest area in this family, and the gap is real rather than a search failure: the primary method developers have not put algorithm lectures online.

Title	Speaker	Length	Link
Structure determination of biomolecules by NMR	Antonio Rosato (CERM Florence)	1:08:16	▶
Biomolecular NMR for Protein Structure and Dynamics (L03)	Bruce Donald (Duke)	1:50:01	▶
NMR, RDCs and Protein Structure Calculation (L04)	Bruce Donald (Duke)	1:54:38	▶
Automated High-Resolution Structure Determination Using RDCs	Bruce Donald	54:29	▶
Residual Dipolar Couplings: Theory and Applications, Part 1	Ad Bax (NIH)	45:52	▶
Residual Dipolar Couplings, Part 2	Ad Bax	1:03:13	▶
The NOE for distance measurement	AMPERE Encyclopedia	24:14	▶
Paramagnetic NMR and the MAXOCC portal	Antonio Rosato	1:03:49	▶
Introduction to Paramagnetic NMR Spectroscopy	ANZMAG	1:14:53	▶
Protein Structure Determination Using Paramagnetic NMR	Alireza Bahramzadeh	58:32	▶
The important role of dynamics	Lewis Kay (Toronto)	1:18:31	▶
Making the Invisible Visible: Molecular Machines in Motion	Lewis Kay	1:16:53	▶

The Bruce Donald lectures are the find here. He is a computer scientist, and he treats NMR structure determination as a constraint-satisfaction and search problem with complexity arguments — including RDC-EXACT, polynomial-time structure determination from sparse RDCs, deliberately contrasted with the simulated-annealing paradigm everyone else uses. That framing exists nowhere else and it is exactly the posture this curriculum is trying to teach.

Kay's relaxation-dispersion work recovers the structure, population *and* exchange rate of a state that is spectroscopically invisible. It is the cleanest experimental ground truth available for any excited-state prediction method, and it is what would falsify an ensemble generator.

Gaps: no algorithm lectures by Güntert (CYANA), Schwieters (XPLOR-NIH), Clore, or on CS-Rosetta. Read Güntert et al. 1997, Herrmann et al. 2002, Schwieters et al. 2003 and Shen et al. 2008 instead.

J.4 SAXS, SANS and low-resolution methods

Title	Speaker	Length	Link
WeNMR Lecture on SAXS, Part I	Alexey Kikhney (EMBL Hamburg)	1:51:33	▶
WeNMR SAXS Part II (DAMMIN/DAMMIF)	Alexey Kikhney	1:02:51	▶
WeNMR SAXS Part III (rigid-body and hybrid modeling)	Alexey Kikhney	1:01:23	▶
SAXS Part I: Introduction to Biological Small Angle Scattering	SBGrid	49:18	▶
SAXS Part II: Advanced Applications	SBGrid	51:53	▶
Analyzing Flexible and Disordered Macromolecules with SAXS	BioCAT (APS)	44:55	▶
DENSS: Ab initio electron density maps from SAXS	Thomas Grant (SBGrid)	1:03:36	▶
SEC-SAXS and Advanced SAXS Analysis	BioCAT	1:10:19	▶
Small Angle Neutron Scattering: a complementary technique	BioCAT	49:50	▶
Deuteration for biological neutron scattering	LINXS (Lund)	1:05:55	▶

The Ensemble Optimization Method in the BioCAT flexible-molecules lecture is a **genetic-algorithm selection of a sub-ensemble whose averaged profile fits the data** — the only honest way to treat a flexible system with an averaging observable. It is the same sparse-selection problem as qFit and as ensemble reweighting in Atlas K, arrived at independently a third time. When the same problem shape appears three times in three communities, that is the signature of something worth generalizing.

J.5 Crosslinking-MS, HDX and native MS

Title	Speaker	Length	Link
Developing crosslinking mass spectrometry	Francis O'Reilly (TU Berlin)	17:40	▶
Elucidating large molecular architectures by crosslinking and MS	Nir Kalisman (HUJI)	35:05	▶
Crosslinking MS (search algorithms and FDR)	Jürgen Cox (MaxQuant)	35:50	▶
Advancing crosslinking MS to elucidate cellular networks	ASBMB	50:32	▶
Structural Insights from HDX Mass Spectrometry	Trajan Scientific	1:14:30	▶
FPOP Tutorial Series: Introduction	The Sharp Lab	33:16	▶
Native Mass Spectrometry for Structural Biology	The Protein Society	2:19:08	▶

Note the recurring theme: **HDX restrains modeling only through a forward model relating structure to exchange rate**. Every one of these data types does. The quality of a structural inference from sparse data is bounded by the quality of its forward model, and forward models are where almost no machine learning has been applied. That observation is row six in the unexchanged-ideas table.

J.6 BUILD — Atlas J

1. **Implement 2D single-particle alignment and averaging** from scratch on simulated noisy projections. Add a CTF. Watch your average degrade as SNR falls, and find the particle count where it recovers.
2. **Implement the FSC** and reproduce the relationship between particle number and resolution. Then deliberately over-mask and watch the FSC lie to you.
3. **Take one structure you use in training data**. Find its validation report, its resolution, its Rfree, its occupancies and its alternate conformations. Write one page on which parts of that coordinate file are measurement and which are prior. Do this for a 1.2 Å structure and a 3.5 Å structure and compare.
4. **The carry exercise**. Read the qFit selection formulation. Write down how you would use the same mixed-integer formulation to select a sub-ensemble from RFdiffusion samples. Two pages. That is a paper outline.

Derivation checkpoints due: 43, 44, 45, 46.

J.7 Paired reading — Atlas J

Watch this	Then read this	Hold this question
Scheres, <i>Image refinement in 2D and 3D</i>	Scheres 2012, <i>J Mol Biol</i> 415:406; <i>J Struct Biol</i> 180:519	What is marginalized over, and what would break if you estimated instead?
Sigworth, <i>EM Image Formation</i>	Sigworth 1998, <i>J Struct Biol</i> 122:328	Write the likelihood for one particle image. What are the nuisance parameters?
Henderson, <i>past present and future</i>	Rosenthal & Henderson 2003	Derive the particle-number vs resolution relation. What is its design analogue?
Sorzano, <i>Resolution</i>	Scheres & Chen 2012	Why 0.143? What would the right threshold be if you derived it?
Clarke, <i>Practical validation</i>	Any Einstein-from-noise analysis	List every failure mode. Which has a generative-design equivalent?
Bepler, <i>Topaz</i>	Bepler et al. 2019, <i>Nat Methods</i> 16:1153	Positive-unlabeled learning. Where else are your labels incomplete rather than negative?
Punjani, <i>3DVA</i>	Punjani & Fleet 2021, <i>J Struct Biol</i> 213:107702	3DVA is probabilistic PCA over volumes. What is the analogue of a TICA coordinate here?
Punjani, <i>3DFlex</i>	Punjani & Fleet 2023, <i>Nat Methods</i> 20:860	A learned deformation field versus a mixture of volumes. When does each win?
Zhong, <i>cryoDRGN-AI</i>	Zhong et al. 2021, <i>Nat Methods</i> 18:176	What does the latent space mean, and how would you validate that interpretation?
Zivanov, <i>Bayesian particle polishing</i>	Zivanov et al. 2019, <i>IUCrJ</i> 6:5	A spatial-coherence prior plus damage-weighted B-factors. What is the prior doing?
Singer, ICM plenary	Any sample-complexity result for cryo-EM	How many images does the inverse problem require at a given SNR?
Terwilliger + DeepEMhancer	Terwilliger et al. 2020, <i>Nat Methods</i> 17:923; Sanchez-Garcia et al. 2021	Principled versus learned map improvement. How would you detect hallucination?
Jamali, <i>ModelAngelo</i>	Jamali et al. 2024, <i>Nature</i> 628:450	It identifies the protein by searching a proteome. How does that relate to Foldseek?
Read, <i>Information content in MR</i>	Oeffner, McCoy & Read 2013	Expected LLG predicts success before you run. What predicts design success?
Cowtan, <i>Density modification</i>	Cowtan 1994; Wang 1985	How do you combine phase information without double-counting?
SBGrid, *CC**	Karplus & Diederichs 2012, <i>Science</i> 336:1030	Model quality bounds data quality. What does that imply for filtering training data?
Murshudov, <i>Refinement</i>	Murshudov, Vagin & Dodson 1997	Write the ML refinement target. Where do restraints enter, and with what weight?
ThornLab, <i>Refinement</i>	Brünger 1992, <i>Nature</i> 355:472	What does Rfree detect that Rwork cannot? What is the ML analogue?
qFit talks	van den Bedem et al. 2009; Riley et al. 2021	Sparse selection by mixed-integer programming. Apply it to generative samples.
Fraser, <i>TSRC Protein Dynamics</i>	Keedy et al. 2015, <i>eLife</i> 4:e07574	Cryocooling erases substates. What fraction of the PDB is a cryo artifact?

Watch this	Then read this	Hold this question
Wankowicz & Fraser, CASP SIG	Any ensemble-scoring proposal	How would you score a predicted ensemble? Propose a metric and its failure mode.
Donald, L03–L04	Wang & Donald 2004	Structure determination as constraint satisfaction. What becomes tractable?
Bax, <i>RDCs Part 1</i>	Tjandra & Bax 1997, <i>Science</i> 278:1111	Orientational versus distance restraints. What does each underdetermine?
Kay, <i>The important role of dynamics</i>	Korzhnev et al. 2004, <i>Nature</i> 430:586	An invisible state with a measured population. Could a generative model predict it?
Kikhney, SAXS II	Svergun 1999; Franke & Svergun 2009	DAMMIN regularizes by compactness. What does that prior exclude?
BioCAT, <i>Flexible molecules</i>	Bernadó et al. 2007; Tria et al. 2015	Sub-ensemble selection by genetic algorithm. Same problem as qFit. Generalize it.
O'Reilly, <i>XL-MS</i>	O'Reilly & Rappsilber 2018, <i>Nat Struct Mol Biol</i> 25:1000	A crosslink is an upper-bound distance restraint. What is its false-positive rate?
HDX lecture	Engen 2009, <i>Anal Chem</i> 81:7870	Write the forward model from structure to protection factor. Could it be learned?

Atlas K — Integrative and Hybrid Modeling

Problem. You have cryo-EM at 8 Å, three crosslinks, a SAXS curve, two homology models and a hypothesis. Build a structure and say honestly how much of it you know.

This is the family with the best-developed epistemology in structural biology. Its central question — *what did the data determine and what did the prior determine* — is the question every AlphaFold user should be asking and mostly is not.

K.1 The integrative modeling platform

Resource	What it is	Link
IMP tutorials	Sali lab's integrative modeling platform, full worked examples	integrativemodeling.org
PDB-Dev / PDB-IHM	The archive for integrative structures, with its own validation	pdb-ihm.org
wwPDB Integrative/Hybrid Methods task force reports	The standards documents	—
HADDOCK information-driven docking	→ Atlas M.2	—

Papers: Russel et al. 2012, *PLoS Biol* 10:e1001244 (Putting the pieces together: integrative modeling platform software) · Alber et al. 2007, *Nature* 450:683 (the nuclear pore complex — the landmark application) · Rout & Sali 2019, *Cell* 177:1384 · Sali et al. 2015, *Structure* 23:1156 (outcome of the first wwPDB hybrid/integrative methods task force) · Schneidman-Duhovny et al. 2014, *Curr Opin Struct Biol* 28:96.

The nuclear pore paper is the one to study. Thirty-plus proteins, no crystal structure of the assembly, and a model with explicit uncertainty bounds. Read how they report it: not as “the structure,” but as an ensemble consistent with the data, with the resolution stated per region.

Derivation checkpoint 40. Write the integrative modeling problem as Bayesian inference: what is the likelihood for each data type, what is the prior, and what does the sampled posterior mean? Then answer the hard part — **how would you detect that your model is determined by the prior rather than by the data?** The IMP answer is sampling exhaustiveness tests; make sure you can state why those are necessary and what they do not catch.

K.1b The recorded literature, filled

The second edition had four videos in this family. Here is the rest.

Sali, IMP, and the integrative modeling platform

Talk	Speaker	Len	Link
Integrative modeling of assemblies and pathways, Lecture 1	Andrej Sali (UCSF)	1:05:24	▶
Lecture 2 — the four-stage IMP protocol and scoring	Andrej Sali	56:55	▶
Session 3 — sampling and validation	Andrej Sali	18:37	▶
Session 4 — Q&A	Andrej Sali	14:16	▶
Faculty Research Lecture — the career retrospective, including the nuclear pore in full	Andrej Sali	1:32:58	▶
Why the PDB had to change its data model	Andrej Sali (PDB50)	32:25	▶
The nuclear pore complex, visualized	Sali Lab	5:06	▶
IMP architecture	Daniel Russel (UCSF, lead IMP author)	8:40	▶
PDB-IHM: deposition, curation, validation	CCPBioSim	41:18	▶
PDB-Dev, the predecessor	BioExcel #45	1:02:47	▶
Depositing IHM data to the PDB	RCSB	33:45	▶
Assemblin — the Kosinski-lab alternative	SBGrid	49:22	▶
Integrative modeling of structure and dynamics	Dina Schneidman-Duhovny (HUJI)	15:16	▶
Data-driven docking and restraint quality	Ezgi Karaca (IBG Izmir)	48:28	▶
Scoring-function construction	Riccardo Pellarin (Institut Pasteur)	35:05	▶
Integrative modeling of complexes, Part 1	BioExcel	45:38	▶
Integrative modeling of complexes, Part 2	BioExcel	43:15	▶
Solving 3D puzzles by physics and AI-based integrative modelling	Alexandre Bonvin (Utrecht)	1:13:39	▶

Bonvin's 2026 talk is the most current here — how AlphaFold changed, and did not change, restraint-driven modeling. **Sali's Faculty Research Lecture is the best single Sali talk**, because the nuclear pore work is told as a whole.

Bayesian inference and ensemble reweighting

Talk	Speaker	Len	Link
Reweighting simulations with explicit-solvent SAXS restraints	Mattia Bernetti (SISSA)	33:07	▶
Molecular Simulation Meets Cryo-Electron Tomography	Gerhard Hummer (MPI Biophysics)	42:11	▶
TCBG seminar — the SARS-CoV-2 glycan shield	Gerhard Hummer	1:05:31	▶
Shoot First, Ask Questions Later (ICTP colloquium)	Gerhard Hummer	1:04:18	▶
Concepts of Probability in the Sciences — why inference, not fitting	Gerhard Hummer	38:16	▶
Bayesian inference of chromatin structure from Hi-C	Simeon Carstens (Pasteur, Habeck-trained)	1:05:42	▶
Bayesian nonparametrics for single molecule	Steve Pressé (ASU)	27:06	▶
Bayesian inference of force fields	Open Force Field	51:06	▶
CASP SIG on Modeling Conformational Ensembles	Wodak & Feig	59:16	▶
PLUMED Masterclass 22.1 part II — metainference machinery	PLUMED	2:16:35	▶

Hummer's *Concepts of Probability in the Sciences* is the talk that states the thesis of this family directly: structure determination is inference, and the output should be a posterior.

Honest gap, and it is significant. There is **no recorded Bonomi, Habeck or Nilges lecture** on Bayesian structure determination, and **no BioEn talk**. Metainference and inferential structure determination are represented only indirectly, through the PLUMED masterclasses and the Carstens interview. **Read instead:** Rieping, Habeck & Nilges 2005, *Science* 309:303 · Bonomi et al. 2016, *Sci Adv* 2:e1501177 (metainference) · Hummer & Köfinger 2015, *JCP* 143:243150 and Köfinger et al. 2019, *JCTC* 15:3390 (BioEn) · Cesari, Reißer & Bussi 2018, *Computation* 6:15 · Bottaro & Lindorff-Larsen 2018, *Science* 361:355.

Fitting models into density

Talk	Speaker	Len	Link
Introduction to Molecular Dynamics Flexible Fitting	Ryan McGreevy (UIUC/TCBG)	30:01	▶
Interactive MDFF demo	TCBG UIUC	7:41	▶
Rosetta for CryoEM-Guided Structure Determination (full workshop)	Frank DiMaio (UW)	3:38:48	▶
Cryo-EM refinement with density-guided simulations in GROMACS	BioExcel #82	59:01	▶
Density guided simulations	Christian Blau (KTH)	1:06:55	▶
Cryo-EM densities vs stereochemistry — the overfitting problem, short	Christian Blau (KTH)	12:39	▶
Validation of atomic models fitted in cryo-EM maps	Agnel Joseph (CCP-EM)	16:38	▶
Recent developments in model fitting and validation (TEMPy)	Agnel Joseph (CCP-EM)	29:56	▶
Atomic flexible fitting — a full methods seminar	SBBf Biophysics	1:34:40	▶
Flexible fitting into cryo-EM maps, L20	IISER Thiruvananthapuram	1:07:15	▶
Flexible fitting into cryo-EM maps, L21	IISER Thiruvananthapuram	49:08	▶
Restraints in Phenix	Phenix developers	26:28	▶

Blau's twelve-minute talk is the one to watch first — it is directly about how density pull destroys geometry if the weighting is wrong, which is derivation checkpoint 41 in compressed form. *Note: the DiMaio workshop has embedding disabled; it plays normally on YouTube.*

Prediction plus sparse restraints

Integrative structural modeling using SAXS data, Dina Schneidman-Duhovny (HUJI), 1:02:50 ([▶](#)) — the definitive SAXS-driven modeling lecture, by the author of FoXS · **Analyzing Flexible and Disordered Macromolecules with SAXS**, BioCAT, 44:55 ([▶](#)) — the ensemble-from-SAXS problem done carefully · Rigid-body modeling with SAXS ([▶](#)) · **Cross-linking MS from the data-generation side**, Şule Yilmaz-Rumpf (MaxQuant Summer School), 36:24 ([▶](#)) — essential, because most modelers badly overtrust crosslink restraints · NMR-driven integrative structural biology, Hugo van Ingen (Utrecht) ([▶](#)).

Papers: Schneidman-Duhovny et al. 2016, *NAR* 44:W424 (FoXS/MultiFoXS) · Bernadó et al. 2007, *JACS* 129:5656 (EOM) · Leitner et al. 2016, *Trends Biochem Sci* 41:20 (XL-MS) · **Stahl, Graef & Hummer 2024, *Nat Methods* 21:2224 (AlphaFold with experimental restraints).**

Honest gap: HDX-guided modeling has no methods lecture — every search returned vendor content. Read Bradshaw et al. 2021, *Protein Sci* 30:2300 and Lee et al. 2022, *Nat Methods* 19:1500.

K.2 Fitting models into maps

Topic	Resource	Link
Rigid and flexible fitting	ChimeraX fitmap ; Situs; MDFF tutorials	▶
Rosetta density-guided refinement	Frank DiMaio's work; RosettaES	→ Atlas J.1
Cryo-EM model building with deep learning	→ Atlas J.1	—
EMReady and map post-processing	—	—

Papers: Trabuco et al. 2008, *Structure* 16:673 (**MDFF — molecular dynamics flexible fitting**) · DiMaio et al. 2015, *Nat Methods* 12:361 (**Rosetta density refinement**) · Wriggers 2010, *Biophys Rev* 2:21 (Situs) · Pintilie et al. 2010, *J Struct Biol* 170:427 (Segger).

The overfitting question is the whole game here. A flexible fitting that drives a model into a 6 Å map will produce a model that fits the map. Whether it is *right* is a different question, answered by cross-validation against held-out data (half-maps, omitted restraints) and by checking that the stereochemistry did not degrade. **Derivation checkpoint 41: design the cross-validation for a flexible fit and state what it does not protect against.**

K.3 Combining prediction with sparse experimental data

This is the most active and most design-relevant corner of Atlas K.

Talk	Speaker / host	Length	Link
Inverse problems with experiment-guided AlphaFold	Vedula & Sellam, Technion (Valence)	47:55	▶
Learning millisecond protein dynamics from what is missing in NMR spectra	El Nesr & Wayment-Steele (ML4PE)	47:43	▶
same, longer	El Nesr (BPDMC)	56:40	▶
Exploring structural heterogeneity through cryoEM, cryoET and deep learning	Joey Davis, MIT (BPDMC)	1:05:57	▶
AF3-style guidance with experimental restraints	→ Atlas K.3	—	

Papers: Hummer & Köfinger 2015, *JCP* 143:243150 (**Bayesian inference of ensembles — the correct way to combine simulation with averaged data**) · Bonomi et al. 2017, *Curr Opin Struct Biol* 42:106 (metainference) · Rieping, Habeck & Nilges 2005, *Science* 309:303 (**inferential structure determination — the paper that made structure determination explicitly Bayesian**) · Stevens et al. 2024 (AF2 with experimental restraints).

Rieping, Habeck & Nilges is the philosophical centerpiece of this family and one of the best papers in structural biology. The argument: structure determination is inference, the output should be a posterior, and the single-structure convention is a lossy summary we adopted for historical reasons. Everything in Atlas J and Atlas P is downstream of whether you accept that.

The design-relevant version of the same point. When you guide AlphaFold with a restraint — a crosslink, a known contact, a steric constraint — you are doing integrative modeling with a learned prior. The question Hummer and Köfinger answer for simulation is exactly the question nobody has answered for AF-style models: *what is the correct way to combine a learned prior with averaged experimental data without double-counting?* **That is Capstone V.**

K.4 Paired reading — Atlas K

Watch this	Then read this	Hold this question
IMP tutorials	Russel et al. 2012, <i>PLoS Biol</i> 10:e1001244	Write the problem as Bayesian inference. Name every likelihood.
—	Alber et al. 2007, <i>Nature</i> 450:683	How is per-region uncertainty reported, and why does that matter?
PDB-IHM validation docs	Sali et al. 2015, <i>Structure</i> 23:1156	What does “validated” mean for a model with no single right answer?
MDFF tutorials	Trabuco et al. 2008, <i>Structure</i> 16:673	Design the cross-validation for a flexible fit.
DiMaio talks (Atlas J)	DiMaio et al. 2015, <i>Nat Methods</i> 12:361	When does density refinement improve a model and when does it bake in error?
Vedula & Sellam, <i>Experiment-guided AF</i>	Their paper + Rieping et al. 2005, <i>Science</i> 309:303	A learned prior plus sparse data. Where is the double-counting?
El Nesr & Wayment-Steele, <i>NMR</i>	Hummer & Köfinger 2015, <i>JCP</i> 143:243150	Averaged data, ensemble answer. What is the maximum-entropy argument?
Davis, <i>cryoEM heterogeneity</i>	Bonomi et al. 2017, <i>Curr Opin Struct Biol</i> 42:106	One map or many states? What decides?

Atlas L — Classical Structure Prediction

Problem. Predict a fold without a neural network. Fragment assembly, threading, homology modeling, and the energy functions that made them work.

Why this family is not historical trivia. Every component AlphaFold replaced is a component you need to understand in order to say what AlphaFold replaced it *with*. The packing problem, the fragment library, the rotamer approximation, the decoy discrimination problem — all of these are still inside modern methods, sometimes explicitly, sometimes as the thing the network learned to do implicitly.

L.1 The classical course

The IPD “Intro to Protein Structure Prediction and Design” lecture series — 12 lectures, enumerated in full in Appendix A.7 (#25–36). This is the canonical treatment. It predates deep learning entirely, which is its value.

#	Lecture	Len	Link
1	Intro to Protein Structure Prediction and Design, incl. structure review	41:06	▶
2	Protein Geometry; Molecular Energies and Forces (part 1)	1:09:31	▶
3	Molecular Energies and Forces (part 2)	58:49	▶
4	Molecular Energies and Forces (part 3)	30:00	▶
5	Ab-Initio Protein Structure Prediction (part 1)	1:11:07	▶
6	Ab-Initio Protein Structure Prediction (part 2)	33:23	▶
7	Refinement of Protein Structures	1:06:11	▶
8	Rotamer Libraries and Side-chain Packing (Brian Weitzner)	1:05:04	▶
9	Protein-Protein Docking	54:42	▶
10	Loop Modeling	50:46	▶
11	Non-Protein Molecules in Rosetta (Jason Labonte)	44:22	▶
12	Modeling Membrane Protein Structure (Julia Koehler Leman)	45:06	▶

These twelve lectures are also Jeffrey Gray’s JHU course — same recordings, cross-posted; see Appendix B.1. Plus (playlist [PLHn7WmALbthnAwBJ4mWw5gk8dgqsjRL87](https://www.youtube.com/playlist?list=PLHn7WmALbthnAwBJ4mWw5gk8dgqsjRL87)) for the classical half, and the **DL4Proteins notebooks** (github.com/Graylab/DL4Proteins-notebooks) for the modern half. Gray’s BPDMC 2026 talk — *Antibody language*

models vs. biology; protein docking diffusion models vs. physics (1:25:10, ▶) — is the bridge between the two halves and should be watched after both.

L.2 Fragment assembly and the Rosetta energy function

Topic	Resource	Link
PyRosetta Score Functions	RosettaCommons, 43:27	▶
Refining structures with minimization and Monte Carlo	RosettaCommons, 39:13	▶
Designing with PyRosetta: Packing and Relaxing	RosettaCommons, 37:22	▶
Introduction to Constraints in Rosetta	RosettaCommons, 29:36	▶
Fold trees explained	RosettaCommons, 36:53	▶
Rosetta 101 / Rosetta 201 (IPD TV)	58:11 / 1:15:08	▶ ▶

Papers: Simons et al. 1997, *J Mol Biol* 268:209 (fragment assembly — the founding paper) · Rohl et al. 2004, *Methods Enzymol* 383:66 (protein structure prediction using Rosetta) · Alford et al. 2017, *JCTC* 13:3031 (the Rosetta energy function REF15 — read the whole thing) · Park et al. 2016, *JCTC* 12:6201 (energy function optimization) · Leaver-Fay et al. 2011, *Methods Enzymol* 487:545 · Dunbrack & Karplus 1993, *J Mol Biol* 230:543 (backbone-dependent rotamer library) · Shapovalov & Dunbrack 2011, *Structure* 19:844.

Derivation checkpoint 20. The side-chain packing problem is NP-hard. Rosetta solves it with simulated annealing over a discrete rotamer library and a precomputed pairwise energy graph. ProteinMPNN replaced the whole apparatus with one forward pass. **State precisely what guarantee was lost.** The answer is not “accuracy” — MPNN is often better. It is that the annealing searched a defined space against an explicit objective, and the network does neither.

Checkpoint 19: explain why the Rosetta energy function is called a *scoring* function rather than a force field, and what that implies about whether its minima are physically meaningful.

L.3 Threading, homology modeling and the older servers

Method	Note	Paper
MODELLER	Satisfaction of spatial restraints; still the homology workhorse	Šali & Blundell 1993, <i>J Mol Biol</i> 234:779
I-TASSER	Threading plus fragment reassembly; dominated CASP 7–12	Zhang 2008, <i>BMC Bioinformatics</i> 9:40; Yang et al. 2015, <i>Nat Methods</i> 12:7
Phyre2	Profile-profile threading	Kelley et al. 2015, <i>Nat Protoc</i> 10:845
HHpred / HHsearch	Profile HMM comparison — still essential for remote homology	Söding 2005, <i>Bioinformatics</i> 21:951; Steinegger et al. 2019, <i>BMC Bioinformatics</i> 20:473
SWISS-MODEL	Automated homology modeling	Waterhouse et al. 2018, <i>NAR</i> 46:W296
Robetta	Rosetta’s server, now DL-backed	—

HHblits/HHsearch did not become obsolete. AlphaFold’s MSA is built by exactly this machinery, and the quality of that search is a substantial part of AF2’s accuracy — which the ColabFold work made visible by swapping in MMseqs2. **If you want to improve a structure predictor today, improving the alignment is still a live option**, and it is less crowded than improving the architecture.

L.4 Contact prediction — the bridge that was crossed

This short subsection is historically the most important in Atlas L, because it is where evolutionary couplings met machine learning and produced AlphaFold.

Step	Paper
Correlated mutations observed	Göbel et al. 1994, <i>Proteins</i> 18:309
The phantom-correlation problem solved	Weigt et al. 2009, <i>PNAS</i> 106:67 (direct coupling analysis)
Contacts enable folding	Marks et al. 2011, <i>PLoS ONE</i> 6:e28766
Pseudolikelihood makes it practical	Ekeberg et al. 2013, <i>PRE</i> 87:012707
Complexes too	Hopf et al. 2014, <i>eLife</i> 3:e03430
Deep learning on contact maps	Wang et al. 2017, <i>PLoS Comput Biol</i> 13:e1005324 (RaptorX)
Distance distributions, not contacts	Senior et al. 2020, <i>Nature</i> 577:706 (AlphaFold 1)
Direct from the MSA, end to end	Jumper et al. 2021, <i>Nature</i> 596:583 (AlphaFold 2)

Read these seven papers in order in one week. It is the cleanest intellectual lineage in computational biology: a statistical-physics idea (inverse Potts), a biological observation (coevolution), and a learning method, each useless without the others. **Derivation checkpoint 18** asks you to explain why direct coupling analysis needs the inverse-Potts step at all — what exactly is wrong with raw mutual information between columns.

L.5 Paired reading — Atlas L

Watch this	Then read this	Hold this question
IPD L5–6, <i>Ab-initio prediction</i>	Simons et al. 1997, <i>J Mol Biol</i> 268:209	Why fragments? What prior do they encode?
RosettaCommons, <i>Score Functions</i>	Alford et al. 2017, <i>JCTC</i> 13:3031	Term by term: which are physical, which are statistical, which are fudge?
IPD L8, <i>Rotamer libraries and packing</i>	Dunbrack & Karplus 1993, <i>J Mol Biol</i> 230:543	Packing is NP-hard. What does annealing guarantee that MPNN does not?
IPD L7, <i>Refinement</i>	Park et al. 2016, <i>JCTC</i> 12:6201	Why is refinement the step that never worked well?
IPD L10, <i>Loop modeling</i>	Mandell et al. 2009, <i>Nat Methods</i> 6:551 (KIC)	Kinematic closure is exact. Why did that matter?
—	Šali & Blundell 1993, <i>J Mol Biol</i> 234:779	Homology modeling as restraint satisfaction. Compare to Atlas K.
—	Söding 2005, <i>Bioinformatics</i> 21:951	Profile HMMs. Why is this still in the AF2 pipeline?
—	Weigt et al. 2009, <i>PNAS</i> 106:67 then Marks et al. 2011, <i>PLoS ONE</i> 6:e28766	Why does raw mutual information fail, and what does DCA fix?
—	Wang et al. 2017 then Senior et al. 2020 then Jumper et al. 2021	Three steps in four years. What changed at each?
Gray, <i>LMs vs biology; diffusion vs physics</i>	Both halves of his course	What is his actual claim, and do you agree?

Atlas M — Docking and Virtual Screening

Problem. What binds where, and how tightly? Two subproblems that are routinely conflated and should not be. **Pose prediction** — where does this ligand sit — is largely solved for well-behaved cases. **Affinity prediction** — how tightly does it bind — is not, and the scoring functions that do the first job well do the second badly. Understanding why is the single most useful thing in this family.

M.1 Protein-ligand docking

Title	Speaker	Length	Link
AutoDock Introduction	CCPBioSim	55:00	▶
AutoDock Specialized Docking Methods	CCPBioSim	32:00	▶
AutoDock CrankPep (peptide docking)	CCPBioSim	48:00	▶
How to Interpret Your Docking Scores: A Medicinal Chemistry Perspective	CCPBioSim	42:00	▶
Finding New Drugs Using Computers	Stefano Forli (Scripps)	52:00	▶
Structure-Based Virtual Screening on the Cheap	David Koes (Pittsburgh)	50:00	▶
Intro to GOLD	CCDC	33:00	▶
BioExcel #80: Protein and ligand flexibility in molecular docking	BioExcel CoE	58:00	▶
Protein Ligand Interactions and Docking	ROSTLAB (TU München)	1:24:00	▶
Covalent Docking Screening Webinar	MolSoft	45:00	▶
Ligand Docking Introduction (Rosetta Workshop 2020)	Meiler Lab	24:00	▶
The Rosetta Scorefunction	Meiler Lab	21:00	▶
Flexible Induced Fit Docking and Screening	MolSoft	55:00	▶

“How to Interpret Your Docking Scores” is the single most useful item in this family. It is an explicit argument that docking scores rank *poses* well and *affinities* badly — the distinction the field most often elides and the one that explains most disappointed expectations.

CrankPep is the peptide-docking extension and is directly relevant to your cyclic-peptide work, where standard small-molecule docking assumptions fail.

The Rosetta scorefunction lecture belongs here as a cultural contrast: it is an *empirical* energy function with terms that have no physical referent, sitting next to a family of methods that at least aspire to physics. Both work. Neither works for the reason its authors claim.

M.2 Protein-protein and information-driven docking

Title	Speaker	Length	Link
Basics of docking and introduction to HADDOCK	BioExcel CoE	1:07:00	▶
WeNMR lecture on information-driven docking with HADDOCK	Alexandre Bonvin (Utrecht)	1:38:00	▶
HADDOCK lecture, CECAM BImBS2019	Alexandre Bonvin	1:48:00	▶
BioExcel #46: The HADDOCK 2.4 server	BioExcel CoE	1:03:00	▶
BioExcel #67: Introducing HADDOCK3, modular integrative pipelines	BioExcel CoE	47:00	▶
HADDOCK3 antibody-antigen tutorial	BioExcel CoE	1:35:00	▶
BioExcel #1: Integrative modelling of complexes with HADDOCK	BioExcel CoE	57:00	▶
BioExcel #36: Prediction of protein-protein interactions in CAPRI	BioExcel CoE	1:21:00	▶
BioExcel #7: PRODIGY, predicting binding affinities in complexes	BioExcel CoE	44:00	▶
Protein-Protein Docking Walkthrough	Meiler Lab	42:00	▶
Protein-Protein Docking	Jeffrey Gray (JHU)	54:43	▶
pyDock	Juan Fernández-Recio (CSIC)	30:00	▶

Bonvin's 98-minute WeNMR lecture is the definitive statement of information-driven docking: the idea that ambiguous restraints from *any* experiment — chemical shift perturbation, mutagenesis, crosslinks, coevolution — should bias the sampling rather than filter the output. That is the conceptual bridge from docking to Atlas family K, and the ambiguous interaction restraint is a construction worth stealing for other problems.

The CAPRI webinar is the only honest accounting of how well any of this works on blind targets. Watch it before trusting a docking result, including your own.

M.3 Virtual screening and cheminformatics

The **Strasbourg Summer School in Chemoinformatics** is effectively a free graduate course taught by the people who built the field.

Title	Speaker	Length	Link
Strasbourg 2022	Jürgen Bajorath (Bonn)	1:24:00	▶
Strasbourg 2024	Alexander Tropsha (UNC)	1:07:00	▶
Strasbourg 2022	Thierry Langer (Vienna)	42:00	▶
Strasbourg 2024	Didier Rognan (Strasbourg)	44:00	▶
Strasbourg 2024	Artem Cherkasov (UBC)	40:00	▶
Strasbourg 2024	Matthias Rarey (Hamburg)	36:00	▶
Strasbourg 2022	Connor Coley (MIT)	43:00	▶
Chemical screening library growth: Is bigger better?	John Irwin (UCSF)	49:00	▶
Where the Rubber Hits the Road: ML on Drug Discovery Projects	Pat Walters (Relay)	42:00	▶
Applying Active Learning in Drug Discovery	Walters & Thompson (Valence)	52:00	▶
Drug Discovery in Readily Available Chemical Space (V-SYNTHES)	Chemspace / Katritch	1:16:00	▶
State of the toolkit (RDKit UGM)	Greg Landrum	34:00	▶
Clustering, sampling and filtering ultra-large chemical databases	Roger Sayle (RDKit UGM)	43:00	▶

Tropsha is the source of modern QSAR validation discipline — y-randomization, applicability domain, and the fact that a good cross-validated r^2 is not evidence of anything. Those are the same lessons Atlas S teaches for protein design, learned a decade earlier in a different field and largely unread by the protein community.

Irwin's talk is the deliberate skeptic: *is bigger actually better?* Watch it before committing a cluster to a billion-compound screen. Cherkasov's Deep Docking is the active-learning trick that makes such screens tractable — train a surrogate on a docked subset, then triage the rest.

M.4 Binding free energy in practice

Title	Speaker	Length	Link
SAMPL6 logP virtual workshop	David Mobley (UC Irvine)	3:00:00	▶
Free Energy Calculations to Guide Pharmaceutical Lead Optimization	David Mobley	1:00:00	▶
Accurate Calculation of Protein-Ligand Binding Energies	Chris Chipot (UIUC/CNRS)	1:40:00	▶
Free Energy Calculations from Butane to COVID-19	William Jorgensen (Yale)	36:00	▶
Free energies: What we've learned about how to estimate them	Michael Shirts (Colorado)	30:00	▶
Pushing the Boundaries of Free Energy Calculations	Lingle Wang (Schrödinger)	35:00	▶
Tight integration of FE, ML and de novo design	Robert Abel (Schrödinger)	1:11:00	▶
Attach-pull-release for pose refinement and ligand ranking	Germano Heinzelmann (UFSC)	30:00	▶
Binding free energy calculations for protein-ligand systems	Vytautas Gapsys (MPI-BPC)	52:00	▶
BioExcel #63: GROMACS/pmx large-scale alchemical screening	BioExcel CoE	54:00	▶
BioExcel #4: Mutation free energy calculations with pmx	BioExcel CoE	1:02:00	▶
Study ligand dissociation kinetics using SILCS	Wenbo Yu	30:00	▶

Lingle Wang and Robert Abel are FEP+ from inside Schrödinger, which is the only way to see a production workflow without a licence. Gapsys and the pmx webinars are the free, open equivalent — and BioExcel #63 is specifically the **screening-mode, large-scale** application, which is the one relevant to ranking a series rather than computing a single number carefully.

M.5 BUILD — Atlas M

1. **Implement a simple grid-based docking scoring function** and a rigid-body search. Dock a ligand with a known crystal pose. Measure your RMSD.
2. **Then do the thing that matters.** Take twenty complexes with measured affinities. Compute your score for each. Plot score against measured ΔG . The correlation will be poor. **Quantify how poor, and write one page on why pose prediction succeeds where affinity prediction fails.**
3. **Run HADDOCK with and without an ambiguous interaction restraint** on a complex where you know the interface. Measure how much information the restraint is worth, in units of success rate.
4. **The cross-family exercise.** Compare your docking score, an MM-GBSA estimate, and a proper alchemical free energy for the same three ligands. Three methods, three numbers, increasing cost. Decide where the cost stops

being worth it and defend the decision.

Derivation checkpoints due: 27, 52.

M.6 Paired reading — Atlas M

Watch this	Then read this	Hold this question
CCPBioSim, <i>AutoDock Introduction</i>	Trott & Olson 2010, JCC 31:455 (Vina)	Separate the search from the scoring. Which one limits performance?
CCPBioSim, <i>Interpret Your Docking Scores</i>	Any docking-power vs scoring-power assessment	Why do scores rank poses well and affinities badly? Write the functional-form argument.
Koes, <i>Virtual Screening on the Cheap</i>	Koes et al. 2013 (smina); McNutt et al. 2021 (gnina)	A CNN replaces an empirical score. What does it learn that the empirical one cannot?
CCPBioSim, <i>CrankPep</i>	Zhang & Forli 2019, Bioinformatics	Peptides break small-molecule docking assumptions. Name three ways.
Bonvin, <i>Information-driven docking</i>	Dominguez, Boelens & Bonvin 2003, JACS 125:1731	Derive the ambiguous interaction restraint. What does the effective distance do?
BioExcel, <i>CAPRI</i>	Lensink & Wodak CAPRI assessment papers	What fraction of blind targets are solved, and did AlphaFold move it?
Gray, <i>Protein-Protein Docking</i>	Gray et al. 2003, JMB 331:281 (RosettaDock)	Centroid then full-atom. Why two stages?
Tropsha, <i>Strasbourg 2024</i>	Tropsha 2010, Mol Inform 29:476	Apply y-randomization and applicability domain to a protein design benchmark.
Langer, <i>Strasbourg 2022</i>	Any pharmacophore methods paper	A pharmacophore is a low-dimensional summary. What does it discard?
Cherkasov, <i>Strasbourg 2024</i>	Gentile et al. 2020, ACS Cent Sci 6:939 (Deep Docking)	A surrogate triages a billion compounds. What is the recall at a given budget?
Irwin, <i>Is bigger better?</i>	Lyu et al. 2019, Nature 566:224	Ultra-large screening works. What is the actual hit rate, and against what baseline?
Walters, <i>Active Learning in Drug Discovery</i>	Any active-learning screening paper	How many oracle calls, and what acquisition function?
Shirts, <i>Free energies</i>	Bennett 1976; Shirts & Chodera 2008 (MBAR)	State the overlap condition. How would you detect its failure from data alone?
Mobley, <i>SAMPL6</i>	Mobley & Gilson 2017, Annu Rev Biophys 46:531	Blind challenges keep the field honest. What is the design-side equivalent?
Gapsys / BioExcel #63	Gapsys et al. 2020, Chem Sci 11:1140	Large-scale alchemy at screening throughput. What accuracy survives the speedup?
Heinzelmann, <i>Attach-pull-release</i>	Boresch et al. 2003, JPCB 107:9535	Absolute binding free energy needs restraint corrections. Derive them.
Abel, <i>FE + ML + de novo design</i>	Any FEP+ retrospective	Where in a design pipeline does a rigorous free energy actually change a decision?

Atlas N — Sequence, Evolution and Phylogeny

Problem. Every learned model in Part III is trained on a database that is a biased, non-independent sample of one evolutionary history. This family is about what that means.

Skip this family and you will misinterpret every benchmark you run. The training examples are related by descent, which breaks the i.i.d. assumption underlying every statistical guarantee you will cite.

N.1 Alignment and search

Topic	Resource	Link
Computational Genomics — full course	Ben Langmead (JHU)	→ Module B.3
Dynamic programming, BLAST, profile HMMs	Langmead's lectures	▶
MMseqs2 and Foldseek	Exploring the Protein Universe , Steinegger (Broad MIA), 1:17:41	▶
End-to-end learning of MSAs with differentiable Smith-Waterman	Samantha Petti (ML4PE), 50:41	▶

Papers: Needleman & Wunsch 1970; Smith & Waterman 1981 · **Eddy 1998, *Bioinformatics* 14:755 (profile HMMs) and Eddy 2011, *PLoS Comput Biol* 7:e1002195 (HMMER3) · Söding 2005, *Bioinformatics* 21:951 · Steinegger & Söding 2017, *Nat Biotechnol* 35:1026 (MMseqs2) · van Kempen et al. 2024, *Nat Biotechnol* 42:243 (Foldseek) · Mirdita et al. 2022, *Nat Methods* 19:679 (ColabFold — the MSA is the bottleneck).**

Petti's talk is the quiet gem here. Making alignment differentiable means the MSA stops being a fixed preprocessing step and becomes part of the model. Almost nobody has followed up. That is an opening.

N.2 Phylogenetics and the non-independence problem

Topic	Resource	Link
Learning, using, and extending variational distributions for phylogenetics	Erik Matsen (MLCB 2019), 14:07	▶
Phylogenetic inference methods	Matsen group materials; RevBayes tutorials	revbayes.github.io
Ancestral sequence reconstruction	GRASP talk, Gabriel Foley (ML4PE), 1:03:40	▶
Evolutionary Dynamics	Martin Nowak (Broad MIA), 54:22	▶

Papers: Felsenstein 1981, *J Mol Evol* 17:368 (ML phylogeny) · **Felsenstein 1985, *Evolution* 39:783 (phylogenies and the comparative method — the non-independence paper)** · Yang & Rannala 2012, *Nat Rev Genet* 13:303 · Thornton 2004, *Nat Rev Genet* 5:366 (ancestral sequence resurrection) · Harms & Thornton 2013, *Nat Rev Genet* 14:559.

Felsenstein 1985 is the paper every machine learning practitioner in this field should have read and almost none have. Sequences in your training set are not independent draws; they are tips on a tree. Two sequences at 90% identity are one observation, not two. **Every cross-validation split that does not account for phylogeny over-reports performance**, which is exactly the failure the CARE benchmark (Atlas Q.6) rediscovered empirically forty years later.

Derivation checkpoint 47. Write down the effective sample size of a protein family given its phylogeny. Then state what a clustering-based split at 30% identity does and does not fix.

N.3 Coevolution and statistical models of families

Topic	Resource
Direct coupling analysis	→ Atlas L.4
Potts models and inverse statistical mechanics	Weigt, Marks, Hopf papers in L.4
Sectors and statistical coupling analysis	→ Atlas I (Ranganathan)
Fitness landscapes and epistasis	Sparsity, Epistasis, and Models of Fitness Functions , Aghazadeh & Brookes (Broad MIA), 1:36:47 ►

Papers: Morcos et al. 2011, *PNAS* 108:E1293 · Figliuzzi et al. 2016, *Mol Biol Evol* 33:268 (DCA for fitness) · Russ et al. 2020, *Science* 369:440 (an evolution-based model for designing chorismate mutase enzymes — a Potts model designed working enzymes) · Poelwijk et al. 2019, *Nat Commun* 10:4213 (sparse epistasis in a Fourier basis) · Starr & Thornton 2016, *Protein Sci* 25:1204 (epistasis review).

Russ et al. 2020 is the strongest argument for the sequence-only culture that exists. A Potts model — no structure, no neural network — generated functional chorismate mutases at a respectable rate. Read it next to ProGen and ESM3 and ask what the extra capacity in a transformer actually buys. The honest answer may be “less than you think, on this task.”

N.4 Databases and their biases

Resource	What to know
UniProt / UniRef	Clustering levels (50/90/100) determine your effective dataset
Pfam, InterPro	Domain definitions; the labels most function benchmarks use
BFD, MGnify, metagenomic sets	Where AF2's deep MSAs come from
AFDB	200M+ predicted structures; Barrio-Hernandez et al. 2023, <i>Nature</i> 622:637
PDB	The structural training set, and the source of every bias in Part III

The PDB bias is not a footnote, it is a confound. It over-represents soluble, stable, crystallizable, bacterially-expressible, human-disease-relevant proteins. Membrane proteins, IDPs, large assemblies and

anything that resists crystallization are under-represented. Every model in Atlas P and R inherits this. When a method “fails on membrane proteins,” the first hypothesis should be the training set, not the architecture.

Derivation checkpoint 48. Quantify one PDB bias. Pick a property — transmembrane fraction, pI, length, organism — and compare the PDB distribution to UniProt’s. Then state what that implies for a benchmark you actually use.

N.5 Paired reading — Atlas N

Watch this	Then read this	Hold this question
Langmead, alignment lectures	Eddy 1998 and Eddy 2011	What is a profile HMM, and why does it beat BLAST on remote homology?
Steinegger, <i>Protein Universe</i>	Steinegger & Söding 2017; Mirdita et al. 2022	MSA generation is the bottleneck. What would improve it?
Petti, <i>Differentiable Smith-Waterman</i>	Her paper	If alignment is learnable, what else in the pipeline is?
Matsen, <i>Variational phylogenetics</i>	Felsenstein 1985, <i>Evolution</i> 39:783	Compute the effective sample size of a protein family.
Foley, <i>GRASP</i>	Thornton 2004, <i>Nat Rev Genet</i> 5:366	Ancestral sequences are often more stable than extant ones. Why?
Nowak, <i>Evolutionary Dynamics</i>	Nowak 2006 book, ch. 1–4	What does a fitness landscape mean when the population is finite?
Aghazadeh & Brookes, <i>Epistasis</i>	Poelwijk et al. 2019, <i>Nat Commun</i> 10:4213	Sparse in a Fourier basis. What does that license you to measure?
—	Russ et al. 2020, <i>Science</i> 369:440	A Potts model made working enzymes. What does a transformer add?
—	Barrio-Hernandez et al. 2023, <i>Nature</i> 622:637	200M predicted structures clustered. What is newly askable?

Atlas O — Geometric Deep Learning for Molecules

Problem. A molecule is a set of atoms in $\mathbb{R}^3$. Rotate it and it is the same molecule. Build a network for which that is true by construction rather than by training, and understand what you paid for it.

This family is the hinge between Part III and Part III. The mathematics comes from the machine learning literature; the constraints come from physics. It is also where the field's current orthodoxy — equivariance is necessary — is being seriously challenged, which makes it an unusually good place to have an opinion.

O.1 The framework

Resource	Instructors	Scale	Link
AMMI Geometric Deep Learning, 2nd ed. (enumerated in Appendix B.5)	Bronstein, Bruna, Cohen, Veličković	18 videos, ~1:15 each	▶
AMMI Geometric Deep Learning, 1st ed. (2021)	same four	13 lectures	▶
Course hub with slides	—	—	geometricdeeplearning.com
Group Equivariant Deep Learning (all 21 lectures in Appendix B.6)	Erik Bekkers	21 videos, 10–50 min	▶
Stanford CS224W: ML with Graphs	Jure Leskovec	60 videos	▶
A Gentle Introduction to Graph Neural Networks	Sanchez-Lengeling (Broad MIA)	44:53	▶
Geometric deep learning for functional protein design	Michael Bronstein (Broad MIA)	1:09:06	▶
Harnessing Geometric ML for Molecular Design	Bronstein (ML4DD Day 3)	1:08:23	▶
Learning Geometry & 3D Symmetries	Mario Geiger, NVIDIA (ML4DD Day 1)	45:18	▶

Papers: Bronstein et al. 2021, arXiv:2104.13478 (Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges) · Gilmer et al. 2017, ICML (message passing neural networks) · Battaglia et al. 2018, arXiv:1806.01261 (relational inductive biases).

The Erlangen-programme framing is the one to carry. Every architecture is characterized by the symmetry group it respects: CNNs are translation equivariant on grids, GNNs are permutation equivariant on graphs, and $E(3)$ -equivariant networks are rotation-translation equivariant in space. **Once you see architectures as choices of symmetry group, designing a new one becomes a question about your problem's symmetries rather than about taste.**

Derivation checkpoint 50. State precisely the difference between *invariance* and *equivariance*, and say which one you want at which layer of a structure predictor. Getting this backwards — invariant intermediate features in a network that must output coordinates — is a real and common bug.

O.2 Equivariant architectures for atoms

Talk	Speaker / host	Length	Link
An Orientation in Symmetry-Aware ML Methods	Tess Smidt (MIT)	1:01	▶
Euclidean neural networks to understand and design atomistic systems	Tess Smidt (Harvard CMSA)	55:00	▶
Symmetry-Aware Neural Networks with e3nn (MRS tutorial)	Smidt, Geiger et al.	6 parts	▶
Symmetries in Inference and Learning	Max Welling (TUM AI)	57:00	▶
Equivariant Networks (NeurIPS 2020 tutorial)	Taco Cohen & Risi Kondor	~2:00	▶
NequIP / Allegro	Batzner & Musaelian (Valence)	1:26:43 / 1:09:17	▶ ▶
MACE	Valence	1:22:55	▶
Is Distance Matrix Enough for Geometric Deep Learning?	Zian Li (Valence)	1:04:15	▶
Learning 3D Representations of Molecular Chirality	Keir Adams (MIT, Valence)	1:09:59	▶
ChiENN: Molecular Chirality with Graph Neural Networks	Gaiński (Valence)	50:33	▶
Torsional Diffusion for Molecular Conformer Generation	Corso & Jing (Valence)	1:36:42	▶
Protein Representation Learning by Geometric Structure Pretraining	Zhang (Valence)	53:20	▶

Papers: Thomas et al. 2018, arXiv:1802.08219 (tensor field networks) · Fuchs et al. 2020, NeurIPS (SE(3)-Transformers) · Satorras et al. 2021, ICML (EGNN — equivariance without spherical harmonics) · Batzner et al. 2022, *Nat Commun* 13:2453 (NequIP) · Batatia et al. 2022, NeurIPS (MACE) · Jumper et al. 2021 supplement, Algorithm 22 (Invariant Point Attention) · Köhler et al. 2020, ICML (equivariant flows).

Read the IPA algorithm box in the AlphaFold supplement next to the SE(3)-Transformer paper. They solve the same problem differently: IPA predicts points in a local frame and transforms them, avoiding spherical harmonics entirely. **Derivation checkpoint 51: show that IPA is equivariant, and identify where the equivariance would break if you changed the frame construction.**

Chirality is the sharp edge of this family. A fully E(3)-equivariant network cannot distinguish a molecule from its mirror image, because reflection is in E(3). Biology is chiral everywhere — L-amino acids, D-sugars,

and the D-residues in your own cyclic peptides. **SE(3) is the group you want, not E(3)**, and a surprising number of published models get this wrong or leave it ambiguous. The two chirality talks above are the ones that take it seriously.

O.3 The challenge to the orthodoxy

This subsection exists because the consensus is actively under attack and you should know it before you build your next model.

Talk	Speaker / host	Length	Link
Does equivariance matter at scale?	Johann Brehmer (Valence)	—	▶
Transformers Discover Molecular Structure Without Graph Priors	Tobias Kreiman (Berkeley, Valence)	1:04:09	▶
UMA: A Family of Universal Models for Atoms	Brandon Wood (Valence)	—	▶
AlphaFold3 review (note the architecture change)	Ovchinnikov (BPDMC)	1:12:58	▶

Papers: Brehmer et al. 2024, arXiv:2410.23179 (**Does equivariance matter at scale?**) · Abramson et al. 2024, *Nature* 630:493 (**AlphaFold3 — which dropped the equivariant frame machinery for a diffusion module with data augmentation**) · Wang et al. 2024 (**on the surprising effectiveness of non-equivariant models given enough data**).

The argument, stated plainly. Equivariance is a hard constraint that buys data efficiency. With enough data and augmentation, an unconstrained model can learn the symmetry and is often faster and more expressive. **AlphaFold3 is the highest-profile piece of evidence for this position:** it replaced AF2’s explicitly equivariant structure module with a diffusion module trained with augmentation, and it got better.

The counter-argument. Low-data regimes are the normal case in biology. Your cyclic-peptide dataset is not going to have a million examples. Where data is scarce, equivariance is not a stylistic preference — it is the difference between learning and not learning.

Derivation checkpoint 52. Design the experiment that would settle this for *your* problem: train equivariant and non-equivariant versions of the same architecture across a data-size sweep, and find the crossover. **Nobody has published that curve for protein structure, and it would be cited constantly.** That is Capstone II.

O.4 BUILD — Atlas O

1. **Implement an E(3)-equivariant message-passing layer from scratch** — EGNN is about forty lines and does not need spherical harmonics. Verify equivariance numerically by rotating the input and checking the output transforms correctly to machine precision.

2. **Break it deliberately.** Add a term that uses absolute coordinates. Watch the equivariance test fail. This is how you will debug every geometric model you ever write.
3. **Run the data-scaling sweep.** Same task, equivariant and non-equivariant, at 100, 1,000, 10,000 and 100,000 examples. Plot the crossover.
4. **Write the one page.** At what dataset size does the inductive bias stop paying for itself on your task?

Derivation checkpoints due: 50, 51, 52.

O.5 Paired reading — Atlas O

Watch this	Then read this	Hold this question
AMMI GDL lectures 1–4	Bronstein et al. 2021, arXiv:2104.13478	Name the symmetry group of every architecture you use.
Bekkers, <i>Group Equivariant DL</i>	Cohen & Welling 2016, ICML (G-CNNs)	Build a G-CNN for a finite group by hand.
Smidt, <i>Orientation in Symmetry-Aware ML</i>	Thomas et al. 2018, arXiv:1802.08219	Invariant, equivariant, or augmented? Choose for your problem and defend it.
Smidt, <i>Euclidean NNs for atomistic systems</i>	Batzner et al. 2022, <i>Nat Commun</i> 13:2453	Spherical harmonics as a basis. What does order ℓ buy?
Geiger, <i>Learning Geometry & 3D Symmetries</i>	Satorras et al. 2021, ICML (EGNN)	EGNN has no spherical harmonics. What is traded?
Welling, <i>Symmetries in Inference and Learning</i>	His argument for data efficiency	Quantify: how many examples is a symmetry worth?
—	AlphaFold2 supplement, Algorithm 22 (IPA)	Prove IPA is equivariant. Where would it break?
Adams, <i>Molecular Chirality</i>	Adams et al. 2022, ICLR	E(3) cannot see chirality. Check your own model.
Kreiman, <i>Transformers without graph priors</i>	Brehmer et al. 2024, arXiv:2410.23179	Run the data-scaling sweep. Where is the crossover?
Ovchinnikov, <i>AlphaFold3 review</i>	Abramson et al. 2024, <i>Nature</i> 630:493	AF3 dropped the equivariant module. Was that a concession or a result?

Atlas P — Learned Structure Prediction

Problem. Sequence in, coordinates out, with no explicit energy function. AlphaFold2 solved the monomer case well enough that the question changed; what remains is complexes, ligands, nucleic acids, ensembles, and — the hardest and least solved — **knowing when the answer is wrong.**

Every entry below is a talk paired with the paper it narrates. Watch first, read second, write the gap.

P.1 The architecture, properly

Method	Talk	Speaker	Length	Link
AlphaFold1	AlphaFold: improved structure prediction using potentials from deep learning	IPD, 2019	1:02:40	▶
AlphaFold2	CASP-16: From AlphaFold1 to AlphaFold2	John Jumper	55:15	▶
AlphaFold2	Kendrew Lecture: Highly accurate protein structure prediction	John Jumper (MRC LMB)	58:49	▶
AlphaFold2	AlphaFold and its implications for understanding biology	John Jumper (NGBS2022)	34:50	▶
AlphaFold2	ML for protein structure prediction, Part 2: AlphaFold2 architecture	Nazim Bouatta (Harvard CMSA)	1:18:32	▶
AlphaFold2	Part 1: Algorithm space	Nazim Bouatta	1:30:25	▶
AlphaFold2	Part 3: AlphaFold2 and OpenFold	Nazim Bouatta	1:32:35	▶

Papers: Senior et al. 2020, *Nature* 577 (AF1) · Jumper et al. 2021, *Nature* 596 — and the **Supplementary Information, which is the real paper.**

The critical talks — watch these too

Talk	Speaker / venue	Length	Link
Initial reaction to AlphaFold2 right after CASP14	BPDMC, Dec 2020	1:17:30	▶
AlphaFold 2: Is Protein Folding Solved?	AISC, 2021	59:57	▶

The BPDMC session is the field processing the result in real time, two weeks after CASP14, with no hindsight and no press office. There is no better record of what a paradigm shift feels like from inside.

P.2 OpenFold — what retraining revealed

Talk	Speaker	Length	Link
OpenFold: Lessons and insights from rebuilding and retraining AlphaFold2	Mohammed AlQuraishi (IPAM)	48:13	▶
OpenFold (Valence Labs version)	Mohammed AlQuraishi	39:28	▶
OpenFold, long technical version	OpenBioML	1:03:55	▶
OpenFold (practical)	SBGrid	1:01:48	▶
OpenFold3 and AQAffinity	SandboxAQ	56:50	▶
OpenFold3 for Biological Structure Prediction	Vecura AI4LIFE	42:44	▶

Paper: Ahdriz et al. 2024, *Nature Methods* 21 — **[landmark]**. The only systematic account of what AF2 actually learns, in what order, and how much training data it needs. Retraining is the one experiment that turns a black box into an object of study.

P.3 The RoseTTAFold line

Method	Talk	Speaker	Length	Link
RoseTTAFold	CASP AI-SIG	Minkyung Baek	55:57	▶
RoseTTAFold2	CASP15 predictor talk	Minkyung Baek	45:18	▶
RF All-Atom	Generalized Biomolecular Modeling and Design with RFAA	Rohith Krishna (Valence)	1:04:54	▶
RF All-Atom	Molecular ML Reading Group: RoseTTAFold All-Atom	MaomLab (critical)	51:00	▶
RF3 / AtomWorks	Accelerating Biomolecular Modeling with AtomWorks and RF3	ML4PE	51:17	▶
—	ML4PE Early Career Talk	Rohith Krishna	56:22	▶

Papers: Baek et al. 2021, *Science* 373 · Baek et al. 2023, bioRxiv (RF2) · Krishna et al. 2024, *Science* 384 (RFAA).

P.4 AlphaFold3 and the co-folding era

Method	Talk	Speaker	Length	Link
AlphaFold3	Towards Rational Drug Design with AlphaFold 3	Max Jaderberg (Isomorphic)	50:28	▶
AlphaFold3	Review and discussion of AlphaFold3	Sergey Ovchinnikov (BPDMC)	1:12:58	▶
AlphaFold3	Lessons from implementing AlphaFold3 in the wild	Arda Goreci (Ligo Biosciences)	25:52	▶
AlphaFold3	What's new and why it matters (practical)	RosettaCommons	24:46	▶
Boltz-1	Democratizing Biomolecular Interaction Modeling	Wohlwend & Corso (Proxima)	40:59	▶
Boltz-1	Boltz-1 and the Future of Biomolecular Foundation Models	Corso & Wohlwend (BPDMC)	1:09:45	▶
Boltz-2	Towards Accurate and Efficient Binding Affinity Prediction	Valence Labs	59:27	▶
Chai-1	Chai Discovery's Bitter Lesson: Drug Design Is Another Scaling Problem	Meier & Dent (Sequoia)	47:23	▶
Chai-1	No Priors Ep. 121	Dent & Meier	49:28	▶
SeedFold	Scaling Biomolecular Structure Prediction	Zhou & Lu (Valence)	59:59	▶
AlphaFold-Multimer	AlphaFold workshop Part 2	Roland Dunbrack (Utah)	1:05:56	▶

Papers: Abramson et al. 2024, *Nature* 630 (AF3) **[landmark]** · Evans et al. 2021, bioRxiv (AF-Multimer) · Wohlwend et al. 2024 (Boltz-1) · Passaro et al. 2025 (Boltz-2) · Chai Discovery Team 2024.

Goreci’s twenty-six minutes are the most useful item in this subsection. Reimplementing a paper from its description is the strongest possible reproducibility test, and his account of what was underspecified is worth more than three admiring reviews. **You run Boltz-2 in production — watch this one before you trust an affinity number.**

No talk exists for Protenix (ByteDance 2025) or HelixFold3 (Liu et al. 2024, arXiv:2408.16975). Read the papers.

P.5 Older lineages and single-sequence prediction

Method	Talk	Speaker	Length	Link
trRosetta	CASP15 predictor talk	Jianyi Yang	22:12	▶
I-TASSER	Protein Structure Prediction	Yang Zhang	1:14:36	▶
D-I-TASSER	CASP-16 predictor talk	Wei Zheng	14:24	▶
MULTICOM	CASP-16 predictor talk	Jianlin Cheng	14:43	▶
ESMFold	Structure Prediction with ESMFold	RosettaCommons	10:32	▶
ESMFold2	Biomolecular Structure Prediction with ESMFold2	Biohub	56:59	▶
RGN2	Predicting protein structures from single sequences	Nazim Bouatta (BPDMC)	1:33:06	▶
RGN	MIA: End-to-end differentiable learning of protein structure	Mohammed AlQuraishi (Broad)	56:12	▶
ColabFold	Making protein folding accessible to all	BPDMC	1:46:09	▶
—	Structure Prediction with AlphaFold2 and OpenFold	RosettaCommons	1:39:36	▶

Papers: Yang et al. 2020, *PNAS* 117 (trRosetta) · Yang et al. 2015, *Nat Methods* 12 (I-TASSER) · Lin et al. 2023, *Science* 379 (ESMFold) · Chowdhury et al. 2022, *Nat Biotechnol* 40 (RGN2) · AlQuraishi 2019, *Cell Systems* 8 (RGN) · Mirdita et al. 2022, *Nat Methods* 19 (ColabFold).

P.6 Conformational ensembles — the frontier

This subsection is the live research front and the one that most directly concerns you. A structure predictor returns one structure. A protein is an ensemble. Everything here is an attempt to close that gap, and none of it has converged.

Method	Talk	Speaker	Length	Link
AF-Cluster	Understanding fold-switching proteins by combining AF2 and sequence clustering	Hannah Wayment-Steele (BPDMC)	59:18	▶
AF-Cluster	Predicting and discovering proteins with multiple conformational states	Hannah Wayment-Steele (ML4PE)	47:49	▶
AFsample	CASP15 predictor talk	Björn Wallner	21:07	▶
AlphaFlow / ESMFlow	AlphaFold Meets Flow Matching for Generating Protein Ensembles	Bowen Jing (Valence)	56:30	▶
BioEmu	Scalable Emulation of Protein Equilibrium Ensembles	Valence Labs	1:17:02	▶
BioEmu	Emulation of protein equilibrium ensembles	Jiménez-Luna & Xie (Proxima)	53:45	▶
BioEmu	third treatment	Andrew Foong	58:12	▶
DiG	Towards Predicting Equilibrium Distributions for Molecular Systems	Shuxin Zheng (Valence)	1:12:14	▶
Boltzmann generators	Deep Generative Learning for Physics Many-Body Systems	Frank Noé (IPAM)	1:01:56	▶
—	Advancing molecular simulation with deep learning	Frank Noé	58:32	▶
PepFlow	Direct Conformational Sampling From Peptide Energy Landscapes	Osama Abdin (Valence)	55:42	▶
—	Accelerating Cryptic Pocket Discovery Using AlphaFold and MSMs	Valence Labs	31:32	▶
—	ML4PE Early Career Talk	Soojung Yang	58:36	▶

Papers: Wayment-Steele et al. 2024, *Nature* 625 [**landmark**] · Wallner 2023, *Bioinformatics* 39 · Jing et al. 2024, ICML (AlphaFlow) · **Lewis et al. 2025, *Science* 389 (BioEmu)** · Zheng et al. 2024, *Nat Mach Intell* 6 (DiG) · **Noé et al. 2019, *Science* 365 (Boltzmann generators)** · Abdin & Kim 2024, *Nat Mach Intell* 6 (PepFlow).

No talk found: Str2Str (Lu et al. 2024, ICLR) · subsampled-MSA ensembles (Monteiro da Silva et al. 2024, *Nat Commun* 15) · MDGen (Jing et al. 2024, NeurIPS).

The question that organizes this subsection. BioEmu and AlphaFlow emulate an equilibrium distribution; a Boltzmann generator *samples* one with an exactness guarantee; an MSM *estimates* one from trajectories. Three different epistemic statuses, routinely compared on the same plots. Atlas J.2’s Wankowicz and Fraser

talk asks how you would even score a predicted ensemble. **Nobody has answered that, and it gates the whole subsection.**

P.7 Co-folding, ligands, and the validity critique

Method	Talk	Speaker	Length	Link
DiffDock	Diffusion Steps, Twists, and Turns for Molecular Docking	Valence Labs / M2D2	56:43	▶
DiffDock	long version with discussion	BPDMC	1:36:23	▶
DiffDock	practical	SBGrid	43:41	▶
DiffDock-PP	Protein Docking with DiffDock-PP	RosettaCommons	21:19	▶
PoseBusters	AI-based docking methods fail to generate physically valid poses	Martin Buttenschoen (Valence)	34:28	▶
PLINDER	New benchmarks for molecular interactions	Cao, Durairaj, Kovtun (Proxima)	59:55	▶
PLINDER	second treatment	Vladas Oleinikovas (RDKit UGM)	35:08	▶
Umol	Molecular ML Reading Group	MaomLab	54:07	▶
NeuralPLexer	Dynamic-backbone protein-ligand complex prediction	Zhuoran Qiao (ML4PE)	42:49	▶
Flow-matching docking	Harmonic Self-Conditioned Flow Matching for Multi-Ligand Docking	Hannes Stärk (Valence)	59:47	▶

Papers: Corso et al. 2023, ICLR (DiffDock) · **Buttenschoen, Morris & Deane 2024, *Chem Sci* 15 (PoseBusters) [landmark]** · Durairaj et al. 2024, ICML ML4LMS (PLINDER) · Bryant et al. 2024, *Nat Commun* 15 (Umol) · Qiao et al. 2024, *Nat Mach Intell* 6 · Stärk et al. 2024, ICML.

PoseBusters is the most important thirty-four minutes in this family. It shows that deep docking methods produce poses that are not physically valid — wrong stereochemistry, clashes, impossible geometry — while scoring well on RMSD. That is a benchmark measuring the wrong thing, caught by someone who looked at the molecules. It is the clearest existing template for Capstone III, and PLINDER is the data-side companion on memorization and leakage.

P.8 Protein-peptide and specialist regimes

Talk	Speaker / venue	Length	Link
Interpreting state-of-the-art structure predictors for protein-peptide complexes	BPDMC, 2025	58:12	▶
Cyclic peptide structure prediction and design using AlphaFold	ML4PE	1:07:20	▶
Disorder Challenge Virtual Discussion	CASP16, Kretsch & Oas	39:05	▶
CASP15 disorder assessment	Damiano Piovesan	20:21	▶

The peptide-complex talk is directly load-bearing for any cyclic-peptide and cyclic-peptide work. **Paper:** Rettie et al. 2023, *Nat Commun* 15 (cyclic peptide design with AlphaFold).

P.9 Where this goes next

Talk	Speaker	Length	Link
What could AlphaFold 4 look like?	Sergey Ovchinnikov (Owl Posting)	2:06:40	▶
Inverting protein structure prediction models to solve problems in biology	Sergey Ovchinnikov	1:29:16	▶
Using AI for protein structure modeling and design	Sergey Ovchinnikov (Harvard CMSA)	47:32	▶
Protein Structure Prediction in a Post-AlphaFold2 World	Mohammed AlQuraishi	54:16	▶
The State of Protein Structure Prediction and Friends	Mohammed AlQuraishi (Simons)	1:05:31	▶
Lecture 11: Protein Language Models (MLCB24)	Manolis Kellis (MIT)	1:23:08	▶

Two hours of Ovchinnikov forecasting is the single highest-yield item in this Atlas for your open-problems file. Mine it.

P.10 BUILD — Atlas P

1. **Implement triangle multiplicative update and triangle attention.** Verify numerically that they reduce triangle-inequality violations in a distance matrix. Checkpoint 60 in code.
2. **MSA depth ablation.** Ten targets, MSA subsampled to 1/4/16/64/256 sequences. Plot pLDDT against true accuracy at each depth. **Does pLDDT track accuracy or does it track MSA depth?** This single experiment teaches more than ten papers.
3. **Run PoseBusters-style validity checks on your own Boltz-2 outputs.** What fraction are physically valid? Report the number.

4. **The ensemble exercise.** Take one protein with a known conformational change. Generate ensembles with AF-Cluster, AFsample and a short MD run. Compare the three distributions. Then propose a metric that would say which is right, and explain why it is hard.

Derivation checkpoints due: 59, 60.

P.11 Paired reading — Atlas P

Watch this	Then read this	Hold this question
Jumper, CASP-16 AF1→AF2	Jumper et al. 2021 SI, <i>Algorithms</i> 1–32	Trace one residue pair through every block. Where does geometry first enter?
Bouatta Part 2	Jumper et al. 2021	Why recycling? What would break without it?
BPDMC, post-CASP14 reaction	The CASP14 abstracts	What did the community believe before the paper existed? What were they wrong about?
AlQuraishi, OpenFold	Ahdritz et al. 2024, <i>Nat Methods</i> 21	In what order do structural concepts emerge? What does that suggest about curriculum learning?
Krishna, RFAA	Krishna et al. 2024, <i>Science</i> 384	All-atom means ligands and nucleic acids. What had to change architecturally?
Jaderberg, AF3	Abramson et al. 2024, <i>Nature</i> 630	The structure module became a diffusion head. What did that buy and cost?
Ovchinnikov, AF3 review	Abramson et al. 2024	Which criticisms are about the model and which about the release?
Goreci, Implementing AF3 in the wild	Abramson et al. 2024	List everything underspecified. That list is a reproducibility finding.
Wohlwend & Corso, Boltz-1	Wohlwend et al. 2024	What did open reproduction reveal that the AF3 paper did not state?
Boltz-2	Passaro et al. 2025	How was the affinity head trained, and what is its applicability domain?
Bouatta, RGN2	Chowdhury et al. 2022, <i>Nat Biotechnol</i> 40	Single-sequence prediction with no MSA. Where does the information come from?
Wayment-Steele, AF-Cluster	Wayment-Steele et al. 2024, <i>Nature</i> 625	MSA clustering reveals alternate states. Are the populations meaningful?
Jing, AlphaFlow	Jing et al. 2024, ICML	Flow matching on top of AF2. What distribution is being matched?
BioEmu	Lewis et al. 2025, <i>Science</i> 389	“Emulation” of equilibrium. What is the guarantee, and how is it validated?
Noé, Boltzmann generators	Noé et al. 2019, <i>Science</i> 365	Compare the guarantee here to BioEmu’s. Which is stronger and why?
Zheng, DiG	Zheng et al. 2024, <i>Nat Mach Intell</i> 6	Predicting equilibrium distributions directly. What is the training signal?
Corso, DiffDock	Corso et al. 2023, ICLR	Docking as diffusion over SE(3) plus torsions. What is the product manifold?
Buttenschoen, PoseBusters	Buttenschoen, Morris & Deane 2024, <i>Chem Sci</i> 15	List every validity check. Apply all of them to your own pipeline’s output.
PLINDER	Durairaj et al. 2024	How do you build a leakage-free protein-ligand split? Why is it hard?
Qiao, NeuralPLexer	Qiao et al. 2024	Dynamic backbone co-folding. What does state-specificity require?

Watch this	Then read this	Hold this question
BPDMC, protein-peptide predictors	Verify current peptide-complex benchmark papers	Why do peptides break these models specifically?
Rettie, cyclic peptides with AF	Rettie et al. 2023, <i>Nat Commun</i> 15	AF2 never saw cyclic backbones. Why does the offset trick work at all?
Ovchinnikov, AlphaFold 4	—	Extract every open problem named, with the date.

Atlas Q — Sequence and Language Models

Problem. Learn a distribution over sequences from evolution alone, with no structures and no labels, and then use it to predict function, score variants, or generate new sequences.

This family is the purest expression of the learning-first culture. It also contains the field's sharpest internal argument — **whether these models learn biology or learn the database** — and that argument has a clean experimental answer you will meet in Q.6.

Q.1 The ESM line

Talk	Speaker / host	Length	Link
Evolutionary Scale Language Models	Alexander Rives (MLCB 2023)	54:32	▶
ESM3: Simulating 500 million years of evolution	Roshan Rao (ML4PE)	1:05:08	▶
Learning to read and write protein evolution	Brian Hie (Broad MIA)	1:02:35	▶
Efficient evolution of human antibodies from general PLMs	Brian Hie (ML4PE)	54:11	▶
Early Career Seminar #3	Brian Hie (ML4PE)	49:35	▶
Multimodal Deep Learning for Protein Engineering	Kevin K. Yang (Valence)	1:02:31	▶
Protein Language Models (tutorial)	RosettaCommons	33:49	▶
How to Make the Most of Your Masked LM for Protein Engineering	Calvin McCarter, BigHat (ML4PE)	53:10	▶
Scaling down: MSA Pairformer	Yo Akiyama, MIT (ML4PE)	37:59	▶

Papers: Rives et al. 2021, *PNAS* 118:e2016239118 (ESM-1b) · Lin et al. 2023, *Science* 379:1123 (ESM-2 / ESMFold) · Hayes et al. 2025, *Science* 387:850 (ESM3) · Hie et al. 2024, *Nat Biotechnol* 42:275 (antibody evolution) · Meier et al. 2021, NeurIPS (ESM-1v zero-shot variant effects).

Q.2 Generative protein language models

Model	Talk	Speaker / host	Length	Link
ProtGPT2	A Deep Unsupervised Language Model for Protein Design	Noelia Ferruz (ML4PE)	1:05:20	▶
ProtGPT2 / ZymCTRL	Early Career Seminar #2	Noelia Ferruz (ML4PE)	56:15	▶
ProGen3	Scaling Unlocks Broader Generation and Deeper Functional Understanding	Aadyot Bhatnagar, Profluent (ML4PE)	1:07:20	▶
PoET-2	Understanding protein function with a multimodal retrieval-augmented foundation model	Tristan Bepler (ML4PE)	54:12	▶
Dayhoff Atlas	Scaling sequence diversity for improved protein generation	Alex Lee, UCSF/MSR (ML4PE)	57:46	▶
DPLM	Diffusion Language Models Are Versatile Protein Learners	Zaixiang Zheng, ByteDance (ML4PE)	1:13:19	▶
EvoDiff	Protein generation with evolutionary diffusion	Kevin Yang, MSR (ML4PE)	59:46	▶
Discrete diffusion	Discrete diffusion models for generative protein design	Sarah Alamdari (BPDMC)	1:00:13	▶
Walk-jump	Protein Discovery with Discrete Walk-Jump Sampling	Nathan Frey (Valence)	55:11	▶
Guided discrete diffusion	Protein Design with Guided Discrete Diffusion	Stanton & Gruver (ML4PE)	58:05	▶
Raygun	Template-based protein editing	Kapil Devkota, Duke (ML4PE)	40:51	▶
Concept bottlenecks	Concept Bottleneck Language Models for Protein Design	Abdelsalam & Frey (ML4PE)	51:46	▶
Reasoning models	Proteo-R1: Reasoning Foundation Models for De Novo Protein Design	ML4PE	52:36	▶
The CRISPR case	Design of highly functional genome editors by modeling the universe of CRISPR-Cas sequences	Ruffolo & Nayfach, Profluent (ML4PE)	1:00:14	▶

Papers: Ferruz et al. 2022, *Nat Commun* 13:4348 (ProtGPT2) · Madani et al. 2023, *Nat Biotechnol* 41:1099 (**ProGen**) · Nijkamp et al. 2023, *Cell Systems* 14:968 (ProGen2) · Truong & Bepler 2023, *NeurIPS* (**PoET**) · Alamdari et al. 2023, *bioRxiv* (EvoDiff) · Ruffolo et al. 2024, *Nature* 637:1176 (OpenCRISPR-1) · Elnaggar et al. 2022, *IEEE TPAMI* 44:7112 (ProtTrans).

OpenCRISPR-1 is the strongest single existence proof in this family. A language model trained on CRISPR-Cas sequences generated a gene editor that was hundreds of mutations from any natural protein and that worked in human cells. If you want one answer to “do PLMs learn biology,” it is this talk — and then Q.6 is the counterargument.

Q.3 Fitness prediction and variant effects

Method	Talk	Speaker / host	Length	Link
EVE / DeepSequence	Structure & fitness from genomic sequences	Debora Marks (Broad MIA)	1:48:55	▶
Alignment-free	Alignment-free models for protein and antibody design	Marks & Kollasch (Broad MIA)	1:40:20	▶
Variant interpretation	Interpreting gene variants	Dias, Frazer & Shin (Broad MIA)	1:37:29	▶
TranceptEVE	Hybrid protein language models for fitness prediction	Pascal Notin (ML4PE)	52:58	▶
METL	Mutational Effect Transfer Learning for Protein Design	Sam Gelman (ML4PE)	59:13	▶
Evolutionary + experimental	Learning Protein Fitness Models from Evolutionary and Experimental Data	Chloe Hsu (ML4PE)	56:51	▶
Few-shot	Rapid protein evolution by few-shot learning with a PLM	Kaiyi Jiang, MIT (ML4PE)	58:18	▶
Site-wise effects	Site-wise mutation effects enable combinatorial protein variant design	David Ding (ML4PE)	51:49	▶
AlphaMissense	AlphaMissense	Jun Cheng, DeepMind (Broad MIA)	1:22:31	▶
3D CNNs	Engineering Proteins with 3D Convolutional Neural Networks	Danny Diaz, UT Austin (ML4PE)	1:12:19	▶
Epistasis theory	Sparsity, Epistasis, and Models of Fitness Functions	Aghazadeh & Brookes (Broad MIA)	1:36:47	▶
Distribution shift	Conformal prediction under feedback covariate shift for biomolecular design	Clara Wong-Fannjiang (ML4PE)	58:45	▶
Distribution shift	Beyond the training set: detecting distribution shift	Farhan Damani (MLCB 2023)	24:14	▶
Bayesian optimization	Bayesian Optimization of Antibodies Informed by a Generative Model of Evolving Sequences	Alan Amin, NYU (ML4PE)	39:53	▶
Bayesian optimization	Pre-trained Ensembles for Bayesian Optimization of Protein Sequences	Ziyue Yang (Valence)	49:01	▶

Papers: Riesselman et al. 2018, *Nat Methods* 15:816 (DeepSequence) · Frazer et al. 2021, *Nature* 599:91 (EVE) · Notin et al. 2022, ICML (Tranception); Notin et al. 2023, NeurIPS (**ProteinGym**) · Cheng et al. 2023, *Science* 381:eadg7492 (AlphaMissense) · Hopf et al. 2017, *Nat Biotechnol* 35:128 (Evcouplings) · Fannjiang & Listgarten 2020, NeurIPS (design by adaptive sampling); Fannjiang et al. 2022, NeurIPS (conformal under FCS).

ProteinGym is the benchmark that disciplines this subsection. When a talk claims a new state of the art on variant effect prediction, the first question is which ProteinGym split, and the second is whether the improvement survives the MSA-depth stratification. Many do not.

The two distribution-shift talks are the theoretical core of the whole family, and they are the same argument Listgarten makes in Atlas R.10. Design queries a model outside its training distribution; conformal prediction under feedback covariate shift is the only entry here that tries to *quantify* that honestly. **Derivation checkpoint 45 lives in this pair.**

Q.4 Interpretability of protein models

Talk	Speaker / host	Length	Link
InterPLM: Discovering Interpretable Features in PLMs via Sparse Autoencoders	Elana Simon, Stanford (ML4PE)	59:42	▶
Decomposing protein and genomic language models to understand what they learn	Sergey Ovchinnikov (MLCB 2024)	56:09	▶
Small But Mighty: What an LSTM Reveals About Protein Design and About ML	Joanna Slusky (BPDMC)	1:14:04	▶
Towards biophysically interpretable sequence models	Maria Chikina (MLCB 2024)	51:41	▶
Singular value decomposition of protein sequences	Gina El Negr (ML4PE)	36:17	▶
Jointly Embedding Protein Structures and Sequences through Residue Level Alignment	Foster Birnbaum (BPDMC)	1:05:21	▶
Adaptive Protein Tokenization	Rohit Dilip, Caltech (Valence)	1:00:04	▶
CHEAP: Tokenized and Continuous Embedding Compressions	Amy Lu, Berkeley (Valence)	1:27:22	▶
The Continuous Language of Protein Structure	Ben Murrell, Karolinska (Valence)	55:56	▶

Papers: Simon & Zou 2024, bioRxiv (InterPLM) · Rao et al. 2021, ICML (MSA Transformer) · Vig et al. 2021, ICLR (BERTology meets biology — attention recovers contacts) · Bhattacharya et al. 2022, *Bioinformatics* 38:ii119.

This is the direct bridge to Part III. InterPLM applies Anthropic’s sparse autoencoder method to a protein model; Ovchinnikov’s MLCB talk is a structural biologist doing mechanistic interpretability. Watch them alongside D.7,

and note that the *evaluation* problem is easier here: a protein feature can be checked against a known binding site or a Pfam domain, which is a luxury language interpretability does not have. That asymmetry is **Capstone VIII**.

Q.5 Genomic and RNA language models

Model	Talk	Speaker / host	Length	Link
Evo	Sequence modeling and design from molecular to genome scale	Nguyen & Hie (ML4PE)	1:06:46	▶
Evo 2	Genome modeling and design across all domains of life	Garyk Brixi, Stanford (ML4PE)	34:46	▶
Genome design	Generative design of novel bacteriophages with genome language models	Samuel King (ML4PE)	40:23	▶
Genome design	Generative Genome Design	King, primer by Hie (Broad MIA)	2:00:11	▶
Nucleotide Transformer	The Nucleotide Transformer: building and evaluating robust foundation models	Thomas Pierrot, InstaDeep (MLCB)	19:05	▶
gLM	Genomic language model predicts protein co-regulation and function	Yunha Hwang (MLCB 2023)	19:52	▶
gLM	Shedding light on functional dark matter with genomic language modeling	Yunha Hwang (BPDMC)	46:39	▶
OMG dataset	An Open MetaGenomic corpus for mixed-modality genomic language modeling	Andre Cornman (ML4PE)	20:38	▶
Enformer / Borzoi lineage	Personal genome interpretation with sequence-to-function models	Nilah Ioannidis, Berkeley (MLCB)	51:35	▶
	Sequence basis of transcription initiation in the human genome	Jian Zhou, UTSW (MLCB)	47:59	▶
	Fine-tuning on Personal Genomes Improves Expression prediction	Shiron Drusinsky, UCSF (MLCB)	19:53	▶
Benchmark critique	DART-Eval: a comprehensive benchmark for DNA language models	Wang, Patel & Singhal, Stanford (MLCB)	19:36	▶
RNA structure	RNA Function, Design, and Modeling	Silvi Rouskin, HMS (BPDMC)	59:33	▶
RNA structure	AI-Driven RNA Structure Prediction	Mile Šikić (MLSB 2025)	24:37	▶

Model	Talk	Speaker / host	Length	Link
RNA design	gRNAde: Geometric Deep Learning for 3D RNA inverse design	Chaitanya Joshi, Cambridge (Valence)	1:06:02	▶
RNA design	RNA-FrameFlow: Flow Matching for RNA backbone generation	Rishabh Anand, NUS (MLCB)	23:44	▶
RNA geometry	Beyond Sequence: Impact of Geometric Context for RNA Property Prediction	Valence	53:32	▶
RNA foundation	A long-context RNA foundation model for predicting transcriptome architecture	Ali Saberi (Valence)	1:29:27	▶
RNA structure code	Deciphering the RNA structural code	Khoroshkin & Karimzadeh (Broad MIA)	1:42:39	▶
DNA design	Designing DNA with Tunable Regulatory activity	Anirban Sarkar, CSHL (MLCB)	20:08	▶
DNA design	Deep exploration networks for engineering functional DNA sequences	Johannes Linder, UW (MLCB 2019)	13:24	▶
Hybrid design	A high-level design language for generative biology with Proto	Aditi Merchant, Stanford/Arc (ML4PE)	58:56	▶

Papers: Nguyen et al. 2024, *Science* 386:eado9336 (Evo) · Brixi et al. 2025, bioRxiv (Evo 2) · Nguyen et al. 2023, NeurIPS (HyenaDNA) · Dalla-Torre et al. 2025, *Nat Methods* 22:287 (Nucleotide Transformer) · **Avsec et al. 2021, *Nat Methods* 18:1196 (Enformer)** · Linder et al. 2025, *Nat Methods* 22:2422 (Borzoï) · **Avsec et al. 2025, *Nature* (AlphaGenome)** · Townshend et al. 2021, *Science* 373:1047 (ARES, RNA) · Wayment-Steele et al. 2022, *Nat Methods* 19:1234 (EternaFold).

Rhiju Das's thread runs through here. ARES is his, EternaFold comes from Eterna, and the RNA design entries are where the Atlas comes closest to his program. Note the structural fact worth holding onto: **RNA structure prediction remains visibly behind protein structure prediction**, and the CASP and RNA-Puzzles results say so plainly. That is one of the most honest open problems in the entire Atlas — and it is the subject of Capstone IX.

Q.6 Function prediction and search

Method	Talk	Speaker / host	Length	Link
CLEAN	Enzyme function prediction using contrastive learning	Tianhao Yu, UIUC (ML4PE)	54:16	▶
Foldseek	Exploring the Protein Universe	Martin Steinegger, SNU (Broad MIA)	1:17:41	▶
Domain-PFP	Protein function prediction using function-aware domain embeddings	Nabil Ibtehaz, Purdue (ML4PE)	57:46	▶
CAFA	Kaggle 1st place: CAFA 5 Protein Function Prediction	GoCurator, Fudan (ML4PE)	35:49	▶
Substrate prediction	A general model to predict small molecule substrates of enzymes	Alexander Kroll (ML4PE)	57:47	▶
CARE benchmark	A Benchmark Suite for the Classification and Retrieval of Enzymes	Jason Yang, Caltech (ML4PE)	47:02	▶
ProTrek	Navigating the Protein Universe through Tri-Modal Contrastive Learning	Jin Su, Westlake (ML4PE)	48:35	▶
Multimodal	Multimodal protein language models for deciphering protein function	Zitnik Lab (Broad MIA)	1:30:33	▶
PTMs	PTM-Mamba: A PTM-Aware Protein Language Model	Zhangzhi Peng, Duke (ML4PE)	46:47	▶
Fold recognition	Protein Fold Recognition with Recurrent Kernel Networks	Dexiong Chen, Inria (MLCB)	11:54	▶

Papers: Yu et al. 2023, *Science* 379:1358 (CLEAN) · van Kempen et al. 2024, *Nat Biotechnol* 42:243 (Foldseek) · Steinegger & Söding 2017, *Nat Biotechnol* 35:1026 (MMseqs2) · Radivojac et al. 2013, *Nat Methods* 10:221 (CAFA) · Barrio-Hernandez et al. 2023, *Nature* 622:637 (AFDB clustering).

The CARE benchmark talk is the one that closes the argument opened in Q.2. It shows that when enzyme function prediction is evaluated with splits that enforce genuine sequence novelty, performance drops sharply — these models are substantially doing retrieval. Hold that against OpenCRISPR-1, which clearly generalized. Both are true. **Reconciling them is derivation checkpoint 46**, and it is the single most interesting unresolved question in Atlas Q.

Q.7 BUILD — Atlas Q

1. **Score a DMS dataset** from ProteinGym with ESM-2 (zero-shot, masked-marginal), an MSA-based model (EVE or EVcouplings), and a simple site-independent frequency baseline. Report Spearman for each.
2. **Stratify by MSA depth.** Re-plot the three Spearman's against the depth of the target's alignment. The ordering usually inverts somewhere.
3. **Construct the retrieval test.** Hold out sequences by cluster identity at 30%, 50% and 90%, retrain or re-evaluate, and plot performance against identity threshold. This is the CARE experiment on your own data.
4. **Write the one page.** At what sequence identity to the training set does your model stop knowing anything? Name the number.

Derivation checkpoints due: 44, 45, 46.

Q.8 Paired reading — Atlas Q

Watch this	Then read this	Hold this question
Rives, <i>Evolutionary Scale LMs</i>	Rives et al. 2021, <i>PNAS</i> 118:e2016239118	What does “unsupervised” mean when the data is a curated database?
Rao, <i>ESM3</i>	Lin et al. 2023, <i>Science</i> 379:1123; Hayes et al. 2025, <i>Science</i> 387:850	ESMFold is fast and less accurate than AF2. Where exactly is the trade?
Ferruz, <i>ProtGPT2</i>	Ferruz et al. 2022, <i>Nat Commun</i> 13:4348	Generated sequences look natural. What fraction folded?
Ruffolo & Nayfach, <i>OpenCRISPR</i>	Ruffolo et al. 2024, <i>Nature</i> 637:1176	Hundreds of mutations from anything natural, and it works. What generalized?
Bepler, <i>PoET-2</i>	Truong & Bepler 2023, NeurIPS	Retrieval-augmented, explicitly. Is that a concession or a design?
Marks, <i>Structure & fitness from sequences</i>	Hopf et al. 2017, <i>Nat Biotechnol</i> 35:128; Riesselman et al. 2018, <i>Nat Methods</i> 15:816	Couplings are statistical. When do they mean contact and when not?
Notin, <i>TranceptEVE</i>	Notin et al. 2023, NeurIPS (ProteinGym)	Look at the per-assay table, not the mean. Where does it fail?
Cheng, <i>AlphaMissense</i>	Cheng et al. 2023, <i>Science</i> 381:eadg7492	Clinical variant classification. What is the circularity risk in the labels?
Wong-Fannjiang, <i>Conformal prediction under FCS</i>	Fannjiang et al. 2022, NeurIPS	Write the feedback loop formally. Why does ordinary conformal fail?
Aghazadeh & Brookes, <i>Sparsity and epistasis</i>	Their paper + Poelwijk et al. 2019, <i>Nat Commun</i> 10:4213	Fitness landscapes are sparse in a Fourier basis. What does that license?
Simon, <i>InterPLM</i>	Simon & Zou 2024, bioRxiv	SAE features in a protein model. How were they validated, and could language do that?
Ovchinnikov, <i>Decomposing protein and genomic LMs</i>	Vig et al. 2021, ICLR	What is actually in the attention maps?
Nguyen & Hie, <i>Evo</i>	Nguyen et al. 2024, <i>Science</i> 386:ead09336	A single model across DNA, RNA, protein. What does the shared representation buy?
Brix, <i>Evo 2</i>	Brix et al. 2025, bioRxiv	Scaling to whole genomes. What broke, and what was redesigned?
Ioannidis, <i>Personal genome interpretation</i>	Avsec et al. 2021, <i>Nat Methods</i> 18:1196 (Enformer)	Long-range regulation from sequence. How far does the receptive field reach?
MLCB, <i>DART-Eval</i>	The DART-Eval paper	Do DNA language models beat a well-tuned CNN? On which tasks?
Šikić, <i>RNA structure prediction</i>	Townshend et al. 2021, <i>Science</i> 373:1047 (ARES)	Why is RNA harder than protein? Name three structural reasons.
Joshi, <i>gRNAde</i>	The gRNAde paper	Inverse folding for RNA. What changes versus ProteinMPNN?
Yu, <i>CLEAN</i>	Yu et al. 2023, <i>Science</i> 379:1358	Contrastive learning for EC numbers. What does the embedding space organize by?
Yang, <i>CARE benchmark</i>	The CARE paper	Enforce novelty in the split. How much performance is retrieval?

Watch this	Then read this	Hold this question
Steinegger, <i>Exploring the Protein Universe</i>	van Kempen et al. 2024, <i>Nat Biotechnol</i> 42:243	A 3Di structural alphabet. Why does discretizing structure make search tractable?

Atlas R — Generative Design

Problem. Produce a molecule that does not yet exist and that does something. This is the destination family, and it is the one where the three cultures of Section 0.2 are most visibly in tension, because here the ground truth is unambiguous: the protein works in the tube or it does not.

That clarity makes practitioners unusually candid about failure rates. **Listen for the numbers.** When a speaker says a campaign had a 1% success rate, that is the most informative sentence in the talk and it will almost never be on a slide.

Every entry is a talk paired with its paper. Where no talk exists — and there are nine such methods — the paper is given alone, because the gap is itself information about which groups give seminars.

R.1 Inverse folding and sequence design

Method	Talk	Speaker / host	Length	Link
ProteinMPNN	Robust deep learning based protein sequence design using ProteinMPNN	Justas Dauparas (ML4PE)	53:42	▶
ProteinMPNN	Sequence Design with ProteinMPNN (hands-on)	RosettaCommons	33:41	▶
ProteinMPNN	MPNN: ML for protein sequence design	RosettaCommons	25:31	▶
ProteinMPNN	Inverse folding using ProteinMPNN	RosettaCommons	29:56	▶
LigandMPNN	Sequence Design with LigandMPNN	RosettaCommons	17:39	▶
ESM-IF	Adapting protein language models for structure-conditioned design	ML4PE	55:47	▶
ThermoMPNN	ThermoMPNN Tutorial: Finding Stabilizing Protein Mutations	RosettaCommons	42:39	▶
Stability at scale	Mega-scale experimental analysis of protein folding stability	ML4PE	50:57	▶
Stability	Early Career Seminar #7	Gabe Rocklin (ML4PE)	1:00:26	▶
Stability + aggregation	Predicting folding stability and aggregation from large-scale experiments	BPDMC	1:00:30	▶
Rosetta fixbb	Sequence Design Introduction	RosettaCommons	13:46	▶
—	Computational Protein Design Then & Now (1988–2024)	Baker, DeGrado, Mayo, Kuhlman (IPD)	59:50	▶
Ensemble-conditioned	Ensemble-conditioned protein sequence design with Caliby	ML4PE	34:00	▶

Papers: Dauparas et al. 2022, *Science* 378:49 (ProteinMPNN) [landmark] · Dauparas et al. 2025, *Nat Methods* 22:717 (LigandMPNN) · Hsu et al. 2022, ICML (ESM-IF) · Dieckhaus et al. 2024, *PNAS* 121:e2314853121 (ThermoMPNN) · Tsuboyama et al. 2023, *Nature* 620:434 [landmark] · Kuhlman & Baker 2000, *PNAS* 97:10383 · Kuhlman et al. 2003, *Science* 302:1364 (Top7).

The metric critique — watch immediately after Dauparas

Talk	Host	Length	Link
Protein sequence design by explicit energy landscape optimization	BPDMC	1:30:36	▶
Beyond sequence recovery: improved modeling of the sequence-energy landscape	BPDMC	33:23	▶

Sequence recovery — how often the model reproduces the native residue — is the headline metric for inverse folding, and these two talks argue it is the wrong objective. **That is derivation checkpoint 61 argued in public.** Maximizing the likelihood of a sequence given a backbone is not the same as maximizing the probability that the sequence folds to that backbone.

No talk exists for ProteinSolver (Strokach et al. 2020, Cell Systems* 11:402), PiFold (Gao et al. 2023, ICLR), or ProstT5 (Heinzinger et al. 2024, NARGAB 6:lqae150).*

R.2 Backbone generation

Method	Talk	Speaker / host	Length	Link
RFdiffusion	Accurate protein design using structure prediction and diffusion	Watson & Juergens (ML4PE)	43:52	▶
RFdiffusion	Molecular ML Reading Group: RFdiffusion (critique)	MaomLab, Michigan	1:00:21	▶
RFdiffusion	Backbone Generation with RFdiffusion (hands-on)	RosettaCommons	25:35	▶
RFdiffusion	Fundamentals of protein binder design with RFdiffusion	RosettaCommons	20:32	▶
RFdiffusion-AA	Backbone Generation with RFdiffusion All-Atom	RosettaCommons	13:41	▶
RFAA	Generalized Biomolecular Modeling and Design with RFAA	Rohith Krishna (Valence)	1:04:54	▶
RFdiffusion2	Atom level enzyme active site scaffolding	Ahern & Yim (ML4PE)	51:56	▶
RFdiffusion2	long form	Yim & Ahern (Valence)	1:12:41	▶
RFdiffusion3	De novo Design of All-atom Biomolecular Interactions	ML4PE	48:07	▶
RFdiffusion3	Protein Binder Design with RFdiffusion3	RosettaCommons	30:59	▶
RFdiffusion3	Simple Motif Scaffolding Guided Tutorial	RosettaCommons	10:21	▶
Chroma	Illuminating protein space with a programmable generative model	Grigoryan & Costello (Proxima)	1:03:38	▶
FrameDiff	Diffusion probabilistic modelling of protein backbones in 3D	Yim & Trippe (Valence)	1:02:39	▶
FrameFlow	Sequence-Augmented SE(3)-Flow Matching	ML4PE	48:47	▶
FoldFlow	SE(3)-Stochastic Flow Matching for Protein Backbone Generation	Bose & Tong (Proxima)	58:53	▶
FoldFlow-2	Flow Matching For Protein Backbone Generation	Huguet & Vuckovic (Proxima)	1:00:30	▶
Genie 2	Designing and Scaffolding Proteins at the Scale of the Structural Universe	ML4PE	53:11	▶

Method	Talk	Speaker / host	Length	Link
ProtDiff / SMCDiff	Diffusion modeling of backbones for motif-scaffolding	ML4PE	1:03:59	▶
—	Framework for conditional diffusion models in motif scaffolding	ML4PE	1:01:57	▶
—	Protein Structure and Sequence Generation with Equivariant DDPMs	ML4PE	55:19	▶
Protpardelle	ML in Drug Discovery Symposium 2023	Possu Huang (Broad)	21:48	▶
ProteinGenerator	Multistate and functional design using sequence space diffusion	Gershon, Lisanza & Tipps (ML4PE)	54:47	▶
ProteinGenerator	alternate venue	Computational Intelligence Group	58:46	▶
Multiflow	Protein structure and sequence co-generation	Yim & Campbell (Valence)	54:54	▶
Multiflow	Generative Flows on Discrete State-Spaces (theory)	Campbell & Yim (Proxima)	52:12	▶
Proteina	Scaling Flow-based Protein Structure Generative Models	ML4PE	1:09:24	▶
Proteina	first authors	Kreis & Geffner (Proxima)	1:02:02	▶
La-Proteina	Atomistic Protein Generation via Partially Latent Flow Matching	ML4PE	1:05:13	▶
—	Programming Biomolecular Interactions with All-Atom Generative Model	ML4PE	54:24	▶
EvoDiff	Protein generation with evolutionary diffusion: sequence is all you need	ML4PE	59:46	▶
—	Discrete diffusion models for generative protein design	BPDMC	1:00:13	▶
Symmetric design	RFdiffusion for Symmetric Applications	RosettaCommons	23:09	▶
Nanoparticles	ML-based design of nanoparticle vaccines	Neil King (AI Proteins)	53:20	▶
Survey	Diffusion models for protein structure generation and design	RosettaCommons	44:43	▶

Method	Talk	Speaker / host	Length	Link
Survey	Recent methods for protein structure generation and design	BPDMC	1:21:22	▶
Benchmark critique	MotifBench: a standardized benchmark for motif-scaffolding	ML4PE	54:48	▶

Papers: [Watson et al. 2023, *Nature* 620:1089 \(RFdiffusion\) \[landmark\]](#) · [Krishna et al. 2024, *Science* 384:eadl2528 \(RFAA\)](#) · [Ahern et al. 2026, *Nat Methods* 23:96 \(RFdiffusion2\)](#) · [Butcher et al. 2025, bioRxiv \(RFdiffusion3, preprint\)](#) · [Ingraham et al. 2023, *Nature* 623:1070 \(Chroma\)](#) · [Yim et al. 2023, ICML \(FrameDiff\)](#) · [Bose et al. 2024, ICLR \(FoldFlow\)](#) · [Lin & AlQuraishi 2023, ICML](#); [Lin et al. 2024, arXiv \(Genie\)](#) · [Trippe et al. 2023, ICLR \(SMCDiff\)](#) · [Chu et al. 2024, *PNAS* 121:e2311500121 \(Protpardelle\)](#) · [Lisanza et al. 2025, *Nat Biotechnol* 43:1288](#) · [Campbell et al. 2024, ICML \(Multiflow\)](#) · [Geffner et al. 2025, arXiv \(La-Proteina\)](#) · [Fallas et al. 2017, *Nat Chem* 9:353](#) · [Bale et al. 2016, *Science* 353:389](#); [Walls et al. 2020, *Cell* 183:1367](#).

Watch the RFdiffusion first-author talk and the Michigan critique back to back, in one sitting. The gap between how authors present a method and how a sharp external group reads it is the single most useful calibration available, and it is rarely this cleanly paired.

No talk exists for ProteinSGM (Lee et al. 2023, Nat Comput Sci 3:382).**

R.3 Hallucination and inverting predictors

Method	Talk	Speaker / host	Length	Link
Hallucination	Inverting protein structure prediction models to solve problems in biology	Sergey Ovchinnikov	~1 hr	▶
AfDesign / ColabDesign	Tutorial on using structure prediction methods for protein design	Ovchinnikov & Jue Wang (BPDMC)	2:04:03	▶
Motif scaffolding	Scaffolding protein functional sites using deep learning	Tischer & Juergens (ML4PE)	52:05	▶
Classical grafting	Motif Grafting and Scaffolding using Rosetta	Meiler Lab	16:59	▶
Classical grafting	Motif Grafting walkthrough	Meiler Lab	1:05:34	▶
BindCraft	One-shot design of functional protein binders	Valence Labs	1:36:09	▶
BindCraft	second account	BPDMC	1:15:23	▶
BoltzDesign1	BoltzDesign and Protein Hunter: Inverting Structure Prediction Models	ML4PE	48:46	▶
BoltzDesign1	How AF3-style models can be used for design	BPDMC	1:10:35	▶
Protein Hunter	Exploiting structure hallucination within diffusion	Yehlin Cho (Valence)	1:14:12	▶

Papers: Anishchenko et al. 2021, *Nature* 600:547 (hallucination) · Wang et al. 2022, *Science* 377:387 (motif scaffolding — note Wang is first author, not Tischer) · Pacesa et al. 2025, *Nature* 646:483 (BindCraft) · Cho et al. 2025, bioRxiv (BoltzDesign1, preprint) · Goverde et al. 2023, *Protein Sci* 32:e4653.

A citation lesson worth teaching. AfDesign has no paper of its own. The BindCraft *Nature* paper, which is built on AfDesign, does not cite ColabDesign at all — it cites Anishchenko and Goverde instead. Cite the software through Zenodo (doi:10.5281/zenodo.13309080). How a field handles uncited infrastructure is a real thing to have an opinion about, and this is a clean case to form one on.

No talk exists for BoltzGen (Stark et al. 2025, bioRxiv) or PocketGen (Zhang et al. 2024, Nat Mach Intell* 6:1382).*

R.4 Binder design — and the honest hit rates

Method	Talk	Speaker / host	Length	Link
Minibinders, pre-DL	Learning to design mini-proteins that bind specific targets	BPDMC	1:12:12	▶
Target-agnostic binders	Binder Design Methods Overview	RosettaCommons	8:08	▶
DL-improved binders	Fundamentals of protein binder design with RFdiffusion	RosettaCommons	20:32	▶
GPCR binders	De novo design of miniprotein agonists and antagonists for family B GPCRs	BPDMC	51:20	▶
Helical-peptide binders	De novo design of high-affinity binders of bioactive helical peptides	Vázquez Torres (Valence)	37:28	▶
Adaptyv competition	Optimizing Cetuximab: Winning Adaptyv's Protein Design Competition	Cradle	53:53	▶
Scaling + test-time compute	Scaling Atomistic Protein Binder Design	ML4PE	55:28	▶
—	long form	NVIDIA (Valence)	1:38:57	▶
Neosurfaces	Targeting protein–ligand neosurfaces with a generalizable DL tool	Anthony Marchand (Valence)	1:11:03	▶
Neosurfaces	ML4PE account	ML4PE	52:30	▶
Drug-binding proteins	Zero-shot design via neural iterative selection-expansion	ML4PE	36:24	▶
Therapeutic application	De novo design of miniprotein-based NK cell engagers	BPDMC	59:56	▶
Production pipeline	BinderFlow: batch-based binder design with live monitoring	RosettaCommons	16:23	▶

The three corrective talks

Talk	Host	Length	Link
Why designs fail, and how they move	BPDMC	1:09:30	▶
Creativity at Scale: Problem Formulation as the Frontier of Protein Design	BPDMC	1:29:26	▶
Protein design roundtable: real challenges and practical solutions	RosettaCommons	23:55	▶

Papers: [Chevalier et al. 2017, *Nature* 550:74](#) · [Cao et al. 2022, *Nature* 605:551](#) · [Bennett et al. 2023, *Nat Commun* 14:2625](#) · [Muratspahić et al. 2026, *Nature* 656:1044](#) · [Vázquez Torres et al. 2024, *Nature* 626:435](#) · [Cotet et al. 2025, *bioRxiv* \(Adaptyv EGFR competition\)](#).

“Why designs fail, and how they move” is the single best corrective to published hit rates in this entire Atlas. Watch it immediately after any advocacy talk in this subsection. The Adaptyv competition talk is the other half: a real campaign scored against a *public wet-lab benchmark*, which is the closest thing design has to CASP — and a standing argument for why Capstone VII matters.

No talk exists for AlphaProteo ([Zambaldi et al. 2024, *arXiv:2409.08022*](#)).

R.5 Enzyme design — the field's most honest corner

Method	Talk	Speaker / host	Length	Link
Serine hydrolases	Computational design of serine hydrolases	Lauko & Pellock (ML4PE)	41:14	▶
Overview	AI-driven de novo Enzyme Design	ML4PE	35:36	▶
RFdiffusion3 enzymes	Enzyme design with RFdiffusion3	RosettaCommons	12:47	▶
Classical de novo enzymes	Computational Protein Design Then & Now	IPD panel	59:50	▶
PROSS / FuncLib	Designing COVID-19 Countermeasures	Sarel Fleishman (Weizmann)	1:03:00	▶
Metalloenzymes	Zero-shot design of a de novo metalloenzyme	ML4PE	51:08	▶
Generative enzymes	Therapeutic enzyme engineering using a generative neural network	BPDMC	1:18:46	▶
Parallel design	Massively Parallel Protein Design for Next-Generation Chemotherapy	BPDMC	1:02:51	▶
Design + screening	Engineering nuclease enzymes by ML and high-throughput screening	ML4PE	58:00	▶
Design + evolution	AI-redesigned starting points enhance protein evolution	Nicholas Krasnow (ML4PE)	53:07	▶

Papers: Lauko, Pellock et al. 2025, *Science* 388:eadu2454 [landmark] · Ahern et al. 2026, *Nat Methods* 23:96 · Röthlisberger et al. 2008, *Nature* 453:190 (Kemp eliminase) · Jiang et al. 2008, *Science* 319:1387 (retro-aldol) · Siegel et al. 2010, *Science* 329:309 (Diels-Alderase) · Goldenzweig et al. 2016, *Mol Cell* 63:337 (PROSS) · Khersonsky et al. 2018, *Mol Cell* 72:178 (FuncLib) · El Nesr et al. 2026, bioRxiv; Kim et al. 2026, *Nature* 649:246.

The essential pairing. Assign the classical de novo enzyme papers alongside **Blomberg et al. 2013, *Nature* 503:418**, which showed that in the most celebrated case the catalysis came from *directed evolution*, not from the design. There is no talk for that paper, which is itself telling. It is the most important negative result in this family and the reason enzyme design remains the honest corner of the field.

Note on attribution: the zero-shot metalloenzyme work is Possu Huang's lab at Stanford with Röthlisberger at EPFL — not Baker.

No talk exists for theozyme/RosettaMatch (Zanghellini et al. 2006, *Protein Sci** 15:2785; Richter et al. 2011, *PLoS ONE* 6:e19230).*

R.6 Peptides and macrocycles

Method	Talk	Speaker / host	Length	Link
AfCycDesign	Cyclic peptide structure prediction and design using AlphaFold	Rettie (ML4PE)	1:07:20	▶
RFpeptides	Deep Learning Enabled Design of Functional Cyclic Peptides	Peptide Drug Hunting Consortium	23:02	▶
Hyperstable peptides	A computational approach to design structured peptides	BPDMC	52:16	▶
Tertiary motifs	Tertiary motifs as building blocks for protein-binding peptides	BPDMC	1:03:00	▶
Fragment-based	Fragment-based backbone sampling and NN potentials for peptide design	BPDMC	1:19:48	▶
Experimental	Massively parallel discovery of peptides to inhibit protein interactions	BPDMC	1:12:44	▶
PepTune	Multi-Objective-Guided Discrete Diffusion for therapeutic peptides	Valence Labs	1:12:13	▶
Sequence-only	Structure-Independent Peptide Binder Design via Generative LMs	Pranam Chatterjee (Valence)	1:00:55	▶
Critical	Interpreting structure predictors for protein-peptide complexes	BPDMC	58:12	▶

Papers: Rettie et al. 2025, *Nat Commun* 16:4730 (AfCycDesign) · **Rettie et al. 2025, *Nat Chem Biol* 21:1948 (RFpeptides)** · Bhardwaj et al. 2016, *Nature* 538:329 · Hosseinzadeh et al. 2017, *Science* 358:1461 (not 2018) · Renfrew et al. 2012, *PLoS ONE* 7:e32637 (non-canonical amino acids in Rosetta) · Mandal et al. 2012, *PNAS* 109:14779 (D-protein, mirror-image).

This is the subsection that matters most for macrocycle work. The Renfrew and Mandal papers have no talks, and they are exactly the two you need for D-residue and non-canonical design — which is precisely what cyclic-peptide pipelines rely on.

R.7 Antibodies and TCRs

Method	Talk	Speaker / host	Length	Link
Germinal	Efficient generation of epitope-targeted de novo antibodies	ML4PE	57:35	▶
Germinal	long form	Valence Labs	1:09:18	▶
IgFold / IgLM / AntiBERTy	Designing proteins with language models	Jeff Ruffolo (ML@Berkeley)	1:16:16	▶
ABlooper / ABodyBuilder	Building the toolkit for computational antibody design	Charlotte Deane (AIRR)	34:53	▶
AF-Multimer for antibodies	Computational design of antibody repertoires	BPDMC	1:10:02	▶
Experimental scale	mBER: controllable de novo antibody design with million-scale screening	BPDMC	1:42:10	▶
AbDiffuser	Full-Atom Generation of In Vitro Functioning Antibodies	ML4PE	1:10:25	▶
Seq-structure co-design	Iterative Refinement GNN for Antibody Co-design	Wengong Jin (Valence)	1:27:05	▶
—	Structured Refinement Network for Antibody Design	Wengong Jin (Valence)	58:36	▶
Bayesian flow networks	Flexible Antibody Design with Multimodal Bayesian Flow Networks	ML4PE	44:20	▶
Classical	RosettaAntibodyDesign (RABD) walkthrough	Meiler Lab	1:14:42	▶
Classical	RosettaAntibody Introduction	Meiler Lab	12:41	▶
Primer	Introduction to Antibody Structural Biology	Meiler Lab	18:25	▶
Critical	FLAb2: How Well Can Protein AI Predict Developability?	RosettaCommons	31:46	▶
TCR	A structure-informed DL framework for TCR-peptide-HLA interactions	Broad Institute	14:08	▶
CAR design	Determinants of efficacious de novo chimeric antigen receptors	ML4PE	46:34	▶

Papers: Mille-Fragoso et al. 2026, *Nat Biotechnol* (Germinal) · **Bennett et al. 2026, *Nature* 649:183 (RFantibody)** · Ruffolo et al. 2023, *Nat Commun* 14:2389 (IgFold) · Shuai et al. 2023, *Cell Systems* 14:979

(IgLM) · Abanades et al. 2023, *Commun Biol* 6:575 (ImmuneBuilder) · **Bradley 2023, *eLife* 12:e82813 (TCRdock)** · Yin et al. 2023, *NAR* 51:W569 (TCRmodel2) · Olsen et al. 2022, *Bioinform Adv* 2:vbac046 (AbLang).

FLab2 is the antibody equivalent of PoseBusters — it asks whether these models predict developability, which is what actually determines whether an antibody becomes a drug. Watch it after any antibody design advocacy talk.

No first-author talk exists for TCRdock or TCRmodel2, despite TCR-pMHC being directly relevant to any TCR-pMHC modelling work. Read Bradley 2023.

R.8 Function, dynamics and the hard design problems

Method	Talk	Speaker / host	Length	Link
Multistate design	Multistate and functional design via sequence space diffusion	ML4PE	54:47	▶
Conformational switches	De novo design of small-molecule-induced conformational change	BPDMC	1:03:21	▶
Conformational bias	Computational design of conformation-biasing mutations	ML4PE	51:45	▶
De novo luciferase	The Coming of Age of De Novo Protein Design	David Baker (NIH)	54:33	▶
Small-molecule binders	Designing ligand-binding proteins from scratch	BPDMC	1:18:23	▶
Small-molecule binders	Design of small molecule binding proteins using deep learning	BPDMC	1:00:53	▶
Channels and transporters	De Novo Designed Voltage-gated Anion Channels Suppress Neuron Firing	Chen Zhou (ML4PE)	28:49	▶
Proteolysis-gated	Autoinhibitory domains for a protease-activated PD-L1 antagonist	BPDMC	1:11:19	▶
Protease substrates	Deep learning guided design of protease substrates	ML4PE	52:49	▶
Glycoproteins	De novo Glycan Modeling and Design	BPDMC	1:10:06	▶
Cell engineering	Designer proteins for stem cell engineering	BPDMC	1:18:23	▶
Therapeutic	Designing cKIT Receptor Inhibitors for Bone Marrow Transplant	BPDMC	27:16	▶

Papers: Praetorius et al. 2023, *Science* 381:754 (two-state hinges) · Boyken et al. 2019, *Science* 364:658 · Langan et al. 2019, *Nature* 572:205 (LOCKR) · Glasgow et al. 2019, *Science* 366:1024 (Kortemme biosensor) · Yeh et al. 2023, *Nature* 614:774 (luciferase) · Polizzi & DeGrado 2020, *Science* 369:1227 · An et al. 2024, *Science* 385:276 · Xu et al. 2020, *Nature* 585:129 (transmembrane pores) · Lu et al. 2018, *Science* 359:1042 (multipass membrane proteins) · Ambroggio & Kuhlman 2006, *JACS* 128:1154 · Leaver-Fay et al. 2011, *PLoS ONE* 6:e20937 (multistate).

The Kortemme biosensor paper and LOCKR both lack talks, and they are the two clearest examples of *designing a function that requires motion*. That gap is the same one Atlas family R has throughout: nearly every method here generates a single static backbone, which is exactly what Section A.4 row five says is unexchanged.

R.9 Protein-nucleic acid and co-design

Method	Talk	Host	Length	Link
RNA and nucleoprotein design	RNA Function, Design, and Modeling	BPDMC	59:33	▶
Protein-ligand co-design	Generalized Biomolecular Modeling and Design with RFAA	Rohith Krishna (Valence)	1:04:54	▶
Infrastructure	Accelerating Biomolecular Modeling with AtomWorks and RF3	ML4PE	51:17	▶
Infrastructure	first author	Nathaniel Corley (Valence)	52:10	▶

Papers: Glasscock et al. 2025, *Nat Struct Mol Biol* 32:2252 (DNA-binding design — NSMB, not *Nature*) · Favor et al. 2025, bioRxiv (RNA and nucleoprotein design, preprint).

R.10 The framing talks and the skeptics

Talk	Speaker / host	Length	Link
Design of new protein functions using deep learning	David Baker (Allen School)	59:14	▶
Protein design using deep learning	David Baker (Broad)	52:07	▶
Design of New Protein Functions Using Deep Learning	David Baker (NCATS NIH)	25:09	▶
Nobel Lecture: De Novo Protein Design	David Baker	38:33	▶
Part 1: Introduction to Protein Design	David Baker (iBiology)	21:22	▶
Some thoughts on ML-based protein engineering	Jennifer Listgarten (Simons)	~62 min	▶
ML-based Design of Proteins and Small Molecules	Jennifer Listgarten (Simons)	45:48	▶
ML-Based Design Of Proteins	Jennifer Listgarten (Simons)	31:56	▶
The State of Protein Structure Prediction and Friends	Mohammed AlQuraishi (Simons)	1:05:31	▶
Review and discussion of AlphaFold3	Sergey Ovchinnikov (BPMC)	1:12:58	▶

Listgarten’s argument, stated as plainly as it can be: prediction is interpolation; design is extrapolation. A model trained on natural proteins is asked, at design time, to score molecules drawn from a region of sequence space where it has no training data and no calibrated uncertainty. Every method in this family does this. The field’s success rates are what they are because of it. If you take one idea from Atlas R, take that one — it is also a direct statement of an open problem, which is why it reappears as Capstone III.

R.11 BUILD — Atlas R

1. **Generate 1,000 backbones** for a binder target with RFdiffusion. Design sequences with ProteinMPNN at three temperatures. Refold and compute self-consistency RMSD, interface pAE, and pLDDT.
2. **Find the filter that does no work.** Plot every filter metric’s distribution and its correlation with every other. In most published pipelines at least one filter is nearly redundant and one is nearly uncorrelated with anything that matters. Identify both.
3. **Run PoseBusters-style physical validity checks** on your designs. Report the fraction that pass.
4. **Write the one page.** What fraction of 1,000 would you actually order, and defend the number against derivation checkpoint 63 — construct a concrete case where a design passes self-consistency with high confidence and still fails experimentally.

Derivation checkpoints due: 61, 62, 63.

R.12 Paired reading — Atlas R

Watch this	Then read this	Hold this question
Dauparas, <i>ProteinMPNN</i>	Dauparas et al. 2022, <i>Science</i> 378:49	Sequence recovery is the headline. What moved experimentally, and by how much?
BPDMC, <i>Beyond sequence recovery</i>	Compare to Dauparas 2022	Write down why likelihood-given-backbone $\neq$ probability-of-folding.
ML4PE, <i>Mega-scale stability</i>	Tsuboyama et al. 2023, <i>Nature</i> 620:434	500,000 measured stabilities. What can you now train that you could not before?
Watson & Juergens, <i>RFdiffusion</i>	Watson et al. 2023, <i>Nature</i> 620:1089	List every filter between generation and ordering. What fraction survives?
MaomLab, <i>RFdiffusion critique</i>	Same paper	Which claims does an external reading weaken?
Ahern & Yim, <i>RFdiffusion2</i>	Ahern et al. 2026, <i>Nat Methods</i> 23:96	What failed in v1 that motivated atom-level generation?
Grigoryan & Costello, <i>Chroma</i>	Ingraham et al. 2023, <i>Nature</i> 623:1070	Conditioners are classifier guidance. What constraint class does that admit?
Yim & Trippe, <i>FrameDiff</i>	Yim et al. 2023, <i>ICML (SE(3) diffusion)</i>	Backbone frames live on SE(3). What breaks under a Euclidean treatment?
Bose & Tong, <i>FoldFlow</i>	Bose et al. 2024, ICLR	Flow matching vs diffusion on the same manifold. What is traded?
Campbell & Yim, <i>Multiflow</i>	Campbell et al. 2024, ICML	Discrete and continuous jointly. How is the discrete flow defined?
ML4PE, <i>EvoDiff</i>	Alamdari et al. 2023, bioRxiv	“Sequence is all you need.” What does that claim give up?
ML4PE, <i>MotifBench</i>	The MotifBench paper	Are published scaffolding numbers comparable? Why not?
Ovchinnikov, <i>Inverting predictors</i>	Anishchenko et al. 2021, <i>Nature</i> 600:547	What does invertibility say about what a predictor represents?
Valence, <i>BindCraft</i>	Pacesa et al. 2025, <i>Nature</i> 646:483	One-shot binders. What is the experimental hit rate and against what baseline?
RosettaCommons, <i>Binder Design Overview</i>	Cao et al. 2022, <i>Nature</i> 605:551	Target structure alone. What information is actually being used?
BPDMC, <i>Why designs fail</i>	Any binder paper’s supplement	Count the designs ordered versus designs reported.
Cradle, <i>Adaptiv competition</i>	Cotet et al. 2025, bioRxiv	A public wet-lab benchmark. Why does design not have a standing one?
Lauko & Pellock, <i>Serine hydrolases</i>	Lauko et al. 2025, <i>Science</i> 388:eadu2454	What kcat was achieved, and how far from a natural enzyme?
IPD panel, <i>Then & Now</i>	Röthlisberger 2008; Jiang 2008; Siegel 2010; then Blomberg et al. 2013, <i>Nature</i> 503:418	Where did the catalysis actually come from?
Fleishman, <i>PROSS/FuncLib</i>	Goldenzeig 2016; Khersonsky 2018	Evolutionary data plus atomistic design. Which culture is this?

Watch this	Then read this	Hold this question
Rettie, <i>RFpeptides</i>	Rettie et al. 2025, <i>Nat Chem Biol</i> 21:1948	Macrocycles with a cyclic constraint. How is the constraint imposed?
BPDMC, <i>Structured peptides</i>	Bhardwaj et al. 2016, <i>Nature</i> 538:329; Hosseinzadeh et al. 2017, <i>Science</i> 358:1461	What makes a macrocycle designable where a linear peptide is not?
ML4PE, <i>Geminal</i>	Mille-Fragoso et al. 2026, <i>Nat Biotechnol</i>	Epitope-targeted de novo antibodies. What is the hit rate per epitope?
RosettaCommons, <i>FLAb2</i>	The FLAb2 paper	Developability, not affinity, kills antibodies. Can any model predict it?
Broad, <i>TCR-pMHC framework</i>	Bradley 2023, <i>eLife</i> 12:e82813	Why do general methods underperform on TCR-pMHC specifically?
BPDMC, <i>Conformational change design</i>	Praetorius et al. 2023, <i>Science</i> 381:754; Langan et al. 2019, <i>Nature</i> 572:205	Designing motion. What objective replaces a single backbone?
BPDMC, <i>Ligand-binding from scratch</i>	Polizzi & DeGrado 2020, <i>Science</i> 369:1227; An et al. 2024, <i>Science</i> 385:276	A binding site is a geometric arrangement. What else must be true?
Listgarten, <i>ML-based protein engineering</i>	Kapoor & Narayanan 2023	Formalize the distribution shift between training and design-time input.
Baker, Nobel Lecture	Kuhlman et al. 2003 (Top7); Watson et al. 2023	Twenty years apart. Which parts of the 2003 pipeline survived?

Atlas S — Benchmarking and Evaluation

Problem. Decide whether a method is better. This is harder than any individual method in this Atlas, and it is the family where a careful person can make the largest contribution with the least compute.

The thesis of this family in one sentence: most reported improvements in computational biology are benchmark artifacts, and the ones that are not become obvious in a blind assessment.

S.1 The blind assessments

These are the field's crown jewels and the reason structure prediction is trustworthy in a way most of computational biology is not.

Assessment	What it tests	Resource
CASP	Protein structure prediction, blind, biennial since 1994	predictioncenter.org
CAPRI	Protein-protein docking and complexes	ebi.ac.uk/pdbe/capri
CAFA	Protein function prediction	biofunctionprediction.org
RNA-Puzzles	RNA structure prediction	rnapuzzles.org
CACHE	Hit-finding for a given target, with experimental follow-up	cache-challenge.org
SAMPL	Solvation, host-guest binding, partition coefficients	samplchallenges.github.io
D3R	Pose and affinity prediction, blind	drugdesigndata.org
Adaptyv	De novo binder design, scored in the wet lab	adaptyvbio.com

Talk	Speaker / host	Length	Link
Critical Assessment of Structure Prediction	John Moult & Krzysztof Fidelis (MLSB 2025)	36:17	▶
Initial reaction to AlphaFold2 right after CASP14	AlQuraishi & Ovchinnikov (BPDMC)	1:17:30	▶
Optimizing Cetuximab: Winning Adaptyv's Competition	Cradle	53:53	▶
MotifBench: a standardized benchmark for motif-scaffolding	ML4PE	54:48	▶
FLAb2: How Well Can Protein AI Predict Developability?	RosettaCommons	31:46	▶
CARE: Benchmark Suite for Classification and Retrieval of Enzymes	Jason Yang (ML4PE)	47:02	▶
PoseBusters: docking methods fail to generate physically valid poses	Buttenschoen (Valence)	34:28	▶
DART-Eval: a benchmark for DNA language models	Stanford (MLCB)	19:36	▶
Panel — Foundation models for biology, when are they useful?	MLCB 2024	51:23	▶
Quantitative Affinity Data at Scale: the Data Bottleneck	Murakowska (BPDMC)	47:16	▶

Papers: Moult et al. 1995, *Proteins* 23:ii (CASP's founding) · Kryshtafovych et al. 2021, *Proteins* 89:1607 (CASP14 assessment) · Buttenschoen et al. 2024, *Chem Sci* 15:3130 (PoseBusters) · Notin et al. 2023, NeurIPS (ProteinGym) · Radivojac et al. 2013, *Nat Methods* 10:221 (CAFA) · Cotet et al. 2025, bioRxiv (Adaptyv EGFR).

Why CASP worked and most benchmarks do not. Targets are *not yet solved* when predictions are submitted, assessment is by third parties, and the assessors publish what failed. Every one of those is costly and every one is load-bearing. **A retrospective benchmark on a public test set has none of them**, which is why a method can be state-of-the-art on CASP-era data and useless on next year's structures.

Adaptyv is the most important entry in this table, because it is the first standing blind assessment for *design*: you submit sequences, they synthesize and measure binding, and everyone sees the numbers. Design has had no CASP for thirty years. It now has the beginning of one.

S.2 The failure modes, enumerated

Learn these by name. Each has cost someone a retraction.

1. **Train-test leakage by homology.** The canonical failure in this field. A 30%-identity split is not sufficient when structural similarity survives below that. Foldseek-based structural splits are better and are rarely used.

2. **Temporal leakage.** Evaluating on structures deposited before the model's training cutoff. This is why CASP's "not yet solved" criterion matters.
3. **Thresholded metrics manufacturing discontinuity.** Schaeffer's argument (D.7.6) applied to biology: "fraction under 4 Å" is a step function over a continuous quantity. Report the continuous quantity.
4. **Means over heterogeneous splits.** A single ProteinGym number hides that performance inverts with MSA depth.
5. **Physical invalidity passing numeric checks.** PoseBusters: poses with good RMSD and impossible bond geometry.
6. **The missing baseline.** Compare to a nearest-neighbor retrieval baseline. In function prediction and in docking it is often not beaten.
7. **The anti-symmetry failure.** $\Delta\Delta G(A \rightarrow B) \neq -\Delta\Delta G(B \rightarrow A)$ in most predictors.
8. **Selection on the test set.** Tuning filters while looking at held-out results, which is what every design pipeline does by construction.
9. **Publication asymmetry in the benchmark itself.** The data were generated by experimentalists who report destabilizing mutations more often than neutral ones.
10. **Evaluating the wrong thing entirely.** FLaB2's point: antibody affinity prediction is not what determines whether an antibody becomes a drug.

Derivation checkpoint 60. Take a benchmark you have used and audit it against all ten. Write one paragraph per item. **Most benchmarks fail three or four, and finding out which ones is more valuable than another model.**

S.3 Statistics and experimental design

Talk	Speaker / host	Length	Link
Data thinning to avoid double dipping	Lucy Gao (Broad MIA)	1:31:46	▶
Knockoffs for Finding Variables with Statistical Guarantees	Lucas Janson (Broad MIA)	1:36:18	▶
Bayesian methods for adaptive experimental design	Martin Jankowiak (Broad MIA)	50:41	▶
Conformal prediction under feedback covariate shift	Wong-Fannjiang (ML4PE)	58:45	▶
Beyond the training set: detecting distribution shift	Damani (MLCB)	24:14	▶
Experimental Design and Data Annotation	Graham Neubig (CMU Advanced NLP 9)	1:17:22	▶
Predictable Noise in LLM Benchmarks	Sida Wang (Berkeley MOOC)	44:04	▶
Six Principles for Evaluating Cognitive Capabilities in AI	Melanie Mitchell (SFI)	48:41	▶
Scaling Up “Vibe Checks” for LLMs	Shreya Shankar (MLSys #97)	55:01	▶
The Future of Language Models: A Perspective on Evaluation	Simons Institute	1:05:41	▶
CS336 2026 Lecture 12: Evaluation	Stanford	1:18:34	▶

Papers: Fannjiang et al. 2022, *NeurIPS* (conformal under feedback covariate shift) · Barber & Candès 2015, *Ann Stat* 43:2055 (knockoffs) · Neyshabur et al. 2021 (on the importance of a proper baseline) · **Kapoor & Narayanan 2023**, *Patterns* 4:100804 (leakage and the reproducibility crisis in ML-based science).

Kapoor & Narayanan is the paper to put on your wall. They audited 294 papers across 17 scientific fields and found data leakage in a large fraction of them. Their taxonomy of leakage types is the checklist in S.2 stated more rigorously. **If you read one paper from Atlas S, read that one.**

Sida Wang’s Predictable Noise in LLM Benchmarks is the under-appreciated one. Benchmark differences of a few points are frequently within run-to-run noise, and almost nobody reports error bars on a benchmark number. That criticism applies verbatim to every design hit rate in Atlas R.

S.4 BUILD — Atlas S

1. **Pick a published comparison** in your own area where method A beat method B. Reproduce both on the stated benchmark.
2. **Re-split structurally** using Foldseek rather than sequence identity. Re-evaluate.
3. **Add the dumb baseline** — nearest neighbor by sequence, or by structure, or the training-set mode. Report it.
4. **Put error bars on everything** by bootstrapping over test examples and, if possible, over training seeds.

5. Write the one page. Does the original conclusion survive? Report the answer whichever way it comes out.

Derivation checkpoint due: 60.

This BUILD is the most likely of the nineteen to become a publication, and it needs no new method and almost no compute. A careful re-evaluation that overturns or confirms a widely-cited comparison is a real contribution, and the field has very few people willing to do it.

S.5 Paired reading — Atlas S

Watch this	Then read this	Hold this question
Moult & Fidelis, <i>Critical Assessment</i>	Moult et al. 1995, <i>Proteins</i> 23:ii	What makes CASP's design work? Which parts are essential?
AlQuraishi & Ovchinnikov, <i>post-CASP14</i>	Kryshtafovych et al. 2021	What convinced experts, in real time, that something had changed?
Buttenschoen, <i>PoseBusters</i>	Buttenschoen et al. 2024, <i>Chem Sci</i> 15:3130	Good RMSD, impossible chemistry. What else is your metric blind to?
ML4PE, <i>MotifBench</i>	The paper	Are published scaffolding results comparable? Why not?
RosettaCommons, <i>FLAb2</i>	The paper	Are you measuring the quantity that decides the outcome?
Yang, <i>CARE</i>	The paper	Enforce novelty in the split. How much performance was retrieval?
Wong-Fannjiang, <i>Conformal under FCS</i>	Fannjiang et al. 2022, <i>NeurIPS</i>	Design queries the model off-distribution. How do you get a valid interval?
Gao, <i>Data thinning</i>	Her paper	How do you use the same data twice without invalidating inference?
Wang, <i>Predictable Noise in Benchmarks</i>	The paper	Put error bars on the last benchmark number you quoted.
Mitchell, <i>Six Principles</i>	Her paper	Does your benchmark measure the capability or a correlate?
—	Kapoor & Narayanan 2023, <i>Patterns</i> 4:100804	Audit your own last project against their leakage taxonomy.
Cradle, <i>Adaptiv competition</i>	Cotet et al. 2025, <i>bioRxiv</i>	What would a standing CASP for design cost, and who would run it?

Atlas T — RNA Structure, Folding and Design

Why this family is new in the third edition. Rhiju Das is one of the ten scientists this curriculum is built around, and the second edition mentioned him three times. That was a real failure, and this family is the repair.

The thread to follow in Das's own work is that he keeps rebuilding the same argument at larger scale: *if the bottleneck is data rather than architecture, go and manufacture the data*. Eterna, OpenVaccine, Ribonanza and OpenKnot are four passes at that one idea. Section T.10 answers, with numbers, why that is the right bet.

One sourcing note. The richest standing seminar in this field is the **CASP RNA SIG** channel, a monthly series running since 2023. Twenty of its talks appear below. Nothing else in RNA comes close as a lecture series.

T.1 Rhiju Das, in his own words

Talk	Len	Link
BioML Seminar — the RNA Folding Problem	1:23:58	▶
Ribonanza: big data for RNA structure prediction	1:01:58	▶
CASP15 RNA assessment	29:12	▶
Kaggle RNA 3D Folding Challenge: template-based prediction wins	58:55	▶
RNA modelling and design (MRC LMB)	35:54	▶
Early results from the OpenKnot challenge	41:12	▶
EteRNA: RNA Nanoengineering through Crowd Science (Stanford)	55:03	▶
EteRNA: A Videogame and a Massive Open Lab	26:48	▶
StepWise Monte Carlo for modeling and design	28:20	▶
Atomic accuracy and blind predictions (ISMB 2011)	20:48	▶
Crowdsourced design of stabilized COVID-19 mRNAs	11:41	▶
RNA Worlds panel with Cech, Krainer and Chin	2:28:55	▶
State of Eterna (with Adrien Treuille)	36:05	▶

Start with the BioML seminar — it runs Eterna through OpenVaccine to Ribonanza as one continuous argument, which is the only way the program makes sense. **Then the CASP15 assessment**, which is the canonical statement of where RNA 3D prediction actually stands and which he delivered as assessor rather than as advocate.

Das lab and alumni: Shujun He on Ribonanza architecture (▶) · **Hannah Wayment-Steele on EternaFold** — the only talk anywhere explaining the conditional log-linear model and how it is retrained on crowdsourced chemical mapping, 20:19 (▶) · *Superfolder* thermostable mRNA (▶) · Rachael Kretsch on RNA cryo-EM (▶) and the **CASP16 RNA assessment** (▶).

Papers: He et al. 2024, *bioRxiv* (**Ribonanza**) · Das et al. 2023, *Proteins* (CASP15 RNA) · Lee et al. 2014, *PNAS* 111:2122 (**Eterna**) · Wayment-Steele et al. 2022, *Nat Methods* 19:1234 (EternaFold) · Wayment-Steele et al. 2021, *NAR* 49:10604 (mRNA stabilization) · Kappel et al. 2020, *Nat Methods* 17:699 (Ribosolve) · Watkins et al. 2018, *Sci Adv* 4:eaar5316 (StepWise Monte Carlo).

T.2 Secondary structure prediction — the dynamic programming lineage

This is the algorithmic core of the family, and the talks that actually *derive* the recursions turn out to be in algorithms courses, not bioinformatics channels.

The Nussinov recursion, derived: Luay Nakhleh (Rice), 21:39 (▶) · **Dan Gusfield (UC Davis), two lectures** that set up the interval DP, fill the table, prove the ordering and walk the traceback (▶ ▶) · Steven Skiena (Stony Brook), 1:20:08, moving from base-pair maximization into the Zuker energy model (▶).

The Sebastian Wild unit (Marburg) — the best derivational sequence found. Treat as one 2.5-hour module; it derives Nussinov, the Zuker loop decomposition and the SCFG/inside-outside material in order.

#	Lecture	Len	Link
8-1	Noncoding RNA	15:08	▶
8-2	RNA Secondary Structure	20:41	▶
8-3	Pseudoknot-free structures — Nussinov, with the non-crossing condition shown to be what makes the decomposition valid	18:19	▶
8-4	Refined energy models — the Zuker-Stiegler step, hairpin/bulge/internal/multibranch, the V/W split	8:58	▶
8-6	Probabilistic Context-Free Grammars	23:23	▶
8-7	Probabilistic Parsing — CYK and inside-outside, the structural twin of McCaskill	20:00	▶

Vienna RNA from its authors — the highest-value find in this family. Ivo Hofacker, 1:29:43, derives the energy model, the MFE recursions *and* the McCaskill partition function and base-pair probability matrix in one sitting (▶). **Ronny Lorenz**, 1:29:12, goes past the textbook model into soft constraints, SHAPE-directed folding and the ensemble quantities RNAfold actually reports (▶).

Comparative methods and the covariation argument: Sebastian Will on Sankoff simultaneous alignment-and-folding, two lectures (▶ ▶) · **Elena Rivas on why covariation evidence, not free energy, is the real arbiter of a claimed structure** — the statistical counterweight to the whole MFE enterprise, 37:04 (▶).

Course lectures: Christopher Burge (MIT 7.91J), 1:22:40 (▶) · Manolis Kellis (MIT 6.047) (▶) · Matthew Macauley's combinatorial derivation (▶).

Linear-time methods — Liang Huang's line: LinearFold (▶) · LinearPartition, which is linear-time McCaskill (▶) · the ISMB keynote on the whole Linear* family, including why beam pruning is often *more* accurate on long sequences (▶) · Parsing Algorithms for COVID-19, Huang to an NLP audience, which is the clearest statement anywhere of why RNA folding is parsing (▶).

Grammars, homology and pseudoknots: Rfam and Infernal covariance models (EMBL-EBI, ▶) · Anne Condon on which pseudoknot classes are polynomial and which are NP-hard (▶) · Hosna Jabbari on Knotty and HFold (▶) · Henri Orland's statistical-physics genus expansion (▶) · Yann Ponty defending the DP tradition against the deep-learning wave (▶).

Papers: Nussinov & Jacobson 1980, *PNAS* 77:6309 · Zuker & Stiegler 1981, *NAR* 9:133 · McCaskill 1990, *Biopolymers* 29:1105 · Sankoff 1985, *SIAM J Appl Math* 45:810 · Eddy & Durbin 1994, *NAR* 22:2079 · Rivas, Clements & Eddy 2017, *Nat Methods* 14:45 · Rivas & Eddy 1999, *JMB* 285:2053 · Do, Woods & Batzoglou 2006, *Bioinformatics* 22:e90 (CONTRAfold) · Lorenz et al. 2011, *Algorithms Mol Biol* 6:26 (ViennaRNA) · Huang et al. 2019, *Bioinformatics* 35:i295 (LinearFold).

Derivation checkpoint 71. Derive the Nussinov recursion and prove the non-crossing condition is what makes the interval decomposition valid. Then state exactly what the Zuker loop model adds, and why the partition function needs a *different* recursion rather than the same one with a max replaced by a sum. Wild 8-3, 8-4 and 8-7 are the three lectures that get you there.

T.3 Thermodynamics and nearest-neighbour parameters

This is the weakest-covered topic in the family relative to its importance, and the gap is specific: the people who built the parameter sets left no lectures.

Talk	Speaker	Len	Link
Nearest Neighbor Method	Albert Courey (UCLA)	24:25	▶
Using Energy Values in Eterna — the Turner rules made operational	Eterna community	1:03:27	▶
RNA, Mechanics, and the Partition Function	Jennifer Pearl (Eterna)	55:40	▶
Folding engines, models and thermodynamics	Eterna community	39:07	▶
High-throughput RNA tertiary contact thermodynamics	Joseph Yesselman (Nebraska)	56:54	▶
Thermodynamics of RNA structures by Wang-Landau sampling	Peter Clote (Boston College)	29:13	▶

The Courey lecture is the one real nearest-neighbour derivation on YouTube — thermodynamic relationships, duplex ΔH and ΔS , a worked calculation, the T_m equation, and how the parameters were measured.

Honest gap: Doug Turner, David Mathews and John SantaLucia have no recorded lectures. These are the people whose measurements every RNA folding program depends on. **Read instead:** SantaLucia 1998, *PNAS* 95:1460 · Mathews et al. 1999, *JMB* 288:911 and 2004, *PNAS* 101:7287 · Turner & Mathews 2010, *NAR* 38:D280 (the NNDB). **Derivation checkpoint 6** in Module B.4 is the exercise.

T.4 Chemical probing – what the data actually constrains

Talk	Speaker	Len	Link
Chemically probing RNA-protein networks inside cells	Chase Weidmann (Michigan, Weeks alum)	1:00:53	▶
Mapping protein interaction networks on RNA (ISMB keynote)	Chase Weidmann	28:14	▶
DMS mapping of higher-order structure and folding thermodynamics	Anthony Mustoe (Baylor)	58:03	▶
Low-energy RNA fluctuations measured by chemical probing (PRIME)	Edric Choi (Lucks lab)	59:32	▶
Drugging RNA Structure	Silvi Rouskin (HMS)	56:31	▶
Landscape and variation of RNA secondary structures in the human transcriptome	Yue Wan (A*STAR)	30:40	▶
In-cell ensemble mapping of structural switches	Danny Incarnato (Groningen)	1:10:34	▶
Automated recognition of functional RNA elements by SHAPE signatures	Sharon Aviran (UC Davis)	46:58	▶
Sparse Reconstruction of RNA Structure Landscapes	Sharon Aviran (UC Davis)	48:30	▶
Visualizing RNA Structural Dynamics Using NMR	Hashim Al-Hashimi (Columbia)	20:52	▶
NMR contributions to RNA structural biology and viral genomes	Harald Schwalbe (Frankfurt)	1:01:32	▶
Hydroxyl radical footprinting	Schlatterer (Brenowitz lab)	13:40	▶

Aviran's *Sparse Reconstruction* talk is the sharpest statement of what probing data actually constrains: one averaged reactivity vector, many possible underlying populations. It is an inverse problem and it is ill-posed. Everyone who uses SHAPE restraints should watch it.

Choi's PRIME result is the number to remember: base pairs dynamically open at **0.5–3 kcal/mol**, roughly half the commonly inferred values. RNA samples open states far more accessible than the models assume.

Papers: Merino et al. 2005, *JACS* 127:4223 (SHAPE) · Deigan et al. 2009, *PNAS* 106:97 · Siegfried et al. 2014, *Nat Methods* 11:959 (SHAPE-MaP) · Zubradt et al. 2017, *Nat Methods* 14:75 (DMS-MaPseq) · Spitale et al. 2015, *Nature* 519:486 (icSHAPE) · Tomezsko et al. 2020, *Nature* 582:438 (DREEM) · Dethoff et al. 2012, *Nature* 491:724 (excited states by NMR).

The most surprising absence in this entire curriculum: Kevin Weeks has no recorded lecture anywhere.

SHAPE is his, and the search was exhaustive. Teach the lineage through his descendants — Weidmann, Mustoe, Choi — and read the papers. **Sarah Woodson is likewise absent**, with only a three-minute interview; read Woodson 2010, *Annu Rev Biophys* 39:61.

T.5 Ribozymes and catalytic RNA

Talk	Speaker	Len	Link
Discovering Ribozymes	Tom Cech (iBiology)	10:06	▶
Ribozymes, Telomerase, Aging, and Cancer	Tom Cech	1:16:44	▶
The Magic of RNA: From CRISPR to Coronavirus Vaccines	Tom Cech	1:06:18	▶
Ribonuclease P: A Small Step in the RNA World	Sidney Altman (Yale)	51:01	▶
How does RNA act like an enzyme?	David Lilley (Dundee)	38:28	▶
The extent of RNA catalysis — are there any limits?	David Lilley (MRC LMB)	34:20	▶
From Structure and Function of Ribosomes to New Antibiotics	Thomas Steitz (Yale)	51:04	▶
The Amazing Ribosome and its Origin	Ada Yonath (Weizmann)	1:34:41	▶
Ribosomal RNA: The Kernel of Life	Harry Noller (UCSC)	39:58	▶
The Story of Deciphering the Ribosome	Venki Ramakrishnan (MRC LMB)	1:06:42	▶
ASBMB Plenary Lecture (group II introns)	Anna Marie Pyle (Yale)	38:44	▶
Pyle Part 1: RNA Structure (iBiology)	Anna Marie Pyle	23:04	▶
Pyle Part 2: Inside an RNA Splicing Machine	Anna Marie Pyle	28:55	▶
Pyle Part 3: RNA Helicases	Anna Marie Pyle	32:16	▶
Evolution of Nucleic Acids Under Abiological Conditions	Gerald Joyce (Scripps)	23:38	▶
Self-replicating synthetic systems	Gerald Joyce	34:43	▶
RNA polymerase ribozyme program	Philipp Holliger (MRC LMB)	1:02:23	▶
The Origin of Cellular Life on Earth	Jack Szostak (Harvard)	54:40	▶
CRISPR systems (Rosalind Franklin Lecture)	Jennifer Doudna (Berkeley)	1:15:53	▶

Lilley's *How does RNA act like an enzyme?* is the best mechanism lecture on the nucleolytic ribozymes — general acid-base catalysis across hammerhead, hairpin and glmS. Pyle Part 1 is the best free introduction to RNA tertiary structure anywhere.

A note on the Doudna choice. Her actual Nobel Lecture contains zero mentions of ribozymes or the group I intron. This lecture narrates the ribozyme-to-CRISPR arc, which is the intellectually honest lineage: Cate et

al. 1996, *Science* 273:1678 leads to Jinek et al. 2012, *Science* 337:816.

Papers: Kruger et al. 1982, *Cell* 31:147 · Guerrier-Takada et al. 1983, *Cell* 35:849 · Ban et al. 2000 and Nissen et al. 2000, *Science* 289:905 and 289:920 · Noller, Hoffarth & Zimniak 1992, *Science* 256:1416 · Toor et al. 2008, *Science* 320:77 · Lincoln & Joyce 2009, *Science* 323:1229 · Johnston et al. 2001, *Science* 292:1319.

T.6 Riboswitches and RNA regulation

Ron Breaker — *The RNA World*, 58:09 (▶) · **Ancient RNA Relics and Modern Drug Discovery** (NIH Director's Lecture), 1:10:23 (▶) · ASBMB-Merck Award Lecture (▶) · **the comparative-genomics discovery method** behind the new riboswitch and ribozyme classes (▶).

Also: Beatrix Suess on engineered riboswitches — the engineering counterpart to Breaker's discovery work (▶) · **Joan Steitz**, Mendel Lecture on viral and cellular noncoding RNAs, 1:14:35 (▶) and the short iBiology *SNURPs and Serendipity* (▶).

Papers: Winkler, Nahvi & Breaker 2002, *Nature* 419:952 · Weinberg et al. 2010, *Genome Biol* 11:R31 · Lerner & Steitz 1979, *PNAS* 76:5495.

T.7 Three-dimensional structure prediction

Talk	Speaker	Len	Link
Unlocking the RNA Universe (ARES)	Raphael Townshend (Atomic AI)	1:00:10	▶
CASP15 RNA assessment	Eric Westhof (Strasbourg)	37:26	▶
CASP16 RNA assessment	Eric Westhof	22:45	▶
Computer-based Predictions of RNA Structures: Where do we stand?	Eric Westhof	1:01:32	▶
CASP15 assessment	Marta Szachniuk (Poznań)	16:19	▶
CASP15 (SimRNA)	Janusz Bujnicki (IIMCB)	19:55	▶
DeepFoldRNA	Robin Pearce (Michigan)	59:28	▶
trRosettaRNA	Yang & Wang (Shandong)	1:00:39	▶
RoseTTAFoldNA	Frank DiMaio (IPD)	1:04:00	▶
Limits of deep-learning-based RNA prediction	Marko Ludaic	31:09	▶
Has AlphaFold3 achieved success for RNA?	Clément Bernard (Évry)	24:12	▶
Ab initio RNA ensembles with RNAnneal	Pratyush Tiwary (Maryland)	1:00:19	▶
RNA with water, Mg ²⁺ ions and chemical probes	Giovanni Bussi (SISSA)	33:31	▶
Simulations of RNA and RNP; fitting cryo-EM	Karissa Sanbonmatsu (LANL)	55:56	▶
A Bottom-Up Approach to RNA Architecture	Bohdan Schneider (Czech Acad.)	1:01:24	▶
X3DNA-DSSR: structural bioinformatics of nucleic acids	Xiang-Jun Lu (Columbia)	58:43	▶
Non-canonical base pair interactions	Jérôme Waldispühl (McGill)	38:02	▶
Hidden errors in cryo-EM and crystal structure models	Grzegorz Chojnowski (EMBL)	1:04:05	▶

Townshend's ARES talk is the single most important data point in T.10: an accurate scoring function learned from only **eighteen** known RNA structures. **Pair it immediately with Ludaic's *Limits of deep-learning-based RNA prediction***, which is the explicit skeptic's talk.

Chojnowski's talk belongs here and is easy to skip: the ground truth is not as solid as the benchmarks assume. Read it before trusting any RNA RMSD.

Papers: Townshend et al. 2021, *Science* 373:1047 (ARES) · Das et al. 2023, *Proteins* (CASP15) · Westhof et al. 2026, *Proteins* (CASP16) · Pearce et al. 2022, *Nat Commun* 13:5093 · Wang et al. 2023, *Nat Commun* 14:7266 · Baek et al. 2024, *Nat Methods* 21:117 (RoseTTAFoldNA) · Leontis & Westhof 2001, *RNA* 7:499 (base-pair classification) · Bernard et al. 2025, *Acta Cryst D* 81 (Has AlphaFold3 achieved success for RNA?).

T.8 RNA design

Talk	Speaker	Len	Link
gRNAd — geometric deep learning for 3D RNA inverse design	Chaitanya Joshi (Cambridge)	1:03:01	▶
3D RNA Design with Deep Learning	Chaitanya Joshi (Eternacon)	26:35	▶
RNA design in theory and practice (inverse folding, 90 min)	Sven Findeiß (Leipzig/Vienna)	1:31:18	▶
RNA Design: From Discrete to Continuous Optimization	Liang Huang	24:56	▶
LinearDesign: optimized mRNA design	Liang Huang	19:13	▶
RNA origami	Ewan McRae (Houston Methodist)	49:18	▶
Crowdsourced design of RNA molecular switches	Stohr & Townley (Eterna)	47:12	▶
Predicting mutations that stabilize the ribosome	Kate Shulgina (Harvard)	28:31	▶
Scalable platforms for RNA sensors and controllers	Christina Smolke (Stanford)	1:02:27	▶
Aptamers and SELEX	Paloma Giangrande (OTS)	31:57	▶

LinearDesign solves joint codon-and-structure optimization as a lattice parsing problem — the mRNA stability question answered by an algorithm rather than by screening. **Findeiß’s ninety minutes is the inverse-folding lecture.**

Papers: Joshi et al. 2025, ICLR (gRNAd) · **Zhang et al. 2023, *Nature* 621:396 (LinearDesign)** · Geary, Rothmund & Andersen 2014, *Science* 345:799 (RNA origami) · **Tuerk & Gold 1990, *Science* 249:505** and Ellington & Szostak 1990, *Nature* 346:818 (SELEX).

T.9 Biophysics — the ion atmosphere

This is the thinnest cluster, and the thinness is itself the finding.

Lois Pollack on time-resolved scattering to watch RNA fold, 53:34 ([▶](#)) · Wide-angle X-ray scattering of structured RNA, including the ill-posedness of 1D → 3D ([▶](#)) · **Bussi’s talk is the closest thing to a dedicated ion-atmosphere lecture that exists** ([▶](#)) · Al-Hashimi on transient and excited states ([▶](#)) · Felix Ritort on single-molecule pulling ([▶](#)).

Papers: Draper 2004, *RNA* 10:335 · Lipfert, Doniach, Das & Herschlag 2014, *Annu Rev Biochem* 83:813 · Liphardt et al. 2001, *Science* 292:733.

T.10 Why RNA structure prediction trails protein structure prediction

The question answered with evidence rather than folklore. Six causes recur, and they are not equally weighted.

1. The training data is roughly a hundredfold smaller, and more redundant than it looks. A direct RCSB PDB query on 2026-10-04 returns **255,456 entries containing protein, 10,475 containing RNA, and only 2,378 that are RNA-only.** That RNA-only set is dominated by redundant tRNAs, ribozymes and riboswitches, so the effective number of independent folds is smaller still. Townshend et al. state the consequence plainly: ARES was “*trained with only 18 known RNA structures,*” and the architectural contribution is framed as a way to work around having no data. **This is why Das’s program is a data-manufacturing program rather than an architecture program.**

2. Deep learning did not win the first fair contest. The CASP15 assessment is explicit: the top three groups “*did not use deep learning,*” and “*predictions from deep learning approaches were significantly worse.*” That is the exact inverse of CASP13 and CASP14 for proteins. **It held in 2026 too** — the Kaggle RNA 3D Folding Challenge was won by template-based methods.

3. The failure is in base-pair geometry, not the global fold — and protein-derived metrics hide this. CASP15 found that models “*correctly predicted the global fold*” while “*challenges remain in modeling fine details such as noncanonical pairs.*” The CASP16 assessment sharpens it: good secondary structure “*is insufficient to guarantee chemical precision or to correctly identify residues involved in non-Watson-Crick interactions.*” Proteins have one backbone torsion problem and twenty side chains. RNA has four bases but **twelve geometric base-pair families** in the Leontis-Westhof classification, a six-torsion backbone, and sugar pucker. GDT and IDDT were built for proteins.

4. RNA is an ensemble, and so is the experimental ground truth. CASP15 lists “*prediction of multiple structures resolved by cryo-EM or crystallography*” as an open problem — the target is not a single structure. PRIME quantifies the floor: base pairs open at 0.5–3 kcal/mol, about half commonly inferred values.

5. The ion atmosphere is a real physical term that essentially no predictor models. RNA is a dense polyanion that cannot fold without counterions, Mg^{2+} especially. Predictors output coordinates in a vacuum while the folded state’s stability depends on an ion distribution they never compute. **Protein folding has no comparable term.**

6. There was no CASP-grade assessment until 2022. RNA-Puzzles launched in 2011 but was smaller and less adversarial. CASP15 was, in the assessors’ words, “*the first CASP exercise that involved RNA structure modeling.*” Protein prediction had fourteen CASP rounds before AlphaFold2. **RNA has had two.**

The synthesis worth carrying. Causes 1 and 6 are contingent and are being fixed — by Ribonanza-style data manufacturing and by CASP’s new RNA category. **Causes 3, 4 and 5 are structural facts about the molecule and will not be fixed by scale alone.** That is the honest read, and it is also the argument for why Das’s data-first strategy is the right bet on the fixable half.

A caution on the general-model claim. AlphaFold3 asserts “*much higher accuracy for protein-nucleic acid interactions compared with nucleic-acid-specific predictors.*” That was tested independently and did not hold cleanly: Bernard et al. 2025 benchmark AF3 across five test sets against ten methods and conclude the limitations for RNA “*are unclear,*” because the AF3 paper reported only the CASP-RNA set. **Capstone IX is this question.**

T.11 BUILD — Atlas T

1. **Implement Nussinov from scratch**, then the Zuker loop model, then McCaskill's partition function. Compare your base-pair probabilities to RNAfold's on the same sequence.
2. **Add SHAPE data as a soft constraint** and re-fold. Measure how much the prediction changes and whether it changes toward the known structure.
3. **Run the ensemble-deconvolution experiment**. Generate a synthetic mixture of two known structures, compute the population-averaged reactivity, then try to recover the two components. This is Aviran's ill-posed inverse problem, and doing it once will permanently change how you read a reactivity profile.
4. **Write the one page**. Given a reactivity vector, what can you and cannot you conclude? Be specific about where the information runs out.

Derivation checkpoints due: 71, 72, 73.

T.12 Paired reading — Atlas T

Watch this	Then read this	Hold this question
Das, <i>BioML Seminar</i>	He et al. 2024 (Ribonanza)	Four passes at one idea. What is the idea?
Das, <i>CASP15 assessment</i>	Das et al. 2023, <i>Proteins</i>	Deep learning lost. What does that tell you about data?
Wild 8-3 and 8-4	Nussinov & Jacobson 1980; Zuker & Stiegler 1981	Why does non-crossing make the decomposition valid?
Hofacker, <i>RNA secondary structures</i>	McCaskill 1990, <i>Biopolymers</i> 29:1105	Why does the partition function need a different recursion?
Lorenz, <i>Improving prediction</i>	Lorenz et al. 2011	What does a soft constraint do to the recursion?
Rivas, <i>Evolutionary conservation</i>	Rivas et al. 2017, <i>Nat Methods</i> 14:45	Covariation or free energy — which is the arbiter?
Huang, <i>Parsing Algorithms for COVID-19</i>	Huang et al. 2019	RNA folding is parsing. What does that buy algorithmically?
Condon, <i>Computational Challenges</i>	Rivas & Eddy 1999, <i>JMB</i> 285:2053	Which pseudoknot classes are tractable, and why does the hierarchy stop?
Courey, <i>Nearest Neighbor Method</i>	SantaLucia 1998, <i>PNAS</i> 95:1460	Compute a T_m by hand. Why is RNA harder than DNA?
Aviran, <i>Sparse Reconstruction</i>	Her papers	One reactivity vector, many populations. What is determined?
Choi, <i>PRIME</i>	The PRIME paper	Base pairs open at half the assumed energy. What breaks?
Townshend, <i>ARES</i>	Townshend et al. 2021, <i>Science</i> 373:1047	Eighteen structures. How did that possibly work?
Ludaic, <i>Limits of deep learning for RNA</i>	Compare to the above	Which of the two is right, and on what evidence?
Bernard, <i>Has AlphaFold3 succeeded for RNA?</i>	Bernard et al. 2025, <i>Acta Cryst D</i> 81	A general model's claim, independently tested. What held?
Bussi, <i>RNA is not alone</i>	Draper 2004; Lipfert et al. 2014	Quantify the ion-atmosphere contribution to stability.
Lilley, <i>How does RNA act like an enzyme?</i>	Lilley 2011, <i>Phil Trans R Soc B</i> 366:2910	General acid-base catalysis without side chains. How?
Pyle, <i>RNA Structure</i> (Part 1)	Leontis & Westhof 2001, <i>RNA</i> 7:499	Learn the twelve base-pair families. They are the vocabulary.
Findeiß, <i>RNA design</i>	Lorenz et al. 2011	Inverse folding against an energy model. Where does it fail?
Huang, <i>LinearDesign</i>	Zhang et al. 2023, <i>Nature</i> 621:396	Codon choice and structure jointly. What made it tractable?

Part III – Upstream AI Research

D.0 Why this part exists

The rest of this document trains you to be excellent at applying artificial intelligence to molecules. This part is about contributing to artificial intelligence itself.

These are different ambitions with different reward structures, and it is worth being honest about the tension before describing the bridge. The AI research community rewards generality, benchmark movement on tasks many people care about, and speed. The molecular community rewards domain insight, experimental validation, and correctness. A person who splits attention evenly between them tends to be out-competed in both: the AI people have read more papers this month than you have, and the biology people have a better wet-lab collaborator.

So the goal here is not to be a part-time AI researcher. It is something more specific and more defensible.

D.1 The bridge thesis

Molecular problems are an unusually good forcing function for genuine AI contributions, because they break assumptions that language and vision let you keep.

This is not a consolation prize. It is the observation that a surprising fraction of core machine learning advances came out of scientific domains, and they came out of those domains precisely because the domain refused to let the researcher take the usual shortcut.

AI contribution	Came from	The assumption the domain refused to allow
Graph neural networks	Chemistry (molecular property prediction)	That inputs have a grid or sequence structure
Equivariant networks, tensor field networks, e3nn	Physics and chemistry	That you can augment your way out of a symmetry
Neural ODEs, continuous-depth models	Dynamical systems	That depth is discrete
Riemannian and discrete flow matching	Geometry and molecular structure	That your data lives in Euclidean space
GFlowNets	Molecule and drug design	That you want the mode rather than a diverse sample from the reward
Boltzmann generators, flow-based samplers	Statistical physics	That generation and sampling a known distribution are different problems
Machine-learned interatomic potentials	Quantum chemistry	That approximate function fitting is good enough
Triangle attention and geometric consistency layers	Structural biology	That attention needs no structural prior
Conformal prediction uptake in science	High-stakes prediction	That a softmax output is a probability
Active learning at scale, closed-loop experimentation	Experimental science	That data is free and i.i.d.

Read that table alongside the one in Section 0.2. The same move appears in both: a constraint that one community treats as an obstacle turns out, carried elsewhere, to be a research program.

The five assumptions molecular work forces you to drop

These are the specific places where your domain hands you an AI research problem that the mainstream does not have to solve.

- 1. Data is not abundant.** Language and vision research operates in a regime where more data is usually available. Protein design operates in a regime where the relevant labeled data is a few thousand examples, expensive, and biased toward what was easy to measure. Everything about low-data learning — inductive bias, transfer, meta-learning, active learning, uncertainty — is *load-bearing* here and merely interesting there.
 - 2. Symmetry is not optional.** You cannot augment your way out of SE(3). A model that assigns different energies to the same molecule in two orientations is not slightly worse; it is wrong. This makes molecular work one of the few areas where equivariant architecture research is forced rather than fashionable.
 - 3. The test distribution is deliberately different from the training distribution.** This is the deepest one. In most of machine learning, distribution shift is a nuisance to be minimized. In design it is *the goal* — you are explicitly asking the model to score molecules that do not exist and were therefore not in training. Almost no mainstream ML method is built for this, and the theory of it is thin. If you want one durable AI research question to own, this is the best candidate in the document.
 - 4. Errors are expensive and uncertainty must be calibrated.** A wrong recommendation costs months of wet-lab time. That makes uncertainty quantification a first-class requirement rather than a reviewer's afterthought, and it is why conformal prediction and proper calibration matter more in your setting than in most.
 - 5. There is a ground truth that is not the dataset.** Physics exists independently of the PDB. This is a luxury other ML fields do not have, and it enables a whole class of method — physics-informed losses, differentiable simulation, hybrid energy-and-learning models, generative models constrained to be physically normalized — that has no analogue in a domain where the only arbiter is held-out data.
-

D.2 What to actually do with this part

Three honest postures, and you should pick one rather than drifting.

Posture A — Informed consumer, deep in one AI sub-area. Follow the field broadly through Section D.5 and the standing archives, but go genuinely deep in exactly one technical area that your molecular work needs: most plausibly generative modeling, equivariant architectures, or uncertainty quantification. Publish AI-venue papers occasionally, molecular papers mainly. This is the highest expected value for most people and the lowest risk.

Posture B — Dual-track researcher. Treat molecular problems as a source of AI research questions and publish into both communities deliberately, with the same work framed differently for each. Requires more time and a tolerance for being slightly foreign in both rooms. The people who pull it off — the geometric deep learning community is full of them — tend to have outsized influence because they are the translation layer.

Posture C — Switch. Decide that core AI is the real interest and molecules were the entry point. Legitimate, common, and worth naming so that you choose it deliberately rather than by drift. If this is the path, the biology becomes a domain you understand unusually well rather than your subject.

The material below serves all three. What differs is depth: Posture A takes Sections D.3 and D.6 seriously and skims the rest; Posture B does all of it; Posture C inverts the weighting of this document.

D.3 How this part is organized

D.4 Vision and perception research. The field where most modern architecture ideas were first tested, and the source of the 3D and generative machinery that molecular work now runs on.

D.5 Deep learning theory. Why any of this works. Generalization, implicit bias, infinite-width limits, scaling laws, and the phenomena — grokking, double descent, edge of stability — that current theory only partly explains.

D.6 Architecture, training and language model research. What actually matters inside a model, how training at scale really works, and the live architecture arguments.

D.7 Interpretability and the science of models. Treating a trained network as an object of empirical study. The fastest-moving area in AI with the most room for a careful newcomer.

D.8 Foundational arguments and research taste. The field's live disagreements, stated by the people having them — scaling versus architecture, whether language models reason, world models, embodiment, the bitter lesson and its critics.

D.9 The bridge, concretely. Specific open AI research problems that your molecular work is unusually well positioned to attack.

D.4 A warning about this part specifically

AI research video has a worse signal-to-noise ratio than anything else in this document. There is an enormous volume of talks that are repackaged press releases, and a genre of confident explainer content that is wrong in ways that are hard to detect without reading the paper.

Three defenses, and they matter more here than in Part II.

Prefer first-author talks at academic venues over keynotes at industry events. The former are accountable to an audience that has read the paper.

Apply the pairing principle without exception. In the molecular sections you can occasionally get away with watching a talk alone. Here you cannot — the gap between what a talk claims and what the paper establishes is systematically wider, because the incentive to overclaim is stronger.

Date everything. A talk from 2021 about what transformers can and cannot do may describe a settled question or a long-refuted guess, and nothing in the video will tell you which. Each entry below carries its date for this reason.

D.2 Foundations — The Courses Everything Else Assumes

Read the caveat before the list. This section is deliberately organized by *course*, not by individual lecture, and that is a departure from the rest of this document. The reason is that these are the few places where watching the whole thing in order is actually correct — a course is a sequenced argument, and cherry-picking lecture 11 of CS236 will not teach you what lecture 11 of CS236 teaches someone who watched the first ten.

Where the Atlas enumerates, this section sequences. Appendix A.7 enumerates the IPD classical course lecture by lecture because you will dip into it. CS231n you should watch.

What to skip. If you already train models daily, skip §1 entirely except Karpathy's *Let's reproduce GPT-2* and the NYU energy-based-model lectures. The sections that will actually add something are §2 (probabilistic ML — the weakest area for most people coming from structural biology), §3 (geometric deep learning, which is the theory behind every equivariant protein model you use), and §4 (diffusion and flow matching derived properly rather than assembled from blog posts).

1. Core deep learning courses with full recorded lectures

MIT 6.S191: Introduction to Deep Learning (2026) | Alexander Amini & Ava Amini, MIT | YouTube + course site | https://www.youtube.com/playlist?list=PLtBw6njQRU-rwp5__7C0oIVt26ZgiG9NI and <https://introtodeeplearning.com/> | 90 videos in the channel playlist across editions; the 2026 edition is 9 lectures of ~45–60 min, including “AI for Science” and “Secrets to Massively Parallel Training” | Fastest route to a working mental model of the whole stack, and the 2026 edition explicitly covers scientific applications and large-scale training.

Stanford CS231n: Convolutional Neural Networks for Visual Recognition (Spring 2017) | Fei-Fei Li, Justin Johnson, Serena Yeung | YouTube | <https://www.youtube.com/playlist?list=PLC1qU-LWwrF64f4QKQT-Vg5Wr4qEE1Zxk> | 16 lectures, ~1h15 each | Still the clearest derivation of backprop, conv architectures and training dynamics, which is the substrate every structure-prediction network sits on.

Stanford CS231N: Deep Learning for Computer Vision (Spring 2025) | Stanford Online | YouTube | <https://www.youtube.com/playlist?list=PLoROMvodv4rOmsNzYBMe0gJY2XS8AQg16> | 18 lectures | The modernized version, with lectures on transformers, generative models and large-scale distributed training that the 2017 course predates.

Stanford CS224N: NLP with Deep Learning (Spring 2024) | Christopher Manning, Stanford | YouTube | <https://www.youtube.com/playlist?list=PLoROMvodv4rOaMFbaqxPDOLWjDaRAp9D> | 23 videos, ~1h20 each | Sequence modeling, attention and pretraining taught properly; protein language models are this machinery applied to a 20-letter alphabet. (2023 edition also live: <https://www.youtube.com/playlist?list=PLoROMvodv4rMFqRtEuo6SGjY4XbRIVRd4>)

Deep Learning Course (NYU, Spring 2020) | Yann LeCun & Alfredo Canziani, NYU | YouTube | <https://www.youtube.com/playlist?list=PL80I41oVxglKcAHllsU0txr3OuTTaWX2v> | 32 videos | LeCun's energy-

based-model framing of learning is unusually well-suited to thinking about protein conformational landscapes. Later editions with full notes: <https://atcold.github.io/NYU-DLSP21/> and <https://atcold.github.io/NYU-DLFL22/>

Practical Deep Learning for Coders (Part 1) | Jeremy Howard, fast.ai | course.fast.ai | <https://course.fast.ai/> | 9 lessons, ~90 min each | Top-down, code-first counterweight to the theory courses; gets you shipping experiments quickly.

Practical Deep Learning for Coders Part 2: Deep Learning Foundations to Stable Diffusion | Jeremy Howard, fast.ai | course.fast.ai | <https://course.fast.ai/Lessons/part2.html> | ~30 hours | Builds a diffusion model from scratch line by line, which is exactly the skill needed to modify RFDiffusion-class models rather than just call them.

Neural Networks: Zero to Hero | Andrej Karpathy | YouTube | <https://www.youtube.com/playlist?list=PLAqIrjkbxWI23v9cThsA9GvCAUhRvKZ> and <https://karpathy.ai/zero-to-hero.html> | 10 videos, 1–4 h each | Builds autograd and a transformer from nothing; the single best preparation for writing a novel architecture instead of fine-tuning someone else's.

Let's build GPT: from scratch, in code, spelled out | Andrej Karpathy | YouTube | <https://www.youtube.com/watch?v=kCc8FmEb1nY> | 1h56 | The attention mechanism derived in code, which demystifies every pair-representation and triangle-attention block in structure models.

Let's reproduce GPT-2 (124M) | Andrej Karpathy | YouTube | <https://www.youtube.com/watch?v=l8pRSuU81PU> | 4h01 | Covers the engineering of a real training run end to end: mixed precision, throughput, LR schedules, gradient accumulation.

Deep Learning for Computer Vision (EECS 498-007 / 598-005) | Justin Johnson, University of Michigan | YouTube | <https://www.youtube.com/playlist?list=PL5-TkQAFzFbzxjBHtzdVCWE0Zbhomg7r> | 22 lectures, ~1h15 each | Johnson's rewrite of CS231n with better coverage of 3D vision and generative models; the 3D lectures transfer directly to coordinate-based protein models.

Neural networks (3Blue1Brown) | Grant Sanderson | YouTube | https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi | 10 videos | Geometric intuition for backprop and attention; worth watching before the formal courses.

DeepMind x UCL Deep Learning Lecture Series 2021 | DeepMind researchers / UCL | YouTube | <https://www.youtube.com/playlist?list=PLqYmG7hTraZDVH599EItIEWsUOsJbAodm> | 13 lectures | Research-level survey from people building the models, including generative modeling and attention lectures.

2. Probabilistic ML and Bayesian methods

Probabilistic Machine Learning 2025 | Philipp Hennig, University of Tübingen | YouTube | <https://www.youtube.com/playlist?list=PL05umP7R6ij0hPfU7Yuz8J9WXjlb3MFjm> | 25 lectures, ~1h20 each | The most rigorous free treatment of Gaussian processes, inference and uncertainty; essential if you want calibrated confidence on designed sequences rather than bare point predictions. Earlier editions: 2023 https://www.youtube.com/playlist?list=PL05umP7R6ij2YE8rRJSb-olDNbntAQ_Bx and 2021 (29 lectures) <https://www.youtube.com/playlist?list=PL05umP7R6ij1tHaOFY96m5uX3J21a6yNd>

Numerics of Machine Learning (Winter 2022/23) | Philipp Hennig and colleagues, Tübingen | YouTube | <https://www.youtube.com/playlist?list=PL05umP7R6ij2lwDdj7IkuHoP9vHlEch0s> | 14 lectures, ~1h20 each | Covers ODE/PDE solvers, Monte Carlo, Bayesian quadrature and uncertainty in deep learning — the numerical toolkit behind diffusion samplers and MD-coupled models.

Stanford CS236: Deep Generative Models (2023) | Stefano Ermon, Stanford | YouTube (Stanford Online) | <https://www.youtube.com/playlist?list=PLoROMvodv4rPOWA-omMM6STXaWW4FvJT8> | 18 lectures, ~1h20 each | The canonical unified treatment of autoregressive models, VAEs, normalizing flows, EBMs, GANs and score-based diffusion; this is the taxonomy you need to place any new protein generative model.

Variational Inference: Foundations and Modern Methods (NIPS 2016 tutorial) | David Blei, Rajesh Ranganath, Shakir Mohamed | YouTube | https://www.youtube.com/watch?v=ogdv_6dbvVQ | 1h53 | The reference tutorial for the ELBO, reparameterization and black-box VI, which underpins every latent-variable protein model.

The Four Pillars of Machine Learning (Distinguished Colloquium) | Kevin Murphy, Google DeepMind | YouTube (Princeton CS) | <https://www.youtube.com/watch?v=uhcw5rvqqE> | 1h04 | Murphy's own unifying frame for his two-volume *Probabilistic Machine Learning* books; a fast map of the territory the books cover in 2,000 pages.

Gaussian Process Summer School 2024 | Neil Lawrence and colleagues, GPSS | YouTube | https://www.youtube.com/playlist?list=PLZ_xn3EIbXZEoWLm9y6OizFkontrhA6G | 8 lectures, ~1h30 each | Deep dive on GPs, deep GPs and multi-task GPs — the surrogate models that drive Bayesian optimization over sequence space.

Math for Deep Learning (MaDL) | Andreas Geiger, University of Tübingen | YouTube | <https://www.youtube.com/playlist?list=PL05umP7R6ij0bo4UtMdZtEJ6TiLOqj4ZCm> | 18 lectures | Guided tour of the linear algebra and probability theory assumed by everything above; useful remediation if the Hennig lectures feel steep.

3. Geometric deep learning and equivariance

AMMI Geometric Deep Learning Course, Second Edition (2022) | Michael Bronstein (Oxford), Joan Bruna (NYU), Taco Cohen (Qualcomm), Petar Veličković (DeepMind) | YouTube | https://www.youtube.com/playlist?list=PLn2-dEmQeTfSLXW8yXP4q_Ii58wFdx3C | 18 videos (12 lectures + tutorials + seminars), ~1h15 each | The “Erlangen programme” framing of grids, groups, graphs, geodesics and gauges is the single most useful theoretical lens for inventing new protein and RNA architectures.

AMMI Geometric Deep Learning Course, First Edition (2021) | same four instructors | YouTube | <https://www.youtube.com/playlist?list=PLn2-dEmQeTfQ8YVUHB0vAhUlnIPYxkeu3> | 13 lectures | Different emphasis and pacing from the 2022 edition; Cohen's geometric-priors lectures are especially good here. Course hub with slides: <https://geometricdeeplearning.com/lectures/>

Group Equivariant Deep Learning (UvA, 2022) | Erik Bekkers, University of Amsterdam | YouTube | <https://www.youtube.com/playlist?list=PL8FnQMH2k7jzPrxqdYuf0iYVHim8PyZWd> | 21 videos, 10–50 min each | The most constructive course on building G-CNNs and steerable/3D-equivariant networks, including 3D steerable graph NNs — this is the mathematics behind e3nn, NequIP and SE(3)-Transformers.

An Orientation in Symmetry-Aware ML Methods | Tess Smidt, MIT (Materials Project Seminars) | YouTube | <https://www.youtube.com/watch?v=R3N6BocbknM> | 1h01 | Smidt's own taxonomy of how to choose among invariant, equivariant and data-augmented approaches; directly applicable to deciding what symmetry your protein model should enforce.

Applications of Euclidean neural networks to understand and design atomistic systems | Tess Smidt, MIT (Harvard CMSA) | YouTube | <https://www.youtube.com/watch?v=Iah-YIFdmbs> | 55 min | Current state of e3nn-

based design on real atomistic systems, including symmetry breaking.

Symmetry-Aware Neural Networks for the Material Sciences with e3nn (MRS 2021 Fall tutorial) | Tess Smidt, Mario Geiger and others | YouTube | <https://www.youtube.com/watch?v=q9EwZsHY1sk> | 6-part tutorial, ~45 min for part 1 | Hands-on introduction to the e3nn library itself. Companion material: https://blondegeek.github.io/e3nn_tutorial/ and <https://e3nn.org/>

Symmetries in Inference and Learning | Max Welling, University of Amsterdam (TUM AI Lecture Series) | YouTube | https://www.youtube.com/watch?v=YihnfamwA_s | 57 min | Welling's argument for why equivariance buys data efficiency — the core justification for using it in low-data protein regimes.

Equivariant Networks (NeurIPS 2020 tutorial) | Taco Cohen & Risi Kondor | neurips.cc | <https://neurips.cc/virtual/2020/tutorial/16650> | ~2h | Cohen and Kondor's joint treatment of equivariant convolutions and their representation theory, from the two people who formalized the field.

Stanford CS224W: Machine Learning with Graphs | Jure Leskovec, Stanford | YouTube | <https://www.youtube.com/playlist?list=PLoROMvodv4rPLKxIpqhjhPgdy7imNkDn> | 60 videos, ~1h each | Proteins and RNA are graphs before they are coordinates; this is the most complete free GNN course. Site: <https://cs224w.stanford.edu/>

4. Diffusion models and flow matching

MIT 6.S184: Generative AI with Stochastic Differential Equations / Flow Matching and Diffusion Models (2025) | Peter Holderrieth & Ezra Erives, MIT CSAIL | YouTube + course site | <https://www.youtube.com/playlist?list=PL57nT7tSGAAUDnli1LhTOoCxlEPGS19vH> and <https://diffusion.csail.mit.edu/> | 6 lectures plus full lecture notes and 3 labs | Derives diffusion and flow matching from the SDE/ODE perspective and ends on molecular design applications — the most direct bridge from theory to protein generative models.

Flow Matching for Generative Modeling (NeurIPS 2024 tutorial) | Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu (Meta FAIR) | neurips.cc / SlidesLive | <https://neurips.cc/virtual/2024/tutorial/99531> and <https://slideslive.com/39031675/flow-matching-for-generative-modeling> | ~2h | Includes the non-Euclidean (Riemannian) and discrete generalizations that matter for backbone frames and sequences. Companion library and docs: https://facebookresearch.github.io/flow_matching/

Denoising Diffusion-based Generative Modeling: Foundations and Applications (CVPR 2022 tutorial) | Arash Vahdat, Karsten Kreis, Ruiqi Gao (NVIDIA / Google) | YouTube | <https://www.youtube.com/watch?v=cS6JQpEY9cs> | 3h46 | The most thorough single-sitting derivation of DDPM, score matching and the SDE/ODE unification.

Denoising Diffusion Models: A Generative Learning Big Bang (CVPR 2023 tutorial) | Jiaming Song, Chenlin Meng, Arash Vahdat | YouTube (CVF) | <https://www.youtube.com/watch?v=1d4r19GEVos> | 3h04 | The follow-up, covering fast samplers, guidance and conditioning — the levers you pull when conditioning a backbone generator on a binding site.

Diffusion and Score-Based Generative Models | Yang Song, Stanford (MIT CBMM) | YouTube | <https://www.youtube.com/watch?v=wMmqCMwuM2Q> | 1h32 | Song presenting his own score-SDE framework, explicitly motivated by molecular structure generation.

A personal journey to diffusion models (ICBS 2025) | Yang Song | YouTube (BIMSA) | <https://www.youtube.com/watch?v=s647Nz0r4r0> | 1h03 | How the ideas actually developed, including dead

ends — unusually instructive if your goal is to invent methods rather than apply them.

Diffusion models for image and video generation (ML in PL 2025) | Sander Dieleman, Google DeepMind | YouTube | <https://www.youtube.com/watch?v=qFIT3mSwWEk> | 51 min | Practitioner's view of latent diffusion: autoencoders, noise schedules, guidance, distillation — the design decisions papers usually omit.

Hugging Face Diffusion Models Course | Hugging Face | huggingface.co | <https://huggingface.co/learn/diffusion-course/en/unit0/1> | 4 units, theory plus 2 notebooks each | Hands-on `diffusers` training, fine-tuning, guidance and custom pipelines; the fastest path from theory to a model you have actually trained.

Stanford CME296: Diffusion & Large Vision Models | Stanford Online | YouTube | <https://www.youtube.com/playlist?list=PLoROMvodv4rNdy8rt2rZ4T2xM0OjADnFu> | 8 lectures | Recent course covering diffusion foundations, score matching and flow matching together in one modern syllabus.

5. Transformers and attention in depth

Stanford CS25: Transformers United (V1–V6) | Stanford, rotating guest researchers | YouTube | https://www.youtube.com/playlist?list=PLoROMvodv4rNiJRchCzutFw5ItR_Z27CM and <https://web.stanford.edu/class/cs25/> | 50 videos, ~1h15 each | Seminar series with the authors of the architectures themselves, including sessions on biology applications; the best single feed for what is actually new.

Stanford CS25 V6: Overview of Transformers | Stanford Online | YouTube | <https://www.youtube.com/watch?v=bHSDPgZYie0> | 1h16 | The 2026 refresher lecture; the right single video if you want current framing rather than the 2017 one.

Stanford CME295: Transformers and Large Language Models (Autumn 2025) | Afshine & Shervine Amidi, Stanford | YouTube | <https://www.youtube.com/playlist?list=PLoROMvodv4rOCXd21gf0CF4xr35yINeOy> | 9 lectures, ~1h40 each | A structured course rather than a seminar: attention internals, tokenization, positional encodings, efficiency.

Structured State Space Models for Deep Sequence Modeling | Albert Gu, CMU (LxMLS) | YouTube | <https://www.youtube.com/watch?v=WC9tqkCpq4s> | 1h34 | Gu's full derivation of S4/Mamba from first principles — relevant because long-sequence linear-time models are an open frontier for whole-genome and long-RNA modeling.

On the Tradeoffs of State Space Models | Albert Gu, CMU (Simons Institute) | YouTube | https://www.youtube.com/watch?v=ksRp_DIHWj4 | 49 min | The honest comparison of where SSMs beat attention and where they do not.

Stanford CS25 V6: On the Tradeoffs of State Space Models and Transformers | Stanford Online | YouTube | <https://www.youtube.com/watch?v=OyimE74UMF8> | 1h17 | The seminar-length version of the same argument, more recent.

6. Reinforcement learning and optimization for design

DeepMind x UCL: Introduction to Reinforcement Learning (2015) | David Silver, DeepMind / UCL | YouTube | <https://www.youtube.com/playlist?list=PLqYmG7hTraZDM-OYHWgPebj2MfCFzFObQ> | 10 lectures, ~1h30

each | Still the clearest exposition of MDPs, value functions and policy gradients; the vocabulary for any RL-driven sequence design loop.

DeepMind x UCL RL Lecture Series (2021) | Hado van Hasselt, Diana Borsa, Matteo Hessel, DeepMind | YouTube | <https://www.youtube.com/watch?v=TCCjZe0y4Qc> | 13 lectures, ~1h30–2h each | The modern replacement for the 2015 course, with deep RL and function approximation treated properly.

CS 185/285: Deep Reinforcement Learning (Spring 2026) | Sergey Levine, UC Berkeley | YouTube | <https://www.youtube.com/playlist?list=PLKq1TCpsv3Y4> and <https://rail.eecs.berkeley.edu/deeprlcourse/> | 27 videos, ~1h00–1h56 each | The graduate standard, including offline RL and model-based RL — directly relevant to learning design policies from fixed experimental datasets.

Stanford CS224R: Deep Reinforcement Learning | Chelsea Finn, Stanford | YouTube | https://www.youtube.com/playlist?list=PLoROMvodv4rPwxEOONYRa_itZFdaKCylL | 19 lectures | Strong complement to CS285 with more meta-learning and few-shot adaptation, which maps onto low-data protein design.

INFORMS TutORial: Bayesian Optimization | Peter Frazier, Cornell | YouTube | https://www.youtube.com/watch?v=c4KKvyWW_Xk | 1h27 | GP regression plus expected improvement, entropy search and knowledge gradient — the exact machinery behind active-learning loops that pick which variants to assay next. Written companion: [arXiv:1807.02811](https://arxiv.org/abs/1807.02811).

(See also Gaussian Process Summer School 2024 under section 2, which supplies the surrogate-model theory this tutorial assumes.)

7. ML systems practicalities

Stanford CS336: Language Modeling from Scratch (Spring 2025) | Percy Liang & Tatsunori Hashimoto, Stanford | YouTube | https://www.youtube.com/playlist?list=PLoROMvodv4rOY23Y0BoGoBGgQ1zmU_MT_ and <https://stanford-cs336.github.io/spring2025/> | 17 lectures, ~22 hours total | Includes GPU programming, Triton kernels, and data/tensor/pipeline parallelism; the best free course on actually scaling a model you wrote yourself.

Stanford CS336 (Spring 2026) | same instructors | YouTube | <https://www.youtube.com/playlist?list=PLoROMvodv4rMqXOcazWaTUhq-yemBLCV> | 18 lectures | The updated edition.

MIT 6.5940: TinyML and Efficient Deep Learning Computing (EfficientML.ai, Fall 2023) | Song Han, MIT | YouTube | <https://www.youtube.com/playlist?list=PL80kAHvQbh-pT4lCkDT53zT8DKmhE0idB> and <https://efficientml.ai/> | 46 videos, ~1h10 each | Pruning, quantization, NAS, distributed training and on-device training; how to make a large structure model fit and run on the hardware you actually have. Fall 2026 edition in progress: <https://www.youtube.com/playlist?list=PLH7PIPKvCm38>

Stanford MLSys Seminars | Stanford, rotating industry and academic speakers | YouTube | https://www.youtube.com/playlist?list=PLSrTvUm384I9PV10koj_cqit9OfbJXEkq | 100 episodes, ~1h each | Covers automated model parallelism (Alpa), distributed ML, deployment and monitoring — the practical counterpart to the research courses.

Stanford CS231N Spring 2025, Lecture 11: Large Scale Distributed Training | Stanford Online | YouTube | <https://www.youtube.com/watch?v=9MvD-XsowsE> | 1h12 | A single self-contained lecture on data/model/pipeline parallelism if you do not want the full CS336 commitment.

Intro to JAX: Accelerating Machine Learning research | Google / DeepMind | YouTube | <https://www.youtube.com/watch?v=WdTeDXsOSj4> | 10 min | `jit`, `grad`, `vmap`, `pmap` in one sitting; AlphaFold, Boltz-adjacent and many equivariant codebases are JAX-native.

Keynote: PyTorch Technical Deep Dive | Alban Desmaison, Peng Wu, Mark Saroufim, Edward Yang (Meta), PyTorch Conference | YouTube | <https://www.youtube.com/watch?v=XdORM2pkyH8> | 43 min | Current internals from the core developers: compile stack, autograd, distributed — what you need when your custom equivariant op is slow.

Flagged as unverifiable

UNVERIFIED — search term: “Stanford CS329S Machine Learning Systems Design lecture videos public” — The course site (<https://stanford-cs329s.github.io/>) is live and all slides and lecture notes are public, but recordings were released only on Canvas to enrolled students, and no public video set exists. Only scattered guest-lecture clips and the 2022 Demo Day are on YouTube. The content was expanded into Chip Huyen’s book *Designing Machine Learning Systems* (O’Reilly, 2022). Use the Stanford MLSys Seminars playlist above as the video substitute.

Two notes on what I could not do. My web-search budget ran out partway through, so the second half of the discovery work was done by fetching YouTube’s search and channel pages and course websites directly with curl, which is actually stricter verification than search snippets since every URL above returned HTTP 200 with the title and video count I have quoted. And I deliberately dropped several third-party re-uploads of CS236 and David Silver that ranked highly in search, in favor of the official Stanford Online and Google DeepMind playlists, because re-uploads get taken down.

D.3 Computer Vision

Why vision, for someone who works on molecules. Because vision is where the ideas are invented and biology is where they arrive two years later, and you would rather be at the first place.

The list is not short: convolutional architectures, batch normalization, residual connections, data augmentation, self-supervised pretraining, vision transformers, diffusion, classifier-free guidance, flow matching, neural fields. **Every one of those was developed on images and then imported into molecular modeling largely unchanged.** RFdiffusion is a denoising diffusion model with SE(3) frames. Chroma's conditioners are classifier guidance. The self-supervised objectives in protein language models are masked modeling from BERT by way of MAE.

Section 6 is the one to read if you read only one. 3D vision — neural radiance fields, signed distance functions, point-cloud networks, coordinate based representations — is the part of vision whose objects are closest to yours. A protein is a set of atoms in $\mathbb{R}^3$ with a learned field over it, which is what a NeRF is, and the two literatures have barely spoken.

The honest warning. Vision has far more hype per unit result than structural biology, and far more papers. The entries below were chosen for talks by the people who did the work, not by explainers, and the gaps where no author talk exists are flagged rather than filled with secondhand summaries.

1. Modern vision architectures — ViT, ConvNeXt, Swin, the convolution debate

#	Title	Speaker / Institution	URL	Dur.	Why it matters	Paper
1	Industry talk — Google AI Brain	Alexey Dosovitskiy (Google Brain), GCPR/VMV/VCBM 2020	https://www.youtube.com/watch?v=KxFF03EI9M	36:03	The ViT first author speaking the same season ViT was submitted — you hear the “can we delete the convolution entirely” bet being placed before it was a consensus	Dosovitskiy et al., <i>An Image is Worth 16×16 Words</i> , ICLR 2021
2	Alexey Dosovitskiy — From pixels to nucleotides	Alexey Dosovitskiy (Inceptive), ML in PL 2025	https://www.youtube.com/watch?v=ZjROgep6jv0	51:51	The ViT author explaining how vision architecture thinking transfers to biological sequence/structure — directly relevant if your target domain is molecular	ViT (ICLR 2021) + Inceptive RNA work
3	Lucas Beyer — Computer Vision in the Age of LLMs	Lucas Beyer (Google DeepMind), ML in PL 2024	https://www.youtube.com/watch?v=kxO6ARgI_SU	49:53	The single best “where did vision architecture research actually land” talk; Beyer co-authored ViT scaling, SigLIP and PaliGemma	Zhai et al., <i>Scaling Vision Transformers</i> , CVPR 2022
4	Lucas Beyer: Vision in the Age of LLMs [ETHZ Robot Learning 2026]	Lucas Beyer (OpenAI, ex-DeepMind), ETH Zürich	https://www.youtube.com/watch?v=0XB7fNS_ONg	1:09:04	The 2026 update of the above — the most current authoritative read on whether vision encoders still matter	—
5	Lucas Beyer Learning General Visual Representations	Lucas Beyer , London ML Meetup	https://www.youtube.com/watch?v=X5Rhm__OxvA	1:03:56	BiT → ViT → transfer-learning methodology, told as a research program rather than a paper	Kolesnikov et al., <i>Big Transfer (BiT)</i> , ECCV 2020
6	Lucas Beyer (Google DeepMind) — Convergence of Vision & Language	Lucas Beyer , The AI Epiphany	https://www.youtube.com/watch?v=en1Ha3tw6d4	55:08	Why the two modalities converged on one architecture — the core inductive-bias argument	ViT; LiT (Zhai et al., CVPR 2022)
7	Architectures Beyond CNNs and Visual Scaling Laws — Tutorial (1/3)	Neil Houlsby (Google Brain), ECCV 2022 CVinW Workshop	https://www.youtube.com/watch?v=JHsFUhGPff4	30:44	ViT co-author giving the explicit “do we still need convolutions” verdict plus visual scaling laws	Zhai et al., <i>Scaling ViT</i> , CVPR 2022; Tolstikhin et al., <i>MLP-Mixer</i> , NeurIPS 2021
8	Cambridge Ellis Unit Seminar — Neil Houlsby (29 Oct 2021)	Neil Houlsby (Google Brain)	https://www.youtube.com/watch?v=YUDR5Oq04Vg	56:25	Longer-form version with Q&A; adapters/parameter-	Houlsby et al., <i>Parameter-Efficient Transfer</i>

#	Title	Speaker / Institution	URL	Dur.	Why it matters	Paper
					efficient transfer lineage included	<i>Learning</i> , ICML 2019
9	Scaling Vision and Language Learning with Vision Transformers — Tutorial (2/3)	Xiaohua Zhai (Google Brain), ECCV 2022 CVinW	https://www.youtube.com/watch?v=BO4iossuTIs	31:32	The ViT-scaling-laws author on compute/data/parameter tradeoffs — the empirical backbone of modern vision scaling	Zhai et al., CVPR 2022; Dehghani et al., <i>ViT-22B</i> , ICML 2023
10	Scalable approaches to localization & dense prediction — Tutorial (3/3)	Matthias Minderer (Google Brain), ECCV 2022 CVinW	https://www.youtube.com/watch?v=p6x9XDDe440	31:58	Shows that the ViT recipe extends to detection without hand-designed priors	Minderer et al., <i>OWL-ViT</i> , ECCV 2022
11	New vision architectures beyond CNNs	Alexander Kolesnikov (Google Brain), IARAI Research	https://www.youtube.com/watch?v=kD3LqIFzzY8	1:18:41	The deepest free lecture on the ViT/MLP-Mixer/convolution tradeoff space, by a co-author of all three	Tolstikhin et al., <i>MLP-Mixer</i> , NeurIPS 2021; BiT, ECCV 2020
12	A ConvNet for the 2020s	Zhuang Liu (Meta AI/FAIR), CVPR 2022 author video	https://www.youtube.com/watch?v=QzCjXqFnWPE	4:54	The rebuttal to ViT, from the first author — the methodological lesson (ablate training recipe before crediting architecture) is the real content	Liu, Mao, Wu, Feichtenhofer, Darrell, Xie, CVPR 2022
13	ConvNeXt V2: Co-designing and Scaling ConvNets with Masked Autoencoders CVPR 2023	Sanghyun Woo (KAIST/Meta AI), CVPR 2023 author video	https://www.youtube.com/watch?v=wXuC7iDZI2M	7:58	Shows masked pretraining is architecture-agnostic — kills the “MAE needs transformers” claim	Woo et al., <i>ConvNeXt V2</i> , CVPR 2023
14	Stanford CS25: V1 — Transformers in Vision: Tackling problems in Computer Vision	Stanford CS25, Stanford Online	https://www.youtube.com/watch?v=BP5CM0YxbP8	1:08:37	Graduate seminar framing of the full ViT-era landscape	ViT; DETR; Swin
15	CV Study Group: Swin Transformer	Hugging Face CV Study Group	https://www.youtube.com/watch?v=Ngikt-K1Ecc	46:05	Careful walkthrough of hierarchical/shifted-window attention — the argument that <i>some</i> locality prior is worth re-introducing	Liu et al., <i>Swin Transformer</i> , ICCV 2021 (Marr Prize)
16	Towards Generic Vision Transformers for Supervised	Computer Vision Talks reading group (<i>speaker not stated in title</i>)	https://www.youtube.com/watch?v=fk-6JdRjLPw	1:18:42	Reading-group depth on unifying supervised/SSL ViT training	ViT; DeiT (Touvron et al., ICML 2021)

#	Title	Speaker / Institution	URL	Dur.	Why it matters	Paper
	and Self-Supervised Representation Learning					
17	Large Scale Visual Representation Learning	Computer Vision Talks reading group (<i>speaker not stated in title</i>)	https://www.youtube.com/watch?v=lZcCT2JTlb8	1:16:09	Scale-first representation learning, pre-foundation-model framing	BiT; ViT
18	【EP3】 Large-Scale Visual Representation Learning with Vision Transformers	The AI Talks	https://www.youtube.com/watch?v=ucoNlnV9R0Q	1:03:21	Companion long-form treatment of ViT-scale representation learning	Zhai et al., CVPR 2022

2. Self-supervised visual learning — contrastive, clustering, masked, JEPA

#	Title	Speaker / Institution	URL	Dur.	Why it matters	Paper
19	Contrastive Self-Supervised Learning and Potential Limitations	Ting Chen (Google Brain), ELLIS UCL CSML Seminar	https://www.youtube.com/watch?v=IEiytaXnggI	55:12	SimCLR's first author on where contrastive learning <i>fails</i> — the most research-useful framing available	Chen, Kornblith, Norouzi, Hinton, <i>SimCLR</i> , ICML 2020
20	Contrastive Learning with SimCLR V1/V2 and Some Intriguing Properties	Computer Vision Talks reading group	https://www.youtube.com/watch?v=-6jb1v1v0vc	1:02:34	Covers the semi-supervised distillation result of SimCLRv2, usually skipped	Chen et al., <i>SimCLRv2</i> , NeurIPS 2020
21	SimCLR: A Simple Framework for Contrastive Learning of Visual Representations	Stanford Contrastive & SS Learning Group (<i>embedding disabled; public</i>)	https://www.youtube.com/watch?v=wySLC4nszv8	36:46	Stanford reading group — the ablation-by-ablation read	Chen et al., ICML 2020
22	Momentum Contrast for Unsupervised Visual Representation Learning	Kaiming He / Haoqi Fan / Yuxin Wu / Saining Xie / Ross Girshick (FAIR), CVPR 2020 author video	https://www.youtube.com/watch?v=4VVGtYPM8JE	4:53	Canonical author presentation of MoCo — the queue + momentum-encoder idea that decoupled batch size from negatives	He et al., <i>MoCo</i> , CVPR 2020
23	Exploring Simple Siamese Representation Learning and Beyond	Computer Vision Talks reading group (SimSiam)	https://www.youtube.com/watch?v=icqUfBzklkc	1:14:14	The collapse-avoidance question — why stop-gradient alone works, and what that means theoretically. Central to understanding BYOL too	Chen & He, <i>SimSiam</i> , CVPR 2021; Grill et al., <i>BYOL</i> , NeurIPS 2020
24	An Empirical Study of Training Self-Supervised Vision Transformers	Xinlei Chen (FAIR), ICCV 2021 author video	https://www.youtube.com/watch?v=LHhu11kOA-Y	10:11	MoCo v3 — the instability analysis of SSL ViT training is a model of empirical rigor	Chen, Xie, He, ICCV 2021
25	Xinlei Chen: Self-supervised learning: two known paradigms and a less-known observation	Xinlei Chen (FAIR/Meta)	https://www.youtube.com/watch?v=Ox_L-c2oIpY	44:26	The clearest contrastive-vs-masked synthesis by someone who co-authored on both sides	MoCo; SimSiam; MAE

#	Title	Speaker / Institution	URL	Dur.	Why it matters	Paper
26	Self-Supervised Learning of Visual Representations with Online Clustering (SwAV)	Computer Vision Talks reading group	https://www.youtube.com/watch?v=zffVe7TTx2I	45:31	Clustering/prototypes as a third paradigm beside contrastive and masked	Caron et al., SwAV, NeurIPS 2020
27	Self-Supervised Learning of Image Features with SwAV (with author Mathilde Caron)	Mathilde Caron (FAIR/Inria), Lightning AI	https://www.youtube.com/watch?v=7QmsTleiRLs	36:45	The SwAV/DINO author in conversation — design rationale and failure modes	Caron et al., NeurIPS 2020
28	Mathilde Caron — VGL group seminar, 27 May 2021	Mathilde Caron (FAIR/Inria), Vision Graphics & Learning Group, York	https://www.youtube.com/watch?v=rKF9WSNQ_BA	57:53	Delivered the month DINO appeared; the emergent-segmentation finding presented by its author	Caron et al., <i>DINO</i> , ICCV 2021
29	DINO: Emerging Properties in Self-Supervised Vision Transformers	Stanford Contrastive & SS Learning Group	https://www.youtube.com/watch?v=Y3XsNNlhrik	52:32	Paper-depth reading-group treatment of DINO’s attention maps and k-NN evaluation	Caron et al., ICCV 2021; Oquab et al., <i>DINOv2</i> , TMLR 2024
30	Masked Autoencoders Are Scalable Vision Learners	Xinlei Chen (FAIR), CVPR 2022 author video	https://www.youtube.com/watch?v=weokLYcY6xk	4:58	The MAE author video — asymmetric encoder/decoder + 75% masking, stated by the people who found it	He, Chen, Xie, Li, Dollár, Girshick, CVPR 2022
31	Deep Learning Bootcamp: Kaiming He	Kaiming He (MIT EECS), MIT Schwarzman College of Computing (<i>embedding disabled; public</i>)	https://www.youtube.com/watch?v=D_jt-xO_RmI	1:15:46	Kaiming He’s own survey of his line of work — ResNet → MoCo → MAE — framed as research strategy, not results	He et al., ResNet CVPR 2016; MoCo CVPR 2020; MAE CVPR 2022
32	Multi-view Invariance and Grouping for Self-Supervised Learning	Ishan Misra (FAIR), Oxford VGG	https://www.youtube.com/watch?v=gbziPln9uDI	36:32	The “what invariance are we actually imposing” question — the conceptual core of SSL pretext design	Misra & van der Maaten, <i>PIRL</i> , CVPR 2020
33	Self-supervised learning for images, video, and 3D	Ishan Misra (Meta AI), GHOST Day AMLC 2022	https://www.youtube.com/watch?v=YpETv9AzW48	1:04:11	Cross-modality generalization of SSL — the bridge to	Omnivore (CVPR 2022); data2vec

#	Title	Speaker / Institution	URL	Dur.	Why it matters	Paper
					3D/molecular settings	
34	General purpose visual recognition systems: beyond a single modality and a task	Ishan Misra (Meta AI), ECCV 2022 CVinW	https://www.youtube.com/watch?v=TIXAgtoUcN8	29:03	Where SSL meets multi-task/multi-modal foundation models	Omnivore; OmniMAE
35	Broaden Your Views for Self-Supervised Video Learning	Computer Vision Talks reading group (BraVe)	https://www.youtube.com/watch?v=BotznJtzMn0	1:04:47	Temporal SSL — narrow-view/broad-view asymmetry, a design pattern that recurs in V-JEPA	Recasens et al., <i>BraVe</i> , ICCV 2021
36	Yann LeCun Self-Supervised Learning, JEPA, World Models, and the future of AI	Yann LeCun (Meta/NYU), Harvard CMSA	https://www.youtube.com/watch?v=yUmDRxV0krq	1:08:37	The primary-source statement of the joint-embedding-predictive thesis and the argument against generative pixel prediction	LeCun, <i>A Path Towards Autonomous Machine Intelligence</i> (2022); Assran et al., <i>I-JEPA</i> , CVPR 2023
37	Yann LeCun: World Models: Enabling the next AI revolution	Yann LeCun , Computer Vision and Geometry Group, ETH Zürich	https://www.youtube.com/watch?v=72Xj8k5WQX4	58:54	The most recent (2026) LeCun JEPA/world-model talk to a vision audience	Bardes et al., <i>V-JEPA</i> (2024); <i>V-JEPA 2</i> (2025)

3. Vision-language and multimodal

#	Title	Speaker / Institution	URL	Dur.	Why it matters	Paper
38	Alec Radford, OpenAI: CLIP — Learning Transferable Visual Models From Natural Language Supervision	Alec Radford (OpenAI) (<i>re-uploaded on channel “feather”</i>)	https://www.youtube.com/watch?v=3X3EY2Fgp3g	49:54	The single most important vision-language talk available free. Radford on why language supervision beats label supervision, and the zero-shot evaluation philosophy	Radford et al., <i>CLIP</i> , ICML 2021
39	ALIGN: Scaling Up Visual and Vision-Language Representation Learning With Noisy Text Supervision	Stanford Contrastive & SS Learning Group	https://www.youtube.com/watch?v=X-S3YV2TmuY	29:11	The “scale beats curation” counterpart to CLIP — 1.8B noisy pairs, no filtering	Jia et al., <i>ALIGN</i> , ICML 2021
40	ALIGN: Scaling Up Visual and Vision-Language Representation Learning With Noisy Text Supervision	Microsoft Research	https://www.youtube.com/watch?v=kejLJ0kbIGM	57:53	Longer MSR seminar version with full Q&A	Jia et al., ICML 2021
41	Lucas Beyer — Sigmoid Loss for Language Image Pre-Training	Lucas Beyer (Google DeepMind), Cohere For AI	https://www.youtube.com/watch?v=Nk9YnMHB6hU	59:57	SigLIP author explaining why removing the softmax/global-batch coupling changes the economics of contrastive pretraining — a genuinely transferable loss-design lesson	Zhai, Mustafa, Kolesnikov, Beyer, <i>SigLIP</i> , ICCV 2023
42	Antoine Miech — Flamingo: a Visual Language Model for Few-Shot Learning	Antoine Miech (DeepMind), Columbia Vision Seminar	https://www.youtube.com/watch?v=5bMkjdeMapM	1:04:46	Flamingo co-author on gated cross-attention into a frozen LM — the architectural ancestor of every modern VLM	Alayrac et al., <i>Flamingo</i> , NeurIPS 2022

#	Title	Speaker / Institution	URL	Dur.	Why it matters	Paper
43	Flamingo: a Visual Language Model for Few-Shot Learning	Samuel Albanie (Cambridge/DeepMind)	https://www.youtube.com/watch?v=H82s6BrJduM	1:35:38	Exhaustive 95-minute technical dissection — the deepest free Flamingo treatment	Alayrac et al., NeurIPS 2022
44	The AI Multimodal Revolution with Junnan Li and Dongxu Li of BLIP & BLIP2	Junnan Li, Dongxu Li (Salesforce Research), Cognitive Revolution	https://www.youtube.com/watch?v=zTr5vDjEy2I	1:21:21	BLIP/BLIP-2 first authors on the Q-Former and bootstrapped captioning — includes what they would do differently	Li et al., <i>BLIP</i> , ICML 2022; <i>BLIP-2</i> , ICML 2023
45	[CVPR24 Vision Foundation Model Tutorial] Large Multimodal Models	Chunyuan Li (ByteDance, ex-Microsoft Research)	https://www.youtube.com/watch?v=S0CpenMvG48	50:19	LLaVA's co-author giving the canonical survey of visual instruction tuning	Liu, Li, Wu, Lee, <i>Visual Instruction Tuning (LLaVA)</i> , NeurIPS 2023
46	[CVPR24 Vision Foundation Models Tutorial] Multimodal LLM Pre-training	Zhe Gan (Apple)	https://www.youtube.com/watch?v=OznJmSQerBE	50:17	Pretraining-recipe-level detail for MLLMs (data mixtures, resolution, connector choice)	Gan et al., MM1 (2024)
47	[CVPR24 Vision Foundation Model Tutorial] Vision in LMMs	Jianwei Yang (Microsoft Research)	https://www.youtube.com/watch?v=bDVbs-fZGUg	56:30	Argues the vision encoder is the bottleneck in MLLMs — a live open research question	Yang et al., Florence-2; SEEM
48	[CVPR2023 Tutorial Talk] Recent Advances in Vision Foundation Models	VLP Tutorial (CVPR 2023)	https://www.youtube.com/watch?v=hE135guhTQo	44:31	Compact taxonomy of the whole VFM landscape	Li et al., <i>Multimodal Foundation Models survey</i> (2023)
49	CVPR #18558 — Recent Advances in Vision Foundation Models (full tutorial)	ComputerVisionFoundation Videos, CVPR 2024	https://www.youtube.com/watch?v=ZknT4EuhhGw	3:27:12	The complete 3.5-hour official CVF recording — the most comprehensive single free	—

#	Title	Speaker / Institution	URL	Dur.	Why it matters	Paper
					resource in this section	
50	Stanford CS25: Transformers United V6 — From Language Models to Native Multimodal Intelligence	Stanford CS25, Stanford Online (<i>embedding disabled; public</i>)	https://www.youtube.com/watch?v=NDdc39KYqDU	1:04:40	2026 state of native (non-bolted-on) multimodality	—
51	Yevgen Chebotar: RT-2 — Vision-Language Action Models Transfer Web Knowledge to Robotic Control	Yevgen Chebotar (Google DeepMind), Montreal Robotics	https://www.youtube.com/watch?v=o5ONDdbReAA	1:10:36	RT-2 co-author on treating actions as tokens in a VLM — the founding VLA talk	Brohan et al., <i>RT-2</i> , CoRL 2023
52	OpenVLA: LeRobot Research Presentation #5	Moo Jin Kim (Stanford), Hugging Face	https://www.youtube.com/watch?v=-0s0v3q7mBk	1:18:40	OpenVLA first author — the open reproduction, with honest ablations of what actually mattered	Kim et al., <i>OpenVLA</i> , CoRL 2024

4. Segmentation, detection, dense prediction

#	Title	Speaker / Institution	URL	Dur.	Why it matters	Paper
53	DETR — End to end object detection with transformers (ECCV 2020)	Nicolas Carion (FAIR/NYU), ECCV 2020 author video	https://www.youtube.com/watch?v=utxbUlo9CyY	9:35	The first author on bipartite-matching set prediction — deleting NMS and anchors. The set-prediction formulation is the reusable idea	Carion, Massa, Synnaeve, Usunier, Kirillov, Zagoruyko, ECCV 2020
54	Alexander Kirillov — 6th BMTT Workshop, ICCV 2021	Alexander Kirillov (FAIR, later Meta/OpenAI), ICCV 2021	https://www.youtube.com/watch?v=7WcRfHXY_lw	32:53	The author of Panoptic Segmentation, PointRend, MaskFormer and SAM on unifying segmentation tasks — recorded 18 months before SAM shipped	Kirillov et al., <i>Panoptic Segmentation</i> , CVPR 2019; Cheng, Schwing, Kirillov, <i>MaskFormer</i> , NeurIPS 2021
55	Invited Talk: From Pixels to Regions: Towards Universal Image Segmentation	Humphrey Shi lab (SHI Labs / UIUC-Georgia Tech)	https://www.youtube.com/watch?v=vBfUFi4Jskg	14:50	The “one architecture for semantic + instance + panoptic” argument in compact form	Cheng et al., <i>Mask2Former</i> , CVPR 2022; Jain et al., <i>OneFormer</i> , CVPR 2023
56	Xiuye Gu: Open-Vocabulary Detection and Segmentation	Xiuye Gu (Google Research), Learning with Limited and Imperfect Data workshop	https://www.youtube.com/watch?v=_6bWNNa7h8Q	27:24	ViLD's first author — distilling CLIP into a detector is the template for open-vocabulary dense prediction	Gu, Lin, Kuo, Cui, <i>ViLD</i> , ICLR 2022
57	Open-Vocabulary Visual Perception upon Frozen Vision and Language Models	Yin Cui (Google Research), ECCV 2022 CVinW	https://www.youtube.com/watch?v=LAesxhjebDA	32:24	The frozen-backbone design pattern — large implications for compute-constrained research	Kuo et al., <i>F-VLM</i> , ICLR 2023
58	Lecture 20 — OWLv2: Scaling	UCF CRCV graduate	https://www.youtube.com/watch?v=5A02sTC5-0M	24:35	Self-training to 1B+ pseudo-	Minderer, Gritsenko,

#	Title	Speaker / Institution	URL	Dur.	Why it matters	Paper
	Open-Vocabulary Object Detection	seminar			boxes — how open-vocab detection scales	Houlsby, OWLv2, NeurIPS 2023
59	SAM 3: The Eyes for AI — Nikhila Ravi & Pengchuan Zhang (Meta Superintelligence Labs), ft. Joseph Nelson	Latent Space	https://www.youtube.com/watch?v=sVo7SC62voA	1:15:04	SAM 2 and SAM 3 leads on the data engine, promptable segmentation, and the memory architecture for video. The data-engine discussion is the most research-transferable part	Kirillov et al., SAM, ICCV 2023; Ravi et al., SAM 2, ICLR 2025; SAM 3 (2025)

Also see Stanford CS231n 2025 Lecture 9 (§8) for the detection/segmentation survey, and §6 entry 85 (Charles Qi) for 3D detection.

5. Generative vision — GANs, diffusion, autoregressive, video, evaluation

#	Title	Speaker / Institution	URL	Dur.	Why it matters	Paper
60	Ian Goodfellow: Generative Adversarial Networks (NIPS 2016 tutorial)	Ian Goodfellow (OpenAI)	https://www.youtube.com/watch?v=HGYYEUSm-0Q	1:55:54	The definitive GAN tutorial from the inventor. The game-theoretic framing and the enumerated open problems are still the right way to think about adversarial objectives	Goodfellow et al., <i>GANs</i> , NeurIPS 2014; Goodfellow, <i>NIPS 2016 Tutorial: GANs</i> (arXiv)
61	BayLearn 2017 Keynote — Ian Goodfellow	Ian Goodfellow (Google Brain)	https://www.youtube.com/watch?v=Zd9kYgUjgSU	59:14	Mid-era reassessment: what worked, what didn't, and why mode collapse resisted fixes	—
62	Ian Goodfellow: Adversarial Machine Learning (ICLR 2019 invited talk)	Ian Goodfellow	https://www.youtube.com/watch?v=sucqskXRkss	43:06	The adversarial-robustness half of the story — essential for understanding what “distribution” means in vision	Goodfellow, Shlens, Szegedy, <i>Explaining and Harnessing Adversarial Examples</i> , ICLR 2015
63	Tero Karras — Training Generative Adversarial Networks with Limited Data	Tero Karras (NVIDIA Research), FCAI	https://www.youtube.com/watch?v=hOx9NBwDkHY	44:54	StyleGAN's author on adaptive discriminator augmentation — a masterclass in diagnosing <i>why</i> a model overfits rather than patching it	Karras et al., <i>StyleGAN2-ADA</i> , NeurIPS 2020; <i>StyleGAN</i> , CVPR 2019
64	Miika Aittala: Alias-Free Generative Adversarial Networks	Miika Aittala (NVIDIA Research), FCAI	https://www.youtube.com/watch?v=MU00DgkI97g	45:07	StyleGAN3 — signal-processing reasoning applied to a generator. The equivariance argument transfers directly to any continuous-signal model	Karras, Aittala, Laine, Härkönen, Hellsten, Lehtinen, Aila, <i>StyleGAN3</i> , NeurIPS 2021
65	Miika Aittala: Elucidating the Design Space of Diffusion-Based Generative Models	Miika Aittala (NVIDIA Research), FCAI	https://www.youtube.com/watch?v=T0Qxzf0eaio	52:46	The EDM talk. Disentangling the sampler, the noise schedule and the network preconditioning is the most reusable	Karras, Aittala, Aila, Laine, <i>EDM</i> , NeurIPS 2022

#	Title	Speaker / Institution	URL	Dur.	Why it matters	Paper
					methodological contribution in diffusion	
66	Stable Diffusion and Friends — Generative Modeling in Latent Space	Robin Rombach (Stability AI / LMU), heidelberg.ai	https://www.youtube.com/watch?v=7W4aZOBNucl	57:58	Latent diffusion from its first author — why moving to a perceptually-compressed latent space is the whole ballgame	Rombach, Blattmann, Lorenz, Esser, Ommer, <i>High-Resolution Image Synthesis with Latent Diffusion Models</i> , CVPR 2022
67	TL#006 Robin Rombach — Taming Transformers for High Resolution Image Synthesis	Robin Rombach (LMU Munich), Transfer Learning seminar	https://www.youtube.com/watch?v=fy153-yXSQk	45:39	VQGAN — the prequel to latent diffusion, and the clearest account of learned discrete visual tokenizers	Esser, Rombach, Ommer, <i>VQGAN</i> , CVPR 2021
68	MAS.S61 presents Andreas Blattmann / Robin Rombach , Stable Diffusion	Blattmann & Rombach (Stability AI), MIT MAS.S61	https://www.youtube.com/watch?v=GgW98il4Lbo	1:03:31	Both LDM authors together, with the video-diffusion extension discussed	Blattmann et al., <i>Align Your Latents</i> , CVPR 2023; <i>Stable Video Diffusion</i> (2023)
69	TUM AI Lecture Series — FLUX: Flow Matching for Content Creation at Scale	Robin Rombach (Black Forest Labs)	https://www.youtube.com/watch?v=nrKKLJXBSw0	1:06:11	The rectified-flow/flow-matching transition told by the person who shipped it at scale — why the industry left DDPM-style diffusion	Esser et al., <i>Scaling Rectified Flow Transformers (SD3)</i> , ICML 2024; Liu, Gong, Liu, <i>Rectified Flow</i> , ICLR 2023
70	TUM AI Lecture Series — The multimodal future: Why visual representation still matters	Saining Xie (NYU)	https://www.youtube.com/watch?v=hnu-mRLebhc	1:04:21	DiT's co-author (and ConvNeXt's) on why representation quality gates generation quality — includes the REPA line of work	Peebles & Xie, <i>Scalable Diffusion Models with Transformers (DiT)</i> , ICCV 2023; Yu et al., <i>REPA</i> , ICLR 2025
71	Stanford CS25: V5 — Transformers in Diffusion	Stanford CS25, Stanford Online	https://www.youtube.com/watch?v=vXtapCFctTI	1:14:32	Graduate-seminar treatment of DiT/MMDiT and the scaling	Peebles & Xie, <i>ICCV</i> 2023

#	Title	Speaker / Institution	URL	Dur.	Why it matters	Paper
	Models for Image Generation and Beyond				behavior that replaced the U-Net	
72	Yang Song: A personal journey to diffusion models	Yang Song (OpenAI), ICBS 2025	https://www.youtube.com/watch?v=s647Nz0r4r0	1:03:35	Score matching $\rightarrow$ NCSN $\rightarrow$ score SDE $\rightarrow$ consistency models, narrated as a research trajectory by the person who built it. Exceptional on research taste	Song & Ermon, NeurIPS 2019; Song et al., <i>Score-Based Generative Modeling through SDEs</i> , ICLR 2021; Song et al., <i>Consistency Models</i> , ICML 2023
73	Learning to Generate Data by Estimating Gradients of the Data Distribution	Yang Song (Stanford), hosted by Yingzhen Li	https://www.youtube.com/watch?v=nv-WTeKRLl0	1:03:15	The technical lecture version of the above, with the SDE/ODE derivation done properly	Song et al., ICLR 2021 (Outstanding Paper)
74	Diffusion and Score-Based Generative Models	MIT CBMM (<i>speaker not named in title; content is the Song-style score-SDE tutorial</i>)	https://www.youtube.com/watch?v=wMmqCMwuM2Q	1:32:01	The longest free rigorous treatment of the score/SDE formalism	Song et al., ICLR 2021
75	Tutorial on Denoising Diffusion-based Generative Modeling: Foundations and Applications	Arash Vahdat, Karsten Kreis, Ruiqi Gao (NVIDIA / Google), CVPR 2022 tutorial	https://www.youtube.com/watch?v=cS6JQpEY9cs	3:46:15	The canonical ~ 4 -hour diffusion tutorial. If you read only one thing on diffusion theory, watch this instead	Ho, Jain, Abbeel, <i>DDPM</i> , NeurIPS 2020; Song et al., ICLR 2021
76	CVPR #18546 — Denoising Diffusion Models: A Generative Learning Big Bang	ComputerVisionFoundation Videos, CVPR 2024 tutorial	https://www.youtube.com/watch?v=1d4r19GEVos	3:04:32	The 2024 refresh of the above — covers distillation, consistency, flow matching, video	Song et al., <i>Consistency Models</i> , ICML 2023; Lipman et al., <i>Flow Matching</i> , ICLR 2023
77	MIT 6.S184: Flow Matching and Diffusion Models — Lecture 01: Generative AI with SDEs (2025)	Peter Holderrieth (MIT)	https://www.youtube.com/watch?v=GCoP2w-Cqtg	1:25:12	A graduate course that derives flow matching and diffusion from one framework — the cleanest free mathematical treatment	Lipman et al., ICLR 2023; Liu et al., <i>Rectified Flow</i> , ICLR 2023

#	Title	Speaker / Institution	URL	Dur.	Why it matters	Paper
78	MIT 6.S184: Flow Matching and Diffusion Models — Lecture 01: Flow and Diffusion Models (2026)	Peter Holderrieth (MIT)	https://www.youtube.com/watch?v=9eJQQVrUUoI	1:18:03	The 2026 re-recording; use whichever edition you prefer	as above
79	[GCV @ CVPR25] Kaiming He — Towards End-to-End Generative Modeling	Kaiming He (MIT), CVPR 2025 GCV Workshop	https://www.youtube.com/watch?v=4VwXBrMoC0E	42:34	He's current research direction — collapsing the two-stage tokenizer+diffusion pipeline. This is an open research frontier stated as such	Li, Tian, Li, Deng, He, <i>MAR</i> , NeurIPS 2024; <i>JiT</i> (2025)
80	[GCV @ CVPR25] Björn Ommer — Bitter Lesson 2.0: Boosting the Efficiency & Control of Generative Models	Björn Ommer (LMU Munich)	https://www.youtube.com/watch?v=toat_cXdoac	36:47	The senior author of Stable Diffusion arguing against pure scale — the most useful contrarian position in generative vision	Rombach et al., CVPR 2022
81	[GCV @ CVPR23] Björn Ommer — Why This is Not the End of Research in Generative AI	Björn Ommer (LMU Munich)	https://www.youtube.com/watch?v=pQ-JE66Wv9M	31:12	Explicitly about where the open problems are — a research-agenda talk	—
82	[GCV @ CVPR23] Phillip Isola — Generative Models as Data++	Phillip Isola (MIT)	https://www.youtube.com/watch?v=YuRAeQsTS08	31:41	Reframes generative models as queryable datasets rather than samplers. One of the genuinely original conceptual contributions in the section	Isola et al., <i>pix2pix</i> , CVPR 2017; Huh et al., <i>Platonic Representation Hypothesis</i> , ICML 2024
83	[GCV @ CVPR 2026] Hila Chefer — Is Scale All You Need? A Case for Native Generative Representation Learning	Hila Chefer (Google DeepMind / Tel Aviv U.)	https://www.youtube.com/watch?v=wuNvKl4-NtI	27:16	2026 frontier talk directly on the scale-vs-inductive-bias question for generative models	Chefer et al., VideoJAM (2025)

#	Title	Speaker / Institution	URL	Dur.	Why it matters	Paper
84	Ishan Misra (Meta) — Emu Video Generation	Ishan Misra (Meta AI/GenAI)	https://www.youtube.com/watch?v=dLcsreHRF1s	55:26	A Sora-class video model explained by its lead, with the factorized text→image→video design and honest evaluation discussion	Girdhar et al., <i>Emu Video</i> (2023)
85	OpenAI Sora 2 Team: How Generative Video Will Unlock Creativity and World Models	Sora 2 team (OpenAI), Sequoia Capital	https://www.youtube.com/watch?v=w9oTtvbyLP8	1:00:26	The closest thing to a Sora research talk that exists publicly; useful for the world-model framing, light on technical detail	Brooks et al., <i>Video generation models as world simulators</i> (OpenAI tech report, 2024)
86	Reliable Fidelity and Diversity Metrics for Generative Models (ICML 2020)	Muhammad Ferjad Naeem (TU Munich / ETH) et al.	https://www.youtube.com/watch?v=_XwsGkryVpk	15:41	The FID critique: density & coverage, and a demonstration that FID conflates fidelity with diversity. Required before you report FID in any paper	Naeem, Oh, Uh, Choi, Yoo, <i>Reliable Fidelity and Diversity Metrics</i> , ICML 2020; cf. Heusel et al., <i>FID</i> , NeurIPS 2017
87	Visual Autoregressive Modeling — Part 1	West Coast Machine Learning reading group	https://www.youtube.com/watch?v=8RQWNgfGNal	1:27:37	Deep reading-group treatment of VAR's next-scale prediction. No author talk for VAR could be verified — see flag below	Tian, Jiang, Yuan, Peng, Wang, <i>VAR</i> , NeurIPS 2024 (Best Paper)

6. 3D vision and geometry — *directly transferable to molecular structure*

The transfer is concrete: SE(3) equivariance, coordinate-based implicit fields, differentiable rendering as a differentiable forward model, and point-set permutation invariance are the same mathematics used in protein structure prediction, cryo-EM reconstruction and docking. Treat SIREN/NeRF as “learned continuous density fields” and cryo-EM map modeling becomes the same problem class.

#	Title	Speaker / Institution	URL	Dur.	Why it matters	Paper
88	Multiple View Geometry — Lecture 1	Prof. Daniel Cremers (TU Munich)	https://www.youtube.com/watch?v=RDkwklFGMfo	1:27:30	The rigorous classical foundation: rigid-body motion, SE(3), Lie groups. This is the same SE(3) machinery as equivariant molecular networks	Hartley & Zisserman, <i>Multiple View Geometry in Computer Vision</i> , CUP 2004
89	Multiple View Geometry — Lecture 2	Prof. Daniel Cremers (TU Munich)	https://www.youtube.com/watch?v=6VbbYXpBIqA	1:24:33	Representing rigid-body motion — rotation groups, exponential map, twists	Hartley & Zisserman (2004); Ma, Soatto, Kořecká, Sastry, <i>An Invitation to 3-D Vision</i>
90	Computer Vision — Lecture 3.1 (Structure-from-Motion: Preliminaries)	Prof. Andreas Geiger (U. Tübingen)	https://www.youtube.com/watch?v=nrlHi1-a85s	26:18	Modern graduate SfM formulation — epipolar geometry, bundle adjustment framing	Hartley & Zisserman (2004)
91	Structure-from-Motion Revisited	Johannes Schönberger (ETH Zürich / Microsoft), CVPR 2016 author video	https://www.youtube.com/watch?v=PmXqdfBQxfQ	4:39	COLMAP's author — the system that every NeRF/3DGS paper silently depends on for camera poses	Schönberger & Frahm, CVPR 2016
92	ROB 2018 — Johannes Schönberger: COLMAP	Johannes Schönberger , Robust Vision Challenge 2018	https://www.youtube.com/watch?v=gTFSIg5ZFL4	13:34	Practical failure modes of SfM — essential if you ever build on reconstructed geometry	Schönberger & Frahm, CVPR 2016
93	NeRF: Neural Radiance Fields	Matthew Tancik (UC Berkeley), ECCV 2020 author video	https://www.youtube.com/watch?v=JuH79E8rdKc	4:15	The original NeRF author video. The key insight — positional encoding to overcome spectral bias in coordinate MLPs — is the one most reusable in scientific modeling	Mildenhall, Srinivasan, Tancik, Barron, Ramamoorthi, Ng, <i>NeRF</i> , ECCV 2020 (Best Paper Honorable Mention)
94	3DV 2024 Keynote — Ben Mildenhall	Ben Mildenhall (Google Research), 3DV 2024 (<i>embedding disabled; public</i>)	https://www.youtube.com/watch?v=dx7qJRkC1y4	1:09:18	NeRF's first author giving a full retrospective keynote four years on — what held up and what didn't	Mildenhall et al., ECCV 2020
95	L13b Neural Radiance Fields	Ben Mildenhall , in	https://www.youtube.com/watch?v=Z96YMktT_T4	1:04:24	Full lecture-format NeRF derivation by its author,	Mildenhall et al., ECCV 2020

#	Title	Speaker / Institution	URL	Dur.	Why it matters	Paper
	— Guest Lecturer Ben Mildenhall	Pieter Abbeel's Berkeley deep-learning course			built for students	
96	Jon Barron — Understanding and Extending Neural Radiance Fields	Jonathan T. Barron (Google Research), MIT Vision & Graphics Seminar	https://www.youtube.com/watch?v=HfJpQCBTqZs	54:43	The best single talk on <i>why</i> NeRF works, with the aliasing/scale analysis that led to mip-NeRF	Barron et al., <i>Mip-NeRF</i> , ICCV 2021
97	TUM AI Lecture Series — Understanding and Extending Neural Radiance Fields	Jonathan T. Barron (Google Research)	https://www.youtube.com/watch?v=nRyOzHpcr4Q	1:02:05	Longer TUM version with extended Q&A	Barron et al., ICCV 2021; <i>Mip-NeRF 360</i> , CVPR 2022
98	Zip-NeRF: Anti-Aliased Grid-Based Neural Radiance Fields — ICCV 2023 Talk	Jonathan T. Barron (Google Research)	https://www.youtube.com/watch?v=Sk3wU-VMoCI	5:00	How to combine hash-grid speed with anti-aliasing rigor — a model of careful incremental research	Barron, Mildenhall, Verbin, Srinivasan, Hedman, <i>Zip-NeRF</i> , ICCV 2023 (Best Paper Honorable Mention)
99	Radiance Fields and the Future of Generative Media	Jonathan T. Barron (Google DeepMind)	https://www.youtube.com/watch?v=hFlF33JZbA0	48:39	Barron's current view on where radiance fields and generative models merge	—
100	Matthew Tancik: Neural Radiance Fields for View Synthesis	Matthew Tancik (UC Berkeley), hosted by Andreas Geiger	https://www.youtube.com/watch?v=dPWLybp4LL0	49:11	Full-length NeRF talk including the Fourier-features theory	Tancik et al., <i>Fourier Features Let Networks Learn High Frequency Functions</i> , NeurIPS 2020
101	MAS.S61 presents Matthew Tancik , Co-Author of NeRF and Nerfstudio	Matthew Tancik (UC Berkeley / Luma AI), MIT	https://www.youtube.com/watch?v=isKbsNKArJU	1:02:02	The research-infrastructure side — how Nerfstudio was built and why tooling accelerated the field	Tancik et al., <i>Nerfstudio</i> , SIGGRAPH 2023
102	Bernhard Kerbl: 3D Gaussian Splatting — Real-Time Radiance Fields and their Applications	Bernhard Kerbl (TU Wien / Inria), ICBS 2025	https://www.youtube.com/watch?v=OXrZOHLoeO	1:04:09	3DGS first author. The explicit-primitive-plus-differentiable-rasterizer design is a direct template for differentiable molecular density modeling	Kerbl, Kopanas, Leimkühler, Drettakis, <i>3D Gaussian Splatting</i> , SIGGRAPH

#	Title	Speaker / Institution	URL	Dur.	Why it matters	Paper
						2023 (Best Paper)
103	Gaussian Splatting with Bernhard Kerbl	Bernhard Kerbl , View Dependent podcast	https://www.youtube.com/watch?v=cT1mQ_ityfE	1:05:39	Long-form interview — the design decisions and dead ends behind 3DGS, rarely written down	Kerbl et al., SIGGRAPH 2023
104	TUM AI Lecture Series — The 3D Gaussian Splatting Adventure: Past, Present, Future	George Drettakis (Inria)	https://www.youtube.com/watch?v=DjOqkVIEGY	1:04:20	The senior author’s full research-program framing, from point-based rendering history to future directions	Kerbl et al., SIGGRAPH 2023
105	TUM AI Lecture Series — Implicit Neural Scene Representations	Vincent Sitzmann (MIT)	https://www.youtube.com/watch?v=__F9CCqbWQk	1:10:59	SIREN/SRN author on coordinate-based neural fields as a general representation — the most directly transferable idea in this whole section for molecular work	Sitzmann et al., <i>SIREN</i> , NeurIPS 2020 (Oral); <i>Scene Representation Networks</i> , NeurIPS 2019
106	Vincent Sitzmann: Implicit Neural Scene Representations	Vincent Sitzmann (MIT), hosted by Andreas Geiger	https://www.youtube.com/watch?v=Or9J-DCDGko	56:42	Alternate-venue version with a different emphasis on generalization across scenes	Sitzmann et al., NeurIPS 2019/2020
107	Implicit Neural Representations with Periodic Activation Functions	Stanford Computational Imaging Lab (Wetzstein group), NeurIPS 2020 author video	https://www.youtube.com/watch?v=Q2fLWGBeaIl	10:20	The SIREN author video — periodic activations for representing signals <i>and their derivatives</i> , which matters whenever you need a differentiable field (PDEs, densities)	Sitzmann, Martel, Bergman, Lindell, Wetzstein, NeurIPS 2020
108	PointNet: Deep Learning on Point Sets for 3D Classification and Segmentation	Charles R. Qi (Stanford), CVPR 2017 author video	https://www.youtube.com/watch?v=Cge-hot0Oc0	11:24	Permutation invariance via symmetric functions — the direct ancestor of every atom-set molecular network	Qi, Su, Mo, Guibas, <i>PointNet</i> , CVPR 2017
109	Charles Qi — 6th BMTT Workshop, ICCV 2021	Charles R. Qi (Waymo), ICCV 2021	https://www.youtube.com/watch?v=jaVIINKSYvI	22:52	PointNet’s author on 3D detection at production scale — how the set abstraction idea evolved	Qi et al., <i>PointNet++</i> , NeurIPS 2017; <i>VoteNet</i> , ICCV 2019
110	Vincent Leroy: From CroCo to MAST3R — A Paradigm Change in 3D Vision	Vincent Leroy (NAVER LABS Europe), Montreal Robotics	https://www.youtube.com/watch?v=OJzj7uCCYaM	53:50	DUST3R/MASt3R co-author. Replacing the entire SfM pipeline with a feed-forward pointmap regressor is the most important recent idea in geometric vision	Wang, Leroy, Cabon, Chidlovskii, Revaud, <i>DUST3R</i> , CVPR 2024; Leroy et al., <i>MASt3R</i> ,

#	Title	Speaker / Institution	URL	Dur.	Why it matters	Paper
						ECCV 2024; Weinzaepfel et al., <i>CroCo</i> , NeurIPS 2022
111	Impossible Image Matching of DUS _t 3R & MAS _t 3R with Jérôme Revaud and Vincent Leroy	Revaud & Leroy (NAVER LABS Europe), View Dependent	https://www.youtube.com/watch?v=iTRBU80QuBc	1:14:01	Both authors, long form — where it fails and why, which the papers don't say	Wang et al., CVPR 2024
112	Zachary Teed — Optimization Inspired Neural Networks for Multiview 3D	Zachary Teed (Princeton), MIT Vision & Graphics Seminar	https://www.youtube.com/watch?v=ul6pXRGKmco	58:51	RAFT/DROID-SLAM author on embedding classical optimization structure inside networks — the best available argument for architecture-as-algorithm	Teed & Deng, RAFT, ECCV 2020 (Best Paper); <i>DROID-SLAM</i> , NeurIPS 2021
113	Physics-based differentiable rendering (CVPR 2021 tutorial)	Shuang Zhao , Ioannis Gkioulekas , Tzu-Mao Li (UC Irvine / CMU / UCSD)	https://www.youtube.com/watch?v=Tou8or1ed6E	3:10:42	The rigorous treatment of differentiating through a physical forward model — the exact problem structure of differentiable cryo-EM / scattering simulation	Li, Aittala, Durand, Lehtinen, <i>Differentiable Monte Carlo Ray Tracing</i> , SIGGRAPH Asia 2018
114	Neural Rendering (CVPR 2020) — Afternoon Session	CVPR 2020 Neural Rendering tutorial (Tewari, Zollhöfer, Niessner et al.)	https://www.youtube.com/watch?v=JlyGNvbGKB8	3:00:01	Full tutorial on the analysis-by-synthesis paradigm that unified graphics and vision	Tewari et al., <i>State of the Art on Neural Rendering</i> , Eurographics 2020
115	[GCV @ CVPR23] Andrea Tagliasacchi — Neural fields for 3D Vision (a MAP perspective)	Andrea Tagliasacchi (Google / SFU), CVPR 2023	https://www.youtube.com/watch?v=iiTyIrlzydg	30:05	Recasts neural fields as MAP inference — a probabilistic framing that generalizes cleanly to scientific inverse problems	Xie et al., <i>Neural Fields in Visual Computing</i> , Eurographics STAR 2022
116	[GCV @ CVPR 2026] Georgia Gkioxari — Beyond Image and Language: Building 3D Perception Systems	Georgia Gkioxari (Caltech)	https://www.youtube.com/watch?v=rKv2SE4tP60	33:01	2026 frontier statement on 3D perception as a first-class problem, not a 2D add-on	Gkioxari et al., <i>Mesh R-CNN</i> , ICCV 2019

#	Title	Speaker / Institution	URL	Dur.	Why it matters	Paper
117	Andreas Geiger: Is 3D Reconstruction ready for the Physical World? (ECCV 2026)	Andreas Geiger (U. Tübingen), ECCV 2026	https://www.youtube.com/watch?v=1wheKcIbTk4	24:46	Current honest assessment of where feed-forward reconstruction actually stands	Geiger et al., KITTI; Mescheder et al., <i>Occupancy Networks</i> , CVPR 2019
118	Andreas Geiger: Opening the Black Box of Generation and Reconstruction (ECCV 2026)	Andreas Geiger (U. Tübingen), ECCV 2026	https://www.youtube.com/watch?v=hJvgQreDjAA	26:00	The generation/reconstruction convergence, stated by one of the field's most careful methodologists	—

7. Research seminars, keynotes, and research taste

#	Title	Speaker / Institution	URL	Dur.	Why it matters	Paper
119	We are (still!) not giving Data enough credit — Keynote at ACVSS 24	Alexei (Alyosha) Efros (UC Berkeley)	https://www.youtube.com/watch?v=dV6sgUiGv-U	1:38:21	The best talk on research taste in computer vision, full stop. 98 minutes of “the model is not the contribution, the data is” — with the historical receipts	Efros & Leung, ICCV 1999; Hays & Efros, <i>Scene Completion</i> , SIGGRAPH 2007
120	Computer Vision after the Victory of Data	Alexei Efros (UC Berkeley), Simons Institute	https://www.youtube.com/watch?v=a13aqr07tJ4	1:18:06	Same thesis, delivered to a theory audience — forces the argument to be made more precisely	—
121	Self Supervision for Learning from the Bottom Up — Invited Talk, ICLR 2021	Alexei Efros (UC Berkeley), ICLR 2021	https://www.youtube.com/watch?v=F5yb5pSfb3E	1:01:11	Efros’s case for non-parametric/bottom-up learning against top-down categorical supervision	Doersch, Gupta, Efros, <i>Unsupervised Visual Representation Learning by Context Prediction</i> , ICCV 2015
122	Alexei Efros: In the end, it’s all about the Data	Alexei Efros (UC Berkeley), hosted by Andreas Geiger	https://www.youtube.com/watch?v=M1VHu1d4sGQ	1:27:31	The extended version with a long Q&A — the Q&A is where the research-strategy advice is	—
123	Alexei Efros — Lecture: “Self-Supervised Visual Learning and Synthesis”	Alexei Efros (UC Berkeley), Strelka Institute	https://www.youtube.com/watch?v=0zCMiOQ8C1U	51:29	Ties self-supervision and synthesis together as one problem — the thread that leads to today’s generative pretraining	Zhang, Isola, Efros, <i>Colorful Image Colorization</i> , ECCV 2016
124	Jitendra Malik — The Hilbert Problems of Computer Vision	Jitendra Malik (UC Berkeley), ILSVRC/ImageNet workshop	https://www.youtube.com/watch?v=QaF2kkez5XU	40:45	An explicit list of the open problems from the field’s most senior figure. This is how to choose a dissertation topic	Malik et al., <i>The Three R’s of Computer Vision</i> , PRL 2016
125	Jitendra Malik: Vision and Robotics for Embodied AI	Jitendra Malik (UC Berkeley / Meta), ETH Zürich CVG Group	https://www.youtube.com/watch?v=HvDU7Vvk4pbc	50:45	Malik’s current position: perception is not separable from action	Malik et al., <i>The Three R’s</i> ; RMA (Kumar et al., RSS 2021)
126	RI Seminar: Jitendra Malik — Robot	Jitendra Malik (UC Berkeley),	https://www.youtube.com/watch?v=ry8itipzBFE	1:07:14	Developmental-learning framing of representation	—

#	Title	Speaker / Institution	URL	Dur.	Why it matters	Paper
	Learning, With Inspiration From Child Development	CMU Robotics Institute			acquisition — a genuinely different source of research hypotheses	
127	What next in Computer Vision + Compositionality in Tasks	Jitendra Malik (UC Berkeley), CICV 2020	https://www.youtube.com/watch?v=wmhDEBqmPCg	59:55	Compositionality as the unsolved problem — directly relevant to anyone working on structured/molecular domains	—
128	TUM AI Lecture Series — The Moon Camera	William T. (Bill) Freeman (MIT)	https://www.youtube.com/watch?v=Ytkkl917paM	1:04:45	Freeman at his best: an apparently absurd problem (use the moon as a lens), solved with careful physics and inverse modeling. A masterclass in problem selection	Freeman group, accidental/non-line-of-sight imaging line of work
129	AIR Distinguished Speaker — Bill Freeman	Bill Freeman (MIT), Hariri Institute, Boston University	https://www.youtube.com/watch?v=C1L61UYAfQA	44:34	Recent (2026) Freeman talk — current directions	—
130	LMSS @ Cornell Tech: Bill Freeman — Learning from Sight and Sound	Bill Freeman (MIT), Cornell Tech	https://www.youtube.com/watch?v=hvH_UUNxFCI	55:33	Audio-visual self-supervision — the “free supervision from physics” idea	Owens et al., <i>Ambient Sound Provides Supervision</i> , ECCV 2016; <i>Visually Indicated Sounds</i> , CVPR 2016
131	The Platonic Representation Hypothesis	Phillip Isola (MIT), MIT CBMM	https://www.youtube.com/watch?v=V7AyriUcXZQ	1:09:11	The claim that models trained on different modalities converge to the same representation. One of the few genuinely new <i>ideas</i> (as opposed to results) in recent vision	Huh, Cheung, Wang, Isola, <i>The Platonic Representation Hypothesis</i> , ICML 2024
132	Dr Kaiming He — 2023 Future Science Prize Laureates Lecture	Kaiming He (MIT), Future Science Prize	https://www.youtube.com/watch?v=jEeL5Gf4vkk	1:15:51	A career-retrospective lecture — ResNet’s origin story and the reasoning behind each subsequent bet	He, Zhang, Ren, Sun, <i>ResNet</i> , CVPR 2016 (Best Paper); <i>Mask R-CNN</i> , ICCV 2017
133	[GCV @ CVPR 2026] Alan Yuille — Vision	Alan Yuille (Johns Hopkins)	https://www.youtube.com/watch?v=3fwVICnB4q4	38:23	A senior vision theorist’s structured critique of current	Yuille & Liu, <i>Deep Nets: What have they ever</i>

#	Title	Speaker / Institution	URL	Dur.	Why it matters	Paper
	Language Models need 3D				VLMs — useful as a source of falsifiable research claims	<i>done for Vision?</i> , IJCV 2021

Additional seminar channels worth subscribing to (all verified as active, enumerate yourself): - **Computer Vision Talks** reading group — <https://www.youtube.com/@computervisiontalks7222/videos> (15 hour-long SSL/representation talks) - **GCV @ CVPR workshops** (Adam Kortylewski) — <https://www.youtube.com/@adamkortylewski1099/videos> (CVPR '23/'25/'26 generative-vision keynotes: Kaiming He, Isola, Ommer, Yuille, Gkioxari, Chelsea Finn, Matthias Niessner, Sitzmann, Angjoo Kanazawa, Gordon Wetzstein) - **Vision & Graphics Seminar at MIT** — <https://www.youtube.com/@visiongraphicsseminaratmit9073/videos> (30 hour-long seminars: Barron, Teed, Snavely, Fragkiadaki, Krähenbühl, Russakovsky, Tulsiani) - **ECCV 2022 Computer Vision in the Wild workshop** — 13 invited talks (Houlsby, Zhai, Minderer, Misra, Yin Cui, Kate Saenko, Stella Yu, Xiaolong Wang) - **Stanford Contrastive & SS Learning Group** — <https://www.youtube.com/@stanfordcontrastivesslearn3141/videos>

8. Full modern vision courses (playlists enumerated to individual videos)

8a. Stanford CS231n – Deep Learning for Computer Vision, Spring 2025 (18 lectures, Stanford Online)

Lecture	Title	URL	Dur.
1	Introduction	https://www.youtube.com/watch?v=2fq9wYslV0A	1:02:53
2	Image Classification with Linear Classifiers	https://www.youtube.com/watch?v=pdqofxJeBN8	1:07:02
3	Regularization and Optimization	https://www.youtube.com/watch?v=dyNGd06MWn4	1:08:39
4	Neural Networks and Backpropagation	https://www.youtube.com/watch?v=25zD5qJHYsk	1:16:47
5	Image Classification with CNNs	https://www.youtube.com/watch?v=f3g1zGdxptI	1:08:04
6	CNN Architectures	https://www.youtube.com/watch?v=aVJy4O5TOk8	1:11:07
7	Recurrent Neural Networks	https://www.youtube.com/watch?v=kG2lAPBF7zA	1:11:48
8	Attention and Transformers	https://www.youtube.com/watch?v=RQowiOF_FvQ	1:06:31
9	Object Detection, Image Segmentation, Visualizing	https://www.youtube.com/watch?v=PTypu6GqEd4	1:13:44
10	Video Understanding	https://www.youtube.com/watch?v=wElqklprhPE	1:08:45
11	Large Scale Distributed Training	https://www.youtube.com/watch?v=9MvD-XsowsE	1:12:53
12	Self-Supervised Learning	https://www.youtube.com/watch?v=4howBU7THbM	1:14:42
13	Generative Models 1	https://www.youtube.com/watch?v=zbHXQRUNIH0	1:12:31
14	Generative Models 2	https://www.youtube.com/watch?v=Edr4uZFh4EE	1:12:09
15	3D Vision	https://www.youtube.com/watch?v=7lxrKDKtykM	1:11:50
16	Vision and Language	https://www.youtube.com/watch?v=mQOKOMfyrkk	1:09:54
17	Robot Learning	https://www.youtube.com/watch?v=XSfmOH_xVSU	1:18:33
18	Human-Centered AI	https://www.youtube.com/watch?v=g8UaBfj6Sh8	1:05:16

Playlist for reference only: [PLoROMvody4r0msNzYBMe0gJY2XS8A0q16](https://www.youtube.com/playlist?list=PLoROMvody4r0msNzYBMe0gJY2XS8A0q16) . **Why it matters:** this is the current canonical graduate vision course; lectures 6, 8, 9, 12, 13–16 are the research-relevant core and are a strict upgrade over the 2017 edition.

Johnson), 22 lectures, Michigan Online

Lec	Title	URL	Dur.
1	Introduction to Deep Learning for Computer Vision	https://www.youtube.com/watch?v=dJYGatp4SvA	57:56
2	Image Classification	https://www.youtube.com/watch?v=0nqvO3AM2Vw	1:02:15
3	Linear Classifiers	https://www.youtube.com/watch?v=qcSEP17uKKY	1:02:06
4	Optimization	https://www.youtube.com/watch?v=YnQJTfbwBM8	1:03:07
5	Neural Networks	https://www.youtube.com/watch?v=g6InpdhUblE	1:02:07
6	Backpropagation	https://www.youtube.com/watch?v=dB-u77Y5a6A	1:11:16
7	Convolutional Networks	https://www.youtube.com/watch?v=ANyxBVxmdZ0	1:08:53
8	CNN Architectures	https://www.youtube.com/watch?v=XaZIIvRiO-Q	1:12:03
9	Hardware and Software	https://www.youtube.com/watch?v=oXPX8GIOiU4	1:12:22
10	Training Neural Networks I	https://www.youtube.com/watch?v=lGbQlr1Ts7w	1:12:14
11	Training Neural Networks II	https://www.youtube.com/watch?v=WUazOtlti0g	1:19:14
12	Recurrent Networks	https://www.youtube.com/watch?v=dUzLD91Sj-o	1:13:27
13	Attention	https://www.youtube.com/watch?v=YAgjfMR9R_M	1:11:53
14	Visualizing and Understanding	https://www.youtube.com/watch?v=G1hGwHVykDU	1:12:04
15	Object Detection	https://www.youtube.com/watch?v=TB-fdISzphQ	1:12:32
16	Detection and Segmentation	https://www.youtube.com/watch?v=9AyMR4IhSWQ	1:10:07
17	3D Vision	https://www.youtube.com/watch?v=S1_nCdLUQQ8	1:12:34
18	Videos	https://www.youtube.com/watch?v=A9D6NXBJdwU	1:15:21
19	Generative Models I	https://www.youtube.com/watch?v=Q3HU2vEhD5Y	1:11:13
20	Generative Models II	https://www.youtube.com/watch?v=igP03FXZqgo	1:12:47
21	Reinforcement Learning	https://www.youtube.com/watch?v=Qex3XzcFKP4	1:11:45

Lec	Title	URL	Dur.
22	Conclusion	https://www.youtube.com/watch?v=s3Ky_Ls4YSY	1:13:28

Playlist for reference only: [PL5-TkQAFaZfbzxjBHTzdVCWE0Zbhong7r](https://www.youtube.com/playlist?list=PL5-TkQAFaZfbzxjBHTzdVCWE0Zbhong7r). **Why it matters:** Justin Johnson (NeRF-adjacent, ex-FAIR, CS231n's original lead TA) lectures with more derivation depth than CS231n. Given the stated CS231n background, **watch lectures 13–20 selectively rather than the whole course** — 14 (visualizing), 17 (3D) and 19–20 (generative) are the ones that add material CS231n 2025 does not cover as well.

8c. Supplementary graduate courses (verified, enumerate individually as needed)

- **Multiple View Geometry — Prof. Daniel Cremers, TU Munich** (channel [cvprtum](#), ~14 lectures, each ~85 min). Entries 88–89 above are lectures 1–2. This is the rigorous geometry course and has no modern free equal.
- **Computer Vision — Prof. Andreas Geiger, U. Tübingen** (channel “Tübingen Machine Learning”, ~80 short lecture segments across 12 units, including 9.1 Coordinate-based Networks/INRs, 11.2 Self-Supervised Learning, 3.1–4.5 SfM & stereo). Entry 90 is one segment.
- **MIT 6.S184 — Flow Matching and Diffusion Models** (Peter Holderrieth, 2025 and 2026 editions). Entries 77–78.
- **MIT 6.801 Machine Vision** (Berthold Horn, MIT OCW, Fall 2020) — classical machine vision, playlist [PLU14u3cNGP63pfpS1gV5P9tDxxL_e4W80](https://www.youtube.com/playlist?list=PLU14u3cNGP63pfpS1gV5P9tDxxL_e4W80). Lecture 1: <https://www.youtube.com/watch?v=tY2gczObpfU> (1:19:53).

Flags — could not verify

- **UNVERIFIED — search term: “MIT 6.8300 Advances in Computer Vision lecture recordings” / “MIT 6.8301 vision lecture 2025”** — no public recordings of MIT 6.8300/6.8301 exist on YouTube. Only MIT 6.801 (Horn, machine vision) and MIT 6.S191 are recorded. Course materials are web-only.
- **UNVERIFIED — search term: “CMU 16-824 Visual Learning and Recognition lecture recordings Deepak Pathak”** — no lecture recordings found; only student project presentations. CMU 16-824 slides are public but video is not.
- **UNVERIFIED — search term: “Keyu Tian VAR Visual Autoregressive Modeling NeurIPS 2024 oral talk”** — no author talk for VAR (NeurIPS 2024 Best Paper) is on YouTube. Entry 87 is the best reading-group substitute.
- **UNVERIFIED — search term: “Alexey Dosovitskiy ViT ICLR 2021 oral presentation”** — the ICLR 2021 oral itself is behind SlidesLive, not on YouTube. Entries 1–2 are the closest free author talks.
- **UNVERIFIED — search term: “Lumiere Omer Bar-Tal Inbar Mosseri research talk”** — only the 1:55 Google Research announcement ([f9ThAzZs32M](#)) exists; no research-depth Lumiere talk. Use entry 84 (Emu Video, Ishan Misra) as the video-generation research talk instead.
- **UNVERIFIED — search term: “William Peebles DiT Scalable Diffusion Models with Transformers ICCV 2023 talk”** — no Peebles talk on YouTube. Entries 70–71 (Saining Xie, Stanford CS25) cover DiT from the co-author side.
- **Speaker attribution unconfirmed** for entries 16, 17, 55 and 74 — the title and channel are verified, but the individual presenter is not stated on the page and I did not fabricate one. (My web-search budget was exhausted mid-task, so these were left flagged rather than guessed.)

Totals: 133 individually verified video URLs — 93 standalone research talks/keynotes/tutorials (§1–7) plus 40 enumerated course lectures (§8a–8b), exceeding the 55–75 target. No files were written; nothing in this task touched a git repository, so there is no commit to report.

D.4 Architecture Research

Why this comes first in Part III. Every model in Atlas P, Q and R is an architecture decision stack: what the attention does, how position is encoded, where the normalization goes, what the tokens are. Those decisions were made in the language-modeling literature and inherited wholesale, usually without re-examination for the molecular setting.

The opportunity is specific. Most of the architecture choices in protein models were imported from text, where the sequence is one-dimensional, causally ordered, and discretely tokenized. None of those three things is true of a protein in 3D. **Every place where that mismatch is unexamined is a paper.**

D.4.1 State space models — the strongest single thread here

Watch these in order. The claims change visibly across the five years, and seeing a researcher become *more modest* as the evidence comes in is itself the lesson.

Talk	Speaker	Year	Len	Link
Efficiently Modeling Long Sequences with Structured State Spaces (S4)	Albert Gu (Stanford MLSys #46)	2021	57:18	▶
Structured State Space Models for Deep Sequence Modeling	Albert Gu (CMU)	2023	1:04:27	▶
Structured State Space Models (the long version)	Albert Gu (LxMLS Lisbon)	2024	1:34:12	▶
EPFL AI Center — SSMs for Deep Sequence Modeling	Albert Gu (EPFL)	2024	1:01:10	▶
On the Tradeoffs of State Space Models	Albert Gu (Simons Institute)	2024	49:05	▶
CS25 V6 — On the Tradeoffs of SSMs and Transformers	Albert Gu (CMU, Cartesia)	2026	1:17:07	▶
Computational Benefits and Limitations of Transformers and SSMs	Eran Malach (Kempner, Simons)	2024	50:52	▶
Hardware-aware Algorithms for Sequence Modeling	Tri Dao (Stanford MLSys #87)	2024	1:19:06	▶
Model Architecture Design for Modern Hardware	Tri Dao (Kempner, Harvard)	2025	1:08:38	▶

Papers: Gu et al. 2021, arXiv:2111.00396 (S4) · Fu et al. 2022, arXiv:2212.14052 (H3) · **Gu & Dao 2023, arXiv:2312.00752 (Mamba)** · Dao & Gu 2024, arXiv:2405.21060 (**Mamba-2, state space duality**).

The molecular connection is direct and it is already being made. Evo and Evo 2 (Atlas Q.5) are built on this line — StripedHyena, then a Mamba-adjacent architecture — precisely because genomes are long and attention is quadratic. If you want to understand *why* genomic language models have a different architecture from protein language models, this subsection is the answer.

Watch the 2021 talk and the 2026 talk back to back. The 2021 talk claims SSMs are a general replacement for attention. The 2026 talk, by the same person, is careful about what a fixed-size state loses relative to a KV cache and argues for hybrids. That is what intellectual honesty looks like across five years, and it is a better lesson than either talk alone.

D.4.2 Linear attention and efficient attention

Talk	Speaker	Year	Len	Link
Linear Attention and Beyond (interactive tutorial)	Songlin Yang (MIT), hosted by Sasha Rush	2025	1:27:08	▶
Understanding and Improving Efficient Language Models	Simran Arora (Stanford, Simons)	2024	46:36	▶
Monarch Mixer: Making Foundation Models More Efficient	Dan Fu (Stanford MLSys #86)	2023	56:32	▶
Dynamic Short Convolutions Improve Transformers	Oliver Sieberling (Cohere Labs)	2026	43:33	▶
ML Perf Reading Group 18: Kimi Delta Attention	EleutherAI	2025	1:22:27	▶

Papers: Yang et al. 2023, arXiv:2312.06635 (Gated Linear Attention) · Yang et al. 2024, arXiv:2406.06484 (DeltaNet, parallelized delta rule) · Katharopoulos et al. 2020, ICML (transformers are RNNs) · Arora et al. 2024 (Based: the recall-throughput tradeoff).

Arora’s recall-throughput tradeoff is the one to internalize. Linear attention buys throughput and pays in associative recall. For a protein MSA — where the model must retrieve the homologous residue in a distant sequence — recall is exactly the operation that matters. **Nobody has carefully measured what linear attention costs an MSA-based model.** That is a clean, tractable experiment and nobody has run it.

D.4.3 Positional encoding, normalization, tokenization

Talk	Speaker	Year	Len	Link
ALiBi enables transformers to handle longer inputs	Ofir Press (author)	2022	46:57	▶
CS336 2026 Lecture 3: Architectures	Liang / Hashimoto (Stanford)	2026	1:29:14	▶
CS336 2026 Lecture 4: Attention Alternatives	Stanford	2026	1:26:20	▶
CS336 2026 Lecture 1: Overview, Tokenization	Stanford	2026	1:19:22	▶
Meta's MEGABYTE with Lili Yu	Lili Yu (author, Cognitive Revolution)	2023	1:32:22	▶

Papers: Su et al. 2021, [arXiv:2104.09864 \(RoPE\)](#) · Press et al. 2021, [arXiv:2108.12409 \(ALiBi\)](#) · Peng et al. 2023, [arXiv:2309.00071 \(YaRN\)](#) · Zhang & Sennrich 2019, NeurIPS (RMSNorm) · Xiong et al. 2020, ICML (pre-LN vs post-LN) · Yu et al. 2023, [arXiv:2305.07185 \(MEGABYTE\)](#).

Honest gap, reported as found. There is no author talk for RoPE, for YaRN, or for normalization placement. The CS336 2026 Lecture 3 is the substitute and it is a good one — a research-grade treatment of exactly these choices with the ablations. **CS336 Lectures 3 and 4 together are the single best “what actually matters in an architecture” resource that exists**, and they are current.

The molecular question this raises. RoPE encodes relative position along a 1D sequence. A protein's relevant “position” is a position in 3D space, which is why AlphaFold uses pair representations and invariant point attention instead. **What is the right relative positional encoding for a structure?** Nobody has a clean answer, and the question is well-posed.

D.4.4 Mixture of experts

Talk	Speaker	Year	Len	Link
Switch Transformers: Scaling to Trillion Parameter Models	Barret Zoph (KUIS AI)	2021	55:54	▶
Sparse Expert Models — with the authors	Zoph & Fedus	2022	58:22	▶
MIAI Deeptails Seminar	Zoph & Fedus (Google Brain)	2022	1:11:39	▶
CS25 V1 — MoE and the Switch Transformer	Stanford (speaker not in metadata)	2022	1:05:44	▶
CS25 V4 — Demystifying Mixtral of Experts	Albert Jiang (Mistral/Cambridge)	2024	1:04:31	▶
CS336 2025 Lecture 4: Mixture of experts	Stanford	2025	1:22:04	▶
ML Perf Reading Group 15: Megablocks	EleutherAI	2025	1:00:30	▶
ML Perf Reading Group 17: MXFP8 Training for MoEs	EleutherAI	2025	37:12	▶

Papers: Fedus et al. 2022, JMLR 23:120 (Switch Transformer) · Zoph et al. 2022, arXiv:2202.08906 (ST-MoE) · Jiang et al. 2024, arXiv:2401.04088 (Mixtral) · Shazeer et al. 2017, ICLR (the original sparsely-gated MoE) · Gale et al. 2023, MLSys (MegaBlocks).

The Zoph & Fedus author interview is the valuable one, because they describe the fine-tuning instabilities that are not in the papers. A field’s oral tradition — what reviewers never see — is most accessible in exactly this format.

Why a protein person should care. A protein model is asked to handle enzymes, membrane proteins, antibodies and IDPs with one set of weights. That is the textbook case for conditional computation, and essentially nobody has tried routing by structural class. **The experiment is sitting there.**

D.4.5 Long context

Talk	Speaker	Year	Len	Link
The Frontier between Retrieval-augmented and Long-context LMs	Danqi Chen (Princeton, Simons)	2025	59:30	▶
ML Perf Reading Group 4: Ring Attention	EleutherAI	2025	48:20	▶
GPU MODE Lecture 13: Ring Attention	GPU MODE	2024	1:15:35	▶
ML Perf Reading Group 20: Native Sparse Attention	EleutherAI	2026	59:24	▶
CMU Advanced NLP (13): Long Sequence Models	Graham Neubig	2024	57:11	▶

Papers: Liu et al. 2023, arXiv:2310.01889 (Ring Attention) · Yuan et al. 2025, arXiv:2502.11089 (Native Sparse Attention) · Beltagy et al. 2020 (Longformer) · Child et al. 2019 (sparse transformers).

Ring attention is directly relevant here. AF3-style models hit memory walls on large complexes — which is the entire reason the chromatin work needed chunked triangle multiplication. Ring attention solves the same class of problem by sharding the sequence across devices with overlapped communication. **Whether the triangle operations admit a ring decomposition is an open engineering question with a real payoff.**

D.4.6 Diffusion language models

Talk	Speaker	Year	Len	Link
How to Build a Modern Diffusion Language Model	Volodymyr Kuleshov (Cornell, Simons)	2026	48:40	▶
Advancing Diffusion Models for Text Generation	Kilian Weinberger (Cornell, Simons)	2025	1:01:50	▶
Unlocking Lossless Speedups via Discrete Diffusion	Subham Sahoo (Cohere Labs)	2026	1:09:04	▶
Diffusion LLM & Why the Future Won't Be Autoregressive	Stefano Ermon (Stanford/Inception)	2026	49:18	▶
Discrete diffusion by estimating ratios of the data distribution	Generative Memory Lab	2024	1:20:34	▶

Papers: Lou et al. 2024, ICML, arXiv:2310.16834 (SEDD — score entropy) · Sahoo et al. 2024, NeurIPS (MDLM) · Austin et al. 2021, NeurIPS (D3PM) · Campbell et al. 2022, NeurIPS (continuous-time discrete diffusion).

This subsection is the clearest two-way bridge in Part III. EvoDiff, DPLM and the discrete-diffusion half of Multiflow (Atlas Q.2, R.2) are the *same mathematics* applied to amino acids instead of word pieces. The molecular field adopted it early and in some respects pushed it further, because protein sequences are short, the vocabulary is 20 symbols, and there is no left-to-right causality to give up.

The idea that has not been carried back: molecular diffusion models routinely condition on a 3D structure. Text diffusion has no equivalent of a structural conditioner. What would it mean to condition a language diffusion model on a *plan* in the way RFDiffusion conditions on a motif? That question is open in both directions.

D.4.7 RWKV, Hyena, xLSTM — the other attention alternatives

Talk	Speaker	Year	Len	Link
How RWKV-7 “Goose” and Its Linear Inference Work	Eugene Cheah (author, Oxen AI)	2025	1:11:16	▶
Beyond Transformers — Intro to RWKV	Cheah & Vanderbyl (Linux Foundation)	2023	34:16	▶
xLSTM: New Architectures for LLMs	Sepp Hochreiter (SCIoI)	2024	1:25:38	▶
LSTM: The Comeback Story?	Hochreiter (MLST)	2025	1:07:02	▶
HyenaDNA: Long-Range Genomic Sequence Modeling	Eric Nguyen (Valence)	2023	1:20:38	▶
Tri Dao and Michael Poli on the future of LLM architectures	Interconnects	2023	36:51	▶

Papers: Peng et al. 2023, arXiv:2305.13048 (RWKV) · Peng et al. 2025, arXiv:2503.14456 (RWKV-7) · Beck et al. 2024, arXiv:2405.04517 (xLSTM) · Poli et al. 2023, ICML (Hyena) · **Nguyen et al. 2023, NeurIPS, arXiv:2306.15794 (HyenaDNA).**

HyenaDNA is the single entry in this subsection that is already biology. Watch it to see an architecture idea cross from language to genomics in under a year, and note what had to change: single-nucleotide resolution, no tokenizer, and a context length chosen by the biology rather than by the benchmark.

D.4.8 Paired reading — D.4

Watch this	Then read this	Hold this question
Gu 2021, <i>S4</i>	arXiv:2111.00396	HiPPO gives the initialization. What is being approximated?
Gu 2026, <i>Tradeoffs of SSMs</i>	arXiv:2405.21060 (Mamba-2)	What exactly does a fixed-size state lose? Name the task class.
Dao, <i>Hardware-aware algorithms</i>	arXiv:2312.00752 (Mamba)	Selectivity breaks the convolution. How is the scan made fast anyway?
Yang, <i>Linear Attention and Beyond</i>	arXiv:2312.06635; arXiv:2406.06484	Write the delta rule as a linear attention update.
Arora, <i>Efficient LMs</i>	The Based papers	Measure the recall-throughput tradeoff. Then ask what it costs an MSA model.
Press, <i>ALiBi</i>	arXiv:2108.12409 then arXiv:2104.09864 (RoPE)	What is the right relative position for a 3D structure?
CS336 2026 L3 + L4	The ablation tables in each	Which architecture choices survive controlled comparison?
Zoph & Fedus, <i>Sparse Expert Models</i>	Fedus et al. 2022, JMLR 23:120	Design an MoE routed by structural class. What is the auxiliary loss?
Danqi Chen, <i>Retrieval vs long context</i>	Her paper	When does long context subsume retrieval? Apply the answer to MSAs.
Kuleshov, <i>Modern diffusion LM</i>	Lou et al. 2024 (SEDD)	Compare the loss to EvoDiff's. What is the same and what differs?
Nguyen, <i>HyenaDNA</i>	arXiv:2306.15794	An architecture crossed domains in a year. What had to change?
Hochreiter, <i>xLSTM</i>	arXiv:2405.04517	The person who invented LSTM is still working on it. What is the claim?

D.5 Training at Scale, Scaling Laws and Data

Why this is in a biology curriculum. Three reasons, and they are all practical.

First, **every compute decision you make is a scaling-law question** whether you frame it that way or not: how long to train, how big a model, how much data to collect. The language field has an explicit, quantitative theory of this. Yours does not, and importing it is a contribution.

Second, **the numerics and memory tricks are directly reusable.** FlashAttention is in AlphaFold forks. Hydrogen mass repartitioning and mixed precision are the same kind of idea in different clothes.

Third, **the data chapter is the one that transfers most cleanly.** Deduplication, contamination, and the measurement of what a training set actually contains are solved problems in language and unsolved problems in biology.

D.5.1 FlashAttention and IO-aware algorithms

Talk	Speaker	Year	Len	Link
FlashAttention	Tri Dao (Stanford MLSys #67)	2023	58:58	▶
Hardware-aware Algorithms for Language Modeling	Tri Dao (ETH SPCL)	2024	1:05:15	▶
Optimizing attention for modern hardware	Tri Dao (Princeton/Together)	2025	35:45	▶
GPU MODE Lecture 12: Flash Attention	GPU MODE	2024	1:12:14	▶
GPU MODE Lecture 36: CUTLASS and Flash Attention 3	Jay Shah (FA-3 co-author)	2024	1:49:15	▶
GPU MODE Lecture 80: How FlashAttention 4 Works	Charles Frye	2025	1:15:09	▶
FlashAttention-4	Ted Zadouri (GPU MODE)	2026	46:23	▶
ML Perf Reading Group 2: Flash Attention	EleutherAI	2024	1:14:21	▶
ML Perf Reading Group 24: Flash Attention 4	EleutherAI	2026	1:05:20	▶
ThunderKittens goes live	Stanford MLSys / Hazy Research	2024	1:14:55	▶
Notes on AI Hardware	Benjamin Spector (Stanford MLSys #88)	2024	1:16:48	▶

Papers: Dao et al. 2022, NeurIPS, arXiv:2205.14135 (FlashAttention) · Dao 2023, arXiv:2307.08691 (FA-2) · Shah et al. 2024, arXiv:2407.08608 (FA-3) · Rabe & Staats 2021 (self-attention does not need $O(n^2)$ memory) · Milakov & Gimelshein 2018 (online softmax).

Start with Spector's Notes on AI Hardware if you have never thought about arithmetic intensity. Without it, the whole IO-aware argument is invisible — FlashAttention computes exactly the same function as standard attention and is faster entirely because of where the data sits. **That is the template for the most portable idea in this entire Part: performance is often about memory movement, not operations.**

It already transferred. Fast AlphaFold implementations use FlashAttention in the Evoformer. The triangle operations do not yet have an equivalent treatment, and writing one is a concrete, high-value project — it is what makes large-complex prediction feasible on consumer hardware.

D.5.2 Parallelism and distributed training

Talk	Speaker	Year	Len	Link
Training LLMs at Scale	Deepak Narayanan (Stanford MLSys #83)	2023	55:59	▶
Efficient Large-Scale Training with Megatron-LM	Jared Casper (NVIDIA)	2023	24:03	▶
GPU MODE Lecture 48: The Ultra Scale Playbook	Nouamane Tazi (Hugging Face)	2025	3:03:48	▶
CS25 V6 — The Ultra-Scale Talk	Nouamane Tazi (HF)	2026	1:01:48	▶
The Ultra-Scale Playbook	Thom Wolf (HF)	2025	1:25:56	▶
ML Perf RG 8: Megatron-LM	EleutherAI	2025	1:09:20	▶
ML Perf RG 6: Zero Bubble Pipeline Parallelism	EleutherAI	2025	1:17:13	▶
ML Perf RG 11: Async Tensor Parallelism	EleutherAI	2025	1:07:38	▶
ML Perf RG 13: Unified Sequence Parallelism	EleutherAI	2025	1:18:41	▶
ML Perf RG 9: Reducing Activation Recomputation	EleutherAI	2025	1:15:01	▶
ML Perf RG 1: GPU Architecture, CUDA, NCCL	EleutherAI	2024	47:39	▶
Alpa: Automated Model-Parallel Deep Learning	Zhuohan Li (Stanford MLSys #59)	2022	55:07	▶
ML Perf RG 3: ZeRO	EleutherAI	2024	1:08:16	▶
Large Model Training with DeepSpeed	Samyam Rajbhandari (ZeRO author)	2023	36:23	▶
Trillion Parameter Training and Inference with DeepSpeed	Rajbhandari & Rasley (REFAI)	2023	1:06:52	▶

Papers: Narayanan et al. 2021, SC'21, [arXiv:2104.04473 \(Megatron-LM\)](#) · Rajbhandari et al. 2020, SC'20, [arXiv:1910.02054 \(ZeRO\)](#) · Zhao et al. 2023 (PyTorch FSDP) · Zheng et al. 2022, OSDI (Alpa).

Tazi's three-hour Ultra Scale Playbook is the most complete free resource on distributed training that exists, and the 2026 CS25 condensation is the right entry point. Watch the short one, then use the long one as reference.

The honest assessment for your situation. You have a handful of machines with 1–8 GPUs each. Tensor and pipeline parallelism are not your problem; **data parallelism, activation checkpointing and memory are**. Watch the ZeRO material and the activation-recomputation session, skip the thousand-GPU material unless

you want it for its own sake. The yash RAM ceiling you already hit is exactly the constraint ZeRO is designed around.

D.5.3 Numerics, precision and optimizers

Talk	Speaker	Year	Len	Link
FP8 Training From Hopper To Blackwell	Luca Wehrstedt (Meta, PyTorch Conf)	2026	25:24	▶
GPU MODE Lecture 52: Scaling Laws for Low Precision	Tanishq Kumar (author)	2025	53:42	▶
Trinity: Training a 400B MoE from Scratch	Arcee	2026	32:59	▶
Tuning Large Neural Networks via Zero-Shot Hyperparameter Transfer	Greg Yang (MSR)	2022	1:12:52	▶
Tuning GPT-3 on a Single GPU via Zero-Shot HP Transfer	Greg Yang	2022	1:33:53	▶
EI Seminar (the tight version)	Greg Yang	2022	53:52	▶
Large N Limits: Random Matrices & Neural Networks	Greg Yang (Cartesian Cafe)	2023	3:01:28	▶
Metrized Deep Learning	Jeremy Bernstein (MIT, Cohere)	2024	1:34:03	▶
Depths of First Order Optimization	Jeremy Bernstein (Cohere)	2025	47:31	▶
Scalable second order optimization for deep learning	Rohan Anil (JAX meetup)	2022	1:28:15	▶
From the Broximal Point Method to Efficient LLM Training	Peter Richtárik (KAUST, Simons)	2026	52:25	▶
Muon and Kimi K-2	Latent Space paper club	2025	1:03:06	▶

Papers: Yang et al. 2022, [arXiv:2203.03466 \(Tensor Programs V, \$\mu\$ P\)](#) · Bernstein et al. 2024, [arXiv:2405.14813 \(modular norm\)](#) · Bernstein & Newhouse 2024, [arXiv:2409.20325 \(Old Optimizer, New Norm\)](#) · Jordan et al. 2024 (Muon) · Liu et al. 2025, [arXiv:2502.16982 \(Muon is scalable\)](#) · Gupta et al. 2018 (Shampoo) · Kumar et al. 2024, [arXiv:2411.04330 \(scaling laws for precision\)](#).

μ P is the single most directly importable idea in this entire Part. Tune hyperparameters on a small model, transfer them zero-shot to a large one. Every structure-prediction group retunes at every scale, by hand, at enormous cost. **Nobody has published a μ P analysis of an Evoformer or a triangle-attention stack.** The

parameterization is nontrivial because of the pair representation — which is exactly why it would be a real contribution rather than an application.

Bernstein’s *Metrized Deep Learning* is the intellectual foundation of Muon and the deepest talk in this subsection. Steepest descent is only defined relative to a norm; choosing the norm is choosing the optimizer. Once you see that, the optimizer zoo organizes itself.

D.5.4 Scaling laws

Talk	Speaker	Year	Len	Link
Scaling Laws and Their Implications	Jared Kaplan (JHU/Anthropic, Harvard CMSA)	2022	1:10:40	▶
Neural Scaling Laws and GPT-3	Jared Kaplan	2020	1:15:20	▶
Neural Scaling Laws and GPT-3 (Physics Meets ML)	Jared Kaplan	2020	1:35:49	▶
Understanding the Origins and Taxonomy of Neural Scaling Laws	Yasaman Bahri (Simons)	2023	1:05:24	▶
Dynamics and scaling laws in deep learning	Yasaman Bahri (Yale YINS)	2021	1:07:08	▶
The Large Learning Rate Phase of Deep Learning	Yasaman Bahri	2020	36:06	▶
(Mis)Fitting: A Survey of Scaling Laws	Sneha Kudugunta (Cohere Labs)	2026	54:55	▶
CS224N Guest Lecture: Scaling Language Models	Stanford	2022	1:14:49	▶
Predicting and optimizing the behavior of large ML models	Simons Institute	2025	1:04:05	▶
CS336 2026 Lecture 9: Scaling Laws	Stanford	2026	1:17:57	▶
CS336 2026 Lecture 11: Scaling Laws	Stanford	2026	1:17:04	▶

Papers: Kaplan et al. 2020, arXiv:2001.08361 · Hoffmann et al. 2022, NeurIPS (Chinchilla) · Bahri et al. 2021, arXiv:2102.06701 (explaining neural scaling laws) · McCandlish et al. 2018, arXiv:1812.06162 (critical batch size) · Lewkowycz et al. 2020, arXiv:2003.02218 (the catapult phase).

Kudugunta’s *(Mis)Fitting* is the most methodologically useful talk here and the one to assign if you only assign one. It is a survey of how scaling laws get fit *badly* — wrong functional forms, too few points, extrapolating past the fitted range — which is precisely the error you would make on your first attempt.

The structural fact about biology that this reveals. Language has effectively unlimited data and a clean loss; biology has a fixed PDB and a loss that is a proxy for a wet-lab outcome. **A scaling law for protein**

models is therefore a different object, and nobody has written down what it should look like. That is Capstone X territory: “if we could measure 10× more structures, how much would prediction improve, and what is the functional form?” The answer is worth knowing before anyone spends the money.

D.5.5 Data — the chapter that transfers most cleanly

Talk	Speaker	Year	Len	Link
DataComp-LM: In Search of the Next Generation of Training Sets	Ludwig Schmidt (Stanford)	2025	48:31	▶
A data-centric view on reliable generalization	Ludwig Schmidt (MLSys #71)	2023	58:41	▶
Creating a large dataset for pretraining LLMs (FineWeb)	Guilherme Penedo (HF)	2025	24:05	▶
Why Data Holds the Key to AI Reasoning Breakthroughs	Penedo (GOSIM)	2025	23:40	▶
Better Data is All You Need	Ari Morcos (Datology, Latent Space)	2025	1:18:43	▶
Poisoning Web-Scale Training Datasets	Nicholas Carlini (MLSys #75)	2023	58:06	▶
OLMo leads on the secrets of training language models	Groeneveld, Lo, Soldaini (AI2)	2025	1:12:43	▶
Open Model Pretraining Masterclass	Elie Bakouch (HF, Latent Space)	2025	1:03:39	▶
Data-distributional Approaches for Generalizable LMs (DoReMi, DSIR)	Sang Michael Xie (Stanford)	2024	1:10:57	▶
Scaling Data-Constrained Language Models	Sasha Rush (Simons)	2023	1:03:59	▶
Scaling Data-Constrained Language Models	Niklas Muennighoff (author)	2023	1:04:05	▶
Open Pretrained Transformers (the OPT-175B logbook)	Susan Zhang (MLSys #77)	2023	1:00:05	▶
CS25 V4 — Behind the Scenes of LLM Pre-training: StarCoder	Loubna Ben Allal (HF)	2024	1:01:36	▶
CS336 2026 Lecture 13: Data (Sources, Datasets)	Stanford	2026	1:22:02	▶
CS336 2026 Lecture 14: Data	Stanford	2026	1:24:46	▶

Papers: Li et al. 2024, arXiv:2406.11794 (DataComp-LM) · Penedo et al. 2024, arXiv:2406.17557 (FineWeb) · Lee et al. 2022, ACL (deduplicating training data makes LMs better) · Sorscher et al. 2022, arXiv:2206.14486 (beating power-law scaling via data pruning) · Xie et al. 2023, arXiv:2305.10429 (DoReMi) · Muennighoff et al. 2023, arXiv:2305.16264 (data-constrained scaling) · Carlini et al. 2023, arXiv:2302.10149.

Susan Zhang’s OPT-175B logbook talk is the most honest hour about what large training runs are actually like, and the closest analogue in this curriculum is Gabe Rocklin’s *Why designs fail*. Both are someone describing, in public, what the paper left out. **Watch them in the same week.** The parallel is the point: two cultures, two kinds of hidden failure, one shared norm worth adopting.

The transferable technique. Deduplication, contamination detection and data-mixture reweighting are mature in language and essentially absent in protein modeling. The analogues are obvious — sequence-identity deduplication exists, but *structure*-level deduplication, contamination between training and CASP targets, and principled mixture weights over Pfam families mostly do not. **DoReMi applied to protein family mixtures is an afternoon’s reimplementation and a real result.**

D.5.6 Paired reading — D.5

Watch this	Then read this	Hold this question
Spector, <i>Notes on AI Hardware</i>	Any roofline-model tutorial	Compute the arithmetic intensity of triangle multiplication.
Dao, <i>FlashAttention</i>	arXiv:2205.14135	Same function, 3× faster. Where did the time go?
Shah, <i>CUTLASS and FA-3</i>	arXiv:2407.08608	What does implementing the idea actually cost?
Tazi, <i>Ultra-Scale</i> (CS25 version)	The Ultra-Scale Playbook (HF)	Which parallelism applies to a 4-GPU node? Which does not?
Rajbhandari, <i>DeepSpeed</i>	arXiv:1910.02054 (ZeRO)	Partition states, not compute. What does each stage buy?
Yang, <i>Zero-Shot HP Transfer</i>	arXiv:2203.03466	What is μP ’s parameterization, and what would it be for pair representations?
Bernstein, <i>Metriized Deep Learning</i>	arXiv:2405.14813; arXiv:2409.20325	Choosing the norm is choosing the optimizer. Which norm fits your problem?
Kaplan, <i>Scaling Laws</i>	arXiv:2001.08361 then Hoffmann et al. 2022	Why was Kaplan’s compute-optimal ratio wrong, and what was the fix?
Bahri, <i>Origins of scaling laws</i>	arXiv:2102.06701	Four regimes. Which one is a protein model in?
Kudugunta, <i>(Mis)Fitting</i>	The survey	List five ways to fit a scaling law badly. Check your own plan against them.
Penedo, <i>FineWeb</i>	arXiv:2406.17557	Every filter is an ablation. What is the protein equivalent of a filter?
Morcos, <i>Better Data</i>	Sorscher et al. 2022, arXiv:2206.14486	Pruning can beat power-law scaling. What would you prune from the PDB?
Xie, <i>DoReMi and DSIR</i>	arXiv:2305.10429	Reimplement DoReMi over Pfam families. What mixture does it find?
Zhang, <i>OPT-175B logbook</i>	Then re-watch Rocklin, <i>Why designs fail</i>	Two cultures, two hidden-failure literatures. What norm should both adopt?
Carlini, <i>Poisoning training data</i>	arXiv:2302.10149	Could the PDB be poisoned? What would it look like?

D.6 Post-Training and Reinforcement Learning

Why a protein designer should care about RLHF. Because design *is* a post-training problem and nobody says so.

A protein language model is pretrained on evolution. Then you want it to produce sequences that are stable, soluble, expressible and active — properties no pretraining objective contains. That is exactly the structure of RLHF: a pretrained distribution, a separately learned reward, and an optimization that moves the policy toward the reward while staying near the prior.

Every pathology the RLHF literature documents therefore applies to you. Reward hacking is a design filter being gamed. KL regularization is the stay-near-natural-sequences constraint. Over-optimization is your 1% hit rate. This subsection is a vocabulary for problems you already have.

D.6.1 The mechanics, and the over-optimization result

Talk	Speaker	Year	Len	Link
Proxy Objectives in RLHF (ICML 2023 invited talk)	John Schulman	2023	1:01:06	▶
RLHF: Progress and Challenges	John Schulman (Berkeley EECS Colloquium)	2023	1:03:32	▶
Reinforcement Learning from Human Feedback	Nathan Lambert (UCL DARK)	2023	47:15	▶
AI Safety, RLHF, and Self-Supervision	Jared Kaplan (MLSys #79)	2023	1:02:37	▶
CS25 V4 — Aligning Open Language Models	Nathan Lambert (AI2)	2024	1:16:21	▶

Papers: Ouyang et al. 2022, NeurIPS, arXiv:2203.02155 (InstructGPT) · Gao, Schulman & Hilton 2023, ICML, arXiv:2210.10760 (Scaling Laws for Reward Model Overoptimization) · Christiano et al. 2017, NeurIPS · Stiennon et al. 2020, NeurIPS · Bai et al. 2022, arXiv:2204.05862.

Gao, Schulman & Hilton is the paper to read twice. It measures what happens as you optimize harder against a learned reward: true performance rises, peaks, and then *falls*, while the proxy reward keeps climbing. They fit the curve.

Now map it. Your learned reward is pLDDT, or interface pAE, or a self-consistency RMSD threshold. Your optimization is the design loop. The paper predicts that past some point, pushing harder on your filters makes your designs *worse* in the tube while looking better on screen. **Nobody in protein design has measured that curve.** Measuring it is Capstone III, and it would be among the most useful things anyone could publish about design pipelines.

D.6.2 The Post-Training Course — Nathan Lambert, AI2, 2026

A complete 14-lecture course, released alongside *The RLHF Book*, and the best-organized systematic treatment of post-training that exists. **It is current to 2026**, which matters in a subfield that reorganizes annually.

#	Lecture	Len	Link
0	ML Foundations (prerequisites) for Post-Training	23:53	▶
1	Post-Training and RLHF Overview	46:10	▶
2	RLHF Foundations, IFT, Reward Modeling, Rejection Sampling	49:49	▶
3	Understanding Policy Gradient Algorithms for RL on LLMs	57:35	▶
4	Implementing RL Algorithms for LLMs	53:36	▶
5	The Rise of Reasoning Models	45:20	▶
6	Direct Preference Optimization (DPO) and Friends	42:44	▶
7	On-Policy Distillation & Synthetic Data in Post-Training	49:40	▶
8	Preference Data: The Most Opaque Part of Post-Training	35:53	▶
9	Over-Optimization and RLHF's Bad Reputation	23:34	▶
10	Regularization in RL, Why RL Generalizes, and Why SFT Forgets	31:33	▶
11	How Language Models Use Tools and the Path to Agents	37:47	▶
12	How Evaluation Has Evolved with Frontier Model Progress	32:11	▶
13	An Introduction to Character Training	32:56	▶
—	From Academic Research to a Frontier LLM: A Case Study in DPO	1:06:30	▶

Lectures 9 and 10 are the two to watch first if you are short on time. Lecture 10 in particular — why RL generalizes and SFT forgets — is a claim with a direct protein analogue that nobody has tested: does fine-tuning a PLM on a narrow family cause catastrophic forgetting of the broader distribution, and would a KL-regularized RL objective avoid it?

D.6.3 DPO, PPO and the algorithmic debate

Talk	Speaker	Year	Len	Link
CS234 Guest Lecture on DPO — all three co-authors	Rafailov, Sharma & Mitchell (Stanford)	2024	1:18:44	▶
CS224N Lecture 10 — Post-training	Archit Sharma (Stanford)	2025	1:19:42	▶
DPO has given us alignment without the overhead	Sharma & Rafailov	2024	20:19	▶
An update on DPO vs PPO for LLM alignment	Nathan Lambert	2024	13:23	▶
DPO Debate: Is RL needed for RLHF?	Nathan Lambert	—	26:55	▶
Asynchronous RLHF: Faster Off-Policy RL for LMs	Michael Noukhovitch (Mila, Cohere)	2025	51:11	▶
Efficient Policy Optimization Techniques for LLMs	Kianté Brantley (Harvard, Simons)	2025	45:35	▶
Coherence in RLHF Preference Data	Shuo Li Liu (Cohere Labs)	2026	48:39	▶

Papers: Rafailov et al. 2023, *NeurIPS*, [arXiv:2305.18290 \(DPO\)](#) · Schulman et al. 2017, [arXiv:1707.06347 \(PPO\)](#) · Ethayarajh et al. 2024 (KTO) · Meng et al. 2024 (SimPO) · Xu et al. 2024, *ICML* (**Is DPO superior to PPO?**).

The DPO derivation is the most elegant result in this whole Part and you should be able to reproduce it. The claim: the optimal KL-regularized policy has a closed form in terms of the reward, so the reward can be *eliminated* and the preference likelihood written directly in terms of the policy. No reward model, no RL loop.

Derivation checkpoint 58. Derive it. Then answer the protein question: your preference data is a pair of designs where one expressed and one did not. Can you DPO a protein language model on that? **What is the Bradley-Terry assumption buying, and is it true for wet-lab outcomes?** The Cohere talk on preference-data coherence is the right companion — it asks exactly when the Bradley-Terry model is valid, and the answer for noisy assay data is not obviously yes.

D.6.4 RLVR — reinforcement learning with verifiable rewards

This is the subsection with the most direct protein analogue, because a verifier is exactly what design pipelines have: a structure predictor that says whether the sequence folds back.

Talk	Speaker	Year	Len	Link
Tulu 3: Frontiers in Open Language Model Post-Training	Nathan Lambert (AI2)	2024	1:01:45	▶
Open Training Recipes: LLM Reasoning	Hanna Hajishirzi (UW/AI2, Berkeley MOOC)	2025	1:20:53	▶
Open Training Recipes for Reasoning	Hajishirzi	2025	46:35	▶
Experimenting with RLVR	Nathan Lambert (USC ISI)	2025	47:13	▶
Spurious Rewards: Rethinking Training Signals in RLVR	Stella Li (Cohere Labs)	2025	1:02:11	▶
Early stages of the RL era of language models	Nathan Lambert (UCSC)	2025	59:31	▶
GRPO's new variants and implementation secrets	Nathan Lambert	2025	22:23	▶
Random Sparse Subnetworks suffice for RLVR	EleutherAI PRA Reading Group	2026	41:24	▶
CS336 2026 Lecture 16: Post-Training — RLVR	Stanford	2026	1:15:50	▶

Papers: Lambert et al. 2024, arXiv:2411.15124 (Tulu 3) · Shao et al. 2024 (GRPO / DeepSeekMath) · DeepSeek-AI 2025 (R1).

Assign the *Spurious Rewards* talk as a critical-reading exercise. The finding: some models improve on math under *random or incorrect* reward signals. If that holds, RLVR is sometimes eliciting a capability that was already in the pretrained model rather than teaching anything. **That is the sharpest open question in post-training**, and its protein analogue is sharp too: when a design pipeline’s filters “improve” results, is the generator getting better or is the filter just selecting from what was already there?

Derivation checkpoint 59 asks you to design the experiment that distinguishes elicitation from learning, in both settings.

D.6.5 Reward modeling, reward hacking and supervision

Talk	Speaker	Year	Len	Link
Introducing RewardBench	Nathan Lambert (AI2)	2024	16:50	▶
Supervising AI on hard tasks	Jan Leike (Anthropic)	2024	33:34	▶
Forecasting and Aligning AI	Jacob Steinhardt (Berkeley)	2022	54:35	▶
Understanding and Steering Generative AI Systems	Jacob Steinhardt (Simons)	2024	1:03:30	▶
Oversight of Foundation Models	Jacob Steinhardt (IPAM/UCLA)	2026	56:33	▶
Prover-Verifier Games Improve Legibility of LLM Outputs	Yining Chen (OpenAI, Simons)	2024	40:49	▶
Interactive Proofs, Debate, and AI Safety	Jonah Brown-Cohen (DeepMind, Simons)	2024	53:35	▶

Papers: Lambert et al. 2024, arXiv:2403.13787 (RewardBench) · Burns et al. 2023, arXiv:2312.09390 (weak-to-strong generalization) · Pan, Bhatia & Steinhardt 2022, ICLR, arXiv:2201.03544 (the effects of reward misspecification) · Kirchner et al. 2024, arXiv:2407.13692.

Pan, Bhatia & Steinhardt is the reward-hacking paper with the cleanest experiments, and it generalizes immediately: a misspecified proxy produces *phase transitions* in behavior as capability increases, not gradual degradation. A more capable design model does not game your filters slightly more — it games them suddenly and completely. If you have ever seen a design campaign produce a cluster of nearly identical, high-scoring, useless sequences, you have seen this.

D.6.6 Distillation and synthetic data

Talk	Speaker	Year	Len	Link
Distilling the Knowledge in a Neural Network	Geoffrey Hinton	—	54:41	▶
Learning by Distilling Context	Charlie Snell (Berkeley, Cohere)	2022	53:16	▶
CMU Advanced NLP (11): Distillation, Quantization, Pruning	Graham Neubig	2024	1:04:21	▶
Instruction Tuning of LLMs (Self-Instruct)	Yizhong Wang (UW, JHU CLSP)	2023	48:22	▶
Lessons from the Alpaca Project	Tatsu Hashimoto (Stanford)	2024	23:36	▶
Textbooks Are All You Need	Sébastien Bubeck	2023	37:15	▶
Small Language Models	Sébastien Bubeck	2024	1:04:03	▶
Synthetic Data / Smol Models	Loubna Ben Allal (HF)	2024	28:07	▶
Self-Play Fine-Tuning (SPIN)	Zixiang Chen (UCLA, first author)	2024	1:02:31	▶
Self-Play for LMs on Programming Puzzles	Adam Tauman Kalai (MSR, Simons)	2023	1:03:54	▶

Papers: Hinton et al. 2015, arXiv:1503.02531 · Wang et al. 2023, ACL, arXiv:2212.10560 (Self-Instruct) · Gunasekar et al. 2023, arXiv:2306.11644 (Textbooks Are All You Need) · Chen et al. 2024, arXiv:2401.01335 (SPIN) · Shumailov et al. 2024, Nature 631:755 (model collapse).

Model collapse is the one to think hardest about, because biology is already doing it. The AFDB contains 200 million *predicted* structures. Models are now trained on them. Shumailov et al. show that recursive training on generated data degrades the distribution’s tails — which, in a structural context, means the unusual folds. **Nobody has measured whether AFDB-trained models have lost tail diversity.** That is a well-defined, important, and currently unanswered question.

D.6.7 Paired reading — D.6

Watch this	Then read this	Hold this question
Schulman, <i>Proxy Objectives in RLHF</i>	Gao et al. 2023, arXiv:2210.10760	Plot the over-optimization curve for pLDDT as a design reward.
Lambert, Post-Training L9–10	The RLHF Book, corresponding chapters	Why does RL generalize where SFT forgets? Does that hold for PLMs?
CS234, <i>DPO with all three authors</i>	Rafailov et al. 2023, arXiv:2305.18290	Derive DPO. Then ask whether Bradley-Terry fits wet-lab outcomes.
Liu, <i>Coherence in RLHF Preference Data</i>	The paper	When is pairwise preference learning valid? Check your assay.
Li, <i>Spurious Rewards in RLVR</i>	The paper	Design the experiment separating elicitation from learning.
Lambert, <i>Experimenting with RLVR</i>	Shao et al. 2024 (GRPO)	A verifier-based reward. What is your verifier, and is it gameable?
Steinhardt, <i>Forecasting and Aligning AI</i>	Pan et al. 2022, arXiv:2201.03544	Reward hacking is a phase transition. Have you seen one?
Leike, <i>Supervising AI on hard tasks</i>	Burns et al. 2023, arXiv:2312.09390	Can a weak verifier supervise a strong generator? What is the protein version?
Wang, <i>Self-Instruct</i>	arXiv:2212.10560	Bootstrapping data from the model. What is the protein analogue, and its risk?
Ben Allal, <i>Synthetic data</i>	Shumailov et al. 2024, <i>Nature</i> 631:755	Measure whether AFDB-trained models lost tail fold diversity.

D.7 Reasoning, Inference-Time Compute, In-Context Learning

Why this matters to a methods researcher in biology. Two ideas here are directly transferable and one argument is directly instructive.

The first idea is **inference-time compute as a second scaling axis**. You already do this — generating a thousand backbones and filtering is inference-time search — but nobody in design has framed it as a compute allocation problem with an optimal policy. The language field has.

The second is **verification**. A design pipeline is generate-then-verify, and so is every reasoning system here. The literature on when verification works, when it is gamed, and how to allocate budget between generation and verification is directly importable.

The argument is **whether reasoning is real**, and it is instructive because it is the same argument as “do protein language models learn biology.” Watching a neighboring field have it carefully will sharpen how you have yours.

D.7.1 Chain-of-thought and the theory behind it

Talk	Speaker	Year	Len	Link
CS25 V5 — Large Language Model Reasoning	Denny Zhou (DeepMind)	2025	1:06:07	▶
LLM Reasoning (earlier version)	Denny Zhou (Berkeley MOOC)	2024	1:04:02	▶
Chain of Thought Empowers Transformers to Solve Inherently Serial Problems	Zhiyuan Li (first author)	2024	55:19	▶
Iterated Models: Expressive Power, Learning, and Chain of Thought	Nati Srebro (TTIC, Simons)	2024	54:08	▶
Towards Revealing the Mystery behind Chain of Thought	—	2023	47:38	▶
Implicit Chain-of-Thought: Internalizing Reasoning	Yuntian Deng (Cohere Labs)	2025	1:21:38	▶
Do Large Language Models Perform Latent Reasoning?	Mor Geva (Tel Aviv, Simons)	2024	32:15	▶
LLMs Cannot Self-Correct Reasoning Yet	Jie Huang (UIUC, Cohere)	2024	49:01	▶
Transformers can learn compositional functions	Simons Institute	2025	1:03:56	▶

Papers: Wei et al. 2022, NeurIPS (chain-of-thought prompting) · Wang et al. 2023, ICLR (self-consistency) · Li et al. 2024, arXiv:2402.12875 (CoT empowers transformers — the circuit-complexity result) · Feng et al. 2023, arXiv:2305.15408 · Deng et al. 2024, arXiv:2405.14838 (implicit CoT) · Huang et al. 2024, ICLR, arXiv:2310.01798.

Li et al. 2024 is the theoretical result that makes this subsection respectable. A fixed-depth transformer is in a bounded circuit class and cannot compute certain serial functions. Chain-of-thought lets it use the output as working memory, which provably escapes that bound. **This is not a prompting trick, it is a complexity-theoretic statement about architecture.**

Derivation checkpoint 56. State the circuit class a fixed-depth transformer falls in, and the class CoT reaches. Then ask the molecular version: a structure predictor recycles its output three times. **Is recycling chain-of-thought?** What serial computation does it enable that a single pass cannot do? Nobody has written this down, and it is exactly the kind of question that makes a theory paper.

D.7.2 Test-time scaling

Talk	Speaker	Year	Len	Link
Scaling LLM Test-Time Compute	Charlie Snell (first author, Berkeley)	2024	53:31	▶
Optimally Scaling Test-Time Compute & Predicting Emergence	Charlie Snell	2025	36:45	▶
From Decoding to Meta-Generation: Inference Time Algorithms	Sean Welleck (CMU)	2024	1:14:44	▶
Beyond Decoding: Meta-Generation Algorithms	Sean Welleck (Simons)	2024	1:21:55	▶
The Key Ingredients of Optimizing Test-Time Compute	Aviral Kumar (CMU, Simons)	2025	1:02:15	▶
Inference Scaling: A New Frontier for AI Capability	Azalia Mirhoseini (Stanford, Simons)	2025	1:07:45	▶
Inference-Time Techniques for LLM Reasoning	Xinyun Chen (DeepMind, Berkeley)	2025	1:21:32	▶
Test Time Scaling Small LMs to o1 level	Isha Puri (Cohere Labs)	2025	53:55	▶
Inverse Scaling in Test-Time Compute	EleutherAI PRA Reading Group	2026	51:06	▶

Papers: [Snell et al. 2024, arXiv:2408.03314](#) · [Welleck et al. 2024, arXiv:2406.16838 \(the survey\)](#) · [Puri et al. 2025, arXiv:2502.01618](#) (particle-based Monte Carlo with a PRM) · [Brown et al. 2024](#) (large language monkeys — repeated sampling).

Welleck’s survey is the one that reframes design as search. It treats generation as a search problem with a budget, and catalogues the algorithms: best-of-N, beam search over reasoning steps, MCTS, particle filtering. **Every one of these has a protein design analogue and most have not been tried.**

Best-of-N with a learned verifier is what every design pipeline already does. **Particle filtering with a process reward model is not** — but it is exactly what you would want for a sequential generation process like diffusion, where you could reweight partially-denoised structures by a partial-quality estimate rather than waiting until the end to filter. **Capstone III lives here.**

And watch the *Inverse Scaling* session as the corrective: more inference compute sometimes makes things worse. In design, more aggressive filtering sometimes selects for filter-gaming rather than quality. Same phenomenon.

D.7.3 The o1/R1 line

Talk	Speaker	Year	Len	Link
Learning to Reason with LLMs	Noam Brown (OpenAI, Simons)	2024	52:03	▶
Learning to Reason with LLMs	Jason Weston (Meta, Berkeley MOOC)	2025	1:16:47	▶
Speculations on Test-Time Scaling (o1)	Sasha Rush (Cornell)	2024	47:55	▶
How DeepSeek changes the LLM story	Sasha Rush (Simons)	2025	1:11:09	▶
DeepSeek-R1 Thoughtology	Siva Reddy (Mila/McGill, Simons)	2025	59:41	▶
s1: Simple test-time scaling	Niklas Muennighoff (first author)	2025	50:59	▶
s1 (TWIML interview)	Muennighoff	2025	48:59	▶
Scaling Test Time Compute to Multi-Agent Civilizations	Noam Brown (Latent Space)	2025	1:17:47	▶
ML Perf Reading Group 7: DeepSeek V3	EleutherAI	2025	53:54	▶

Papers: DeepSeek-AI 2025, arXiv:2501.12948 (R1) · Muennighoff et al. 2025, arXiv:2501.19393 (s1) · OpenAI o1 system card.

Rush’s *Speculations on Test-Time Scaling* is a masterclass in a skill you need: reconstructing how a closed system probably works from public papers alone. He is explicit about what is inference and what is evidence. That is exactly the discipline required when you read an industrial structure-prediction paper with no code and no weights.

s1 is the result that should change what you attempt. A thousand curated examples and a simple budget-forcing trick reproduced much of the reasoning behavior. **The lesson for a small lab: careful data and a simple inference-time idea can substitute for enormous training compute.** That is directly applicable to your situation and it is the most encouraging single finding in Part III.

D.7.4 Process versus outcome supervision

Talk	Speaker	Year	Len	Link
Mathematical Reasoning in Language Models	Karl Cobbe (OpenAI)	2023	51:17	▶
Noam Brown, Ilge Akkaya and Hunter Lightman on o1	Sequoia Training Data	2024	45:22	▶

Papers: Cobbe et al. 2021, arXiv:2110.14168 (GSM8K and verifiers) · Lightman et al. 2024, ICLR, arXiv:2305.20050 (Let’s Verify Step by Step).

This is the subsection with the most under-exploited protein analogue in the entire curriculum.

Outcome supervision scores the final answer; process supervision scores each intermediate step, and Lightman et al. show process wins decisively.

A diffusion trajectory is a process. RFdiffusion takes 50 steps and is scored only at the end. Nobody has built a process reward model for intermediate denoising states — a model that says “this partially-denoised backbone is on a bad trajectory, resample now.” The data to train it is obtainable: run the pipeline, record intermediate states, label by final outcome. **This is a concrete, fundable, novel project and it is stated here in three sentences.**

D.7.5 In-context learning and induction heads

Talk	Speaker	Year	Len	Link
Induction Heads (BlackboxNLP keynote)	Catherine Olsson (Anthropic)	2022	37:57	▶
CS25 V1 — Transformer Circuits, Induction Heads, ICL	Anthropic	2022	59:34	▶
A Walkthrough of In- Context Learning and Induction Heads	Neel Nanda w/ Charles Frye	2022	1:03:51	▶
Induction Heads and Phase Transitions	Rohan Hitchcock (SLT Summit)	2023	58:06	▶
In-context Language Learning and N-gram Heads	Ekin Akyürek (MIT)	2024	53:47	▶
What Learning Algorithm is In-Context Learning?	Jacob Andreas (MIT, Harvard CMSA)	2023	50:16	▶
Toward Understanding In- context Learning	Tengyu Ma (Stanford, Simons)	2024	1:29:35	▶
In-Context Learning: Simple Function Classes	Gregory Valiant (Stanford, Simons)	2023	1:03:40	▶
Pretraining Task Diversity and Non- Bayesian ICL	Surya Ganguli (Stanford, Simons)	2023	1:05:52	▶
Uncovering Mesa- Optimization Algorithms in Transformers	Nino Scherrer (IVADO)	2026	37:33	▶
Associative memories as a building block in Transformers	Alberto Bietti (Flatiron, Simons)	2024	38:37	▶
Using Algorithms to Understand Transformers	Vatsal Sharan (USC, Simons)	2024	48:04	▶

Papers: Olsson et al. 2022, Transformer Circuits (induction heads) · Akyürek et al. 2023, ICLR, arXiv:2211.15661 (ICL implements gradient descent) · Garg et al. 2022, NeurIPS, arXiv:2208.01066 · Xie et al. 2022, ICLR, arXiv:2111.02080 (ICL as implicit Bayesian inference) · von Oswald et al. 2023, ICML.

The direct molecular connection, and it is not a stretch. An MSA Transformer attends across sequences in an alignment to find the residue at the corresponding position in a homolog. **That is an induction head doing structural biology.** Olsson’s prefix-matching + copying description is a mechanistic account of exactly the operation the MSA axis performs.

Nobody has looked for induction heads in an MSA Transformer. The methodology exists, the model is open, and the hypothesis is sharp. **That is Capstone VIII**, and it is the single most tractable bridge project in this entire document.

D.7.6 Emergence — and the argument about it

Talk	Speaker	Year	Len	Link
CS25 V2 — Emergent Abilities and Scaling in LLMs	Jason Wei (author)	2023	1:07:48	▶
Emergence and reasoning in large language models	Jason Wei (JHU CLSP)	2022	52:04	▶
Investigating emergent abilities and challenging dominant research ideas	Rylan Schaeffer (Imbue)	2024	1:03:16	▶
A Theory for Emergence of Complex Skills in Language Models	Sanjeev Arora (Princeton, Simons)	2023	1:04:45	▶
Emergence of Complex Skills in LLMs	Sanjeev Arora (FAR.AI)	2024	6:05	▶
Progress Measures for Grokking via Mechanistic Interpretability	Neel Nanda w/ Lawrence Chan	2023	43:10	▶
CS25 V6 — Distinct Modes of Generalization from Parameters and Context	Andrew Lampinen (Anthropic)	2026	1:12:30	▶
You Know It Or You Don't: Compositionality and Phase Transitions	Simons Institute	2025	57:00	▶
ICML 2024 Tutorial: Physics of Language Models	Zeyuan Allen-Zhu	2024	1:53:43	▶

Papers: Wei et al. 2022, TMLR (emergent abilities) · Schaeffer et al. 2023, NeurIPS, arXiv:2304.15004 (Are Emergent Abilities a Mirage?) · Arora & Goyal 2023, arXiv:2307.15936 · Nanda et al. 2023, ICLR, arXiv:2301.05217 (grokking).

Watch Wei and then Schaeffer, in that order, in one sitting. This is the best-constructed disagreement in Part III. Wei documents capabilities appearing abruptly with scale. Schaeffer argues the abruptness is manufactured by discontinuous metrics — exact-match accuracy is a step function over a continuously improving quantity — and that smooth metrics show smooth improvement.

Then apply it to yourself. “Success rate above 4 Å RMSD” is a thresholded metric. “Fraction of designs that express” is a thresholded metric. **Every apparent discontinuity in a protein benchmark should be checked against Schaeffer’s argument before it is believed,** and most never are.

Allen-Zhu’s *Physics of Language Models* tutorial is the methodology lesson of this whole Part. Controlled synthetic data, clean ablations, falsifiable claims — doing science on models rather than reporting benchmark numbers. If you want a template for how to run a rigorous empirical study of a model, this two-hour tutorial is it.

D.7.7 Is reasoning real? — the standing debate

Talk	Speaker	Year	Len	Link
Debate: Sparks versus Embers	Bubeck, McCoy, Izmailov, Moitra (Simons)	2024	1:22:10	▶
LLMs can’t plan (..but they can help you in planning)	Subbarao Kambhampati (ICAPS keynote)	2023	1:12:44	▶
LLM-Modulo Frameworks	Kambhampati (ICRA workshop)	2024	30:51	▶
Do you think that ChatGPT can reason?	Kambhampati (MLST)	2024	1:42:28	▶
Pattern Recognition vs True Intelligence	François Chollet (MLST)	2024	2:42:55	▶
Chollet on OpenAI o-models and ARC	François Chollet (MLST)	2025	1:26:47	▶
Investigating Abstract Reasoning in Humans and Machines	Melanie Mitchell (SFI)	2026	56:59	▶
Six Principles for Evaluating Cognitive Capabilities in AI Models	Melanie Mitchell (SFI)	2026	48:41	▶
Failures in Reasoning Models	EleutherAI PRA Reading Group	2025	1:03:03	▶

Papers: McCoy et al. 2024, *PNAS* 121:e2322420121 (Embers of Autoregression) · Bubeck et al. 2023, arXiv:2303.12712 (Sparks of AGI) · Valmeekam et al. 2023, NeurIPS (planbench) · Kambhampati et al. 2024, arXiv:2402.01817 (LLM-Modulo) · Chollet 2019, arXiv:1911.01547 (ARC).

The Sparks-versus-Embers debate is the single most useful thing in this section, because it is two careful people disagreeing in public with the evidence on the table. McCoy’s position — that LLM failures are

predicted by the *probability of the task under the pretraining distribution*, not by its logical difficulty — is a falsifiable, mechanistic claim, and it has an exact protein analogue: **protein model failures should be predicted by the rarity of the fold in the PDB, not by the protein's biophysical difficulty**. Someone should test that. It is a one-figure paper and it would settle a lot of arguments.

Mitchell's *Six Principles* is the methodology talk to keep. It is about how to design an evaluation that measures a capability rather than a correlate — which is the question behind every benchmark in Atlas S.

Watch the two Chollet talks in order, before and after o3's ARC result. He updates in public. That is rarer and more valuable than being right the first time.

D.8 Deep Learning Theory

The “why does any of this work” layer

Why this module exists

Everything else in an AI curriculum teaches you to *build*. This module is the only one that teaches you to *explain*, and explanation is where original contributions to artificial intelligence itself actually come from. The field has a large and embarrassing gap between what works and what anyone can justify: networks with more parameters than data points generalize, gradient descent on a wildly nonconvex loss finds good minima, test error goes down after it has already gone up, and abilities appear at scale that nobody predicted. Each of those is an open problem with a recorded literature of people arguing about it on stage.

A warning specific to this module. **Deep learning theory is the one area where the recorded talks are strictly better than the papers for learning purposes.** The papers compress away the failed attempts; the talks keep them in. Nati Srebro telling you which capacity measure he tried first and why it did not work is worth more than the theorem. Peter Bartlett saying out loud that uniform convergence cannot be the answer is a thing he writes much more carefully in print. This is why the module is built almost entirely out of the Simons Institute and IAS archives rather than out of survey papers.

A structural note on sources. Two institutions have, between them, recorded essentially the entire modern history of this field: the **Simons Institute for the Theory of Computing** at Berkeley and the **Institute for Advanced Study** at Princeton. The Simons programs are listed in §DLT-9 as coherent archives, because watching one program end to end is a different and better experience than watching ten talks from ten places. Individual talks are pulled forward into the topic sections where they belong.

Conventions

Every entry below is an **individual video**, verified to exist and play, with its real duration and the year the talk was given or posted. All 291 video links in this document were checked programmatically against YouTube; nothing here is a bare playlist, and where a playlist is named it is also enumerated into individual lectures. Where a talk has a canonical paper, it is paired. Where I could not confirm something, it is flagged **UNVERIFIED** with the search term that would find it, rather than guessed at.

Durations are given as H:MM:SS. Years are the year the talk was delivered where the hosting institution publishes a schedule, and the upload year otherwise; the two differ by a few months in some Simons entries, where the talk date is used.

Entries marked **[CORE]** are the ones to watch if you only do a third of this.

DLT-1 Generalization theory for deep networks

The founding question: a network with 60 million parameters trained on 50,000 images, which can fit random labels perfectly, nevertheless generalizes. Every classical bound is vacuous by many orders of magnitude. What is the real explanation?

1.1 The canonical lecture series

This four-part sequence from the Simons Deep Learning Boot Camp is the single best structured introduction that exists, and it is taught by the two people who wrote most of the relevant theorems.

1. Generalization I [CORE] Peter Bartlett (UC Berkeley) + Sasha Rakhlin (MIT) · 2019 · 1:16:56 ·

https://www.youtube.com/watch?v=Ntl_WNW8yGc Sets up the classical uniform-convergence and VC machinery, then shows precisely where it breaks for overparameterized networks. Start here; everything downstream is a reaction to this lecture. *Paper*: Bartlett, Harvey, Liaw, Mehrabian, “Nearly-tight VC-dimension bounds for piecewise linear neural networks” (arXiv:1703.02930)

2. Generalization II [CORE] Peter Bartlett + Sasha Rakhlin · 2019 · 1:23:06 · https://www.youtube.com/watch?v=QO8HOqQ_6BQ

Builds Rademacher complexity and the norm-based route to capacity control — the first serious attempt to replace parameter counting with something that tracks reality. *Paper*: Bartlett, Foster, Telgarsky, “Spectrally-normalized margin bounds for neural networks” (arXiv:1706.08498)

3. Generalization III Peter Bartlett + Sasha Rakhlin · 2019 · 1:18:23 · <https://www.youtube.com/watch?v=8hZD5bMBbY4>

Margin-based bounds in full detail. This is the technical core of spectrally-normalized margin theory. *Paper*: same as above (arXiv:1706.08498)

4. Generalization IV [CORE] Peter Bartlett + Sasha Rakhlin · 2019 · 1:22:14 · <https://www.youtube.com/watch?v=n5Zxi22801Q>

Closes on interpolation and benign overfitting, which is the bridge from this section to §DLT-2. *Paper*: Bartlett, Montanari, Rakhlin, “Deep learning: a statistical viewpoint” (arXiv:2103.09177)

1.2 Implicit regularization and the implicit bias of gradient descent

The most important single idea in this module: the hypothesis class does not constrain the solution, **the optimizer does**. Gradient descent silently picks a minimum-norm or maximum-margin solution out of the infinite set of zero-training-error solutions, and that choice is the whole explanation.

5. Implicit Regularization I [CORE] Nati Srebro (TTIC) · 2019 · 1:16:51 · <https://www.youtube.com/watch?v=7uRVR9hsF0g>

The canonical lecture on why the algorithm, not the loss, picks the solution. If you watch one hour of this module, make it this one. *Paper*: Neyshabur, Tomioka, Srebro, “In search of the real inductive bias” (arXiv:1412.6614)

6. Implicit Regularization II Nati Srebro (TTIC) · 2019 · 1:22:54 · <https://www.youtube.com/watch?v=lkuOmf7ey14>

Characterizes the implicit bias concretely for matrix factorization and linear convolutional networks — where it can actually be computed. *Paper*: Gunasekar, Lee, Soudry, Srebro, “Implicit bias of gradient descent on linear convolutional networks” (arXiv:1806.00468)

7. Tutorial: Implicit Bias I [CORE] Nati Srebro (TTIC), Simons Deep Learning Theory Workshop & Summer School · 2022 · 1:20:40 · <https://www.youtube.com/watch?v=NeTU5TJOiY>

The rebuilt, three-years-wiser version of #5. If you only do one Srebro tutorial in 2026, do this pair instead of the 2019 pair. *Paper*: Gunasekar et al., “Characterizing implicit bias in terms of optimization geometry” (arXiv:1802.08246)

8. Tutorial: Implicit Bias II Nati Srebro (TTIC) · 2022 · 1:36:30 · <https://www.youtube.com/watch?v=8mO1CFfpbRE>

Extends to homogeneous networks, max-margin convergence, and the rich-versus-kernel transition that §DLT-3 depends on. *Paper*: Lyu & Li, “Gradient descent maximizes the margin of homogeneous neural networks” (arXiv:1906.05890)

9. Optimization’s Untold Gift to Learning: Implicit Regularization Nati Srebro (TTIC), Simons · 2017 · 1:01:19

· https://www.youtube.com/watch?v=YwX44phI_gc The original talk that put “norm, not size, controls capacity”

on the map. Historically the most important entry in this subsection. *Paper*: Neyshabur, Tomioka, Srebro, “Norm-based capacity control in neural networks” (arXiv:1503.00036)

10. Implications of the implicit bias in neural networks Gal Vardi (Weizmann/TTIC), One World Theoretical ML Seminar · 2022 · 0:58:28 · <https://www.youtube.com/watch?v=t9ZRPpVKt4c> The consequences nobody wanted: the same implicit bias that gives you generalization also gives you training-data reconstruction attacks and adversarial non-robustness. *Paper*: Vardi, “On the implicit bias in deep-learning algorithms” (arXiv:2208.12591)

1.3 Uniform convergence and its limits

This is a genuine live argument, and the two sides are both on video. Watch them back to back as one session.

11. Uniform convergence may be unable to explain generalization in deep learning (NeurIPS 2019 oral) Vaishnavh Nagarajan & Zico Kolter (CMU) · 2019 · 0:02:52 · <https://www.youtube.com/watch?v=o3GfnEjTdIQ> Three minutes, by the authors. The claim: *no* uniform-convergence bound, however clever, can explain this — the proof technique itself is the problem. *Paper*: Nagarajan & Kolter (arXiv:1902.04742)

12. In Defense of Uniform Convergence [CORE] Daniel M. Roy (Toronto), IAS · 2020 · 1:21:46 · <https://www.youtube.com/watch?v=6mdFQR0-WT0> The full-length rebuttal: derandomization recovers meaningful uniform-convergence bounds, and the impossibility result is narrower than it sounds. The best hour on YouTube for learning how theorists actually argue. *Paper*: Negrea, Dziugaite, Roy, “In defense of uniform convergence” (arXiv:1912.04265)

1.4 Margin theory

13. Margins, perceptrons, and deep networks [CORE] Matus Telgarsky (UIUC / Courant), IAS Theoretical ML Seminar · 2020 · 1:22:27 · <https://www.youtube.com/watch?v=yfk-IPlr6M0> Telgarsky’s own account, from the perceptron through to spectrally-normalized margin bounds. The cleanest single narrative of the margin thread. *Papers*: Bartlett, Foster, Telgarsky (arXiv:1706.08498); Ji & Telgarsky, “Gradient descent aligns the layers of deep linear networks” (arXiv:1810.02032)

14. A Primal-dual Analysis of Margin Maximization by Steepest Descent Methods Matus Telgarsky (NYU), Simons “Frontiers of Deep Learning” · 2019 · 0:41:42 · <https://www.youtube.com/watch?v=Rc76ZBP6fHM> The technical follow-through: *why* steepest descent maximizes margin, in duality language. *Paper*: Ji, Telgarsky, “Directional convergence and alignment in deep learning” (arXiv:2006.06657)

15. Optimisation and Generalization of deep networks Matus Telgarsky (UIUC), SMILES Summer School · 2019 · 1:17:16 · <https://www.youtube.com/watch?v=rv-vzl5-oIs> Summer-school pacing, covering optimization and generalization jointly. Good if the IAS talk moves too fast. *Paper*: Soudry et al., “The implicit bias of gradient descent on separable data” (arXiv:1710.10345)

16. A Perceptron Trio Matus Telgarsky (Courant/NYU), IPAM · 2024 · 0:50:45 · <https://www.youtube.com/watch?v=b5bYObH8rKs> The most current Telgarsky entry: margin/perceptron arguments reframed for shallow ReLU feature learning. *Paper*: Telgarsky, “Feature selection and low test error in shallow low-rotation ReLU networks” (arXiv:2208.02789)

17. Feature Selection with Gradient Descent on Two-layer Networks in Low-rotation Regimes Matus Telgarsky (NYU), Simons Deep Learning Theory Summer School · 2022 · 0:57:45 · <https://www.youtube.com/watch?v=bSJZQVNiLP0> The research-talk version of #16, with more proof detail. *Paper*: arXiv:2208.02789

1.5 PAC-Bayes for neural networks

The only framework that has produced **non-vacuous** generalization bounds for real networks on real data. That is a low bar and it is still the record.

18. Studying Generalization in Deep Learning via PAC-Bayes [CORE] Gintare Karolina Dziugaite (Element AI), Simons “Frontiers of Deep Learning” · 2019 · 0:44:39 · https://www.youtube.com/watch?v=_bYT2VOOxU0 The author on the first bounds for real deep networks that were not vacuous. This is the entry point. *Paper*: Dziugaite & Roy, “Computing nonvacuous generalization bounds for deep (stochastic) neural networks with many more parameters than training data” (arXiv:1703.11008)

19. Nonvacuous Generalization Bounds for Deep Neural Networks via PAC-Bayes Gintare Karolina Dziugaite, Borealis AI · 2018 · 0:26:02 · <https://www.youtube.com/watch?v=dHUH0hmKvs8> The earliest recorded version, closest to the paper. Use as a 26-minute primer. *Paper*: arXiv:1703.11008

20. PAC-Bayesian approaches to understanding generalization in deep learning Gintare Karolina Dziugaite, IAS · 2019 · 0:31:34 · <https://www.youtube.com/watch?v=MrUqeSj9Pgk> Condensed overview of the whole PAC-Bayes programme. *Paper*: arXiv:1703.11008

21. Distribution-dependent generalization bounds for noisy, iterative learning algorithms Gintare Karolina Dziugaite, MTL MLOpt Seminar · 2021 · 1:05:17 · <https://www.youtube.com/watch?v=iZC5H9dYVeI> The information-theoretic / SGLD line that succeeds naive PAC-Bayes. This is where the field actually went next. *Paper*: Negrea, Haghighat, Dziugaite, Khisti, Roy (arXiv:1911.02151)

22. Size of Teachers as a Measure of Data Complexity: PAC-Bayes Excess Risk Bounds and Scaling Laws Dan Roy (Toronto), IPAM · 2024 · 0:56:32 · <https://www.youtube.com/watch?v=pJqV2YNlkWc> The newest item here, and the one that connects PAC-Bayes directly to δ DLT-5 scaling laws. An underexplored seam. *Paper*: Roy et al., NeurIPS 2024

1.6 Overparameterization, compression, and reality checks

23. Toward Theoretical Understanding of Deep Learning (ICML 2018 tutorial) [CORE] Sanjeev Arora (Princeton/IAS) · 2018 · 2:13:56 · <https://www.youtube.com/watch?v=rcR6P5O8CpU> The two-hour tutorial that frames generalization, optimization and expressivity as one problem. Best single survey assignment in the module. *Paper*: Zhang, Bengio, Hardt, Recht, Vinyals, “Understanding deep learning requires rethinking generalization” (arXiv:1611.03530)

24. Why do deep nets generalize, that is, predict well on unseen data Sanjeev Arora (Princeton/IAS), DeepMath · 2018 · 0:56:16 · <https://www.youtube.com/watch?v=xscvWCC-y6U> The compression-based route to bounds — a genuinely distinct third approach alongside margins and PAC-Bayes. *Paper*: Arora, Ge, Neyshabur, Zhang, “Stronger generalization bounds for deep nets via a compression approach” (arXiv:1802.05296)

25. Is Optimization the Right Language to Understand Deep Learning? Sanjeev Arora (Princeton), DeepMath · 2020 · 1:03:01 · <https://www.youtube.com/watch?v=HMDJd2minAI> Arora arguing that the trajectory, not the endpoint, is the object of study. A useful corrective to the whole “find the minimum” framing. *Paper*: Arora, Cohen, Hazan, “On the optimization of deep networks: implicit acceleration by overparameterization” (arXiv:1802.06509)

26. Opening the black box: Toward mathematical understanding of deep learning Sanjeev Arora (Princeton), Harvard CMSA · 2020 · 0:57:58 · <https://www.youtube.com/watch?v=DbpJ6wTM5vQ> A broader, better-paced survey than #25 for a first pass.

27. Toward a Causal Analysis of Generalization in Deep Learning [CORE] Behnam Neyshabur (Google), IAS · 2019 · 0:33:17 · <https://www.youtube.com/watch?v=HafQ-Q8pn0g> The empirical reality check that should be watched immediately after any theory talk here: most proposed generalization measures fail when tested causally at scale. *Paper*: Jiang, Neyshabur, Mobahi, Krishnan, Bengio, “Fantastic generalization measures and where to find them” (arXiv:1912.02178)

28. The mystery of over-parametrization in neural networks Behnam Neyshabur, IAS · 2017 · 0:17:33 · <https://www.youtube.com/watch?v=4Mga1cERS34> Seventeen minutes stating the puzzle cleanly. Good pre-reading before the Srebro tutorials. *Paper*: Neyshabur et al., “Towards understanding the role of over-parametrization in generalization” (arXiv:1805.12076)

29. Learning and Generalization in Over-parametrized Neural Networks, Going Beyond Kernels Yuanzhi Li (Stanford), Simons · 2019 · 0:49:42 · <https://www.youtube.com/watch?v=NNPCk2gvTnI> The first serious results showing networks provably beat their own tangent kernel. The technical bridge to §DLT-3. *Paper*: Allen-Zhu, Li, Liang, “Learning and generalization in overparameterized neural networks, going beyond two layers” (arXiv:1811.04918)

30. Size-free Generalization Bounds for Convolutional Neural Networks Hanie Sedghi (Google Brain), Simons · 2019 · 0:34:34 · <https://www.youtube.com/watch?v=jRK3W48cO90> Norm-based bounds for convnets that do not grow with parameter count. *Paper*: Long & Sedghi, “Generalization bounds for deep convolutional neural networks” (arXiv:1905.12600)

31. Understanding Generalization from Pre-training Loss to Downstream Tasks Tengyu Ma (Stanford), Simons Modern Paradigms in Generalization Boot Camp · 2024 · 1:17:51 · https://www.youtube.com/watch?v=rFp0tVQW_q0 The modern reframing: in the foundation-model era the quantity to bound is downstream transfer, not in-distribution test error. This is where the open problems now are.

32. Training on the Test Set and Other Heresies Benjamin Recht (UC Berkeley), Simons · 2019 · 0:49:37 · <https://www.youtube.com/watch?v=NTz4rJS9BAI> The adaptive-overfitting reality check — do the benchmarks we measure generalization *on* still mean anything after a decade of reuse? *Paper*: Recht, Roelofs, Schmidt, Shankar, “Do ImageNet classifiers generalize to ImageNet?” (arXiv:1902.10811)

33. An Observation on Generalization Ilya Sutskever (OpenAI), Simons · 2023 · 0:57:21 · https://www.youtube.com/watch?v=AKMuA_TVz3A Compression-as-generalization argued from the practitioner side by someone who built the systems. Worth watching precisely because it does not sound like the rest of this section.

DLT-2 Double descent, benign overfitting, interpolation

The empirical discovery that broke the bias-variance picture, and the theory that grew up to explain it. The short version: past the interpolation threshold, more capacity makes test error go **down** again, and interpolating noisy data is not always fatal.

34. From Classical Statistics to Modern Machine Learning [CORE] Misha Belkin (UC San Diego), Simons “Frontiers of Deep Learning” · 2019 · 0:49:47 · <https://www.youtube.com/watch?v=OBCCiGnOJV8> Belkin presenting double descent close to its original publication. The founding talk of this section. *Paper*: Belkin, Hsu, Ma, Mandal, “Reconciling modern machine learning practice and the bias-variance trade-off” (arXiv:1812.11118)

35. From Classical Statistics to Modern ML: the Lessons of Deep Learning Mikhail Belkin, IAS · 2019 · 0:37:29 · <https://www.youtube.com/watch?v=5-Kqb80h9rk> Shorter companion to #34, same period.

- 36. Mikhail Belkin: From classical statistics to modern deep learning** Mikhail Belkin (UCSD), Oxford ML and Physics Seminars · 2022 · 0:58:00 · <https://www.youtube.com/watch?v=5-QjjOYfeSI> Three years on, with the interpolation story matured and transparent-kernel results included. *Paper*: Belkin, “Fit without fear: remarkable mathematical phenomena of deep learning through the prism of interpolation” (arXiv:2105.14368)
- 37. The elusive generalization: classical bounds to double descent to grokking [CORE]** Misha Belkin (UCSD), Simons Modern Paradigms in Generalization Boot Camp · 2024 · 1:19:56 · <https://www.youtube.com/watch?v=h2I0Hs2K2KI> The best single arc in this module: VC bounds → interpolation → double descent → grokking, as one continuous story by the person who started it. If you watch one Belkin talk, this is it. *Paper*: arXiv:1812.11118
- 38. Benign Overfitting in Linear Prediction [CORE]** Peter Bartlett (UC Berkeley), Simons · 2019 · 0:47:40 · <https://www.youtube.com/watch?v=GXpP-rXEDpk> The theorem that made “interpolating noise is sometimes harmless” a provable statement rather than an observation. Requires overparameterized linear regression with a specific eigenvalue decay — knowing *which* condition is the real content. *Paper*: Bartlett, Long, Lugosi, Tsigler, “Benign overfitting in linear regression” (arXiv:1906.11300)
- 39. Benign Overfitting** Peter Bartlett (UC Berkeley), TOPML Workshop · 2021 · 1:01:04 · <https://www.youtube.com/watch?v=o0tzfayJfVg> Longer and more self-contained than #38, two years later. *Paper*: arXiv:1906.11300
- 40. Benign Overfitting** (NeurIPS 2021 invited talk) Peter Bartlett (UC Berkeley) · 2021 · 1:30:04 · <https://www.youtube.com/watch?v=zNGm0iXWxYE> The plenary version — broadest framing, aimed at a general ML audience. *Paper*: Bartlett, Montanari, Rakhlin, “Deep learning: a statistical viewpoint” (arXiv:2103.09177)
- 41. Peter Bartlett: Benign overfitting — Lecture 1** Peter Bartlett (UC Berkeley), CIRM · 2022 · 1:08:15 · <https://www.youtube.com/watch?v=MZFXVGGB11E> Mathematics-institute pacing with the proofs actually done. Use this one if you intend to work in the area rather than just know about it.
- 42. Benign, Tempered, or Catastrophic: A Taxonomy of Overfitting [CORE]** Neil Mallinar (UCSD) & Preetum Nakkiran, Simons Deep Learning Theory Summer School · 2022 · 0:57:45 · <https://www.youtube.com/watch?v=cg9s7jpWgck> The necessary correction to #38: most real networks are not benign, they are *tempered*, and the three-way taxonomy is more useful than the binary. *Paper*: Mallinar, Simon, Abedsoltan, Pandit, Belkin, Nakkiran, “Benign, tempered, or catastrophic: a taxonomy of overfitting” (arXiv:2207.06569)
- 43. Is Overfitting Actually Benign? On the Consistency of Interpolating Methods** Simons · 2021 · 0:26:25 · <https://www.youtube.com/watch?v=aZL4mIvuYd4> Short, sharp skeptical take. Pairs with #42.
- 44. The Devil is in the Tails and Other Stories of Interpolation** Niladri Chatterji (OpenAI), Simons Deep Learning Theory Summer School · 2022 · 0:54:40 · <https://www.youtube.com/watch?v=e7Y3hgQlaaE> Where the benign-overfitting conditions come from and how fragile they are.
- 45. Reconsidering Overfitting in the Age of Overparameterized Models** Fanny Yang (ETH Zurich), Simons Modern Paradigms in Generalization Boot Camp · 2024 · 1:18:05 · <https://www.youtube.com/watch?v=g1Gb3UEgpOk> The current state of the question, including the robustness-vs-interpolation tension that is now the live part.
- 46. The Deep Bootstrap Framework: A New Lens to Understand Generalization** Preetum Nakkiran (Harvard), ICTP · 2021 · 0:19:54 · <https://www.youtube.com/watch?v=3hDWN7Ka23I> Twenty minutes that reframe generalization as a statement about *optimization speed* online versus offline. One of the genuinely original reframings of the last decade. *Papers*: “The Deep Bootstrap Framework” (arXiv:2010.08127); “Deep Double Descent” (arXiv:1912.02292)

- 47. An Empirical Theory of Deep Learning** (Gradient Podcast) Preetum Nakkiran · 2022 · 1:37:29 · <https://www.youtube.com/watch?v=EBSLLOJJJuc> Long-form interview. The only place Nakkiran lays out the methodological argument — that deep learning needs an empirical science, not just theorems — at length.
- 48. The Interpolation Phase Transition in Neural Networks: Memorization and Generalization under Lazy Training** Simons · 2020 · 1:06:39 · <https://www.youtube.com/watch?v=FNG8yTnZuOY> Connects interpolation directly to the lazy/NTK regime of §DLT-3 with precise asymptotics. *Paper*: Montanari, Zhong (arXiv:2007.12826)
- 49. Kernel and Deep Regimes in Overparameterized Learning** Suriya Gunasekar (Microsoft Research), Simons · 2019 · 0:46:15 · <https://www.youtube.com/watch?v=fUQCJNSdskA> Separates the kernel regime from the genuinely rich feature-learning regime — which determines which of the above bounds even apply. *Paper*: Woodworth et al., “Kernel and rich regimes in overparametrized models” (arXiv:2002.09277)
- 50. Kernel and Rich Regimes in Deep Learning** Nati Srebro (TTIC), IAS · 2019 · 0:37:01 · <https://www.youtube.com/watch?v=ZjTMiYhW4XY> Srebro’s short version of the same distinction. *Paper*: arXiv:2002.09277
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DLT-3 Neural tangent kernel and infinite-width limits

The one place deep learning becomes exactly solvable. Take width to infinity under the standard parametrization and the network becomes a linear model in a fixed kernel — trainable, analyzable, and **unable to learn features**. That last clause is the whole drama of this section: the tractable limit is the wrong limit, and finding a different scaling that keeps feature learning (muP) turned out to have enormous practical consequences.

3.1 The NTK and the wide limit

- 78. Neural Tangent Kernel: Convergence and Generalization in Neural Networks** (STOC 2021) Arthur Jacot, Franck Gabriel, Clément Hongler (EPFL) · 2021 · 0:22:25 · <https://www.youtube.com/watch?v=Yc5EMw3wDwg> The original authors on the founding paper. Twenty-two minutes, and it is the primary source — watch it first, then go to the longer lectures. *Paper*: Jacot, Gabriel, Hongler (arXiv:1806.07572)
- 79. The Wide limit of Neural Networks: NNGP and NTK [CORE]** Jascha Sohl-Dickstein (Google Brain), Understanding Deep Learning lecture series · 2021 · 1:16:51 · <https://www.youtube.com/watch?v=fcpI5z9q91A> The best self-contained lecture on both correspondences — infinite-width networks at initialization are Gaussian processes, and under gradient descent they evolve as linear models. If you learn NTK from one video, learn it from this one. *Papers*: Lee et al., “Deep neural networks as Gaussian processes” (arXiv:1711.00165); Lee et al., “Wide neural networks of any depth evolve as linear models under gradient descent” (arXiv:1902.06720)
- 80. Understanding overparameterized neural networks** Jascha Sohl-Dickstein (Google Brain), Oxford ML and Physics · 2021 · 1:18:22 · <https://www.youtube.com/watch?v=6C1hKqLe2Tc> Same material with more emphasis on *why* width makes networks analytically tractable at all. *Paper*: arXiv:1902.06720
- 81. Towards an Understanding of Wide Neural Networks** Yasaman Bahri (Google Brain), DeepMath · 2020 · 1:11:37 · https://www.youtube.com/watch?v=tECNnHxJH_M Frames the wide-network programme as a statistical-mechanics research agenda rather than a single theorem. Good for seeing where the open problems are.
- 82. Computation in Very Wide Neural Networks** Yasaman Bahri (Google Brain), Simons · 2019 · 0:48:40 · <https://www.youtube.com/watch?v=EWLQoO0SGvI> The compact research-talk version.
- 83. Neural Tangent Kernel** (Stanford CS229M Lecture 13) Tengyu Ma (Stanford) · 2022 · 1:29:30 · <https://www.youtube.com/watch?v=btphvvnad0A> The derivation done properly at blackboard pace, inside a

graduate course. Use this if you want to be able to *reproduce* the result rather than cite it.

84. On the Connection between Neural Networks and Kernels: a Modern Perspective Simon Du (University of Washington), IAS · 2019 · 0:30:46 · <https://www.youtube.com/watch?v=HvEGJUwQEO8> Thirty minutes on the optimization consequence: NTK is why gradient descent provably converges.

85. The Catapult phase of Neural Networks Guy Gur-Ari (Google), Understanding Deep Learning lecture series · 2021 · 1:07:20 · <https://www.youtube.com/watch?v=aPY-M4epEz4> Where the lazy/NTK picture visibly breaks: at large learning rate the loss spikes and the network lands somewhere the kernel theory cannot describe. The cleanest empirical wedge into the rich regime. *Paper*: Lewkowycz, Bahri, Dyer, Sohl-Dickstein, Gur-Ari (arXiv:2003.02218)

3.2 Lazy training versus feature learning, and mean-field limits

86. Analysis of Gradient Descent on Wide Two-Layer ReLU Neural Networks [CORE] Lénaïc Chizat (CNRS / Paris-Saclay), One World Theoretical ML · 2020 · 1:07:29 · <https://www.youtube.com/watch?v=cE-Q5jPRRik> The definitive treatment of the lazy-versus-rich dichotomy, by the person who named lazy training. This is the hinge of the whole section. *Papers*: Chizat, Oyallon, Bach, “On lazy training in differentiable programming” (arXiv:1812.07956); Chizat & Bach (arXiv:2002.04486)

87. Analysis of Gradient Descent on Wide Two-Layer Neural Networks (IHES) Lénaïc Chizat · 2021 · 0:38:55 · <https://www.youtube.com/watch?v=jVOTGfqfFKw> Tighter 39-minute version of #86. Use as the assigned short form.

88. On the Global Convergence of Gradient Descent for Over-parameterized Models using Optimal Transport Francis Bach (INRIA / ENS), Institut Henri Poincaré · 2019 · 1:00:51 · <https://www.youtube.com/watch?v=rQM5uh2EsHA> The optimal-transport formulation that launched the mean-field line in machine learning: training a wide two-layer net is Wasserstein gradient flow on a measure. *Paper*: Chizat & Bach (arXiv:1805.09545)

89. Mean Field Descriptions of Two Layers Neural Networks Andrea Montanari (Stanford), FODSI · 2019 · 1:05:39 · <https://www.youtube.com/watch?v=eMFqg-B0oPE> The distributional-dynamics / PDE view of SGD, from the author of the other founding mean-field paper. *Paper*: Mei, Montanari, Nguyen (arXiv:1804.06561)

90. Self-induced regularization from linear regression to neural networks [CORE] Andrea Montanari (Stanford), Harvard CMSA · 2020 · 1:06:11 · https://www.youtube.com/watch?v=bjRqmlI_SF Connects random-feature asymptotics to benign overfitting and answers directly *when kernels do and do not match real networks*. The bridge between §DLT-2 and §DLT-3. *Papers*: “When do neural networks outperform kernel methods?” (arXiv:2006.13409); “The generalization error of random features regression” (arXiv:1908.05355)

91. A mean-field theory of lazy training in two-layer neural nets Maxim Raginsky (UIUC) · 2020 · 0:49:38 · <https://www.youtube.com/watch?v=oWE9jFIEwUw> Casts lazy training as entropic-regularized stochastic control — the sharpest formal statement of what “lazy” actually means. *Paper*: arXiv:2002.01987

92. Trainability and accuracy of artificial neural networks Eric Vanden-Eijnden (NYU Courant) · 2020 · 1:11:07 · <https://www.youtube.com/watch?v=tcrt9y7ypQc> The interacting-particle-system formulation — the Rotskoff–Vanden-Eijnden half of the mean-field literature. *Paper*: arXiv:1805.00915

93. Mean-Field Theory of Two-Layers Neural Networks: Dimension Free Bounds and Kernel Limit Theodor Misiakiewicz (Stanford), Simons · 2021 · 0:49:55 · <https://www.youtube.com/watch?v=BMnxZaFeWNA> The quantitative, dimension-free version of the mean-field approximation. *Paper*: arXiv:1902.06015

94. A general framework for the mean field limit of multilayer neural networks Huy Tuan Pham (Stanford) · 2020 · 0:56:38 · <https://www.youtube.com/watch?v=3gxDiTEE-bM> Extends mean-field theory past two layers — the gap everyone notices immediately. *Paper:* arXiv:2001.11443

95. Learning Over-parameterized Neural Networks: From Neural Tangent Kernel to Mean-field Analysis Quanquan Gu (UCLA) · 2020 · 0:44:06 · <https://www.youtube.com/watch?v=zlqQ7VRba2Y> Explicitly bridges the two regimes in one lecture. Watch this as the synthesis session after the NTK and mean-field blocks. *Paper:* arXiv:1902.01384

3.3 Tensor Programs, muP, and the parametrization that actually matters

The highest practical payoff in this entire module. The argument is that NTK parametrization provably kills feature learning at infinite width, that a different scaling (maximal update parametrization) preserves it, and that under muP the optimal hyperparameters stop moving with width — so you can tune a small model and transfer to a huge one.

96. Feature Learning in Infinite-Width Neural Networks [CORE] Greg Yang (Microsoft Research), ELLIS UCL CSML Seminar · 2021 · 0:47:37 · <https://www.youtube.com/watch?v=DuBQCBWcq4M> The core theoretical argument, stated cleanly: NTK parametrization cannot learn features, muP can. Watch before the muTransfer talk. *Paper:* Yang & Hu, “Tensor Programs IV: Feature learning in infinite-width neural networks” (arXiv:2011.14522)

97. Tuning Large Neural Networks via Zero-Shot Hyperparameter Transfer [CORE] Greg Yang (Microsoft Research), AutoML Seminars · 2022 · 1:12:52 · <https://www.youtube.com/watch?v=XpU3mDKJOak> muTransfer: the clearest case in the field of infinite-width theory producing a practice-changing engineering recipe. Tune on a 40M-parameter proxy, transfer to 6.7B. *Paper:* Yang, Hu, Babuschkin et al., “Tensor Programs V” (arXiv:2203.03466)

98. The unreasonable effectiveness of mathematics in large scale deep learning Greg Yang (xAI), Sydney Mathematical Research Institute · 2023 · 1:08:48 · https://www.youtube.com/watch?v=_p0S_zZd4NQ The mature overview of the whole Tensor Programs arc, after it had been deployed at scale.

99. Large N Limits: Random Matrices & Neural Networks (Cartesian Cafe) Greg Yang with Timothy Nguyen · 2023 · 3:01:27 · <https://www.youtube.com/watch?v=1aXOXHA7Jcw> Three hours, blackboard-style, going through the actual mathematics with an interlocutor who pushes back. The deepest available treatment, and the only long-form one.

100. Infinite limits and scaling laws of neural networks [CORE] Blake Bordelon (Harvard), IPAM · 2024 · 0:59:29 · <https://www.youtube.com/watch?v=WcWFFiPRslM> The best single bridge from §DLT-3 to §DLT-5: dynamical mean field theory used to *derive* scaling-law exponents rather than fit them. *Papers:* Bordelon, Atanasov, Pehlevan, “A dynamical model of neural scaling laws” (arXiv:2402.01092); “Self-consistent dynamical field theory of kernel evolution in wide neural networks” (arXiv:2205.09653)

101. Infinite Limits and Scaling Laws for Deep Neural Networks Blake Bordelon (Harvard), Harvard CMSA · 2024 · 1:05:38 · <https://www.youtube.com/watch?v=0998FJhPdJ8> Longer version of #100 with the depth-scaling and residual-network material included. *Paper:* Bordelon, Noci, Li, Hanin, Pehlevan, “Depthwise hyperparameter transfer in residual networks” (arXiv:2309.16620)

102. Scaling Insights from Infinite-Width Theory for Next Gen Architectures & Learning Leena Vankadara (Amazon Research), IPAM · 2024 · 0:51:06 · <https://www.youtube.com/watch?v=2QwdaGIOjcw> “Beyond muP” — carries the parametrization analysis into state-space models and newer architectures. The open frontier. *Paper:* “On feature learning in structured state space models” (arXiv:2406.03529)

3.4 Finite-width corrections — the effective-theory programme

103. The Principles of Deep Learning Theory [CORE] Dan Roberts (MIT / Salesforce), IAS · 2021 · 1:20:17 · <https://www.youtube.com/watch?v=YzR2gZrsdJc> The $1/\text{width}$ perturbative expansion: infinite width is the free theory, and real networks are that plus corrections. A genuinely different intellectual approach, imported wholesale from theoretical physics. *Paper*: Roberts, Yaida, Hanin, “The Principles of Deep Learning Theory” (arXiv:2106.10165)

104. The Principles of Deep Learning Theory (Harvard CMSA version) Dan Roberts, Harvard CMSA · 2021 · 1:15:33 · <https://www.youtube.com/watch?v=wXZKoHEzASg> Same content, mathematics-department audience. Pick whichever framing suits you.

105. Random Neural Networks at Finite Width and Large Depth Boris Hanin (Princeton), Rocky Mountain Mathematical Physics Seminar · 2021 · 0:58:43 · https://www.youtube.com/watch?v=QLz_Lu9pOnQ The depth-to-width ratio as the real control parameter, showing the infinite-width Gaussian picture is only a perturbative first term. *Paper*: arXiv:2204.01058

106. Effective Theory of Transformers at Initialization Sho Yaida (Meta FAIR), IAIFI Summer Workshop · 2023 · 0:46:04 · <https://www.youtube.com/watch?v=BhpMsDbOI2c> Applies the effective theory specifically to transformers and derives initialization and hyperparameter scaling rules from it. *Paper*: arXiv:2304.02034

107. Statistical mechanics of deep learning Surya Ganguli (Stanford), IAS · 2019 · 0:29:55 · https://www.youtube.com/watch?v=-QF_jX8L0nw Thirty minutes placing this whole section in the physics tradition it came from.

108. Signal Propagation and Dynamical Isometry in Deep Neural Networks Understanding Deep Learning lecture series · 2021 · 1:06:50 · <https://www.youtube.com/watch?v=IwrpOhfhj9o> Why very deep networks are trainable at all — the random-matrix condition on the input-output Jacobian. (*Speaker is not listed in the video description.*) *Paper*: Pennington, Schoenholz, Ganguli, “Resurrecting the sigmoid in deep learning through dynamical isometry” (arXiv:1711.04735)

109. Disentangling Trainability and Generalization in Deep Neural Networks Lechao Xiao (Google Brain), Understanding Deep Learning lecture series · 2021 · 0:52:09 · <https://www.youtube.com/watch?v=4wiFeJOdro4> Shows the two properties are controlled by *different* spectral quantities — architectures can be trainable and not generalize, or the reverse. *Paper*: arXiv:1912.13053

DLT-4 Optimization theory for deep learning

Gradient descent on a nonconvex, non-smooth, million-dimensional loss should not work. This section is about the three reasons it does anyway: the landscape is not as bad as it looks, the dynamics are not what the textbook describes, and the step size is doing something the theory never modeled.

4.1 Edge of stability — the most important recent result

The textbook says you need a step size below $2/L$ for stability. Real training sits *exactly* at $2/\eta$ with sharpness pinned there, oscillating, and still converging. Every classical convergence proof is therefore inapplicable to actual deep learning. This is the cleanest “the theory is simply wrong” result in the field.

51. Jeremy Cohen (CMU AI Seminar) — Gradient Descent on Neural Networks Typically Occurs at the Edge of Stability [CORE] Jeremy Cohen (Carnegie Mellon) · 2021 · 0:58:18 · <https://www.youtube.com/watch?v=6xeh6gfESuc> The founding empirical paper, presented by its author. Sharpness rises until it hits $2/\eta$ and then

hovers there — “progressive sharpening” followed by stability at the edge. *Paper*: Cohen, Kaur, Li, Kolter, Talwalkar (arXiv:2103.00065)

52. Understanding Optimization in Deep Learning with Central Flows [CORE] Alex Damian (Princeton), One World Theoretical ML · 2026 · 0:55:15 · <https://www.youtube.com/watch?v=WopNCEV5BQ> The current best quantitative theory of edge-of-stability: a deterministic ODE (“central flow”) that actually predicts real training curves and the effect of learning-rate schedules. This is the state of the art as of 2026. *Paper*: Damian, Nichani, Lee et al. (arXiv:2410.24206) (*Harvard CMSA version of the same talk, 2025, 0:50:55*: <https://www.youtube.com/watch?v=04E8r76TetQ>)

53. Gradient Optimization Methods: The Benefits of a Large Step-size Peter Bartlett (UC Berkeley / Google DeepMind), University of Waterloo Distinguished Lecture · 2026 · 1:11:23 · https://www.youtube.com/watch?v=o7BdY9Qd6_g Proves the oscillatory non-monotone phase is a *feature*: large step sizes reach low loss faster and induce a specific implicit bias. *Paper*: Wu, Braverman, Bartlett et al., “Large stepsize gradient descent for logistic loss” (arXiv:2402.15926)

54. Learning threshold neurons via the Edge of Stability Kwangjun Ahn & Felipe Suarez (MIT), hosted by Sébastien Bubeck · 2023 · 0:32:50 · https://www.youtube.com/watch?v=_TpSDM8jG-Y A solvable toy model where edge-of-stability instability is the *mechanism* that creates useful features. Instability is not noise to be suppressed. *Paper*: Ahn, Bubeck, Chewi, Lee, Suarez, Zhang (arXiv:2212.07469)

4.2 Loss landscape geometry and mode connectivity

55. Loss Surfaces, fast ensembling and weight averaging of DNNs Timur Garipov · 2018 · 1:06:00 · <https://www.youtube.com/watch?v=GU2sQgLrTx8> The original mode-connectivity result by its first author: independently trained minima are connected by simple low-loss curves, which means the landscape is not a set of isolated basins. *Papers*: “Loss surfaces, mode connectivity, and fast ensembling of DNNs” (arXiv:1802.10026); SWA (arXiv:1803.05407)

56. Explaining Landscape Connectivity of Low-cost Solutions for Multilayer Nets [CORE] Rong Ge (Duke), Simons · 2019 · 0:45:11 · <https://www.youtube.com/watch?v=0kXDqS2OmeI> The theory behind #55 — dropout stability and noise stability as the conditions that *force* connectivity. Watch these two in order. *Paper*: Kuditipudi, Wang, Lee, Zhang, Li, Hu, Ge, Arora (arXiv:1906.06247)

57. Loss Landscape of Neural Networks Andrew Gordon Wilson (NYU), EPFL Virtual Symposium · 2022 · 0:40:26 · <https://www.youtube.com/watch?v=XHZvLeMkgDM> Mode connectivity and flat-basin averaging recast as a Bayesian statement about generalization. *Paper*: Wilson & Izmailov, “Bayesian deep learning and a probabilistic perspective of generalization” (arXiv:2002.08791)

58. Towards the Science of Deep Learning — The Loss Landscape Geometry Stanislav Fort (Stanford), GoodAI · 2020 · 0:38:53 · <https://www.youtube.com/watch?v=VOPviJBvNpw> The high-dimensional wedge/radial picture, and why deep ensembles explore genuinely different functions while SWA-style averaging does not. *Paper*: Fort, Hu, Lakshminarayanan, “Deep ensembles: a loss landscape perspective” (arXiv:1912.02757)

59. Optimization Landscape and Two-Layer Neural Networks Rong Ge (Duke), IAS · 2019 · 0:58:27 · <https://www.youtube.com/watch?v=dmfyYXl0xUY> Where the landscape is provably benign — all local minima global, strict saddles — and exactly how far that extends.

60. Toward a theory of optimization for deep learning Misha Belkin (UCSD), DeepMath · 2020 · 1:04:55 · <https://www.youtube.com/watch?v=gM-FLiBNg5A> The PL-condition / transition-to-linearity argument: overparameterization makes the landscape effectively convex-like along the trajectory, which is why optimization

is easy at all. *Paper*: Liu, Zhu, Belkin, “Loss landscapes and optimization in over-parameterized non-linear systems” (arXiv:2003.00307)

4.3 Sharpness, flat minima, and implicit bias of SGD

61. Sharpness-Aware Minimization (SAM): Current Method and Future Directions Hossein Mobahi (Google Research), ELLIS UCL CSML Seminar · 2022 · 0:53:56 · <https://www.youtube.com/watch?v=QBiLph-r5Hw> From a SAM co-author: why minimizing a neighborhood-maximum rather than the loss itself buys generalization, and what remains unexplained about it. *Paper*: Foret, Kleiner, Mobahi, Neyshabur (arXiv:2010.01412)

62. Implicit Bias of SGD for Diagonal Linear Networks: a Provable Benefit of Stochasticity Nicolas Flammarion (EPFL) · 2022 · 1:04:30 · <https://www.youtube.com/watch?v=FQ7w3ubKDoU> A clean separation theorem: SGD noise provably biases toward sparser solutions than gradient flow. One of the few places stochasticity is shown to help rather than merely be tolerated. *Paper*: Pesme, Pillaud-Vivien, Flammarion (arXiv:2106.09524)

63. Implicit Regularization effect of the noise (Stanford CS229M Lecture 17) Tengyu Ma (Stanford) · 2022 · 1:32:15 · <https://www.youtube.com/watch?v=60GqpISctCU> The lecture-course treatment of SGD noise as an implicit regularizer, derived rather than asserted.

4.4 Why Adam works, and the preconditioned-optimizer wave

The most practically consequential open question in optimization theory right now. Adam has worked for a decade with no satisfying explanation; Shampoo, SOAP and Muon have reopened the question.

64. Understanding Adam Optimizer via Online Learning of Updates: Adam is FTRL in Disguise [CORE] Kwangjun Ahn (MIT), ICML 2024 · 2024 · 0:14:16 · <https://www.youtube.com/watch?v=AU39SNkkIsA> Fourteen minutes containing the sharpest current answer: Adam’s update rule is exactly Follow-The-Regularized-Leader applied to the sequence of updates. Short — pair with #65. *Paper*: Ahn, Zhang, Singh, Sra (arXiv:2402.01567)

65. Making sense of training large AI models (MIT EECS thesis defense) Kwangjun Ahn (MIT) · 2024 · 0:40:33 · <https://www.youtube.com/watch?v=5rgrB7TGPdc> A single coherent pass over edge-of-stability, SAM, and Adam-as-FTRL from one of the most active people in this area.

66. Scalable second order optimization for deep learning (JAX Meetup) Rohan Anil (Google Brain) · 2022 · 1:28:15 · <https://www.youtube.com/watch?v=YDL8NXIS8hA> The Distributed Shampoo author on making a full-matrix preconditioner run at scale. The direct engineering ancestor of SOAP and Muon. *Paper*: Anil, Gupta, Koren, Regan, Singer (arXiv:2002.09018)

67. Depths of First Order Optimization [CORE] Jeremy Bernstein (MIT CSAIL), Cohere Labs · 2025 · 0:47:31 · <https://www.youtube.com/watch?v=4OAiakkmKQs> Reframes Shampoo, muP and Muon as one idea — steepest descent under a well-chosen norm — and introduces modular duality. This is the conceptual core of the entire modern optimizer wave, and the most intellectually satisfying talk in §DLT-4. *Papers*: “Old optimizer, new norm” (arXiv:2409.20325); “Modular duality in deep learning” (arXiv:2410.21265)

68. Metrized Deep Learning (MIT 6.7960, Lecture 23) Jeremy Bernstein (MIT), MIT OpenCourseWare · course Fall 2024, posted 2026 · 1:07:50 · <https://www.youtube.com/watch?v=zBvsoxC6tAo> The same material at lecture pace with derivations. Use this if #67 moves too fast.

69. Why Muon Is Good but May Not Be Optimal Weijie Su (University of Pennsylvania), CRUNCH Group · 2026 · 1:00:45 · <https://www.youtube.com/watch?v=GBsPpmh1vo4> Separates curvature-anisotropy preconditioning

(Adam) from gradient-anisotropy preconditioning (Muon), and argues orthogonalization is directionally right but not optimal. The most current word on Muon theory. *Papers*: arXiv:2505.21799, arXiv:2511.00674

70. The Muon Optimizer and Non-Euclidean gradient descent Robert Gower (Flatiron Institute), IMPA · 2026 · 0:59:45 · <https://www.youtube.com/watch?v=phM8GzT7ZV4> The convergence-theory view of Muon from the optimization-theory side rather than the deep-learning side.

71. Memory-Efficient Adaptive Optimization Yoram Singer (Princeton/Google), FODSI · 2019 · 0:52:43 · <https://www.youtube.com/watch?v=kN5KMiIqpEw> The SM3 work, and useful historical context for why full preconditioners were abandoned and then revived. *Paper*: Anil, Gupta, Koren, Singer (arXiv:1901.11150)

4.5 Convergence theory for overparameterized networks

72. Recent Developments in Over-parametrized Neural Networks, Part I [CORE] Jason Lee (USC), Simons Deep Learning Boot Camp · 2019 · 1:14:29 · <https://www.youtube.com/watch?v=uC2IGoTE2u4> The tutorial on why gradient descent provably finds global minima in the overparameterized regime. The companion lecture to Srebro’s implicit-bias pair. *Paper*: Du, Lee, Li, Wang, Zhai, “Gradient descent finds global minima of deep neural networks” (arXiv:1811.03804)

73. Recent Developments in Over-parametrized Neural Networks, Part II Jason Lee (USC), Simons · 2019 · 1:13:57 · <https://www.youtube.com/watch?v=NGon2JyjO6Y> Continues into the limits of the kernel argument and what lies beyond it.

74. On the Foundations of Deep Learning: SGD, Overparameterization, and Generalization Jason Lee (USC), Simons · 2019 · 0:45:38 · <https://www.youtube.com/watch?v=l0im8AJAMco> The research-talk condensation of #72–73.

75. Analyzing Optimization and Generalization in Deep Learning via Trajectories of Gradient Descent Nadav Cohen (IAS / Tel Aviv), Simons · 2019 · 0:46:27 · <https://www.youtube.com/watch?v=Lmj2bU9MdwM> The trajectory-based alternative to landscape analysis — and the implicit acceleration result for deep linear networks. *Paper*: Arora, Cohen, Hazan (arXiv:1802.06509)

76. How Over-Parameterization Slows Down Gradient Descent Simon Du (University of Washington) · 2024 · 0:48:34 · <https://www.youtube.com/watch?v=W8mVZzWI5gM> The counterweight: overparameterization is not free, and the rate degradation is quantifiable.

77. The Large Learning Rate Phase of Deep Learning Yasaman Bahri (Google Brain) · 2020 · 0:36:06 · <https://www.youtube.com/watch?v=gBFmS8qyuFQ> Identifies a sharp phase boundary in learning-rate space with qualitatively different training behavior on each side. The link between §DLT-4 and §DLT-3. *Paper*: Lewkowycz, Bahri, Dyer, Sohl-Dickstein, Gur-Ari, “The large learning rate phase of deep learning: the catapult mechanism” (arXiv:2003.02218)

DLT-5 Scaling laws

Loss falls as a power law in parameters, data, and compute, over many orders of magnitude, with exponents that are stable across architectures. Nobody fully knows why. This is simultaneously the most economically consequential empirical regularity in the field and one of its least explained.

Watch this section in the order given: the empirical result, then the compute- optimal correction, then the theory, then the emergence argument — because the emergence debate only makes sense once you know what a scaling law actually claims.

5.1 The empirical result and compute-optimal training

110. Neural Scaling Laws and GPT-3 [CORE] Jared Kaplan (Johns Hopkins / Anthropic), Initiative for the Theoretical Sciences · 2020 · 1:15:20 · https://www.youtube.com/watch?v=sNfkZfVm_xs The original scaling-laws paper by its lead author, with the physicist's framing of why power laws should be expected at all. Start here. *Paper:* Kaplan, McCandlish et al., "Scaling laws for neural language models" (arXiv:2001.08361)

111. Neural Scaling Laws and GPT-3 (Physics Meets ML) Jared Kaplan (Johns Hopkins) · 2020 · 1:35:49 · https://www.youtube.com/watch?v=QMqPAM_knrE The longest version, with extended Q&A covering the data-manifold-dimension derivation — an attempt at an actual mechanism for the exponent. *Paper:* Sharma & Kaplan, "A neural scaling law from the dimension of the data manifold" (arXiv:2004.10802)

112. Scaling Language Models (Stanford CS224N guest lecture) Jared Kaplan (Anthropic) · 2022 · 1:14:49 · <https://www.youtube.com/watch?v=UFem7xa3Q2Q> The best-paced classroom version, post-Chinchilla. If you only assign one Kaplan lecture, assign this one.

113. Scaling laws (Stanford CS336, Lecture 9) [CORE] Stanford CS336 "Language Modeling from Scratch" · 2025 · 1:05:18 · <https://www.youtube.com/watch?v=6Q-ESEmDf4Q> The most rigorous available treatment of Chinchilla — all three of its estimation approaches, and the places where the fits disagree with each other. This is the Chinchilla lecture, since no DeepMind author talk exists on video. *Paper:* Hoffmann, Borgeaud, Mensch et al., "Training compute-optimal large language models" (arXiv:2203.15556)

114. How to train an LLM Samuel L. Smith (Google DeepMind), IPAM · 2024 · 0:53:16 · <https://www.youtube.com/watch?v=GfAT2zkB6-U> A Chinchilla co-author on how compute-optimal reasoning is actually used inside a frontier lab, as opposed to how it reads in the paper. *Paper:* arXiv:2203.15556

115. (Mis)Fitting: A Survey of Scaling Laws [CORE] Sneha Kudugunta (Google DeepMind), Cohere For AI · 2026 · 0:54:55 · <https://www.youtube.com/watch?v=ggz4iaQpzcY> A methodological audit showing how much published scaling-law practice is underspecified or irreproducible. The necessary skeptical companion to everything above, and the most current item in this section. *Paper:* "(Mis)Fitting: A survey of scaling laws" (ICLR 2025)

116. Scaling Data-Constrained Language Models Niklas Muennighoff (Stanford / Contextual AI), Cohere For AI · 2024 · 1:00:01 · <https://www.youtube.com/watch?v=ILV-g-rGPhk> Extends Chinchilla into the repeated-data regime — the constraint that actually binds now that high-quality web text is exhausted. *Paper:* arXiv:2305.16264

117. Scaling LLM Test-Time Compute Charlie Snell (UC Berkeley / Google DeepMind) · 2024 · 0:53:31 · <https://www.youtube.com/watch?v=OXwGp9YeuBg> The inference-compute scaling law, and its tradeoff against pretraining compute. The newest axis in the field and the one underlying reasoning models. *Paper:* Snell, Lee, Xu, Kumar (arXiv:2408.03314)

5.2 Deriving the exponents

118. Explaining Neural Scaling Laws [CORE] Jaehoon Lee (Google Brain), Understanding Deep Learning lecture series · 2021 · 0:56:12 · <https://www.youtube.com/watch?v=A8F4Qga3NaM> The variance-limited versus resolution-limited taxonomy — the foundational theory talk for this section, and the first serious derivation rather than fit. *Paper:* Bahri, Dyer, Kaplan, Lee, Sharma, "Explaining neural scaling laws" (arXiv:2102.06701)

119. Understanding the Origins and Taxonomy of Neural Scaling Laws [CORE] Yasaman Bahri (Google Brain / Stanford), Simons Institute · 2023 · 1:05:24 · <https://www.youtube.com/watch?v=MUVFuZpxLU8> The updated and most complete presentation of the four-regime taxonomy, by the paper's first author. *Paper:* arXiv:2102.06701

120. Dynamics and scaling laws in deep learning Yasaman Bahri (Google Brain), Yale Institute for Network Science · 2021 · 1:07:08 · <https://www.youtube.com/watch?v=WbHfM774bKc> Pairs the scaling-law derivation directly with the wide-network dynamics of §DLT-3.

121. Toward a Theory of Neural Scaling Laws [CORE] Cengiz Pehlevan (Harvard / Kempner Institute), KUIS AI · 2026 · 0:58:59 · <https://www.youtube.com/watch?v=8kax6scOskM> The most current first-principles derivation of scaling exponents from solvable random-feature and kernel models. This is the state of the art as of 2026. *Papers:* Atanasov, Zavatone-Veth, Pehlevan, “Scaling and renormalization in high-dimensional regression” (arXiv:2405.00592); Bordelon, Atanasov, Pehlevan (arXiv:2402.01092)

122. Scaling Neural Networks: Laws and Limits Cengiz Pehlevan (Harvard / Kempner), Principles of Intelligence · 2026 · 0:53:03 · <https://www.youtube.com/watch?v=PC-eaqv5EnI> Companion to #121, extending the solvable-model programme to the emergence of in-context learning.

123. The Quantization Model of Neural Scaling [CORE] Eric Michaud (MIT), SLT Summit · 2023 · 0:45:19 · <https://www.youtube.com/watch?v=qSw75ix83r4> The discrete “quanta” account: smooth aggregate power laws and sharp per-task jumps come out of the *same* mechanism. This single talk unifies §5.1 and §5.3, and it is the most intellectually satisfying item in §DLT-5. *Paper:* Michaud, Liu, Girit, Tegmark (arXiv:2303.13506)

124. The Mathematics of Scaling Laws and Model Collapse in AI Elvis Dohmatob (Meta FAIR), IPAM · 2024 · 1:03:19 · <https://www.youtube.com/watch?v=qZVJuFv-wyw> Derives how scaling laws deform when training on synthetic data. A genuinely new theoretical regime, and increasingly the practical one. *Paper:* Dohmatob, Feng, Yang, Charton, Kempe, “A tale of tails: model collapse as a change of scaling laws” (arXiv:2402.07043)

125. When is Scale Enough? Ethan Dyer (Google Research, Blueshift), Simons Deep Learning Theory Summer School · 2022 · 1:12:25 · <https://www.youtube.com/watch?v=Jpu3kQv39L4> The question posed directly, from inside the group that produced the scaling-law theory papers.

5.3 Emergence — and whether it is real

A clean, well-documented scientific dispute with both sides on video. Watch #126 then #128 back to back; it is the best available case study in how an empirical claim in ML gets contested.

126. Emergence in Large Language Models [CORE] Jason Wei (Google Brain / OpenAI), Generative AI at MIT · 2023 · 0:57:32 · <https://www.youtube.com/watch?v=0SuyDLjNR9g> The canonical statement of the emergent-abilities claim by its first author: some capabilities appear abruptly at scale and are unpredictable from smaller models. *Paper:* Wei, Tay, Bommasani et al., “Emergent abilities of large language models” (arXiv:2206.07682)

127. Scaling unlocks emergent abilities in language models Jason Wei (Google Brain), USC ISI · 2023 · 1:00:11 · https://www.youtube.com/watch?v=Z_Qt737HG-0 Alternative version with more chain-of-thought and instruction-tuning material.

128. Investigating emergent abilities and challenging dominant research ideas [CORE] Rylan Schaeffer (Stanford), Imbue · 2024 · 1:03:16 · <https://www.youtube.com/watch?v=blX2RzZYeJo> The rebuttal by its first author: emergence is largely an artifact of choosing discontinuous metrics, and switching to continuous ones makes the jumps disappear. Teach immediately after Wei. *Paper:* Schaeffer, Miranda, Koyejo, “Are emergent abilities of large language models a mirage?” (arXiv:2304.15004)

129. Are Emergent Behaviors in LLMs an Illusion? (TWIML podcast) Sanmi Koyejo (Stanford) · 2024 · 1:05:25 · https://www.youtube.com/watch?v=3BQ9_b8JAMU The senior author’s framing of the metric critique and what it implies for capability forecasting and AI safety. Podcast format, but substantive. *Paper:* arXiv:2304.15004

DLT-6 Science-of-deep-learning phenomena

This section is different in kind from the rest. These are not theorems but **reproducible empirical phenomena** that any correct theory will have to explain, and several of them were found by people who were simply looking carefully. That is a useful thing to notice if you intend to contribute: the highest-impact results here came from measurement, not proof.

6.1 Grokking and phase transitions in training

130. Progress Measures for Grokking via Mechanistic Interpretability [CORE] Neel Nanda (DeepMind) · 2023 · 0:43:10 · <https://www.youtube.com/watch?v=IHikLL8ULa4> The author reverse-engineering the actual Fourier-multiplication algorithm a grokked network learns, and showing that the apparently sudden generalization is gradual underneath once you measure the right thing. *Paper:* Nanda, Chan, Lieberum, Smith, Steinhardt (arXiv:2301.05217)

131. A solvable model of the grokking transition in neural networks (Part 1) Andrey Gromov (UMD), Leinweber Institute for Theoretical Physics · 2023 · 1:11:10 · <https://www.youtube.com/watch?v=-EJn3xnmJXY> A physicist's analytically solvable two-layer model where the transition can be *computed* rather than observed. *Paper:* Gromov, "Grokking modular arithmetic" (arXiv:2301.02679)

132. A solvable model of the grokking transition (Part 2) Andrey Gromov · 2023 · 1:06:35 · <https://www.youtube.com/watch?v=fCE55iC-HaY> Continues the derivation.

133. Emergence and grokking in "simple" architectures Misha Belkin (UCSD), Simons "Transformers as a Computational Model" · 2024 · 0:52:40 · <https://www.youtube.com/watch?v=dzpA30qZxYw> Grokking in architectures simple enough to analyze, which strongly suggests it is not a transformer-specific phenomenon.

134. You Know It Or You Don't: Compositionality and Phase Transitions in LMs [CORE] Naomi Saphra (Kempner Institute, Harvard), Simons · 2025 · 0:57:00 · <https://www.youtube.com/watch?v=WJ89r5x5hDA> Training is not smooth. Abilities arrive in discrete breakthroughs, and this talk gives methods for detecting them rather than just noticing them afterward. *Paper:* Chen, Saphra et al., "Sudden drops in the loss: syntax acquisition, phase transitions, and simplicity bias in MLMs" (arXiv:2309.07311)

135. Interpreting Training (MIT EI Seminar) Naomi Saphra · 2024 · 1:04:20 · <https://www.youtube.com/watch?v=J0tHAZlFGSc> The broader argument that the training *trajectory*, not the final checkpoint, is the right object of interpretability study.

136. The Structure and Development of Neural Networks Jesse Hoogland (Timaeus), TAIS 2024 · 2024 · 0:32:40 · <https://www.youtube.com/watch?v=lbHDbV53-sk> The developmental-interpretability programme: singular learning theory used to detect discrete developmental stages during transformer training. A genuinely different mathematical toolkit. *Paper:* Hoogland, Wang, Farrugia-Roberts et al., "The developmental landscape of in-context learning" (arXiv:2402.02364)

137. Dynamics of Concept Learning and Emergent Abilities in Neural Networks Ekdeep Singh Lubana (Harvard), TTIC · 2025 · 1:03:31 · <https://www.youtube.com/watch?v=nYGyeJRHqEA> Formal models of how data scaling produces sudden emergence, with in-context learning treated as competition between algorithms the model could run. *Paper:* "Competition dynamics shape algorithmic phases of in-context learning" (arXiv:2412.01003)

6.2 Induction heads and in-context learning theory

138. Transformer Circuits, Induction Heads, In-Context Learning (Stanford CS25) [CORE] Chris Olah (Anthropic) · 2022 · 0:59:34 · https://www.youtube.com/watch?v=pC4zRb_5noQ The primary-source lecture on induction heads and the sharp, visible bump in the loss curve their formation creates. One of the few places where a specific circuit has been tied to a specific capability. *Paper*: Olsson, Elhage, Nanda, Olah et al., “In-context learning and induction heads” (arXiv:2209.11895)

139. A Walkthrough of In-Context Learning and Induction Heads (Part 1 of 2) Neel Nanda, with Charles Frye · 2022 · 1:03:51 · <https://www.youtube.com/watch?v=dCkQQYwPxDM> A line-by-line technical walkthrough of the same paper by a co-author. The ideal companion to #138 if you want to actually understand the evidence rather than the conclusion.

140. What Learning Algorithm is In-Context Learning? [CORE] Jacob Andreas (MIT), Harvard CMSA · 2023 · 0:50:16 · <https://www.youtube.com/watch?v=UNVl64G3BzA> Evidence that in-context learning on linear tasks implements something close to gradient descent or ridge regression *inside the forward pass*. If true, the forward pass is running a learning algorithm. *Paper*: Akyürek, Schuurmans, Andreas, Ma, Zhou (arXiv:2211.15661)

141. In-Context Learning: A Case Study of Simple Function Classes Gregory Valiant (Stanford), Simons · 2023 · 1:03:40 · <https://www.youtube.com/watch?v=DiJsg93zQDc> The controlled-function-class methodology that turned in-context learning from anecdote into measurement. *Paper*: Garg, Tsipras, Liang, Valiant (arXiv:2208.01066)

142. Toward Understanding In-context Learning Tengyu Ma (Stanford), Simons LLM Boot Camp · 2024 · 1:29:35 · <https://www.youtube.com/watch?v=hxrR39mAlR4> The ninety-minute tutorial version, with the theory laid out systematically.

143. Learning Theory of Transformers: Generalization and Optimization of In-Context Learning Taiji Suzuki (University of Tokyo), Simons · 2024 · 0:45:35 · <https://www.youtube.com/watch?v=WyeomuU2vQw> Actual generalization bounds for in-context learning — the formal end of this thread.

144. Associative memories as a building block in Transformers Alberto Bietti (Flatiron Institute), Simons · 2024 · 0:38:37 · <https://www.youtube.com/watch?v=ncAhx70jTic> A mechanistic account of what transformer weights are *storing*, which underwrites the induction-head picture. *Paper*: Bietti, Cabannes, Bouchacourt, Jegou, Bottou, “Birth of a transformer: a memory viewpoint” (arXiv:2306.00802)

6.3 Memorization versus generalization

145. Chasing the Long Tail: What Neural Networks Memorize and Why [CORE] Vitaly Feldman (Apple ML Research), Simons · 2022 · 0:51:40 · https://www.youtube.com/watch?v=w_BUN5tPiuA Proves that memorizing outliers is *necessary* for near-optimal generalization on long-tailed data. Memorization is not a failure mode; it is part of the mechanism. This result should change how you read every privacy and copyright argument in ML. *Papers*: “Does learning require memorization? A short tale about a long tail” (arXiv:1906.05271); Feldman & Zhang, “What neural networks memorize and why” (arXiv:2008.03703)

146. Does Learning Require Memorization? A Short Tale about a Long Tail Vitaly Feldman (Google Research), FODSI · 2019 · 0:49:53 · <https://www.youtube.com/watch?v=YWy2Iwn-1S8> The earlier version, closer to the original theorem.

147. On Memorization of Large Language Models in Logical Reasoning Chiyuan Zhang (Google Research), Simons · 2024 · 0:48:05 · <https://www.youtube.com/watch?v=eULIf02frIw> From the author of “Understanding

deep learning requires rethinking generalization,” now asking the same question about reasoning benchmarks. Directly relevant to whether any reasoning evaluation means what it claims.

148. Understanding the abilities of AI systems: Memorization, generalization, and points in between Tom McCoy (Yale), Simons · 2024 · 0:44:25 · <https://www.youtube.com/watch?v=3c9TiKryTtA> The “embers of autoregression” argument: apparent reasoning ability is strongly modulated by training-distribution probability.

149. Threat Models for Memorization: Privacy, Copyright, and Everything In-Between Michael Aerni (ETH Zürich), Google TechTalks · recorded 2025, posted 2026 · 0:49:30 · <https://www.youtube.com/watch?v=MdZjXUHIq4> The measurement side: how much verbatim training text appears in ordinary, non-adversarial model output. The numbers are higher than most people assume. *Paper*: Aerni, Rando, Carlini, Tramèr (arXiv:2411.10242)

150. Generalization in the representations and computations of frontier language models Joshua Batson (Anthropic), Simons · 2024 · 0:52:05 · <https://www.youtube.com/watch?v=2xb3mhIqjLw> What generalization looks like when you can open the model up and look — the interpretability perspective on this section’s question.

151. Weak-to-Strong Generalization Pavel Izmailov (Anthropic), Simons · 2024 · 0:43:25 · <https://www.youtube.com/watch?v=VViyQRGxSKo> A strong model supervised by a weak one outperforms its supervisor. Nobody has a satisfying theory for this, and it is load-bearing for alignment plans. *Paper*: Burns, Izmailov, Kirchner et al. (arXiv:2312.09390)

6.4 Lottery tickets, neural collapse, loss of plasticity

152. The Lottery Ticket Hypothesis: On Sparse, Trainable Neural Networks [CORE] Jonathan Frankle (MIT), ELLIS UCL CSML Seminar · 2020 · 0:54:19 · <https://www.youtube.com/watch?v=dYKiDwUEbCM> The author’s own full account of winning tickets, rewinding, and what sparsity reveals about what training actually requires. The best single version. *Paper*: Frankle & Carbin (arXiv:1803.03635)

153. The Lottery Ticket Hypothesis (MIT EI Seminar) Michael Carbin (MIT) · 2020 · 1:05:47 · https://www.youtube.com/watch?v=0cU8r6dgD_A The co-author’s version, with more on the stability-and-rewinding follow-up work.

154. Neural Collapse (MIT 9.520, Class 25) [CORE] Vardan Papayan, X. Y. Han, David Donoho · 2020 · 1:44:55 · <https://www.youtube.com/watch?v=D5eP0CWta8k> All three original authors presenting the simplex-equianangular-tight-frame collapse of last-layer features in the terminal phase of training. The definitive source, and a strikingly clean empirical regularity. *Paper*: Papayan, Han, Donoho, PNAS 2020 (arXiv:2008.08186)

155. Principles of Deep Representation Learning via Neural Collapse Qing Qu (Michigan) · 2022 · 1:09:28 · <https://www.youtube.com/watch?v=IxzRVhQr3kl> The landscape-analysis follow-up that explains *why* collapse occurs, via the unconstrained-features model.

156. Maintaining Plasticity in Deep Continual Learning [CORE] Richard Sutton (Alberta / DeepMind), CoLLAs · 2022 · 1:11:20 · https://www.youtube.com/watch?v=p_zknyfV9fY Backpropagation demonstrably loses the ability to learn over long training runs — networks go dead. This is one of the most under-appreciated results in the field and directly limits continual and online learning. *Paper*: Dohare, Hernandez-Garcia, Rahman, Mahmood, Sutton, “Loss of plasticity in deep continual learning” (arXiv:2306.13812; Nature 2024)

157. Maintaining Plasticity in Deep Continual Learning (Amii seminar) Shibhansh Dohare (Alberta) · 2023 · 0:54:25 · https://www.youtube.com/watch?v=oA_XLqh4Das The first author’s version, with more experimental detail and the continual-backprop algorithm.

158. Why do neural networks lose plasticity? Clare Lyle (DeepMind), CoLLAs · 2023 · 1:01:29 · <https://www.youtube.com/watch?v=1EwKYesnKAA> The mechanistic diagnosis — curvature collapse, dead units, and which interventions actually restore plasticity. *Paper*: Lyle, Zheng, Nikishin et al., “Understanding plasticity in neural networks” (arXiv:2303.01486)

DLT-7 Expressivity and architecture theory

What a given architecture *can* represent, independent of whether training finds it. This is the oldest and most mathematically settled part of the module, and also the one that has recently become urgent again: the question “can a transformer do this at all?” turns out to have crisp answers in circuit complexity, and those answers explain real failures.

A channel worth knowing about. Most of the transformer-expressivity entries below come from the **Formal Languages and Neural Networks Seminar** (<https://www.youtube.com/channel/UCrp8k-nSuMKHM4sSUvIPdAw>), which holds roughly 120 hour-long talks, nearly all given by the first author of the paper. It is the single best-concentrated archive for this subfield and is largely unknown outside it.

7.1 Universal approximation and approximation rates

159. Deep Neural Networks: From Approximation to Expressivity [CORE] Gitta Kutyniok (LMU Munich), Isaac Newton Institute · 2025 · 0:58:10 · <https://www.youtube.com/watch?v=Co5jT2FME6g> The cleanest modern survey: classical universal approximation, then optimal approximation rates with sparse connectivity, then where expressivity stops being the binding constraint. *Papers*: Bölcskei, Grohs, Kutyniok, Petersen (arXiv:1705.01714); “The modern mathematics of deep learning” (arXiv:2105.04026)

160. Approximation with neural networks of minimal size Dmitry Yarotsky (Skoltech), HSE Computer Science Colloquium · 2022 · 1:17:46 · <https://www.youtube.com/watch?v=xkYd411hdCs> The definitive long-form treatment of how many weights a ReLU network actually needs, including the strange super-expressive regimes. *Paper*: Yarotsky, “Error bounds for approximations with deep ReLU networks” (arXiv:1610.01145)

161. Optimal approximation of continuous functions by very deep ReLU networks (COLT 2018) Dmitry Yarotsky · 2018 · 0:10:33 · <https://www.youtube.com/watch?v=Rsw1lQdCs1o> Ten minutes stating the depth-versus-width rate tradeoff precisely. Watch before #160. *Paper*: arXiv:1802.03620

162. Expressive Power of Narrow Networks Sejun Park (KAIST), DeepMath · 2020 · 0:17:03 · <https://www.youtube.com/watch?v=NHCfKFz3IeU> Closes the *width* side of universality: the exact minimum width for universal approximation. *Paper*: Park, Yun, Lee, Shin, “Minimum width for universal approximation” (arXiv:2006.08859)

163. Approximation Power [CORE] Matus Telgarsky (NYU), Simons Deep Learning Boot Camp · 2019 · 1:13:54 · <https://www.youtube.com/watch?v=KU6IaE37B9o> The boot-camp tutorial tying approximation theory to everything else in this module. The best single lecture if you want one.

164. Approximation power of deep networks Matus Telgarsky, SMILES Summer School · 2019 · 1:22:56 · <https://www.youtube.com/watch?v=6Ss9kFTUS-Y> Longer, slower version of #163.

7.2 Depth separation

165. Benefits of depth in neural networks (COLT 2016) [CORE] Matus Telgarsky · 2016 · 0:09:06 · <https://www.youtube.com/watch?v=ssaXJqG9Dz4> Nine minutes, by the author, on the sawtooth construction —

the canonical proof that depth buys exponential expressive power. One of the highest insight-per-minute items in the whole module. *Paper*: Telgarsky, “Benefits of depth in neural networks” (arXiv:1602.04485)

166. The Power of Depth for Feedforward Neural Networks (COLT 2016) Ronen Eldan & Ohad Shamir · 2016 · 0:12:37 · https://www.youtube.com/watch?v=Ue_hR6x0B-U The radial-function three-versus-two-layer separation — the complementary half of the depth story. *Paper*: arXiv:1512.03965

167. On the benefits of deep and narrow neural networks Gilad Yehudai (Weizmann), HUJI Machine Learning Club · 2022 · 1:01:17 · <https://www.youtube.com/watch?v=NzmDUI3gNFo> Modern synthesis on when width can and cannot substitute for depth. *Paper*: Vardi, Yehudai, Shamir, “Width is less important than depth in ReLU neural networks” (arXiv:2202.03841)

168. Is Deeper Better Only When Shallow Is Good? Shai Shalev-Shwartz (Hebrew University of Jerusalem), Simons “Emerging Challenges in Deep Learning” · 2019 · 0:44:48 · <https://www.youtube.com/watch?v=I8KOeXuCLm4> The uncomfortable caveat: many depth-separation constructions only separate on functions a shallow network was never going to learn anyway.

169. On Expressiveness and Optimization in Deep Learning Nadav Cohen (IAS), IAS · 2018 · 1:03:26 · <https://www.youtube.com/watch?v=F079b2dwcAg> The tensor-decomposition view of expressivity for convolutional architectures, and why expressivity and optimization interact.

7.3 What transformers can and cannot compute

This is the most active area in this section, and the results are sharp enough to be predictive. The headline: a fixed-depth transformer lives in TC^0 , a very small complexity class, and therefore provably cannot do certain serial computations — which is exactly what chain-of-thought buys back.

170. Transformers are Uniform Constant Depth Threshold Circuits [CORE] William Merrill (NYU), FLANN Seminar · 2022 · 0:34:32 · <https://www.youtube.com/watch?v=WU9RSiT4R8> The foundational TC^0 upper bound. Start the transformer unit here. *Paper*: Merrill, Sabharwal, Smith, “Saturated transformers are constant-depth threshold circuits” (arXiv:2106.16213)

171. The Parallelism Tradeoff: Understanding Transformer Expressivity Through Circuit Complexity [CORE] Will Merrill (NYU), Simons · 2024 · 0:45:13 · https://www.youtube.com/watch?v=7GVesfXD6_Q The polished Simons version of the whole log-precision circuit-complexity programme. The argument that parallelizability — the thing that made transformers trainable — is exactly what limits them. *Paper*: Merrill & Sabharwal (arXiv:2207.00729)

172. Transformers, parallel computation, and logarithmic depth Daniel Hsu (Columbia), Simons · 2024 · 0:57:21 · <https://www.youtube.com/watch?v=spxJnEhs1qI> The positive direction: logarithmic depth suffices to simulate substantial parallel computation. *Paper*: Sanford, Hsu, Telgarsky (arXiv:2402.09268)

173. Transformer Expressivity and Formal Logic David Chiang (Notre Dame), Simons · 2024 · 0:45:34 · <https://www.youtube.com/watch?v=hR3G5jsuOHs> The logic side: first-order logic with counting as an *exact* characterization, not just an upper bound. *Paper*: Chiang, Cholak, Pillay (arXiv:2301.10743)

174. Thinking Like Transformers [CORE] Gail Weiss (Technion), FLANN Seminar · 2022 · 1:07:11 · <https://www.youtube.com/watch?v=t5LjgczaS80> RASP, by its author, at full length: a small programming language whose programs compile to transformers. The constructive counterpart to all the impossibility results, and the thing to read before trying to prove a transformer can’t do something. *Paper*: Weiss, Goldberg, Yahav (arXiv:2106.06981)

- 175. Masked Hard-Attention Transformers and B-RASP Recognize Exactly the Star-Free Languages** Andy Yang (Notre Dame), FLaNN Seminar · 2024 · 0:35:08 · <https://www.youtube.com/watch?v=gKzyfqrZvkI> A clean three-way exact equivalence between an architecture, a programming language, and a language class. *Paper*: arXiv:2310.13897
- 176. The Expressive Power of Transformers with Chain of Thought [CORE]** Will Merrill (NYU), FLaNN Seminar · 2024 · 0:39:40 · <https://www.youtube.com/watch?v=30MhUdapqc8> Exactly how many chain-of-thought steps buy how much additional computational power. This is the theoretical explanation for why reasoning models work. *Paper*: Merrill & Sabharwal (arXiv:2310.07923)
- 177. Chain of Thought Empowers Transformers to Solve Inherently Serial Problems** Zhiyuan Li (TTIC), FLaNN Seminar · 2024 · 0:55:19 · https://www.youtube.com/watch?v=_tmzV4ZRwVs The complementary result: constant depth plus chain of thought escapes TC^0 into P/poly. *Paper*: Li, Liu, Zhou, Ma (arXiv:2402.12875)
- 178. Iterated Models: Expressive Power, Learning, and Chain of Thought** Nati Srebro (TTIC), Simons · 2024 · 0:54:08 · https://www.youtube.com/watch?v=NRyIPyO_IWY Learning-theoretic framing of iteration and chain of thought, rather than the complexity-theoretic one.
- 179. Representational Strengths and Limitations of Transformers** Clayton Sanford (Columbia), FLaNN Seminar · 2023 · 0:51:27 · <https://www.youtube.com/watch?v=7hYIN7Q-YRQ> Communication-complexity lower bounds — a different and quite powerful proof technique from circuits. *Paper*: Sanford, Hsu, Telgarsky (arXiv:2306.02896)
- 180. Transformers Learn Shortcuts to Automata [CORE]** Bingbin Liu (CMU), FLaNN Seminar · 2023 · 0:48:49 · <https://www.youtube.com/watch?v=ni9jCjhRUyY> Shallow transformers simulate long automaton runs via Krohn–Rhodes shortcuts. The key bridge from “what is representable” to “what training actually finds.” *Paper*: Liu, Ash, Goel, Krishnamurthy, Zhang (arXiv:2210.10749)
- 181. What Algorithms can Transformers Learn? A Study in Length Generalization** Hattie Zhou (Mila), FLaNN Seminar · 2024 · 0:53:49 · <https://www.youtube.com/watch?v=koo5Bo0k9Wc> The RASP-L conjecture: length generalization happens roughly when a short RASP program for the task exists. A rare *predictive* theory of a training outcome. *Paper*: arXiv:2310.16028
- 182. A Formal Framework for Understanding Length Generalization in Transformers** Xinting Huang (ETH / Saarland), FLaNN Seminar · 2025 · 0:38:23 · https://www.youtube.com/watch?v=3G_4VYGhgvQ The current state of the art on predicting which tasks length-generalize. *Paper*: arXiv:2410.02140
- 183. The emergence of clusters in self-attention dynamics** Philippe Rigollet (MIT), Simons · 2024 · 0:48:48 · <https://www.youtube.com/watch?v=ZrsQGhG0su0> Attention as an interacting-particle system; tokens provably cluster as depth increases. A completely different mathematical lens on the same architecture. *Paper*: Geshkovski, Letrouit, Polyanskiy, Rigollet (arXiv:2305.05465)
- 184. Exact solutions to the geometric dynamics of signal propagation through transformers predict their trainability** Surya Ganguli (Stanford), Simons · 2024 · 0:44:20 · https://www.youtube.com/watch?v=_THXU94rarw Signal-propagation theory specialized to transformers, with trainability predictions that hold up.

7.4 Weisfeiler–Leman hierarchy and GNN expressivity

- 185. Theoretical Foundations of Graph Neural Networks [CORE]** Petar Veličković (DeepMind) · 2021 · 1:12:20 · <https://www.youtube.com/watch?v=uF53xsT7mjc> The best single lecture deriving message passing from permutation invariance as a first principle. Start here even if you know GNNs. *Paper*: Bronstein, Bruna, Cohen, Veličković, “Geometric deep learning” (arXiv:2104.13478)

- 186. Representation and Learning in Graph Neural Networks [CORE]** Stefanie Jegelka (MIT), Tübingen Machine Learning · 2020 · 0:41:38 · <https://www.youtube.com/watch?v=RrOpBFGcnjw> GIN and the 1-WL upper bound, by a co-author of the paper that established the result. This is the “message-passing GNNs are exactly as powerful as 1-WL” talk. *Paper*: Xu, Hu, Leskovec, Jegelka, “How powerful are graph neural networks?” (arXiv:1810.00826)
- 187. A Deep Dive into the Weisfeiler-Leman Algorithm** Martin Grohe (RWTH Aachen) · 2023 · 0:56:28 · <https://www.youtube.com/watch?v=ymRBPnDVw0I> The algorithm itself, in depth, from the logic-and-combinatorics side. Essential if you want to understand *why* the hierarchy is the right yardstick.
- 188. The Descriptive Complexity of Graph Neural Networks** Martin Grohe (RWTH Aachen), FLANN Seminar · 2023 · 0:52:41 · <https://www.youtube.com/watch?v=1VHIMszFGnk> GNN expressivity restated in logic and circuit terms rather than as “equals 1-WL.” *Paper*: arXiv:2303.04613
- 189. Subgraph-based networks for expressive, efficient, and domain-independent graph learning** Haggai Maron (Technion / NVIDIA), CIRM · 2022 · 0:49:41 · <https://www.youtube.com/watch?v=QICDaP2RU9A> How subgraph GNNs climb above 1-WL, and the symmetry group that explains why they work. *Paper*: Frasca, Bevilacqua, Bronstein, Maron (arXiv:2206.11140)
- 190. On Computational Hardness with Graph Neural Networks** Joan Bruna (NYU), IPAM · 2018 · 0:55:31 · <https://www.youtube.com/watch?v=ZSIZSVRvd4Q> The hardness side — what GNNs cannot do for reasons other than WL.
- 191. Boot camp on generalization theory for graph learning** Simons “Graph Learning Meets TCS” · 2025 · 1:39:50 · <https://www.youtube.com/watch?v=AhgY4ErqXDo> Current tutorial-level treatment of generalization specifically for graph models, which has its own subtleties.

7.5 State space model theory

- 192. On the Tradeoffs of State Space Models [CORE]** Albert Gu (CMU), Simons · 2024 · 0:49:05 · https://www.youtube.com/watch?v=ksRp_DIHwj4 Gu’s own framing of the SSM-versus-transformer tradeoff, delivered to a theory audience rather than a systems one. *Paper*: Gu & Dao, “Mamba” (arXiv:2312.00752)
- 193. Structured State Space Models for Deep Sequence Modeling** (LxMLS tutorial) Albert Gu (CMU) · 2024 · 1:34:12 · <https://www.youtube.com/watch?v=WC9tqkCpq4s> The full ninety-minute tutorial from S4 through Mamba and the SSD duality with linear attention. *Paper*: Dao & Gu, “Transformers are SSMs” (arXiv:2405.21060)
- 194. The Illusion of State in State-Space Models [CORE]** Will Merrill (NYU), FLANN Seminar · 2024 · 0:45:42 · <https://www.youtube.com/watch?v=4VXe1yPDjk> The negative result that matters: despite the name, SSMs are no more expressive than transformers for state tracking — both are stuck in TC^0 . Recurrence in form is not recurrence in power. *Paper*: Merrill, Petty, Sabharwal (arXiv:2404.08819)
- 195. The Expressive Capacity of State Space Models: A Formal Language Perspective** Yash Sarrof (Saarland), FLANN Seminar · 2024 · 0:48:38 · <https://www.youtube.com/watch?v=-CBUWqvmVVU> The formal-language counterpart to #194. *Paper*: arXiv:2405.17394
- 196. Computational Benefits and Limitations of Transformers and State-Space Models** Eran Malach (Kempner Institute, Harvard), Simons · 2024 · 0:50:52 · <https://www.youtube.com/watch?v=sbViSPM3IVE> Direct head-to-head comparison of the two architectures’ computational classes.
- 197. Scaling Insights from Infinite-Width Theory for Next Gen Architectures** Leena Vankadara (Amazon Research), IPAM · 2024 · 0:51:06 · <https://www.youtube.com/watch?v=2QwdaGIOjcw> μP -style parametrization analysis carried into state-space models — where $\S DLT-3$ and $\S DLT-7$ meet. *Paper*: arXiv:2406.03529

DLT-8 Full courses with complete recordings

Four complete graduate courses, enumerated lecture by lecture. These are the spine of the module — the talks above are commentary on material that gets taught properly here.

If you do only one, do **CS229M**. It is the only course that goes all the way from Hoeffding's inequality to the neural tangent kernel and implicit regularization in a single coherent sequence, taught by someone actively publishing in it.

8.1 Stanford CS229M / STATS214 — Machine Learning Theory

Tengyu Ma (Stanford) · recorded Fall 2021, posted 2022 · 19 lectures, ≈ 27 hours Course notes:

<https://web.stanford.edu/class/stats214/>

#	Lecture	Link	Length
1	Overview, supervised learning, empirical risk minimization	https://www.youtube.com/watch?v=I-tmjGFaaBg	1:04:07
2	Asymptotic analysis, uniform convergence, Hoeffding inequality	https://www.youtube.com/watch?v=Fx3xldCEfsM	1:20:14
3	Finite hypothesis class, discretizing infinite hypothesis space	https://www.youtube.com/watch?v=io-YFfXbIXk	1:14:13
4	Advanced concentration inequalities	https://www.youtube.com/watch?v=fKM6fcOkXuk	1:31:16
5	Rademacher complexity, empirical Rademacher complexity	https://www.youtube.com/watch?v=tkJd2B98hII	1:23:58
6	Margin theory and Rademacher complexity for linear models	https://www.youtube.com/watch?v=echF7IWE05c	1:22:41
7	Challenges in DL theory, generalization bounds for neural nets	https://www.youtube.com/watch?v=kVkMRDZ5fcU	1:25:43
8	Refined generalization bounds for neural nets, kernel methods	https://www.youtube.com/watch?v=gwKfeDRCvSg	1:27:11
9	Covering number approach, Dudley's theorem	https://www.youtube.com/watch?v=wDfardbL50I	1:26:23
10	Generalization bounds for deep nets	https://www.youtube.com/watch?v=P5-VV11qLxA	1:23:35
11	All-layer margin	https://www.youtube.com/watch?v=GeXBfyrKfM4	1:29:24
13	Neural tangent kernel	https://www.youtube.com/watch?v=btphvvnad0A	1:29:29
14	NTK, implicit regularization of gradient descent	https://www.youtube.com/watch?v=xpT1ymwCk9w	1:33:00
15	Implicit regularization effect of initialization	https://www.youtube.com/watch?v=l-CR_TLihdg	1:24:13
16	Implicit regularization in classification problems	https://www.youtube.com/watch?v=mham4hHpo7A	1:29:51
17	Implicit regularization effect of the noise	https://www.youtube.com/watch?v=60GqpISCtCU	1:32:14
18	Unsupervised learning, mixture of Gaussians, moment methods	https://www.youtube.com/watch?v=4xDEsLUkdG4	1:22:08
19	Mixture of Gaussians, spectral clustering	https://www.youtube.com/watch?v=E6rZeGIKdRY	1:30:30
20	Spectral clustering	https://www.youtube.com/watch?v=UYBRLG64oSQ	1:28:25

Lecture 12 was never posted — the sequence runs 11 → 13 on YouTube. The gap is covered in the course notes.

Lectures 7–17 are the heart of this entire module. Lectures 1–6 are standard statistical learning theory you can get in many places; lectures 7 onward are Tengyu Ma deriving the deep-learning-specific results at blackboard pace, and there is no substitute for them anywhere else on video.

8.2 MIT 9.520 / 6.860 — Statistical Learning Theory and Applications

Tomaso Poggio & Lorenzo Rosasco (MIT) · Fall 2019 · 25 classes, ≈33 hours Channel: MITCBMM · no playlist exists; URLs below are the Fall 2019 set specifically

A caution worth stating up front: **the MITCBMM channel hosts several years of this course with identical generic titles** (“Class 1”, “Class 2”, ...). Mixing years gives you a scrambled course. Every link below was date-checked to fall between 2019-09-05 and 2019-12-10, so this is one coherent offering.

Class	Link	Length
1	https://www.youtube.com/watch?v=4nPABKIuYKo	1:21:55
2	https://www.youtube.com/watch?v=kNWONiLbVs	1:18:18
3	https://www.youtube.com/watch?v=MiyPgGqEPpQ	1:20:06
4	https://www.youtube.com/watch?v=_hOZw7SsTXc	1:17:27
5	https://www.youtube.com/watch?v=bISZ605iJ8Q	1:15:46
6	https://www.youtube.com/watch?v=WvscTPTEqow	1:18:45
7	https://www.youtube.com/watch?v=k8hUi6xYgtQ	1:14:16
8	https://www.youtube.com/watch?v=0Tw036csfhI	1:20:25
9	https://www.youtube.com/watch?v=M3lTpmTjZ0Q	1:19:54
10	https://www.youtube.com/watch?v=emnYADN_73o	1:21:20
11	https://www.youtube.com/watch?v=KU5cSQpPP5Y	1:15:51
12	https://www.youtube.com/watch?v=BKemj2t5HRU	1:16:44
13	https://www.youtube.com/watch?v=4leS-ga_B3w	1:19:51
14	https://www.youtube.com/watch?v=awkA8oQxxog	1:20:53
15	https://www.youtube.com/watch?v=LLTyPRWPtjE	1:20:17
16	https://www.youtube.com/watch?v=UJK1O01UyMc	1:21:18
17	https://www.youtube.com/watch?v=8wjQIZ8bm1o	1:23:49
18	https://www.youtube.com/watch?v=u7dRrT5HX1w	1:22:33
19	https://www.youtube.com/watch?v=WX3oIjupleo	1:25:31
20	https://www.youtube.com/watch?v=CTzRZGMgGIE	1:24:17
21	https://www.youtube.com/watch?v=sgwcKA6ej6Q	1:25:57
22	https://www.youtube.com/watch?v=UAE9JgBm5Yo	1:24:17
23	https://www.youtube.com/watch?v=fqVQ48_G1-8	1:15:59
25	https://www.youtube.com/watch?v=I4JJYnkUQ80	1:15:12
26	https://www.youtube.com/watch?v=iiFynQczYiM	1:13:17

Class 24 from Fall 2019 was never uploaded. The Fall 2018 Class 24 (<https://www.youtube.com/watch?v=-v992r3IiyY>, 1:11:42) is a reasonable substitute.

Use 9.520 as the **kernel-methods and RKHS foundation** that CS229M assumes. Poggio's own deep-learning-theory material appears in the later classes.

8.3 Mathematics of Deep Learning (minicourse)

Boris Hanin (Princeton) · April 2026 · 5 lectures, ≈ 4.9 hours University of Chicago Department of Mathematics
Playlist: <https://www.youtube.com/playlist?list=PLXOaY9trlJU0E1u1Ru7nxu87qFrXuIlDJ>

#	Link	Length
1	https://www.youtube.com/watch?v=4SVWo5kspR8	1:08:52
2	https://www.youtube.com/watch?v=nz5I9Nze6CM	0:56:22
3	https://www.youtube.com/watch?v=2rSi7iYuhRU	0:57:29
4	https://www.youtube.com/watch?v=PlQs3IdWBmg	1:10:55
5	https://www.youtube.com/watch?v=4mtlk_HnwV8	1:01:16

The most current complete treatment available, posted April 2026. Five hours rather than thirty, aimed at mathematicians, and it reflects what the field believes now rather than in 2019. If your time is short, this plus CS229M lectures 7–17 is a defensible minimum.

8.4 Neural Network Theory (APS 2025 Summer School)

Matus Telgarsky (NYU Courant) · 2025 · 3 lectures, ≈ 4.4 hours Playlist: <https://www.youtube.com/playlist?list=PLVsDasYggo92OhNlrmfKjjR196uVLh1dLZ>

#	Link	Length
1	https://www.youtube.com/watch?v=OjyPVFX-fAY	1:26:20
2	https://www.youtube.com/watch?v=rLFNU9c1hr4	1:29:48
3	https://www.youtube.com/watch?v=Tp8XV_u1vxw	1:27:44

Telgarsky's own course-length treatment of approximation, optimization and generalization together, updated to 2025. His earlier SMILES 2019 pair splits the same material differently: “Approximation power of deep networks” (<https://www.youtube.com/watch?v=6Ss9kFTUS-Y>, 1:22:56) and “Optimisation and Generalization of deep networks” (<https://www.youtube.com/watch?v=rv-vzl5-oIs>, 1:17:16).

8.5 Understanding Deep Learning lecture series

Data ICMC (University of São Paulo) · 2021 · 10 lectures + panel, ≈ 11 hours

A genuinely underknown gem: a lecture series where Google Brain and academic researchers each taught one full hour on their own result. Several lectures cited elsewhere in this module live here.

#	Lecture	Speaker	Link	Length
1	Understanding Generalization Requires Rethinking Deep Learning	Boaz Barak & Gal Kaplun (Harvard)	https://www.youtube.com/watch?v=wi9mjnDfS7Y	1:15:41
2	The Wide Limit of Neural Networks: NNGP and NTK	Jascha Sohl-Dickstein (Google Brain)	https://www.youtube.com/watch?v=fcpI5z9q91A	1:16:51
3	The Catapult Phase of Neural Networks	Guy Gur-Ari (Google)	https://www.youtube.com/watch?v=aPY-M4epEz4	1:07:20
4	Signal Propagation and Dynamical Isometry	<i>not listed</i>	https://www.youtube.com/watch?v=IwrpOhfhj9o	1:06:50
5	Neural Network Loss Landscape in High Dimensions	Stanislav Fort (Stanford/Google)	https://www.youtube.com/watch?v=stTzg8iUaXM	1:02:22
6	Disentangling Trainability and Generalization	Lechao Xiao (Google Brain)	https://www.youtube.com/watch?v=4wiFeJOdro4	0:52:09
7	Explaining Neural Scaling Laws	Jaehoon Lee (Google Brain)	https://www.youtube.com/watch?v=A8F4Qga3NaM	0:56:12
8	Progress Towards Understanding Generalization	Gintare Karolina Dziugaite (Element AI)	https://www.youtube.com/watch?v=Et_v-ElxqUs	1:02:25
9	Information-Theoretic Generalization Bounds for SGD	Gergely Neu (UPF)	https://www.youtube.com/watch?v=jJrOLPZJxSI	1:05:37
10	Backprop as a Functor	Brendan Fong (MIT / Topos Institute)	https://www.youtube.com/watch?v=N9zZeACcV98	1:08:41
—	Panel: The Many Paths to Understanding Deep Learning	—	https://www.youtube.com/watch?v=NfvYufQwA_o	1:02:40

8.6 Francis Bach — optimization for machine learning

Francis Bach (INRIA / ENS) · no single recorded course, but three verified multi-part series:

Large-scale Machine Learning and Convex Optimization (Hausdorff Center, 2016, 4 lectures)

<https://www.youtube.com/watch?v=V7lBkV9-kgc> (1:07:34) · <https://www.youtube.com/watch?v=4C65WnxoWPg> (1:22:01) · <https://www.youtube.com/watch?v=naXDNOMNazo> (1:04:38) · https://www.youtube.com/watch?v=KGx23IxC_Fk (1:15:59)

MLSS 2020 Tübingen — Optimization (2 parts) <https://www.youtube.com/watch?v=0MeNygohD6c> (1:38:50) · <https://www.youtube.com/watch?v=hg2h53bU5ic> (1:42:35)

CIRM — Large-scale machine learning and convex optimization (2 parts) <https://www.youtube.com/watch?v=RPIfP00emcs> (1:22:17) · <https://www.youtube.com/watch?v=nM4lK9ORQx8> (1:28:41)

UNVERIFIED — search term: "Francis Bach learning theory from first principles lecture" — that title appears to exist only as his book (free PDF on his INRIA page), not as a recorded course. The MLSS 2020 pair is the closest video equivalent.

8.7 Princeton ORFE Deep Learning Theory Summer School

Organized by Boris Hanin (Princeton) · 2021 · 7 full days, ≈ 30 hours Posted as whole unsegmented days on Hanin's own channel, which makes it harder to assign but it is the largest single block of deep-learning-theory lecture content freely available.

Day 1 <https://www.youtube.com/watch?v=PHcodnoOlgI> (5:22:38) · Day 2 <https://www.youtube.com/watch?v=PRyEApLxcSQ> (2:42:47) · Day 3 <https://www.youtube.com/watch?v=OBrjyorRCxo> (5:50:04) · Day 4 <https://www.youtube.com/watch?v=sS4vvt4dp5U> (2:57:17) · Day 5 https://www.youtube.com/watch?v=0BUl_5cMilM (4:17:27) · Day 6 <https://www.youtube.com/watch?v=2rRUfmo4Ays> (3:13:55) · Day 7 <https://www.youtube.com/watch?v=dM1PDZ8Bgkw> (5:58:25)

DLT-9 The archives: Simons Institute and IAS programs

Both institutions post complete video archives of multi-month programs. Watching one program end to end is a different experience from watching scattered talks: you see the same people disagree across a week, and you see which questions the community considered open at that moment. For a would-be contributor that situational knowledge is worth as much as the technical content.

Below, each program is named with its playlist, then **the specific talks worth watching** with individual URLs.

9.1 Simons: Foundations of Deep Learning (Summer 2019)

The program that organized this field. Two archives.

Deep Learning Boot Camp · May 28–31, 2019 · tutorial-level, four lecturers Already enumerated in §DLT-1 and §DLT-4; the full set is:

Talk	Speaker	Link	Length
Generalization I	Bartlett & Rakhlin	https://www.youtube.com/watch?v=Ntl_WNW8yGc	1:16:56
Generalization II	Bartlett & Rakhlin	https://www.youtube.com/watch?v=QO8HOqQ_6BQ	1:23:06
Generalization III	Bartlett & Rakhlin	https://www.youtube.com/watch?v=8hZD5bMBbY4	1:18:23
Generalization IV	Bartlett & Rakhlin	https://www.youtube.com/watch?v=n5Zxi22801Q	1:22:14
Approximation Power	Matus Telgarsky (NYU)	https://www.youtube.com/watch?v=KU6IaE37B9o	1:13:54
Over-parametrized Neural Networks I	Jason Lee (USC)	https://www.youtube.com/watch?v=uC2IGoTE2u4	1:14:29
Over-parametrized Neural Networks II	Jason Lee (USC)	https://www.youtube.com/watch?v=NGon2JyjO6Y	1:13:57
Implicit Regularization I	Nati Srebro (TTIC)	https://www.youtube.com/watch?v=7uRVR9hsF0g	1:16:51
Implicit Regularization II	Nati Srebro (TTIC)	https://www.youtube.com/watch?v=IkuOmf7ey14	1:22:54

This nine-lecture set is effectively a short course and is the best single entry point to the entire module.

Frontiers of Deep Learning · July 15–18, 2019 · research talks Playlist: https://www.youtube.com/playlist?list=PLgKuh-lKre11ekU7g-Z_qsvjDD8cT-hi9

Must-watch eight:

1. **Benign Overfitting in Linear Prediction** — Peter Bartlett (UC Berkeley) · 0:47:40 · <https://www.youtube.com/watch?v=GXpP-rXEDpk>
2. **From Classical Statistics to Modern Machine Learning** — Misha Belkin (UCSD) · 0:49:47 · <https://www.youtube.com/watch?v=OBCciGnOJVs>
3. **Learning and Generalization in Over-parametrized Neural Networks, Going Beyond Kernels** — Yuanzhi Li (Stanford) · 0:49:42 · <https://www.youtube.com/watch?v=NNPCK2gvTnI>
4. **Analyzing Optimization and Generalization in Deep Learning via Trajectories of Gradient Descent** — Nadav Cohen (IAS) · 0:46:27 · <https://www.youtube.com/watch?v=Lmj2bU9MdwM>
5. **Explaining Landscape Connectivity of Low-cost Solutions for Multilayer Nets** — Rong Ge (Duke) · 0:45:11 · <https://www.youtube.com/watch?v=0kXDqS2OmeI>
6. **Computation in Very Wide Neural Networks** — Yasaman Bahri (Google Brain) · 0:48:40 · <https://www.youtube.com/watch?v=EWLQoO0SGvI>
7. **Studying Generalization in Deep Learning via PAC-Bayes** — Gintare Karolina Dziugaite (Element AI) · 0:44:39 · https://www.youtube.com/watch?v=_bYT2VOOxU0
8. **Kernel and Deep Regimes in Overparameterized Learning** — Suriya Gunasekar (MSR) · 0:46:15 · <https://www.youtube.com/watch?v=fUQCJNSdskA>

Also strong: **Training on the Test Set and Other Heresies** — Ben Recht · 0:49:37 ·

<https://www.youtube.com/watch?v=NTz4rJS9BAI> · **A Primal-dual Analysis of Margin Maximization** — Telgarsky · 0:41:42 · <https://www.youtube.com/watch?v=Rc76ZBP6fHM> · **Size-free Generalization Bounds for CNNs** — Hanie Sedghi · 0:34:34 · <https://www.youtube.com/watch?v=jRK3W48cO90>

Emerging Challenges in Deep Learning · 2019–2020 Playlist: <https://www.youtube.com/playlist?list=PLgKuh-lKre10BpafDrv0fg2VNUweWXWVd> Mostly RL, robustness and fairness; the one theory item worth pulling out is **Is Deeper Better Only When Shallow Is Good?** — Shai Shalev-Shwartz (Hebrew University) · 0:44:48 · <https://www.youtube.com/watch?v=I8KOeXuCLm4>

9.2 Simons: Deep Learning Theory Workshop and Summer School (August 1–5, 2022)

The best-balanced single week in the archive — tutorials in the morning, research talks in the afternoon, and the first serious airing of the “benign overfitting is not what usually happens” correction.

Must-watch seven:

1. **Tutorial: Statistical Learning Theory and Neural Networks I** — Spencer Frei (UC Davis) · 0:59:25 · <https://www.youtube.com/watch?v=pb9LQV3fytE>
2. **Tutorial: Statistical Learning Theory and Neural Networks II** — Spencer Frei · 1:02:20 · <https://www.youtube.com/watch?v=PqentqYpUXk>
3. **Tutorial: Implicit Bias I** — Nati Srebro (TTIC) · 1:20:40 · <https://www.youtube.com/watch?v=NeTUt5TJOiY>
4. **Tutorial: Implicit Bias II** — Nati Srebro · 1:36:30 · <https://www.youtube.com/watch?v=8mO1CFfpbRE>
5. **Benign, Tempered, or Catastrophic: A Taxonomy of Overfitting** — Neil Mallinar (UCSD) & Preetum Nakkiran · 0:57:45 · <https://www.youtube.com/watch?v=cg9s7jpWgck>

6. **Feature Selection with Gradient Descent on Two-layer Networks in Low-rotation Regimes** — Matus Telgarsky (NYU) · 0:57:45 · <https://www.youtube.com/watch?v=bSJZQVNiLP0>
7. **When is Scale Enough?** — Ethan Dyer (Google Research) · 1:12:25 · <https://www.youtube.com/watch?v=Jpu3kQv39L4>

Also: **The Devil is in the Tails and Other Stories of Interpolation** — Niladri Chatterji (OpenAI) · 0:54:40 · <https://www.youtube.com/watch?v=e7Y3hgQlaaE> · **A Theoretical Framework of Convolutional Kernels on Image Datasets** — Song Mei (UC Berkeley) · 1:00:05 · <https://www.youtube.com/watch?v=DD0RBODEE78> · **Understanding the Robustness of Deep Learning** — Aditi Raghunathan (Stanford) · 1:00:23 · <https://www.youtube.com/watch?v=pToDHIS2kcE>

9.3 Simons: Modern Paradigms in Generalization (Fall 2024)

The 2024 restatement of the whole question, five years after the 2019 program. Useful precisely as a before-and-after comparison: watch a 2019 talk and its 2024 counterpart to see what actually got settled.

Modern Paradigms in Generalization Boot Camp · August 27–30, 2024 Playlist: https://www.youtube.com/playlist?list=PLgKuh-lKre10_AXZufHVP1qZRfnDt5RFD

Must-watch six:

1. **Overview of Statistical Learning Theory, Part 1** — Nati Srebro (TTIC) · 1:16:40 · <https://www.youtube.com/watch?v=BxQxsuRjoR8>
2. **Overview of Statistical Learning Theory, Part 2** — Nati Srebro · 1:02:44 · <https://www.youtube.com/watch?v=Mntd78eCrgk>
3. **Modern paradigms of generalization, the heliocentric model of Aristarchus, gradient descent is a lazy scientist, and other stories** — Matus Telgarsky (Courant, NYU) · 1:09:38 · https://www.youtube.com/watch?v=-TIZPe_YaiU
4. **The elusive generalization: classical bounds to double descent to grokking** — Misha Belkin (UCSD) · 1:19:56 · <https://www.youtube.com/watch?v=h2IOHs2K2KI>
5. **Understanding Generalization from Pre-training Loss to Downstream Tasks** — Tengyu Ma (Stanford) · 1:17:51 · https://www.youtube.com/watch?v=rFp0tVQW_q0
6. **Reconsidering Overfitting in the Age of Overparameterized Models** — Fanny Yang (ETH Zurich) · 1:18:05 · <https://www.youtube.com/watch?v=g1Gb3UEgpOk>

Also: **Prediction, Generalization, Complexity: Revisiting the Classical View from Statistics, Part 1 / Part 2** — Ryan Tibshirani (UC Berkeley) · 1:18:06 / 1:01:38 · <https://www.youtube.com/watch?v=mwnoEZgsZpA> · <https://www.youtube.com/watch?v=-PWYupkL6Kg>

Unknown Futures of Generalization · December 2–6, 2024 Playlist: <https://www.youtube.com/playlist?list=PLgKuh-lKre10arpoJoYeh2B0QIB3OjuLe>

Must-watch five:

1. **Learning Theory of Transformers: Generalization and Optimization of In-Context Learning** — Taiji Suzuki (University of Tokyo) · 0:45:35 · <https://www.youtube.com/watch?v=WyeomuU2vQw>
2. **Weak-to-Strong Generalization** — Pavel Izmailov (Anthropic) · 0:43:25 · <https://www.youtube.com/watch?v=VViyQRGxSKo>
3. **On Memorization of Large Language Models in Logical Reasoning** — Chiyuan Zhang (Google Research) · 0:48:05 · <https://www.youtube.com/watch?v=eULIf02frIw>

4. **Generalization in the representations and computations of frontier language models** — Joshua Batson (Anthropic) · 0:52:05 · <https://www.youtube.com/watch?v=2xb3mhIqjLw>
5. **Understanding the abilities of AI systems: Memorization, generalization, and points in between** — Tom McCoy (Yale) · 0:44:25 · <https://www.youtube.com/watch?v=3c9TiKryTtA>

Also in this program: **Emerging Generalization Settings** playlist — https://www.youtube.com/playlist?list=PLgKuh-lKre12gP84pKqe_1n4FLk6JD-Eo

9.4 Simons: Special Year on Large Language Models and Transformers (2023–2024)

Where theory met the actual systems. Three archives.

Part 1 Boot Camp · September 4–6, 2024 Playlist: https://www.youtube.com/playlist?list=PLgKuh-lKre13qQH5G4jpudE3_TYcqySVE

1. **Introduction to Transformers (How Do Transformers Work, Part 1)** — Ankur Moitra (MIT) · 1:10:24 · <https://www.youtube.com/watch?v=9L3i4rqcGG8>
2. **How Do Transformers Work? (Part 2)** — Daniel Hsu (Columbia) · 1:15:41 · https://www.youtube.com/watch?v=uIsej_SIIQU
3. **Toward Understanding In-context Learning** — Tengyu Ma (Stanford) · 1:29:35 · <https://www.youtube.com/watch?v=hxrR39mAlR4>
4. **On large language models and transformers: perspectives from physics, neuroscience, and theory** — Surya Ganguli (Stanford) · 1:32:41 · <https://www.youtube.com/watch?v=qXEuY6qLISE>

Transformers as a Computational Model · September 23–27, 2024 Playlist: <https://www.youtube.com/playlist?list=PLgKuh-lKre11RuxGM038u0OSxVdCicIMF>

This workshop is the single densest source of expressivity theory that exists. Must-watch eight:

1. **Transformers, parallel computation, and logarithmic depth** — Daniel Hsu (Columbia) · 0:57:21 · <https://www.youtube.com/watch?v=spxJnEhs1qI>
2. **The Parallelism Tradeoff: Understanding Transformer Expressivity Through Circuit Complexity** — Will Merrill (NYU) · 0:45:13 · https://www.youtube.com/watch?v=7GVesfXD6_Q
3. **Transformer Expressivity and Formal Logic** — David Chiang (Notre Dame) · 0:45:34 · <https://www.youtube.com/watch?v=hR3G5jsuOHs>
4. **Iterated Models: Expressive Power, Learning, and Chain of Thought** — Nati Srebro (TTIC) · 0:54:08 · https://www.youtube.com/watch?v=NRyIPyO_IWY
5. **Computational Benefits and Limitations of Transformers and State-Space Models** — Eran Malach (Kempner, Harvard) · 0:50:52 · <https://www.youtube.com/watch?v=sbViSPM3IVE>
6. **On the Tradeoffs of State Space Models** — Albert Gu (CMU) · 0:49:05 · https://www.youtube.com/watch?v=ksRp_DIHWj4
7. **The emergence of clusters in self-attention dynamics** — Philippe Rigollet (MIT) · 0:48:48 · <https://www.youtube.com/watch?v=ZrsQGhG0su0>
8. **Emergence and grokking in “simple” architectures** — Misha Belkin (UCSD) · 0:52:40 · <https://www.youtube.com/watch?v=dzpA30qZxYw>

Also: **Associative memories as a building block in Transformers** — Alberto Bietti (Flatiron) · 0:38:37 · <https://www.youtube.com/watch?v=ncAhx70jTic> · **Exact solutions to the geometric dynamics of signal**

propagation through transformers — Surya Ganguli (Stanford) · 0:44:20 · https://www.youtube.com/watch?v=_THXU94rarw

Large Language Models and Transformers (Aug 2023 workshop) Playlist: <https://www.youtube.com/playlist?list=PLgKuh-lKre12qVTl88k2n2N37tT-BpmHT> Pull out: **Understanding the Origins and Taxonomy of Neural Scaling Laws** — Yasaman Bahri · 1:05:24 · <https://www.youtube.com/watch?v=MUvFuZpxLU8> · **In-Context Learning: A Case Study of Simple Function Classes** — Gregory Valiant (Stanford) · 1:03:40 · <https://www.youtube.com/watch?v=DiJsg93zQDc> · **An Observation on Generalization** — Ilya Sutskever (OpenAI) · 0:57:21 · https://www.youtube.com/watch?v=AKMuA_TVz3A

Also: **The Future of Transformers and LLMs** — <https://www.youtube.com/playlist?list=PLgKuh-lKre13F6duXbU7e8dJqXnRpJuxy>

9.5 Simons: Graph Learning Meets Theoretical Computer Science (2025)

Playlist: <https://www.youtube.com/playlist?list=PLgKuh-lKre13j2lBO8AeVUCu0TQrY9GhE> The current reference archive for GNN theory. Four to start with:

1. **Boot Camp on Graph Learning** · 1:38:30 · https://www.youtube.com/watch?v=ENW8T59-_eE
2. **Boot camp on generalization theory for graph learning** · 1:39:50 · <https://www.youtube.com/watch?v=AhgY4ErqXDo>
3. **Boot camp on logic and graph learning** · 1:30:19 · https://www.youtube.com/watch?v=G_8csFllabM
4. **Boot camp on invariances in graph learning** · 1:29:00 · <https://www.youtube.com/watch?v=jeXww0ClIzY>

Plus **Homomorphism Indistinguishability** · 0:57:40 · <https://www.youtube.com/watch?v=8spy7FJgOjc> — the sharpest modern restatement of what the Weisfeiler–Leman hierarchy is actually measuring.

9.6 Simons: Theoretical Foundations — From the Early Days of Neural Networks to the Modern Deep Learning Era

Playlist: <https://www.youtube.com/playlist?list=PLgKuh-lKre13Qgeo9Yf4pMIqwpLRPa1Lk> Short historical talks from the people who were there. Of particular value: **Talk by Matus Telgarsky (Courant, NYU)** · 0:31:06 · <https://www.youtube.com/watch?v=2QmxxsxEjLU> · **Talk by Nati Srebro (TTIC)** · 0:23:09 · <https://www.youtube.com/watch?v=-A6LFTLmnYg> · **Talk by Misha Belkin (UC San Diego)** · 0:15:46 · <https://www.youtube.com/watch?v=xgaZPTG7hVo>

9.7 IAS: Special Year on Optimization, Statistics, and Theoretical Machine Learning (2019–2020)

The Institute for Advanced Study ran a full special year on this subject under Sanjeev Arora, and posted the seminar series. The talks are shorter than Simons' (25–45 minutes is typical) and more sharply focused on a single result, which makes them good for sampling breadth.

Must-watch eight:

1. **In Defense of Uniform Convergence** — Daniel M. Roy (Toronto) · 2020 · 1:21:46 · <https://www.youtube.com/watch?v=6mdFQR0-WT0>
2. **The Principles of Deep Learning Theory** — Dan Roberts (MIT/Salesforce) · 2021 · 1:20:17 · <https://www.youtube.com/watch?v=YzR2gZrsdJc>
3. **Margins, perceptrons, and deep networks** — Matus Telgarsky · 2020 · 1:22:27 · <https://www.youtube.com/watch?v=yfk-IPlr6M0>

4. **Toward a Causal Analysis of Generalization in Deep Learning** — Behnam Neyshabur (Google) · 2019 · 0:33:17 · <https://www.youtube.com/watch?v=HafQ-Q8pn0g>
5. **Kernel and Rich Regimes in Deep Learning** — Nati Srebro (TTIC) · 2019 · 0:37:01 · <https://www.youtube.com/watch?v=ZjTMiYhW4XY>
6. **From Classical Statistics to Modern ML: the Lessons of Deep Learning** — Mikhail Belkin · 2019 · 0:37:29 · <https://www.youtube.com/watch?v=5-Kqb80h9rk>
7. **Optimization Landscape and Two-Layer Neural Networks** — Rong Ge (Duke) · 2019 · 0:58:27 · <https://www.youtube.com/watch?v=dmfyYXl0xUY>
8. **Statistical mechanics of deep learning** — Surya Ganguli (Stanford) · 2019 · 0:29:55 · https://www.youtube.com/watch?v=-QF_jX8L0nw

Also worth it: **On the Connection between Neural Networks and Kernels: a Modern Perspective** — Simon Du · 2019 · 0:30:46 · <https://www.youtube.com/watch?v=HvEGJUwQEO8> · **Towards a theoretical foundation of neural networks** — Jason Lee · 2019 · 0:24:39 · <https://www.youtube.com/watch?v=vQsuF21tGQk> · **PAC-Bayesian approaches to understanding generalization** — Gintare Karolina Dziugaite · 2019 · 0:31:34 · <https://www.youtube.com/watch?v=MrUqeSj9Pgk> · **Designing explicit regularizers for deep models?** — Tengyu Ma · 2019 · 0:33:55 · <https://www.youtube.com/watch?v=QjxzY8W-caY> · **The mystery of over-parametrization in neural networks** — Behnam Neyshabur · 2017 · 0:17:33 · <https://www.youtube.com/watch?v=4Mga1cERS34> · **On Expressiveness and Optimization in Deep Learning** — Nadav Cohen · 2018 · 1:03:26 · <https://www.youtube.com/watch?v=F079b2dwcAg> · **Understanding Deep Neural Networks: From Generalization to Interpretability** — Gitta Kutyniok · 2020 · 1:02:04 · <https://www.youtube.com/watch?v=qVZksJt1WQo>

9.8 Formal Languages and Neural Networks Seminar

<https://www.youtube.com/channel/UCrp8k-nSuMKHM4sSUvlpdAw>

Not an institute program, but functionally an archive: roughly 120 hour-long talks on transformer and SSM expressivity, nearly all delivered by the first author of the paper being discussed. Ten of §DLT-7's entries come from here. If you intend to work on architecture theory, this is the channel to subscribe to.

DLT-10 How to sequence this

There are 197 numbered entries above. Counting the enumerated course lectures and the archive listings in §DLT-9, the module contains **291 distinct videos, 330 hours in total**. Nobody should watch all of it linearly. Three paths.

The twelve-week path (≈ 10 hrs/week). Weeks 1–3: the Simons Deep Learning Boot Camp nine-lecture set (§9.1) plus CS229M lectures 1–6. Week 4: double descent and benign overfitting (§DLT-2, items 34, 37, 38, 42, 46). Weeks 5–6: NTK and the lazy/rich distinction (§DLT-3, items 78, 79, 85, 86, 90) plus CS229M 13–14. Week 7: muP and Tensor Programs (items 96, 97, 100). Week 8: optimization (§DLT-4, items 51, 52, 56, 64, 67). Week 9: scaling laws (§DLT-5, items 110, 113, 118, 121, 123). Week 10: emergence debate and phenomena (items 126, 128, 130, 138, 145). Week 11: expressivity (§DLT-7, items 165, 170, 171, 174, 176, 186). Week 12: the 2024 restatement — watch §9.3 and compare against week 1.

The six-week compressed path. Boris Hanin's five-lecture minicourse (§8.3), then CS229M lectures 7–17, then every item marked **[CORE]** in §DLT-2 through §DLT-7. That is roughly 60 hours and covers the live questions

without the historical build-up.

The research-entry path. If you already know the material and want to find a problem: watch §9.3 and §9.4 end to end, both from 2024, and write down every sentence beginning “we don’t know” or “it’s open whether.” Then watch items 115 ((Mis)Fitting), 27 (Neyshabur’s causal analysis), 42 (the overfitting taxonomy) and 194 (The Illusion of State), which are the four talks in this module that most directly say *the published consensus is wrong about something specific*. Those four are where contributions are currently cheapest.

Where the open problems actually are

Based on what the 2024–2026 talks say out loud rather than what the 2019 talks hoped:

- **Generalization bounds remain vacuous for real models.** PAC-Bayes is still the only approach that has produced non-vacuous bounds, and only for small stochastic networks (items 18, 22). Nobody has a bound for a frontier model.
- **The feature-learning regime has no clean theory.** NTK is solved and wrong; mean-field works for two layers; muP is an empirical scaling prescription with partial theory. The gap between items 86 and 97 is the gap in the field.
- **Why Adam works is genuinely unsettled** and has become urgent again because Muon and SOAP changed the practical landscape (items 64, 67, 69).
- **Scaling-law methodology is weaker than it looks** (item 115), and the theoretical derivations (items 118, 121, 123) do not yet reproduce the measured exponents from first principles.
- **Loss of plasticity** (items 156–158) is a hard obstacle to continual learning that has attracted far less theoretical attention than it deserves.
- **Expressivity theory is now ahead of learning theory for transformers.** We know what they can compute (§7.3); we have much less idea what gradient descent on them will find. Item 180 is one of the few bridges.

Flagged as unverifiable

Everything listed above was confirmed to exist and play, with its real title, duration and year. These are the items I looked for and could **not** confirm:

- UNVERIFIED – search term: "Nitish Keskar large batch training generalization gap talk" — no recorded talk by Keskar on the large-batch/sharp-minima result appears to exist.
- UNVERIFIED – search term: "Ziming Liu grokking Omnigrok MIT talk" — no standalone talk found; Gromov (items 131–132) covers the physics-of-grokking angle.
- UNVERIFIED – search term: "Jordan Hoffmann training compute optimal large language models talk" — no DeepMind-author Chinchilla talk on video. Items 113 and 114 are the substitutes.
- UNVERIFIED – search term: "Utkarsh Sharma Jared Kaplan scaling laws data manifold dimension" — no dedicated talk; the result is covered inside item 111.
- UNVERIFIED – search term: "Alexander Maloney random feature model scaling laws" — no dedicated talk; Pehlevan (items 121–122) covers the same mathematics.
- UNVERIFIED – search term: "Christopher Morris Weisfeiler Leman graph neural networks talk" — Morris appears only inside the Simons 2025 graph boot camp, not as a standalone lecture.
- UNVERIFIED – search term: "Denny Zhou chain of thought expressivity talk" — no standalone talk; item 177 (Zhiyuan Li) covers the same theorem.
- UNVERIFIED – search term: "Francis Bach learning theory from first principles lecture" — appears to exist only as his book, not as a recorded course. §8.6 lists the closest recorded equivalents.

- **Stanford CS229M Lecture 12** was never posted to YouTube; the course notes cover it.
- **MIT 9.520 Fall 2019 Class 24** was never uploaded; the Fall 2018 recording is the only version.

A closing note on how to use the talks

The single highest-value habit in this module is to **watch pairs that disagree**. The field's real content lives in the disagreements, and this module is deliberately stocked with them:

- Nagarajan & Kolter (item 11) against Roy (item 12) on uniform convergence.
- Bartlett's benign overfitting (item 38) against Mallinar & Nakkiran's taxonomy (item 42) on whether interpolation is actually harmless.
- Wei (item 126) against Schaeffer (item 128) on whether emergence is real.
- Arora (item 25) against the entire landscape-analysis literature on whether optimization is even the right language.
- Gu's SSM tradeoffs (item 192) against Merrill's illusion-of-state (item 194) on whether recurrence buys anything.

Watch each pair back to back in one sitting and write down which argument you find stronger and why. That exercise — not the lecture count — is what converts this module into the ability to produce an original result.

D.9 Interpretability and the Science of Models

This is the section with the most obvious two-way traffic, and the traffic currently runs one way.

Mechanistic interpretability asks what computation a trained network is actually performing. The methods — activation patching, sparse autoencoders, circuit analysis, probing — are domain-general, and InterPLM (Atlas Q.4) has already shown they transfer to protein language models.

What has not been noticed is that the transfer is advantageous in the other direction. Language interpretability's hardest problem is *validation*: when a sparse autoencoder surfaces a feature, there is no ground truth for what that feature should mean, so the field falls back on human inspection of top activations. A protein model does not have that problem. A feature can be checked against a catalytic triad, a Pfam domain boundary, a transmembrane helix, a disulfide — thousands of independently-established structural and functional labels that nobody had to invent for the purpose.

Molecular models are therefore a better testbed for interpretability methods than language models are, and essentially nobody in the interpretability community has noticed. That observation is the whole of Capstone VIII, and it is the single clearest opportunity in this document.

1. Mechanistic interpretability

Circuits, feature visualization, Chris Olah

1. **CVPR'20 iMLCV tutorial: Introduction to Circuits in CNNs by Chris Olah** | Chris Olah (OpenAI→Anthropic), hosted by Bolei Zhou | 2020 | 44:54 | https://www.youtube.com/watch?v=gXsKyZ_Y_i8 — The original circuits program before it moved to transformers; curve detectors, high-low frequency detectors, equivariance. Papers: *Zoom In: An Introduction to Circuits* (Distill 2020); *Feature Visualization* (Distill 2017).
2. **Chris Olah - Looking Inside Neural Networks with Mechanistic Interpretability** | Chris Olah (Anthropic), FAR.AI Alignment Workshop | 2023 | 40:59 | <https://www.youtube.com/watch?v=2Rdp9GvcYOE> — The clearest single statement of the superposition framing and why it makes interpretability hard. Paper: *Toy Models of Superposition* (Elhage et al. 2022).
3. **Catherine Olsson - Mechanistic Interpretability: Getting Started** | Catherine Olsson (Anthropic), Cohere For AI | 2022 | 54:10 | <https://www.youtube.com/watch?v=ll0oduwDEwI> — An actual research-practice talk on how to begin, from a co-author of the induction-heads paper.
4. **What the hell is going on inside neural networks?** | Chris Olah | Chris Olah (Anthropic), 80,000 Hours | 2023 | 3:09:20 | https://www.youtube.com/watch?v=k_QVDwhR8FU — Long-form, but the best articulation of the research taste behind the whole agenda. Optional depth.

Induction heads and the transformer-circuits line

5. **Stanford CS25: V1 I Transformer Circuits, Induction Heads, In-Context Learning** | Chris Olah (Anthropic) | 2022 | 59:34 | https://www.youtube.com/watch?v=pC4zRb_5noQ — The canonical lecture on induction heads and the phase change in the loss curve. Papers: *A Mathematical Framework for Transformer Circuits* (2021); *In-context Learning and Induction Heads* (2022).

6. **A Walkthrough of A Mathematical Framework for Transformer Circuits** | Neel Nanda | 2022 | 2:50:13 | <https://www.youtube.com/watch?v=KV5gbOmHbjU> — Line-by-line reading of the hardest and most foundational paper in the field. Paper: Elhage et al. 2021.
7. **A Walkthrough of In-Context Learning and Induction Heads Part 1 of 2 (w/ Charles Frye)** | Neel Nanda with Charles Frye | 2022 | 1:03:51 | <https://www.youtube.com/watch?v=dCkQQYwPxdM> — Pairs directly with Olsson et al. 2022.
8. **What is a Transformer? (Transformer Walkthrough Part 1/2)** | Neel Nanda | 2023 | 1:03:00 | <https://www.youtube.com/watch?v=bOYE6E8JrtU> — The architecture as a mech-interp researcher needs to see it (residual stream as a communication channel).
9. **Implementing GPT-2 From Scratch (Transformer Walkthrough Part 2/2)** | Neel Nanda | 2023 | 1:19:25 | <https://www.youtube.com/watch?v=dsjUDacBw8o> — The hands-on companion; this is effectively the TransformerLens onboarding.

Superposition, SAEs, dictionary learning

10. **Towards Monosemanticity: Decomposing Language Models Into Understandable Components** | Arize AI paper-reading group | 2023 | 43:39 | <https://www.youtube.com/watch?v=hlCxSqWS6Rw> — Walkthrough of the paper that started the SAE wave. Paper: Bricken et al. 2023.
11. **Scaling interpretability** | Anthropic interpretability team | 2024 | 53:18 | <https://www.youtube.com/watch?v=sQar5NNGbw4> — The engineering story behind scaling SAEs to Claude 3 Sonnet; unusually candid about what broke. Paper: *Scaling Monosemanticity* (Templeton et al. 2024).
12. **Hoagy Cunningham — Finding distributed features in LLMs with sparse autoencoders [TAIS 2024]** | Hoagy Cunningham (Anthropic) | 2024 | 28:49 | <https://www.youtube.com/watch?v=HPLlI9ZOpuQ> — From the author of the independent SAE paper. Paper: *Sparse Autoencoders Find Highly Interpretable Features in Language Models* (2023).
13. **Sparse Autoencoders: Progress & Limitations with Joshua Engels** | Joshua Engels (MIT), NDIF | 2025 | 1:03:47 | <https://www.youtube.com/watch?v=eVlGeHA2Cnw> — The negative results: not all features are linear, SAEs miss things. Paper: *Not All Language Model Features Are Linear* (2024).
14. **What Happened With Sparse Autoencoders?** | Neel Nanda (DeepMind) | 2025 | 2:49:15 | <https://www.youtube.com/watch?v=Tgq7E4YcPKQ> — A field leader explaining why the SAE hype cycle deflated. Essential antidote to the two talks above.
15. **Bayesian and Dynamical Transitions in a Toy Model of Superposition - SLT Seminar 53** | metauni / Timaeus | 2024 | 1:14:31 | https://www.youtube.com/watch?v=yL4ZkDCe2_c — Superposition analyzed with singular learning theory; the mathematically serious treatment.
16. **Introduction to Sparse AutoEncoders | ML@P Reading Group | Jinen Setpal** | Jinen Setpal, ML Purdue | 2025 | 1:03:33 | https://www.youtube.com/watch?v=_vUKIPYOaJw — Implementation-level tutorial.

Attribution graphs and circuit tracing

17. **Stanford CS25: V5 I On the Biology of a Large Language Model, Josh Batson of Anthropic** | Josh Batson (Anthropic) | 2025 | 1:12:32 | <https://www.youtube.com/watch?v=vRQs7qfIDaU> — The single best talk on attribution graphs: multi-step reasoning, planning in poetry, unfaithful chain-of-thought. Papers: *Circuit Tracing* and *On the Biology of a Large Language Model* (Anthropic 2025).
18. **Attribution Graphs for Dummies - 1. What are Attribution Graphs?** | Neuronpedia | 2025 | 49:09 | https://www.youtube.com/watch?v=ruLcDtr_cGo — Mechanics of transcoders and replacement models.

19. **Attribution Graphs for Dummies - 2. Building and Testing a Circuit** | Neuronpedia | 2025 | 1:23:41 | <https://www.youtube.com/watch?v=hdi1a9MjwDs> — The hands-on half: build and validate a circuit yourself.
20. **Anthropic: Circuit Tracing + On the Biology of a Large Language Model** | Latent Space, with Anthropic authors | 2025 | 56:27 | <https://www.youtube.com/watch?v=ig5RNJJJaFJE> — Author Q&A on the same two papers.
21. **Interpretability: Understanding how AI models think** | Josh Batson, Emmanuel Ameisen, Jack Lindsey (Anthropic) | 2025 | 59:02 | <https://www.youtube.com/watch?v=fGKNUvivvnc> — Panel on hallucination and sycophancy mechanisms.

Neel Nanda / getting started / open problems

22. **An Introduction to Mechanistic Interpretability – Neel Nanda** | IASEAI 2025 | Neel Nanda (DeepMind) | 2025 | 25:12 | <https://www.youtube.com/watch?v=0704iLc55Fs> — The tightest 25-minute orientation available.
23. **Neel Nanda – Mechanistic Interpretability: A Whirlwind Tour** | Neel Nanda, FAR.AI | 2024 | 21:31 | <https://www.youtube.com/watch?v=veT2VI4vHyU>
24. **Open Problems in Mechanistic Interpretability: A Whirlwind Tour** | Neel Nanda, Google TechTalks | 2023 | 55:27 | <https://www.youtube.com/watch?v=ZSg4-H8L6Ec> — Explicitly a research-problem list. Start here for picking a thesis topic.
25. **Concrete Open Problems in Mechanistic Interpretability: Neel Nanda at SERI MATS** | Neel Nanda | 2023 | 1:26:48 | <https://www.youtube.com/watch?v=FnNTbqSG8w4> — Longer and more concrete than #24. Paper: *200 Concrete Open Problems in Mechanistic Interpretability*.
26. **The Story of Mech Interp** | Neel Nanda | 2025 | 2:06:05 | <https://www.youtube.com/watch?v=kkfLHmujzO8> — Intellectual history of the field, which is exactly what you need to avoid redoing solved work.
27. **What Matters Right Now In Mechanistic Interpretability?** | Neel Nanda | 2025 | 59:21 | https://www.youtube.com/watch?v=XXZ_CFFvGlc — Current research-direction triage.
28. **How Reasoning Models Break Mechanistic Interpretability Techniques** | Neel Nanda | 2025 | 42:21 | <https://www.youtube.com/watch?v=s3HgyprprhM> — The frontier problem: reasoning models invalidate existing tooling.
29. **Neel Nanda: Mechanistic Interpretability (HAAISS 2024)** | Neel Nanda, Alignment of Complex Systems | talk 2024, posted 2025 | 1:18:02 | https://www.youtube.com/watch?v=yG3TxLPO_Uc
30. **19 - Mechanistic Interpretability with Neel Nanda** | AXRP (Daniel Filan) | 2023 | 3:52:47 | <https://www.youtube.com/watch?v=3YbE7zybc5k> — Deepest available interview on methodology and epistemics.

Activation patching, causal intervention, causal scrubbing

31. **A Walkthrough of Interpretability in the Wild Part 1/2: Overview (w/ authors Kevin, Arthur, Alex)** | Neel Nanda with Kevin Wang, Arthur Conmy, Alexandre Variengien | 2022 | 57:20 | <https://www.youtube.com/watch?v=gzwj0jWbvbo> — The IOI circuit, the canonical worked example of path patching. Paper: *Interpretability in the Wild* (Wang et al. 2022).
32. **A Walkthrough of Interpretability in the Wild Part 2/2: Deep Dive (w/ authors Kevin, Arthur & Alex)** | same | 2022 | 1:46:05 | <https://www.youtube.com/watch?v=b9xfYBKlaX4>
33. **Causal Scrubbing | Intro to Neural Network Interpretability** | Cadenza Labs | 2023 | 7:27 | https://www.youtube.com/watch?v=_oCWsaFBY1M — Short but the only clear video explainer of Redwood's

causal scrubbing. Paper: Chan et al., *Causal Scrubbing* (2022).

34. **Uncovering and Inducing Interpretable Causal Structure in Deep Learning Models** | Atticus Geiger | Atticus Geiger (Pr(Ai)²R), Valence Labs | 2024 | 56:13 | <https://www.youtube.com/watch?v=EbVsoR7cjf0> — The formal theory (causal abstraction, DAS) under activation patching. Paper: *Causal Abstraction: A Theoretical Foundation for Mechanistic Interpretability*.
35. **A Walkthrough of Aligning Causal Variables and Distributed Representations w/ Atticus Geiger (1/3)** | Neel Nanda with Atticus Geiger | 2023 | 30:43 | https://www.youtube.com/watch?v=iZm0_l2H2CQ — Distributed Alignment Search explained by its author.
36. **Atticus Geiger - State of Interpretability & Ideas for Scaling Up [Alignment Workshop]** | Atticus Geiger, FAR.AI | 2024 | 21:14 | <https://www.youtube.com/watch?v=eqZ1iEoor5s>
37. **Formalizing Explanations of Neural Network Behaviors** | Paul Christiano (Alignment Research Center), Simons Institute | 2023 | 59:17 | <https://www.youtube.com/watch?v=u0619QrWxQc> — The critique that informal mech interp does not scale, and the heuristic-arguments alternative. Deeply relevant to anyone wanting to contribute foundationally.
38. **40 - Jason Gross on Compact Proofs and Interpretability** | AXRP | 2025 | 2:36:04 | <https://www.youtube.com/watch?v=ZzEj7jPk5fc> — Proof-length as a rigorous metric for whether an explanation is real.

Grokking, modular addition, ARENA

39. **A Walkthrough of Progress Measures for Grokking via Mechanistic Interpretability: What? (Part 1/3)** | Neel Nanda with Lawrence Chan | 2023 | 43:10 | <https://www.youtube.com/watch?v=IHikLL8ULa4> — Paper: Nanda et al., ICLR 2023.
40. **A Walkthrough of Reverse-Engineering Modular Addition: Model Training (Part 1/3)** | Neel Nanda | 2023 | 34:37 | <https://www.youtube.com/watch?v=ob4vuiqG2Go> — The full reverse-engineering done live, Fourier features and all.
41. **ARENA Lecture, Week 1 Day 1: Transformers: Building, Training, Sampling** | ARENA – AI Safety Education | 2025 | 34:09 | <https://www.youtube.com/watch?v=11Z50mi8dSg> — Entry point to the ARENA curriculum.
42. **ARENA Lecture, Week 1 Day 2: Introduction to Mechanistic Interpretability** | ARENA | 2025 | 37:52 | <https://www.youtube.com/watch?v=lfwT79Hgytc> — Pair with the ARENA notebooks at arena.education.
43. **Vision Mechanistic Interpretability - MATS Talk Summer 2024** | Sonia Joseph | 2024 | 58:31 | <https://www.youtube.com/watch?v=gQbh-RZtsq4> — Mech interp applied to ViTs; Prisma library.
44. **What do models learn during finetuning? A model diffing paper walkthrough w/ Clement & Julian** | Neel Nanda with Clément Dumas, Julian Minder | 2025 | 2:54:52 | https://www.youtube.com/watch?v=VQ_7zLXHf3s — Model diffing, one of the newest and most open subfields.
45. **NEURAL NETWORKS ARE WEIRD! - Neel Nanda (DeepMind)** | MLST | 2024 | 3:42:37 | <https://www.youtube.com/watch?v=YpFaPKOeNME> — Research-epistemics-heavy; good on how to not fool yourself.

2. Representation analysis

46. **Probing** | Stanford CS224U Natural Language Understanding | Spring 2021 | Christopher Potts (Stanford) | recorded 2021, posted 2022 | 11:29 | <https://www.youtube.com/watch?v=ELDtqhqv5ZE> — Compact and rigorous framing of probing as a method.

47. **Talk 4: Probing** | Zining Zhu (Toronto) | 2021 | 47:23 | <https://www.youtube.com/watch?v=hJ45rJgGyRw> — Full treatment including the critique literature.
48. **INFORMATION-THEORETIC PROBING WITH MINIMUM DESCRIPTION LENGTH** | Elena Voita, USC ISI | 2020 | 1:03:08 | https://www.youtube.com/watch?v=CakeVH_svdo — The principal answer to “your probe just memorized the labels.” Paper: Voita & Titov, EMNLP 2020. See also Hewitt & Liang, *Designing and Interpreting Probes with Control Tasks* (2019).
49. **Interpreting Natural Language Processing Models** | Yonatan Belinkov (Technion), Simons Institute | 2020 | 1:06:48 | <https://www.youtube.com/watch?v=vKCvcG6SHWc> — Paper: Belinkov, *Probing Classifiers: Promises, Shortcomings, and Advances* (CL 2022). This is the critique you must read before using a probe.
50. **Yonatan Belinkov - Causal Mediation Analysis for Interpreting Neural NLP: The Case of Gender Bias** | Yonatan Belinkov, UMass ML & Friends | 2020 | 1:10:29 | <https://www.youtube.com/watch?v=ew-P4vU-2yI> — The bridge from correlational probing to causal methods. Paper: Vig et al., *Investigating Gender Bias in Language Models Using Causal Mediation Analysis* (NeurIPS 2020).
51. **Feature learning & the linear representation hypothesis for steering & monitoring LLMs** | Mikhail Belkin (UC San Diego), Broad Institute Schmidt Center Symposium | 2026 | 29:52 | <https://www.youtube.com/watch?v=0eL69dWZ1lA> — A learning-theorist’s take on why representations go linear.
52. **Linear Structure of High-Level Concepts in Text-Controlled Generative Models** | Victor Veitch | Victor Veitch (Chicago), Valence Labs | 2024 | 1:07:59 | <https://www.youtube.com/watch?v=6yUuWknqSdM> — The formal statement of the linear representation hypothesis. Paper: Park, Choe & Veitch, *The Linear Representation Hypothesis and the Geometry of Large Language Models* (2023).
53. **Oskar John Hollinsworth — Linear Representations of Sentiment in Large Language Models [TAIS 2024]** | Oskar Hollinsworth | 2024 | 29:10 | <https://www.youtube.com/watch?v=8QNGVSVWukw> — A concrete worked case of a concept direction.
54. **Andy Zou – Top-Down Interpretability for AI Safety [Alignment Workshop]** | Andy Zou (CMU), FAR.AI | 2024 | 19:07 | <https://www.youtube.com/watch?v=ub1livlmzSc> — From the lead author of representation engineering. Paper: Zou et al., *Representation Engineering: A Top-Down Approach to AI Transparency* (2023).
55. **The Platonic Representation Hypothesis** | Phillip Isola (MIT), Simons Institute | 2024 | 44:27 | https://www.youtube.com/watch?v=1_xH2mUFpZw — Paper: Huh, Cheung, Wang & Isola, ICML 2024.
56. **Kathleen Creel (Northeastern University): ‘Against the Platonic Representation Hypothesis’** | Kathleen Creel, LSE Philosophy | 2026 | 43:05 | <https://www.youtube.com/watch?v=W1jIpcWgpcw> — The rebuttal. Pair with #55; the disagreement is where the research is.
57. **Pekka Marttinen: How to better compare representations learned by neural networks** | Pekka Marttinen (Aalto), FCAI | 2023 | 41:45 | <https://www.youtube.com/watch?v=dA7jKK3TNqc> — Directly on CKA/CCA and why existing similarity measures misbehave. Paper: Kornblith et al., *Similarity of Neural Network Representations Revisited* (ICML 2019).
58. **Contributed Talks: “Representational Alignment” - CCN 2025** | Yiqing Bo, Itamar Avitan, Imran Thobani et al., CCN Amsterdam | 2025 | 57:22 | <https://www.youtube.com/watch?v=vT-3kV89Rhk> — Includes “Evaluating Representational Similarity Measures from the Lens of Functional Correspondence” and “Linear Probing Fails to Identify the Ground-Truth Model.” This is the current state of the art on the validity of similarity metrics.
59. **Inspecting Neural Networks with CCA - A Gentle Intro (Explainable AI for Deep Learning)** | Jay Alammur | 2021 | 19:28 | https://www.youtube.com/watch?v=u7Dvb_a1D-0 — Visual primer on SVCCA before the harder material.

50. **Interpretability via Symbolic Distillation** | Miles Cranmer (Cambridge/Flatiron), Simons Institute | 2023 | 51:04 | <https://www.youtube.com/watch?v=XHBJJ2N-kUc> — A genuinely different route: distill the network into symbolic form. Paper: Cranmer et al., *Discovering Symbolic Models from Deep Learning with Inductive Biases* (2020).

3. Classical interpretability and attribution

51. **Quantitative Testing with Concept Activation Vectors (TCAV) – Been Kim (Google) - 2018** | Been Kim (Google Brain), JHU CLSP | talk 2018, posted 2023 | 56:09 | <https://www.youtube.com/watch?v=wBcrPDPUTrE> — Paper: Kim et al., *Interpretability Beyond Feature Attribution: Quantitative Testing with Concept Activation Vectors*, ICML 2018.
52. **Interpretability Beyond Feature Attribution** | Been Kim, MLconf | 2018 | 27:56 | <https://www.youtube.com/watch?v=Ff-Dx79QEEY> — Shorter conference version of the same work.
53. **Interpretability - now what?** | Been Kim (Google Brain), Simons Institute “Frontiers of Deep Learning” | 2019 | 47:37 | https://www.youtube.com/watch?v=5w_rgBbwQHW — Where she presents the saliency-map failure results. Paper: Adebayo et al., *Sanity Checks for Saliency Maps* (NeurIPS 2018); also Kindermans et al., *The (Un)reliability of Saliency Methods*.
54. **SecML18: Been Kim on Interpretability for when NOT to use machine learning** | Been Kim | 2019 | 26:53 | <https://www.youtube.com/watch?v=t1rJ9gIrYKM>
55. **Been Kim wants interpretability for everyone** | Been Kim, OATML Oxford | 2021 | 1:05:18 | <https://www.youtube.com/watch?v=06hIoM-cLVM> — Covers the human-subject studies showing saliency maps do not help people.
56. **Alignment and Interpretability: How We Might Get It Right | Been Kim (Google DeepMind)** | FAR.AI | 2024 | 33:21 | <https://www.youtube.com/watch?v=JVoYzS2RcRc> — Her current position; pairs with *Beyond Interpretability: Towards a Machine-Centric Vocabulary*.
57. **Feature Attribution | Stanford CS224U Natural Language Understanding | Spring 2021** | Christopher Potts (Stanford) | recorded 2021, posted 2022 | 15:32 | <https://www.youtube.com/watch?v=RFE6xdfJvag> — Covers integrated gradients and its axioms cleanly. Paper: Sundararajan, Taly & Yan, *Axiomatic Attribution for Deep Networks* (ICML 2017).
58. **Scott Lundberg, Microsoft Research - Explainable Machine Learning with Shapley Values** | Scott Lundberg (MSR), H2O World | 2019 | 18:34 | <https://www.youtube.com/watch?v=ngOBhhINWb8> — From the SHAP author. Paper: Lundberg & Lee, NeurIPS 2017.
59. **Explainable AI explained! | #4 SHAP** | DeepFindr | 2021 | 15:50 | <https://www.youtube.com/watch?v=9haIOplEIGM> — The mechanics, if the above is too high-level.
70. **Stop Explaining Black Box Machine Learning Models - Cynthia Rudin** | Cynthia Rudin (Duke), Caltech | 2021 | 30:15 | <https://www.youtube.com/watch?v=4oXFEDoEcAk> — Paper: Rudin, *Stop Explaining Black Box Machine Learning Models for High Stakes Decisions and Use Interpretable Models Instead* (Nature MI 2019).
71. **Stop explaining black box machine learning models for high stakes decisions and..... - Cynthia Rudin** | Cynthia Rudin, Institute for Advanced Study | 2022 | 1:04:02 | <https://www.youtube.com/watch?v=HY2NoQQzRic> — The long technical version with the optimal-sparse-decision-tree work.
72. **Interpretable Machine Learning for High-Stakes Decisions - Cynthia Rudin** | Chennai Mathematical Institute | 2022 | 1:05:24 | <https://www.youtube.com/watch?v=2wj1Ztym5qY>
73. **Allen School Colloquium: Pang Wei Koh (Stanford University)** | Pang Wei Koh | 2022 | 58:49 | <https://www.youtube.com/watch?v=5tidNjeVG8s> — Paper: Koh & Liang, *Understanding Black-box Predictions*

via *Influence Functions* (ICML 2017 best paper).

- 74. **Interpreting Deep Neural Networks (DNNs)** | Bin Yu (UC Berkeley), Simons Institute | 2019 | 46:48 | <https://www.youtube.com/watch?v=mCDv3kpzqwQ> — The statistician’s framing: stability and the PCS framework.
- 75. **25. Interpretability** | Peter Szolovits (MIT 6.S897) | recorded 2019, posted 2020 | 1:18:41 | <https://www.youtube.com/watch?v=wDLzLN1tArA> — Clean survey of the classical toolkit.
- 76. **#047 Interpretable Machine Learning - Christoph Molnar** | MLST | 2021 | 1:40:22 | <https://www.youtube.com/watch?v=0LIACHcxpHU> — Author of the *Interpretable Machine Learning* book; strong on method pitfalls.

4. Science of model behavior

- 77. **Datamodels: Predicting Predictions with Training Data** | Andrew Ilyas (MIT, Mądry Lab), Norbert Wiener Center | 2022 | 1:00:23 | <https://www.youtube.com/watch?v=1djD0kK2uik> — Paper: Ilyas, Park, Engstrom, Leclerc & Mądry, *Datamodels* (ICML 2022).
- 78. **MedAI #51: Datamodels - Predicting Predictions from Training Data** | Andrew Ilyas | Stanford MedAI | 2022 | 56:56 | https://www.youtube.com/watch?v=dgo7nRQcw_E — Shorter, more accessible version.
- 79. **ML Robustness & Engineering - Andrew Ilyas (MIT)** | MLST | 2024 | 1:28:01 | <https://www.youtube.com/watch?v=9BC9BsZRCKQ> — Covers the TRAK line and the broader “models are a function of their data” program. Paper: Park, Georgiev, Ilyas, Leclerc & Mądry, *TRAK: Attributing Model Behavior at Scale* (ICML 2023).
- 30. **What Do Our Models Learn? - Aleksander Mądry** | Aleksander Mądry (MIT), Institute for Advanced Study | 2020 | 1:27:40 | https://www.youtube.com/watch?v=_Osf6F7zZUM — The lab’s foundational framing. Paper: *Adversarial Examples Are Not Bugs, They Are Features* (2019).
- 31. **Scalable Extraction of Training Data from (Production) Language Models** | Nicholas Carlini (Google DeepMind), Simons Institute | 2024 | 49:11 | <https://www.youtube.com/watch?v=adCLAhoQhOQ> — Paper: Nasr, Carlini et al. (2023), the divergence attack on ChatGPT.
- 32. **TrustML Seminar: Nicholas Carlini on “Extracting training data from neural networks”** | Nicholas Carlini | 2021 | 1:00:50 | https://www.youtube.com/watch?v=2Xl2B2R7_1M — Paper: Carlini et al., *Extracting Training Data from Large Language Models* (USENIX Security 2021).
- 33. **USENIX Security ’19 - The Secret Sharer: Evaluating and Testing Unintended Memorization in Neural Networks** | Nicholas Carlini | 2019 | 20:36 | <https://www.youtube.com/watch?v=U9XbFtCWEdE> — The origin of quantitative memorization measurement (exposure metric).
- 34. **Jesse Hoogland: Singular Learning Theory, Developmental interpretability (HAAISS 2024)** | Jesse Hoogland (Timaheus) | talk 2024, posted 2025 | 1:17:19 | <https://www.youtube.com/watch?v=dtcRhrOgqKU> — Papers: *Towards Developmental Interpretability*; Lau et al., *The Local Learning Coefficient*.
- 35. **31 - Singular Learning Theory with Daniel Murfet** | AXRP | 2024 | 2:32:06 | <https://www.youtube.com/watch?v=hdB9gIwD6x4> — The mathematical foundation (Watanabe’s SLT) at depth.
- 36. **ALIGN Webinar Series #12 Jesse Hoogland Singular Learning Theory for AI Safety** | AI Alignment Network | 2025 | 1:02:17 | <https://www.youtube.com/watch?v=jgcatt6eZyY>
- 37. **38.2 - Jesse Hoogland on Singular Learning Theory** | AXRP | 2024 | 18:18 | <https://www.youtube.com/watch?v=P528XdjWvZg> — Quick orientation if the above are too long.

38. **EI Seminar - Naomi Saphra - Interpreting Training** | Naomi Saphra (Harvard/Kempner), MIT | 2024 | 1:04:20 | <https://www.youtube.com/watch?v=J0tHAZlFGSc> — The argument that interpretability should study training dynamics, not just final checkpoints.
39. **Training Is Nothing Like Learning with Naomi Saphra (Harvard)** | The Information Bottleneck | 2026 | 1:11:34 | <https://www.youtube.com/watch?v=2WqvUQY1nlg> — Her current, sharper position.
40. **Naomi Saphra: Linear Connectivity Reveals Generalization Strategies** | Formal Languages and Neural Networks Seminar | 2022 | 54:23 | https://www.youtube.com/watch?v=zxtD9rW_3Bo — Loss-landscape connectivity as an interpretability signal. Paper: Juneja, Bansal, Cho, Sedoc & Saphra (ICLR 2023).
41. **“Dynamics of Concept Learning and Emergent Abilities in Neural Networks” - Ekdeep Lubana** | Ekdeep Singh Lubana (Harvard/Michigan), TTIC | 2025 | 1:03:31 | <https://www.youtube.com/watch?v=nYGyeJRHqEA> — Toy models that explain emergence as a dynamics phenomenon, not a scaling mystery.
42. **Explaining emergence in NN with model systems analysis - Ekdeep Singh Lubana (PIBSS Speaker Series)** | Principles of Intelligence | 2024 | 1:24:56 | <https://www.youtube.com/watch?v=bCEXKmDGUzQ> — Methodological: how to build toy “model systems” like a biologist.
43. **The Surprising Simplicity of the Early-Time Learning Dynamics of Neural Networks in High Dimension** | Wei Hu (Princeton), Simons Institute | 2020 | 47:10 | <https://www.youtube.com/watch?v=2ytGG5qtYvE> — Theory side of training-dynamics analysis.

5. Evaluation as a science

44. **Keynote Talk: Sanmi Koyejo - Beyond Benchmarks; Building a Science of AI Measurement** | Sanmi Koyejo (Stanford), Deep Learning Indaba | 2025 | 53:03 | <https://www.youtube.com/watch?v=xXYvsDJS9vw> — The single best framing talk for this whole section. Paper: Schaeffer, Miranda & Koyejo, *Are Emergent Abilities of Large Language Models a Mirage?* (NeurIPS 2023 best paper).
45. **CITP Special Event: Sayash Kapoor: Final Public Oral (FPO) - The Missing Science of AI Evaluation** | Sayash Kapoor (Princeton) | 2026 | 1:17:59 | <https://www.youtube.com/watch?v=8SGdYYtmmwI> — A full dissertation defense on exactly this topic. Papers: *AI Agents That Matter*; *Leakage and the Reproducibility Crisis in ML-based Science*.
46. **Building and evaluating AI Agents — Sayash Kapoor, AI Snake Oil** | AI Engineer | 2025 | 19:59 | <https://www.youtube.com/watch?v=d5ElXhbcfA> — Compact version with the cost-control argument.
47. **The Challenge of Valid Evaluations** | Amanda Coston (UC Berkeley), Simons Institute | 2026 | 32:40 | <https://www.youtube.com/watch?v=WUitKvhjn90> — Construct validity and counterfactual evaluation done rigorously.
48. **The Inadequacy of Offline LLM Evaluations: A Need to Account for Personalization in Model Behavior** | Angelina Wang (Cornell Tech), Simons Institute | 2026 | 31:57 | <https://www.youtube.com/watch?v=q4ilyQs3nEY> — Why stateless benchmark inference misrepresents deployed behavior.
49. **Leaderboardism in NLP: panel with Kawin Ethayarajh, Jesse Dodge, Rachael Tatman & Anna Rogers** | Anna Rogers (chair) | 2020 | 1:10:10 | <https://www.youtube.com/watch?v=VauPmCJSIH8> — The definitive critique of leaderboard culture, by the people who wrote it up. Paper: Ethayarajh & Jurafsky, *Utility is in the Eye of the User* (EMNLP 2020).
50. **Rasa Reading Group: What Will it Take to Fix Benchmarking in Natural Language Understanding (Part 2)** | Rasa | 2021 | 1:08:02 | <https://www.youtube.com/watch?v=UDK6ZTUIvig> — Paper: Bowman & Dahl, NAACL 2021. The four criteria any benchmark must meet.

01. **ACL 2022 Talk – The Dangers of Underclaiming** | Sam Bowman (NYU/Anthropic) | 2022 | 12:19 | <https://www.youtube.com/watch?v=fPnX374CaFQ> — The other half of the evaluation-rigor argument: over-correction is also a failure mode.
02. **Learning Machines Seminar @ Cornell: Sam Bowman (NYU)** | Sam Bowman | 2019 | 58:13 | https://www.youtube.com/watch?v=Ei_EKCDPYwQ — GLUE/SuperGLUE from the person who built them and then argued they were saturated.
03. **Naman Jain - “LiveCodeBench: Holistic and contamination free evaluation of LLMs for code”** | Naman Jain (UC Berkeley) | 2024 | 56:47 | <https://www.youtube.com/watch?v=HGqhS9-rnQM> — Contamination as a first-class design constraint. Paper: Jain et al., LiveCodeBench (ICLR 2025).
04. **Context for Interpreting Benchmark Performances** | Kenneth Church (Baidu/Northeastern) | 2021 | 24:35 | <https://www.youtube.com/watch?v=3Lo-iqo1PjQ> — Short and sharp on what a benchmark number does and does not license.
05. **Rishi Bommasani – Holistic Evaluation of Language Models** | Rishi Bommasani (Stanford CRFM) | 2023 | 37:47 | <https://www.youtube.com/watch?v=A0kD00WdlKY> — Paper: *HELM* (Liang, Bommasani et al. 2022).
06. **Stanford CME295 Transformers & LLMs | Autumn 2025 | Lecture 8 - LLM Evaluation** | Stanford Online | 2025 | 1:49:25 | <https://www.youtube.com/watch?v=8fNP4N46RRo> — Current, technical, covers LLM-as-judge bias directly.
07. **LLM-as-a-Judge / Autoraters • Guest Lecture at Northeastern University • March 24, 2026** | Aman Chadha | 2026 | 56:39 | https://www.youtube.com/watch?v=N_DwZR-XCc — Position bias, self-preference, agreement-with-humans calibration.
08. **Joelle Pineau: Reproducibility, Reusability, and Robustness in Deep Reinforcement Learning ICLR 2018** | Joelle Pineau (McGill/Meta) | 2018 | 49:22 | <https://www.youtube.com/watch?v=Vh4H0gOwdIg> — The statistical-rigor talk: seed variance, confidence intervals, the ML reproducibility checklist. Paper: Henderson et al., *Deep RL That Matters* (AAAI 2018).

6. NeuroAI and biological inspiration

09. **Reverse engineering visual intelligence - James DiCarlo** | Jim DiCarlo (MIT), Stanford | 2018 | 40:59 | <https://www.youtube.com/watch?v=Sc8-3XD3bDw> — Paper: Yamins & DiCarlo, *Using goal-driven deep learning models to understand sensory cortex* (Nat Neuro 2016).
10. **Jim DiCarlo, MIT: Reverse engineering visual intelligence** | CCBM | 2018 | 40:54 | <https://www.youtube.com/watch?v=5kq7M6pcQ5g> — Alternate recording; covers Brain-Score.
11. **Stony Brook University Mind/Brain Lecture 2026 with Dr. James DiCarlo** | Jim DiCarlo | 2026 | 1:03:05 | <https://www.youtube.com/watch?v=WVvZaMQLS6U> — His current position, including model-driven neural control.
12. **BI 075 Jim DiCarlo: Reverse Engineering Vision** | Brain Inspired podcast | 2020 | 1:16:04 | <https://www.youtube.com/watch?v=H7LzjJ2J7qw> — Interview format, good on the epistemics of model-brain comparison.
13. **Deep Learning and the Brain 2019 – Prof. Daniel Yamins** | Dan Yamins (Stanford), ELSC Jerusalem | 2019 | 59:57 | <https://www.youtube.com/watch?v=eq7abzaktM4> — Paper: Yamins et al., PNAS 2014.
14. **A Fruitful Reciprocity: The Neuroscience-AI Connection** | Dan Yamins (Stanford), MIT CBMM | 2023 | 1:10:35 | <https://www.youtube.com/watch?v=6NOFtwKU3sA> — Argues the exchange runs both ways; the best single NeuroAI framing talk.

15. **CSL seminar: Dan Yamins** | Dan Yamins, MIT Improbable AI | 2021 | 1:01:05 | <https://www.youtube.com/watch?v=7pTH0cY01AU> — Unsupervised and developmentally plausible models of cortex.
16. **On the Neural Machinery of Faces** | Doris Tsao (Caltech/Berkeley), MIT CBMM | 2018 | 1:03:18 | https://www.youtube.com/watch?v=n-C0YdJu_A — Paper: Chang & Tsao, *The Code for Facial Identity in the Primate Brain* (Cell 2017). The cleanest example anywhere of a decoded neural code; the direct biological analogue of a “feature basis.”
17. **Doris Tsao: 2010 Allen Institute for Brain Science Symposium** | Doris Tsao | 2010 | 22:36 | <https://www.youtube.com/watch?v=wpVm1QiNQEc> — The earlier face-patch work.
18. **Surya Ganguli - Deep Learning Theory: From Generalization to the Brain** | Surya Ganguli (Stanford), DeepMath | 2020 | 1:00:22 | <https://www.youtube.com/watch?v=UJ6FTAg7Ks> — Paper: Saxe, McClelland & Ganguli, *Exact solutions to the nonlinear dynamics of learning in deep linear networks* (2014).
19. **Neural networks and the brain: from the retina to semantic cognition - Surya Ganguli** | Stanford | 2018 | 36:37 | <https://www.youtube.com/watch?v=FKi6sWK9Qo0>
20. **Surya Ganguli, A theory of neural dimensionality: McGovern Institute Symposium** | McGovern/MIT | 2016 | 40:08 | https://www.youtube.com/watch?v=3Kfei1_hAiY — Dimensionality of neural data as a measurable quantity; underrated methodologically.
21. **Konrad Kording - Why neuroscience needs deep learning theory** | Konrad Kording (UPenn), NAS Colloquia | 2019 | 14:16 | <https://www.youtube.com/watch?v=GgzRdQ5nm0U> — Short and provocative. Paper: Jonas & Kording, *Could a Neuroscientist Understand a Microprocessor?* (PLoS CB 2017) — arguably the most important methodological critique in this entire curriculum.
22. **Michigan Biostatistics presents a seminar with Konrad Kording, PhD** | Konrad Kording | 2021 | 52:03 | <https://www.youtube.com/watch?v=2mRJrXKUT1M> — Causality in neuroscience and ML.
23. **DLRLSS 2019 - Biological DL - Blake Richards** | Blake Richards (McGill/Mila), Amii | 2019 | 1:20:56 | https://www.youtube.com/watch?v=yqQe_Qcamwo — The full tutorial on biologically plausible credit assignment. Paper: Richards et al., *A deep learning framework for neuroscience* (Nat Neuro 2019).
24. **Fundamentals of deep learning in neuroscience** | Blake Richards, BrainHack School | 2020 | 1:35:12 | <https://www.youtube.com/watch?v=kpNI6or-qJs> — Inductive-bias framing; a good lecture-style entry point.
25. **Deep Learning with Ensembles of Neocortical Microcircuits - Dr. Blake Richards** | Blake Richards | 2018 | 50:14 | <https://www.youtube.com/watch?v=YUVLgccVi54> — Paper: Guerguiev, Lillicrap & Richards, *Towards deep learning with segregated dendrites* (eLife 2017).
26. **Backpropagation and Deep Learning in the Brain** | Timothy Lillicrap (DeepMind), Simons Institute | 2018 | 59:43 | <https://www.youtube.com/watch?v=-kHLKLLxIF4> — Paper: Lillicrap, Santoro, Marris, Akerman & Hinton, *Backpropagation and the brain* (Nat Rev Neuro 2020). The central reference for this debate.
27. **Stanford Seminar - Can the brain do back-propagation? Geoffrey Hinton** | Geoffrey Hinton (Toronto/Google) | 2016 | 1:25:13 | <https://www.youtube.com/watch?v=VIRCybGgHts> — Hinton’s own early framing of the problem.
28. **Does the brain do backpropagation? CAN Public Lecture - Geoffrey Hinton - May 21, 2019** | Canadian Association for Neuroscience | 2019 | 1:22:03 | <https://www.youtube.com/watch?v=qIEfJ6OBGj8> — Updated version with temporal derivatives as error signals.
29. **MoroccoAI Conference 2022 Honorary Keynote Prof. Geoffrey Hinton - The Forward-Forward Algorithm** | Geoffrey Hinton | talk Dec 2022, posted 2023 | 49:48 | https://www.youtube.com/watch?v=_5W5BvKe_6Y — Paper: Hinton, *The Forward-Forward Algorithm: Some Preliminary Investigations* (2022).

30. **Geoffrey Hinton Unpacks The Forward-Forward Algorithm** | Eye on AI | 2023 | 58:55 | https://www.youtube.com/watch?v=NWqy_b1OvwQ — Interview; covers mortal computation, which the paper only gestures at.
31. **Nobel Prize lecture: John J. Hopfield, Nobel Prize in Physics 2024** | John Hopfield (Princeton) | lecture Dec 2024, posted 2025 | 40:01 | <https://www.youtube.com/watch?v=8SffhDk4mdU> — Paper: Hopfield, PNAS 1982.
32. **Nobel Prize lecture: Geoffrey Hinton, Nobel Prize in Physics** | Geoffrey Hinton | lecture Dec 2024, posted 2025 | 31:53 | <https://www.youtube.com/watch?v=XDE9DjpcSdI> — Papers: Ackley, Hinton & Sejnowski, *A Learning Algorithm for Boltzmann Machines* (1985).
33. **2024 Nobel Prize lectures in physics | John Hopfield and Geoffrey Hinton** | Nobel Prize | 2024 | 1:18:57 | <https://www.youtube.com/watch?v=lPIVl5eBPh8> — Both lectures in one stream, with the laudatio.
34. **Modern Hopfield Networks - Dr Sepp Hochreiter** | Sepp Hochreiter (JKU Linz), IARAI | 2020 | 1:26:33 | <https://www.youtube.com/watch?v=bsdPZJKOlQs> — Paper: Ramsauer et al., *Hopfield Networks is All You Need* (ICLR 2021) — the attention/associative-memory equivalence.
35. **Hopfield Networks in 2021 - Fireside chat between Sepp Hochreiter and Dmitry Krotov | NeurIPS 2020** | IARAI | 2020 | 1:15:53 | <https://www.youtube.com/watch?v=k3YmWrK6wxo> — Two of the principals arguing it out.
36. **Dmitry Krotov | Modern Hopfield Networks for Novel Transformer Architectures** | Dmitry Krotov (IBM/MIT-IBM), Harvard CMSA | 2023 | 1:05:07 | <https://www.youtube.com/watch?v=5LXiQUsnHrI> — Paper: Krotov & Hopfield, *Dense Associative Memory for Pattern Recognition* (NeurIPS 2016).
37. **Dmitry Krotov - Generative AI models through the lens of Dense Associative Memory - IPAM at UCLA** | IPAM | 2024 | 1:10:15 | <https://www.youtube.com/watch?v=lzYwzIkO89Q> — Energy-based reading of modern generative models.
38. **Energy-based Approaches to Representation Learning - Yann LeCun** | Yann LeCun (NYU/Meta), Institute for Advanced Study | 2019 | 39:54 | <https://www.youtube.com/watch?v=m17B-cXcZFI> — Paper: LeCun et al., *A Tutorial on Energy-Based Learning* (2006).
39. **Yann LeCun | May 18, 2021 | The Energy-Based Learning Model** | Harvard Mathematical Picture Language | 2021 | 1:15:52 | <https://www.youtube.com/watch?v=4lthJd3DNTM> — Longer; leads into the JEPa line.
40. **Debate: “Does Hierarchical Predictive Coding Explain Perception?” (Clark, Heeger, Melloni, Rescorla)** | NYU Center for Mind, Brain and Consciousness | 2018 | 2:00:33 | <https://www.youtube.com/watch?v=CvAbPtbjxhw> — A real adversarial debate, which is rarer and more useful than another sympathetic overview. Paper: Rao & Ballard, *Nat Neuro* 1999; Heeger, *PNAS* 2017.
41. **The Predictive Brain: Michael Pollan, Celeste Kidd, Christos Papadimitriou, and Bruno Olshausen** | Simons Institute | 2019 | 1:25:39 | <https://www.youtube.com/watch?v=ItU0HeFmsrY> — Olshausen on sparse coding, which is the direct ancestor of dictionary learning in Section 1. Paper: Olshausen & Field, *Nature* 1996.
42. **A New Framework for Modeling Brain Information Processing - Nikolaus Kriegeskorte** | Nikolaus Kriegeskorte (Columbia), Stanford | 2015 | 49:50 | <https://www.youtube.com/watch?v=tyYIuvbV2po> — RSA, the method ML later rediscovered as CKA. Paper: Kriegeskorte, Mur & Bandettini (2008).
43. **BI 163 Ellie Pavlick: The Mind of a Language Model** | Ellie Pavlick (Brown/DeepMind), Brain Inspired | 2023 | 1:21:35 | https://www.youtube.com/watch?v=o8_lZbQ0PEs — The cognitive-science-meets-interpretability perspective.
44. **Ellie Pavlick, (How) Does AI Think?** | Ellie Pavlick, Remarque Institute NYU | 2026 | 1:33:50 | <https://www.youtube.com/watch?v=etWMRit98sA> — Current and unusually careful about what

interpretability claims actually establish.

7. Technical AI safety research

45. **Can We Get Asymptotic Safety Guarantees Based On Scalable Oversight?** | Geoffrey Irving (UK AI Safety Institute), Simons Institute | 2025 | 59:10 | https://www.youtube.com/watch?v=cJ0YPyP-I_o — Complexity-theoretic treatment of debate. The most technical scalable-oversight talk available. Papers: Irving, Christiano & Amodei, *AI Safety via Debate* (2018); *Prover-Estimator Debate* (2025).
46. **Sam Bowman - Adversarial Scalable Oversight for Truthfulness: Work in Progress** | Sam Bowman (NYU/Anthropic), FAR.AI | 2024 | 29:11 | <https://www.youtube.com/watch?v=bkZsjmyF00k> — Paper: Michael et al., *Debate Helps Supervise Unreliable Experts* (2023); Bowman et al., *Measuring Progress on Scalable Oversight* (2022).
47. **Collin Burns - Weak-to-Strong Generalization** | Collin Burns (OpenAI Superalignment), FAR.AI | 2024 | 26:54 | <https://www.youtube.com/watch?v=wBPZNhw1LV4> — From the first author. Paper: Burns et al., *Weak-to-Strong Generalization* (ICML 2024).
48. **Transfer learning for weak-to-strong generalization** | Yuekai Sun (Michigan), Simons Institute | 2024 | 42:10 | <https://www.youtube.com/watch?v=q68VtFoWmFI> — The theory side; when W2S provably works.
49. **Aligning ML objectives with human values** | Paul Christiano (OpenAI), Simons Institute | 2019 | 47:00 | <https://www.youtube.com/watch?v=3fZvahTlPaQ> — Christiano's framing of the problem that ELK later formalized. Paper: Christiano, Cotra & Xu, *Eliciting Latent Knowledge* (ARC 2021).
50. **23 - Mechanistic Anomaly Detection with Mark Xu** | AXRP, Mark Xu (ARC) | 2023 | 2:05:53 | <https://www.youtube.com/watch?v=sEWei02m7qk> — The most substantive public explanation of ELK and the ARC heuristic-arguments agenda.
51. **Dangerous Capability Evals: Basis for Frontier Safety** | Mary Phuong (Google DeepMind) | FAR.AI | 2024 | 15:15 | <https://www.youtube.com/watch?v=pO8IcIqhHuk> — Paper: Phuong et al., *Evaluating Frontier Models for Dangerous Capabilities* (2024).
52. **David Duvenaud | The big picture of LLM dangerous capability evals** | David Duvenaud (Toronto/Anthropic), Schwartz Reisman Institute | 2025 | 1:19:50 | <https://www.youtube.com/watch?v=zvIN7PnVuy4> — Methodologically critical about what these evals can and cannot establish.
53. **Alignment faking in large language models** | Ryan Greenblatt, Monte MacDiarmid, Benjamin Wright, Evan Hubinger (Anthropic/Redwood) | 2024 | 1:30:19 | <https://www.youtube.com/watch?v=9eXV64O2Xp8> — Authors walking through the full experimental design. Paper: Greenblatt et al. (2024).
54. **27 - AI Control with Buck Shlegeris and Ryan Greenblatt** | AXRP | 2024 | 2:56:04 | <https://www.youtube.com/watch?v=gQbCO6zGRiI> — The control agenda as an alternative to alignment. Paper: Greenblatt et al., *AI Control: Improving Safety Despite Intentional Subversion* (ICML 2024).
55. **Interpretability Agents** | Sarah Schwettmann (MIT/Transluce), Simons Institute | 2024 | 42:07 | <https://www.youtube.com/watch?v=9q9deK7RfII> — Automating interpretability with agents. Paper: Shaham, Schwettmann et al., *A Multimodal Automated Interpretability Agent (MAIA)* (ICML 2024).
56. **Sarah Schwettmann - Scalable Oversight and Understanding [Alignment Workshop]** | FAR.AI | posted 2026 | 5:41 | <https://www.youtube.com/watch?v=8oJW7hdbc2I> — Short; the argument that understanding must scale with capability.
57. **Neel Nanda - Our Pivot To Pragmatic Interpretability [Alignment Workshop]** | Neel Nanda (DeepMind), FAR.AI | posted Dec 2025 | 10:20 | <https://www.youtube.com/watch?v=k93o4R145Os> — An important strategic update: the DeepMind team stepping back from ambitious reverse-engineering.

58. “AI Safety Through Interpretable and Controllable Language Models” - Peter Hase, YRSS | Peter Hase (UNC/Anthropic), TTIC | 2024 | 59:18 | <https://www.youtube.com/watch?v=Ao8Gprp9mC0> — Model editing, belief localization, and why localization does not imply editability. Paper: Hase et al., *Does Localization Inform Editing?* (NeurIPS 2023).
59. Yoshua Bengio - Disentangling Agency & Predictive Power Without Solving ELK [Alignment Workshop] | Yoshua Bengio (Mila), FAR.AI | posted 2026 | 30:56 | <https://www.youtube.com/watch?v=ZndiBDmss-w> — The Scientist AI proposal, framed explicitly against ELK.
50. 42 - Owain Evans on LLM Psychology | AXRP, Owain Evans (Truthful AI/Berkeley) | 2025 | 2:14:26 | <https://www.youtube.com/watch?v=3D4pgIKR4cQ> — Introspection, out-of-context reasoning, emergent misalignment. Papers: *Looking Inward*; *Emergent Misalignment* (2025).
51. An Observation on Generalization | Ilya Sutskever (OpenAI), Simons Institute | 2023 | 57:21 | https://www.youtube.com/watch?v=AKMuA_TVz3A — Compression-theoretic account of unsupervised learning. Bonus framing talk for the whole curriculum.
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Flagged as unverifiable — I could not find a real recording and will not invent one:

- A dedicated recorded talk by **Julius Adebayo** on *Sanity Checks for Saliency Maps* — UNVERIFIED. Search terms: Julius Adebayo sanity checks saliency maps NeurIPS 2018 spotlight, Adebayo debugging tests for model explanations MIT. Entry #63 (Been Kim, Simons 2019) covers the same results. The paper is Adebayo, Gilmer, Muelly, Goodfellow, Hardt & Kim (NeurIPS 2018).
- A dedicated recorded talk by **Mukund Sundararajan** on integrated gradients — UNVERIFIED. Search terms: Mukund Sundararajan integrated gradients talk Google, axiomatic attribution for deep networks ICML 2017 oral. Entry #67 covers the method.
- A dedicated **TRAK** talk by Sung Min Park or Kristian Georgiev — UNVERIFIED. Search terms: Sung Min Park TRAK attributing model behavior at scale ICML 2023 oral, Kristian Georgiev data attribution MIT talk. Entry #79 (Andrew Ilyas, MLST 2024) covers TRAK within the broader data-attribution program.
- A dedicated **model stitching** talk (Bansal, Nakkiran & Barak, or Lenc & Vedaldi) — UNVERIFIED. Search terms: Yamini Bansal revisiting model stitching NeurIPS 2021, Lenc Vedaldi understanding image representations equivariance equivalence. Entries #57 and #58 cover the adjacent representational-similarity literature.

Two notes on using this list. First, Neel Nanda’s paper walkthroughs (#6, #7, #31, #32, #39, #40, #44) are the closest thing to a graduate seminar in the field, but they are long and assume you have the paper open; budget real time for them rather than treating them as lectures. Second, I deliberately paired several talks with their own rebuttals — #55 with #56 on the platonic representation hypothesis, #11 and #12 with #14 on sparse autoencoders, #24 with #157 on whether ambitious mech interp is tractable. For someone aiming at original contributions, those disagreements are where the open problems actually live.

I did not write any files to the project directory, and since the working directory is not a git repository, there was nothing to commit or push. All search and verification scratch files are in the session scratchpad.

D.10 The Bridge, Concretely

Part III has been a long argument that the AI literature is worth your time. This section makes the argument specific. Below are **fourteen transfers**: seven ideas from AI that have a stated, unexploited application in molecular modeling, and seven places where the molecular setting has something the AI field lacks.

None of these are gestures. Each names the source talk, the source paper, the target problem, and what the first experiment would be. **If you want a thesis topic, it is in this table.**

D.10.1 Seven ideas to carry into molecules

1. μ P — hyperparameter transfer across scale

From: Greg Yang, *Tuning Large Neural Networks via Zero-Shot Hyperparameter Transfer* (D.5.3) · Yang et al. 2022, arXiv:2203.03466. **To:** Every structure-prediction and design model retunes hyperparameters at every scale, by hand, at enormous cost. **First experiment:** Derive the μ P parameterization for a pair-representation network. The non-obvious part is the triangle operations, which have no analogue in a standard transformer. Verify coordinate-scale stability across widths, then tune a small model and transfer. **Why it is a contribution and not an application:** the derivation is genuinely new, because the architecture is.

2. Process reward models over generation trajectories

From: Lightman et al., *Let's Verify Step by Step* (D.7.4), arXiv:2305.20050. **To:** Diffusion-based design scores only the final backbone after 50 denoising steps. There is no intermediate signal. **First experiment:** Run RFDiffusion, record intermediate states, label them by the final design's self-consistency outcome, train a model to score partial trajectories, then resample from bad trajectories early. **Why it matters:** it converts a fixed-cost generator into one where compute is allocated adaptively, and it is the exact structure that made reasoning models work.

3. Over-optimization curves for design filters

From: Gao, Schulman & Hilton (D.6.1), arXiv:2210.10760. **To:** Design pipelines optimize against learned proxies — pLDDT, interface pAE, self-consistency RMSD — with no measurement of where the proxy stops tracking reality. **First experiment:** Take a target with experimental data at several optimization strengths. Plot wet-lab success against filter stringency. Find the peak. **Why it matters:** the curve has a maximum, and nobody knows where it is. Publishing it would immediately change how campaigns are run.

4. Ring attention and IO-aware kernels for triangle operations

From: Dao, *FlashAttention* (D.5.1); Liu et al., *Ring Attention* (D.4.5). **To:** Triangle multiplication is the memory bottleneck in every AF3-class model and the reason large complexes do not fit. **First experiment:** Write a tiled, recomputation-based triangle-multiplicative update, then a ring-sharded version across devices. Measure peak memory against sequence length. **Why it matters:** this is what makes megabase-scale and large-assembly modeling feasible on hardware anyone owns.

5. DoReMi-style data mixture optimization over protein families

From: Sang Michael Xie, *Data-distributional Approaches* (D.5.5), arXiv:2305.10429. **To:** Protein models train on whatever UniRef gives them, with mixture weights nobody chose. **First experiment:** Train small proxy models, run the DoReMi minimax reweighting over Pfam clans, and compare downstream performance against uniform and against size-proportional sampling. **Why it matters:** it is a reimplement, but on data where nobody has done it, and the answer is directly actionable.

6. Sparse autoencoders with real ground truth

From: D.9 (interpretability) and InterPLM (Atlas Q.4). **To:** Protein models have thousands of independently-established labels — catalytic residues, domain boundaries, binding sites, transmembrane spans — that language has no equivalent of. **First experiment:** Train SAEs on an MSA Transformer and evaluate recovered features against Pfam, CATH and catalytic-site annotations. **Report precision and recall of features against real labels**, which is a thing language interpretability literally cannot do. **Why it matters:** this is Capstone VIII and it benefits the AI field more than the biology field.

7. Scaling laws for structure prediction

From: Kaplan, Hoffmann, Bahri, Kudugunta (D.5.4). **To:** Nobody knows the functional form of structure-prediction performance in data, parameters or compute, which means nobody can say what another 10,000 structures would be worth. **First experiment:** Subsample the PDB at many fractions, train a fixed architecture at several sizes, fit the surface. Use Kudugunta's methodology critique as the checklist. **Why it matters:** it is the quantitative answer to "should we fund more structure determination or more compute," and that question is currently decided by opinion.

D.10.2 Seven things molecules have that AI wants

1. Ground truth for interpretability

As above, in reverse: the molecular domain is the best available testbed for validating interpretability methods, and the AI community does not know it.

2. A real distribution shift, with a physical ground truth

Design is extrapolation (Listgarten, Atlas R.10). Unlike most benchmark distribution shifts, this one can be *checked in the wet lab*. **Conformal prediction methods have no better evaluation setting anywhere.**

3. A hard constraint that cannot be learned away

Physical validity — bond lengths, stereochemistry, clashes — is checkable, objective and not negotiable. PoseBusters is a model for the kind of validity check that language generation lacks entirely.

4. Symmetry that is genuinely exact

SE(3) equivariance is not an approximation or a modeling convenience. The equivariance-versus-scale debate (Atlas O.3) can be settled empirically here in a way it cannot be in vision.

5. A blind benchmark with a thirty-year record

CASP. The AI field keeps rediscovering that held-out test sets leak and that benchmarks saturate. **Structural biology solved this in 1994** and almost no one outside it knows how, or why the specific design choices mattered.

6. Long sequences with real long-range dependencies

A genome's regulatory elements act over megabases. A protein contact can span the whole chain. These are not synthetic long-context benchmarks — the dependencies are physical, labeled and verifiable.

7. A culture of negative results, in one corner

Enzyme design publishes its failures (Blomberg 2013; Rocklin's *Why designs fail*). The AI field's analogous literature is thin and recent. **The norm is transferable and the transfer is free.**

D.10.3 How to actually work the bridge

Three practical habits, in descending order of importance.

First, keep a standing diet. One talk a week from Part III, permanently. Simons Institute, Stanford CS25, EleutherAI reading groups, Valence, ML4PE — listed in the archives appendix. The point is not to keep up; it is to keep the vocabulary live so that when a transfer is available you recognize it. **Most transfers are recognized, not derived.**

Second, when you read an AI paper, ask the three questions. What is the object being modeled? What is the symmetry or constraint being exploited? What is the metric, and is it the quantity anyone cares about? **The third question is where most of the transfers in D.10.1 came from**, because the answer in both fields is usually no.

Third, publish in both directions. A protein result submitted to a biology venue reaches biologists. The same result framed as an interpretability or benchmarking contribution, submitted to NeurIPS or ICML, reaches the people who will build on the method. **The people doing the most interesting work in this curriculum — Ovchinnikov, AlQuraishi, Zhong, Noé, Bowman — publish in both**, and that is not an accident of productivity. It is the mechanism.

Part IV — Performance Engineering

E.0 Why this part exists

Most curricula treat making things faster as plumbing — necessary, unglamorous, and not where the ideas are. That is wrong, and the history of this specific field is the proof.

Consider what the last fifteen years of computational biology actually turned on. Anton was not a new force field or a new algorithm; it was an ASIC, and it made a class of question askable that had not been askable before. FlashAttention introduced no new mathematics — it computes exactly the same function as standard attention — and it changed what everyone could train. ColabFold made AlphaFold usable by thousands of labs, and the speedup came from replacing the sequence search with MMseqs2, not from touching the network. GROMACS's dominance is a twenty-year accumulation of SIMD kernels, cluster pair lists and careful load balancing. OpenFold's contribution was partly scientific and substantially a memory-efficiency result that let people fine-tune a model they previously could only run.

None of those are plumbing. All of them are contributions, and several are more consequential than the methods they accelerated.

E.1 The argument for working here

Four reasons this is unusually good territory for one person with a few GPUs.

The problems are well-posed. “Make triangle attention fit in 11 GB” is a statement with an unambiguous success criterion. Compare that to most method papers, where the hard part is arguing that your benchmark means anything. **A performance result is either reproducible on someone else's machine or it is not**, which is a far healthier epistemics than most of Part II enjoys.

The field is thin. The number of people who understand both molecular modeling and GPU kernel programming is small. Appendix B will tell you that several of the most-used tools in structural biology have had their performance work done by two or three individuals. That is not true of, say, binder design, where every improvement is contested by a dozen groups.

Speed changes what is scientifically possible, not just convenient. A 10× speedup is not a 10% better paper. It is the difference between one trajectory and thirty, between a pilot and a campaign, between a method you demonstrate and a method people use. **Every statistical complaint in Atlas B.5 and Atlas S — n=1, no error bars, no replicates — is downstream of things being too slow to do properly.**

It compounds into everything else. The person who can profile a model can also debug it, can tell which of two architectures is actually cheaper, and can evaluate a claim about efficiency without taking the authors' word for it. That last skill is worth having in a literature where “efficient” is a marketing term about as often as it is a measurement.

E.2 The one idea to take from this part

If you internalize nothing else, internalize this: **most scientific code is not limited by arithmetic. It is limited by memory movement.**

Modern accelerators can do far more floating-point operations per second than they can feed with data. The ratio is brutal and getting worse with every hardware generation. A kernel that reads a matrix from global memory, does one multiply, and writes it back is running at a tiny fraction of the chip’s capability, and no amount of reducing the multiply count will help it.

This single observation explains FlashAttention, explains why fused kernels win, explains why GROMACS’s cluster pair algorithm is shaped the way it is, explains why quantization helps inference more than it “should,” and explains why TPUs and GPUs reward different algorithms. **It is the roofline model, and once you can place your own code on a roofline plot you will stop guessing about performance.**

Derivation checkpoint 64. Compute the arithmetic intensity of the triangle multiplicative update in an AlphaFold-class model — floating-point operations per byte of memory traffic — and place it on a roofline for a card you actually own. Then do the same for the pairwise non-bonded force calculation in MD. **The two land in very different places, and that difference is why the two communities optimize completely differently.**

E.3 How this part is organized

Section	Subject
E.4	GPU architecture and CUDA kernel engineering
E.5	Compilers, TPUs and alternative accelerators
E.6	Simulation engine internals and high-performance scientific computing
E.7	Systems engineering for biomolecular machine learning
E.8	The open engineering problems, and what to attempt

A note on sequencing. E.4 and E.6 are the two that pay off immediately — one teaches you why your GPU is idle, the other teaches you why your MD run is slower than the benchmark table claims. E.5 is the most intellectually interesting and the least immediately useful. E.7 is the one closest to your own work and the thinnest in recorded material, which is itself the signal.

On prerequisites. You need C-level thinking for E.4 — not fluency, but the willingness to read a kernel. Module B.3 covers the computer science this assumes, and MIT’s performance engineering course enumerated in E.6 is the bridge if you want one.

E.4 GPU Architecture and Kernel Engineering

This is the section that will make the most immediate difference to your daily work, because it answers the question you have never asked: *why is my GPU idle?* The answer is almost always memory movement, and almost never the thing you assumed.

Two structural facts about this literature, both discovered in compiling it. First, **GPU MODE is the richest free archive in this entire curriculum** — 116 numbered lectures, enumerated below in full. Second, **NVIDIA systematically does not put its deep GTC sessions on YouTube.** Stephen Jones’s canonical architecture talks, Micikevicius’s mixed-precision sessions and the CUTLASS deep-dives live behind the free-but-registration-walled NVIDIA On-Demand portal. The community archives below exist largely because of that absence.

E.4.1 Architecture fundamentals

Talk	Speaker	Yr	Len	Link
Notes on AI Hardware (Stanford MLSys #88)	Benjamin Spector (Stanford, ThunderKittens)	2024	1:16:48	▶
CUDA + ThunderKittens, but increasingly drunk	Benjamin Spector	2024	4:25:26	▶
GPU Programming Fundamentals + ThunderKittens	Brandon (Anthropic) & Arora (Stanford)	2025	2:39:22	▶
Nvidia's H100 GPU (Stanford Seminar)	Jack Choquette (H100 chief architect)	2023	50:05	▶
Computer Architecture Lecture 26: GPU Programming	Onur Mutlu (ETH)	2022	2:32:38	▶
HetSys Lecture 5: GPU Performance Considerations	ETH Zürich	2022	1:23:30	▶
Stanford CS149 L7: GPU Architecture and CUDA Programming	Stanford	2024	1:18:47	▶
Stanford CS149 L8: Data- Parallel Thinking	Stanford	2024	1:17:49	▶
Introduction to SASS and GPU Microarchitecture	Arun Demeure	2024	1:50:40	▶
The One-Decade Task: Putting std::atomic in CUDA	Olivier Giroux (NVIDIA memory-model architect)	2019	1:05:38	▶
Designing (New) C++ Hardware	Olivier Giroux (NVIDIA)	2017	59:57	▶
Hardware for Deep Learning (Hot Chips keynote)	Bill Dally (NVIDIA Chief Scientist)	2023	1:05:28	▶
Trends in Deep Learning Hardware	Bill Dally (Berkeley)	2025	1:13:07	▶
What is the Hardware Lottery? (MLSys #15)	Sara Hooker (Google Brain)	2021	1:00:24	▶
Hardware-aware Algorithms for Sequence Modeling (MLSys #87)	Tri Dao (Princeton)	2024	1:19:06	▶
A Performance Engineer's Guide to NVIDIA Blackwell GPUs	Chris Sullivan (NVIDIA)	2025	35:17	▶
Introduction to GPU Architecture and	Tim Warburton (Virginia Tech)	2018	2:14:04	▶

Talk	Speaker	Yr	Len	Link
Programming Models (ATPESC)				
GPU optimization workshop (OpenAI, NVIDIA, PyTorch)	Saroufim, Chetlur, Tillet	2024	2:33:53	▶

The Stephen Jones sequence — and a caveat about where it lives

Stephen Jones is NVIDIA's CUDA architect and his two most famous talks are the best forty minutes in existence on why GPUs are bandwidth machines rather than FLOP machines. They exist on YouTube *only as third-party re-uploads* — NVIDIA has not posted them, and the canonical versions sit behind the free-but-registration-gated GTC On-Demand portal. The re-uploads below are complete and verified live.

Talk	Yr	Len	Link
How GPU Computing Works — latency hiding and arithmetic intensity from first principles	2021	39:36	▶
How CUDA Programming Works — coalescing, occupancy, caches, and why the hardware rewards specific access patterns	2022	41:15	▶
CUDA: New Features and Beyond, GTC 2024 (<i>official NVIDIA upload</i>)	2024	50:08	▶
CUDA: New Features and Beyond, GTC 2025 (<i>official</i>)	2025	44:04	▶
CUDA: New Features and Beyond, GTC 2026 (<i>official, most current</i>)	2026	44:27	▶
Interview with CUDA Architect Stephen Jones — CUDA's design history	2024	1:01:50	▶

NVIDIA's official CUDA C++ Class is a genuine course rather than marketing: *Accelerating Applications with Parallel Algorithms* (2:05:27, [▶](#)) · *Asynchrony and CUDA Streams* (47:43, [▶](#)) · *Implementing New Algorithms with CUDA Kernels* (1:12:29, [▶](#)).

Start with Spector's Notes on AI Hardware. It is the talk that argues a researcher should learn the hardware, then teaches the memory hierarchy and tensor-core throughput ratios well enough to act on it.

Then Hooker's Hardware Lottery — the argument that which algorithms “win” is partly an accident of what silicon happened to exist, which is a sobering frame for anyone choosing a method.

Giroux's `std::atomic` talk is the only serious free treatment of the GPU memory consistency model — forward progress, scopes, and the Volta scheduling change. If you have ever written a kernel with a race you could not explain, this is why.

Papers: Jia et al., *Dissecting the NVIDIA Volta GPU Architecture via Microbenchmarking*, arXiv:1804.06826 (Turing sequel arXiv:1903.07486) · the Hopper and Blackwell architecture whitepapers · the CUDA C++ Programming

E.4.2 Programming Massively Parallel Processors — the complete course

Izzat El Hajj (American University of Beirut), 23 lectures, ~25 hours, keyed to the Hwu/Kirk/El Hajj textbook. This is the single best free CUDA course that exists and it is almost unknown. **Lectures 4 through 7 are the performance-engineering spine.**

#	Lecture	Len	Link
1	Introduction	42:34	▶
2	Data Parallel Programming	1:19:18	▶
3	Multidimensional Grids and Data	1:11:09	▶
4	GPU Architecture	1:23:57	▶
5	Memory and Tiling	1:24:26	▶
6	Performance Considerations	1:08:05	▶
7	Profiling	1:00:59	▶
8	Convolution	1:06:06	▶
9	Stencil	1:23:20	▶
10	Reduction	1:21:26	▶
11	Scan (Kogge-Stone)	1:12:15	▶
12	Scan (Brent-Kung)	1:26:07	▶
13	Histogram	1:06:00	▶
14	Merge	1:08:36	▶
15	Sort	1:15:39	▶
16	Sparse Matrix Computation (COO, CSR)	1:03:21	▶
17	Sparse Matrix Computation (ELL, JDS)	1:07:28	▶
18	Graph Processing	1:14:12	▶
19	Graph Processing, Part 2	1:11:39	▶
20	Intra-Warp Synchronization	1:04:26	▶
21	Pinned Memory and Streams	1:12:12	▶
22	Dynamic Parallelism	1:01:48	▶
23	Potpourri	50:11	▶
+	Advanced Optimizations for Matrix Multiplication (2026)	1:23:11	▶

The supplementary GEMM lecture is the single best worked example of this discipline anywhere — register tiling, thread coarsening, the full optimization ladder from naive to competitive, in one sitting.

The graduate sequel — UIUC ECE508, Wen-mei Hwu (the PMPP author), on algorithm-level transformation rather than CUDA API: L1 Introduction ▶ · L2 Scatter-to-Gather ▶ · L3 Thread Coarsening and Register Tiling

▶ · L4 Joint Register and Shared Memory Tiling ▶ · L5 Input Binning ▶ · L7 Graph Representation and BFS ▶ · L9 Triangle Counting ▶

E.4.3 GPU MODE — 116 lectures, enumerated

Channel [@GPUMODE](#); slides and code at github.com/gpu-mode/lectures. **All 116 verified present with no gaps.**

One correction worth knowing before you use the repo: the GitHub README's numbering diverges from YouTube at lectures 40 and 41. YouTube — which is self-consistent and authoritative — has **L40 = CUDA Docs for Humans** and **L41 = FlashInfer**; the README swaps them. The numbering below is YouTube's.

★ **marks the priority lectures** for a molecular-simulation person: CUDA fundamentals, memory, profiling, Triton, CUTLASS/CuTe, quantization, fused kernels, sparsity, collectives, PyTorch internals and Hopper/Blackwell.

#	Lecture	Speaker	Len	Link
★1	How to profile CUDA kernels in PyTorch	Mark Saroufim	56:13	▶
★2	Recap Ch. 1–3, PMPP book	Andreas Köpf	52:26	▶
★3	Getting Started With CUDA for Python Programmers	Jeremy Howard	1:17:56	▶
★4	Compute and Memory Basics	Thomas Viehmann	56:55	▶
★5	Going Further with CUDA for Python Programmers	Jeremy Howard	1:17:34	▶
6	Optimizing PyTorch Optimizers	Jane Xu	1:06:13	▶
★7	Advanced Quantization	Charles Hernandez	1:23:26	▶
★8	CUDA Performance Checklist	Mark Saroufim	1:08:10	▶
★9	Reductions	Mark Saroufim	46:42	▶
10	Build a Prod Ready CUDA Library	Oscar Amoros Huguet	1:25:17	▶
★11	Sparsity	Jesse Cai	56:43	▶
★12	Flash Attention	Thomas Viehmann	1:12:14	▶
13	Ring Attention	Andreas Köpf	1:15:35	▶
★14	A Practitioner's Guide to Triton	Umer Adil	1:21:43	▶
★15	CUTLASS	Eric Auld	1:34:24	▶
★16	On-Hands Profiling	Taylor Robie	55:41	▶
★17	GPU Collective Communication (NCCL)	Dan Johnson	59:43	▶
★18	Fusing Kernels	Kapil Sharma	1:23:22	▶
19	Data Processing on GPUs	Devavret Makkar	59:09	▶
20	Scan Algorithm	Izzat El Hajj	1:02:20	▶
21	Scan Algorithm, Part 2	Izzat El Hajj	1:04:41	▶
22	Speculative Decoding in vLLM	Cade Daniel	1:09:25	▶
★23	Tensor Cores	Thakkar & Ramani (NVIDIA)	1:47:50	▶
24	Scan at the Speed of Light	Hemstad & Evtushenko (NVIDIA)	1:06:19	▶
25	Composable Kernel (CK)	Haocong Wang (AMD)	1:30:52	▶
26	SYCL Mode (Intel GPU)	Patric Zhao	1:18:35	▶

#	Lecture	Speaker	Len	Link
27	gpu.cpp — portable compute via WebGPU	Austin Huang	58:12	▶
★28	Liger Kernel — Efficient Triton Kernels	Byron Hsu	1:11:27	▶
★29	Triton Internals	Kapil Sharma	1:04:49	▶
★30	Quantized Training	Thien Tran	1:16:40	▶
31	Beginner's Guide to Metal Kernels	Nikita Shulga	1:32:08	▶
32	Unsloth — LLM Systems Engineering	Daniel Han	1:24:54	▶
★33	BitBLAS	Lei Wang (MSRA)	1:01:48	▶
★34	Low-Bit Triton Kernels	Hicham Badri	1:45:31	▶
35	SGLang Performance Optimization	Yineng Zhang	45:19	▶
★36	CUTLASS and FlashAttention 3	Jay Shah (Colfax)	1:49:16	▶
★37	Introduction to SASS and GPU Microarchitecture	Arun Demeure	1:50:40	▶
38	Low-Bit Kernels for ARM CPU	Scott Roy	1:03:41	▶
★39	TorchTitan (FSDP2 / DTensor internals)	Saroufim & Liu	1:23:38	▶
★40	CUDA Docs for Humans	Charles Frye (Modal)	51:07	▶
41	FlashInfer	Zihao Ye (UW)	1:08:51	▶
42	Mosaic GPU	Adam Paszke (Google)	1:26:25	▶
★43	int8 Tensor-Core Matmul for Turing	Erik Schultheis	1:17:11	▶
★44	NVIDIA Profiling (2h deep dive)	NVIDIA tools engineers	2:07:16	▶
★45	Outperforming cuBLAS on H100	pranjalssh	1:16:01	▶
★46	Distributed GEMM	Ali Hassani	1:27:29	▶
47	KernelBot	—	1:02:23	▶
★48	The Ultra-Scale Playbook	Nouamane Tazi (HF)	3:03:48	▶
49	Low-Bit Metal Kernels	Manuel Candales	1:31:55	▶
50	A Learning Journey: CUDA, Triton, Flash Attention	Umar Jamil	1:20:43	▶
51	Consumer GPU Performance	Jake Cannell	1:16:01	▶

#	Lecture	Speaker	Len	Link
★52	Scaling Laws for Low Precision	Tanishq Kumar (Harvard)	53:43	▶
★53	torch.compile Q&A	Richard Zou (Meta)	1:26:51	▶
54	Small RL Models with LeanRL	—	53:49	▶
55	Mojo	Modular	2:13:12	▶
★56	Kernel Benchmarking Tales	Georgii Evtushenko (NVIDIA)	1:07:57	▶
★57	CuTe	Cris Cecka (NVIDIA, CuTe architect)	1:24:33	▶
58	Disaggregated LLM Inference	Junda Chen	1:15:19	▶
59	FastVideo	—	1:04:11	▶
★60	Optimizing Linear Attention	Songlin Yang (MIT)	1:07:39	▶
61	D-Matrix Corsair	Jain, Arunkumar, Srivastava	1:38:09	▶
62	Exo 2 — Growing a Scheduling Language	Yuka Ikarashi (MIT)	1:08:23	▶
63	Search-Based DL Compilers (Luminal)	Joe Fioti	1:09:19	▶
★64	Multi-GPU Programming	Markus Hrywniak (NVIDIA)	1:15:23	▶
65	Neighborhood Attention	Ali Hassani	1:37:53	▶
66	Game Arena	Lanxiang Hu	53:33	▶
★67	NCCL and NVSHMEM	Jeff Hammond (NVIDIA)	1:40:43	▶
★68	Landscape of GPU-Centric Communication	Didem Unat (Koç)	1:00:07	▶
★69	Quartet — 4-bit Training	Castro & Panferov (ISTA)	1:09:11	▶
★70	PCCL — Fault-Tolerant Collectives	mike64_t	1:05:23	▶
71	[ScaleML] FlexOlmo	Sewon Min (Berkeley)	1:24:51	▶
72	[ScaleML] Efficient Long-Context Modeling	Guangxuan Xiao (MIT)	55:18	▶
★73	[ScaleML] Quantization in Large Models	Chris De Sa (Cornell)	1:18:26	▶
74	[ScaleML] Positional Encodings and PaTH Attention	Songlin Yang	1:40:32	▶
★75	[ScaleML] GPU Fundamentals + ThunderKittens	Brandon & Arora	2:39:22	▶

#	Lecture	Speaker	Len	Link
76	BackendBench	Mark Saroufim	17:54	▶
★77	Domain-Specific Languages for GPU Kernels	Tri Dao	23:39	▶
★78	Iris — Multi-GPU Programming in Triton	Awad, Osama, Potter (AMD)	1:16:38	▶
79	Mirage — Compiling LLMs into Megakernels	Wu & Cheng (CMU)	1:12:52	▶
★80	How FlashAttention 4 Works	Charles Frye (Modal)	1:15:09	▶
81	Futhark	Troels Henriksen	1:11:03	▶
★82	Helion — A High-Level DSL for ML Kernels	Ansel, Ulgen, Feng (Meta)	1:06:20	▶
83	Formalized Kernel Derivation	Vincent Abbott (MIT)	1:40:06	▶
★84	Numerics and AI	Paulius Micikevicius (NVIDIA)	2:39:09	▶
85	Factorio Learning Environment	Jack Hopkins	1:06:12	▶
★86	Getting Started with CuTe DSL	Vicki Wang (NVIDIA)	1:13:19	▶
★87	Low-Latency Communication Kernels with NVSHMEM	Prajwal Singhania	1:07:07	▶
88	TinyTPU	William Zhang	54:32	▶
★89	cuTile	Amini & Roesch (NVIDIA)	1:43:31	▶
90	Building Resilient ML Engineering Skills	Stas Bekman	1:54:07	▶
91	Mega Lecture: RL, Agents and OpenEnv	multiple	3:21:51	▶
92	Smol Training Playbook	Loubna Ben Allal (HF)	1:26:03	▶
93	Cornserve	Jeff Ma	44:19	▶
94	tvm-ffi	Tianqi Chen (CMU)	1:28:31	▶
95	Single-Controller Programming with Monarch	Wang & Taylor (Meta)	2:02:24	▶
★96	TLX — Triton-Like Simplicity, Peak Performance	Hongtao Yu (Meta)	1:29:58	▶
★97	HipKittens	William Hu (Stanford)	1:25:05	▶
★98	GPU Observability	Yusheng Zheng	46:53	▶
99	Distributed ML on Consumer Devices	Matt Beton	1:05:53	▶

#	Lecture	Speaker	Len	Link
100	InferenceX	—	1:12:06	▶
★101	Learning CUTLASS the Hard Way	Kapil Sharma	1:11:17	▶
★102	Quartet v2	Panferov & Schultheis	1:47:17	▶
★103	CuTe Layout Algebra and its Category-Theoretic Interpretation	Carlisle & Jay Shah	2:33:57	▶
★104	Gluon and Linear Layouts	Bell, Zhou, Lezcano	1:57:57	▶
★105	cuDNN mxfp8 Attention	—	1:00:18	▶
★106	Hugging Face Kernels	de Kok & Holz (HF)	1:28:19	▶
107	PithTrain	Lai & Kang	1:05:01	▶
108	One Layer Deeper competition	—	36:35	▶
109	TIRx	Bohan Hou	38:45	▶
★110	The 4-Bitter Lesson — NVFP4 RL stability	Ziang Li	1:01:27	▶
111	Spectral Compute — Compile CUDA Everywhere	Chris Kitching	1:08:33	▶
★112	Production Megakernels for Real-World Inference	Joe Fioti	58:05	▶
★113	Near Speed-of-Light GPU Collectives	—	52:53	▶
★114	PyCuTe	Cris Cecka with Saroufim	1:44:03	▶
115	Proving Kernels Correct Instead of Testing Them	Ben Koska	40:07	▶
116	GPU Kernel Formal Verification	Jubi Taneja	30:35	▶

The GPU MODE IRL 2024 keynotes are posted separately on the Accel channel and are easy to miss: Wenmei Hwu (the PMPP author, ▶) · Andrej Karpathy (▶) · Tri Dao (▶) · Tim Dettmers (▶) · Supriya Rao (▶) · Lily Liu, vLLM (▶).

E.4.4 Triton

The key discovery: OpenAI maintains an official Triton channel (@Triton-openai) carrying the full Developer Conference archive for 2023, 2024 and 2025. It is authoritative, free, and widely unknown.

Philippe Tillet’s own talks: *Triton Today and Beyond* (2025, 51:59, ▶) — the best current state-of-the-project talk · *Keynote* (2024, 15:05, ▶) · *The Triton Language* (2021, 12:11, ▶) — the original pitch for block-level programming · *Maximizing Kernel Development Productivity Under Performance Constraints* (2024, 22:02, ▶) — his core thesis on where Triton deliberately gives up control.

Developer Conference 2025: [Gluon — Tile-Based Programming with Low-Level Control](#) · [Proton — Portable Performance Profiling](#) · [TLX](#) · [Blackwell guide](#) · [Triton on AMD](#) · [Triton-distributed](#) · [Helion](#).

Developer Conference 2024: [Writing an MLIR Pass](#) · [Proton / Interpreter](#) · [Pipelining Persistent Kernels](#) · [Triton on Blackwell](#) · [Mosaic GPU](#) · [Exo](#).

Developer Conference 2023: [Hopper Support in Triton](#) — how WGMMMA and TMA first entered the compiler · [Grouped GEMMs](#) · [PyTorch 2.0 and TorchInductor](#) · [Pallas](#).

The compiler panel — Tillet, Ansel, Pienaar (MLIR), Tianqi Chen (TVM), Zolotukhin and Peng Wu arguing about IR design and autotuning, 35:31, — is excellent seminar material.

Paper: Tillet, Kung & Cox, *Triton: An Intermediate Language and Compiler for Tiled Neural Network Computations*, MAPL 2019.

E.4.5 CUTLASS and CuTe

Cris Cecka — the CuTe architect — has two full-length free lectures, which is unusual and valuable.

Talk	Speaker	Len	Link
CuTe	Cris Cecka (NVIDIA)	1:24:33	▶
PyCuTe	Cris Cecka with Saroufim	1:44:03	▶
CuTe Layout Algebra and its Category-Theoretic Interpretation	Carlisle & Jay Shah	2:33:57	▶
CUTLASS and FlashAttention 3	Jay Shah (Colfax, FA3 co-author)	1:49:16	▶
Tensor Cores (MMA atoms)	Thakkar & Ramani (NVIDIA)	1:47:50	▶
CUTLASS (introduction)	Eric Auld	1:34:24	▶
Getting Started with CuTe DSL	Vicki Wang (NVIDIA)	1:13:19	▶
How FlashAttention 4 Works	Charles Frye (Modal)	1:15:09	▶
Zero to Hero: Programming Hopper Tensor Core with MLIR's NVGPU Dialect	Guray Ozen (NVIDIA)	42:26	▶
CUTLASS Python DSL Infrastructure (LLVM Dev Meeting)	Guray Ozen (NVIDIA)	22:14	▶

Ozen's EuroLLVM talk is the best free single source on TMA, warpgroup MMA and mbarrier — the three Hopper primitives that everything fast on that architecture depends on.

Paper: Shah, Bikshandi, Zhang, Thakkar, Ramani & Dao, *FlashAttention-3*, arXiv:2407.08608. Colfax's written tutorials at research.colfax-intl.com/blog are the real substitute for their missing video channel.

E.4.6 Profiling — finding the actual bottleneck

Roofline, in the right order: the FAU 18-minute derivation (▶) → Pennycook, Yang and Deslippe on performance portability and empirical ceilings (1:07:12, ▶) → SOL analysis in Nsight Compute (▶) → Edward Yang’s LLM-era worked example (▶).

The tool deep dives: GPU MODE Lecture 44 is the most complete free Nsight treatment on YouTube at 2:07:16 (▶) · Memory Analysis with Nsight Compute (▶) · Profiling with NVTX (▶).

Long-form from HPC centres: Introduction to Performance Analysis for NVIDIA GPUs, Dominik Ernst, NHR@FAU, 1:11:39 (▶) · Kernel Performance Analysis with Nsight Compute, ALCF (▶) · HPCToolkit, John Mellor-Crummey (Rice), (▶).

The single most important talk here is the one about occupancy. James Demmel’s SC19 Test of Time Award talk (40:14, ▶), accepting for Volkov and Demmel’s SC08 paper, explains the result that overturned a decade of advice: **lower occupancy with more instruction-level parallelism beat the universal “maximize occupancy” rule**. If you have ever tuned a kernel by chasing occupancy numbers, watch this before you do it again.

Also: *PyTorch Unleashed*, Taylor Robie (PyTorch profiler core dev), 37:48 (▶) — reading real traces to separate launch-bound from memory-bound from compute-bound.

E.4.7 PyTorch internals and compilers

Horace He’s *Building ML Systems for a Trillion Trillion Floating Point Operations* (Jane Street, 2024, 1:03:21, ▶) is the one talk to watch in this subsection. Compute-bound, memory-bound and overhead-bound as three distinct regimes with three different fixes — the diagnostic framework that makes everything else actionable.

Also by He: *Accelerating Generative AI* (▶) · *FlexAttention* (▶) · *GPT-Fast* (▶).

Edward Yang: *torchdynamo deep dive* (1:35:59, ▶) · *torch.compile: The Missing Manual* (▶) · symbolic shapes (▶) · a real debugging session inside the CUDA caching allocator’s private mempool (46:37, ▶) — almost nothing else free covers allocator internals at that level.

Jason Ansel: *PyTorch 2 — Faster ML Through Dynamic Python Bytecode Transformation and Graph Compilation* (Allen School, 1:01:17, ▶) — the full ASPLOS 2024 paper talk; anchor the module on it.

Compiler internals: Deep Dive on TorchDynamo (▶) · Inside torch.compile Guards: How They Work, What They Cost (▶) · CUTLASS backend for Inductor (▶) · CUDAGraph in a Partial Graph World (▶).

Paper: Ansel et al., *PyTorch 2*, ASPLOS 2024.

E.4.8 Numerics and determinism

Talk	Speaker	Len	Link
Numerics and AI	Paulius Micikevicius (NVIDIA)	2:39:09	▶
Deep Learning Determinism	Duncan Riach (NVIDIA)	37:38	▶
Reproducible C++ Floating-Point Reductions in CUB	Shreyas Atre	5:25	▶
Numerical Stability at Extreme Scale and Low Precisions (ICM 2022)	Nicholas Higham (Manchester)	44:30	▶
The Rise of Multiprecision Computations	Nicholas Higham	49:52	▶
DGEMM on Integer Tensor Cores (the Ozaki scheme)	NHR@FAU	35:15	▶
AMP Training in PyTorch	NVIDIA	19:18	▶

Micikevicius's *Numerics and AI* is the flagship of this subsection. The author of *Mixed Precision Training* and *FP8 Formats* giving two hours and forty minutes on FP16, BF16, FP8, MX scaling, loss scaling, and where low precision breaks. **It is the free long-form replacement for his gated GTC sessions, and it is better than any of them because of its length.**

Riach's determinism talk is the one that matters for science. Atomics and non-deterministic reductions mean two runs of the same code give different answers. For a published benchmark comparison, that is a correctness problem, not a curiosity. **Derivation checkpoint 69 is this question applied to force accumulation.**

Papers: Micikevicius et al., *Mixed Precision Training*, arXiv:1710.03740 · *FP8 Formats for Deep Learning*, arXiv:2209.05433 · the OCP Microscaling Formats Specification v1.0 · Kumar et al., *Scaling Laws for Precision*, arXiv:2411.04330.

E.4.9 Multi-GPU and collectives

The entries that matter for a small fleet, in order: **NCCL and NVSHMEM**, Jeff Hammond (NVIDIA), 1:40:43 ([▶](#)) · **GPU Collective Communication**, Dan Johnson, 59:43 ([▶](#)) · **Landscape of GPU-Centric Communication**, Didem Unat, 1:00:07 ([▶](#)) · **Multi-GPU Programming**, Markus Hrywniak (NVIDIA), 1:15:23 ([▶](#)) · NCCL from its author, Sylvain Jeaugey, 41:06 ([▶](#)) · **HOTI keynote on accelerator clusters**, Bill Dally, 57:05 ([▶](#)) — bandwidth tapering and collective cost models, which is why NVLink and NVSwitch exist.

E.4.10 Honest gaps

The largest is structural: NVIDIA does not publish its deep GTC sessions to YouTube. Stephen Jones's *How GPU Computing Works* and *How CUDA Programming Works* exist there only as third-party re-uploads. The same applies to Micikevicius's mixed-precision sessions and the full CUTLASS deep-dives. NVIDIA On-Demand is free but requires registration, and that is where the canonical versions live.

The OLCF CUDA Training Series is not on YouTube at all — Oak Ridge publishes to Vimeo. Slides and exercises are free at github.com/olcf/cuda-training-series.

No Vasily Volkov talk exists. *Better Performance at Lower Occupancy* was never recorded; the Demmel Test-of-Time retrospective is the honest substitute. Pair it with Volkov's 2016 Berkeley thesis, *Understanding Latency Hiding on GPUs*.

Paulius Micikevicius has no YouTube GTC session either, despite writing both canonical numerics papers. **GPU MODE Lecture 84**, at two hours forty, is the free substitute and is better for being unhurried.

There is no talk on batch-invariant determinism in LLM inference. Every search result was AI-generated explainer content. The Thinking Machines blog post is the primary source; Lecture 84 and the Riach talk carry the rigorous version.

No dedicated PyTorch caching-allocator talk exists. Allocator internals appear only incidentally, inside Edward Yang's refcount-bug session and Elias Ellison's CUDA-graph talk. Zachary DeVito's *Understanding GPU Memory* posts on pytorch.org remain the authoritative source.

No Samuel Williams roofline lecture is recorded, and **no William Kahan video exists** — his floating-point lectures circulate as PDFs and audio only. **Colfax Research's YouTube channel is empty**; their written tutorials are the resource.

A methodological warning worth repeating, because it will cost you a day. Sustained `yt-dlp` metadata fetches start failing wholesale under rate-limiting, and those failures look exactly like dead videos. When a batch comes back all-failed, re-check the same IDs through `oEmbed` before dropping anything.

E.4.11 Paired reading — E.4

Watch this	Then read this	Hold this question
Spector, <i>Notes on AI Hardware</i>	Any roofline tutorial	Compute arithmetic intensity for one kernel you own.
Hooker, <i>The Hardware Lottery</i>	Her paper	Which molecular methods won because of hardware rather than merit?
PMPP L4–L7	Hwu, Kirk & El Hajj, <i>PMPP</i> , ch. 4–6	Occupancy, coalescing, bank conflicts: audit one kernel against all three.
PMPP supplementary GEMM lecture	—	Re-derive the optimization ladder from naive to competitive.
GPU MODE L8, <i>CUDA Performance Checklist</i>	—	Run the checklist on a real kernel.
GPU MODE L18, <i>Fusing Kernels</i>	FlashAttention, arXiv:2205.14135	What memory traffic does fusion eliminate?
Giroux, <i>std::atomic in CUDA</i>	CUDA memory-model docs	What is a scope, and when does forward progress fail?
Demmel, <i>SC19 Test of Time</i>	Volkov & Demmel, SC08	Occupancy is not the objective. What is?
He, <i>Trillion Trillion FLOPs</i>	—	Classify your bottleneck: compute, memory, or overhead?
Yang, <i>torchdynamo deep dive</i>	Ansel et al., <i>ASPLOS 2024</i>	What does a guard cost, and when does it recompile?
Cecka, <i>CuTe</i>	CuTe docs 00_quickstart – 0t_mma_atom	What is a layout, algebraically?
Shah, <i>CUTLASS and FA3</i>	arXiv:2407.08608	Warp specialization and async pipelining. What is overlapped?
Micikevicius, <i>Numerics and AI</i>	arXiv:1710.03740 ; arXiv:2209.05433	Where does FP8 break, and what does loss scaling fix?
Riach, <i>Deep Learning Determinism</i>	—	Are your benchmark numbers reproducible bitwise? Test it.
Dally, <i>HOTI accelerator clusters</i>	—	At what message size does your interconnect stop being free?

E.5 Accelerators, Compilers and the Codesign Bargain

The thesis of this section. A processor is a bet about what computation looks like. A GPU bets on throughput over latency. A TPU bets that almost everything is a matrix multiply. **Anton bets that it is a range-limited N-body problem with a fixed interaction pattern** — and that bet is the most successful piece of hardware-software codesign in the history of biology.

This section exists so that you can read those bets rather than accept them. **Section E.5.4 is the centerpiece:** one Hot Chips session in which the architects of Graphcore, Cerebras, SambaNova and Anton 3 present back to back. Four different answers to the same question, in one sitting, by the people who made the decisions.

E.5.1 TPU architecture

#	Talk	Speaker	Yr	Len	Link
1	A Domain-Specific TPU Supercomputer for Training Deep Neural Networks	Norm Jouppi (Google)	2020	1:05:02	▶
2	Evaluation of the Tensor Processing Unit	David Patterson (Berkeley/Google)	2017	56:26	▶
3	Domain Specific Architectures for DNNs: Three Generations of TPUs	David Patterson (Allen School)	2019	1:12:36	▶
4	Computer Architecture Lecture 27: Systolic Arrays	Onur Mutlu (ETH Zürich)	2021	1:04:51	▶
5	Computer Architecture Lecture 28: VLIW and Systolic Array Architectures	Onur Mutlu (ETH Zürich)	2023	1:26:20	▶
6	Episode 8: Systolic Arrays	Happy Hour with Architects	2021	56:05	▶
7	TPU v4 and Trends in Accelerator Hardware	Mike Hutton (Google)	2023	45:46	▶
8	Seminar in Computer Architecture: TPuv4i and Mensa	Mutlu group (ETH)	2022	1:55:13	▶
9	ML and the Implications for Computer System Design (Hot Chips keynote)	Jeff Dean (Google)	2017	58:58	▶
10	Exciting Directions for ML Models and the Implications for Computing Hardware	Jeff Dean + Amin Vahdat	2023	1:03:59	▶
11	TPU SparseCore: Hardware and Software	OpenXLA / Google	2026	22:56	▶
12	Architecting TPU 8t and TPU 8i for frontier AI	Google Cloud Next '26	2026	43:11	▶

Watch #4 first. Mutlu derives systolic arrays from Kung and Leiserson's original idea through to the TPU's matrix unit, and without it the rest of this subsection is a tour of acronyms. **Then #1**, which is Jouppi on why v2 and v3

moved from inference to training and how the 2D torus scales near-linearly to a thousand chips. #7 is the clearest treatment of the optical circuit switch and the reconfigurable 3D torus; #8 teaches you to interrogate the design claims rather than accept them.

Papers: Jouppi et al., *In-Datacenter Performance Analysis of a Tensor Processing Unit*, ISCA 2017, arXiv:1704.04760 · Jouppi et al., *TPU v4: An Optically Reconfigurable Supercomputer with Hardware Support for Embeddings*, ISCA 2023, arXiv:2304.01433.

Why a protein person should care. Triangle attention is not a matrix multiply in the shape a systolic array wants. Evoformer-class models run on TPUs anyway, and understanding *how badly* they fit is the beginning of knowing whether a different decomposition would run better. **Derivation checkpoint 64 is this question made concrete.**

E.5.2 XLA, MLIR and the compiler layer

#	Talk	Speaker	Yr	Len	Link
13	XLA Crash Course	OpenXLA / Google	2026	31:08	▶
14	XLA Architecture (DevLab 2025)	OpenXLA / Google	2025	1:00:32	▶
15	Operator Fusion (DevLab 2025)	OpenXLA / Google	2025	51:46	▶
16	XLA GPU MLIR Codegen	OpenXLA / Google	2024	1:03:16	▶
17	StableHLO and PJRT	OpenXLA Dev Summit	2023	1:06:47	▶
18	MLIR: Multi-Level Intermediate Representation	Shpeisman + Chris Lattner	2019	40:22	▶
19	MLIR Tutorial (LLVM Dev Meeting)	Amini + Riddle (Google)	2020	1:15:09	▶
20	The Golden Age of Compiler Design in an Era of HW/SW Co-design	Chris Lattner (ASPLOS keynote)	2021	52:21	▶
21	TVM: An End to End Deep Learning Compiler Stack	Tianqi Chen	2020	28:16	▶
22	TVM Tutorial: AutoTVM	Eddie Yan (UW SAMPL)	2019	31:20	▶
23	Making a blur faster in Halide	Andrew Adams (Adobe)	2020	41:17	▶
24	ML for ML Compilers (Stanford MLSys #80)	Mangpo Phothilimthana (DeepMind)	2023	58:06	▶

#15 is the one you came for. Fusion is the dominant optimization in every ML compiler, for exactly the reason Section E.2 gives: it eliminates round trips to memory. **#23 is the best intuition-builder anywhere** — Halide’s main author live-scheduling a blur, showing the tiling-versus-locality tradeoff as he makes it. Forty minutes that will change how you read any kernel.

Also verified and worth having: Anso (Lianmin Zheng, OSDI ’20, [▶](#)) · Ragan-Kelley on the algorithm/schedule separation ([▶](#)) · Tillet on Triton ([▶](#)) and **Triton Internals** ([▶](#)) · IREE GPU codegen ([▶](#)) · **Pallas** / **Mosaic** ([▶](#)) · Tianqi Chen’s full MLC course, episode 1 ([▶](#)).

Papers: Chen et al., *TVM*, OSDI 2018, arXiv:1802.04799 · Lattner et al., *MLIR*, CGO 2021, arXiv:2002.11054 · Ragan-Kelley et al., *Halide*, PLDI 2013 · Zheng et al., *Anso*, OSDI 2020.

E.5.3 JAX, and JAX for molecular simulation

#	Talk	Speaker	Yr	Len	Link
25	JAX: accelerated ML research via composable function transformations	Matthew Johnson (Google Brain)	2021	57:05	▶
26	Stanford MLSys Ep. 6: JAX	Roy Frostig (Google Brain)	2020	1:06:58	▶
27	Outgrowing NumPy	Dougal Maclaurin (Google)	2021	13:29	▶
28	Intro to JAX on Cloud TPUs	Skye Wanderman-Milne (Google)	2021	1:57:05	▶
29	Debugging JAX models	Skye Wanderman-Milne	2025	49:22	▶
30	Large scale training techniques and best practices	Google (DevLab 2025)	2025	1:01:45	▶
31	Sharding the Sphere with <code>jax.shard_map</code>	Google	2025	16:09	▶
32	Pallas tutorial (custom kernels in JAX)	Google JAX team	2026	56:50	▶
33	JAX MD: A Framework for Differentiable Atomistic Physics	Samuel Schoenholz (Google Brain)	2021	1:33:38	▶
34	Hybridization of Neural Networks and Numerical Solvers in JAX	Felix Köhler (TU Munich)	2026	1:21:53	▶
35	DiffraX: Numerical Differential Equation Solvers in JAX	Patrick Kidger	2022	24:29	▶

#33 is the one that matters for you. Neighbour lists, energy functions and an end-to-end differentiable molecular dynamics engine — which means you can take gradients *through* a simulation with respect to force field parameters. A near-identical reupload is at [▶](#). **#34** is the best single lecture on coupling a learned component to a differentiable simulator, which is the architecture behind half of Atlas D.6.

Also verified: Rafi Witten’s 10-part *High Performance LLMs in JAX* course, TPU-centric, session 1 at [▶](#) · NERSC *Python on GPUs: JAX* ([▶](#)) · *JAX-GCM, a differentiable atmospheric model* ([▶](#)).

Papers: Schoenholz & Cubuk, *JAX MD*, NeurIPS 2020, [arXiv:1912.04232](#) · Bradbury et al., JAX (2018) · Austin, Douglas, Frostig, Levskaya et al., *How to Scale Your Model*, [jax-ml.github.io/scaling-book](#) — a book, with no recorded lecture; #30 is the closest substitute.

E.5.4 The codesign argument, and four answers to it

Watch entry 40 as a single sitting. It is the most valuable two hours in Part IV.

#	Talk	Speaker	Yr	Len	Link
36	Hennessy and Patterson, 2017 Turing Award Lecture	Hennessy + Patterson	2018	1:19:38	▶
37	A New Golden Age for Computer Architecture	David Patterson (Berkeley)	2018	1:16:09	▶
38	New Golden Age for Computer Architecture	John Hennessy (Stanford)	2018	1:15:20	▶
39	Domain-Specific Architectures for Deep Neural Networks	David Patterson (ScaledML)	2019	1:00:16	▶
40	ML and Computation Platforms — Graphcore + Cerebras + SambaNova + Anton 3, one session	Knowles, Lie, Prabhakar, Butts	2021	2:16:44	▶
41	Cerebras Architecture Deep Dive	Sean Lie (CTO, Cerebras)	2022	27:00	▶
42	Thinking Outside the Die (SAFARI seminar, ETH)	Sean Lie (Cerebras)	2022	1:58:56	▶
43	Groq architecture (Cornell ECE 5545 guest lecture)	Dennis Abts (Chief Architect, Groq)	2023	59:29	▶
44	Domain-Specific Networks for Machine Learning (NOCS keynote)	Dennis Abts (Groq)	2020	38:29	▶
45	Designing Processors for Intelligence	Simon Knowles (CTO, Graphcore)	2017	1:16:22	▶

Read the claims, because they are different claims. Cerebras claims *bandwidth* — on-wafer SRAM and fabric bandwidth orders of magnitude above HBM, with capacity as the acknowledged weakness. Groq claims *latency* — fully deterministic, compiler-scheduled execution with no caches and no reorder buffers, and SRAM-only capacity as the binding constraint. Graphcore claims bandwidth too, via large in-processor memory plus fine-grained MIMD. SambaNova claims *capacity* — tiered memory holding very large or many models, plus kernel fusion to remove memory traffic. **None of these is “faster” in the same sense, and conflating them is the most common error in reading accelerator claims.**

Also verified: Graphcore at Hot Chips 2021 ([▶](#)) · Olukotun, *Let the Data Flow!* ([▶](#)) · **Bill Dally’s GPU-side rebuttal** at Cornell ([▶](#)) and Hot Chips ([▶](#)) · Jim Keller’s DAC keynote ([▶](#)) · Cerebras for scientific computing ([▶](#)).

Papers: Hennessy & Patterson, *A New Golden Age for Computer Architecture*, CACM 62(2):48–60, 2019 ·
Abts et al., *Think Fast: A Tensor Streaming Processor*, ISCA 2020.

E.5.5 Anton — the biology codesign result

This is the most important subsection in Part IV for someone in your field, and it is also the hardest material to source. What follows is everything that exists on free video.

#	Talk	Speaker	Yr	Len	Link
46	Anton: A Special-Purpose Machine That Achieves a Hundred-Fold Speedup in Biomolecular Simulations	David E. Shaw	2012	1:20:34	▶
47	SC23 Test of Time Award Talk — retrospective across Anton 1, 2 and 3	David E. Shaw	2023	54:55	▶
48	Anton 2: A 2nd-Generation ASIC for Molecular Dynamics (Hot Chips 26; segment begins 1:05:53)	J. Adam Butts + D. E. Shaw	2014	34:45	▶
49	The Anton 3 ASIC: a Fire-Breathing Monster for Molecular Dynamics (Hot Chips 33; segment begins 1:40:31)	J. Adam Butts	2021	36:13	▶
50	High Speed Protein Simulations with Anton (UW)	David E. Shaw	2014	58:29	▶
51	Long Molecular Dynamics Simulations: Progress, Problems, and Promise (ISMB keynote)	David E. Shaw	2016	1:35:36	▶
52	MDGRAPE-4A: A Special-Purpose Computer for Molecular Dynamics	Makoto Taiji (RIKEN)	2020	33:03	▶
53	RISC-V Tokyo: special-purpose MD silicon	Makoto Taiji (RIKEN)	2020	28:33	▶
54	Codesign from Semiconductors to AI	Cliff Young (Google) — Anton co-architect and TPU co-designer	2023	31:47	▶

#49 is the only recorded Anton 3 architecture talk in existence, and it sits in the same session as Graphcore, Cerebras and SambaNova. **#47 states the thing worth knowing**: the gap between Anton and general-purpose machines has *grown* to more than $400\times$ for drug-discovery-sized systems. **#52 is the honest counterexample** — a different special-purpose MD machine, explicitly framed against Anton, explaining why attached-accelerator designs hit a host-side strong-scaling wall and had to become systems-on-chip.

#54 is the bridge, and it is why this section is placed here. Cliff Young co-architected Anton 1 and 2, then co-designed the TPU. **Brian Towles** is an author on both the Anton 2 network paper and the TPU v4 paper. The lineage from molecular dynamics silicon to machine learning silicon is not an analogy. **It is literally the same people**, carrying the same codesign discipline across.

Secondary, all verified: PSC *Anton 3 Capabilities and Enhanced Sampling* 2025 (▶) and 2024 (▶) — allocation-oriented, not architectural · Tiankai Tu on fault-tolerant parallel analysis of millisecond trajectories (▶).

Papers, DOIs resolved rather than recalled: Shaw et al., *Anton, a special-purpose machine for molecular dynamics simulation*, ISCA 2007, [10.1145/1250662.1250664](#) · Shaw et al., *Millisecond-scale molecular dynamics simulations on Anton*, SC09 (Gordon Bell Prize), [10.1145/1654059.1654126](#) · Shaw et al., *Anton 2*, SC14, [10.1109/SC.2014.9](#) · Shaw et al., *Anton 3: Twenty Microseconds of Molecular Dynamics Simulation Before Lunch*, SC21, [10.1145/3458817.3487397](#) · *Anton: A Specialized Machine for Millisecond-Scale MD Simulations of Proteins*, IEEE ARITH 2009, [10.1109/arith.2009.33](#) — **the best source on Anton's custom arithmetic, and the direct pair for E.6.3** · Towles et al., *Unifying On-Chip and Inter-Node Switching Within the Anton 2 Network*, ISCA 2014 · Shim et al., *The Specialized High-Performance Network on Anton 3*, HPCA 2022.

Slides, both link-checked: the Anton 2 and Anton 3 Hot Chips decks are at [old.hotchips.org/.../HC26.11.130-Anton-2-Butts-Shaw-Shaw-Res-Search.pdf](#) and [hc33.hotchips.org/assets/program/conference/day2/HC2021.DESRES.AdamButts.v03.pdf](#).

E.5.6 FPGAs and reconfigurable computing

#	Talk	Speaker	Yr	Len	Link
55	Productive Parallel Programming for FPGA with High Level Synthesis (SC'20 tutorial)	Torsten Hoefler + de Fine Licht (ETH)	2021	3:21:54	▶
56	Scientific Applications of FPGAs at the LHC (FPGA 2021 keynote)	Philip Harris (MIT)	2021	1:00:01	▶
57	A Reconfigurable Fabric for Accelerating Datacenter Services (Catapult; segment at 34:05)	Putnam, Caulfield, Chung (Microsoft)	2014	~40:00	▶
58	Reconfigurable Computing at HyperScale	Andrew Putnam (MSR)	2018	54:31	▶
59	Accelerating HPC Applications with Reconfigurable Logic	Microsoft Research	2016	1:09:00	▶
60	Comparing FPGA vs Custom CMOS and the Impact on Processor Architecture	U. Toronto EECG	2013	28:47	▶
61	hls4ml: Co-Design for Scientific Low-Power ML Devices	Javier Duarte (UCSD)	2021	20:31	▶
62	Real-Time ML for Massive LHC Data Streams	Thea Aarrestad (ETH/CERN)	2023	57:52	▶

Watch #60 first, for one number. The FPGA-versus-ASIC tax is $17\text{--}27\times$ in area and $18\text{--}26\times$ in delay for soft logic, but only $2\text{--}7\times$ for hard blocks. **That single fact explains why Anton is an ASIC and Catapult is not**, and it is the whole decision procedure for anyone wondering whether to build custom hardware.

Also verified: AI Engine Architecture (SPCL, [▶](#)) · Doug Burger, *Transitioning from the Era of Multicore to the Era of Specialization* ([▶](#)) · FBLAS: streaming linear algebra on FPGA ([▶](#)).

E.5.7 Paired reading — E.5

Watch this	Then read this	Hold this question
Mutlu, <i>Systolic Arrays</i>	Kung & Leiserson 1979	Why does a systolic array suit matrix multiply and not N-body?
Jouppi, <i>TPU supercomputer</i>	arXiv:1704.04760 then arXiv:2304.01433	Roofline the TPU. Where would an Evoformer land on it?
OpenXLA, <i>Operator Fusion</i>	The XLA fusion documentation	What does fusion eliminate? Connect it to Section E.2.
Adams, <i>Making a blur faster in Halide</i>	Ragan-Kelley et al., PLDI 2013	Separate algorithm from schedule for one MD kernel on paper.
Chen, <i>TVM</i>	arXiv:1802.04799	Learned cost models for kernel search. Could you autotune a force kernel?
Schoenholz, <i>JAX MD</i>	arXiv:1912.04232	Differentiating through a simulation. What becomes possible?
Köhler, <i>NNs and numerical solvers in JAX</i>	The accompanying paper	Where does the gradient through a long trajectory become useless?
Hot Chips session (#40)	Compare the four claims	Bandwidth, latency, or capacity? Classify each vendor correctly.
Lie, <i>Thinking Outside the Die</i>	Cerebras WSE papers	What does wafer-scale buy, and what does it give up?
Shaw, <i>Anton</i>	Shaw et al., SC09, 10.1145/1654059.1654126	What was specialized, and what was deliberately left general?
Butts, <i>Anton 3</i>	Shaw et al., SC21, 10.1145/3458817.3487397	Twenty microseconds before lunch. What changed from Anton 2?
Shaw, <i>SC23 Test of Time</i>	The retrospective	The gap to general-purpose hardware grew to 400×. Why?
Taiji, <i>MDGRAPE-4A</i>	The MDGRAPE papers	Why did attached accelerators have to become SoCs?
Young, <i>Codesign from Semiconductors to AI</i>	Anton ISCA 2007 + TPU ISCA 2017	The same people built both. What transferred, and what did not?
U. Toronto, <i>FPGA vs Custom CMOS</i>	The paper	17–27× area tax. When is an FPGA the right answer?

E.6 Simulation Engines and Scientific HPC

This is the section that pays off tomorrow. You run GROMACS across a small multi-GPU fleet. The material below is why your ns/day is what it is, and what the handful of decisions are that actually move it.

One observation before the lists, and it is the most useful thing in this section. The molecular dynamics community has a decade of *public performance culture* — the BioExcel webinar series runs from #2 in 2016 to #93 in 2026, almost all of it about making simulations faster, presented by the people who write the code.

The structure-prediction community has nothing equivalent. That asymmetry is the subject of E.8.

E.6.1 GROMACS internals and performance

Talk	Speaker	Yr	Len	Link
Peeking into the black box of GROMACS performance	Alekseenko & Páll (KTH)	2026	59:43	▶
Improvements in the GROMACS heterogeneous parallelization	Szilárd Páll (KTH)	2022	54:09	▶
Performance Tuning and Optimization of GROMACS	Mark Abraham (KTH)	2016	1:02:58	▶
Getting good performance in GROMACS	Berk Hess (KTH)	2022	17:41	▶
GROMACS HPC usage best practices and Q&A	Berk Hess (KTH)	2022	54:13	▶
Creating Faster Molecular Dynamics Simulations	Alan Gray (NVIDIA)	2020	1:03:50	▶
A deep dive into GROMACS on GPU	Alan Gray (NVIDIA)	2021	25:45	▶
More bang for your buck — improved use of GPUs	BioExcel	2019	52:33	▶
Optimizing cluster and simulation setup for GROMACS	BioExcel	2016	58:52	▶
NB-LJIB — a performance portable force/energy library	Jordan (KTH) & Keller (CSCS)	2021	53:15	▶
What's new in GROMACS 2025	Berk Hess (KTH)	2025	59:03	▶
What's new in GROMACS 2026	Hess, Müllender, Miletic	2026	57:31	▶
GROMACS Features and Future	Paul Bauer (KTH)	2020	38:30	▶
A walk through .mdp parameter options	BioExcel	2019	1:14:00	▶
MDBenchmark — scaling studies made straightforward	BioExcel	2018	51:47	▶
Could free energy calculations be faster?	Suriñach (Nostrum)	2022	26:43	▶

Watch *Peeking into the black box* first, then Hess's seventeen minutes. The first teaches you to read an `mdrun` log line by line and infer what the engine is doing with your hardware — whether you are PME-

bound, bonded-bound or PCIe-bound. The second is the highest value-per-minute item in this entire Part: the handful of decisions that actually move ns/day, from the person who wrote the algorithms.

Papers: Páll et al. 2020, *J Chem Phys* 153:134110 (heterogeneous parallelization and the GPU-resident path) · Páll et al. 2015, EASC, LNCS 8759:3 · Páll & Hess 2013, *Comput Phys Commun* 184:2641 — the cluster pair algorithm, which is the actual reason GROMACS is fast, and which has no talk at all · Abraham et al. 2015, *SoftwareX* 1–2:19.

E.6.2 OpenMM, NAMD, LAMMPS

Talk	Source	Yr	Len	Link
Introduction to the OpenMM API	Simbios / Stanford	2013	44:21	▶
Customizing Forces and Integrators with OpenMM	Simbios / Stanford	2013	23:14	▶
Validation of OpenMM	Simbios / Stanford	2013	15:07	▶
OpenMM (library overview)	SBGrid	2017	31:05	▶
NAMD and Charm++: What You Should Know	HPC-AI Advisory Council	2020	50:37	▶
Experiences with Charm++ and NAMD on Summit	Jim Phillips (UIUC)	2018	23:25	▶
Improving NAMD Performance on Multi-GPU Platforms	David Hardy (UIUC)	2018	31:28	▶
Lessons from Scaling NAMD	Jim Phillips (UIUC)	2014	29:10	▶
NAMD on Heterogeneous Architectures (the GPU-resident redesign)	Charm++ Workshop	2020	33:00	▶
Accelerating Discovery with NAMD 3	CCPBioSim	2024	1:07:51	▶
Scaling MD on Aurora with NAMD (SYCL)	IXPUG	2025	26:37	▶
Optimizing GPU Performance: the Chain Benchmark in LAMMPS	Stan Moore (Sandia)	2023	44:10	▶
Porting LAMMPS with Kokkos/SYCL to Aurora	IWOCL	2024	19:50	▶
HOOMD-Blue v3.0 — a GPU-native MD engine	Joshua Anderson (Michigan)	2022	31:44	▶

OpenMM's distinctive engineering idea is in talk #2: `CustomNonbondedForce` and `CustomIntegrator` compile user-supplied algebraic expressions into GPU kernels at runtime. That is why OpenMM is the engine people extend. **Talk #3, on validation, is the one nobody watches and everybody should** — it shows how an MD engine is proven numerically correct, which is the question behind every force field comparison in Atlas A.

Honest gap: there is no substantial free talk on the AMBER GPU implementation, despite its influence. Only two-minute NVIDIA marketing clips exist. Read **Salomon-Ferrer, Götz, Poole, Le Grand & Walker 2013, JCTC 9:3878** instead — it defines the SPFP hybrid precision model, which is the most important numerics idea in GPU molecular dynamics.

E.6.3 The algorithms that make MD fast

Topic	Talk	Len	Link
GPU force-loop mapping	TCBG: NAMD — MD on GPU Clusters, Part 1	14:08	▶
Spatial decomposition	TCBG: MD on GPU Clusters, Part 4	14:08	▶
PME on GPUs	TCBG: GPU Particle-Grid Algorithms, Part 1	14:08	▶
PME on GPUs	TCBG: Particle-Grid Algorithms, Part 2	14:08	▶
Ewald derivation	Ewald / PME / PPPM / SPME	21:25	▶
The PME scaling wall	Periodic Coulomb Tree Method	34:00	▶
Constraints	SHAKE, RATTLE, LINCS, SETTLE	12:11	▶
Hydrogen mass repartitioning	HMR in GROMACS (hands-on)	14:45	▶
Multiple timestepping	BioExcel #18: multiple timescales	1:10:03	▶
Load balancing	Dynamic Load-Balancing for Particle Simulations	53:54	▶

The TCBG particle-grid lectures are the only place anyone explains charge spreading and grid interpolation on GPUs at this level of detail — the hard, memory-scatter half of PME that every other treatment skips.

Papers: Essmann et al. 1995, *JCP* 103:8577 (smooth PME) · Hess et al. 1997, *J Comput Chem* 18:1463 (LINCS) and Hess 2008, *JCTC* 4:116 (P-LINCS) · Miyamoto & Kollman 1992, *J Comput Chem* 13:952 (SETTLE) · Hopkins, Le Grand, Walker & Roitberg 2015, *JCTC* 11:1864 (hydrogen mass repartitioning).

Derivation checkpoint 67 lives here. GPU molecular dynamics uses cluster pair lists rather than per-atom Verlet neighbour lists, and the reason is about warps and memory coalescing rather than algorithmic complexity. It is the clearest example in this document of hardware dictating an algorithm — and the paper that explains it, Páll & Hess 2013, has no recorded talk.

E.6.4 HPC fundamentals

The NHR@FAU Parallel Programming series (Georg Hager and colleagues, Erlangen) is the best free university-level treatment of this material anywhere. The lectures that matter for a GPU fleet:

#	Lecture	Len	Link
4	Basic OpenMP	51:14	▶
6	Advanced OpenMP and performance issues	1:10:43	▶
7	ccNUMA and wavefront parallelization — first-touch, pinning	59:07	▶
8	Introduction to MPI	45:02	▶
10	Collective communication and distributed-memory architecture	59:21	▶
11	MPI datatypes, virtual topologies, performance pitfalls	1:00:17	▶
13	Hybrid Programming with MPI and OpenMP	1:03:57	▶

Lecture 13 is exactly what `gmx mdrun -ntmpi -ntomp` is doing, and it explains the tradeoff you are making whenever you choose those numbers. Lecture 7 is why thread pinning is not optional on a multi-socket node.

The memory wall, stated properly:

Talk	Speaker	Len	Link
A very short intro to the Roofline model	NHR@FAU	18:37	▶
Memory Bandwidth and System Balance in HPC Systems	John McCalpin (TACC, author of STREAM)	1:07:58	▶
Main Memory Trends and Importance	Onur Mutlu (ETH)	29:22	▶
Processing-in-Memory	Onur Mutlu (ETH)	1:23:55	▶
Accelerating Genome Analysis — the memory wall on a biological workload	Onur Mutlu (ETH)	2:12:22	▶
Optimizing your HPC applications with LIKWID	HPC.NRW	1:15:54	▶
How to use likwid-pin	NHR@FAU	16:56	▶
Memory & Caches	Matt Godbolt (Compiler Explorer)	1:11:20	▶
What Every Programmer Should Know about How CPUs Work	Matt Godbolt	43:28	▶

McCalpin's talk is the structural reason molecular dynamics is hard to speed up, by the person whose benchmark defines how we measure it. Mutlu's genome analysis lecture is included deliberately — it is the

memory-wall argument worked through on a biological workload rather than on matrix multiply, and it is where the innovation opportunity is most visible.

Papers: Wulf & McKee 1995, *SIGARCH Comput Archit News* 23(1):20 (Hitting the Memory Wall) · Williams, Waterman & Patterson 2009, *CACM* 52(4):65 (Roofline).

Also from the national labs: NUMA and SIMD optimization, NERSC (▶) · ATPESC MPI sessions by Thakur, Raffenetti and Gropp (▶) · **Beyond SMP — NUMA and GPU Programming**, Tim Mattson (▶) · **How learning about GPUs actually made me good at computational science**, Max Katz (▶) · **Mixed-precision arithmetic: hardware, algorithms and analysis**, Theo Mary (▶).

E.6.5 MIT 6.172 — Performance Engineering of Software Systems

Charles Leiserson and Julian Shun, MIT. The twelve most relevant of 23 lectures. The opening lecture alone — a 53,000× speedup on matrix multiply, built step by step — establishes that most “fast” code leaves two orders of magnitude on the table.

#	Lecture	Len	Link
1	Introduction and Matrix Multiplication	1:00:20	▶
2	Bentley Rules for Optimizing Work	1:20:10	▶
4	Assembly Language and Computer Architecture	1:17:34	▶
5	C to Assembly	1:21:30	▶
6	Multicore Programming	1:16:46	▶
9	What Compilers Can and Cannot Do	1:18:46	▶
10	Measurement and Timing	1:21:28	▶
14	Caching and Cache-Efficient Algorithms	1:18:23	▶
15	Cache-Oblivious Algorithms	1:21:47	▶
17	Synchronization Without Locks	1:20:10	▶
21	Tuning a TSP Algorithm (a complete worked optimization)	1:20:52	▶
22	Graph Optimization (irregular access — neighbour lists in disguise)	1:18:39	▶

Lecture 10, *Measurement and Timing*, should be watched before you trust any ns/day comparison you have ever made — including your own. Variance, clock sources, warm-up, statistical significance. Pair it with NHR@FAU's *Experiments and Data Presentation in HPC* (▶).

On library-level optimization: BLIS: A Framework for Rapidly Instantiating BLAS Functionality (▶) is the clearest public explanation of the Goto/Van de Geijn blocked matrix-multiply structure. **Kazushige Goto has no**

public recorded lecture; the paper is Goto & van de Geijn 2008, *ACM TOMS* 34:12.

On performance portability: C++ Performance Portability — A Decade of Lessons Learned, Christian Trott (Sandia, Kokkos lead), [▶](#) · *Kokkos: Getting Lucky By Design* ([▶](#)) · the ATPESC Kokkos tutorial ([▶](#)).

E.6.6 Scientific software engineering and fleet operations

Talk	Source	Len	Link
Getting it Right: System Testing of Scientific Software	IDEAS / ECP	59:00	▶
Writing Clean Scientific Software	IDEAS	1:01:47	▶
Benchpark with Ramble — continuous benchmarking done properly	LLNL	2:34:02	▶
GROMACS: Testing and testing infrastructure	BioExcel	1:22:04	▶
GROMACS: GitLab and version control	BioExcel	47:16	▶
GROMACS: Structures and interfaces	BioExcel	1:34:37	▶
Reproducible Research	CodeRefinery	2:01:59	▶
Automated Testing	CodeRefinery	1:31:26	▶

The three GROMACS “Learn to Code” sessions are the entry point for Capstone XI. If you intend to contribute a speedup upstream, the testing-infrastructure lecture is where you start — not the algorithm.

Fleet operations: Slurm 21.08 and Beyond, Tim Wickberg (SchedMD CTO, [▶](#)) · OCI containers and `scrun` ([▶](#)) · **Container solutions for HPC: Singularity/Apptainer** ([▶](#)) · NERSC container training ([▶](#)) · **Flux Framework — hierarchical scheduling**, Tom Scogland (LLNL, [▶](#)) — nesting a scheduler inside an allocation, which is the right model for keeping many GPUs busy with many small jobs · **Spack** ([▶](#)).

Honest gap, and it is your exact situation: there is no video anywhere on running a small multi-GPU Linux fleet for molecular dynamics. Everything public is either single-workstation tutorials or facility-scale operations at thousands of nodes. The nearest usable substitutes are the Slurm `gres.conf` documentation on cgroup-based GPU binding and the Flux talks above.

E.6.7 Exascale, and what actually scaled

Frontier — The World’s First Exascale Supercomputer, Bronson Messer (OLCF Director of Science, [▶](#)) · Frontier’s Exascale Architecture, Scott Atchley (ORNL, [▶](#)) · **ECP Applications: Lessons Learned**, Erik Draeger (LLNL, [▶](#)) — unusually candid about what failed to scale · ECP Software Technology, Mike Heroux (Sandia, [▶](#)) · **Performance of Particle Applications on Early Exascale Hardware** ([▶](#)) — the most directly relevant exascale session for an MD person · **Characterizing Performance Improvements in ECP Projects** ([▶](#)) — how a large program defines and audits a speedup claim, which is methodology you can borrow.

E.6.8 Honest gaps in this family

Three absences are surprising enough to name, and in all three the paper is the only source.

There is no talk on the AMBER GPU implementation, despite `pmemd.cuda` being one of the most consequential pieces of GPU scientific software ever written. **There is no talk on the RELION GPU port** — note that Erik Lindahl is an author on that paper, so the same engineering culture produced both GROMACS and RELION-2's GPU support. And **there is no talk on SVE or Arm vectorization of MD kernels**; the GROMACS SIMD abstraction layer is documented only in the developer manual.

Most tellingly, the single most important GROMACS performance paper — Páll and Hess 2013 on the cluster pair algorithm, the actual reason GROMACS is fast — has no accompanying talk in the public record.

Three channel-handle traps that will waste your time: `@MolSSI` has no videos tab (MolSSI's software-engineering training is written, at `education.molssi.org`), and `@NERSC`, `@olcf` and `@ALCF` all belong to unrelated accounts. **OLCF has no public YouTube training channel at all** — Oak Ridge publishes to Vimeo.

E.6.9 Paired reading — E.6

Watch this	Then read this	Hold this question
Alekseenko & Páll, <i>Black box of GROMACS performance</i>	Your own <code>md. log</code>	Are you PME-bound, bonded-bound or PCIe-bound? Find out.
Hess, <i>Getting good performance</i>	Páll et al. 2020, <i>JCP</i> 153:134110	Which three settings move ns/day most on your hardware?
Páll, <i>Heterogeneous parallelization</i>	Same paper	What did GPU-resident mode change about the CPU/GPU balance?
—	Páll & Hess 2013, <i>CPC</i> 184:2641	Why cluster pairs and not Verlet lists? (Checkpoint 67.)
TCBG, <i>Particle-Grid Algorithms</i>	Essmann et al. 1995, <i>JCP</i> 103:8577	Derive the PME communication cost. (Checkpoint 68.)
Simbios, <i>Customizing Forces in OpenMM</i>	Eastman et al. 2017, <i>PLoS Comput Biol</i> 13:e1005659	Runtime kernel compilation. What does that enable?
Simbios, <i>Validation of OpenMM</i>	—	How would you prove your own simulation is numerically correct?
Hardy, <i>NAMD on Multi-GPU</i>	Phillips et al. 2020, <i>JCP</i> 153:044130	As GPUs got faster, what became the bottleneck?
NHR@FAU L7, <i>ccNUMA</i>	—	Check thread pinning on your own nodes. Is first-touch respected?
NHR@FAU L13, <i>MPI + OpenMP</i>	—	What are you trading when you set <code>-ntmpi</code> and <code>-ntomp</code> ?
McCalpin, <i>Memory Bandwidth and System Balance</i>	Wulf & McKee 1995	Compute bytes-per-flop for your machine.
Mutlu, <i>Accelerating Genome Analysis</i>	The accompanying papers	The memory wall on a biological workload. Where is the opportunity?
MIT 6.172 L10, <i>Measurement and Timing</i>	—	Re-run your last benchmark with error bars. Does the conclusion hold?
MIT 6.172 L14, <i>Caching</i>	—	Design a neighbour-list layout against the cache model.
Trott, <i>Performance Portability</i>	The Kokkos papers	One source, every accelerator. What does the abstraction cost?
LLNL, <i>Benchpark</i>	—	Set up one reproducible benchmark for your fleet. Track it.

E.7 Systems Engineering for Biomolecular Machine Learning

This is the thinnest recorded literature in the entire curriculum, and that is the finding. The molecular dynamics community has a decade of public performance culture — BioExcel webinar #2 in 2016 through #93 in 2026, almost all of it about making simulations faster. **The structure-prediction community has essentially none.** The most-run model in computational biology has no public profile.

Read this section as a map of where the work has not been done.

E.7.1 AlphaFold, OpenFold and what training actually costs

Talk	Speaker	Yr	Len	Link
OpenFold: Lessons Learned and Insights Gained From Rebuilding and Retraining AlphaFold2	Mohammed AlQuraishi (Columbia)	2023	39:28	▶
OpenFold: retraining AlphaFold2 yields insights on learning mechanisms	OpenFold team (OpenBioML)	2023	1:03:54	▶
OpenFold (IPAM version)	AlQuraishi (UCLA/IPAM)	2023	48:12	▶
OpenFold Keynote	AlQuraishi (OMSF Symposium)	2024	40:40	▶
Insights from retraining OpenFold	Nazim Bouatta (Harvard Medical School)	2023	1:16:29	▶
FastFold: Reducing AlphaFold Training Time from 11 Days to 67 Hours	Shenggan Cheng (NUS, first author)	2023	9:56	▶
Lessons from implementing AlphaFold3 in the wild	Arda Goreci (Ligo Biosciences)	2024	25:51	▶
Boltz-1 and the Future of Biomolecular Foundation Models	Corso & Wohlwend (MIT)	2025	1:09:45	▶
Boltz-2: Accurate and Efficient Binding Affinity Prediction	Wohlwend et al. (MIT/Recursion)	2025	59:26	▶
SeedFold: Scaling Biomolecular Structure Prediction	Zhou & Lu (ByteDance Seed)	2026	59:59	▶
Accelerating Biomolecular Modeling with AtomWorks and RF3	Nathaniel Corley (IPD)	2025	52:10	▶
Model. Package. Deploy. Repeat: DevOps for Biomolecular Scalability	Colby Ford (BPDMC)	2025	1:19:04	▶
What could AlphaFold 4 look like?	Sergey Ovchinnikov (MIT)	2025	2:06:39	▶

The OpenFold cluster is the most important material in this section, because AlphaFold2's training recipe was never released. OpenFold is the only public account of what it actually costs: roughly 50,000 GPU-hours, the initial-training and fine-tuning split, the observation that **the model converges in the first 3% of training**, and the memory optimizations — in-place operations, chunked attention, low-memory recycling — that made a 40 GB A100 sufficient.

Goreci’s talk is the best single account of what breaks when you rebuild AF3 from the paper alone: diffusion-module instabilities, the cost of the atom-level representation, and what the paper omits. It is twenty-five minutes and it is the only talk of its kind in existence.

Papers: Ahdritz et al. 2024, *Nat Methods* 21:1514 (OpenFold) · Cheng et al. 2022, arXiv:2203.00854 (FastFold) · Abramson et al. 2024, *Nature* 630:493 (AF3) · Wohlwend et al., Boltz-1 and Boltz-2.

Alternate recordings of the OpenFold talk, all verified: Chalmers (▶) · Rutgers IQB (▶) · SBGrid (▶).

E.7.2 Triangle attention — the memory wall nobody has written about

Talk	Speaker	Len	Link
AlphaFold Decoded: Evoformer	Kilian Mandon (Bamberg)	24:21	▶
AlphaFold Decoded: Attention	Kilian Mandon	23:51	▶
AlphaFold 3 From Scratch in PyTorch — Introduction	Kilian Mandon	34:21	▶
AlphaFold 3 From Scratch — Feature Extraction	Kilian Mandon	24:22	▶
ML for protein structure prediction, Part 2: AlphaFold2 architecture	Nazim Bouatta (Harvard CMSA)	1:18:32	▶
MSA Pairformer — scaling down deliberately	ML4PE	37:58	▶

Mandon’s series is a hand-implementation course, not an explainer channel — he writes the triangle multiplicative update and triangle self-attention in PyTorch tensor by tensor. **It is the only recorded material that makes the $O(N^3)$ pair-tensor cost concrete rather than asserted.**

Honest gap, and it is the sharpest one in this document: there is no talk anywhere on the performance of triangle attention itself. FlashAttention solved the $O(N^2)$ sequence-attention memory problem by tiling and recomputation, and the whole field adopted it. The pair representation’s $O(N^3)$ triangle multiplication has received exactly one fused kernel — DeepSpeed’s `DS4Sci_EvoformerAttention` — **which has no recorded talk at all.** Everyone else falls back to `chunk_size` loops, which trade memory for launch overhead and leave the GPU idle.

Read instead: the OpenFold source (`openfold/model/triangular_multiplicative_update.py` and the chunking logic in `openfold/utils/chunk_utils.py`), AlphaFold2 Supplementary Algorithms 11–15, and arXiv:2310.04610 for the DeepSpeed kernel. **Derivation checkpoints 64 and 66 are this gap made into exercises.**

E.7.3 Equivariant-network performance

Talk	Speaker	Len	Link
e3nn: Euclidean Neural Networks	Mario Geiger (then EPFL/MIT, now NVIDIA)	1:18:41	▶
Computational Aspects of Equivariant Neural Networks	Risi Kondor (Chicago)	1:09:11	▶
Does equivariance matter at scale?	Johann Brehmer (Qualcomm AI)	1:04:10	▶
Efficient and Expressive 3D Equivariant GNNs	Valence Labs	54:59	▶
Euclidean Neural Networks: learning with 3D geometry and geometric tensors	Tess Smidt (MIT)	1:06:01	▶

Kondor’s talk is the closest thing to a cost model: the computational structure of Clebsch-Gordan products and where the complexity actually sits.

A major gap with a direct bearing on your hardware. There is no recorded talk on NVIDIA cuEquivariance, and none by Mario Geiger about his NVIDIA kernel work. Read the cuEquivariance docs — particularly the segmented-polynomial representation and the fused `TensorProduct` kernels — and **Bharadwaj et al. 2025, *OpenEquivariance: a GPU kernel generator for equivariant deep learning***, which benchmarks fused CG kernels against e3nn and reports the speedups. **This is the single highest-leverage unrecorded topic in Part IV.**

E.7.4 Interatomic potential inference at scale

Talk	Speaker	Len	Link
Learning Local Equivariant Representations for Large-Scale Atomistic Dynamics (Allegro)	Albert Musaelian (Harvard)	1:09:17	▶
NequIP and local equivariant representations	Batzner & Musaelian (Harvard)	1:26:42	▶
Improving Speed and Accuracy of NN Interatomic Potentials	Valence Labs	54:24	▶
Orb-v3: atomistic simulation at scale	Duignan & Vandenhoute (Orbital)	1:13:34	▶
MACE: Higher Order Equivariant Message Passing	Valence Labs	1:22:55	▶
Learning ML Interatomic Potentials	Gianni De Fabritiis (TorchMD author)	51:38	▶
JAX MD: A Framework for Differentiable Atomistic Physics	Samuel Schoenholz	1:27:51	▶
First Principles Molecular Dynamics on a Large Scale	Gábor Csányi (Cambridge)	1:07:45	▶
AquaGen: generative models at MD precision on thousands of atoms	Valence Labs	1:16:26	▶

Watch Batzner first, then Musaelian. NequIP’s story is data efficiency; Allegro’s is that **strict locality removes the message-passing communication that caps NequIP’s scaling**, which is what let them reach 100M+ atoms on Perlmutter. **Orb-v3 is the clearest statement of the production trade:** it deliberately drops strict equivariance to buy inference throughput.

Papers: Musaelian et al. 2023, *Nat Commun* 14:579 (Allegro) and Musaelian et al. 2023, SC’23 — **read this one for the actual scaling numbers** · Batzner et al. 2022, *Nat Commun* 13:2453 (NequIP) · Pelaez et al. 2024, *JCTC* (TorchMD-NET 2.0) · Eastman et al. 2024, *J Phys Chem B* 128:109 (OpenMM 8 with ML potentials) — no talk exists for the last two.

E.7.5 Serving, batching and memory

Read this subsection as transferable mechanism, not as language-model content. The entries: **Fast LLM Serving with vLLM and PagedAttention**, Woosuk Kwon (Berkeley), 32:07 ([▶](#)) · **Exploring the Latency/Throughput and Cost Space for Inference**, Timothée Lacroix (Mistral CTO), 30:25 ([▶](#)) · **Faster and Cheaper Offline Batch Inference with Ray**, 28:04 ([▶](#)) · **Efficient LLM Inference with SGLang** ([▶](#)) · **Disaggregated Inference** ([▶](#)).

The transferable insight is continuous batching. Static batching wastes capacity whenever sequence lengths differ — which is exactly a folding campaign where targets range from 80 to 2,000 residues. Lacroix’s

thirty minutes on choosing an operating point on the latency/throughput frontier maps almost one-to-one onto “I have ten thousand Boltz predictions and four GPUs.”

On memory: ZeRO, Samyam Rajbhandari (author), 1:05:10 (▶) · ZeRO-Offload (▶) · Too Big to Train — FSDP, Sharcnet, 47:33 (▶) and the FSDP2 rewrite (▶) · Slaying OOMs with PyTorch FSDP and torchao, Saroufim & Xu, 49:37 (▶) — the most practically useful hour here, debugging real OOMs with the memory snapshot tool.

Papers: Kwon et al. 2023, SOSP (PagedAttention) · Rajbhandari et al. 2020, SC'20 (ZeRO) · Chen et al. 2016, arXiv:1604.06174 (gradient checkpointing) — the paper OpenFold leans on heavily.

E.7.6 The data pipeline — where the time actually goes

Talk	Speaker	Len	Link
ColabFold — Making protein folding accessible to all	Ovchinnikov, Mirdita & Steinegger	1:46:09	▶
Exploring the Protein Universe	Martin Steinegger (SNU)	1:17:41	▶
MMseqs software suite	Martin Steinegger	1:29:37	▶
MMseqs2 profile/profile searches	Martin Steinegger	14:03	▶
Foldseek	Martin Steinegger (SBGrid)	57:17	▶
Faster AlphaFold predictions using ColabFold	UCSF ChimeraX	10:08	▶
Zarr vs HDF5	Joe Jevnik (PyData)	52:15	▶
HDF5 at the speed of Zarr	Pangeo	25:15	▶

The ColabFold talk is the single most important entry in this section and one of the most important in Part IV. All three authors, one hour and forty-six minutes, explaining the actual engineering: **why MSA generation dominated AlphaFold2 wall-clock**, how MMseqs2 replaced JackHMMER and HHblits at 40–60× less compute, and how the public MSA server absorbs that load.

Derivation checkpoint 70 is this talk as an exercise. Most people assume the network dominates. Measure it before you believe it.

Papers: Mirdita et al. 2022, *Nat Methods* 19:679 (ColabFold) · Steinegger & Söding 2017, *Nat Biotechnol* 35:1026 (MMseqs2) · Steinegger & Söding 2018, *Nat Commun* 9:2542 · van Kempen et al. 2024, *Nat Biotechnol* 42:243 (Foldseek).

E.8 The Open Engineering Problems

This section is the point of Part IV. Eight problems, drawn from what the survey of this literature could *not* find. Several are holes in the field, not just in the recorded record. Each is stated with enough specificity that you could start it this month.

- 1. Nobody has published a profile of a structure-prediction model.** The survey turned up a dozen good GPU profiling talks and zero that point the tooling at AlphaFold-class inference. **A careful Nsight profile of Boltz-2 or OpenFold across sequence lengths** — reporting the split between MSA stack, triangle operations, diffusion sampling and data loading — would be genuinely citable and costs a weekend on hardware you already own. **Nothing else on this list is blocked on it, and everything else gets easier once you have the numbers.** Start here.
- 2. Triangle attention has no public cost model and no general kernel.** FlashAttention solved the $O(N^2)$ problem by tiling and recomputation and the field adopted it universally. The $O(N^3)$ triangle update has one fused kernel, from DeepSpeed, with no talk and limited adoption. **A FlashAttention-style treatment of the triangle multiplicative update is unwritten.**
- 3. There is no serving stack for structure prediction.** PagedAttention pages a KV cache; AF3-class models have no KV cache. The expensive resident object is the pair tensor, whose size grows quadratically in a length that varies ten-fold across a campaign, so static batching wastes most of the allocation on short targets. **A length-bucketed continuous batcher for folding models does not exist publicly**, and for anyone running thousands of predictions on four GPUs it is worth more than any accuracy improvement.
- 4. The MSA bottleneck was solved once, in 2017, and never revisited.** ColabFold cut MSA cost by an order of magnitude by swapping JackHMMER for MMseqs2, and that is still the state of the art. Meanwhile models are moving toward fewer MSAs. **The open question nobody has a talk on is the trade curve: for a given accuracy target, what is the cheapest MSA you can get away with?** Measurable, publishable, and it needs no new model.
- 5. Equivariant kernels are fast now and nobody has said so publicly.** cuEquivariance and OpenEquivariance both report large speedups on Clebsch-Gordan tensor products and neither has a recorded talk. Simultaneously Brehmer argues equivariance's advantage shrinks with scale, and Orb-v3 ships a deliberately non-equivariant model for throughput. **These two facts have never been put in the same room.** If fused kernels close the speed gap, the Orb-style accuracy sacrifice may be unnecessary.
- 6. Precision discipline does not transfer between the two halves of a bio-ML fleet.** Molecular dynamics precision is a conservation-law question with thirty years of careful work behind it — Le Grand's SPFP model uses fixed-point accumulation precisely because naive FP32 breaks energy conservation. Structure prediction precision is an unexamined throughput knob. **Nobody has published where bf16 actually costs accuracy in a folding trunk** — whether the triangle updates, the diffusion sampler or the recycling accumulator is the sensitive stage. It is a weekend's ablation and it would be widely cited precisely because it is unglamorous.
- 7. Memory sharding assumes weights are the problem.** ZeRO and FSDP partition parameters, gradients and optimizer states, which is right when weights dominate. **In a folding model the activation dominates: the pair tensor dwarfs the parameters.** Ring attention is the only talk in the whole survey that shards an activation along a sequence axis, and it does so for a 2D object. Sharding a 3D pair representation, with the triangle update's all-

to-all communication pattern, is unaddressed everywhere. FastFold's Dynamic Axial Parallelism is the one attempt and it got a ten-minute lightning slot.

8. The reimplementation ecosystem has no shared engineering record. Goreci's talk is the only one in existence about what breaks when you rebuild these models, and it is twenty-five minutes. Protenix, Chai-1 and Uni-Fold have no verifiable engineering talks at all. **Each team is rediscovering the same instabilities privately.** OpenFold set the precedent by publishing the retraining recipe; nobody has done the equivalent for the AF3 generation.

Which of these to attempt. Problems 1, 3 and 4 are measurement and systems work where a small fleet does better than a large lab, because they need careful instrumentation rather than compute. Problems 2 and 7 are genuine kernel and distributed-systems research. Problem 6 is the cheapest and would be the most cited. **Capstone XI asks you to do one of them and get the change merged upstream.**

Part V — From Consumption to Contribution

C.1 The derivation checkpoints

Sixty things you must be able to produce on a blank sheet of paper, unaided, by the end of this program. They are chosen because each is load-bearing: a person who can derive all sixty can read any paper in this field and immediately see what is new, what is borrowed, and what is hidden.

This is the real examination. The lectures are support material.

From potential energy functions (Family A)

1. Write down every term in a modern molecular mechanics force field and name the physical effect each stands in for — including the ones standing in for nothing.
2. Derive the Lennard-Jones potential's relationship to dispersion and Pauli repulsion, and explain why the r^{-12} term is chosen for convenience rather than physics.
3. Explain Ewald summation and why particle-mesh Ewald is $O(N \log N)$. State what artifact periodic boundary conditions introduce for a charged solute and how large it is.
4. Explain what a polarizable force field adds, and why the added cost has not been worth it for most biomolecular applications.
5. Derive the functional form of a machine-learned interatomic potential's energy as a sum of local atomic contributions, and state precisely what that locality assumption forbids.

From molecular dynamics (Family B)

6. Show that velocity Verlet is symplectic and time-reversible, and explain what that buys over a higher-order non-symplectic scheme.
7. Derive the Langevin equation's fluctuation-dissipation relation and explain why violating it gives wrong kinetics but can still give right thermodynamics.
8. Explain why constraining bonds to hydrogen permits a larger timestep, and derive what hydrogen mass repartitioning does to the fastest vibrational mode.
9. Show why the Berendsen thermostat does not sample the canonical ensemble, and state what distribution it does sample.
10. Derive the pressure estimator from the virial, and explain why it is noisy.

From Monte Carlo (Family C)

11. State the Metropolis-Hastings acceptance criterion and prove it satisfies detailed balance for an arbitrary symmetric or asymmetric proposal.
12. Explain why detailed balance is sufficient but not necessary for correct sampling, and give one method that is correct without it.
13. Derive Hamiltonian Monte Carlo and explain where the Metropolis correction enters and why it is needed despite the dynamics being deterministic.
14. Explain configurational bias Monte Carlo and derive its acceptance ratio.
15. Derive the Gillespie stochastic simulation algorithm from the chemical master equation.

16. Explain Wang-Landau sampling and state what it converges to.

From enhanced sampling (Family D)

17. Derive the weighted histogram analysis method from maximum likelihood.

18. Explain why well-tempered metadynamics converges where the original did not, and write the asymptotic form of the bias.

19. Derive the scaling of the required replica count with system size in temperature replica exchange, and explain why it is fatal for large systems.

20. Derive the weighted ensemble resampling scheme and show it is unbiased.

21. Explain transition path sampling's shooting move and why it preserves the path ensemble.

22. State and derive the Jarzynski equality, and explain why it is nearly useless in practice without careful attention to the tails.

From free energy (Family E)

23. Derive the free energy perturbation identity and state the overlap condition under which it fails.

24. Derive the Bennett acceptance ratio estimator and show it is the minimum-variance estimator in its class.

25. Explain thermodynamic integration and the endpoint singularity problem, and state the standard fix.

26. Write the thermodynamic cycle for absolute binding free energy and name every correction term.

27. Explain what MM-GBSA actually computes and why its entropy treatment is usually wrong.

From coarse-graining and polymer theory (Families F, H)

28. Derive the multiscale coarse-graining / force-matching objective and state what it is a projection of.

29. Explain the Mori-Zwanzig decomposition and identify which term coarse-grained models normally discard.

30. Explain why coarse-grained models systematically get entropy and dynamics wrong even when they get structure right.

31. Derive the end-to-end distance distribution of an ideal chain and the worm-like chain's force-extension relation.

32. Derive the Flory scaling exponent and explain what it predicts for a disordered protein versus a folded one.

33. Explain the random energy model of protein folding and derive the condition for a funnel-shaped landscape.

From continuum and mesoscale methods (Family G)

34. Write the Poisson-Boltzmann equation and derive the Debye length. State what the linearized form assumes and when it breaks.

35. Explain the generalized Born approximation and what it approximates.

36. Explain the hydrophobic effect in terms of entropy, and explain why implicit solvent models get the enthalpy-entropy decomposition wrong even when the total free energy is approximately right.

37. Derive the Ermak-McCammon Brownian dynamics propagator and state when the overdamped approximation is valid.

38. Explain why hydrodynamic interactions change association rates and write the Rotne-Prager tensor's role.

From dynamics analysis (Family I)

39. Derive the Gaussian network model and show its connection to the contact matrix's pseudo-inverse.

40. Explain why TICA finds slow coordinates where PCA finds high-variance ones.
41. Write the transfer operator for a molecular system, explain the relationship between its eigenvalues and implied timescales, and derive why a Markov state model's lag time must exceed the fastest resolved process.
42. Explain what a Chapman-Kolmogorov test checks and what passing it does not guarantee.

From structure determination (Families J, K)

43. Explain the crystallographic phase problem and how molecular replacement solves it.
44. State the relationship between a B-factor, a cryo-EM local resolution, and a conformational ensemble — and why treating any as a per-atom uncertainty is wrong.
45. Derive the maximum-likelihood objective for single-particle cryo-EM reconstruction and explain what the marginalization is over.
46. Explain Fourier shell correlation and why the 0.143 threshold is a convention rather than a theorem.
47. Write the Bayesian formulation of integrative modeling and state where the forward model for each data type enters.
48. Explain maximum-entropy reweighting of a simulated ensemble against experimental data, and state what it assumes about the prior.

From classical prediction and docking (Families L, M)

49. Derive why side-chain packing with a rotamer library is NP-hard, and explain what dead-end elimination actually eliminates.
50. Explain what a Ramachandran plot is a projection of, and what a backbone-dependent rotamer library encodes that an independent one does not.
51. Explain the loop closure problem and why it has an analytic solution while loop modeling remains unsolved.
52. Explain why docking scoring functions rank poses better than they rank affinities, and what that implies about their functional form.

From sequence and evolution (Family N)

53. Derive the Needleman-Wunsch recursion and state its complexity. Then explain what affine gap penalties change.
54. Derive the forward algorithm for a profile HMM and explain the relationship between HMM emission probabilities and a substitution matrix.
55. Formalize direct coupling analysis as inference in a Potts model, and state precisely what mean-field DCA approximates and what it costs.

From machine learning (Families O, P, Q, R)

56. Derive backpropagation for an arbitrary computational graph; then derive the adjoint method for differentiating through an ODE solver.
57. Derive the evidence lower bound and show where the reparameterization trick enters and why it is necessary.
58. Derive the denoising diffusion objective from the variational bound, show its equivalence to score matching, and state the relationship between the probability-flow ODE and the reverse SDE. Then show that the learned score is a force up to temperature.
59. Define equivariance and invariance formally for $SE(3)$, prove that a network built from scalar products of relative position vectors is $SE(3)$ -invariant, and explain what is lost by taking invariance instead of

equivariance.

50. Derive the triangle inequality constraint on a distance matrix and explain precisely what AlphaFold's triangle multiplicative update enforces — and the sense in which it is only a soft enforcement.

From RNA (Family T)

71. **Derive the Nussinov recursion** and prove that the non-crossing condition is exactly what makes the interval decomposition valid. Then state what the Zuker loop model adds, and explain why the partition function requires a *different* recursion rather than the same one with max replaced by sum.
72. **Write down what a SHAPE reactivity profile determines and what it does not.** Construct a synthetic two-state mixture, compute its population-averaged reactivity, and attempt to recover the components. The recovery is ill-posed; characterize how ill-posed.
73. **Quantify the RNA-versus-protein data gap** from the PDB directly, then decompose the performance gap into the six causes in Atlas T.10 and argue, with evidence, which are contingent and which are structural.

The design-specific trio

51. Formalize inverse folding as a conditional distribution and explain why maximizing sequence likelihood given a backbone is not the same as maximizing the probability that the sequence folds to that backbone.
52. Explain designability: why some backbone folds admit many sequences and others none, and how that relates to the density of states.
53. Explain what self-consistency validation does and does not establish, and construct a concrete case where a design passes with high confidence and still fails experimentally.

If you can do all sixty-three unaided, you are no longer a user of this field's methods.

From performance engineering (Part IV)

These seven are the **cheapest checkpoints in the document to attempt and the ones most likely to change what you do next week**. Every one is answerable on hardware you already own, in an afternoon, and each has a number as its answer.

54. Compute the **arithmetic intensity of the triangle multiplicative update** in an AlphaFold-class model — floating-point operations per byte of memory traffic — and place it on a roofline for a card you own. Do the same for the pairwise non-bonded force kernel in MD. Explain why the two land in such different places and what that implies about how each community optimizes.
55. **Derive the FlashAttention tiling** and prove it computes exactly the same function as standard attention. Then state precisely which term in the memory traffic it eliminates and why the online-softmax trick is what makes it possible.
56. Write down the **memory required to store the pair representation** as a function of sequence length, for a given precision and number of channels. Find the length at which it exceeds each card in your fleet. Then find how much chunking and activation recomputation buy you, and at what cost in time.
57. Explain why GPU molecular dynamics uses **cluster pair lists rather than per-atom Verlet neighbor lists**. The answer is about warps and memory coalescing, not about algorithmic complexity — and it is the clearest example in this document of hardware dictating an algorithm.
58. Derive the **communication cost of the PME FFT** across ranks, and explain the scaling wall it creates. Then explain why separate PME ranks exist and how you would choose their number for a given system and node

count.

59. Work out **why reduced precision is safe in attention but dangerous in force accumulation**. Write the error analysis. This is the single most useful numerics question in computational biology and almost nobody can answer it cleanly.
 70. For one AlphaFold-class prediction on your own hardware, **measure what fraction of wall-clock time is MSA search versus network inference**. Most people assume the network dominates. Measure it before you believe it, then decide which one is worth optimizing.
-

C.2 The capstone projects

Eleven projects, each named and cross-referenced from the family it belongs to. Where a section elsewhere in this document says “that is Capstone VII,” this is the list it means. Begin each as soon as its prerequisite families are done; do not wait until the end.

Every one of these is scoped to the multi-machine GPU fleet you already run, and every one is stated specifically enough that you could write the methods section before you start. That is deliberate: a capstone you cannot specify is a wish.

I — The reimplementation (*after Families B, C, I*)

Build a Markov state model pipeline from scratch. Featurization, TICA, clustering, transition matrix estimation with reversibility, implied timescales, Chapman-Kolmogorov. Validate against PyEMMA on data you already have. *Why:* it forces you to confront, in code, every place the statistics can silently fail. After this you will never trust an MSM figure without asking about the lag time.

II — The equivariance crossover (*after Family O*)

Find the dataset size at which an architectural symmetry constraint stops paying for itself. Train equivariant and non-equivariant versions of one architecture on one task across a data-size sweep, and plot the crossover. *Why:* AlphaFold3 dropped AF2’s explicitly equivariant structure module and got better, while low-data molecular problems clearly still benefit. **Nobody has published that curve for protein structure**, and it would be cited constantly. See Atlas O.3 and O.4.

III — The over-optimization curve (*after Families P, R, S*)

Measure where your design filters stop tracking reality. Take a target with experimental data at several optimization strengths and plot wet-lab success against filter stringency. The curve has a maximum. Find it. *Why:* Gao, Schulman & Hilton did exactly this for learned reward models in language and found true performance peaks and then falls while the proxy keeps climbing. Design pipelines optimize against pLDDT and interface pAE with no equivalent measurement. See D.6.1 and Atlas R.10.

IV — The Boltzmann-generator audit (*after Family D*)

Test whether a learned sampler actually samples the target distribution. Take a system where you can afford converged reference sampling, generate from a flow or diffusion emulator, compute importance weights, and report the effective sample size and the free energy difference against the reference. *Why:* the claim these methods make is Boltzmann sampling, not plausible structures, and the distinction is checkable. See Atlas D.6.

V — The learned prior meets sparse data (*after Family K*)

Work out the correct way to combine a learned structural prior with averaged experimental restraints without double-counting. Hummer and Köfinger solved this for simulation ensembles; nobody has solved it for AlphaFold-class priors, which is what every crosslink-guided and density-guided prediction now does informally. *Why:* it is a well-posed inference question with immediate practical payoff, and it sits directly on top of any restraint-guided prediction work. See Atlas K.3.

VI — The whole-cell parameter gap (*after Family G*)

Take the minimal-cell whole-cell model and ask what it would take to put a designed protein into it. Enumerate every parameter the model needs, mark which are computable today, and quantify how wrong the computable ones are. *Why:* it is the clearest statement of the distance between design and cell biology, and nobody in design has written it down. See Atlas G.2.

VII — The standing assessment (*ambitious*)

Build the blind assessment for de novo design that does not exist. Target set, held out by a third party, experimental readout, published protocol, annual cadence. Adaptiv is the proof it can work; it is not yet CASP. *Why:* this is the single largest piece of missing infrastructure in the field. It is hard because it needs wet-lab partners, which is exactly why nobody has done it. See Atlas R.4 and S.1.

VIII — Interpretability with ground truth (*after Family Q and D.9*)

Train sparse autoencoders on an MSA Transformer and evaluate the recovered features against Pfam domains, CATH folds, catalytic-site annotations and transmembrane spans. Report precision and recall of features against real labels — which language interpretability literally cannot do. *Why:* an MSA Transformer attending across an alignment to find the corresponding residue in a homolog is an induction head doing structural biology, and nobody has looked. **This is the most tractable bridge project in this document, and it benefits the AI field more than the biology field.** See Atlas Q.4, D.7.5, D.9 and D.10.1.

IX — The RNA gap (*after Family Q*)

Characterize, quantitatively, why RNA structure prediction is behind protein structure prediction. Is it data volume, conformational heterogeneity, the ion atmosphere, or the absence of a CASP-grade assessment? Decompose it. *Why:* this is the Rhiju Das question, it is stated honestly nowhere, and the answer determines where effort should go. See Atlas Q.5.

X — Scaling laws for structure prediction (*after D.5*)

Fit the functional form of structure-prediction performance in data, parameters and compute. Subsample the PDB at many fractions, train a fixed architecture at several sizes, fit the surface, and use Kudugunta's methodology critique as the checklist. *Why:* it is the quantitative answer to "should we fund more structure determination or more compute," and that question is currently decided by opinion. See D.5.4.

XI — The speedup (*after Part IV*)

Make one real tool measurably faster and get the change merged. Not a fork, not a benchmark script — an upstream contribution to something people run: GROMACS, OpenMM, OpenFold, Boltz, ColabFold, e3nn, MDAnalysis. Profile it, find the actual bottleneck, fix it, show the number, defend it in review. *Why:* it is the single most verifiable contribution in this entire document — a speedup either reproduces on someone else's machine or

it does not — and it is how you become known to the people who maintain the tools you depend on. It is also the one capstone where the field's gatekeeping is purely technical. **See Part IV, especially E.8.**

C.2b — Four standing commitments, not projects

These were capstones in the first edition. They are better understood as things you do permanently, starting now, rather than things you finish.

The hard target. Pick one biological system nobody has solved and that you already have a head start on. Know it better than anyone. Domain depth is what makes a methods person's benchmarks credible; it is the Phil Bradley lesson.

The infrastructure. Build and release one tool other people use. The Baker and Das lesson is that infrastructure compounds. A narrow, correct, well-documented tool with ten real users is worth more to a methods career than three papers nobody runs.

The negative result. Publish one thing that did not work, properly characterized. The field has almost no record of its failures, which is why people keep repeating them — see Appendix B.3 for how badly the record is skewed. A careful negative result on a method that sounds like it should work is a real contribution and is far easier to get right than a positive one.

The review nobody has written. Write the survey that connects two families in Section A.4 that have never been connected in print. A good review is a research contribution — it creates the frame other people then work inside — and it is the cheapest way to establish that you see the whole field.

C.3 The five generators

Having watched several hundred hours of people describing their own work, the same small number of move-types recur. Named, they can be applied deliberately.

1 — The cross-cultural carry. The highest-yield move, and the reason to be trilingual. Ask continuously: what is routine in one family that another has never heard of? Section A.4 is the current inventory.

2 — Attack the invisible assumption. Every field has a simplifying assumption so universal it has stopped being visible. Live candidates here: that a protein has α structure; that the PDB is a representative sample of protein space; that sequence recovery is a meaningful design metric; that a confidence score estimates accuracy rather than training-set density; that a pairwise decomposable energy is adequate; that static backbone design suffices for anything involving function. Each is false in an interesting way.

3 — Change the level of the stack. The Shaw move. When the limit is the hardware, build hardware. When it is the data, build an assay. When it is the benchmark, build a benchmark. Most people accept the level they were handed.

4 — Take a method where it should not work. RNA is the obvious target — roughly where proteins were twenty years ago, thinner data, harder physics, and the protein methods have not been properly adapted. Disordered proteins are the second. Both are large territories adjacent to your existing skills.

5 — Make it fast enough to change the question. When a calculation goes from a week to a second, people do not do the same science faster; they do different science. A thousandfold speedup is a methods contribution even with no new mathematics in it.

C.4 The permanent seminar diet

After the program ends the practice begins. Roughly three to four hours a week keeps you current in a field that turns over its state of the art about every eighteen months.

- **Weekly:** one ML4PE or Institute for Protein Design seminar.
- **Weekly:** scan new preprints in the relevant bioRxiv and arXiv categories; read two properly, skim twenty.
- **Monthly:** one full conference session from the archives — MLSB, Simons, IPAM, KITP, a CASP meeting, or a Free Energy Workshop.
- **Quarterly:** one historical dive. Pick a method you use daily and watch or read its original presentation. The gap between how a method was originally motivated and how it is now used is reliably instructive and is where a surprising number of new ideas hide in plain sight.
- **Annually:** CASP and CAPRI assessments when they appear. Still the field's only genuinely adversarial evaluation and therefore its most honest few hours.

And the habit that matters most: **maintain the open-problems file**. Every time a speaker says “we don't know how to do X,” or “that's still open,” or “this is the part that doesn't work” — write it down with the date and the speaker. Review it monthly. After a year that file is a more valuable research asset than anything else this curriculum produces, because it is a list of problems that senior people have personally confirmed are both unsolved and worth solving.

That is Hamming's list, and his whole argument is that keeping one is what separates the people who do important work from the people who do competent work.

Part V (continued) — Research Craft

C.5 Why this is in a technical curriculum

Everything else in this document makes you better at the work. This section is about the surrounding skills that determine whether the work matters: choosing problems, writing them up, presenting them, refereeing, and not fooling yourself.

A warning about this genre. It attracts enormous quantities of low-quality motivational content, and the search results for every query here are dominated by productivity influencers and AI-generated summaries. Every entry below is by a practicing researcher with real standing, and where the canonical piece of advice exists only as text, that is said rather than papered over.

C.6 Choosing problems

Talk	Speaker	Yr	Len	Link
You and Your Research	Richard Hamming (Naval Postgraduate School, formerly Bell Labs)	1995	44:02	▶
Creativity	Richard Hamming	1995	1:03:47	▶
Experts	Richard Hamming	1995	44:31	▶
How Do We Know What We Know	Richard Hamming	1995	40:23	▶
Why truly innovative science demands a leap into the unknown	Uri Alon (Weizmann)	2013	15:53	▶
Social Aspects of Science, parts 1–5	Uri Alon	2011	~56:00	▶▶▶▶▶
We have to change the culture of science to do better research	Uri Alon	—	16:16	▶
The Usefulness of Useless Knowledge	Robbert Dijkgraaf (IAS)	2021	1:13:45	▶
Feynman, Shannon, and Darwin Worry About Understanding	David Krakauer (Santa Fe Institute)	2023	28:15	▶
More is Different	Sydney Brenner (Para Limes)	—	1:32:01	▶

Hamming is the single most important item in this section, and the observation that does the work is this: *most scientists never write down their field’s important open problems, and therefore never work on one.* The Method Atlas exists partly to make that list easy for you to write.

The companion text is the 1986 Bellcore transcript (the video is a 1995 restaging) and the book *The Art of Doing Science and Engineering*.

Alon’s “cloud” — the phase where the project is not what you planned and not yet anything else — is the most useful reframing available for a multi-year research program, because it says the disorientation is structural rather than personal. The long-form five-part version has the feasibility-versus-interest selection grid that the TED talk omits. Companion text: Alon, *How To Choose a Good Scientific Problem*, **Mol Cell** 35:726.

Honest gap: Peter Medawar’s *Advice to a Young Scientist* and “Is the Scientific Paper a Fraud?” exist only as text and radio. Read the book; there is no video substitute. Likewise **Michael Nielsen’s *Principles of Effective Research*** and **Karpathy’s *A Survival Guide to a PhD*** are essays with no recorded version.

C.7 Writing

Talk	Speaker	Len	Link
How to Write a Great Research Paper	Simon Peyton Jones (Microsoft Research)	57:39	▶
The Craft of Writing Effectively	Larry McEnerney (University of Chicago)	1:21:51	▶
Writing Beyond the Academy	Larry McEnerney	1:16:54	▶
Scientific Writing from the Readers’ Perspective	Judith Swan (Princeton)	1:15:17	▶
Writing in the Sciences, units 1–8	Kristin Sainani (Stanford)	~3:30 total	▶▶▶▶▶▶▶▶
Linguistics, Style and Writing in the 21st Century	Steven Pinker (Harvard)	53:40	▶
How to Write a Paper to Communicate Your Research	George Whitesides (Harvard)	8:39	▶

McEnerney has the highest ratio of behavior change to runtime in this entire document. His claim — that your paper’s job is not to explain your ideas but to create value for a specific community of readers who have the power to accept or reject them — reframes the introduction, the contribution claim, and the response to reviewers all at once.

Peyton Jones’s rule is the one to adopt immediately: do not wait until the research is done to write the paper. Write it first and let it drive the work.

Companion texts: Gopen & Swan, *The Science of Scientific Writing*, **American Scientist** 78:550 — arguably the best six pages ever written on scientific prose · Whitesides, *Writing a Paper*, **Adv Mater** 16:1375, which is the real artifact; the video is a condensation of it.

C.8 Talks and posters

Talk	Speaker	Len	Link
How to Speak	Patrick Winston (MIT)	1:03:42	▶
How to Give a Great Research Talk	Simon Peyton Jones	58:41	▶
Creating effective slides	Jean-luc Doumont (Stanford CTL)	1:03:34	▶
Designing effective scientific presentations	Susan McConnell (Stanford, iBiology)	42:09	▶
The Assertion-Evidence Approach	Michael Alley (Penn State)	25:56	▶
How to create a better research poster in less time	Mike Morrison	20:53	▶

Winston's lecture was delivered months before his death and is still the genre's benchmark. Peyton Jones's rule for talks is the complement to his rule for papers: **convey one idea, and make them want to read the paper**. Alley's assertion-evidence format — a sentence headline supported by a visual, never a bulleted topic phrase — has actual comprehension data behind it.

Note: Doumont has exactly one full-length free lecture despite his reputation; his other seminars are behind institutional hosting. Do not expect a playlist.

C.9 Refereeing

Talk	Speaker	Len	Link
How CS Conference Reviewing Works and How to Write a Good Review	Jordan Boyd-Graber (Maryland)	17:46	▶
How to write a good review (CVPR tutorial)	Sattler (Chalmers) & Leal-Taixé (TUM)	5:04:04	▶
Custodians of High-Quality Science: Are Editors and Peer Reviewers Good Enough?	David Moher (Ottawa)	30:14	▶
Peer Review: Are We Looking at the Right Things?	Lisa Bero (Colorado)	22:03	▶
Open Collaborative Peer Review	ASAPbio and PRereview	59:01	▶

Start with Boyd-Graber's seventeen minutes — the clearest explanation of the reviewer, area chair and rebuttal pipeline, and what an AC actually reads your review for. Bero's argument is the one worth arguing with: review focuses on novelty and presentation while missing the things that determine whether a finding replicates. The ASAPbio session is the practical route into the skill before anyone invites you.

C.10 Reproducibility and not fooling yourself

This subsection is the most directly load-bearing of the AI part.

Talk	Speaker	Len	Link
Leakage and the Reproducibility Crisis in ML-based Science	Sayash Kapoor (Princeton)	48:10	▶
same, shorter	Sayash Kapoor	35:00	▶
Empirical Research on Research and the Reproducibility Crisis	John Ioannidis (Stanford)	1:26:42	▶
Improving Reproducibility	Brian Nosek (Center for Open Science)	41:40	▶
Culture change toward open, rigorous, reproducible research	Brian Nosek	1:04:32	▶
Pre-Registration and Registered Reports	Daniël Lakens (Eindhoven)	52:11	▶
The Dark Side of Science: Misconduct in Biomedical Research	Elisabeth Bik	1:06:35	▶
12 Ways to Fool the Masses with Irreproducible Results	Lorena Barba (GWU)	37:12	▶
Reproducible, Reusable, and Robust Reinforcement Learning	Joelle Pineau (McGill/Meta)	39:53	▶
Calling Bullshit: Skepticism in a Data-Driven World	Bergstrom & West (UW)	58:45	▶
Does Science Self-Correct?	Ivan Oransky (Retraction Watch)	33:40	▶

Kapoor’s talk is the most directly relevant item in this entire section. Everything else makes you a better scientist in general; this one is about the specific way your work will be wrong. A taxonomy of **eight kinds of leakage, documented across 329 papers in 17 fields, with bioinformatics explicitly among them.** Forty-eight minutes, and it should be watched before your next pipeline rather than after. **Atlas S.2 is this talk turned into a checklist.**

Pineau’s result is the one that should change your benchmarking today: seed variance alone can manufacture an apparent improvement. **Barba’s talk is the computational-science version** — a satirical checklist of the practices that make results unreproducible, by someone who publishes full reproducibility packages for her own work.

Papers: Kapoor & Narayanan 2023, *Patterns* 4:100804 · Ioannidis 2005, *PLoS Med* 2:e124 · Open Science Collaboration 2015, *Science* 349:aac4716.

C.11 Statistics and inference hygiene

Richard McElreath's *Statistical Rethinking*, 2026 edition — 20 lectures, all posted. This is the only free course that teaches causal inference, generative modeling and Bayesian estimation as **one subject**, which is exactly the combination a methods-building computational biologist needs.

#	Lecture	Len	Link	#	Lecture	Len	Link
A01	Introduction to Bayesian Workflow	1:00:49	▶	B01	Multilevel Models	59:36	▶
A02	Garden of Forking Data	59:32	▶	B02	Multilevel Model Expansion	1:11:00	▶
A03	Geocentric Models	1:01:16	▶	B03	Adventures in Covariance	56:57	▶
A04	Categories and Causes	1:03:11	▶	B04	Group-level Confounding	58:46	▶
A05	Estimands and Estiplans	1:02:34	▶	B05	Social Networks II	1:02:54	▶
A06	Elemental Confounds	1:01:31	▶	B06	Gaussian Processes	59:36	▶
A07	Good and Bad Controls	1:00:44	▶	B07	Measurement Models	59:55	▶
A08	MCMC and Item Response Models	1:07:42	▶	B08	Missing and Censored Data	1:02:17	▶
A09	Modeling Events	1:00:00	▶	B09	Generalized Linear Madness	59:53	▶
A10	Hidden Confounds and Sensitivity Analysis	1:00:05	▶	B10	Hidden Markov Models	1:08:21	▶

If you do two lectures from this course, do A02 and A07. A02 shows that Bayesian updating is just counting paths through possibility space. A07 shows that adding a control variable can make your estimate worse in a way no amount of careful coding will reveal. For someone building methods, telling a good control from a bad one is worth more than any individual technique.

Also: Andrew Gelman, *The Statistical Crisis in Science* (57:34, [▶](#)) — the garden of forking paths, and why p-hacking requires no dishonesty, only analytic choices made after seeing the data · *Solve All Your Statistics Problems Using P-Values* ([▶](#)) · the ASA statement on p-values, debated by its authors ([▶](#)).

C.12 Software, grants and career

Scientific software as a career asset: What We Actually Know About Software Development, Greg Wilson (Software Carpentry founder), 1:06:37 (▶) · *Software Carpentry: Lessons Learned* (▶) · **Academic Open Source**, Kathryn Huff (Illinois, later US DOE), 45:26 (▶) — directly on making a released tool legible to hiring and tenure committees · **The Making and Maintenance of our Open Source Infrastructure**, Nadia Eghbal, 1:09:04 (▶) — the economics of maintenance, and why the costs of a successful tool arrive years after the paper · **The Rise of the Research Software Engineer**, Mike Croucher, 39:59 (▶).

Grants and career: How to Write a Great Research Proposal, Simon Peyton Jones, 57:41 (▶) — with the key reframing that behind a badly written proposal there is usually a badly reasoned research plan · **How to Have a Bad Career**, David Patterson, 58:37 (▶) · **A Simplified Mock NIH Peer Review Meeting**, 25:18 (▶) — the clearest twenty-five minutes on how your proposal is actually judged · Inside NIH Study Sections (▶) · NSF CISE CAREER Proposal Writing Workshop (▶).

On the structural realities: Shirley Tilghman, *The Malthusian Dilemma in Biomedical Research* (▶) and *Righting the Ship* (▶).

C.13 The five to do first

1. **Kapoor, *Leakage and the Reproducibility Crisis*** (48 min). Not because it is inspiring but because it is about the specific way your work will be wrong, and the fix is a checklist you can apply this week.
2. **Hamming, *You and Your Research*** (44 min). The only talk here that treats “make original contributions” as a tractable problem with identifiable inputs.
3. **McElreath A02 and A07** (2 hours). Counting paths, and the fact that a control variable can make your estimate worse.
4. **McEnerney, *The Craft of Writing Effectively*** (82 min). The difference between a technically correct manuscript and a cited one.
5. **Peyton Jones on papers, paired with Winston on speaking** (two hours). The same skill pointed at page and podium: write first and let it drive the work; convey one idea and make them want the paper.

Part VI — Appendices

Appendix A — The Standing Archives, Enumerated

The Atlas selects. This appendix does not: it is the complete talk-level index of the archives that matter, so that when you want something the Atlas did not choose, you can find it without re-doing the search.

Why enumerate instead of linking the channel. A channel link is an invitation to scroll. An enumerated archive is a thing you can grep, sort by runtime, and schedule. It also makes the gaps visible — and the gaps turned out to be informative, which is the subject of A.9.

Every entry below came from the live listing (`yt-dlp --flat-playlist` against the channel or playlist), joined against each series' official archive page for speaker and affiliation. Nothing here is reconstructed from memory.

A.1 ML for Protein Engineering Seminar Series — 113 talks

Channel: [@mlforproteinengineeringsem6420](#) · site: [ml4proteinengineering.com](#)

Warning worth knowing before you plan around it: the official schedule page is frozen at **4 November 2025**. Roughly 25 talks have been posted since then with no corresponding schedule entry, and nine of those name no speaker anywhere in the public listing. Those are marked *speaker not named* below rather than attributed to a guessed first author.

#	Talk	Speaker	Len	Link
1	Optimal Design of Stochastic DNA Synthesis Protocols based on Generative Sequence Models	Eli Weinstein (Harvard)	59:50	▶
2	ML-Assisted Protein Engineering with ftMLDE and evSeq	Bruce Wittman (Caltech, Arnold lab)	54:23	▶
3	Learning Protein Fitness Models from Evolutionary and Experimental Data	Chloe Hsu (UC Berkeley)	56:51	▶
4	Systematic molecular evolution using PRANCE	Erika DeBenedictis (UW)	50:45	▶
5	Scaffolding protein functional sites using deep learning	Tischer & Juergens (UW)	52:05	▶
6	Iterative Refinement GNN for Antibody Docking and Design	Wengong Jin (Broad)	1:00:14	▶
7	A Deep Unsupervised Language Model for Protein Design (ProtGPT2)	Noelia Ferruz (Bayreuth)	1:05:20	▶
8	Efficient Evolution of Human Antibodies From General PLMs	Brian Hie (Stanford / Meta AI)	54:11	▶
9	Engineering Proteins with 3D Convolutional Neural Networks	Danny Diaz (UT Austin)	1:12:19	▶
10	Engineering indel and substitution variants of ancient enzymes (GRASP)	Gabriel Foley (Queensland)	1:03:40	▶
11	Protein Structure and Sequence Generation with Equivariant DDPMs	Namrata Anand (Stanford)	55:19	▶
12	Diffusion probabilistic modeling of protein backbones for motif-scaffolding	Trippe (Columbia) & Yim (MIT)	1:03:59	▶
13	Robust deep learning based protein sequence design using ProteinMPNN	Justas Dauparas (UW)	53:42	▶
14	ML-guided engineering of fatty acyl-ACP reductases	Jonathan Greenhalgh (Alpha Bio)	45:31	▶

#	Talk	Speaker	Len	Link
15	Conformal prediction under feedback covariate shift for biomolecular design	Clara Wong-Fannjiang (Berkeley)	58:45	▶
16	Multi-segment preserving sampling for deep manifold sampler	Daniel Berenberg (NYU)	1:00:38	▶

Four moderated panels in 2022 (22 Mar, 17 May, 19 Jul, 27 Sep, 15 Nov) were not recorded.

2023

#	Talk	Speaker	Len	Link
17	Multi-objective engineering of therapeutic antibodies	Emily Makowski (Michigan)	52:40	▶
18	SVD of protein sequences for visualizing sequence and residue space	Gina El Nesr (Stanford)	36:17	▶
19	RFdiffusion: Accurate protein design using structure prediction and diffusion	Watson & Juergens (IPD)	43:52	▶
20	End-to-end learning of MSAs with differentiable Smith-Waterman	Samantha Petti (Harvard)	50:41	▶
21	Dynamic-backbone protein-ligand complex prediction with diffusion models	Zhuoran Qiao (Entos)	42:49	▶
22	Deploying protein machine learning models on the web	Simon Dürr (EPFL)	46:32	▶
23	Hybrid protein language models for fitness prediction (TranceptEVE)	Pascal Notin (Oxford)	52:58	▶
24	SAMPLE: A Self-Driving Laboratory System for Protein Engineering	Jacob Rapp (UW-Madison)	46:35	▶
25	Cyclic peptide structure prediction and design using AlphaFold	Kozlov & Rettie (UW / Harvard)	1:07:20	▶
26	Predicting (and discovering) proteins with multiple conformational states	Hannah Wayment-Steele (Brandeis)	47:49	▶
27	Site-wise mutation effects enable combinatorial protein variant design	David Ding (Harvard / Berkeley)	51:49	▶
28	Mutational Effect Transfer Learning (METL) for Protein Design	Sam Gelman (UW-Madison)	59:13	▶
29	Protein generation with evolutionary diffusion (EvoDiff)	Kevin Yang (Microsoft)	59:46	▶
30	Protein Discovery with Discrete Walk-Jump Sampling	Nathan Frey (Prescient Design)	26:36	▶

#	Talk	Speaker	Len	Link
31	Mega-scale experimental analysis of protein folding stability	Kotaro Tsuboyama (U. Tokyo)	50:57	▶
32	AbDiffuser: Full-Atom Generation of In Vitro Functioning Antibodies	Karolis Martinkus (Prescient)	1:10:25	▶
33	Uncovering Protein Ensembles: Automated Multiconformer Model Building (qFit)	Stephanie Wankowicz (UCSF)	57:53	▶
34	Decoupling catalysis and TS-analog binding in a phosphatase (HT-MEK)	Craig Markin (Stanford)	59:49	▶
35	Protein Design with Guided Discrete Diffusion	Stanton (Prescient) & Gruver (NYU)	58:05	▶

28 Nov 2023 (Zaixiang Zheng, “Structure-informed Language Models Are Protein Designers”) was scheduled but never posted.

#	Talk	Speaker	Len	Link
36	Domain-PFP: protein function prediction using domain embeddings	Nabil Ibtehaz (Purdue)	57:46	▶
37	Enzyme function prediction using contrastive learning (CLEAN)	Tianhao Yu (UIUC)	54:16	▶
38	Kaggle 1st place, CAFA 5 Protein Function Prediction (GoCurator)	GoCurator group (Fudan)	35:49	▶
39	Predicting small molecule substrates of enzymes	Alexander Kroll (Düsseldorf)	57:47	▶
40	PTM-Mamba: A PTM-Aware Protein Language Model	Zhangzhi Peng (Duke)	46:47	▶
41	Sequence modeling and design from molecular to genome scale with Evo	Nguyen & Hie (Stanford)	1:06:46	▶
42	Design of genome editors by modeling the universe of CRISPR-Cas sequences	Ruffolo & Nayfach (Profluent)	1:00:14	▶
43	Diffusion Language Models Are Versatile Protein Learners (DPLM)	Zaixiang Zheng (ByteDance)	1:13:19	▶
44	A framework for conditional diffusion modelling in motif scaffolding	Didi, Mathis & Vargas (Cambridge)	1:01:57	▶
45	Designing and Scaffolding Proteins at the Scale of the Structural Universe (Genie 2)	Lin & Lee (Columbia)	53:11	▶
46	Engineering nuclease enzymes by ML and ultra-high-throughput screening	Thomas & Belanger (Google)	58:00	▶
47	ESM3: Simulating 500 million years of evolution with a language model	Roshan Rao (EvolutionaryScale)	1:05:08	▶
48	CARE: a Benchmark Suite for the Classification and Retrieval of Enzymes	Jason Yang (Caltech)	47:02	▶

#	Talk	Speaker	Len	Link
49	Sequence-Augmented SE(3)-Flow Matching (FoldFlow-2)	Tong (Mila) & Huguet (Montréal)	48:47	▶
50	Rapid protein evolution by few-shot learning with a protein language model	Kaiyi Jiang (MIT)	58:18	▶
51	Adapting protein language models for structure-conditioned design	Jeff Ruffolo (Profluent)	55:47	▶
52	CHEAP: Tokenized and Continuous Embedding Compressions	Amy Lu (UC Berkeley)	53:32	▶
53	Template-based protein editing using Raygun	Kapil Devkota (Duke)	40:51	▶
54	The OMG dataset: an Open MetaGenomic corpus	Andre Cornman (Tatta Bio)	20:38	▶
55	Lessons from implementing AlphaFold3 in the wild	Arda Goreci (Ligo Biosciences)	25:52	▶
56	AI-driven de novo Enzyme Design	Will Hua (McGill)	35:36	▶
57	ProTrek: Navigating the Protein Universe through Tri-Modal Contrastive Learning	Jin Su (Westlake)	48:35	▶
58	Multistate and functional protein design using RF sequence space diffusion	Gershon, Lisanza & Tipps (UW)	54:47	▶

16 Jul 2024 (Francesca-Zhoufan Li, “Feature Reuse and Scaling”) was scheduled but never posted.

#	Talk	Speaker	Len	Link
59	Mapping olfactory receptor-perception coding with deep learning	Seyone Chithrananda (Berkeley)	40:03	▶
60	InterPLM: Interpretable Features in Protein LMs via Sparse Autoencoders	Elana Simon (Stanford)	59:42	▶
61	Concept Bottleneck Language Models for Protein Design	Abdelsalam (Guide Labs) & Frey	51:46	▶
62	ProtComposer: Compositional Protein Structure Generation with 3D Ellipsoids	Jing & Stark (MIT)	53:49	▶
63	Proteina: Scaling Flow-based Protein Structure Generative Models	Karsten Kreis (NVIDIA)	1:09:24	▶
64	Bayesian Optimization of Antibodies Informed by a Generative Model	Alan Amin (NYU Courant)	39:53	▶
65	Targeting protein-ligand neosurfaces with a generalizable deep learning tool	Anthony Marchand (Novate)	52:30	▶
66	Genome modeling and design across all domains of life with Evo 2	Garyk Brixi (Stanford)	34:46	▶
67	MotifBench: a standardized benchmark for motif-scaffolding	Zhuoqi Zheng (SJTU)	54:48	▶
68	Learning millisecond protein dynamics from what is missing in NMR spectra	El Nesr & Wayment-Steele	47:43	▶
69	Computational design of serine hydrolases	Lauko (UW/Xaira) & Pellock (UW)	41:14	▶
70	Atom level enzyme active site scaffolding using RFdiffusion2	Ahern (UW) & Yim (MIT)	51:56	▶
71	Deep learning guided design of protease substrates	Ava Amini (Microsoft Research)	52:49	▶
72	Flexible Antibody Design with Multimodal Bayesian Flow Networks	Guloglu & Barrett (InstaDeep)	44:20	▶

#	Talk	Speaker	Len	Link
73	Functional alignment of protein language models via reinforcement learning	<i>speaker not named</i>	40:50	▶
74	La-Proteina: Atomistic Protein Generation via Partially Latent Flow Matching	Didi (Oxford) & Geffner (NVIDIA)	1:05:13	▶
75	Scaling down and expanding the scope of PLMs with MSA Pairformer	Yo Akiyama (MIT)	37:59	▶
76	Accelerating Biomolecular Modeling with AtomWorks and RF3	Nate Corley (UW)	51:17	▶
77	The Dayhoff Atlas: scaling sequence diversity for improved protein generation	Alex Lee (UCSF / MSR)	57:46	▶
78	Generative design of novel bacteriophages with genome language models	Samuel King (Stanford / Arc)	40:23	▶
79	PoET-2: multimodal retrieval-augmented foundation model	Tristan Bepler (OpenProtein.AI)	54:12	▶
80	ProGen3: Scaling Unlocks Broader Generation and Deeper Functional Understanding	Aadyot Bhatnagar (Profluent)	1:07:20	▶

2026 — posted but not on the schedule page

#	Talk	Speaker	Len	Link
81	BoltzDesign & Protein Hunter: Inverting Structure Prediction Models	Yehlin Cho (MIT)	48:46	▶
82	Ensemble-conditioned protein sequence design with Caliby	Richard Shuai (Stanford)	34:00	▶
83	De novo Design of All-atom Biomolecular Interactions with RFdiffusion3	<i>speaker not named</i> (IPD/UW)	48:07	▶
84	Efficient generation of epitope-targeted de novo antibodies with Germinal	<i>speaker not named</i>	57:35	▶
85	Computational design of conformation-biasing mutations to alter protein function	<i>speaker not named</i>	51:45	▶
86	Proteo-R1: Reasoning Foundation Models for De Novo Protein Design	<i>speaker not named</i>	52:36	▶
87	Zero-shot design of a de novo metalloenzyme (dEVA)	<i>speaker not named</i>	51:08	▶
88	Programming Biomolecular Interactions with All-Atom Generative Model	ByteDance team	54:24	▶
89	Scaling Atomistic Protein Binder Design with Pretraining and Test-Time Compute	NVIDIA team	55:28	▶
90	Language Modeling Materializes a World Model of Protein Biology	<i>speaker not named</i>	1:59:23	▶
91	General Multimodal Protein Design Enables DNA-Encoding of Chemistry	<i>speaker not named</i>	43:27	▶
92	Extensive Gene Evolution on Laboratory Timescale	Chang Liu (UC Irvine)	1:02:46	▶
93	De Novo Designed Voltage-gated Anion Channels Suppress Neuron Firing	Chen Zhou (Westlake)	28:49	▶

#	Talk	Speaker	Len	Link
94	Determinants of efficacious de novo chimeric antigen receptors	Chow & Chu	46:34	▶
95	How to Make the Most of Your Masked LM for Protein Engineering	Calvin McCarter (BigHat)	53:10	▶
96	A high-level design language for generative biology with Proto	Aditi Merchant (Stanford / Arc)	58:56	▶
97	AI-redesigned starting points and outcomes enhance protein evolution	Nicholas Krasnow	53:07	▶
98	Zero-shot design of drug-binding proteins via neural iterative selection—expansion	<i>speaker not named</i>	36:24	▶

ML4PE Early Career Seminar Series — 15 talks

Short research-program overviews by people one career stage ahead of you. If you want to understand what a *lab's agenda* looks like rather than a single result, this is the most efficient 14 hours in the whole curriculum.

#	Speaker	Affiliation	Len	Link
99	Mohammed AlQuraishi	Columbia, Systems Biology	1:05:58	▶
100	Noelia Ferruz	CRG Barcelona	56:15	▶
101	Brian Hie	Stanford / Arc Institute	49:35	▶
102	Wengong Jin	Northeastern / Broad	1:12:24	▶
103	Possu Huang	Stanford	1:16:22	▶
104	Pranam Chatterjee	UPenn, Bioengineering & CIS	1:03:29	▶
105	Gabe Rocklin	Northwestern	1:00:26	▶
106	Rohit Singh	Duke	1:04:52	▶
107	Alex Tong	Mila / Montréal	55:10	▶
108	Alan Amin	NYU Courant	49:42	▶
109	Rohith Krishna	UW / IPD	56:22	▶
110	Eli N. Weinstein	Columbia / Harvard	51:20	▶
111	Soojung Yang	MIT	58:36	▶
112	Sarah Gurev	MIT / Marks lab	59:36	▶
113	<i>(series opener, ungated)</i>	—	—	—

Two of your ten named scientists give their program talk here: Possu Huang (#103) and Pranam Chatterjee (#104). These are the closest thing to a direct answer to “what does this person actually work on” that exists on video.

A.2 Boston Protein Design and Modeling Club — 64 talks

Channel: [@bpdmc](#)

BPDMC is the longest-running seminar in the field and the one most willing to host a talk whose message is “this did not work.” The archive pages carry speakers and dates but **no video links at all**, so the mapping below is the only joined index of the two that exists.

Recording gap worth knowing: the 2018 and 2019 archives list 28 talks that were never recorded — the channel’s oldest upload is January 2021. Seven 2020 entries and two 2021 entries are likewise text-only. Those talks are gone.

2026

#	Talk	Speaker	Len	Link
1	mBER: Controllable de novo antibody design with million-scale screening	Swanson & Nichols	1:42:10	▶
2	Antibody language models vs. biology; protein docking diffusion vs. physics	Jeff Gray (Johns Hopkins)	1:25:10	▶
3	Creativity at Scale: Problem Formulation as the Frontier of Protein Design	Isaac Lutz	1:29:26	▶
4	Computational design of antibody repertoires	Ariel Tennenhouse	1:10:02	▶
5	Beyond sequence recovery: modeling the sequence-energy landscape	Foster Birnbaum	33:23	▶
6	Energy-Based Models of Protein Structure, Dynamics, and Folding	James Roney	1:18:18	▶
7	De novo design of small-molecule-induced conformational change	Jeffrey Chang	1:03:21	▶
8	Quantitative Affinity Data at Scale: the Data Bottleneck in AI-Enabled Design	Natasha Murakowska	47:16	▶

Talk #2 is the one to watch first in this year. Gray argues both halves of the Atlas at once — that antibody language models are not doing biology, and that docking diffusion models are not doing physics. It is the most compressed statement of the three-cultures tension anywhere on this list.

2025

#	Talk	Speaker	Len	Link
9	Design of small molecule binding proteins using deep learning	Ben Fry	1:00:53	▶
10	Boltz-1 and the Future of Biomolecular Foundation Models	Wohlwend & Corso (MIT)	1:09:45	▶
11	From DNA Origami to Protein Design: Symmetry-Guided Principles Across Scales	Seth Fraden (Brandeis)	1:18:28	▶
12	Interpreting state-of-the-art structure predictors for protein-peptide complexes	Lindsey Guan	58:12	▶
13	Designing new and improved functions in natural protein folds	Sarel Fleishman (Weizmann)	1:21:26	▶
14	RNA Function, Design, and Modeling	Silvi Rouskin (HMS)	59:33	▶
15	Model. Package. Deploy. Repeat: A DevOps Approach to Biomolecular Scalability	Colby Ford	1:19:04	▶
16	BindCraft: one-shot design of functional protein binders	Martin Pacesa (EPFL)	1:15:23	▶
17	Empowering Biomedical Discovery with “AI Scientists”	Marinka Zitnik (HMS)	1:25:29	▶
18	Predicting folding stability and aggregation from large-scale experiments	Gabe Rocklin (Northwestern)	1:00:30	▶
19	Learning millisecond protein dynamics from what is missing in NMR spectra	Gina El Nesr (Stanford)	56:40	▶
20	How AF3-Style Models Can Be Used for Design: BoltzDesign and Protein Hunter	Yehlin Cho (MIT)	1:10:35	▶

2024

#	Talk	Speaker	Len	Link
21	Unsupervised early warning of viral antibody escape	Gurev & Youssef (Harvard/MIT)	1:26:30	▶
22	Recent methods for protein structure generation and design	Sergey Ovchinnikov (MIT)	1:21:22	▶
23	Computational Design of non-porous, pH-responsive antibody nanoparticles	Erin Yang	58:58	▶
24	Shedding light on functional dark matter with genomic language modeling	Yunha Hwang (Harvard)	46:39	▶
25	Designer proteins for stem cell engineering	Rubul Mout	1:18:23	▶
26	Review and discussion of AlphaFold3	Sergey Ovchinnikov (MIT)	1:12:58	▶
27	Discrete diffusion models for generative protein design	Sarah Alamdari (MSR)	1:00:13	▶
28	Jointly Embedding Protein Structures and Sequences through Residue Level Alignment	Foster Birnbaum	1:05:21	▶
29	Small But Mighty: What an LSTM Reveals About Protein Design and About ML	Joanna Slusky (Kansas)	1:14:04	▶
30	Physics-aware agentic AI to model, design and discover proteins	Markus Buehler (MIT)	1:29:17	▶
31	Massively parallel discovery of peptides to inhibit cellular protein interactions	Andrew Savinov	1:12:44	▶
32	De novo design of miniprotein-based natural killer cell engagers	Mireia Solà Colom	59:56	▶

2023

#	Talk	Speaker	Len	Link
33	Designing ligand-binding proteins from scratch	Nick Polizzi (Dana-Farber/HMS)	1:18:23	▶
34	Expanding the Universe of Functional Proteins by Computational Design	Bruno Correia (EPFL)	1:17:45	▶
35	DiffDock: Diffusion Steps, Twists and Turns for Molecular Docking	Corso, Stärk & Jing (MIT)	1:36:23	▶
36	Fold-switching proteins by combining AlphaFold2 and sequence clustering	Hannah Wayment-Steele	59:18	▶
37	Building a global data supply chain to improve protein design	Philipp Lorenz (Basecamp)	1:05:28	▶
38	Fragment-based backbone sampling and NN potentials for peptide design	Sebastian Swanson (MIT)	1:19:48	▶
39	Exploration of novel functional sequence space using evolution-informed design	Nick Gauthier	1:16:42	▶
40	Data volume and diversity needed for generalizable antibody $\Delta\Delta G$ prediction	Alissa Hummer (Oxford)	1:14:43	▶
41	Autoinhibitory domains for a protease-activated PD-L1 antagonist	Odessa Goudy	1:11:19	▶
42	Exploring structural heterogeneity through cryoEM, cryoET, and deep learning	Joey Davis (MIT)	1:05:57	▶
43	Bridging Biophysics and AI to Optimize Protein Design	Ava Amini (MSR)	1:28:45	▶
44	Learning transferable protein backmapping from conformational ensembles	Soojung Yang (MIT)	1:03:00	▶

2022

#	Talk	Speaker	Len	Link
45	ColabDesign — Tutorial on using structure prediction methods for design	Sergey Ovchinnikov	2:04:03	▶
46	Learning from natural antibodies for sequence generation and fast structure prediction	Jeff Ruffolo (JHU)	58:51	▶
47	Therapeutic enzyme engineering using a generative neural network	Dousis & Giessel (Moderna)	1:18:46	▶
48	From large to small: protein design for stability and de novo interfaces	Eva-Maria Strauch (Georgia)	1:07:37	▶

#	Talk	Speaker	Len	Link
49	Initial reaction to AlphaFold2 right after CASP14	AlQuraishi & Ovchinnikov	1:17:30	▶
50	The 7 Habits of Highly Successful Protein Therapeutics	Chris Ing	1:01:04	▶
51	De novo design of disulfide-rich miniprotein agonists for family B GPCRs	Ben Meinen	51:20	▶
52	Designing cKIT Receptor Inhibitors for Bone Marrow Transplant	Frances Chu	27:16	▶
53	De novo design of mini-proteins to inhibit bacterial biofilm formation	Andrea Garavito	1:00:01	▶
54	New tools for studying the mechanics of single proteins	Pallav Kosuri (Salk)	1:18:03	▶
55	ColabFold — Making protein folding accessible to all	Sergey Ovchinnikov	1:46:09	▶
56	Predicting protein structures from single sequences (RGN2)	Nazim Bouatta (HMS)	1:33:06	▶
57	Learning to design mini-proteins that bind specific protein targets	Brian Coventry (IPD)	1:12:12	▶
58	How local changes in secondary structure flexibility drive protein function	Anum Glasgow (Columbia)	1:05:03	▶

2020

#	Talk	Speaker	Len	Link
59	Massively Parallel Protein Design for Next-Generation Chemotherapy	Brahm Yachnin (Rutgers)	1:02:51	▶
60	De novo Glycan Modeling and Design	Jared Adolf-Bryfogle	1:10:06	▶
61	Protein sequence design by explicit energy landscape optimization	Sergey Ovchinnikov	1:30:36	▶
62	Why designs fail, and how they move	Gabe Rocklin	1:09:30	▶
63	A computational approach to design structured peptides	Parisa Hosseinzadeh (Oregon)	52:16	▶
64	Tertiary motifs as building blocks for protein binding peptides	Sebastian Swanson (MIT)	1:03:00	▶

Entry #58, Anum Glasgow, is the only recorded talk anywhere in this curriculum by someone from the Kortemme lineage speaking about designed conformational change. Kortemme herself has almost no recorded talks — which is a real gap in the curriculum and the reason her ideas reach you through papers in Atlas I and R.8 rather than through video.

A.3 RosettaCommons — 99 tutorial and bootcamp videos

Channel: [@RosettaCommons](#)

These are institutional recordings with no named speaker in title or description. They are the *hands-on* half of the curriculum: where ML4PE and BPDMC tell you what was done, these show you the commands.

ML for Protein Design Bootcamp (2026) — 13 videos

#	Talk	Len	Link
1	Tools intro: PyMOL & VS Code	13:49	▶
2	Protein structure prediction: history, challenges, AI breakthroughs	10:42	▶
3	AlphaFold2 Structure Prediction	45:48	▶
4	ESMFold Structure Prediction	16:20	▶
5	PyTorch explained: from neural networks to real-world models	33:10	▶
6	AlphaFold3 Structure Prediction: what's new and why it matters	24:46	▶
7	Backbone generation in protein design workflows	22:25	▶
8	Inverse folding using ProteinMPNN	29:56	▶
9	Cutting-edge ML tools for protein design pipelines	33:32	▶
10	Protein design compute: when to use GPU vs CPU	19:25	▶
11	Modern Protein Docking: Traditional Methods vs AI Tools	24:21	▶
12	AI-driven protein design workflow: a practical example	21:12	▶
13	Protein design roundtable: real challenges and practical solutions	23:55	▶

ML Methods for Protein Modeling (2025) — 18 videos

#	Talk	Len	Link
14	Introduction to Protein Design with ML	10:33	▶
15	PyMOL and VS Code (mini-session)	29:40	▶
16	Protein Structure Prediction Introduction	20:22	▶
17	Structure Prediction with AlphaFold2 and OpenFold	1:39:36	▶
18	Version Control with Git (mini-session)	11:31	▶
19	Structure Prediction with ESMFold	10:32	▶
20	Structure Prediction with AlphaFold3	42:51	▶
21	Developing ML models with PyTorch (mini-session)	36:49	▶
22	Backbone Generation Introduction	11:05	▶
23	Backbone Generation with RFdiffusion	25:35	▶
24	Backbone Generation with RFdiffusion All-Atom	13:41	▶
25	CPU vs GPU Hardware for Protein Design (mini-session)	15:36	▶
26	Sequence Design Introduction	13:46	▶
27	Sequence Design with ProteinMPNN	33:41	▶
28	Sequence Design with LigandMPNN	17:39	▶
29	Protein Docking Introduction	11:18	▶
30	Protein Docking with DiffDock-PP	21:19	▶
31	Other Protein ML Tools	25:35	▶

RaMP Bootcamp 2023 — 12 videos (the PyRosetta core)

#	Talk	Len	Link
32	Intro to PyRosetta — Pose Basics	26:32	▶
33	PyRosetta Score Functions	43:27	▶
34	Modeling Macromolecules with RosettaScripts Movers	13:57	▶
35	Refining biomolecular structures with minimization and Monte Carlo	39:13	▶
36	Designing with PyRosetta: Packing and Relaxing	37:22	▶
37	Introduction to Constraints in Rosetta	29:36	▶
38	Machine learning basics for protein modeling and design	1:02:34	▶
39	Protein Language Models	33:49	▶
40	AlphaFold — ML for protein structure prediction	50:40	▶
41	MPNN — ML for protein sequence design	25:31	▶
42	Diffusion models for protein structure generation and design	44:43	▶
43	Hallucinating Protein Backbones and Sequences	12:32	▶

Rosetta++ Bootcamp (2026) — 20 videos

For reading and extending the Rosetta C++ source. Watch this only if you intend to write a mover — which Capstone II asks you to do.

#	Talk	Len	Link
44	C++ object-oriented programming (with Python comparisons)	17:49	▶
45	Introduction to Version Control with Git	48:38	▶
46	Graph algorithms in Rosetta: trees, DAGs, and energy graphs	22:17	▶
47	Rosetta Library Structure 1	33:28	▶
48	Rosetta Library Structure 2	35:52	▶
49	Unit testing in Rosetta	25:10	▶
50	Rosetta Library Structure 3	35:18	▶
51	Fold trees explained: how protein structures are modeled in Rosetta	36:53	▶
52	RosettaScripts explained: building custom protocols	34:56	▶
53	RosettaScripts in action: designing your first workflow	13:53	▶
54	More C++ classes: this pointer, const, function signatures	47:41	▶
55	RosettaScripts Mover subclassing: building your first mover class	22:57	▶
56	Rosetta Library Structure 4	35:51	▶
57	Shell scripting essentials for large-scale data processing	26:51	▶
58	Controlling flexibility in Rosetta	36:28	▶
59	Loop modeling workflows in Rosetta	7:32	▶
60	Rosetta development guide: resources, conventions, best practices	55:18	▶
61	SimpleMetric Subclassing	11:08	▶
62	Adding Code to Rosetta	35:22	▶
63	How to create software that others want to use	41:46	▶

Protein Design Workflows — 14 videos

The pipeline-engineering layer. This is the part of the field that papers never describe and that determines whether your campaign takes a week or a quarter.

#	Talk	Len	Link
64	Protein Design Workflows introduction	4:27	▶
65	Ovo, an Open-Source Ecosystem for De Novo Protein Design	37:56	▶
66	Ovo Tutorial: developing a plugin	1:00:30	▶
67	Automating protein design with ProtFlow	34:45	▶
68	Prosculpt — an open-source pipeline for accessible protein design	27:29	▶
69	PyRosetta, RosettaScripts and RosettaLink	30:23	▶
70	GradioBio: interoperable and portable protein design workflows	28:45	▶
71	Flexible and scalable protein design with pipelines	24:46	▶
72	RC-run: making biomolecular workflows easier to build and run	2:32	▶
73	AI.zymes: a modular platform for evolutionary enzyme design	33:10	▶
74	BinderFlow: batch-based binder design with live monitoring	16:23	▶
75	BioExcel Building Blocks: interoperable biomolecular simulation workflows	36:36	▶
76	BoltzGen: Toward Universal Binder Design	32:45	▶
77	BioPipelines: composable protein and ligand design on HPC and Colab	35:06	▶

#76 is the only talk that exists for BoltzGen — Atlas R.3 flagged the paper as talk-less, and this workflow session is the gap-filler.

Designing Protein-Protein Interactions (2025) — 8 videos

#	Talk	Len	Link
78	PPI Workshop Overview	3:53	▶
79	Intro to Deep Learning for Protein Design	21:29	▶
80	Protein-Protein Interfaces	7:49	▶
81	Binder Design Methods Overview	8:08	▶
82	BindCraft (hands-on)	23:06	▶
83	Symmetric Protein Assemblies	2:18	▶
84	Designing Self-Assembling Protein Nanoparticles	34:59	▶
85	RFdiffusion for Symmetric Applications	23:09	▶

Tech Tea Time — 9 videos

#	Talk	Len	Link
86	Protein Binder Design with RFdiffusion3	30:59	▶
87	Docker containers	13:02	▶
88	Protpardelle-1c Tutorial: motif scaffolding, multi-chain, all-atom design	30:00	▶
89	uv for Python package management	12:14	▶
90	FLAb2: How Well Can Protein AI Predict Developability?	31:46	▶
91	Caliby: Protein Sequence Design for Structurally Complex Backbones	38:32	▶
92	HBDesigner: rapid, versatile hydrogen bond network design	27:21	▶
93	The State of AI Agents in Scientific Coding	33:20	▶
94	Using AI Agents for Scientific Workflows in Industry	16:52	▶

Guided Tutorials — 5 videos

#	Talk	Len	Link
95	Fundamentals of protein binder design with RFdiffusion	20:32	▶
96	RFdiffusion3 installation with Foundry	6:31	▶
97	Binder Design with RFdiffusion3	9:07	▶
98	RFdiffusion3 Simple Motif Scaffolding	10:21	▶
99	Enzyme design with RFdiffusion3	12:47	▶

A.4 Broad Institute MIA — 78 relevant talks of 226

Playlist: **Models, Inference & Algorithms** at the Broad.

MIA's format is the best in the field and nobody copies it: **a 30-minute “primer” by a junior person teaching the required background, then the main talk**. You get the method and the prerequisites in one sitting. The primer speaker is named in the entry where MIA named them.

The channel has 226 talks; roughly 150 are single-cell, spatial-omics, GWAS, EHR or imaging and are excluded here. The 78 below are the protein, structure, molecular-ML and statistical-methods talks. Bracketed numbers are MIA's own episode numbering, which counts backwards from the oldest.

Proteins, structure, molecular design

#	Talk	Speaker (+ primer)	Len	Link
1	Machine Learning & the Life Sciences: Beyond Data Analysis	Ryan Adams (Harvard/Princeton)	53:34	▶
2	Geometry and Gene Regulation	Caroline Uhler (MIT)	1:04:11	▶
3	Unmixing next-gen mass spec proteomics	Peckner; primer Krug	1:45:58	▶
4	Structure & fitness from genomic sequences	Debora Marks (HMS); Ingraham & Riesselman	1:48:55	▶
5	Deep learning chemical space	Rafa Gómez-Bombarelli (MIT)	57:31	▶
6	Continuous directed evolution: advances, applications, opportunities	Ahmed Badran (Broad)	54:49	▶
7	Improving endogenous antigen prediction for cancer vaccines	Sarkizova & Rooney	1:38:35	▶
8	Message passing algorithms for cryo-EM and synchronization	Alex Wein (NYU)	58:29	▶
9	Machine-Learning-Based CRISPR Guide Design	Jennifer Listgarten (Berkeley)	56:19	▶
10	Learning protein structure with a differentiable simulator	John Ingraham (Harvard)	55:13	▶
11	Evolutionary Dynamics	Martin Nowak (Harvard)	54:22	▶
12	Hamilton's rule makes no prediction and cannot be tested empirically	Martin Nowak (Harvard)	45:52	▶
13	Convolutional models of molecular structure	Boomsma & Frelsen	1:47:16	▶
14	End-to-end differentiable learning of protein structure (RGN)	Mohammed AlQuraishi (Harvard)	56:12	▶
15	Alignment-free models for protein and antibody design	Marks; primer Kollasch	1:40:20	▶
16	Rapid mutation & continuous directed evolution in vivo	Chang Liu (UCI); Badran	1:43:18	▶
17	Geometric deep learning for functional protein design	Michael Bronstein	1:09:06	▶

#	Talk	Speaker (+ primer)	Len	Link
18	Machine-guided capsid engineering for gene therapy	Kelsic; primer Sinai	1:42:15	▶
19	ML-based design of proteins (and small molecules and beyond)	Jennifer Listgarten (Berkeley)	57:06	▶
20	Determinants of base editing outcomes from library analysis & ML	Max Shen (MIT)	55:21	▶
21	Interpreting gene variants	Dias & Frazer; primer Shin	1:37:29	▶
22	Generative models of proteins and genomes	Eli Weinstein; primer Amin on PolyA trees	1:46:15	▶
23	Actionable machine learning for drug discovery	Marinka Zitnik (HMS); Li	1:40:40	▶
24	Deciphering the RNA structural code	Khoroshkin & Karimzadeh; Goodarzi	1:42:39	▶
25	Learning to read and write protein evolution	Brian Hie (Stanford/Meta)	1:02:35	▶
26	Computational protein design	Bruno Correia (EPFL); primer Goverde	1:34:43	▶
27	Sparsity, Epistasis, and Models of Fitness Functions	Aghazadeh & Brookes	1:36:47	▶
28	A Gentle Introduction to Graph Neural Networks	Benjamin Sanchez-Lengeling (Google)	44:53	▶
29	Protein design with generative diffusion models	Amini & Alamdari (Microsoft)	1:21:48	▶
30	AlphaMissense	Jun Cheng (DeepMind); primer Avsec	1:22:31	▶
31	Unsupervised viral antibody escape prediction	Gurev & Youssef	1:45:50	▶
32	Multimodal protein language models for deciphering protein function	Zitnik Lab (HMS)	1:30:33	▶
33	Large-scale conformational transition in a membrane transporter	Trebesch / Fakharzadeh (Tajkhorshid lab)	1:34:25	▶
34	ML for reconstructing structural landscapes from cryoEM	Ellen Zhong (Princeton); Raghu	1:41:32	▶

#	Talk	Speaker (+ primer)	Len	Link
35	Automating Chemistry With LLM-Based Agents	Sam Cox; primer Andrew White	1:22:30	▶
36	Exploring the Protein Universe	Martin Steinegger (SNU) ; Cha	1:17:41	▶
37	Predicting small-molecule binding sites using AlphaFold2	Nick Polizzi (Dana-Farber); Fry	1:50:51	▶
38	Generative Genome Design	Samuel King (Stanford/Arc); primer Hie	2:00:11	▶

Statistical and generative-modeling methods

These are the talks that will make you better at *inventing* methods rather than applying them. The Jonas Peters causality series and the Gil Strang linear algebra lecture are the two highest-value items in this block.

#	Talk	Speaker (+ primer)	Len	Link
39	Bayesian structured sparsity	Barbara Engelhardt; Reshef on GPs	1:58:42	▶
40	Scaling & generalizing variational inference	David Blei (Columbia)	1:40:24	▶
41	Compressed experiments	Yaniv Erlich; Peckner on linear codes	1:46:18	▶
42	Automated inference & probabilistic programming	Kucukelbir; Ranganath on PGMs	1:47:27	▶
43	What can persistent homology see?	Chad Giusti; Sizemore on TDA	1:53:40	▶
44	Probabilistic models of diversity: determinantal point processes	Stefanie Jegelka (MIT); Li	1:45:31	▶
45	Composing graphical models with neural networks	Matt Johnson; Linderman	1:47:02	▶
46	Lectures on Causality, Part 1	Jonas Peters (ETH Zürich)	1:44:05	▶
47	Lectures on Causality, Part 2	Jonas Peters	1:53:51	▶
48	Lectures on Causality, Part 3	Jonas Peters	1:55:04	▶
49	Lectures on Causality, Part 4	Jonas Peters	1:31:37	▶
50	Learning developmental landscapes with optimal transport	Schiebinger; Chizat	1:34:26	▶
51	Maximizing submodular functions exponentially faster	Yaron Singer (Harvard); Breuer	1:46:57	▶
52	What might ML learn from probabilistic programming?	Tran & Suter	57:01	▶
53	Structured factor models for interpretable signal in genomic data	Barbara Engelhardt	1:04:42	▶
54	Manifold learning of cellular state space	Krishnaswamy (Yale); van Dijk	1:48:15	▶
55	Knockoffs for finding variables with statistical guarantees	Lucas Janson (Harvard); Wang	1:36:18	▶
56	Random matrix theory for single-cell RNA; TDA	Andrew Blumberg	1:45:05	▶
57	Topology of regularized linear autoencoders	Kunin, Bloom, Goeva, Seed	1:45:27	▶

#	Talk	Speaker (+ primer)	Len	Link
58	Regularized linear autoencoders, Morse homology, and backprop in the brain	Jon Bloom (@ Google Brain)	58:45	▶
59	PCA and stratification in GWAS	Chen; Bloemendal on random matrix theory	1:46:04	▶
60	Interpreting and learning from black box models	Wesley Tansey (MSKCC)	51:59	▶
61	Extracting causal signal from high-dimensional data	Alex D'Amour (Google Brain)	55:10	▶
62	Noise2Self: blind denoising by self-supervision	Josh Batson (CZ Biohub)	54:29	▶
63	Key Ideas in Linear Algebra	Gil Strang (MIT); primers Bloom and Cho	1:45:51	▶
64	Bayesian deep learning	Andrew Gordon Wilson (NYU); Izmailov & Kirichenko	1:39:07	▶
65	It's time to talk about irregularly-sampled time series	David Duvenaud (Toronto)	54:43	▶
66	Feature analysis in 3D shapes	Lorin Crawford (Brown/Microsoft); Winn	1:39:46	▶
67	Bagged posteriors for robust inference	Jonathan Huggins; Stephenson	1:33:38	▶
68	Smoothed nested testing on directed acyclic graphs	Jackson Loper; Bloemendal	1:40:45	▶
69	The kernel interaction trick	Tamara Broderick (MIT); Agrawal on GPs	1:36:18	▶
70	Deep probabilistic programming with Pyro	Obermeyer; Bingham	1:50:53	▶
71	Bayesian methods for adaptive experimental design	Martin Jankowiak	50:41	▶
72	Low-dimensional embeddings of words and documents (UMAP)	Leland McInnes; Cho	1:51:42	▶
73	Probabilistic inference and learning with Stein's method	Lester Mackey (Microsoft); Riabiz	1:38:17	▶
74	A new approach for high-dimensional hierarchical modeling	Brian Trippe	53:03	▶
75	Moscot: optimal transport in single-cell genomics	Lange; primer Marco Cuturi	1:52:42	▶

#	Talk	Speaker (+ primer)	Len	Link
76	Neural Optimal Transport for Cell Perturbation Responses	Charlotte Bunne (ETH); Ursu	1:50:40	▶
77	Data thinning to avoid double dipping	Lucy Gao (UBC); Chen	1:31:46	▶
78	Structured Variational Autoencoders	Lähdesmäki (Aalto); Dumitrescu & Korpela	1:44:19	▶

A.5 MLSB Workshop — 10 talks (complete)

Channel: [@MLSBWorkshop](#)

Flag: only the 2025 edition is on this channel. MLSB has run since 2020 but earlier editions were hosted elsewhere and the recordings are not public. These ten are the whole archive.

#	Talk	Speaker	Len	Link
1	Opening Remarks	Gina El Nesr (Stanford)	4:21	▶
2	Automating Scientific Discovery	Michael Pieler (OpenBioML)	29:12	▶
3	Exploring Novel Molecular Space	Bruno Correia (EPFL)	31:00	▶
4	Ensemble-conditioned protein sequence design with Caliby	Richard Shuai (Stanford)	13:10	▶
5	Big Data for Protein Dynamics	Hannah Wayment-Steele (UW-Madison)	33:47	▶
6	Boltzmann Weighted Ensembles	Pratyush Tiwary (Maryland)	31:12	▶
7	ML for Protein Structure Determination	Ellen Zhong (Princeton)	28:53	▶
8	Biomolecular Structure Predictions and Alternative Conformations	Karson Chrispens (UCSF)	13:30	▶
9	AI-Driven RNA Structure Prediction	Mile Šikić (A*STAR)	24:37	▶
10	Critical Assessment of Structure Prediction	John Moult & Krzysztof Fidelis (CASP)	36:17	▶

Talk #10 is CASP's founders describing what CASP is for. Watch it before Atlas S. Talk #6, Tiwary on Boltzmann-weighted ensembles, is the cleanest single statement of the physics-first objection to generative structure models: they produce *samples*, not samples *from the right distribution*.

A.6 MLCB Conference — 82 talks

Channel: [@mlcbconf](#)

MLCB is where computational-biology ML gets presented to an audience that will push back on the statistics. Several titles below are genuinely truncated with “...” by the uploader — that is the real title on YouTube.

MLCB 2024 — the protein and genomics core

#	Talk	Speaker	Len	Link
1	Introduction	Sara Mostafavi (UW)	5:24	▶
2	Fine-tuning on Personal Genomes Improves Expression...	Shiron Drusinsky (UCSF)	19:53	▶
3	Approximating mutual information of high-dimensional...	Gokul Gowri (Harvard)	23:27	▶
4	Predicting cell cycle stage from 3D single-cell nuclear...	Gang Li (UW)	13:13	▶
5	Regulatory sequence analysis of RNA-seq via parameter-efficient...	Han Yuan (Calico)	19:21	▶
6	RegDiffusion: Fast Gene Regulatory Networks using Diffusion...	Hao Zhu (Tufts)	17:36	▶
7	Decomposing protein and genomic language models to understand...	Sergey Ovchinnikov (MIT)	56:09	▶
8	RNA-FrameFlow: Flow Matching for RNA backbone generation	Rishabh Anand (NUS)	23:44	▶
9	Computational design of target-specific linear...	Haowen Zhao (Imperial)	10:10	▶
10	A deep learning model of tumor cell architecture	Akshat Singhal (UCSD)	19:02	▶
11	Spotlights 2	—	19:34	▶
12	Towards biophysically interpretable sequence...	Maria Chikina (Pittsburgh)	51:41	▶
13	Panel — Foundation models for biology, when are they useful?	—	51:23	▶
14	The Nucleotide Transformer	Thomas Pierrot (InstaDeep)	19:05	▶
15	Personal genome interpretation with sequence-to-function models	Nilah Ioannidis (Berkeley)	51:35	▶
16	Cross-modality Matching and Prediction of Perturbation Responses	Romain Lopez (Genentech)	25:26	▶
17	Sequence basis of transcription initiation in the human...	Jian Zhou (UTSW)	47:59	▶

#	Talk	Speaker	Len	Link
18	Spotlights 1	—	25:05	▶
19	IsoCLR: Contrastive learning for RNA foundation models	Philip Fradkin (Vector)	20:33	▶
20	Significant Latent Factor Interaction Discovery...	Jishnu Das (Pittsburgh)	20:30	▶
21	Bridging biophysics & AI for generative protein design	Ava Amini (MIT / MSR)	44:31	▶
22	DART-Eval: a comprehensive benchmark for DNA language models	Wang, Patel & Singhal (Stanford)	19:36	▶
23	Designing DNA with Tunable Regulatory...	Anirban Sarkar (CSHL)	20:08	▶
24	Multi-marginal Schrödinger Bridges with Iterative Reference Refinement	Yunyi Shen (MIT)	18:10	▶

MLCB 2023 — selected full talks

#	Talk	Speaker	Len	Link
25	ViDa: Visualizing DNA hybridization trajectories	Chenwei Zhang	15:08	▶
26	A Deeper Dive into Single Cell RNA Sequencing Foundation Models	Rebecca Boiarsky (MIT)	16:00	▶
27	Panel — Bridging Academia and Industry	—	1:07:46	▶
28	Probabilistic modeling of dynamics in the tumor microenvironment	Elham Azizi (Columbia)	47:20	▶
29	Towards morphogenomics	Bianca Dumitrascu	45:21	▶
30	Evolutionary Scale Language Models	Alexander Rives (EvolutionaryScale)	54:32	▶
31	Beyond the training set: detecting distribution shift	Farhan Damani	24:14	▶
32	Sequential Optimal Experimental Design of Perturbation Screens	Kexin Huang (Stanford)	23:09	▶
33	MLCB Intro Day 1	—	8:33	▶
34	TCR BERT: learning the grammar of T cell receptors	Kevin Wu (Stanford)	21:23	▶
35	Protein Discovery with Discrete Walk-Jump Sampling	Nathan Frey (Prescient)	16:53	▶
36	Genomic language model predicts protein co-regulation and function	Yunha Hwang (Harvard)	19:52	▶
37	Assessing the limits of zero-shot foundation models in single cell biology	Kasia Kedzierska (Oxford)	20:02	▶
38	Random Subsampling Meets Confounders	Samuel Alber	18:06	▶
39	A Simple But Hard to Beat Foundation Model for Genes and Cells via ChatGPT	Yiqun Chen	22:59	▶
40	Multi-modal biophysical CNN for discovering cis-regulatory code	Alexander Sasse (UW)	21:13	▶
41	Interpreting Cis-Regulatory Interactions	Shushan Toneyan (CSHL)	19:04	▶

#	Talk	Speaker	Len	Link
	from Large-Scale Deep NNs			
42	Modeling cell-type-specific chromatin folding through differentiable loop extrusion	Tina Subic	20:40	▶
43	DragonNNFruit: Dissecting cis and trans regulation of chromatin accessibility	Jacob Schreiber (Stanford)	19:19	▶
44	Cross-cell-type regulatory predictions using representation learning	Nauman Javed	16:06	▶
45	Deciphering cell-cell communication with deep learning	Yutong Wang	16:14	▶
46	Investigating RNA splicing as a source of cellular diversity	Karin Isaev	17:57	▶
47	A Masked Image Modeling Approach to CyCIF Panel Reduction	Zachary Sims (OHSU)	17:50	▶
48	Pseudotime analysis for time-series single-cell data	Gang Li (UW)	15:46	▶
49	Decoding the Genetic Code of Neuronal Type Connectivity	Mu Qiao	16:42	▶

MLCB 2023 also posted 20 one-minute poster flashes (IDs on the channel); they are too short to schedule but useful for scanning what a cohort is working on.

MLCB 2019 — the early-method talks that aged well

#	Talk	Speaker	Len	Link
50	Intro and travel awards	—	3:56	▶
51	Deep Multiple Instance Learning for Taxonomic Classification	Vincent Fortuin (ETH)	13:11	▶
52	Cross-species Transcription Factor Binding Prediction	Kelly Cochran	12:20	▶
53	Deep exploration networks for engineering functional DNA sequences	Johannes Linder (UW)	13:24	▶
54	Efficient variance components analysis across millions of genomes	Sriram Sankararaman (UCLA)	13:34	▶
55	Learning, using, and extending variational distributions for phylogenetics	Erik Matsen (Fred Hutch)	14:07	▶
56	Energy-based models for atomic-resolution protein conformations	Yilun Du (MIT)	12:00	▶
57	Protein Fold Recognition with Recurrent Kernel Networks	Dexiong Chen (Inria)	11:54	▶
58	Learning unsupervised feature representations for single-cell microscopy	Alex Lu (Toronto)	14:14	▶
59	Reconstructing continuous distributions of 3D protein structure (cryoDRGN)	Ellen Zhong (MIT)	15:44	▶
60	Keynote — ML methods for big genomic and proteomic data	William Noble (UW)	48:28	▶

Entry #59 is cryoDRGN announced in fifteen minutes in 2019. Compare it to Ellen Zhong's 2025 MIA talk (A.4 #34) to see six years of a method maturing, and note how much of the 2019 talk is already right.

A.7 Institute for Protein Design — 36 videos

Flag: the [@UWproteindesign](#) channel has only **seven public videos**. The real IPD teaching corpus is embedded on ipd.uw.edu/learn and is pulled from several different channels, which is why it does not surface in a channel listing. All 29 embeds on that page are enumerated below.

The channel itself

#	Talk	Len	Link
1	David Baker explains his Nobel Prize research on protein design	8:52	▶
2	David Baker Nobel Prize press conference (full)	26:33	▶
3	PROTEIN_GENERATOR: Unleashing AI for Protein Design	4:54	▶
4	Coronavirus Research Update from Rosetta@Home	1:42	▶
5	AlphaFold: improved structure prediction using potentials from deep learning	1:02:40	▶
6	Our Audacious Project: Launching the Protein Design Revolution	0:57	▶
7	What is the Institute for Protein Design?	3:30	▶

#5 is AlphaFold 1 presented at UW in August 2019, before CASP14. Watch it immediately before the BPDMC post-CASP14 reaction (A.2 #49) — eighteen months separate them and the field changed completely in between.

Introductory and IPD TV

#	Talk	Speaker	Len	Link
8	5 challenges we could solve by designing new proteins	David Baker (TED)	—	▶
9	The protein folding revolution	—	3:44	▶
10	Design of novel protein nanomaterials for structure-based vaccine design	—	55:46	▶
11	UW Discoveries — Protein Design	—	3:10	▶
12	Part 1: Introduction to Protein Design	David Baker (iBiology)	21:22	▶
13	IPD TV Ep. 1 — Using Design Tracker	Lauren Carter	11:57	▶
14	IPD TV Ep. 2 — Rosetta 101: Basic Concepts in Rosetta Core Functionality	—	58:11	▶
15	IPD TV Ep. 3 — Rosetta 201: Introduction to Rosetta Development	Vikram Mulligan	1:15:08	▶
16	Compute Support at the IPD	Darwin Alonso	3:18	▶

Rosetta Bootcamp, Seattle, January 2016 — 8 videos

#	Talk	Speaker	Len	Link
17	Lecture 02 — Library Structure 2	Hahnbeom Park	49:38	▶
18	Lecture 03 — C++ Classes #2	Vikram Mulligan	45:38	▶
19	Lecture 04 — Graphs	Alex Ford	1:05:42	▶
20	Lecture 05 — Library Structure 4: Simple Protocols	Tom Linsky	1:13:07	▶
21	Lecture 06 — Library Structure 6	Frank DiMaio	1:13:00	▶
22	Lecture 07 — Adding Code to Rosetta	Vikram Mulligan	59:12	▶
23	Lecture 08 — Library Structure 5	Daniel Silva Adriano	48:19	▶
24	Lecture 09 — Library Structure 7: Miscellaneous Protocols	Indigo Chris King	29:43	▶

Lecture 01 is missing from the playlist and from the page — it does not exist publicly.

Intro to Protein Structure Prediction and Design — 12 lectures

This is the classical course referenced throughout Part I and Atlas L. It predates deep learning entirely, which is exactly why it is worth your time: it teaches the energy function, the sampling problem and the packing problem without a neural network anywhere, so you learn what the networks replaced.

#	Lecture	Len	Link
25	Intro to Protein Structure Prediction and Design, incl. review of protein structure	41:06	▶
26	Protein Geometry; Molecular Energies and Forces (part 1)	1:09:31	▶
27	Molecular Energies and Forces (part 2)	58:49	▶
28	Molecular Energies and Forces (part 3)	30:00	▶
29	Ab-Initio Protein Structure Prediction (part 1)	1:11:07	▶
30	Ab-Initio Protein Structure Prediction (part 2)	33:23	▶
31	Refinement of Protein Structures	1:06:11	▶
32	Rotamer Libraries and Side-chain Packing — guest: Brian Weitzner	1:05:04	▶
33	Protein-Protein Docking	54:42	▶
34	Loop Modeling	50:46	▶
35	Non-Protein Molecules in Rosetta — guest: Jason Labonte	44:22	▶
36	Modeling Membrane Protein Structure — guest: Julia Koehler Leman	45:06	▶

Lecture 32 is the single most under-watched hour in this entire curriculum. Side-chain packing is NP-hard, Rosetta solves it with simulated annealing over a discrete rotamer library, and ProteinMPNN replaced that entire machinery with a forward pass. You cannot evaluate what was gained or lost without first understanding what was there — which is **derivation checkpoint 20**.

A.8 Valence Labs / Portal — 110 structure-and-simulation talks of 398

Flag: the handle @ValenceLabs is a dead legacy channel with zero videos. The live channel is [@valence_labs](#), which has 398 videos. The 110 below are the protein, biomolecular-structure, molecular-simulation and molecular-generative talks; the remaining ~288 are pure graph-ML theory, causal representation learning, single-cell omics or general diffusion theory.

Valence runs the **longest** talks in the field — many over 90 minutes, with extended Q&A. When you want the full technical treatment of a method rather than a conference-length summary, check here first.

Structure prediction, design and biomolecular foundation models

#	Talk	Speaker	Len	Link
1	Generative Protein Design at Scale: Structural Space to High-Affinity Binders	—	49:26	▶
2	Adaptive Protein Tokenization	Rohit Dilip (Caltech)	1:00:04	▶
3	General Multimodal Protein Design Enables DNA-Encoding of Chemistry	—	1:22:42	▶
4	Scaling Atomistic Protein Binder Design with Pretraining and Test-time Compute	NVIDIA	1:38:57	▶
5	SeedFold: Scaling Biomolecular Structure Prediction	Zhou & Lu (ByteDance Seed)	59:59	▶
6	SLAE: Strictly Local All-atom Environment for Protein Representation	Yilin Chen	1:04:01	▶
7	Efficient generation of epitope-targeted de novo antibodies (Germinal)	—	1:09:18	▶
8	Protein Hunter: exploiting structure hallucination within diffusion	Yehlin Cho (MIT)	1:14:12	▶
9	Accelerating Biomolecular Modeling with AtomWorks and RF3	Nathaniel Corley (UW/IPD)	52:10	▶
10	ATOMICA: Learning Universal Representations of Intermolecular Interactions	Ada Fang (Harvard)	52:00	▶
11	Boltz-2: Towards Accurate and Efficient Binding Affinity Prediction	—	59:27	▶
12	Atom level enzyme active site scaffolding using RFdiffusion2	Yim (MIT) & Ahern (UW)	1:12:41	▶
13	Inverse problems with experiment-guided AlphaFold	Vedula & Sellam (Technion)	47:55	▶
14	Targeting protein–ligand neosurfaces with a	Anthony Marchand (EPFL)	1:11:03	▶

#	Talk	Speaker	Len	Link
	generalizable deep learning tool			
15	PepTune: Multi-Objective-Guided Discrete Diffusion for therapeutic peptides	—	1:12:13	▶
16	BindCraft: one-shot design of functional protein binders	Martin Pacesa (EPFL)	1:36:09	▶
17	CHEAP: Tokenized and Continuous Embedding Compressions	Amy X. Lu (Berkeley)	1:27:22	▶
18	Generalized Biomolecular Modeling and Design with RoseTTAFold All-Atom	Rohith Krishna (UW/IPD)	1:04:54	▶
19	Towards Rational Drug Design with AlphaFold 3	Max Jaderberg (Isomorphic)	50:28	▶
20	OpenFold: Lessons Learned From Rebuilding and Retraining AlphaFold2	—	39:28	▶
21	AlphaFold2, OpenFold, Protein Language Models and Beyond	Nazim Bouatta (HMS)	1:01:26	▶
22	The Continuous Language of Protein Structure	Ben Murrell (Karolinska)	55:56	▶
23	Multiflow: protein structure and sequence co-generation	Yim (MIT) & Campbell (Oxford)	54:54	▶
24	De novo design of high-affinity binders of bioactive helical peptides	Vázquez Torres (UW/IPD)	37:28	▶
25	SE(3)-Stochastic Flow Matching for Protein Backbone Generation (FoldFlow)	Akhound-Sadegh & Bose (Mila)	1:11:54	▶
26	Diffusion probabilistic modelling of protein backbones in 3D	Yim (MIT) & Trippe (Columbia)	1:02:39	▶
27	Protein Discovery with Discrete Walk-Jump Sampling	Nathan Frey (Prescient)	55:11	▶
28	Multimodal Deep Learning for Protein Engineering	Kevin K. Yang (MSR)	1:02:31	▶

#	Talk	Speaker	Len	Link
29	Protein Representation Learning by Geometric Structure Pretraining	Zuobai Zhang (Mila)	53:20	▶
30	Structure-Aware Protein Self-Supervised Learning	Can Chen (Mila)	30:20	▶
31	Geometric Deep Learning for Protein Understanding	Jian Tang (Mila/HEC)	45:55	▶
32	Structure-Independent Peptide Binder Design via Generative LMs	Pranam Chatterjee (Duke/UPenn)	1:00:55	▶
33	Iterative Refinement GNN for Antibody Sequence-Structure Co-design	Wengong Jin (MIT)	1:27:05	▶
34	Structured Refinement Network for Antibody Design	Wengong Jin (Broad)	58:36	▶
35	Contextual AI Models for Single-cell Protein Biology (PINNACLE)	Michelle Li (Harvard)	32:00	▶

Simulation, force fields, coarse-graining and sampling

This block is the single best collection of ML-meets-molecular-simulation talks anywhere, and it is the direct video support for Atlas families A, B, D, E and F. Nothing else in the curriculum covers neural network potentials this thoroughly.

#	Talk	Speaker	Len	Link
36	AquaGen: Scaling generative models to MD precision on thousands of atoms	—	1:16:27	▶
37	Introducing AquaGen	—	1:08	▶
38	AquaGen — Simulating the Physics of Water	—	2:18	▶
39	Consistent Sampling and Simulation: MD with Energy-Based Diffusion Models	—	43:48	▶
40	Transformers Discover Molecular Structure Without Graph Priors	Tobias Kreiman (Berkeley)	1:04:09	▶
41	Action-Minimization Meets Generative Modeling: Transition Path Sampling	Sanjeev Raja (Berkeley)	1:04:36	▶
42	Orb-v3: atomistic simulation at scale	Duignan & Vandenhoute (Orbital)	1:13:35	▶
43	FEAT: Free energy Estimators with Adaptive Transport	Du (Cornell) & He (Cambridge)	1:00:59	▶
44	Improving Speed and Accuracy of NN Interatomic Potentials Across Chemical Domains	—	54:24	▶
45	Scalable Emulation of Protein Equilibrium Ensembles (BioEmu)	—	1:17:02	▶
46	Learning Equivariant Non-Local Electron Density Functionals	Nicholas Gao (TUM)	48:18	▶
47	Stability-Aware Boltzmann Estimator (StABLE) Training of NN Potentials	Sanjeev Raja	1:22:10	▶
48	Diffusion Models on Sampling Rare Events	Chenru Duan (Deep Principle)	48:08	▶
49	AlphaFold Meets Flow Matching for Generating Protein Ensembles (AlphaFlow)	Bowen Jing (MIT)	56:30	▶
50	MOFDiff: Coarse-grained Diffusion for Metal-Organic Framework Design	Xiang Fu (MIT)	1:13:36	▶

#	Talk	Speaker	Len	Link
51	Timewarp: Transferable Acceleration of MD by Time-Coarsened Dynamics	Klein & Foong	54:35	▶
52	Timewarp (extended session)	—	1:14:35	▶
53	Designing Losses for Data-Free Training of Normalizing Flows on Boltzmann Distributions	—	1:07:25	▶
54	Machine Learning for Multi-Scale Molecular Simulation and Design	Xiang Fu (MIT)	1:01:03	▶
55	PepFlow: Direct Conformational Sampling From Peptide Energy Landscapes	Osama Abdin (Toronto)	55:42	▶
56	Differentiable Simulations for Enhanced Sampling of Rare Events	Martin Šípka	41:28	▶
57	Differentiable Simulations for Enhanced Sampling of Rare Events	Gómez-Bombarelli (MIT)	1:01:15	▶
58	Rigid Body Flows for Sampling Molecular Crystal Structures	Jonas Köhler (FU Berlin)	1:24:24	▶
59	Learning Local Equivariant Representations for Large-Scale Atomistic Dynamics (Allegro)	Albert Musaelian (Harvard)	1:09:17	▶
60	Local Equivariant Representations (earlier session)	Batzner & Musaelian (Harvard)	1:26:43	▶
61	MACE: Higher Order Equivariant Message Passing NNs for Force Fields	—	1:22:55	▶
62	Learning QM Using Embedding, Symmetric Polynomials and Composition	Gábor Csányi (Cambridge)	38:34	▶
63	Benchmark and Critical Evaluation for ML Force Fields with Simulations	Xiang Fu (MIT)	1:00:38	▶
64	Systematic Analysis of Biomolecular	Martin Vögele (Stanford)	50:33	▶

#	Talk	Speaker	Len	Link
	Conformational Ensembles with PENSEA			
65	Accelerating Cryptic Pocket Discovery With Deep Learning (PocketMiner)	Greg Bowman (UPenn)	37:23	▶
66	Accelerating Cryptic Pocket Discovery Using AlphaFold and Markov State Modelling	—	31:32	▶
67	Chemically Transferable Generative Backmapping of Coarse-Grained Proteins	Soojung Yang (MIT)	1:06:07	▶
68	Generative Coarse-Graining of Molecular Conformations	Wujie Wang (MIT)	1:53:41	▶
69	Simulate Time-integrated Coarse-grained MD with Geometric ML	Xiang Fu (MIT)	1:40:25	▶
70	Converging Advances to Accelerate Molecular Simulation	Max Welling (Amsterdam/MSR)	1:14:04	▶
71	ProtMD: Conformation Flexibility for Drug Binding via MD Pretraining	—	43:51	▶
72	Scalable molecular simulation of electrolytes with quantum chemical accuracy	Tim Duignan	1:12:12	▶

Docking, chemistry and generative molecular design

#	Talk	Speaker	Len	Link
73	PoseBusters: AI-based docking methods fail to generate physically valid poses	Martin Buttenschoen (Oxford)	34:28	▶
74	DiffDock: Diffusion Steps, Twists, and Turns for Molecular Docking	—	56:43	▶
75	Equivariant Scalar Fields for Molecular Docking with FFTs	Bowen Jing (MIT)	38:02	▶
76	Harmonic Self-Conditioned Flow Matching for Multi-Ligand Docking	Hannes Stärk (MIT)	59:47	▶
77	Torsional Diffusion for Molecular Conformer Generation	Corso & Jing (MIT)	1:36:42	▶
78	SynCoGen: Synthesizable 3D Molecule Generation	—	1:12:12	▶
79	Equivariant 3D-Conditional Diffusion for Molecular Linker Design (DiffLinker)	Ilia Igashov (EPFL)	1:05:34	▶
80	Integrating structure-based and ligand-based modeling for drug design	Joseph Paggi (Stanford)	51:26	▶
81	Fragment-Based Hit Discovery via Unsupervised Learning of Fragment-Protein Complexes	—	53:50	▶
82	Pre-trained Ensembles for Bayesian Optimization of Protein Sequences	Ziyue Yang	49:01	▶
83	Data-Efficient Graph Grammar Learning for Molecular Generation	Minghao Guo (MIT)	58:01	▶
84	Learning 3D Representations of Molecular Chirality	Keir Adams (MIT)	1:09:59	▶
85	ChiENN: Embracing Molecular Chirality with Graph Neural Networks	Piotr Gaiński (Jagiellonian)	50:33	▶
86	Interpretable Chirality-Aware GNNs for QSAR Modeling	Yunchao Liu (Vanderbilt)	49:17	▶

#	Talk	Speaker	Len	Link
87	Quantum Machine-Learning for Drug-like Molecules	Clemens Isert (ETH)	55:25	▶
88	Examining GNNs for Crystal Structures: Limitations and Periodicity	—	1:00:19	▶
89	Is Distance Matrix Enough for Geometric Deep Learning?	Zian Li	1:04:15	▶
90	Scalable Geometric Deep Learning on Molecular Graphs	Nathan Frey (MIT LL)	59:05	▶
91	Geometric Deep Learning for Drug Discovery	Jian Tang (Mila)	1:03:53	▶
92	How GFlowNets Explore Chemical Space to Find Novel Molecules	—	5:00	▶
93	Introducing SAFE: A New Framework for Molecular Design	—	1:17	▶

Nucleic acids and genome-scale

#	Talk	Speaker	Len	Link
94	gRNAde: Geometric Deep Learning for 3D RNA inverse design	Chaitanya Joshi (Cambridge)	1:06:02	▶
95	Beyond Sequence: Geometric Context for RNA Property Prediction	—	53:32	▶
96	A long-context RNA foundation model for transcriptome architecture	Ali Saberi	1:29:27	▶
97	Geometric deep learning framework for de novo genome assembly	Lovro Vršek	1:15:31	▶

ML4DD Summer School 2024 — 13 sessions

A complete taught course inside the channel. **Day 2 is the coarse-graining and simulation day** and is the most efficient single day of lectures in the Atlas for families B, D and F.

#	Session	Speaker	Len	Link
98	Day 1 — Learning Geometry & 3D Symmetries	Mario Geiger (NVIDIA)	45:18	▶
99	Day 2 — Learning ML Interatomic Potentials	Gianni De Fabritiis (UPF)	51:39	▶
100	Day 2 — Accelerate Atomistic Simulations, Sampling, and Dynamics	Pratyush Tiwary (UMD)	1:01:06	▶
101	Day 2 — Coarse Grained Biological Systems	Jacopo Venturin	57:04	▶
102	Day 2 — ML in Structure Based Drug Discovery	Gabriele Corso (MIT)	57:33	▶
103	Day 3 — Harnessing Geometric ML for Molecular Design	Michael Bronstein (Oxford)	1:08:23	▶
104	Day 3 — Generative Models of Molecular Structures	Camille Bilodeau (UVA)	54:54	▶
105	Day 3 — Synthesizability & Molecular Synthesis	Connor Coley (MIT)	1:04:22	▶
106	Day 5 — Protein Folding & Design	Alex Tong (Mila)	1:05:32	▶
107	Lab 1 — Virtual Screening	—	1:22:40	▶
108	Lab 1 — Virtual Screening (explanation)	—	26:29	▶
109	Lab 2 — Binding Affinity Prediction with ML-Based Docking	—	1:09:46	▶
110	Lab 2 — Binding Affinity Prediction (explanation)	—	26:09	▶

Also on the channel, just outside the structure filter: Molecular Synthesizability and Synthetic Tree Generation (Wenhao Gao), Bayesian Flow Networks (**Alex Graves** — worth it for Part IV), HyenaDNA (Eric Nguyen), Evo 2 (Garyk Brixi, two recordings), Branched Schrödinger Bridge Matching (Tang & Chatterjee), UMA: A Family of Universal Models for Atoms (Brandon Wood), and **Does equivariance matter at scale?** (Johann Brehmer) — that last one is a direct challenge to Atlas O's equivariance orthodoxy and belongs with D.5.

A.9 What the gaps tell you

Enumerating these archives surfaced absences that are more informative than any individual entry. Four are worth stating plainly.

1. The pre-2021 seminar record is simply gone. BPDMC's 2018 and 2019 archives list 28 talks with speakers, titles and dates, and not one was recorded. Seven more from 2020 and two from 2021 are likewise text-only. The field's institutional memory from before the pandemic normalized recording is a list of titles. If you ever run a seminar, record it.

2. Some of the most-cited people give the fewest talks. Tanja Kortemme, one of the ten scientists you named, has almost no recorded talks anywhere; her ideas reach you through papers and through students like Anum Glasgow (A.2 #58). Phil Bradley has no first-author talk on TCRdock despite TCR-pMHC modeling being a field of its own. **Visibility on video and influence in the literature are weakly correlated, and treating a talk archive as a map of the field would systematically mislead you.** The Atlas corrects for this by pairing papers to talks rather than organizing around talks alone.

3. Nine design methods in Atlas R have no talk at all — ProteinSolver, PiFold, ProstT5, ProteinSGM, BoltzGen (until a 2026 workflow session), PocketGen, AlphaProteo, Kortemme’s biosensors, and the classical theozyme/RosettaMatch line. Several are from industry labs (DeepMind’s AlphaProteo) and several are from groups that publish without presenting. The pattern is real: **the seminar circuit over-represents academic groups with strong US presences and under-represents industry and non-US work.**

4. The negative results are the least-recorded of all. Blomberg et al. 2013 — which showed the famous Kemp eliminase’s catalysis came from directed evolution rather than design — has no talk. Neither does the Kortemme biosensor work. Meanwhile every advocacy result has three recordings across three venues. **This asymmetry is a property of how science is communicated, not of what is true,** and compensating for it is the single most useful habit this curriculum can give you. When a method has three enthusiastic talks and no critical one, that is not evidence the method is uncontested. It is evidence that criticism does not get invited.

Appendix B — The Enumerated Courses

The second edition named ten courses as important and left every one of them as a bare playlist link — the exact failure it criticized elsewhere. This appendix is the repair. Every lecture, with its real title, runtime and link.

Three things worth knowing before the tables.

First, a discovery that resolves a long-standing confusion. *Jeffrey Gray’s JHU course and the “IPD lecture series” enumerated in Appendix A.7 are the same twelve recordings.* Same video IDs, same guest lecturers, same order. They are cross-posted, and anyone who has seen both lists has been looking at one course twice. **Those twelve lectures are the whole published set** — graylab.jhu.edu’s own video page confirms the rest of the fourteen-week semester was never recorded.

Second, several of these courses are incomplete and the gaps are listed. Lindahl has re-recorded his course five times and only the 2016 edition is complete; the missing lectures in 2017, 2019 and 2020 are named below rather than papered over.

Third, two counts that look wrong are correct. The Chodera workshop really has exactly 43 videos — and its real title is *Workshop on Free Energy, Kinetics, and MSM in Drug Design 2018*, not anything to do with Alchemistry. The PLUMED Masterclass stops at 2022; there is no 23.x series.

B.1 Jeffrey Gray — Computational Protein Structure Prediction and Design (JHU 540.414/614)

Channel @ProfessorJJGray · playlist PLHn7WmALbthnAwbJ4mWw5gk8dgqsjRL87 . **Twelve lectures. This is the classical curriculum — no deep learning anywhere — which is exactly its value.** It teaches the energy function, the sampling problem and the packing problem so that you can say what the networks in Atlas P and R actually replaced.

#	Lecture	Len	Link
1	Intro to Protein Structure Prediction and Design, incl. review of protein structure	41:07	▶
2	Protein Geometry; Molecular Energies and Forces (part 1)	1:09:32	▶
3	Molecular Energies and Forces (part 2)	58:50	▶
4	Molecular Energies and Forces (part 3)	30:01	▶
5	Ab-Initio Protein Structure Prediction (part 1)	1:11:08	▶
6	Ab-Initio Protein Structure Prediction (part 2)	33:23	▶
7	Refinement of Protein Structures	1:06:12	▶
8	Rotamer Libraries and Side-chain Packing — guest: Brian Weitzner	1:05:04	▶
9	Protein-Protein Docking	54:43	▶
10	Loop Modeling	50:47	▶
11	Non-Protein Molecules in Rosetta — guest: Jason Labonte	44:23	▶
12	Modeling Membrane Protein Structure — guest: Julia Koehler Leman	45:06	▶

Two alternate cuts exist on the channel but are not in the playlist: *Lecture 2: Protein Geometry* (1:09:28, ▶) and *Lecture 4: Molecular Energies and Forces part 3* (30:05, ▶). They are near-duplicates.

Gray's research seminars, which are the modern counterpart to the classical course:

Talk	Len	Link
How Deep Learning is Changing Protein Structure Prediction and Design	1:10:38	▶
Artificial Intelligence Tools for Antibody Engineering and Protein Docking	1:01:09	▶
How Proteins Dance with Their Partners	1:03:10	▶
Antibody language models vs. biology; docking diffusion models vs. physics (BPDMC 2026)	1:25:10	▶

On the DL4Proteins notebooks (github.com/Graylab/DL4Proteins-notebooks, arXiv:2511.02128): there are **no video recordings of this material at all** — a YouTube search returns zero results. It is ten Jupyter chapters, and its own README points learners at Karpathy and Veličković rather than at its own lectures. Work through the notebooks; the videos do not exist. Chapters: NumPy networks, PyTorch networks, CNNs,

language models for Shakespeare and proteins, embeddings and transfer learning, AlphaFold, GNNs for proteins, DDPMs, the RFdiffusion→ProteinMPNN→AlphaFold pipeline, and RFdiffusion All-Atom.

B.2 Erik Lindahl — Biophysical Chemistry / Molecular Biophysics

Channel [@eriklindahl](#), 399 videos. **Five full-length editions exist and only 2016 is complete.** There is no separate molecular dynamics or cryo-EM course — MD is lectures 5–7 of the main course, and cryo-EM appears only as isolated concept clips.

The 2016 edition — the one to watch. Thirteen lectures, ~30 hours.

#	Len	Link	#	Len	Link
1	2:15:32	▶	8	2:13:43	▶
2	2:30:14	▶	9	2:36:32	▶
3	2:12:42	▶	10	1:34:48	▶
4	1:43:59	▶	11	2:34:10	▶
5	2:19:33	▶	12	2:19:44	▶
6	2:27:44	▶	13	2:34:15	▶
7	2:05:21	▶			

Lecture 7's YouTube title says "2006" — that is a typo by the uploader; it is the 2016 lecture 7.

The 2018 edition is the best alternative, missing only lecture 6 and the main lecture 14: L1 ▶ · L2 ▶ · L3 ▶ · L4 ▶ · L5 ▶ · L7 ▶ · L8 ▶ · L9 ▶ · L10 ▶ · L11 ▶ · L12 ▶ · L13 ▶ · L14 follow-up ▶

Incomplete editions, listed so you do not waste time discovering the gaps. 2017 is missing lectures 5, 8, 9, 11 and 13; 2019 is missing 6–11; 2020 is missing 1 and 6–11.

The current "Molecular Biophysics" concept edition is what Lindahl now maintains: thirteen lectures delivered as 339 clips of one to eight minutes each, plus a lipid-protein special topic. It is complete and it is the best version for looking something up rather than sitting through a lecture.

#	Topic	Clips	Playlist
1	From life to molecular biophysics	21	▶
2	Modeling molecules with computers	19	▶
3	Interactions and flexibility in amino acids	16	▶
4	Hydrogen bond formation in a protein	22	▶
5	Amino acids are chiral zwitterions	26	▶
6	From static structures to simulations (MD and MSMs)	24	▶
7	Fibrous, globular and membrane proteins	31	▶
8	The lipid bilayer	22	▶
9	Sequence space and folds	26	▶
10	Denaturation, molten globule, folding kinetics	23	▶
11	Bioinformatics, sequencing, AlphaFold	27	▶
12	Drug discovery, docking, QSAR	30	▶
13	Free energy calculations, protein design, biologicals	11	▶
ST	Lipid-protein interactions (special topic)	41	▶

Full-length companions: *course overview and introduction* (1:13:06, [▶](#)), *complete lecture 2* (1:28:11, [▶](#)), *Structure and Dynamics of Biological Membranes* (2:07:23, [▶](#)).

B.3 PLUMED Masterclass — 43 videos

Channel: **Plumed Org**. Every class is two parts: Lecture I is theory, Lecture II is an instructor working the exercises live. **This is the best hands-on enhanced-sampling resource that exists, and it is free.**

Class	Topic (instructor)	Part I	Part II
21.1	PLUMED syntax and analysis (Bonomi)	▶ 52:46	▶ 1:52:57
21.2	Statistical errors in MD (Tribello)	▶ 42:11	▶ 1:26:41
21.3	Umbrella sampling (Bussi)	▶ 1:04:16	▶ 1:56:06
21.4	Metadynamics (Bonomi)	▶ 45:31	▶ 2:01:46
21.5	Replica exchange methods (Bussi)	▶ 1:01:41	▶ 1:25:51
21.6	Dimensionality reduction (Tribello)	▶ 1:14:19	▶ 1:03:38
21.7	Optimizing PLUMED performance (Bonomi)	▶ 40:09	▶ 1:49:26
22.0	Introduction to Masterclass 22	▶ 3:02	—
22.1	Funnel metadynamics (Limongelli, Raniolo)	▶ 1:19:45	▶ 2:16:36
22.2	Metadynminer output analysis (Spiwok)	▶ 1:01:19	▶ 1:49:55
22.3	OPES (Invernizzi)	▶ 58:17	▶ 1:36:34
22.5	ML collective variables with PyTorch (Bonati)	▶ 1:08:01	▶ 1:26:05
22.6	Experiment-directed simulation (Hocky, White)	▶ 48:25	▶ 42:07
22.7	Information bottleneck / automated sampling (Tiwary)	▶ 54:53	▶ 53:48
22.8	Concentration-driven processes (Salvalaglio)	▶ 50:46	▶ 1:05:06
22.9	Path collective variables (Ensing)	▶ 1:37:40	▶ 2:08:49
22.10	Hamiltonian replica exchange with GROMACS (Bussi)	▶ 55:09	▶ 1:13:23
22.11	Variationally enhanced sampling (Valsson)	▶ 54:44	▶ 1:10:12
22.12	Free energy in crystalline solids (Piaggi)	▶ 54:38	▶ 45:49
22.13	SASA module / implicit solvent (Arsiccio)	▶ 53:27	▶ 55:41
22.15	Mechanical pulling and FISST (Hocky, Stirneman)	▶ 26:09	▶ 50:22
22.17	Bayesian cryo-EM integration into MD (Bonomi, Hoff)	▶ 46:56	▶ 42:50

Two warnings from the verification pass. Masterclass 21.8 was a poster session and was never recorded. [plumed.org](#) links the wrong video for **22.03 Lecture I** — it points at 22-2.1; the correct one is in the table

above. Classes 22.04, 22.14 and 22.16 appear on neither the site nor the playlist.

B.4 Workshop on Free Energy, Kinetics and MSM in Drug Design, 2018 — 43 videos

Channel: **Chodera lab**. Not called “Alchemistry” — that is the wiki, not the workshop. Forty talks plus two welcomes and a session stream. **This single workshop fills most of Atlas E and a good part of Atlas D**, and several of the scientists this curriculum is built around speak here.

Speaker	Len	Link	Speaker	Len	Link
Mark Murcko (keynote)	31:11	▶	Michael Shirts	29:49	▶
Frank Noé	37:32	▶	Lucie Delemotte	31:39	▶
Pratyush Tiwary	27:42	▶	Greg Ross	18:03	▶
Dave Huggins	25:39	▶	Marie Laury	15:27	▶
David Koes	30:33	▶	Cecilia Clementi	27:30	▶
Benoit Roux	33:43	▶	Israel Cabeza de Vaca	22:51	▶
Christina Schindler	28:01	▶	Muneeb Sultan	16:04	▶
Germano Heinzemann	30:27	▶	Dan Zuckerman	33:08	▶
Lingle Wang (FEP+)	35:36	▶	Léa El Khoury	23:48	▶
Greg Bowman	33:23	▶	Christopher Bayly	33:11	▶
Ayşegül Özen	26:00	▶	Simon Olsson	17:03	▶
Charlie Brooks	30:56	▶	Antonia Mey	13:28	▶
John Chodera (welcome)	6:24	▶	Maxwell Zimmerman	17:01	▶
Chris Chipot	32:12	▶	Fabian Paul	15:34	▶
Jana Shen (constant-pH)	31:42	▶	Justin Porter	20:05	▶
Vince Voelz	36:07	▶	Jordi Juarez-Jimenez	13:56	▶
Chia-En Chang	31:45	▶	Bryn Taylor	17:13	▶
Rebecca Wade	32:26	▶	Zhifeng Jing	13:51	▶
Mike Schnieders (welcome)	10:48	▶	Monday session stream	1:21:25	▶
Gianni de Fabritiis	30:10	▶	Jean-Philip Piquemal	29:18	▶
JC Gumbart	22:09	▶	Thomas Simonson	30:44	▶
Purushottam Dixit	16:48	▶			

Shirts, Zuckerman, Bowman, Noé, Clementi and Tiwary in one workshop is the densest concentration of Atlas B, D and E in this document. The same channel hosts free energy workshops for 2020 through 2024 that are not enumerated here.

B.5 AMMI Geometric Deep Learning, 2nd edition (2022)

Twelve lectures plus five seminars. **The playlist also contains an unrelated Ukrainian-language political interview as its eighteenth item** — an erroneous addition from a different channel. The course is seventeen videos.

#	Lecture	Lecturer	Len	Link
1	Introduction	Bronstein	1:14:59	▶
2	Learning in High Dimensions	Bruna	1:19:40	▶
3	Geometric Priors I	Cohen	56:48	▶
4	Geometric Priors II	Bruna	53:37	▶
5	Graphs and Sets	Veličković	1:01:37	▶
6	Graphs and Sets II	Veličković	59:12	▶
7	Grids	Bruna	1:05:21	▶
8	Groups and Homogeneous Spaces	Cohen	1:11:51	▶
9	Manifolds	Bronstein	1:14:01	▶
10	Gauges	Cohen	46:54	▶
11	Beyond Groups	Veličković	1:15:33	▶
12	Applications and Trends	Bronstein	1:11:00	▶
S1	Physics-based GNNs	Di Giovanni	1:12:39	▶
S2	Subgraph GNNs	Frasca	1:00:24	▶
S3	Equivariance in ML	Williamson	1:01:52	▶
S4	Neural Sheaf Diffusion	Bodnar	52:51	▶
S5	AlphaFold	Russ Bates	55:50	▶

B.6 Erik Bekkers — Group Equivariant Deep Learning (UvA 2022), 21 videos

The most constructive course on actually building equivariant networks. Three blocks: regular G-CNNs, steerable G-CNNs, and SE(3)-equivariant graph networks. Block 3 is the mathematics behind e3nn, NequIP and SE(3)-Transformers.

#	Lecture	Len	Link
1.1	Introduction	19:04	▶
1.2	Group theory: product, inverse, representations	23:27	▶
1.3	Regular group convolutional neural networks	20:46	▶
1.4	Example	16:38	▶
1.5	A brief history of G-CNNs	16:08	▶
1.6	Group theory: homogeneous and quotient spaces	16:31	▶
1.7	Group convolutions are all you need	23:36	▶
2.1	Steerable kernels and basis functions	22:28	▶
2.2	Revisiting regular G-convs with steerable kernels	15:20	▶
2.3	Group theory: irreducible representations, Fourier	18:35	▶
2.4	Group theory: induced representations, feature fields	8:57	▶
2.5	Steerable group convolutions	19:22	▶
2.6	Activation functions for steerable G-CNNs	11:33	▶
2.7	Deriving harmonic networks from regular G-convs	20:14	▶
3.1	Motivation for SE(3) equivariant graph NNs	18:43	▶
3.2	Equivariant message passing as non-linear convolution	33:08	▶
3.3	Tensor products as conditional linear layers	12:34	▶
3.4	SO(3) irreps, Wigner-D, Clebsch-Gordan	34:17	▶
3.5	Literature survey: 3D steerable graph NNs	48:43	▶
3.6	Literature survey: regular equivariant graph NNs	43:56	▶
3.7	Gauge equivariant graph NNs	31:40	▶

B.7 Stanford CS224W — Machine Learning with Graphs (2021), 60 videos

Full enumeration available in the playlist; the lectures most relevant to this curriculum are listed here. **Lecture 18 is Marinka Zitnik's 81-minute guest lecture on GNNs in computational biology and is the single most relevant video in the course.**

#	Lecture	Len	Link
1.1	Why Graphs	11:55	▶
2.1–2.3	Traditional feature-based methods (node, link, graph)	27:30 / 16:47 / 20:10	▶ ▶ ▶
3.1–3.3	Node embeddings; random walks; embedding whole graphs	14:44 / 27:07 / 18:04	▶ ▶ ▶
6.1–6.3	Introduction to GNNs; deep learning basics; DL for graphs	10:31 / 29:31 / 35:41	▶ ▶ ▶
7.1–7.3	A general perspective on GNNs; a single layer; stacking layers	5:51 / 40:09 / 18:11	▶ ▶ ▶
8.1–8.3	Graph augmentation; training GNNs; prediction tasks	27:50 / 40:19 / 17:49	▶ ▶ ▶
9.1–9.2	How expressive are GNNs; designing the most powerful GNNs	25:21 / 31:52	▶ ▶
15.1–15.4	Deep generative models for graphs; GraphRNN; scaling; applications	15:44 / 24:52 / 15:29 / 13:35	▶ ▶ ▶ ▶
16.1–16.4	Limitations of GNNs ; position-aware; identity-aware; robustness	11:10 / 12:41 / 20:01 / 22:39	▶ ▶ ▶ ▶
17.1–17.4	Scaling up GNNs; GraphSAGE sampling; Cluster-GCN; simplifying	14:51 / 16:50 / 19:18 / 18:40	▶ ▶ ▶ ▶
18	GNNs in Computational Biology (Zitnik)	1:21:19	▶
19.1–19.3	Pre-training GNNs; hyperbolic embeddings; GNN design space	20:07 / 31:47 / 18:10	▶ ▶ ▶

The remaining lectures — PageRank (4.x), message passing and node classification (5.x), knowledge graphs (10.x–11.x), subgraph matching (12.x), community detection (13.x) and random graph models (14.x) — are in the playlist and are good, but are the least relevant to molecular work.

B.8 Stanford CS236 — Deep Generative Models (Autumn 2023), 18 lectures

The canonical unified treatment: autoregressive models, VAEs, normalizing flows, GANs, energy-based models and score-based diffusion in one taxonomy. This is the course that lets you place any new protein generative model.

#	Lecture	Len	Link
1	Introduction	57:28	▶
2	Background	1:20:09	▶
3	Autoregressive Models	1:21:24	▶
4	Maximum Likelihood Learning	1:23:52	▶
5	VAEs	1:21:02	▶
6	VAEs	1:22:01	▶
7	Normalizing Flows	1:23:02	▶
8	Normalizing Flows	1:22:58	▶
9	GANs	1:18:40	▶
10	GANs	1:27:30	▶
11	Energy Based Models	1:24:48	▶
12	Energy Based Models	1:23:15	▶
13	Score Based Models	1:21:56	▶
14	<i>(titled “Energy Based Models” — almost certainly mislabeled; sits in the score-based block)</i>	1:25:03	▶
15	Evaluation of Generative Models	1:21:34	▶
16	Score Based Diffusion Models	1:09:37	▶
17	Discrete Latent Variable Models	1:13:35	▶
18	Diffusion Models for Discrete Data	1:00:03	▶

Lecture 18 is the direct theoretical foundation for EvoDiff, DPLM and the discrete half of Multiflow in Atlas Q.2 and R.2.

B.9 WESTPA — weighted ensemble

Honest finding: only one official recorded series exists, from 2015. There is no newer WESTPA workshop, bootcamp or tutorial series on YouTube. For a method this important to Atlas D.4 that is a real gap in the public record.

Talk	Speaker	Len	Link
Introduction to WESTPA	Matt Zwier	48:44	▶
Non-spatial systems	Rory Donovan	29:50	▶
Molecular scale systems	Adam Pratt	47:25	▶
Analysis tools	Ali Sinan Saglam	34:29	▶

Individual talks that fill the gap, including two by Zuckerman himself:

Talk	Speaker	Len	Link
Introduction to Weighted Ensemble Simulation	Dan Zuckerman	31:14	▶
Recent Advances in Weighted Ensemble Simulation (2020)	Dan Zuckerman	35:59	▶
Zuckerman seminar (TCBG UIUC, 2020)	Dan Zuckerman	1:08:03	▶
HITS Colloquium on weighted ensemble simulations	Lillian Chong	41:54	▶
Chong seminar (TCBG UIUC, 2020)	Lillian Chong	56:39	▶
Starting your own WE simulation (walkthrough)	Shirley Ahn	27:09	▶

B.10 PyEMMA and Markov state models — 29 videos

Channel: **MarkovModel** (Noé group). Three workshop years, roughly 17.7 hours. **The 2017 edition is the most complete and is the one to work through.**

PyEMMA 2017 — 13 videos:

#	Topic	Len	Link
1	Overview	1:09:18	▶
2	The software	20:08	▶
3	Featurization and TICA	36:16	▶
4	MSM theory	1:00:32	▶
5	MSM estimation	51:04	▶
6	MSM analysis	27:17	▶
7	Coarse graining	20:11	▶
8	Hidden Markov models	31:59	▶
9	Transition path theory	41:23	▶
10	Enhanced sampling (<i>title typo: “PEMMA”</i>)	1:14:18	▶
11	TRAM and MEMM theory	1:27:17	▶
12	TRAM and MEMM hands-on	41:27	▶
13	molPX: interactive molecular projection visualization	21:35	▶

PyEMMA 2018 — 11 videos, with better MSM validation coverage: Introduction [▶](#) · The software [▶](#) · MSM theory [▶](#) · **MSM estimation and validation** [▶](#) · Analysis [▶](#) · **PCCA+ + coarse graining** [▶](#) · The variational approach [▶](#) · **VAMP score model selection** [▶](#) · VAMPnets [▶](#) · HMMs [▶](#) · Multi-ensemble Markov models [▶](#)

PyEMMA 2021 — 5 videos, the most current framing: Introduction [▶](#) · MSM theory [▶](#) · **VAMP theory** [▶](#) · HMMs [▶](#) · VAMPnets [▶](#)

No deeptime workshop recordings exist, despite deeptime being PyEMMA's successor. If you use the current library, the 2021 PyEMMA lectures are the closest recorded instruction available.

Appendix C — The Gaps Register

A list of what this curriculum could not find, stated plainly, because a curriculum that only lists what exists teaches you that the public record is complete. It is not, and the shape of the absence is informative.

C.1 Scientists with no recorded talks

These people are central to the field and essentially unavailable on video. Their ideas reach you through papers, through students, or not at all.

Person	Why it matters	Substitute
Tanja Kortemme	One of your ten named scientists; the biosensor and conformational-change design line	Anum Glasgow, BPDMC 2021 (Appendix A.2 #58); the papers in Atlas R.8
Phil Bradley	One of your ten; TCRdock and the TCR-pMHC modeling line	Bradley 2023, <i>eLife</i> 12:e82813; Broad's TCR framework talk (Atlas R.7)
Dan Herschlag	RNA catalysis and the quantitative-enzymology tradition	His papers; the HT-MEK talk (Appendix A.1 #34)
Jianlin Su	RoPE, now in essentially every transformer	CS336 2026 Lecture 3 (D.4.3)
Jordan Hoffmann	Chinchilla — the scaling result everyone cites	CS336 2026 Lectures 9 and 11 (D.5.4)
Keller Jordan	Muon, now widely deployed	Bernstein's <i>Metrized Deep Learning</i> (D.5.3) is the theory behind it

C.2 Methods with no talk at all

Each of these has a paper and no recorded presentation anywhere. Six of the nine design methods are from industry labs or non-US groups, which is the pattern.

Design (Atlas R): ProteinSolver · PiFold · ProstT5 · ProteinSGM · PocketGen · AlphaProteo (DeepMind) · Kortemme biosensors · theozyme/RosettaMatch · membrane protein design.

Language and architecture (Atlas Q, D.4): YaRN · Byte Latent Transformer · H-Net dynamic chunking · DeepSeek Multi-head Latent Attention · normalization placement as a research question.

Efficiency (D.5, D.9): GPTQ · QuaRot · BitNet b1.58 · Medusa · Wanda pruning · *LoRA Learns Less and Forgets Less*.

Evaluation and reasoning: self-consistency (Xuezhi Wang) · *Is DPO Superior to PPO* · Math-Shepherd · Apple's *Illusion of Thinking* · Keskar on the large-batch generalization gap.

C.3 The negative results are the least-recorded of all

This is the most important entry in this appendix.

Result	What it showed	Talk?
Blomberg et al. 2013, <i>Nature</i> 503:418	The celebrated designed Kemp eliminase's catalysis came from directed evolution, not design	None
Pancotti et al. 2022	Most $\Delta\Delta G$ predictors fail the anti-symmetry test	None
Fu et al. 2023, TMLR	ML force fields with excellent force MAE produce unstable simulations	One (Valence)
Rosta & Hummer 2009	REMD's efficiency gain is smaller than usually claimed	None
Schaeffer et al. 2023	Emergent abilities are partly a metric artifact	One (Imbue)
Kapoor & Narayanan 2023	Data leakage across 294 papers in 17 fields	None found

Compare: RFdiffusion has at least eight recorded talks across four venues. **The asymmetry is a property of how science is communicated, not of what is true.** Compensating for it — deliberately seeking the refutation of every result you are excited about — is the single most useful habit this document can give you, and it is the reason every Atlas family pairs its advocacy talks with their counterpoints rather than collecting skepticism in a separate chapter.

C.4 Courses that exist but are not recorded

- **Princeton COS 597 (Danqi Chen, LLMs)** — slides and readings public, no video. Substitute: her two Simons talks (D.4.5, D.5.5).
- **MIT 6.8300 / 6.8301 Advances in Computer Vision** — materials web-only. Substitute: CS231n 2025 and EECS 498 (D.3 §8).
- **CMU 16-824 Visual Learning and Recognition** — slides public, no lectures.
- **Stanford CS279 / CS273B** — Canvas-only, not public.
- **MIT 7.51, 7.88J, 18.335, 6.441** — no video.
- **Stanford CS229M Lecture 12** and **MIT 9.520 Fall 2019 Class 24** — gaps inside otherwise complete recorded courses.

C.5 Whole eras that were never recorded

- **BPDMC 2018 and 2019** — 28 talks listed with speakers and titles, none recorded. The channel's oldest upload is January 2021.
- **MLSB before 2025** — the workshop ran from 2020 but only the 2025 edition is on its channel.
- **Rosetta Bootcamp Lecture 01 (2016)** — missing from both the playlist and the IPD page.

If you ever run a seminar, record it. The 2018–2019 BPDMC archive is a list of titles describing talks that are simply gone, and that is the state of most of this field's institutional memory from before 2020.

C.6 Where attribution could not be confirmed

Nine ML4PE talks posted after the schedule page froze on **4 November 2025** name no speaker anywhere in the public listing. They are marked *speaker not named* in Appendix A.1 rather than attributed to a guessed first author. Four computer vision entries and several RosettaCommons tutorials are likewise institutional recordings with no named presenter.

Three other things worth knowing about the record itself:

- The @ValenceLabs handle is a dead legacy channel with zero videos. The live channel is @valence_labs.
 - The Institute for Protein Design’s own YouTube channel has **seven** videos. Its actual teaching corpus is embedded on `ipd.uw.edu/learn` and pulled from several other channels — it does not surface in a channel listing.
 - Roughly 22 of the videos in Part III return HTTP 401 from YouTube’s oEmbed endpoint. That means **embedding is disabled**, not that the video is dead; each was confirmed separately. Stanford Online disables embedding on much of its catalogue.
-

Appendix D — The Progress Tracker

This document is years of material. The tracker is one page because a tracker you will not maintain is worse than none.

D.1 The only three things to record

For each item you complete, write **three lines** in a single file:

```
2026-10-11 Atlas B.5 Shirts, error estimation (1:02)
  Claim I can now defend: block averaging gives the error bar; RMSD plateau does not.
  Gap I noticed: nobody reports effective sample size per observable. Checkable on my own
trajectories.
```

The second line is what converts watching into knowing. The third line is where your research program comes from. **After six months you will have a hundred “gap” lines, and five of them will be real.**

D.2 The checkpoint ledger

Seventy-three derivation checkpoints are specified in Part V. Track them as a flat list with three states: *not started*, *attempted*, *can reproduce from blank paper*. Only the third counts.

Family	Checkpoints	Where
B.4 Chemistry	1–6	Module B.4
Atlas B — MD	7–11	Atlas B.7
Atlas A — Potentials	12–17	Atlas A
Atlas L / N — Classical, evolution	18–20, 47–48	Atlas L.5, N.5
Atlas D — Enhanced sampling	21–25	Atlas D.7
Atlas E — Free energy	26–28	Atlas E.5
Atlas F / G — Coarse-graining, mesoscale	29–34	Atlas F, G.1
Atlas H / I — Polymer, dynamics	35–39	Atlas H, I
Atlas K — Integrative	40–41	Atlas K.1, K.2
Atlas J / M — Determination, docking	42–43	Atlas J, M
Atlas Q — Sequence models	44–46	Atlas Q.7
Atlas O — Geometric DL	50–52	Atlas O.4
Part III — AI	56–59	D.6, D.7
Atlas S — Benchmarking	60	Atlas S.4
Atlas R — Design	61–63	Atlas R.11
Part IV — Performance	64–70	Part V, C.1
Atlas T — RNA	71–73	Atlas T.11

D.3 The eleven capstones

Specified in Part V. **Do not start one until you have completed the BUILD block of the family it belongs to**, because every capstone is the BUILD done properly and at publication scale. Three of them — the over-optimization curve (III), the equivariance crossover (II), and the SAE-with-ground-truth study (VIII) — are stated concretely enough in D.10 that you could begin this month.

D.4 The honest review, every six months

Four questions, written answers, thirty minutes:

1. **What can I now derive that I could not six months ago?** If the answer is nothing, you have been watching rather than working.
2. **Which of my recorded gaps is still open?** Check the literature. Some will have been published, and knowing which is itself information about how fast that subfield moves.
3. **What did I believe six months ago that I no longer believe?** If nothing, you have not engaged with the counterpoints.
4. **What would I do with three months and a GPU?** The answer should get more specific every time. When it is specific enough to write a methods section, it is a project.

Appendix E — Index of Methods, Software and Systems

Every named method, model, package and system that appears in at least two places, with the sections that discuss it. **411 entries**. Generated from the assembled text at build time, so it cannot drift out of date.

Section labels: **Atlas A–Atlas S** are the method families in Part II; **B.n** are the foundation modules in Part I; **D.n** are the AI sections in Part III; **DLT-n** are the deep learning theory sections; **E.n** are the performance engineering sections in Part IV.

A

- AbDiffuser — A.1, R.7, Appendix E
- AFDB — D.6.6, D.6.7, N.4, Q.6, Appendix E
- AIMNet2 — A.4, A.6, Appendix E
- AlphaFlow — A.8, D.6, D.8, P6, P11, Appendix E
- AlphaFold — A.0, A.1, A.3, A.7, A.8, B.1, B.2, B.5, C.1, C.2, D.2, D.4.3, D.5, D.5.1, D.6, E.0, E.1, E.2, E.7.1, E.7.2, E.7.6, E.8, H.3, H.6, J.2, K.0, K.1, K.3, L.0, L.3, L.4, M.6, O.2, P1, P4, P6, P8, P9, P11, R.6, Appendix E
- AlphaFold 2 — A.2, A.3, A.4, A.8, E.7.1, E.7.2, E.7.6, O.5, P0, P1, P2, P5, P9, S.1, S.5
- AlphaFold 3 — A.1, A.2, A.3, C.2, E.7.1, O.3, O.5, P4, R.10, S.5
- AlphaMissense — A.4, Q.3, Q.8, Appendix E
- AlphaProteo — A.9, C.2, R.4, Appendix E
- AMBER — A.1, B.3, E.6.2, E.6.8, Appendix E
- AMD — E.4.3, E.4.4, Appendix E
- AMMI — B.5, D.2, O.1, O.5, Appendix E
- AMOEBA — A.1, A.6, B.6, E.1, Appendix E
- ANN — B.3.8, B.3.9, B.3.11, Appendix E
- APBS — A.6, G.6, Appendix E
- APS — J.4, DLT-8, Appendix E
- AquaGen — A.8, B.3, E.7.4, Appendix E
- ARC — D.7.7, D.9, Appendix E
- ARES — Q.5, Q.8, S.5, Appendix E
- ASBMB — I.9, J.5, S.5, Appendix E
- ASIC — E.0, E.5.5, E.5.6, Appendix E
- ASPLOS — E.4.7, E.4.11, E.5.2, Appendix E
- AtomWorks — A.1, A.8, E.7.1, P3, R.9, Appendix E
- ATPESC — E.4.1, E.6.4, E.6.5, Appendix E
- AutoDock — M.1, M.6, Appendix E

B

- BAR — E.2, E.5, E.6, Appendix E
- BDA — C.6, C.8, Appendix E
- BERT — A.6, D.3, Appendix E
- BigHat — A.1, Q.1, Appendix E
- BindCraft — A.2, A.3, A.8, R.3, R.12, Appendix E
- BinderFlow — A.3, R.4, Appendix E
- BioCAT — J.2, J.4, J.7, K.1, Appendix E
- BioEmu — A.8, D.6, D.8, P6, P11, Appendix E
- BLAST — B.3.10, N.1, N.5, Appendix E

- BMC — G.2, L.3, Appendix E
- BoltzDesign — A.1, A.2, R.3, Appendix E
- BoltzGen — A.3, A.9, R.3, Appendix E
- BrownDye — G.1, Appendix E
- ByteDance — A.1, A.8, D.3, E.7.1, P4, Q.2, Appendix E

C

- CACM — E.5.4, E.6.4, Appendix E
- CAFA — A.1, Q.6, S.1, Appendix E
- CALVADOS — G.5, G.7, I.9, Appendix E
- CAPRI — C.4, M.2, M.6, S.1, Appendix E
- CARE — A.1, N.2, Q.6, Q.7, Q.8, S.1, S.5, Appendix E
- CASP — A.2, A.4, A.5, A.7, C.2, C.4, D.5.5, D.10.2, E.1, G.4, I.2, J.2, J.7, K.1, L.3, P1, P3, P5, P11, Q.5, R.4, S.1, S.2, S.5, Appendix E
- CASP15 — P3, P5, P6, P8, S.5, Appendix E
- CASP16 — P8, S.5, Appendix E
- CATH — C.2, D.10.1, Appendix E
- CBMM — D.2, D.3, D.9, Appendix E
- CCDC — I.2, M.1, Appendix E
- CHARMM — A.1, B.1, B.3, B.4, B.6, I.7, Appendix E
- CHARMM36 — A.6, B.3, B.9, Appendix E
- ChatGPT — A.6, D.7.7, D.9, Appendix E
- CHEAP — A.1, A.8, Q.4, Appendix E
- ChiENN — A.8, O.2, Appendix E
- CIRM — DLT-2, DLT-7, DLT-8, Appendix E
- CLEAN — A.1, Q.6, Q.8, Appendix E
- CLSP — D.6.6, D.7.6, D.9, Appendix E
- CME295 — D.2, D.9, Appendix E
- CMOS — E.5.6, E.5.7, Appendix E
- CNRS — E.1, I.1, M.4, DLT-3, Appendix E
- CodeRefinery — B.3.7, B.3.9, B.3.11, E.6.6, Appendix E
- ColabDesign — A.2, R.3, Appendix E
- ColabFold — A.2, C.2, E.0, E.7.6, E.8, L.3, N.1, P5, Appendix E
- COVID — E.1, M.4, R.5, S.5, Appendix E
- CrankPep — M.1, M.6, Appendix E
- CRG — A.1, G.3, I.1, Appendix E
- CRISPR — A.1, A.4, Q.2, S.5, Appendix E
- CRUNCH — F.1, DLT-4, Appendix E
- CryoEM — J.1, K.1, Appendix E
- CSHL — A.6, Q.5, Appendix E
- CSIC — J.1, M.2, Appendix E

- CSML — D.3, DLT-3, DLT-4, DLT-6, Appendix E
- CTF — J.1, J.6, Appendix E
- CTO — E.5.4, E.6.6, E.7.5, Appendix E
- CUDA — B.3.3, B.3.9, B.3.10, D.5.2, E.3, E.4.1, E.4.2, E.4.3, E.4.7, E.4.10, E.4.11, Appendix E
- CuTe — E.4.3, E.4.5, E.4.11, Appendix E
- CUTLASS — D.5.1, D.5.6, E.4, E.4.3, E.4.5, E.4.7, E.4.10, E.4.11, Appendix E
- CVF — D.2, D.3, Appendix E

D

- DAMMIN — J.4, J.7, Appendix E
- DART — A.6, Q.5, Q.8, S.1, Appendix E
- DCA — B.2.1, B.2.7, B.3.11, C.1, L.5, N.3, Appendix E
- DDPM — B.2.7, D.2, D.3, Appendix E
- DEE — B.3.11, C.8, Appendix E
- DeepEMhancer — J.1, J.7, Appendix E
- DeepMath — D.9, DLT-1, DLT-3, DLT-4, DLT-7, Appendix E
- DeepSeek — C.2, D.6.4, D.7.3, Appendix E
- DeepSpeed — D.5.2, D.5.6, E.7.2, E.8, Appendix E
- DeGrado — R.1, R.8, R.12, Appendix E
- DevLab — E.5.2, E.5.3, Appendix E
- DevOps — A.2, E.7.1, Appendix E
- DFT — A.3, A.5, A.6, Appendix E
- DiffDock — A.2, A.3, A.8, P7, P11, Appendix E
- DiMaio — A.7, K.1, K.2, K.4, S.5, Appendix E
- DLT — D.8, DLT-1, DLT-2, DLT-3, DLT-4, DLT-5, DLT-6, DLT-7, DLT-8, DLT-9, DLT-10, Appendix E
- DMS — B.4.7, I.9, Q.7, S.5, Appendix E
- DoReMi — D.5.5, D.5.6, D.10.1, Appendix E
- DPD — F.3, G.6, G.7, Appendix E
- DPLM — A.1, B.8, D.4.6, Q.2, Appendix E
- DPO — C.2, D.6.2, D.6.3, D.6.7, Appendix E
- DSIR — D.5.5, D.5.6, Appendix E
- DSL — E.4.3, E.4.5, Appendix E

E

- ECE408 — B.3.3, B.3.11, Appendix E
- ECE508 — B.3.3, E.4.2, Appendix E
- ECP — E.6.6, E.6.7, Appendix E
- EECS — C.4, D.2, D.3, D.6.1, DLT-4, Appendix E
- EfficientML — B.3.4, D.2, Appendix E
- EGFR — R.4, S.1, Appendix E
- EGNN — O.2, O.4, O.5, Appendix E

- ELLIS — D.3, DLT-3, DLT-4, DLT-6, Appendix E
- EMBL — I.7, J.4, S.5, Appendix E
- ENS — B.2.4, C.1, C.4, H.4, DLT-3, DLT-8, Appendix E
- ESM — Q.1, Q.7, R.1, Appendix E
- ESM3 — A.1, N.3, Q.1, Q.8, Appendix E
- ESMFold — A.3, P5, Q.1, Q.8, Appendix E
- ESS — C.6, C.8, Appendix E
- EternaFold — Q.5, S.5, Appendix E
- EVA — B.1.2, C.6, Appendix E
- EVE — Q.3, Q.7, Appendix E
- EvoDiff — A.1, B.8, D.4.6, D.4.8, Q.2, R.2, R.12, Appendix E
- EvolutionaryScale — A.1, A.6, Appendix E

F

- FAB — C.5, C.8, Appendix E
- FastFold — E.7.1, E.8, Appendix E
- FAU — B.3.3, E.4.6, E.4.8, E.6.4, E.6.5, E.6.9, Appendix E
- FCAI — D.3, D.9, Appendix E
- FCS — Q.3, Q.8, S.5, Appendix E
- FEAT — A.8, E.2, E.6, Appendix E
- FEBS — B.4.6, B.4.7, Appendix E
- FEP — B.2.7, B.4, E.1, E.2, E.3, E.6, M.4, M.6, Appendix E
- FFT — C.1, J.1, Appendix E
- FineWeb — D.5.5, D.5.6, Appendix E
- FlashAttention — B.3.11, C.1, D.5, D.5.1, D.5.6, D.10.1, E.0, E.2, E.4.3, E.4.5, E.4.11, E.7.2, E.8, Appendix E
- FLOP — B.3.11, E.4.1, Appendix E
- FODSI — DLT-3, DLT-4, DLT-6, Appendix E
- FoldFlow — A.1, A.8, R.2, R.12, Appendix E
- FPGA — E.5.6, E.5.7, Appendix E
- FrameDiff — R.2, R.12, Appendix E
- FrameFlow — A.6, Q.5, R.2, Appendix E
- FRET — H.3, H.6, Appendix E
- FSC — J.1, J.6, Appendix E
- FSDP — D.5.2, E.7.5, E.8, Appendix E
- FSDP2 — E.4.3, E.7.5, Appendix E
- FuncLib — R.5, R.12, Appendix E

G

- GBSA — C.1, E.1, E.3, E.5, M.5, Appendix E
- GEMM — E.4.2, E.4.3, E.4.11, Appendix E
- GFlowNets — A.8, D.1, Appendix E

- GNN — A.1, A.8, B.7, D.2, R.7, DLT-7, DLT-9, Appendix E
- GoCurator — A.1, Q.6, Appendix E
- GPCR — I.7, R.4, Appendix E
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